

QuantProf

Complete Problems Collection

848 Problems with Hints & Full Solutions

Probability · Combinatorics · Brain Teasers

Citadel · Jane Street · Jump Trading · Optiver · Two Sigma · and more

quantprof.org

Contents

Questions	2
Hints	160
Solutions	317

Questions

1. Non-empty Intersection

Probability Lvl 7/10 (Hard)

Jane Street, Jump Trading

Five random intervals are generated within the interval $[0, 1]$. Each interval is formed by independently selecting two points uniformly at random from $[0, 1]$, and taking the smaller as the left endpoint and the larger as the right endpoint. What is the probability that all five intervals share a common (non-empty) intersection?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

2. Standing Table II

Probability Lvl 8/10 (Hard)

Citadel, Jane Street

Five points are chosen independently and uniformly at random on the circumference of a circular table. Legs are attached at those five points. What is the probability that the table stands?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

3. Game Of Chords

Brainteasers Lvl 5/10 (Medium)

Citadel

Forty points are placed evenly around the circumference of a circle. Alice and Bob take turns drawing chords between two distinct points. A new chord must not intersect any existing chord in the interior of the circle, though it may meet other chords at points on the circumference. Alice moves first. The player who cannot draw a legal chord on their turn loses. Under optimal play which player has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

4. Unequal Neighbors

Brainteasers Lvl 10/10 (Very Hard)

Susquehanna International Group

Given an $m \times n$ grid of random integers, each cell can have its value increased by at most 1. A cell is considered a neighbor of another if they share a side (not diagonally connected). Determine whether it is possible to modify any grid so that no cell has the same value as any of its neighbors. If possible, explain how this can be achieved. If not, provide an example of a grid where this transformation is impossible.

Enter 1 if it is possible and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

5. Find The Units Digit

Brainteasers Lvl 8/10 (Hard)

Citadel

Consider every 100-digit number (including those with leading zeros), i.e. all integers from 0 to $10^{100} - 1$. For each such number, compute the product of its nonzero digits (and by convention treat the product for the number zero as 1). Sum these products over all 10^{100} numbers. What is the last decimal digit of the sum?

Note: For the numbers 0 to 9, this sum (of product of non-zero digits) is 46 (and not 45), as the number 0 contributes 1 to the total since, by convention, we treat the product for the zero as 1.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

6. Ship Destroyer

Probability Lvl 4/10 (Medium)

Citadel

Sixteen ships occupy the cells of a 4×4 grid. You have four bombs to drop, each landing independently and uniformly at random in one of the 16 cells (with repetition allowed). Whenever a bomb lands in a particular cell, it destroys every ship in that bomb's row and its column. What is the probability that, after all four bombs have fallen, every ship on the grid has been destroyed?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

7. Moving The Piece I

Brainteasers Lvl 7/10 (Hard)

Citadel, Optiver

Given a grid with 183 rows and 181 columns, Alice and Bob take turns moving a piece to an adjacent square (diagonally connected squares are also considered adjacent). Once a square is visited, it cannot be visited again. The piece starts on the top-left square, which cannot be revisited either. The loser is the player that does not have a legal move to make. If Alice goes first, which player has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

8. Even Marbles Everywhere

Probability Lvl 6/10 (Medium)

Citadel, Two Sigma

Each square on an 8×8 grid is randomly filled with a black or white marble. What is the probability that the number of white marbles in EVERY row and EVERY column is even?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

9. Hard Grid Arrangement

Combinatorics Lvl 8/10 (Hard)

Akuna Capital

Consider an $n \times n$ grid where each cell can be filled with either 0 or 1. How many such arrangements are there such that the sum of the cells in every 2×2 sub-grid is equal to 2? Express your answer as a closed-form formula in terms of n .

Enter the answer for $n = 10$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

10. All Odds Before Even

Probability Lvl 3/10 (Easy)

Citadel, IMC Trading, Susquehanna International Group

You are rolling a fair 6-sided die repeatedly. What is the probability that all odd values (1, 3, 5) appear before any even value (2, 4, 6)?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

11. Standing Table I

Probability Lvl 7/10 (Hard)

Akuna Capital, Citadel, Five Rings, Jane Street

A circular disk is supported by three legs attached to its circumference. What is the probability that the table remains upright?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

12. 10 Consecutive Heads

Probability Lvl 5/10 (Medium) *Citadel, IMC Trading, Jane Street, Susquehanna International Group, Two Sigma*

What is the expected number of flips required to obtain 10 consecutive heads when flipping a fair coin?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

13. Chords Intersection I

Probability Lvl 5/10 (Medium)

Five Rings, Hudson River Trading, Jump Trading

If 10 chords are drawn on a circle with their endpoints chosen uniformly at random along the circumference, what is the expected number of intersections?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

14. 1 Leftward Jump

Probability Lvl 6/10 (Medium)

Jane Street

A frog began its journey on the first stone in a sequence of seven stones, numbered 1 through 7 from left to right. With each move, the frog hopped to a stone it hadn't already landed on. What is the probability that after touching all the stones, the frog made precisely one jump to the left during its hops?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

15. Marble Swap II

Probability Lvl 4/10 (Medium)

Citadel

You have 3 marbles: red, blue, and red, placed in that order. In each move, you randomly select two distinct marbles and swap them. After 10 moves, what is the probability that the marbles are back in their original order (red, blue, red)?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

16. Scary Permutation

Combinatorics Lvl 8/10 (Hard)

Citadel

How many permutations of the numbers $\{1, 2, \dots, 9\}$ are there such that every number after the first has at least one number to its left with a difference of exactly 1?

In other words, for each number from the 2nd position onward, there exists at least one number to its left where the absolute difference is 1. This preceding number need not be immediately adjacent; it only needs to appear somewhere earlier in the sequence.

For example, the permutation 4, 5, 6, 7, 3, 8, 2, 1, 9 is valid because:

- 5 has 4 to its left ($|5 - 4| = 1$)
- 6 has 5 to its left ($|6 - 5| = 1$)
- 7 has 6 to its left ($|7 - 6| = 1$)
- 3 has 4 to its left ($|3 - 4| = 1$)
- 8 has 7 to its left ($|8 - 7| = 1$)
- 2 has 3 to its left ($|2 - 3| = 1$)
- 1 has 2 to its left ($|1 - 2| = 1$)
- 9 has 8 to its left ($|9 - 8| = 1$)

Conversely, the permutation 4, 8, 1, 2, 3, 5, 6, 7, 9 is invalid because 8, which appears in the 2nd position, does not have any preceding number with a difference of 1.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

17. Grid Frog I

Combinatorics Lvl 8/10 (Hard)

Citadel

A 2×8 grid has one frog in each of its 16 cells. Simultaneously, every frog must jump to one of its neighboring cells (only cells sharing a side count; diagonal moves are not allowed). After all jumps, exactly one frog must occupy each cell again. In how many different ways can all the frogs move so that no two frogs land in the same square?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

18. At Least Six I

Probability Lvl 4/10 (Medium)

Citadel

You roll a fair six-sided die repeatedly, keeping a running total of the rolls. Let T be the number of rolls needed for this running total to reach or exceed 6. What is the expected value $E[T]$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

19. Game Of Placing Knights

Brainteasers Lvl 5/10 (Medium)

Graviton, Susquehanna International Group

Alice and Bob take turns placing knights on an 8×8 chessboard. A knight cannot be placed on a square that is already occupied, nor can it be placed on a square that is attacked by an existing knight. A player loses if they have no legal move left—meaning there are no safe, unoccupied squares available. Alice moves first. Who has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

20. Same Arc I

Probability Lvl 8/10 (Hard)

Citadel

N points are selected independently and uniformly at random on the circumference of a circle. What is the probability that all N points lie within some semicircular arc of the circle, given that $N \geq 2$?

Enter the answer for $N = 10$ rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

21. Increasing Dice Rolls III

Probability Lvl 1/10 (Easy)

N/A

A fair six-sided die is rolled twice, and the two outcomes are recorded in order. What is the probability that the number obtained on the second roll is strictly larger than the number obtained on the first roll?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

22. Fill In The Blanks

Probability Lvl 4/10 (Medium)

DRW

Consider the sequence 1.2.3.4.5.6.7.8.9, where each blank can be filled with either a + or a − with equal probability ($\frac{1}{2}$). What is the probability that the final result, after performing all the operations, equals 0?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

23. Exponent Perfect Square

Brainteasers Lvl 3/10 (Easy)

Graviton

How many integers p between 100 and 400, inclusive, satisfy the condition that p^p is a perfect square?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

24. Odd Heads II

Probability Lvl 4/10 (Medium)

Akuna Capital, Citadel, Jane Street

You toss 100 coins. 99 of them are biased and show heads with a probability p , where $0 \leq p \leq 1$. The remaining one coin is fair. What is the probability of getting an odd number of heads in terms of p ?

Enter the answer for $p = 0.10$ rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

25. Dice Rolls Product

Probability Lvl 10/10 (Very Hard)

N/A

A fair six-sided die is rolled 10 times. For each face $i \in \{1, 2, \dots, 6\}$, let N_i be the number of times side i appears. Compute the expectation $\mathbb{E}[N_1 N_2 N_3 N_4 N_5 N_6]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

26. Jumping Frog V

Combinatorics Lvl 4/10 (Medium)

DRW, Five Rings, Optiver

A frog begins at point 0 and must reach point 10. The frog can only jump more than 1 point at a time. How many unique ways can the frog travel from point 0 to point 10?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

27. Dot Product

Probability Lvl 5/10 (Medium)

Citadel

Suppose you have two 10-dimensional vectors. Each entry in the first vector is randomly chosen to be 0 or 1 with equal probability (i.e., 50% chance of being 0 and 50% chance of being 1). In the second vector, each entry is 1 with probability $\frac{3}{4}$ and 0 with probability $\frac{1}{4}$, independently. What is the probability that the dot product of these two vectors is odd?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

28. Marble Swap I

Probability Lvl 6/10 (Medium)

Hudson River Trading

You have 3 marbles: red, blue, and green, placed in that order. In each move, you randomly select two distinct marbles and swap them. After 125 moves, what is the probability that the marbles are back in their original order (red, blue, green)?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

29. Monster Escape I

Probability Lvl 7/10 (Hard)

Hudson River Trading

Mario starts at the bottom-left corner of a 100×100 grid, and a monster starts at the bottom-right corner. Every second, both Mario and the monster simultaneously move to one of the four neighboring squares (up, down, left, or right - diagonal moves are not allowed). If, at any point, they occupy the same square, Mario is caught and dies. What is the probability that Mario survives for 1000 seconds without ever landing on the same square as the monster?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

30. Chords Forming Quadrilateral

Probability Lvl 3/10 (Easy)

N/A

Given 8 points on a circle, you draw all possible chords that connect each pair of these points. Then, 4 chords are randomly selected. What is the probability that these 4 chords form a convex quadrilateral?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

31. Monster Escape II

Probability Lvl 5/10 (Medium)

Optiver

Mario starts at $(0,0)$ and a monster starts at $(4,4)$. Every second, Mario moves either up or right with equal probability, but if he reaches the border of the 4×4 grid, he moves in the only direction that keeps him within the border. Similarly, the monster moves down or left, with the same border constraint. What is the probability that Mario and the monster meet before reaching the opposite corners?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

32. Matrix Attack III

Probability Lvl 9/10 (Very Hard)

Hudson River Trading

Let $A = (a_{ij})$ be a 7×7 matrix of integers satisfying:

- Column-permutation condition: Each column contains the numbers $1, 2, \dots, 7$ exactly once.
- Symmetry condition: $a_{ij} = a_{ji}$ for all i, j .

What is the probability that the main diagonal entries $a_{11}, a_{22}, \dots, a_{77}$ also form a permutation of $1, 2, \dots, 7$; that is, the numbers 1 through 7 appear exactly once on the main diagonal?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

33. Bulbs In A Grid

Probability Lvl 5/10 (Medium)

Jane Street

Consider a 3×3 grid of light bulbs, where each bulb is independently turned on with probability $\frac{1}{2}$. Calculate the probability that no two adjacent bulbs (sharing a common side) are turned on simultaneously.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

34. Chess Tournament III

Probability Lvl 5/10 (Medium)

Citadel

Alice and Bob are competing in a chess tournament featuring 32 participants. Each round of the tournament involves randomly pairing the remaining players. In each match, either player has an equal chance of winning, with the loser being eliminated. What is the probability that Alice and Bob will face each other during the course of the tournament?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

35. Count Paths II

Combinatorics Lvl 4/10 (Medium)

Citadel

How many paths lead from $(0, 0)$ to $(8, 10)$, given that only moves of one unit right or one unit up are allowed, and it is prohibited to pass through points (x, y) where both x and y are odd?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

36. Five Light Bulbs

Probability Lvl 4/10 (Medium)

Jane Street

Five light bulbs are initially all off. Every second, one bulb is chosen uniformly at random and flipped (if it's on it turns off; if it's off it turns on). What is the expected time until all five bulbs are off again?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

37. Letter-envelope Pairing

Probability Lvl 8/10 (Hard)

Jane Street

We have 10 pairs of gloves, with each left glove randomly paired with a right glove. The gloves are numbered from 1 to 10. What is the probability that the gloves are paired such that, for every pair, the absolute difference between the numbers of the left and right gloves is at most 1?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

38. Frog Return

Probability Lvl 4/10 (Medium)

IMC Trading

A grasshopper jumps along a line. Its first jump is 1 cm, the second is 2 cm, and so on. Each jump can be to the right or to the left. What is the probability that the grasshopper returns to its starting position on the 25th jump?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

39. Direction Change I

Probability Lvl 5/10 (Medium)

Citadel

Bob travels from $(0, 0)$ to $(16, 16)$ in the coordinate plane by moving one unit in the positive x or y direction at a time. Assume all possible paths are equally likely. What is the probability that he changes his direction an odd number of times in the entire journey?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

40. 3 Dice Minimum

Probability Lvl 7/10 (Hard)

Jane Street

What is the expected value of the smallest number rolled when three fair 100-sided dice are tossed?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

41. Game On Divisors III

Brainteasers Lvl 5/10 (Medium)

Susquehanna International Group

Alice and Bob play with a pile of 2025 stones. On a turn a player may remove any positive number of stones that is a factor of the current number of stones in the pile. The player who takes the last stone loses. Alice goes first. Who has a winning strategy under optimal play?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

42. Jumping Frog IV

Combinatorics Lvl 4/10 (Medium)

Optiver

A frog begins at point 0. The frog is allowed to jump an odd number of points with each move. How many unique ways can the frog reach point 10 from the starting position?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

43. Dice Game III

Probability Lvl 6/10 (Medium)

Citadel

There are 6 players numbered 1 through 6. Player 1 rolls a die. If he rolls a 1, he wins, and the game ends. If he rolls any other number, the die is passed to the player whose number matches the die roll. That player then rolls the die. The game continues with each player passing the die to the player corresponding to the rolled number. If a player rolls his own number, he wins, and the game ends. Calculate the probability that Player 1 wins.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

44. 105 Pebbles

Brainteasers Lvl 8/10 (Hard)

Jane Street

You are given three piles containing 5, 49, and 51 pebbles, respectively. You are allowed to perform two types of operations: (a) merge any two piles into one or (b) divide a pile with an even number of pebbles into two equal piles. Is it possible to perform a sequence of these operations to achieve a total of 105 piles, each containing exactly one pebble?

Enter 1 if it is possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

45. Missing Pairwise Product

Brainteasers Lvl 3/10 (Easy)

Susquehanna International Group

You have four positive real numbers a, b, c , and d . The products of each pair of these numbers form the set $\{9, 10, 12, 27, 30, x\}$. Determine the value of x .

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

46. Constrained Subsets II

Combinatorics Lvl 6/10 (Medium)

Jane Street

How many subsets of the set $\{1, 2, 3, 4, \dots, 30\}$ have the property that the sum of the elements of the subset is greater than 232?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

47. Fifth Card

Brainteasers Lvl 10/10 (Very Hard)

N/A

You and your friend are presented with a challenge. Five cards are drawn from a standard 52-card deck, and you are allowed to see all five cards. You can arrange the cards in any order and reveal the first four cards face up. The challenge is to arrange the cards in such a way that, based on the four revealed cards, your friend can always determine the fifth card. Before the cards are given to you, you and your friend can discuss a strategy.

Find a strategy that achieves this.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

48. Expected Area

Probability Lvl 1/10 (Easy)

N/A

Two integers X and Y are chosen independently and uniformly from the set $\{1, 2, 3, 4, 5\}$. Note that since the selection is independent, X and Y may be equal. These values serve as the coordinates of a right triangle with vertices at $(0, 0)$, $(X, 0)$, and $(0, Y)$. What is the expected area of this triangle?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

49. Difficult Die Constraints

Probability Lvl 8/10 (Hard)

N/A

You roll a fair six-sided die repeatedly until both a 2 and a 3 have appeared at least once. What is the probability that exactly two 6's occur during this entire process?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

50. Game Of Discarding Cards

Probability Lvl 3/10 (Easy)

Jane Street

A casino offers a card game using a standard 52-card deck. Players turn over two cards at a time: if both cards are black, they go to the dealer's pile; if both are red, they go to the player's pile; if one card is red and the other is black, both are discarded. This process is repeated for 26 pairs until all cards are exhausted. If the player ends up with more cards than the dealer, they win \$10; otherwise, including ties, they receive nothing. What is the fair value of this game?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

51. Normal Inequality IV

Probability Lvl 3/10 (Easy)

Akuna Capital, Five Rings, Hudson River Trading

Let X and Y be independent random variables, each following a standard normal distribution. Determine the probability $P(Y > 4X)$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

52. Ants On A Circle I

Probability Lvl 4/10 (Medium)

Citadel

Nine ants are placed at equal spacing around a circle. Each ant independently chooses clockwise or counterclockwise and then moves at constant speed so that, if uninterrupted, each would make exactly one full revolution in one minute. When two ants meet they instantly reverse direction and continue at the same speed. The ants are indistinguishable, so we only care about which geometric points around the circle are occupied, not which ant is which. What is the probability that after one minute the set of occupied points is exactly the same set of nine starting points?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

53. Ants On A Circle II

Probability Lvl 10/10 (Very Hard)

Citadel

Nine ants are placed at equal spacing around a circle. Each ant independently chooses clockwise or counterclockwise and then moves at constant speed so that, if uninterrupted, each would make exactly one full revolution in one minute. When two ants meet they instantly reverse direction and continue at the same speed. The ants are distinguishable. What is the probability that after one minute every ant is exactly at its own starting point? Here, since the ants are distinguishable, we require every ant to return to its own original position.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

54. Ants On A Circle III

Probability Lvl 10/10 (Very Hard)

N/A

Ten ants are placed at equal spacing around a circle. Each ant independently chooses clockwise or counterclockwise and then moves at constant speed so that, if uninterrupted, each would make exactly one full revolution in one minute. When two ants meet they instantly reverse direction and continue at the same speed. The ants are distinguishable. What is the probability that after one minute every ant is exactly at its own starting point? Here, since the ants are distinguishable, we require every ant to return to its own original position.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

55. Game Of Evaluation

Brainteasers Lvl 6/10 (Medium)

N/A

The numbers 1 through 100 are written in a row. Alice and Bob take turns alternately. On each turn, a player fills an empty space between two consecutive numbers with either a +, -, or x symbol. Once all spaces are filled, the resulting expression is evaluated according to the standard order of operations (i.e., multiplications are evaluated before additions and subtractions). If the final value of the expression is odd, Alice wins, otherwise, Bob wins. If Alice goes first, who has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

56. Ant On A Cube I

Probability Lvl 5/10 (Medium)

Akuna Capital

An ant starts at one corner of a cube and can only move along the cube's edges. The ant is disoriented and chooses a random edge to follow each time. What is the expected number of edges the ant will traverse before reaching the vertex directly opposite its starting point?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

57. Dice Game VII

Probability Lvl 6/10 (Medium)

DRW, Hudson River Trading, Optiver

You roll a fair six-sided die once and record that number. Then you continue rolling the same die until you obtain a result different from the first roll. Your payout is the sum of all the numbers rolled, including the first roll and the final roll that differs. What is the expected value of your payout?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

58. Duck Escape

Brainteasers Lvl 5/10 (Medium)

N/A

A duck is sitting at the center of a circular lake. A fox is waiting at the shore, unable to swim, wishing to eat the duck. The fox can move around the whole lake at a speed four times the speed at which the duck can swim. The duck can fly, but only once it reaches the shore of the lake—it can't fly from the water directly. Can the duck always reach the shore without being eaten by the fox? Note that the duck cannot submerge in the water.

Enter 1 if it is possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

59. Count Regions I

Combinatorics Lvl 5/10 (Medium)

Jane Street

Determine the maximum number of regions into which a plane can be divided using n straight lines. For example, a single line divides the plane into 2 regions.

Enter the answer for $n = 10$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

60. Count Regions III

Combinatorics Lvl 10/10 (Very Hard)

Jane Street

Determine the maximum number of regions into which 3-dimensional space can be divided using n planes.

Enter the answer for $n = 10$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

61. Square Intersection I

Probability Lvl 2/10 (Easy)

N/A

Consider the unit square with vertices at $(0,0)$, $(1,0)$, $(0,1)$, and $(1,1)$. Let $A = (0,0)$ and $B = (1,0)$. Two points are chosen uniformly at random within the square. Find the probability that the line passing through these two points (extended infinitely in both directions) intersects the segment AB .

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

62. Couple I

Probability Lvl 5/10 (Medium)

N/A

Ten married couples, consisting of 10 husbands and 10 wives, are randomly seated around a circular table with 20 seats. The seating pattern ensures that men and women alternate seats. Find the average number of couples who end up sitting next to each other.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

63. Even Ratio

Probability Lvl 10/10 (Very Hard)

N/A

Let X and Y be independent $\text{Uniform}(0,1)$ random variables. Define $S = \lceil \frac{Y}{X} \rceil$. Compute the probability that S is even. You may use the fact that the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$ converges to $\ln(2)$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

64. 2 Eggs

Brainteasers Lvl 6/10 (Medium)

Akuna Capital, Optiver

You are given two identical eggs and are situated in a building with 100 floors. An egg will break if dropped from any floor equal to or above a certain unknown threshold floor X ; otherwise, it will remain intact. What is the minimum number of drops needed to find the threshold floor X in the worst-case scenario?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

65. Heads Run I

Probability Lvl 5/10 (Medium)

N/A

A fair coin is tossed 10 times. What is the probability that the sequence never contains three heads in a row (i.e., there is no occurrence of "HHH" anywhere in the sequence)?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

66. Smallest Set

Brainteasers Lvl 9/10 (Very Hard)

N/A

Let's define S as the smallest group of integers that includes 0. Also, if a number x is in S , then both $3x$ and $3x + 1$ are also in S . How many non-negative integers in S are less than 2008?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

67. Variable Game

Probability Lvl 4/10 (Medium)

N/A

You are playing a game where the outcome of each round depends on the result of the previous round. In the first round, you win with probability $\frac{1}{2}$. For every subsequent round, the probability of winning is $\frac{1}{2}$ if you lost the previous round, and $\frac{1}{3}$ if you won the previous round. What is the probability that you win in the 5th round?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

68. Chess Tournament II

Probability Lvl 6/10 (Medium)

Five Rings

In a knockout chess tournament with 128 participants, each player has a unique rating. The higher-rated player always defeats a lower-rated opponent, and the victor advances to the next round. The tournament progresses through 7 rounds, resulting in the final match. What is the probability that the player with the top rating and the player with the third-highest rating will face each other in the final match?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

69. Game Of Smaller Number

Probability Lvl 5/10 (Medium)

Jane Street

Alice and Bob play a game where Alice randomly selects an integer uniformly from the set $\{1, 2, \dots, 100\}$, inclusive. Bob can choose any integer from the same range. The player with the larger number pays the smaller number to the other player. If both select the same number, no payment occurs. Assuming Bob plays optimally, what is his expected payout?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

70. Missing Sequence

Combinatorics Lvl 4/10 (Medium)

N/A

Consider the sequence: 3, 4, 5, a , b , 30, 40, 50. How many distinct ordered pairs of integers (a, b) satisfy the conditions that the sequence remains strictly increasing and no four numbers in the sequence form an arithmetic progression in any order?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

71. Passing The Coin II

Probability Lvl 3/10 (Easy)

N/A

Five players, labeled 1 through 5, are playing a game with a biased coin that lands heads with probability $\frac{1}{3}$ and tails with probability $\frac{2}{3}$. The game begins with player 1 holding the coin. On their turn, a player flips the coin. If it lands heads, that player wins immediately. If it lands tails, they pass the coin to the next player in order: player 1 passes to player 2, player 2 to player 3, and so on, with player 5 passing it back to player 1. This cycle continues until some player eventually flips heads and wins the game. What is the probability that player 3 wins the game?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

72. Remove The Subtree

Probability Lvl 7/10 (Hard)

Graviton

Alice is working with a tree that has 7 nodes. The tree consists of one root, two children of the root, and each of those children has two children of their own, making up the 7 nodes in total. In each step, she randomly picks a node and removes it, including the entire subtree under it. This continues until every node is removed. What is the expected number of steps needed to complete this process?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

73. Interesting Summation III

Brainteasers Lvl 7/10 (Hard)

N/A

Evaluate the sum $\sum_{k=0}^{\infty} \frac{F_k}{10^{k+1}}$, where F_k denotes the k -th Fibonacci number defined by the recurrence relation $F_0 = 0$, $F_1 = 1$, and $F_k = F_{k-1} + F_{k-2}$ for $k \geq 2$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

74. Interesting Summation V

Probability Lvl 10/10 (Very Hard)

N/A

Consider all integers from 0 through 99999 (inclusive) whose five-digit decimal representations (with leading zeros allowed) contain no two consecutive 1's. What is the sum of all such numbers?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

75. Pair Partition

Combinatorics Lvl 9/10 (Very Hard)

N/A

Consider the set $\{1, 2, 3, \dots, 20\}$. In how many ways can you partition this set into 10 disjoint pairs (each element used exactly once) such that, for every pair (a, b) , the absolute difference $|a - b|$ is either 1 or 10?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

76. Last One Off

Probability Lvl 7/10 (Hard)

N/A

Twenty-five ants are randomly placed on a meter-long rod, with the thirteenth ant from the west end designated as Ant Alice. Each ant has an equal probability of facing east or west. They move forward at a speed of 1 cm/sec, and upon colliding, they reverse directions. What is the probability that Alice is the last ant to fall off the rod?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

77. Strictly Within

Probability Lvl 9/10 (Very Hard)

N/A

Let a, b, c , and d be integers chosen independently and uniformly at random from the set $S_n = \{-n, -n + 1, \dots, n - 1, n\}$, which contains all integers between $-n$ and n , inclusive. Using these values, define two closed intervals: $I_1 = [\min(a, b), \max(a, b)]$ and $I_2 = [\min(c, d), \max(c, d)]$. Let $p(n)$ be the probability that interval I_2 is strictly contained within interval I_1 , which occurs when $\min(a, b) < \min(c, d) \leq \max(c, d) < \max(a, b)$. Find an expression for $p(n)$ as a function of n .

Enter the answer for $n = 10$ rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

78. Get King And Ace

Probability Lvl 6/10 (Medium)

N/A

Bob has a standard deck of 52 cards shuffled uniformly at random placed face down on a table. The deck contains four aces and four kings (no card is both an ace and a king). He now begins to flip over the cards one by one, leaving a card face up once he has flipped it over. He continues until the set of cards he has flipped over contains at least one ace and at least one king, at which point he stops. What is the expected value of the number of cards he flips over?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

79. Max Equals Min

Probability Lvl 3/10 (Easy)

N/A

Consider two integers, a and b , chosen independently and uniformly at random from the set $S = \{-10, -9, \dots, 9, 10\}$. Determine the probability that $\max\{0, a\} = \min\{0, b\}$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

80. Uniform Sum Probability

Probability Lvl 7/10 (Hard)

N/A

Can two non-fair 6-sided dice, numbered 1 to 6, produce a uniform probability distribution for their sum?

Enter 1 if it is possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

81. Stop Or Continue

Probability Lvl 3/10 (Easy)

Jane Street, Jump Trading, Susquehanna International Group

You play a game where you roll a fair 10-sided die labeled 1 through 10. On each roll, if you get a number between 1 and 9, that number is added to your score, and you may choose to roll again. However, if you roll a 10, your entire score is lost, and the game ends immediately with zero payout. At any point before rolling a 10, you may choose to stop and cash out your current score. What is the minimum score at which you should stop in order to maximize your expected earnings?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

82. Greedy Pirates I

Brainteasers Lvl 3/10 (Easy)

Susquehanna International Group

Five pirates need to divide 100 Gold Coins. Pirates have a hierarchy, from Level 1 to Level 5. The highest-level pirate is the leader. The leader proposes a plan to distribute the gold, and all the pirates vote on it (including the leader). If at least 50% of the pirates agree on the plan, the gold is split according to the proposal. If not, Level-5 pirate is kicked from the ship, and the Level-4 pirate now proposes a new plan. This process continues until a proposal is accepted. All pirates are extremely smart and extremely greedy. What is the maximum number of gold coins the level 5 pirate can get?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

83. Coin Toss Game V

Probability Lvl 6/10 (Medium)

Graviton

Alice, Bob, and Charlie are each flipping fair coins until they get their first heads. Let A , B , and C denote the number of flips it takes for Alice, Bob, and Charlie to get heads, respectively. Determine the probability that $A < B < C$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

84. Expected Ones In Binary

Probability Lvl 9/10 (Very Hard)

N/A

If a and b are selected uniformly from $\{0, 1, \dots, 511\}$ with replacement, what is the expected number of 1's in the binary representation of $a + b$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

85. Points On The Same Face

Probability Lvl 5/10 (Medium)

N/A

Let C be a cube. Let P , Q , and R be random vertices of C , chosen uniformly and independently from the set of vertices of C . (Note that P , Q , and R might be equal.) Compute the probability that some face of C contains P , Q , and R .

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

86. Product Of Cards

Brainteasers Lvl 8/10 (Hard)

Jane Street

You have all 13 clubs from a standard deck, where you can select 2 cards and receive a payout equal to the product of their values. Face cards (Jack, Queen, King) have a value of 0, and the Ace is worth 1. You can

pay \$1 to reveal the difference between any two cards, and you may use this option as many times as you wish. Ultimately, you must choose two cards, and your payout will be based on their product. What is the maximum guaranteed profit you can secure under an optimal strategy? Note that the revealed difference isn't the absolute difference. For example, if you point to card a , followed by card b , you would be given the value of $a - b$ and not $|a - b|$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

87. Gold Auction

Probability Lvl 5/10 (Medium)

Susquehanna International Group

A person is auctioning gold, and the bids are uniformly distributed between \$500 and \$1000. The auction shop charges \$5 per recruited bidder. Determine the optimal number of bidders that maximizes the expected payout for the person auctioning the gold and calculate this expected payout.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

88. Cutting Cube

Brainteasers Lvl 4/10 (Medium)

N/A

A cubic block of wood with dimensions $3 \times 3 \times 3$ is to be divided into 27 smaller cubes, each measuring $1 \times 1 \times 1$. The block and the resulting pieces can be rearranged after each cut. Determine the minimum number of cuts required to achieve this.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

89. Three Pegs

Brainteasers Lvl 9/10 (Very Hard)

N/A

Three pegs are placed in an infinite square grid. The coordinates of the three pegs are $(1, 0)$, $(0, 1)$ and $(1, 1)$. At any moment, one peg can jump over another, landing the same distance away on the opposite side. For example, if one peg is at $(0, 0)$ and another peg is at $(1, 0)$, the first peg can jump over the second peg and land at $(2, 0)$. The peg that is jumped over remains in place and is not removed. Is it possible to move one of the pegs to $(0, 0)$?

Enter 1 if it is possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

90. Chocolate Problem

Probability Lvl 7/10 (Hard)

N/A

A bag contains 3 large chocolates, each weighing 40 grams, and 6 small chocolates, each weighing 20 grams. In each turn, you randomly draw one chocolate from the bag. If you pick a small chocolate, you eat it immediately. If you pick a large chocolate, you split it into two (equal sized) small chocolates, eat one of them, and return the other to the bag. You repeat this process until only small chocolates remain in the bag. What is the expected total quantity of chocolate you consume?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

91. Fixed Points I

Probability Lvl 3/10 (Easy)

Citadel

Given a random function $f : S_n \rightarrow S_n$ where $S_n = \{1, 2, \dots, n\}$, let F_n denote the number of fixed points of f . Find the limit:

$$\lim_{n \rightarrow \infty} \mathbb{P}(F_n = 5)$$

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

92. Reaching 50

Brainteasers Lvl 4/10 (Medium)

Susquehanna International Group

Alice and Bob are playing a game where they take turns picking an integer from 1 to 10 and adding it to a running total. The first person to make the total exactly 50 wins. Alice gets to go first. What number should she choose to guarantee a win?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

93. Game On Divisors I

Brainteasers Lvl 8/10 (Hard)

Jane Street

Consider the set of integers $S = 2, 3, \dots, 30$. Alice and Bob engage in a game where they alternately select integers from S . A player loses if they choose an integer that shares a common factor with any previously selected integer. Alice can decide whether to play first or second. Should she choose to go first or second? Additionally, what are Alice's potential moves for her first turn (not necessarily the first move of the game) that would ensure victory?

Enter 1 if she should go first and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

94. At Least Six II

Probability Lvl 8/10 (Hard)

Citadel, Jump Trading

You roll a fair n -sided die (faces $1, 2, \dots, n$ equally likely) repeatedly, keeping a running total of the rolls. Let $T = \min\{k : X_1 + X_2 + \dots + X_k \geq n\}$ be the number of rolls needed for the sum to reach or exceed n . Find $E[T]$.

Enter the answer for $n = 18$ rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

95. Painted Cube III

Probability Lvl 4/10 (Medium)

N/A

Each face of a cube is painted either red or blue, each with probability $\frac{1}{2}$. The color of each face is determined independently. What is the probability that the painted cube can be placed on a horizontal surface so that the four vertical faces are all the same color?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

96. Sufficient Sum I

Probability Lvl 9/10 (Very Hard)

Akuna Capital, IMC Trading

Let $n \geq 1$ be a fixed integer and let $X_1, \dots, X_n$ be independent random variables, each uniformly distributed on $[0, 1]$. Compute the probability $P(X_1 + X_2 + \dots + X_n \leq 1)$ for $n = 5$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

97. Interesting Summation II

Brainteasers Lvl 4/10 (Medium)

N/A

Determine the sum of all numbers in the form $0.abcabcabc...$ where a, b , and c are distinct digits from 0 to 9.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

98. Random Pairing

Probability Lvl 4/10 (Medium)

N/A

Three couples (6 individuals) are randomly paired. Determine the probability that at least one couple is paired together.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

99. Odd Heads I

Probability Lvl 2/10 (Easy)

Akuna Capital

If 150 coins are tossed, what is the probability that the number of heads is odd?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

100. Tetrahedral Die II

Probability Lvl 8/10 (Hard)

N/A

Maryam has a fair tetrahedral die, with the four faces of the die labeled 1 through 4. At each step, she rolls the die and records which number is on the bottom face. She stops when the current number is $\geq$ the previous number. (In particular, she takes at least two steps.) What is the expected number (average number) of steps that she takes?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

101. Greedy Pirates II

Brainteasers Lvl 10/10 (Very Hard)

Susquehanna International Group

A pirate ship has 2009 treasure chests (all chests are closed). Each chest contains some amount of gold and some amount of silver. The captain is going to decide first how many chests he wants to keep and tell that number to the rest of the pirates. Then he is going to open all the chests and decide which ones he wants to keep. The captain wants to make sure he can keep at least half of the total gold and half of the total silver. However, he wants to say the smallest possible number to keep the rest of the pirates as happy as he can. What number should the captain say? Note: The amount of gold and silver in each chest may be different.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

102. Horse Racing

Brainteasers Lvl 6/10 (Medium)

IMC Trading, Optiver

Given 25 horses, each with a unique and constant speed, and a track with five lanes allowing a maximum of five horses to race simultaneously, determine the minimum number of races required to uniquely identify the top three fastest horses. Note that the races provide only relative rankings without the use of a stopwatch.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

103. 3 Heads 3 Tails I

Probability Lvl 5/10 (Medium)

Susquehanna International Group

A coin is tossed till 3 heads or 3 tails occur consecutively. What is the expected number of tosses?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

104. Going In Circles

Probability Lvl 6/10 (Medium)

N/A

You stand at point 1 on a circle of 100 points labeled 0 through 99. Each step you move one point clockwise or one point counterclockwise with equal probability. You stop as soon as you land on point 0. What is the expected number of steps to reach 0?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

105. Even Majority

Probability Lvl 4/10 (Medium)

N/A

Three fair 6-sided dice are rolled. If it's known that at least two of the dice display an even number, what is the probability that the total sum of the numbers rolled is no more than 7?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

106. Candy Distribution

Brainteasers Lvl 6/10 (Medium)

Susquehanna International Group

2022 boys are seated around a circular table, with a total of 2023 candies distributed among them. Bob is seated at the table but currently possesses no candies. In a move, a boy who holds at least two candies gives one to the person seated to his left and one to the person to his right. The process continues until Bob acquires a candy. Is it certain that these moves will eventually conclude, resulting in Bob receiving a candy? You also need to formally prove your claim, just intuition doesn't count.

Enter 1 if it is certain and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

107. Roll Till Even

Probability Lvl 1/10 (Easy)

IMC Trading

You roll a fair 12-sided die till you roll an even number. What is the expected number of rolls?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

108. Triangle Inside Circle I**Probability Lvl 6/10 (Medium)**

N/A

1153 points are evenly distributed on the circumference of a circle. Find the probability of randomly picking 3 points that constitute the vertices of an obtuse triangle.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

109. Lily Pad**Brainteasers Lvl 2/10 (Easy)***Akuna Capital*

A pond starts with just one lily pad, which doubles in size each day until it covers the entire pond. It takes 10 days for the pond to be fully covered starting with a single lily pad. Suppose you start with 8 lily pads on the first day. How many days will it take to cover the whole pond in this scenario?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

110. Passing The Bottle**Probability Lvl 8/10 (Hard)**

N/A

Moor and nine friends are seated around a circular table. Moor starts out holding a bottle, and whoever holds the bottle passes it to the person on his left or right with equal probability until everyone has held the bottle. Compute the expected distance between Moor and the last person to receive the bottle, where distance is the fewest number of times the bottle needs to be passed in order to go back to Moor.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

111. Jumping Frog I**Probability Lvl 6/10 (Medium)***DRW, Five Rings*

A frog starts at position 0 and jumps along the positive number line. At each jump, it moves forward by either 1 or 2 units with equal probability. Define p_k as the probability that the frog will reach position k where $k > 0$. What is the maximum value of p_k ?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

112. Penguin Walk II**Probability Lvl 4/10 (Medium)**

N/A

Imagine a penguin starting at position 0 on a number line. With each step, it moves to the right (to position $i + 1$) with a probability of $\frac{1}{3}$, and to the left (to position $i - 1$) with a probability of $\frac{2}{3}$. What is the probability that the penguin eventually reaches position 1?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

113. Interview I**Probability Lvl 9/10 (Very Hard)**

N/A

You are interviewing 6 candidates for a position, each with a distinct and unknown rating — higher ratings indicate better candidates. The candidates are interviewed one by one in a random order. After each interview,

you learn the candidate's relative ranking among all those interviewed so far. Immediately after each interview, you must decide whether to hire that candidate or reject them permanently — you cannot return to a previously rejected candidate. You are allowed to select only one candidate in this process. Your goal is to select the single best candidate out of all six. What is the optimal strategy to maximize the probability of choosing the best candidate, and what is this maximum probability?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

114. Free Throws

Probability Lvl 9/10 (Very Hard)

Citadel, Optiver

You're practicing 100 free throws. After successfully making the first shot but missing the second, the probability of making each subsequent shot is based on the ratio of successful throws to total attempts. For example, if you've made 12 out of the first 25 throws, the probability of making the 26th throw is $\frac{12}{25}$. After all 100 attempts, what is the probability of making exactly 66 successful throws?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

115. Starting With Heads

Probability Lvl 5/10 (Medium)

Five Rings, Graviton

If a fair coin is tossed 10 times, and no two heads are consecutive, what is the probability that the first toss resulted in heads?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

116. Circular Table I

Probability Lvl 6/10 (Medium)

N/A

Six boys and six girls are seated randomly around a circular table. Determine the probability that there is a group of at least four consecutive girls at some point around the table.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

117. Gold Thief

Brainteasers Lvl 1/10 (Easy)

Jane Street

A thief possesses five gold bars. The weights of all possible pairs of these gold bars are 21, 22, ..., and 30 pounds. What is the total weight of all the gold bars?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

118. Hanging Chain

Brainteasers Lvl 4/10 (Medium)

N/A

You are given an 80-foot long chain suspended at its two ends, which are fastened to a wall. The points where the ends of the chain are attached are at the same height. Under the influence of gravity, the center portion of the chain sags downward, forming a curve. The vertical distance from the lowest point of the chain to the line connecting the tacks (the horizontal axis) is 40 feet. Determine the distance between the two tacks.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

119. Coin Sequence Guess

Probability Lvl 6/10 (Medium)

N/A

Your friend flips a fair coin 100 times in a row and notes down the sequence. You can ask your friend just one Yes/No question to figure out the exact sequence. What is the highest probability you can have of correctly identifying the sequence with your question?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

120. Higher Roll I

Probability Lvl 4/10 (Medium)

N/A

Alice rolls a 20-sided die once. Bob rolls a 15-sided die twice and keeps the higher of the two results. What is the probability that Alice's roll is strictly greater than Bob's maximum?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

121. Dice Game IV

Probability Lvl 5/10 (Medium)

DRW

You play a game by repeatedly rolling a fair six-sided die according to the following rules: each time you roll a 1, 2, or 3, you receive \$1 and continue rolling; if you roll a 4 or 5, you immediately collect whatever you've earned so far and the game ends; if you roll a 6, you lose all your accumulated money and the game ends. What is the fair value of the game?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

122. Half Right Hats

Brainteasers Lvl 6/10 (Medium)

Graviton

One hundred prisoners have the opportunity to win their freedom through a game. In the dark, each prisoner will be given either a red or black hat based on a fair coin flip. When the lights turn on, each prisoner will see the colors of the hats worn by the others but will not be able to see their own. They are not allowed to communicate with each other. Each prisoner must write down their guess about the color of their own hat. If at least half of the prisoners guess correctly, they will be freed. The prisoners can discuss a strategy beforehand. Devise a plan that ensures all of them can be freed.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

123. Pppower

Brainteasers Lvl 4/10 (Medium)

N/A

How many integers p with $1 \leq p \leq 2000$ have the property that the units digit of p^p is 1?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

124. Grid Arrangement II

Combinatorics Lvl 5/10 (Medium)

N/A

How many distinct 4×4 matrices, consisting solely of entries 1 and -1 , satisfy the condition that the sum of the entries in each row is 0 and the sum of the entries in each column is 0?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

125. Distant Points

Probability Lvl 6/10 (Medium)

N/A

Three points are chosen independently and uniformly at random on the circumference of a circle. What is the probability that no two of these points lie within 90° of each other along the circle?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

126. Probabilistic Strategy IV

Probability Lvl 6/10 (Medium)

Hudson River Trading

Six mathematicians stand around a tree in a circle. Each has a hat whose color is randomly either red or blue. They cannot see the color of their own hat, nor the color of the hat of the person across from them, but they can see the hats of the four other mathematicians. If they can pick their strategy beforehand, what is the maximum chance they could have for everyone silently guessing their hat color right?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

127. Optimal Card Guessing

Probability Lvl 6/10 (Medium)

Five Rings

There are 3 aces and 3 jacks in a deck. On each turn, a card is drawn without putting it back. Alice receives \$1 for each correct guess of the card drawn. She plays 6 rounds using the best possible strategy. How much money can Alice expect to win?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

128. k-inversions

Probability Lvl 6/10 (Medium)

N/A

Let $a_1, a_2, \dots, a_n$ be a sequence of real numbers. A k-inversion in the sequence is a collection of indices $i_1 < i_2 < \dots < i_k$ such that the corresponding values strictly decrease, i.e., $a_{i_1} > a_{i_2} > \dots > a_{i_k}$. Given a random permutation of the set $\{1, 2, \dots, 10\}$, compute the expected number of 6-inversions in the permutation.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

129. Five In Numerator

Probability Lvl 3/10 (Easy)

N/A

Let S be the set of all rational numbers in lowest terms lying between 0 and 1 such that if $\frac{a}{b} \in S$, then $a \times b = 10!$, where "!" denotes factorial. If a fraction is chosen uniformly at random from S , what is the probability that its numerator is divisible by 5?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

130. W-permutation

Probability Lvl 5/10 (Medium)

N/A

A permutation of a finite set S is a one-to-one function from S to S . A permutation P of the set $\{1, 2, 3, 4, 5\}$ is called a W-permutation if $P(1) > P(2) < P(3) > P(4) < P(5)$. A permutation of the set $\{1, 2, 3, 4, 5\}$ is selected at random. Compute the probability that it is a W-permutation.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

131. Prisoner Hats

Brainteasers Lvl 6/10 (Medium)

N/A

One hundred prisoners are lined up, each wearing either a red or blue hat, assigned randomly. Each prisoner can see the hats in front of them but not behind. Starting from the back of the line, each prisoner must say either "red" or "blue." If their answer matches the color of their own hat, they are freed; if not, they remain imprisoned. Prisoners can hear all previous answers but do not know whether those answers were correct. A guard informs them the night before, giving them time to devise a strategy. Devise a strategy that guarantees at least 99 prisoners to be freed.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

132. Contiguous THTH

Probability Lvl 6/10 (Medium)

Jane Street

You flip a fair coin eleven times and observe exactly four heads and seven tails in the sequence. Given this information, what is the probability that the contiguous pattern "THTH" appears at least once among those eleven flips?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

133. 50 Wise Men

Brainteasers Lvl 8/10 (Hard)

N/A

Imagine a scenario where a sultan has imprisoned 50 wise men. A glass, initially placed bottom down, sits in the room, and each minute, one randomly selected wise man is summoned. He has the option to either flip the glass from its current position or leave it unchanged. This process repeats until one wise man confidently declares that all 50 men have been called at least once, thus securing their freedom. If the claim is incorrect, they all face execution. The wise men are allowed to communicate just once before they are isolated. Devise a strategy that will ensure their freedom.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

134. Rolling Everything I

Probability Lvl 5/10 (Medium) *Akuna Capital, DRW, Hudson River Trading, IMC Trading, Optiver, Two Sigma*

How many rolls of a fair 6-sided die are needed, on average, to roll all six sides at least once?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

135. Socks Remaining I

Probability Lvl 7/10 (Hard)

Jump Trading, Jane Street

You have 60 red socks and 40 blue socks in a drawer. You draw socks uniformly at random until all socks of one color have been drawn. What is the expected number of socks remaining in the drawer?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

136. Consecutive Integral Envelopes

Probability Lvl 6/10 (Medium)

Optiver

Alice is presented with three sealed envelopes, each containing a different consecutive number: $x + 1$, $x + 2$, and $x + 3$. The value of x is unknown. Alice opens one envelope at random and sees the number inside. Based on this, she must decide whether to accept this number as her payout or move on to open a second envelope. After revealing the number in the second envelope, she faces the same choice: accept that number as her payout or open the third and final envelope, which will then determine her payout. What is the expected value of Alice's payout in terms of x under optimal strategy?

Enter the answer for $x = 10$ rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

137. Trailing Zeros I

Brainteasers Lvl 3/10 (Easy)

DRW, Susquehanna International Group

How many trailing zeros does $100!$ have?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

138. Expected Minimum Till 5

Probability Lvl 5/10 (Medium)

N/A

You roll a die repeatedly until you get a 5. What is the expected value of the minimum number rolled?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

139. Palindrome Summation

Brainteasers Lvl 4/10 (Medium)

DRW

A five-digit palindrome is a positive integer with respective digits $abcba$, where a is non-zero. What is the sum of all 5-digit palindromes?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

140. Matrix Attack II

Brainteasers Lvl 10/10 (Very Hard)

Graviton

Consider an 8×8 matrix composed of zeros and ones. You are permitted to repeatedly select any 3×3 or 4×4 submatrix and flip all the bits within that submatrix (i.e., convert zeros to ones and ones to zeros). Is

it possible to always eliminate all the ones from the matrix, regardless of the initial configuration of zeros and ones?

Enter 1 for possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

141. Consecutive 1s

Probability Lvl 2/10 (Easy)

N/A

You roll a die 5 times, noting the result of each roll. What is the probability of getting at least 3 consecutive ones in those 5 rolls?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

142. Ball With Maximum Value

Probability Lvl 6/10 (Medium)

N/A

An urn contains 20 balls, each labeled from 1 to 20. You randomly select 10 balls from this urn. What is the expected highest value among the 10 balls you have drawn?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

143. Digit Multiply I

Brainteasers Lvl 5/10 (Medium)

N/A

What is the smallest positive integer for which the product of its digits equals 10,000?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

144. Regular Octagon

Probability Lvl 6/10 (Medium)

N/A

Select two distinct diagonals at random from a regular octagon. What is the probability that the two diagonals intersect at a point strictly within the octagon?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

145. Payment

Probability Lvl 4/10 (Medium)

N/A

A bag contains a large number of \$10 notes. Half of these notes are real, and half are fake. You want to make a payment of \$50. To do this, you provide the notes one by one randomly. If a note is fake, all your previously provided notes are returned, and the process restarts. What is the expected number of notes required to complete the payment? Assume the probability of drawing a real or fake note is always $\frac{1}{2}$ (as there are a large number of notes).

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

146. Rock Paper Scissors

Brainteasers Lvl 3/10 (Easy)

N/A

Alice and Bob engage in 10 rounds of a Rock–Paper–Scissors game. Each round resulted in a winner. It is known that Alice used Rock 3 times, Paper once, and Scissors 6 times. On the other hand, Bob used Rock twice, Paper 4 times, and Scissors 4 times. How many rounds did Alice win?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

147. Party Groups I

Probability Lvl 8/10 (Hard)

Jane Street

There are 50 guests at a party, and they form groups for a game by placing their names in a hat. Each guest randomly picks a name from the hat. A guest joins the group of the person whose name they pull. If a guest picks their own name, they form a group by themselves. If Guest A picks Guest B's name, Guest B picks Guest C's name, and Guest C picks Guest A's name, they form a closed group. No one else can join this group. What is the expected number of groups?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

148. Min, Median, Max Guess

Probability Lvl 9/10 (Very Hard)

N/A

Three independent random variables X_1, X_2, X_3 are drawn uniformly from $[0, 1]$. You observe only X_1 , and must guess whether it is the smallest, the middle, or the largest of the three, without seeing X_2 or X_3 . What strategy maximizes your probability of being correct, and what is that probability?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

149. Normal Inequality III

Probability Lvl 6/10 (Medium)

Akuna Capital

Let N_1, N_2 , and N_3 be independent standard normal random variables. What is the probability that $N_1^2 + N_2^2 > N_3^2$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

150. Permutation Variance

Probability Lvl 8/10 (Hard)

N/A

Let a permutation of the numbers $(1, 2, 3, \dots, 1000)$ be chosen uniformly at random. A fixed point in a permutation is an index i such that the element in position i is also equal to i ; in other words, the number at position i has not moved from its original place. For example, in the permutation $(2, 1, 3)$, the number 3 is a fixed point because it appears in the 3rd position. Let X be the number of fixed points in the permutation, and let Y be the number of non-fixed points (i.e., positions i where the element is not equal to i). Find the variance of the random variable $X - Y$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

151. Probabilistic Strategy II

Probability Lvl 4/10 (Medium)

Graviton

Three wizards are seated in a circular room. A magician will place one hat on each wizard's head, where each hat is either black or white, chosen uniformly at random. Each wizard cannot see their own hat but can see the hats of the other two wizards. Upon hearing a bell, all wizards must react simultaneously by either announcing the color of their own hat or remaining silent. If at least one wizard makes an announcement and all announcements are correct, the wizards win the game collectively. Wizards can confer and devise a strategy beforehand. What is the maximum probability of winning if they play optimally?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

152. 3D Tic-tac-toe

Combinatorics Lvl 4/10 (Medium)

Jump Trading

In a standard 3×3 tic-tac-toe game, a player wins by placing three of their symbols consecutively in a straight line—either in a row, a column, or along a diagonal. Now consider a $5 \times 5 \times 5$ cube (a 3D version of the game). How many distinct straight-line paths of exactly 5 consecutive cells exist in this cube? A valid path must consist of 5 cells that are connected in a straight line—whether along the same row, the same column, or diagonally in any direction.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

153. Color And Roll

Probability Lvl 2/10 (Easy)

Jane Street

Jane paints the outer faces of a $5 \times 5 \times 5$ cube green and then divides the cube into 125 smaller $1 \times 1 \times 1$ cubes. A single cube is then selected at random and rolled. Calculate the probability that the upward face of the selected cube is green.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

154. Expected 2s

Probability Lvl 4/10 (Medium)

Citadel

You have two urns. Urn A contains 8 pieces of paper with values $1, 2, \dots, 8$. Urn B has 8 pieces of paper with values either 1 or 2, with each appearing on 4 pieces of paper. You select an urn at random and draw a piece of paper uniformly from it, which shows the value 2. After replacing the paper, you draw from the same urn 40 more times, with replacement. What is the expected number of times you will draw a 2 in these 40 draws?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

155. Drunk Passenger I

Probability Lvl 8/10 (Hard)

IMC Trading

A line of 100 airline passengers is waiting to board a plane. They each hold a ticket to one of the 100 seats on that flight. For convenience, let's say that the n th passenger in line has a ticket for seat number ' n '. Being drunk, the first person in line picks a random seat (equally likely for each seat). All of the other passengers are sober, and will go to their assigned seats unless it is already occupied; if it is occupied, they will then find a free seat to sit in, at random. What is the probability that the last (100th) person to board the plane will sit in their own seat (100)?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

156. Broken Calculator

Brainteasers Lvl 8/10 (Hard)

N/A

We have a calculator displaying an integer x and two buttons. Pressing the first button replaces x with $\lfloor \frac{x}{2} \rfloor$, and pressing the second button replaces x with $4x + 1$. Initially, the display shows $x = 0$. How many integers less than or equal to 2014 can be achieved by pressing the buttons in any sequence? The displayed number is allowed to exceed 2014 during the sequence. (Here, $\lfloor y \rfloor$ represents the greatest integer less than or equal to y .)

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

157. Square Fitting II

Probability Lvl 6/10 (Medium)

N/A

Nine points are chosen uniformly at random inside the square with corners $(\pm 1, \pm 1)$. Determine the expected area of the smallest square, centered at the origin, that encloses the four closest points (out of the nine selected) to the origin.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

158. Increasing Dice Rolls II

Probability Lvl 7/10 (Hard)

Graviton, Hudson River Trading, IMC Trading

Suppose we roll a fair die 10 times. What is the probability that the sequence of rolls is non-decreasing (i.e., the next roll is never less than the current roll)?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

159. Direction Change II

Probability Lvl 6/10 (Medium)

Citadel, Graviton, Two Sigma

Alice travels from $(0, 0)$ to $(8, 8)$ in the coordinate plane by moving one unit in the positive x - or y -direction at a time. If all paths are equally likely, what is the probability that she changes his direction exactly 4 times?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

160. Close Factors

Brainteasers Lvl 3/10 (Easy)

Optiver

Find positive integers x_0 and y_0 such that $x_0 y_0 = 1234567890$, and $|x_0 - y_0|$ is minimized among all pairs (x, y) with $xy = 1234567890$.

Enter their absolute (positive) difference as the answer.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

161. Non-absorbing Random-walk

Probability Lvl 4/10 (Medium)

N/A

A frog moves on the integer nodes $1, 2, \dots, n$. From any node i strictly between 1 and n , it jumps to $i - 1$ or $i + 1$ with equal probability. When it is at node 1, it always jumps to node 2, and when it is at node n , it always jumps to node $n - 1$. Starting from node 1, what is the expected number of steps the frog takes before it reaches node n for the first time?

Enter the answer for $n = 10$ rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

162. Equal Logarithms

Probability Lvl 3/10 (Easy)

N/A

Two real numbers x and y are chosen independently and uniformly at random from the interval $(0, 1)$. Find the probability that $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

163. Mathematical Expression III

Combinatorics Lvl 3/10 (Easy)

IMC Trading

How many arithmetic expressions using only $*$ (multiplication) and $+$ (addition) can be constructed that include the digits 123456789 in exactly that order? A few valid expressions are $1 + 23 * 45678 + 9$ and 123456789.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

164. Dice Multiple I

Probability Lvl 5/10 (Medium)

N/A

You repeatedly roll a fair 6-sided die until the total sum of the rolled numbers is a multiple of 6. Determine the expected number of rolls required.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

165. Self-avoiding Paths

Combinatorics Lvl 8/10 (Hard)

Jane Street

Consider paths on the two-dimensional integer lattice that start at the origin $(0, 0)$ and consist of n steps. At each step, you may move: - one unit to the right $(+1, 0)$, - one unit up $(0, +1)$, or - one unit down $(0, -1)$, subject to the condition that the path never visits the same lattice point more than once (i.e., it is self-avoiding). Find the number of such paths of length n (in terms of n).

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

166. SOS Game

Brainteasers Lvl 10/10 (Very Hard)

N/A

On a piece of paper, there is a row of n empty boxes. Alice and Bob take turns writing either an "S" or an "O" in a previously blank box, with Alice going first. The winner is the player who forms the sequence "SOS" in consecutive boxes. Which player has a winning strategy for $n = 11$?

Enter 1 for Alice and 2 for Bob.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

167. Moving Ants I

Probability Lvl 1/10 (Easy)

N/A

Three ants are positioned on the three vertices of an equilateral triangle. Each ant randomly chooses one of the two adjacent vertices to move to, with equal probability. What is the probability that two ants will meet at the same vertex?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

168. Moving Ants III

Probability Lvl 7/10 (Hard)

Hudson River Trading

A thousand ants, each traveling at a constant rate of one meter per second, are randomly placed along a meter-long line. When two ants collide, they both instantly reverse direction but continue traveling at the same speed. On average, how long will it take for all the ants to fall off the line?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

169. Constrained Subsets I

Combinatorics Lvl 3/10 (Easy)

N/A

How many nonempty subsets of $\{1, 2, 3, \dots, 12\}$ have the property that the sum of the largest and smallest elements is 13?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

170. Sufficient Sum IV

Probability Lvl 10/10 (Very Hard)

Akuna Capital

Let $X_1, X_2, \dots$ be i.i.d. $\text{Unif}(0, 1)$ random variables, and for $x > 0$ define $N_x = \min\{n : X_1 + \dots + X_n > x\}$. Compute $E[N_2]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

171. Numerator I

Brainteasers Lvl 4/10 (Medium)

N/A

Every rational number $0 < q < 1$ can be expressed as the sum of unit fractions. For example, $\frac{4}{5} = \frac{1}{2} + \frac{1}{4} + \frac{1}{20}$. Determine the largest denominator in the unit fraction representation of $\frac{179}{720}$ that minimizes the number of terms and maximizes the largest possible denominator.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

172. Predicting King Draws

Probability Lvl 6/10 (Medium)

N/A

A dealer is using a shuffled deck made up of 10 regular card decks, meaning there are 40 cards of each rank and a total of 520 cards. The dealer begins flipping over cards one by one from the shuffled deck. The game ends when the fifth king is revealed. To win, you must predict how many cards will be turned over (including the fifth king) before the game ends. What number of cards should you guess to maximize your chances of winning?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

173. 40-sided And 60-sided Die

Probability Lvl 2/10 (Easy)

DRW

You are given a fair 40-sided die and a fair 60-sided die. If both are rolled, determine the probability that the number shown on the 60-sided die is strictly greater than the number on the 40-sided die.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

174. Count Bitonic Permutations

Combinatorics Lvl 5/10 (Medium)

Graviton, IMC Trading

Count the number of bitonic permutations of the numbers $1, 2, \dots, 9$. A permutation is considered bitonic if it first increases to a peak and then decreases. For example, $1, 3, 5, 7, 9, 8, 6, 4, 2$ is a bitonic permutation. Also, $1, 2, 3, 4, 5, 6, 7, 8, 9$ and $9, 8, 7, 6, 5, 4, 3, 2, 1$ are both considered bitonic permutations.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

175. Uniform Number Bet I

Probability Lvl 9/10 (Very Hard)

N/A

A game is played as follows: a random number is generated uniformly from the interval $(0, 1)$. The player may either accept this number as their cash payout or pay a fee of 0.045 to generate a new random number, also uniformly from $(0, 1)$. This process can be repeated as many times as desired, with the player paying 0.045 for each additional number generated after the first. The final payout is the maximum of all numbers generated. What is the expected payout if the player follows an optimal strategy?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

176. Asset Price

Probability Lvl 2/10 (Easy)

N/A

An asset's price changes daily, increasing or decreasing by 20% of its value from the previous day, each with equal probability. What is the probability that the asset's price remains unchanged after 10 days?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

177. Prism Count

Combinatorics Lvl 3/10 (Easy)

N/A

Consider a $5 \times 5 \times 5$ cube. Determine the total number of rectangular prisms contained within this cube.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

178. Bulb Factory

Probability Lvl 2/10 (Easy)

Jane Street

A factory has a conveyor belt that delivers bulbs continuously. Each bulb is either good or bad. If a bulb is good, the next one is also good with probability $\frac{2}{5}$. If a bulb is bad, the next one is also bad with probability $\frac{3}{5}$. You discover the first bulb is bad. Find the expected number of bulbs that pass until the next bad one.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

179. Moving Checker

Probability Lvl 5/10 (Medium)

Citadel, Graviton

Consider a 5×5 checkerboard. You place a checker randomly within the central 3×3 area. The checker moves randomly in one of two directions: diagonally up-left or diagonally up-right. If the checker is on the edge of the board, it can only move within valid bounds. What is the probability that the checker will end up in either the top-left or top-right corner of the board?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

180. XXYYYYZZ

Combinatorics Lvl 4/10 (Medium)

N/A

Alice begins with the string XXYYYYZZ. A move involves one of the following transformations: replacing an occurrence of XY with YX, replacing an occurrence of YZ with ZY, or replacing an occurrence of ZX with XZ. Determine the total number of distinct strings that Alice can obtain after executing zero or more of these transformations.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

181. Factorial Factor

Probability Lvl 4/10 (Medium)

Optiver

Determine the probability that a randomly chosen divisor of $20!$ is divisible by 2000.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

182. Interesting Division

Brainteasers Lvl 3/10 (Easy)

N/A

Determine the number of positive integers n less than 100 such that n^2 divides $n!$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

183. Bulb Toggle III

Brainteasers Lvl 7/10 (Hard)

Citadel, Susquehanna International Group

Given 100 light bulbs numbered from 1 to 100, all initially ON, each person k toggles the state of every bulb that is a multiple of the k -th positive prime number. Determine how many bulbs remain ON at the end of the process.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

184. Broken Stick I

Probability Lvl 5/10 (Medium)

Citadel

A meter stick is broken into 3 parts. It is known that the first breaking point is to the left of the mid-point and the second breaking point is to the right of the mid-point. What is the probability that the length of the middle part is more than $\frac{1}{3}$ meter?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

185. Factorial Vs Exponent

Brainteasers Lvl 4/10 (Medium)

Jump Trading

Which number is greater: $99!$ or 50^{99} ?

Enter 1 for $99!$ and 2 for 50^{99} .

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

186. Expected Arc Length III

Probability Lvl 8/10 (Hard)

N/A

Three points are chosen independently and uniformly on the circumference of the unit circle, cutting it into three random arcs. Given that two fixed points $(1, 0)$ and $(0, 1)$ lie in the same one of those arcs, what is the expected length of that arc?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

187. Heavy Group

Probability Lvl 3/10 (Easy)

IMC Trading

Six weights, with masses of 201 g, 202 g, 203 g, 204 g, 205 g, and 206 g, are randomly divided into two groups of three and placed on either side of a scale. What is the probability that the group containing the 206 g weight is heavier?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

188. First And Last Flip

Probability Lvl 6/10 (Medium)

N/A

A fair coin is repeatedly flipped until 20 consecutive coin flips are the same. Compute the probability that the first and last flips of the coin come up differently.

If the answer is x , enter the value $10000x$ rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

189. Integral Angles

Combinatorics Lvl 3/10 (Easy)

N/A

Determine the number of values of n for which a regular n -gon has interior angles that are integer values in degrees.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

190. Perfect Ratio

Probability Lvl 10/10 (Very Hard)

N/A

Let X and Y be independent $\text{Uniform}(0, 1)$ random variables. Define $S = \lceil \frac{X}{Y} \rceil$. Compute the probability that S is a perfect square. You may use the following sum:

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

191. Winning Odds

Probability Lvl 3/10 (Easy)

N/A

Alice and Bob are playing a game, both starting with 0 points. In each round, Alice has a 30% chance of winning, while Bob has a 70% chance. The winner of each round earns 1 point, and the loser loses 1 point. The first player to reach 2 points wins the game. Determine the probability that Alice wins the game.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

192. Biased Coin Game I

Probability Lvl 5/10 (Medium)

N/A

Alice and Bob play a game involving coins, where each chooses the probability of getting heads for their respective coin. After selecting their probabilities, they flip their coins. If both coins land heads, Alice pays Bob \$6. If both land tails, Alice pays Bob \$4. In all other outcomes, Bob gives Alice \$5. However, Alice must reveal her chosen probability p_1 for heads before Bob picks his p_2 . The task is to find the range of values for p_1 such that, regardless of Bob's choice, he does not have a positive expected outcome. This range is written as $[a, b]$, where a and b are rational numbers, and you need to determine the value of $b - a$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

193. 3 Marbles Remaining

Probability Lvl 3/10 (Easy)

N/A

Deepali has a bag containing 10 red marbles and 10 blue marbles (and nothing else). She removes a random marble from the bag. She keeps doing so until all of the marbles remaining in the bag have the same color. Compute the probability that Deepali ends with exactly 3 marbles remaining in the bag.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

194. Infinite Notes

Brainteasers Lvl 5/10 (Medium)

N/A

You have an infinite number of notes with values 5 and 21. Determine the largest value that cannot be expressed as a combination of these notes.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

195. Polynomial Coefficient

Brainteasers Lvl 9/10 (Very Hard)

Jump Trading

Alice is permitted to select any polynomial $p(x)$ with nonnegative integer coefficients, of any degree. Bob aims to determine the coefficients of $p(x)$ using evaluations of the polynomial. In one evaluation, Bob can choose a real number a and Alice provides the value of $p(a)$. Determine the minimum number of evaluations required for Bob to successfully infer all the coefficients of $p(x)$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

196. Game On Prime Numbers

Brainteasers Lvl 8/10 (Hard)

N/A

Alice and Bob take turns naming prime numbers. Alice starts with the number 2. Each turn, a player must add a positive integer to the current number to get another prime number, adding no more than twice what is necessary to do so. In other words, if the current number is n and the next prime number is p , the player may not add more than $2(p - n)$. Both players want to be the one who says 4567. Who has a winning strategy? Enter 1 for Alice and 2 for Bob.

Alice and Bob take turns naming prime numbers. Alice starts with the number 2. Each turn, a player must add a positive integer to the current number to get another prime number, adding no more than twice what is necessary to do so. In other words, if the current number is n and the next prime number is p , the player may not add more than $2(p - n)$. Both players want to be the one who says 4567. Who has a winning strategy? Enter 1 for Alice and 2 for Bob.

Enter 1 for Alice and 2 for Bob.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

197. Double Headed Coin I

Probability Lvl 1/10 (Easy)

Akuna Capital, DRW, Five Rings, Hudson River Trading, Two Sigma

A set of coins consists of 98 fair coins, one double-headed coin, and one double-tailed coin. A coin is chosen at random, flipped, and it shows heads. What is the probability that the chosen coin is the double-headed one?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

198. Tic Tac Toe

Probability Lvl 4/10 (Medium)

N/A

Three marbles are tossed onto a tic-tac-toe board at the same time. Each marble has an equal chance of landing in any of the nine squares. What is the probability that the marbles will end up forming a winning line, where all three marbles align either horizontally, vertically, or diagonally?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

199. Dominant Flip

Probability Lvl 6/10 (Medium)

Citadel

Consider flipping a fair coin 100 times. We say a particular outcome (heads or tails) is dominant after n flips if it occurs more than $\frac{n}{2}$ times within the first n flips. Find the probability that the result of the first flip is dominant after 100 flips.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

200. Queen Before King

Probability Lvl 2/10 (Easy)

N/A

Alice has a well-shuffled standard deck of 52 cards. She earns \$100 if the last queen appears before the last king. Alice is informed that there are exactly two queens before the first king. What is Alice's expected earnings?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

201. Expected Arc Length II

Probability Lvl 8/10 (Hard)

N/A

A unit circle centered at the origin is cut into three arcs by choosing three points independently and uniformly at random on its circumference. A fixed point at $(1, 0)$ lies on exactly one of these arcs. What is the expected length of the arc containing this point?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

202. Normal Inequality II

Probability Lvl 5/10 (Medium)

N/A

Let $X_1, X_2, \dots, X_{10}$ be independent and identically distributed random variables, each following a standard normal distribution $N(0, 1)$. What is the probability that $X_1 > X_2 \cdot X_3 \cdots X_{10}$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

203. Draw A Spade

Probability Lvl 5/10 (Medium)

N/A

Jack, Jill, and John play a game in which each randomly picks and then replaces a card from a standard 52-card deck until a spade card is drawn. What is the probability that Jill draws the spade? (Jack, Jill, and John draw in that order, and the game repeats if no spade is drawn.)

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

204. Median Roll

Probability Lvl 3/10 (Easy)

N/A

Three fair 6-sided dice are rolled simultaneously. Calculate the expected value of the median of the three rolled dice.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

205. Permutation Return

Probability Lvl 6/10 (Medium)

N/A

Consider the permutation $1, 2, 3, \dots, 100$. In one operation, you randomly select two distinct elements and swap their positions. What is the probability that after performing 25 such swaps, the permutation returns to its original order?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

206. Getting Singular Matrix

Probability Lvl 6/10 (Medium)

N/A

You begin with a 2×2 identity matrix. At each step, one of the four elements is chosen randomly. If the chosen element is 1, it is flipped to 0, and if it is 0, it is flipped to 1. Determine the expected number of steps required to make the matrix singular.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

207. Chess Tournament V

Probability Lvl 5/10 (Medium)

N/A

In a chess tournament with 50 players numbered from 1 to 50, a series of 49 matches determines the winner. The tournament begins with two randomly selected players competing in the first match. In each subsequent match, the loser is eliminated, and the winner plays against another randomly selected player who has not yet been eliminated. Suppose that the player with the larger number always defeats the player with the smaller number. What is the probability that player 49 is still in the tournament after the first 10 matches?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

208. Gambler's Ruin II

Probability Lvl 7/10 (Hard)

Akuna Capital

A gambler begins with an initial fortune of i dollars. Each time he plays, he has the possibility of winning 1 dollar with a probability of p (where $0 < p < 1$) or losing 1 dollar with a probability of $q = 1 - p$. The gambler will only stop playing if he either accumulates N dollars or loses all of his money. What is the likelihood that he will end up with N dollars?

Compute for $p = \frac{1}{3}$, $i = 7$, and $N = 10$ rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

209. Sum Of Divisor Ratio

Brainteasers Lvl 3/10 (Easy)

N/A

Bob calculates the sum of all positive divisors of 10000 that are multiples of 100 and records this result on a blackboard. He also calculates the sum of all positive divisors of 100 and notes this value below the previous result. Determine the ratio of the sum of the divisors of 10000 that are multiples of 100 to the sum of all positive divisors of 100.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

210. No Intersections

Probability Lvl 4/10 (Medium)

Five Rings

Suppose 6 points are chosen uniformly at random along the perimeter of a circle. These 6 points are then paired into 3 pairs, and a chord is drawn between each pair. What is the probability that none of the chords intersect?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

211. Expected Area Of Intersection

Probability Lvl 7/10 (Hard)

N/A

Let H be a regular hexagon with area 360. Three distinct vertices X, Y , and Z are picked randomly, with all possible triples of distinct vertices equally likely. Let A, B , and C be the unpicked vertices. What is the expected value of the area of the intersection of $\triangle ABC$ and $\triangle XYZ$?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

212. Blue, Ugly And Special

Probability Lvl 5/10 (Medium)

N/A

A bag contains nine blue marbles, ten ugly marbles, and one special marble. Ryan picks marbles randomly from this bag with replacement until he draws the special marble. He notices that none of the marbles he drew were ugly. Given this information, what is the expected value of the number of total marbles he drew?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

213. Card Picking III

Probability Lvl 4/10 (Medium)

Optiver, Two Sigma

A thoroughly shuffled deck is dealt card by card from the top. The 27th card turned over happens to be the first ace. Determine the probability that the next card, the 28th, is the 2 of hearts.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

214. Probability Of Subset

Probability Lvl 8/10 (Hard)

Hudson River Trading

Two subsets X and Y are chosen independently and uniformly at random from all subsets of the set $\{1, 2, 3, \dots, 10\}$, including the empty set. What is the probability that $X \subseteq Y$?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

215. Circle Fitting II

Probability Lvl 5/10 (Medium)

N/A

Five points are chosen independently and uniformly at random from the unit disk centered at the origin. A concentric circle is drawn whose radius is just large enough to contain the two points closest to the center. What is the expected area of this smallest concentric circle?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

216. Triangle Inside Circle II

Probability Lvl 4/10 (Medium)

Graviton

1153 points are evenly distributed on the circumference of a circle. Find the probability of randomly picking 3 points that constitute the vertices of an obtuse triangle.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

217. Game Of Subtracting Digits

Brainteasers Lvl 5/10 (Medium)

N/A

The number 1234 is written on a blackboard. Two players, Alice and Bob, play a game alternating turns. On a move, a player chooses a nonzero digit of the current number and subtracts that digit from the number, replacing the written number by the difference. The player who writes the number 0 wins. Alice goes first. Which player has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

218. Count Rankings

Combinatorics Lvl 4/10 (Medium)

Graviton

A competition consists of 10 players $P_1, P_2, \dots, P_{10}$ competing in a ladder-style tournament. Player P_{10} plays a game with P_9 : the loser is ranked 10th, while the winner plays P_8 . The loser of that game is ranked 9th, while the winner plays P_7 . They keep repeating this process until someone plays P_1 : the loser of that final game is ranked 2nd, while the winner is ranked 1st. How many different rankings of the players are possible?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

219. Count Rectangles I

Combinatorics Lvl 3/10 (Easy)

N/A

Count the number of non-degenerate rectangles in an 8×8 grid built from 7 parallel horizontal segments and 7 parallel vertical segments (so each rectangle's sides must lie on those grid lines). Non-degenerate means the rectangle has positive area.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

220. Line Splits I

Probability Lvl 5/10 (Medium)

Susquehanna International Group

Three points are randomly chosen on a line segment of length 3 units. What is the probability that the middle point lies between 1 and 2 units on the segment?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

221. Sharing A Secret

Combinatorics Lvl 6/10 (Medium)

N/A

A team of 5 individuals needs to secure a confidential document in a safe. They want to ensure that only a majority of at least 3 people can access the safe in the future. To achieve this, they plan to install several locks on the safe, where all locks must be opened to gain access. Each lock can have multiple keys, but each key unlocks only one specific lock. What is the minimum number of locks needed to meet these requirements?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

222. Falling Ants

Brainteasers Lvl 4/10 (Medium)

IMC Trading

Twenty-five ants are randomly placed on a meter-long rod, with the thirteenth ant from the west end being Ant Alice. Each ant faces either east or west with equal probability and moves at a speed of 1 cm/sec. When two ants collide, they reverse their directions. What is the minimum time required for us to be certain that Alice has left the rod?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

223. Random Segments

Probability Lvl 6/10 (Medium)

N/A

Leo is drawing line segments on a plane using the following rules:

He starts at the origin and draws a segment of length 1 along the positive x-axis.

After drawing each segment, he rotates his direction by 120° —either clockwise or counterclockwise with equal probability—and then draws the next segment of length 1 from the end of the previous one.

Leo continues this process—randomly turning by 120° left or right and drawing a new segment—until the new segment either intersects or touches any previously drawn segment, including at their endpoints.

For example, if Leo draws the first segment along the x-axis, then turns counterclockwise and draws a second segment, and then again turns counterclockwise and draws a third, the third segment will touch the first segment at the origin and thus counts as a hit.

What is the expected number of segments Leo will have drawn (including the final one that causes a hit) by the time this first intersection or contact occurs?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

224. Even Squares

Brainteasers Lvl 4/10 (Medium)

N/A

On a 9×9 chessboard (with 81 unit squares), consider every possible axis-aligned rectangle whose edges lie on the grid lines. For each such rectangle, mark every unit square it contains. After marking with all possible rectangles, how many unit squares have been marked an even number of times?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

225. Triangle Formation II

Probability Lvl 6/10 (Medium)

Akuna Capital

A rod is broken into three parts. What is the probability that the three parts can be arranged to form a triangle?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

226. Get Heads

Probability Lvl 4/10 (Medium)

Susquehanna International Group

Alice and Bob each flip fair coins until they get heads. What is the probability that Alice will need more coin tosses than Bob to get her first head?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

227. Distinct Faces

Probability Lvl 5/10 (Medium)

Five Rings

Determine the average number of distinct faces you would see when rolling a fair 6-sided die until you roll a face that you have already observed.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

228. Initial Roll

Probability Lvl 3/10 (Easy)

N/A

Alice rolls a fair die and notes the outcome. She then keeps rolling the die until she rolls a number that is greater than or equal to her initial roll. Define N as the number of additional rolls she makes after the first one. Determine $\mathbb{E}[N]$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

229. Digit Sum IV

Brainteasers Lvl 4/10 (Medium)

N/A

What is the sum of the digits of all numbers from 1 to 1 million, inclusive? For instance, the sum of the digits of 45 is 9.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

230. Baseball Series

Probability Lvl 3/10 (Easy)

N/A

The Red Sox play the Yankees in a best-of-seven series that ends as soon as one team wins four games. Suppose that the probability that the Red Sox win Game n is $\frac{n-1}{6}$. What is the probability that the Red Sox will win the series?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

231. Collinear Integer Points

Brainteasers Lvl 5/10 (Medium)

Graviton, Susquehanna International Group

Alice and Bob are engaged in a game on a 2D coordinate plane, where an integer point is defined as a point with integer coordinates. Bob challenges Alice to demonstrate that, given any five randomly chosen integer points on this plane, it is always possible to find two among these five points such that the line segment connecting them contains at least one additional integer point. Can Alice identify five such points that would disprove Bob's claim?

Enter 1 if it is possible and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

232. Gambler's Ruin I

Probability Lvl 4/10 (Medium)

Hudson River Trading

Player M has 1 dollars and player N has 2 dollars. Each game gives the winner 1 dollar from the other. As a better player, M wins $\frac{2}{3}$ of the games. They play until one of them is bankrupt. What is the probability that M wins?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

233. Local Maxima

Probability Lvl 6/10 (Medium)

Five Rings, Jane Street

Fourteen pieces of paper, labeled 1 through 14, are randomly arranged in a sequence. A position i is considered a local maximum if the number at the i -th position is strictly greater than both of its adjacent numbers. Determine the expected number of local maxima in this sequence. For instance, in the arrangement 513247 of six numbers, the first, third, and last positions are local maxima, yielding a total of 3 local maxima.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

234. Expected Collisions I

Probability Lvl 7/10 (Hard)

DRW

Twenty-five ants are randomly positioned on a 1-meter rod. They all move forward at 1 cm/sec in the direction they are facing, and upon collision with another ant, they reverse their direction. During the process, what is the expected number of collisions that take place on the rod?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

235. Only Consecutive Heads

Probability Lvl 4/10 (Medium)

N/A

If a coin is flipped 10 times, what is the probability that any heads occur only in consecutive pairs? For example, THHT, HHHH, and TTTT are valid but HTTT and THHH are invalid.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

236. Constrained Digits

Combinatorics Lvl 3/10 (Easy)

N/A

How many 5-digit numbers $abcde$ exist such that the digits b and d satisfy $b = a + c$ and $d = c + e$ while ensuring that each digit a, b, c, d, e is between 0 and 9, and the leading digit a is at least 1?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

237. Max Fill

Brainteasers Lvl 5/10 (Medium)

Jump Trading

We have an 8×8 square grid of unit cells. We fill some cells with distinct integers so that every filled cell is allowed to have a value greater than at most one of its neighboring filled cells. Two cells are neighbors when they share an edge or a corner, so diagonal adjacency counts. What is the maximum possible number of filled cells under this restriction?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

238. Up, Right And Diagonal

Probability Lvl 6/10 (Medium)

N/A

Tim lives on the origin of the Cartesian plane. Every second, Tim moves 1 unit up with probability $\frac{1}{9}$, 1 unit right with probability $\frac{1}{3}$, 1 unit up and 1 unit right with probability $\frac{1}{3}$, and he doesn't move with probability $\frac{2}{9}$. After 100 seconds, Tim ends up on the point (A, B) . What is the expected value of $A \cdot B$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

239. Product Of Solutions

Combinatorics Lvl 10/10 (Very Hard)

N/A

Let S denote the set of all triples (i, j, k) of positive integers where $i + j + k = 17$. Compute

$$\sum_{(i,j,k) \in S} i \cdot j \cdot k$$

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

240. Ranged Guess

Probability Lvl 10/10 (Very Hard)

N/A

You and a friend take turns guessing a secret integer chosen uniformly at random from $\{1, 2, \dots, 100\}$. You go first. After each incorrect guess — regardless of whose turn it is — both you and your friend are told whether the secret number is higher or lower than the guess. Then the other person takes their turn. Under optimal play by both, what is your probability of winning?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

241. Circular Regions

Probability Lvl 2/10 (Easy)

Susquehanna International Group

Four random points P_1, P_2, P_3 , and P_4 are chosen on the circumference of a circle. A chord is drawn between P_1 and P_2 , and another chord is drawn between P_3 and P_4 . What is the expected number of separate regions into which the circle is divided?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

242. Integer Selection

Probability Lvl 4/10 (Medium)

N/A

Seven individuals each randomly pick an integer from the set $\{1, 2, \dots, 7\}$ with equal likelihood. If it is known that no one has chosen the number 7, determine the probability that the number 7 is the only unselected number.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

243. Double Headed Coin II

Probability Lvl 2/10 (Easy)

Akuna Capital, DRW, Five Rings

A bag contains 19 fair coins and 1 double-headed coin. A coin is randomly selected and tossed. What is the minimum number of consecutive heads that must appear to be at least 95% certain that the selected coin is the double-headed one?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

244. M,P,F,G

Probability Lvl 1/10 (Easy)

N/A

Alex has a bag containing 4 buttons with distinct letters M, P, F, G on them (one letter per button). She picks buttons randomly, one at a time, without replacement, until she picks the button with letter G. What is the probability that she has AT LEAST three picks and her third pick is the button with letter M?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

245. Tetrahedral Dice Game

Probability Lvl 5/10 (Medium)

DRW

Two tetrahedral dice, one red and one blue, each labeled 1 – 4, are rolled simultaneously in a game. If the blue die shows a value greater than the red, you receive the difference as a payout. If the values are equal, the game ends immediately with no payout. Otherwise, the game continues with another roll. What is the fair value of this game?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

246. Probabilistic Strategy I

Probability Lvl 3/10 (Easy)

Jump Trading

Alice and Bob are each placed in separate, distant rooms. Alice will flip a coin, and Bob will also flip a coin. Alice must guess the outcome of Bob's coin toss, while Bob must guess the outcome of Alice's coin toss. They will win if both guesses are correct. Before the game, they are allowed to discuss and devise a strategy. What is the probability of their winning, assuming they use an optimal strategy?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

247. Coin Toss Game I

Probability Lvl 5/10 (Medium)

Akuna Capital, Citadel, Susquehanna International Group

Alice and Bob are playing a game with a fair coin. Alice wins if the sequence HTH appears before the sequence HHT. Otherwise, Bob wins. What is the probability that Bob will win?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

248. Adding 1 Or 2

Probability Lvl 5/10 (Medium)

N/A

Starting with the number 0, Alice performs an infinite sequence of moves as follows: she chooses a number from $\{1, 2\}$ at random (each with probability $\frac{1}{2}$) and adds it to the current number. Let p_m be the probability that Alice ever reaches the number m . Find $p_{20} - p_{15}$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

249. Positive Product

Probability Lvl 6/10 (Medium)

N/A

Find the probability that the product of 2020 real numbers chosen independently and uniformly at random from the interval $[-1, 2]$ is positive.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

250. Not That Easy

Probability Lvl 3/10 (Easy)

Akuna Capital, Hudson River Trading

There are three coins: one has heads on both sides, another has tails on both sides, and the third is a standard coin with heads on one side and tails on the other. You randomly select a coin and observe one side, which turns out to be heads. What is the probability that the chosen coin is the double-headed one?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

251. Measuring Everything I

Brainteasers Lvl 6/10 (Medium)

Susquehanna International Group

What is the minimum number of weights needed to measure any integer weight from 1 to 100 grams on a standard balance with two pans, where these weights can only be placed on the left pan?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

252. Count Solutions

Combinatorics Lvl 3/10 (Easy)

N/A

Determine the number of non-negative odd integer solutions to the equation $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 96$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

253. Dice Pattern Game I

Probability Lvl 5/10 (Medium)

Susquehanna International Group

Solve the following problem without using pen and paper. We roll a fair 6-sided die repeatedly and stop when two specific numbers appear consecutively. The payoff is the total number of rolls. Should we prefer: Case 1: 4 followed by 5, Case 2: 4 followed by 4, Case 3: It doesn't matter? If the answer is Case x, enter the value of x.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

254. Power Distance I

Probability Lvl 6/10 (Medium)

N/A

A particle starts at the origin and moves one step left or right each second with equal probability. Let d_n be the distance from the origin after n seconds. Find the expected value of d_{30}^2 .

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

255. Power Factor

Probability Lvl 4/10 (Medium)

Akuna Capital

What is the probability that a randomly chosen positive divisor of 10^{99} is divisible by 10^{85} ?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

256. Training Sessions

Combinatorics Lvl 3/10 (Easy)

IMC Trading

A company plans to assign 5 employees to attend 3 training sessions each over the next month. There are 15 available training slots, but one employee is unavailable for the first slot. How many distinct ways can the training sessions be scheduled?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

257. Same Cards

Probability Lvl 5/10 (Medium)

N/A

A deck contains 12 cards-4 kings, 4 queens, and 4 jacks. The cards are distributed equally among 4 players. What is the probability that each player receives 1 king, 1 queen, and 1 jack?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

258. Ant On A Cube II

Probability Lvl 5/10 (Medium)

Akuna Capital

An ant begins at a vertex of a 3D cube and at each step moves randomly to one of the three neighboring vertices with equal chance. Determine the expected number of steps required for the ant to return to its starting point.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

259. 2 Or 11

Probability Lvl 4/10 (Medium)

Susquehanna International Group

Imagine you are rolling a fair six-sided die repeatedly. You continue rolling until you either roll a 2 for the first time or you roll two consecutive 1s. What is the expected number of rolls you will make?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

260. Geometric Progression

Probability Lvl 7/10 (Hard)

N/A

Three distinct integers are chosen uniformly at random from the set $\{1, 2, \dots, 100\}$. What is the probability that those three numbers, when arranged in increasing order, form a geometric progression? (Here the common ratio of the progression may be any rational number, not necessarily an integer.)

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

261. Sensor Signal

Probability Lvl 5/10 (Medium)

N/A

Two sensors detect events modeled by independent Poisson processes with intensity parameters 6 and 10, respectively. Each detected event triggers a signal with probabilities $\frac{1}{6}$ and $\frac{1}{5}$, independently. Find the expected time between signals.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

262. Uniform Random Variables II**Probability** Lvl 2/10 (Easy)

N/A

Consider $P_1, P_2, \dots, P_{20}$ as IID random variables each uniformly distributed between 0 and 1. Find the probability that P_1 is greater than the maximum of P_2, P_{10} , and P_{15} .

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

263. Russian Roulette I**Probability** Lvl 3/10 (Easy)

Jane Street

You're playing Russian Roulette with a friend using a six-chamber revolver loaded with one bullet. The cylinder is spun randomly before each turn. You and your friend alternate pulling the trigger, and the game ends when someone fires the bullet. If your friend goes first, what is the probability that you win?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

264. Drunk Passenger II**Probability** Lvl 9/10 (Very Hard)

N/A

A line of 100 airline passengers is waiting to board a plane. They each hold a ticket to one of the 100 seats on that flight. For convenience, let's say that the n -th passenger in line has a ticket for seat number ' n '. Being drunk, the first person in line picks a random seat (equally likely for each seat). All of the other passengers are sober, and will go to their assigned seats unless it is already occupied; if it is occupied, they will then find a free seat to sit in, at random. What is the expected number of passengers that sit on their own seat?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

265. Binary Cards Removal**Brainteasers** Lvl 9/10 (Very Hard)

Jump Trading

On a table lie 1025 cards numbered from 0 to 1024. Bob and Alice take turns removing cards. Bob begins by removing 2^9 cards, then Alice removes 2^8 cards, then Bob removes 2^7 cards, and so on — each player removing half as many cards as the previous move. This continues until only two cards remain, numbered a and b . Alice must then pay Bob $|a - b|$ dollars. What is the maximum amount of money Bob can guarantee to win, regardless of how Alice plays?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

266. Early 5s**Probability** Lvl 5/10 (Medium)

N/A

A player has a fair 6-sided die with faces numbered 1 to 6. The player rolls the die repeatedly, recording the outcomes in the order they occur. The player continues rolling until either two 5s appear (not necessarily consecutive) or both a 4 and a 6 have been rolled. Determine the probability that the player stops rolling due to rolling two 5s.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

267. Bacteria Population

Probability Lvl 2/10 (Easy)

N/A

Consider a bacterial colony beginning with a single cell. In each generation, every cell has an independent chance of either dying without leaving offspring or producing a random positive integer number of offspring. The probability of extinction for a colony starting with one cell is $\frac{1}{3}$. Determine the probability of extinction for a colony that starts with three cells.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

268. Cube Fitting I

Brainteasers Lvl 3/10 (Easy)

N/A

How many cubes with dimensions $3 \times 3 \times 3$ can fit inside a larger cube measuring $20 \times 20 \times 20$?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

269. Nine Children

Brainteasers Lvl 6/10 (Medium)

N/A

A man has nine children. The children were born with a fixed interval of time between them. The sum of the squares of their ages equals the square of the man's age. What is the man's age, given that it does not exceed 90?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

270. Unit Digit Probability II

Probability Lvl 5/10 (Medium)

N/A

Let m be randomly chosen from the set $\{111, 133, 155, 177, 199\}$ and n be randomly selected from the set $\{2004, 2005, \dots, 2023\}$. Determine the probability that the units digit of m^n is 1.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

271. Grid Paint I

Brainteasers Lvl 7/10 (Hard)

N/A

We have a grid with 50 rows and c columns. In each column, exactly 10 of the 50 squares are painted blue. Find the smallest integer c such that no matter how those 10 blues in each column are placed, there is always at least one row containing 15 blue squares.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

272. Dice Game V

Probability Lvl 4/10 (Medium) *Akuna Capital, Citadel, DRW, Optiver, Susquehanna International Group*

Alice is playing a game with a fair six-sided die. After each roll, she sees the number on the die and can choose to roll again at a cost of \$1 per roll. When she decides to stop rolling, she gets a reward equal to the number on the last die she rolled. There is no maximum number of rolls. What is the expected value of her final payout?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

273. No Rocks Allowed

Probability Lvl 4/10 (Medium)

Jane Street

You play rock-paper-scissors with an opponent who cannot choose rock. Each win earns you \$1, each loss costs you \$1, and draws result in no change. Assuming both players use optimal strategies, what is your expected profit per round?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

274. Expected Remainder

Probability Lvl 8/10 (Hard)

N/A

Let a, b, c, d, e, f be integers selected from the set $\{1, 2, \dots, 100\}$, uniformly and at random with replacement. Set $M = a + 2b + 4c + 8d + 16e + 32f$. What is the expected value of the remainder when M is divided by 64?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

275. Coin Inside Square

Probability Lvl 3/10 (Easy)

Graviton, Jump Trading, Optiver, Two Sigma

A tablecloth is made up of 2 cm by 2 cm squares. A coin with a diameter of 1 cm is randomly tossed onto the tablecloth. Find the probability that the coin lands entirely within a single square.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

276. Coin Toss Game IV

Probability Lvl 4/10 (Medium)

Akuna Capital

Alice and Bob are flipping coins repeatedly until each gets their first heads. What is the probability that Alice will need at least four times as many flips as Bob?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

277. Infectious Ants

Probability Lvl 9/10 (Very Hard)

N/A

Twenty-five ants are randomly positioned on a 1-meter rod, with the thirteenth ant from the west being our friend, Ant Alice. Each ant faces either east or west, with equal probability. They all move forward at 1 cm/sec in the direction they are facing, and upon collision with another ant, they reverse their direction. Suppose that Alice has a cold, which is transmitted from ant to ant instantly upon collision. How many ants will be infected, on average, before the rod is cleared?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

278. Unique Ids

Combinatorics Lvl 4/10 (Medium)

N/A

Each employee ID in a company is formed using all 10 distinct digits $a_1, a_2, \dots, a_{10}$. These digits must satisfy the constraints $a_1 > a_2 > a_3 > a_4 > a_5$ and $a_5 < a_6 < a_7 < a_8 < a_9 < a_{10}$. Determine the total number of such unique IDs.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

279. Functional Equation

Brainteasers Lvl 4/10 (Medium)

N/A

Let $f(x)$ be a polynomial function which satisfies

$$f(1) + f(2) + \dots + f(n) = n^3$$

Find $f\left(\frac{3}{2}\right)$. Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

280. Expected Wait Time I

Probability Lvl 1/10 (Easy)

Jane Street

A bus arrives every 10 minutes on a fixed schedule. If you arrive at a random time, what is the expected time until the next bus arrives?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

281. Weighing Coins I

Brainteasers Lvl 3/10 (Easy)

Citadel, IMC Trading, Susquehanna International Group

You have 8 coins, one of which is slightly heavier than the others. What is the minimum number of times you need to use a balance scale to identify the heavier coin?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

282. Coin Toss Game III

Probability Lvl 4/10 (Medium)

Graviton

Alice and Bob are playing yet another coin game. In this game, a coin is flipped repeatedly. Alice wins if the sequence "HHH" occurs before "THH" and Bob wins otherwise. What is the probability that Alice wins?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

283. Card Flipping Game

Brainteasers Lvl 6/10 (Medium)

N/A

Arrange ten playing cards in a row, all face up. Alice, the first player, begins by turning down any one card. After Alice's move, Bob can choose to turn down either a single card or two adjacent cards. The game continues in this manner, with Alice and Bob alternating turns. Alice must always turn down one card on her first move, but from Bob's turn onwards, each player can choose to turn down either one card or two adjacent cards. The player who turns down the last card wins the game. Which player has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

284. Divisible By 5

Brainteasers Lvl 4/10 (Medium)

N/A

The numbers 1 through 21 are written on a blackboard. Alice and Bob take turns alternately. On each turn, a player erases one available number. The game continues until only two numbers remain. Alice wins if the sum of the final two numbers is divisible by 5; otherwise, Bob wins. If Alice goes first, who has a winning strategy under optimal play?

Enter 1 for Alice and 2 for Bob.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

285. Get More Heads

Probability Lvl 8/10 (Hard)

Akuna Capital, Citadel, Hudson River Trading

Two gamblers are playing a coin toss game. Gambler A has $n + 1$ fair coins, B has n fair coins. What is the probability that A will have more heads than B if both flip all their coins?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

286. To Infinity And Beyond I

Probability Lvl 5/10 (Medium)

N/A

A random sequence $(X_n)_{n \geq 1}$ is defined as follows. First pick X_1 uniformly from the interval $(0, \frac{1}{2})$. For every $n \geq 2$, given $X_{n-1} = x$, choose X_n uniformly on the interval $(x, x + 2^{-n})$ (independently of earlier uniform draws except through the conditioning on X_{n-1}). Find the expected value of the limit $\lim_{n \rightarrow \infty} X_n$.

Enter your answer rounded to 5 decimal places. If the limit does not exist, enter -1.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

287. Tails Removal

Probability Lvl 6/10 (Medium)

N/A

Thirty-one fair coins are placed in a row and flipped once. After each flip, any coin that lands tails is removed. This process continues, with the remaining coins flipped again, until either no coins remain or all of the remaining coins show heads. Determine the probability that the game concludes with all of the coins showing heads.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

288. 1 Dot Removed

Probability Lvl 1/10 (Easy)

Jane Street

A fair 6-sided die, with faces numbered 1 through 6, has one face randomly chosen and reduced by 1. Determine the probability that an even number is rolled on the modified die.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

289. Even Head Stop II

Probability Lvl 4/10 (Medium)

Five Rings, IMC Trading

You toss a coin till you get 2 consecutive heads. What is the probability that the number of tosses is even?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

290. Normal Inequality I

Probability Lvl 6/10 (Medium)

Akuna Capital, Hudson River Trading, Jump Trading

Let X and Y be independent and identically distributed standard normal random variables. Determine the conditional probability $P(Y > \sqrt{3}X \mid Y > 0)$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

291. Contained Projection

Probability Lvl 9/10 (Very Hard)

N/A

Four points A, B, C, D are chosen at random on the circumference of a circle, each independently and uniformly distributed. We consider the triangle ABC and look at each of its three sides in turn. For a given side—say the segment AB —we drop a perpendicular from D to the infinite line passing through A and B , and we ask whether the foot of that perpendicular lies between A and B . What is the expected number of sides of triangle ABC for which this perpendicular projection of D falls in the interior of the corresponding segment?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

292. Boat And Steamer

Brainteasers Lvl 1/10 (Easy)

Hudson River Trading

Points A and B are located along a river. A boat, stationary with respect to the water, drifts from A to B in 6 seconds. A steamer, moving downstream at 1 meter per second relative to the water, covers the same distance in 4 seconds. Determine the distance between A and B in meters.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

293. King, Queen, Ace I

Probability Lvl 6/10 (Medium)

N/A

A standard deck of 52 cards is shuffled, and cards are drawn one by one without replacement. Find the expected number of cards drawn until we have observed either a king, or both a queen and an ace.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

294. Coin Toss Game VII

Probability Lvl 3/10 (Easy)

N/A

In a coin-flipping game, you and your opponent take turns flipping a coin. If the first player gets heads, the second player owes them 30 dollars. If tails, the coin is passed to the second player. If the second player gets heads, they win 30 dollars from the first player. If tails, the game continues with the same rules. What is the maximum amount you would pay to go first?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

295. Interview II

Probability Lvl 9/10 (Very Hard)

N/A

You are interviewing n candidates for a position, each with a distinct and unknown rating — higher ratings indicate better candidates. The candidates are interviewed one by one in a random order. After each interview, you learn the candidate's relative ranking among all those interviewed so far. Immediately after each interview, you must decide whether to hire that candidate or reject them permanently — you cannot return to a previously rejected candidate. You are allowed to select only one candidate in this process. Your goal is to select the single best candidate out of all n . What is the optimal strategy to maximize the probability of choosing the best candidate, and what is this maximum probability in the limit as $n \rightarrow \infty$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

296. Plus One, Minus One

Brainteasers Lvl 7/10 (Hard)

Graviton, IMC Trading

Given an $n \times n$ grid where each cell at row i and column j contains an integer a_{ij} , you can perform the following operation: choose two adjacent cells and either add 1 to both or subtract 1 from both. Determine the necessary and sufficient condition on the values a_{ij} for all cells in the grid to be made equal to 0.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

297. Largest Sum Subset

Brainteasers Lvl 8/10 (Hard)

Jane Street

Consider the set $S = \{1, 2, 3, \dots, 30\}$. We seek a subset $T \subseteq S$ in which every pair of elements is coprime. The objective is to find the subset that maximizes the sum of its elements. What is the total sum of all the elements in T ?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

298. Careful Soldiers

Brainteasers Lvl 9/10 (Very Hard)

Graviton

An odd number of soldiers are positioned in a field such that all pairwise distances between them are distinct. Each soldier is instructed to monitor the nearest adjacent soldier. Prove that at least one soldier is not being observed.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

299. Dice Difference III

Probability Lvl 8/10 (Hard)

Graviton

What is the expected number of rolls needed to obtain two consecutive rolls with a difference no greater than 1?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

300. Two Coins Remaining I

Brainteasers Lvl 3/10 (Easy)

Two Sigma

You have 32 coins placed in a row. Initially, all coins are heads. You can perform the following operation any number of times: remove a coin that is not at one of the ends of the row, and flip its neighbors. The removed

coin must be heads. For example, HTHTH becomes HHHH if you remove the middle coin. Can you reach the state "HT"?

Enter 1 if it is possible and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

301. Conditional Expected Rolls

Probability Lvl 6/10 (Medium)

Susquehanna International Group

A fair 6-sided die is rolled repeatedly until the number 6 appears. Assuming the first occurrence of 6 happens before the first occurrence of 5, determine the expected total number of rolls.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

302. Maximising Heads II

Probability Lvl 2/10 (Easy)

Hudson River Trading, Susquehanna International Group

You flip 8 fair coins. After the initial flip, you may choose to reflip any subset of the coins, but each coin can only be flipped one additional time. You earn \$1 for every coin that shows heads. What is the fair value of this game if you play optimally?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

303. Green Ball Draw

Probability Lvl 2/10 (Easy)

N/A

In a bin with 4 green balls and 11 red balls, balls are drawn one at a time without replacement. Determine the probability that the 4th green ball is selected on the 9th draw.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

304. Same Values On Dice

Probability Lvl 5/10 (Medium)

N/A

Three fair 6-sided dice are rolled, and the numbers on their faces are noted. What is the probability that when these dice are rolled again, they show the same numbers as before?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

305. Meeting Probability

Probability Lvl 4/10 (Medium)

N/A

Two individuals, each arriving at a station at a random time between 5:00 AM and 6:00 AM (with their arrival times uniformly distributed), will each stay for exactly five minutes before departing. What is the probability that they will meet on a given day?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

306. Knight Return

Probability Lvl 8/10 (Hard)

N/A

An 8×8 chessboard is given and a knight is placed on the bottom left corner. At each move the knight chooses uniformly at random one of its legal knight moves and moves there. What is the expected number of moves until the knight first returns to the bottom left corner?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

307. Rolling Everything II

Probability Lvl 7/10 (Hard)

N/A

You have two fair six-sided dice. In each move, you roll both dice simultaneously. What is the expected number of moves until all 6 sides appear?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

308. No Empty Bucket

Probability Lvl 5/10 (Medium)

N/A

You repeatedly toss a fair coin until the first head appears. If the first head occurs on the k -th toss, you take k identical marbles and place each marble into one of three buckets, with each bucket being equally likely for every marble. What is the probability that, after this process, no bucket is empty?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

309. Painted Cube II

Probability Lvl 4/10 (Medium)

Citadel, Jane Street

A $3 \times 3 \times 3$ cube is painted red on its surface and then cut into 27 $1 \times 1 \times 1$ smaller cubes. One of these cubes is selected uniformly at random and rolled. Given that the face showing is red, find the probability that the selected cube is a corner cube.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

310. Random Point On Disc

Probability Lvl 2/10 (Easy)

N/A

A point (P, Q) is chosen randomly with uniform distribution from within the unit disc. Compute $E[PQ]$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

311. 14 Marbles

Probability Lvl 6/10 (Medium)

Graviton

You have 14 marbles, with 2 marbles of each of 7 different colors. You arrange these marbles in a row at random. What is the probability that the number of marbles between every pair of marbles of the same color is even?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

312. Heaven Or Hell

Probability Lvl 10/10 (Very Hard)

N/A

Contessa is taking a random lattice walk in the plane, starting at $(1, 1)$. (In a random lattice walk, one moves up, down, left, or right 1 unit with equal probability at each step.) If she lands on a point of the form $(6m, 6n)$ for $m, n \in \mathbb{Z}$, she ascends to heaven, but if she lands on a point of the form $(6m + 3, 6n + 3)$ for $m, n \in \mathbb{Z}$, she descends to hell. What is the probability that she ascends to heaven?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

313. Largest Area Expectation

Probability Lvl 7/10 (Hard)

N/A

Consider a 3×3 grid where each of the 9 squares is independently painted either red or blue, with equal probability. What is the expected area of the largest square (of any size) that is entirely painted red?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

314. No Fives

Probability Lvl 2/10 (Easy)

N/A

You roll a fair 6-sided die till you get a 6. What is the probability that you never roll a 5 in this process?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

315. Maximum Rolled Value

Probability Lvl 4/10 (Medium)

N/A

Alice rolls 3 fair dice. What is the probability that the highest value rolled is a four?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

316. Random Knight Movement

Probability Lvl 7/10 (Hard)

N/A

A knight starts at the origin on an infinite two-dimensional grid. Every second, it makes one random knight move, meaning it moves 2 units in one direction and 1 unit perpendicular to that direction. Each of the 8 possible knight moves is equally likely. After 100 seconds, what is the expected value of the square of the knight's distance from the origin?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

317. Game On 1-10

Brainteasers Lvl 4/10 (Medium)

IMC Trading

We begin with the numbers 1 through 10 displayed on a board. You select a number, and then the dealer takes all numbers that divide your chosen number exactly. This process continues until no numbers remain. However, you can only select numbers that still have at least one factor that hasn't yet been taken. If any numbers are left unclaimed at the end, the dealer takes them. Your score is the sum of the numbers you collect. Given an optimal strategy, what is the highest possible score you can achieve?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

318. Easy Card Expectation

Probability Lvl 3/10 (Easy)

Akuna Capital, Optiver

A deck contains 35 cards, each labeled with a unique number from 1 to 35. A player draws four cards at random WITHOUT replacement, and for each card drawn, they receive a payout equal to five times the value of the number on the card. What is the expected total payout for the four cards drawn?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

319. Bot Move

Probability Lvl 6/10 (Medium)

Citadel

You and a friend are playing a game where each player selects an integer between 1 and 100. The player who chooses the strictly higher number wins \$1 from the other player. Additionally, a player can win by selecting a number that is exactly 2 less than the higher number. For example, if you choose 64 and your friend chooses 66, you win. In the event of a tie, there is no outcome. In this scenario, both players design a bot that randomly selects a value X from the range $[1, 100]$ according to a specific distribution that the player can decide. Both players can observe the strategy of the other player's bot before deploying their own. What is $\text{Var}(X)$ for the optimal distribution?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

320. Rigged Game

Probability Lvl 4/10 (Medium)

N/A

A game is played as follows: The player first pays d dollars to the dealer, where d is a positive integer less than 100. The dealer then flips a coin d times. If two heads appear consecutively during the flips, the game stops, and the player wins 100 dollars. If no such pair of consecutive heads appears, the player loses and wins nothing. A player pays 29 dollars to participate, and before the game begins, the dealer adjusts the probability of landing heads to ensure the player's expected net profit is zero or less. The player is unaware of this adjustment. Determine the maximum probability the dealer should use.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

321. Maximising Heads I

Probability Lvl 2/10 (Easy)

N/A

We toss four fair coins once, obtaining an initial configuration of heads and tails. After seeing this outcome, we may perform the following operation any number of times (including zero): choose exactly two coins and invert their states, meaning each selected coin changes from heads to tails or from tails to heads. Your goal is to maximize the number of heads. What is the expected number of heads under optimal play?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

322. Distinct Number Count

Probability Lvl 5/10 (Medium)

N/A

Given the set of numbers from 1 to n : $\{1, 2, 3, \dots, n\}$, we randomly draw n numbers from this set with replacement, where each number is drawn with uniform probability. What is the expected number of distinct values obtained when $n = 4$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

323. Non-linearity Of Expectation

Probability Lvl 7/10 (Hard)

N/A

We have 6 boxes numbered 1 to 6. We perform 6 moves. In the i -th move, we place a ball uniformly at random into one of the first i boxes. After all 6 moves, we define the score as the sum of the squares of the number of balls in each box. For example, if the final distribution of balls is: box 1 has 2 balls, box 2 has 1 ball, box 3 has none, and boxes 4, 5, and 6 have 1 ball each, then the score would be $2^2 + 1^2 + 0^2 + 1^2 + 1^2 + 1^2 = 8$. What is the expected value of this score?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

324. Noodle Loop

Probability Lvl 8/10 (Hard)

Citadel, Hudson River Trading, IMC Trading, Jane Street

You have 100 noodles in a bowl, and you randomly pick two ends at a time to tie together, creating longer noodles. You repeat this process until all the ends are connected. On average, how many loops (closed circles) will be formed by the time you're done?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

325. Dice Sum Parity

Probability Lvl 4/10 (Medium)

Hudson River Trading, Optiver

A fair 6-sided die is rolled until a 6 appears. What is the probability that the sum of all rolled numbers is even?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

326. Nearest Diagonal I

Probability Lvl 3/10 (Easy)

N/A

A point is chosen uniformly at random from the rectangle with vertices $(0, 0), (2, 0), (2, 1), (0, 1)$. What is the expected perpendicular distance from this point to the line joining $(0, 0)$ and $(2, 1)$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

327. No Adjacent Standing Pair

Probability Lvl 8/10 (Hard)

N/A

Eight individuals are arranged in a circle, each with a fair coin. They simultaneously flip their coins, with a result of heads causing the person to stand and tails causing them to stay seated. Determine the probability that no two standing individuals are directly next to one another.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

328. Odd Heads III

Probability Lvl 5/10 (Medium)

N/A

You have 10 coins, each of which lands on heads with probability $\frac{1}{3}$ and tails with probability $\frac{2}{3}$. When you flip all these coins, what is the probability that you end up with an odd number of heads?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

329. Handshakes On A Chessboard

Probability Lvl 5/10 (Medium)

N/A

Imagine 64 people positioned on an 8×8 chessboard, each randomly facing one of their neighboring squares. If two individuals end up facing one another, they will shake hands. What is the expected number of handshakes that will occur?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

330. Erase

Brainteasers Lvl 3/10 (Easy)

N/A

Erase 10 digits from the number 1234512345123451234512345 so that the remaining number is as large as possible.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

331. Permutation Blocks

Probability Lvl 9/10 (Very Hard)

N/A

A permutation of $\{1, 2, \dots, 7\}$ is chosen uniformly at random. A partition of the permutation into contiguous blocks is correct if, when each block is sorted independently, the entire permutation becomes sorted. For example, the permutation $(3, 4, 2, 1, 6, 5, 7)$ can be partitioned correctly into the blocks $[3, 4, 2, 1]$ and $[6, 5, 7]$, since when these blocks are sorted, the permutation becomes $(1, 2, 3, 4, 5, 6, 7)$. Find the expected value of the maximum number of blocks into which the permutation can be partitioned correctly.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

332. Count Paths III

Combinatorics Lvl 8/10 (Hard)

N/A

How many paths can be taken from $(0, 0)$ to $(5, 3)$, moving only one unit up or right at each step, without crossing the line $y = x$ but allowing contact with it?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

333. Tigers And The Sheep

Brainteasers Lvl 2/10 (Easy)

Citadel

Hundred tigers and one sheep are put on a magic island that only has grass. Tigers can live on grass, but they want to eat sheep. If a tiger bites the sheep, then it will become a sheep itself. If 2 tigers attack a sheep, only the first tiger to bite converts into a sheep. Tigers don't mind being a sheep, but they have a risk of getting eaten by another tiger. All tigers are intelligent and want to survive. Will the sheep survive?

Enter 1 for YES and 2 for NO.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

334. Unique Birthdays Expectation

Probability Lvl 6/10 (Medium)

N/A

Let there be n individuals. Let $p(n)$ denote the expected number of days in a year on which exactly one person has a birthday. Given that there are exactly 2 values of n that maximize $p(n)$, find the values of n that maximize $p(n)$.

If the answers are x and y , enter the value of $x+y$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

335. Simpson's Paradox

Brainteasers Lvl 4/10 (Medium)

Optiver

In a football match, Team X completes a higher percentage of its passes than Team Y in the first half, and again completes a higher percentage than Team Y in the second half. Could it happen that Team Y ends up with a higher overall pass-completion percentage for the entire game? If no, why? If yes, give an example.

Enter 1 for Yes and 2 for No.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

336. 2014 Stones

Brainteasers Lvl 5/10 (Medium)

N/A

Alice and Bob are playing a game with a pile of 2014 stones. Alice and Bob alternate turns, with Alice going first. On each turn, Alice must take a number of stones that is a power of 2 (such as 1, 2, 4, 8, etc.), while Bob must take a number of stones that is a power of 4 (such as 1, 4, 16, 64, etc.). The player who takes the last stone wins the game. Which player has a winning strategy under optimal play.

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

337. Chess Tournament IV

Probability Lvl 6/10 (Medium)

N/A

There are 32 teams, ranked 1 through 32, entering a chess tournament that proceeds as follows: the teams are randomly paired, and the losing team from each pair is eliminated. The remaining 16 teams are randomly paired, and this process continues until a winner is determined. A higher-ranked team always wins against a lower-ranked team. What is the probability that the team ranked 3 (the third-best team) is one of the last four teams remaining?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

338. Interesting Summation I

Brainteasers Lvl 7/10 (Hard)

N/A

Determine the value of the sum

$$\sum_{k \in T} \frac{1}{k^3}$$

where T represents all positive integers such that the decimal expansion of $\frac{1}{k}$ is finite. For example, $2 \in T$ because $\frac{1}{2} = 0.5$ terminates, while $7 \notin T$ since $\frac{1}{7} = 0.\overline{142857}$ is non-terminating.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

339. Chess Tournament I

Probability Lvl 6/10 (Medium)

Five Rings

In a single-elimination (knockout) chess tournament with 128 players, each competitor has a unique rating. In every match, the higher-rated player wins and advances to the next round. The tournament is structured in 7 rounds, resulting in a final match. What is the probability that the two highest-rated players meet in the final round?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

340. How Many 5s

Probability Lvl 6/10 (Medium)

N/A

You roll a fair 6-sided die until you roll a 6. Calculate the expected number of times you roll a 5 before getting your first 6.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

341. Sphere Variance

Probability Lvl 5/10 (Medium)

Graviton, Two Sigma

Bob selects a random point (x, y, z) uniformly from the surface of the unit sphere. Determine the variance of the z -coordinate.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

342. B Before O I

Combinatorics Lvl 4/10 (Medium)

N/A

Determine the number of distinct anagrams of BOOLAHUBBOO in which all the B's appear before the first O.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

343. Count Regions II

Combinatorics Lvl 5/10 (Medium)

N/A

Determine the maximum number of regions into which a plane can be divided using n circles. For example, one circle divides the plane into 2 regions: the region inside the circle and the region outside it.

Enter the answer for $n = 10$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

344. Square Fitting I

Probability Lvl 4/10 (Medium)

Jump Trading

Nine points are chosen uniformly at random inside the square with corners $(\pm 1, \pm 1)$. Find the expected area of the largest square, centered at the origin, that contains none of the nine points in its interior.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

345. Ant On An Icosahedron

Probability Lvl 8/10 (Hard)

N/A

An ant begins at a vertex of a convex regular icosahedron (a figure with 20 triangular faces and 12 vertices). The ant moves along one edge at a time. Each time the ant reaches a vertex, it randomly chooses to next walk along any of the edges extending from that vertex (including the edge it just arrived from). Find the probability that after walking along exactly six (not necessarily distinct) edges, the ant finds itself at its starting vertex.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

346. Count 9s

Brainteasers Lvl 3/10 (Easy)

Akuna Capital

How often does the digit 9 appear when counting from 1 to 1000? For instance, in the number 199, the digit 9 is counted twice as it appears in two positions.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

347. Grid Travel I

Combinatorics Lvl 5/10 (Medium)

N/A

On the Cartesian grid, Johnny wants to travel from $(0, 0)$ to $(5, 1)$, and he wants to pass through all twelve points in the set $S = \{(i, j) \mid 0 \leq i \leq 1, 0 \leq j \leq 5, i, j \in \mathbb{Z}\}$. Each step, Johnny may go from one point in S to another point in S by a line segment connecting the two points. How many ways are there for Johnny to start at $(0, 0)$ and end at $(5, 1)$ so that he never crosses his own path?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

348. Jumping Frog III

Combinatorics Lvl 3/10 (Easy)

N/A

A frog starts at point 0 and wants to reach point 10. The frog can jump either 1 or 2 points at each step. How many unique ways can the frog get to point 10?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

349. Seven Colors I

Probability Lvl 2/10 (Easy)

N/A

You have 140 balls placed uniformly at random in a row: 20 of each colour - violet, indigo, blue, green, yellow, orange, red. You are told that among the first eight positions there are exactly six more red balls than yellow balls. Given this information, find the probability that the ball in position 71 is violet.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

350. Triangle Angle I

Probability Lvl 1/10 (Easy)

N/A

A right triangle is formed with legs A and B , where both A and B are independently and identically distributed (i.i.d.) random variables, each following a uniform distribution over $[0, 1]$. Let θ represent the angle opposite to side A . Determine the probability that $\theta > \frac{\pi}{4}$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

351. Ant On A Cube III

Probability Lvl 6/10 (Medium)

N/A

A mouse walks along the edges of a unit cube whose vertices are given by (x, y, z) with $(x, y, z) \in \{0, 1\}$. The mouse starts at vertex $A = (0, 0, 0)$. There is a mouse trap at vertex $E = (1, 1, 0)$ and a piece of cheese at vertex $H = (1, 1, 1)$. On each move, the mouse chooses to move to one of the adjacent vertices with equal probability. What is the probability that the mouse reaches the cheese before it encounters the trap?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

352. Odd Sum

Probability Lvl 4/10 (Medium)

N/A

Four different positive integers less than 10 are chosen randomly. What is the probability that their sum is odd?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

353. Coin Equalizer

Probability Lvl 7/10 (Hard)

Citadel

We flip a biased coin repeatedly until the number of heads seen equals the number of tails for the first time. The coin shows heads with probability $p = \frac{1}{3}$. What is the probability that this first time of equality occurs exactly on the 10th flip?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

354. Combining Turtles

Probability Lvl 3/10 (Easy)

N/A

There are 2017 turtles in a room. Every second, two turtles are chosen uniformly at random and combined to form one super-turtle. (Super-turtles are still turtles.) What is the probability that after 2015 seconds (meaning when there are only two turtles remaining) there is some turtle that has never been combined with another turtle?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

355. Roll Die Till 65**Probability** Lvl 4/10 (Medium)

N/A

What is the expected number of rolls of a 6-sided die until the sequence "6 followed by 5" first appears?

[\[Hint ↓ | Solution ↓ | ↑ \]](#)

356. Assigning Numbers To Dice**Combinatorics** Lvl 4/10 (Medium)

DRW

In how many unique ways can you assign numbers to the faces of a blank 6-sided die, considering that dice identical under rotation are counted as the same?

[\[Hint ↓ | Solution ↓ | ↑ \]](#)

357. Diagonal Cut I**Brainteasers** Lvl 4/10 (Medium)*Susquehanna International Group*

We have a rectangular grid consisting of 199×991 small squares. If a diagonal is drawn from one corner to the opposite corner, how many squares does the diagonal go through?

[\[Hint ↓ | Solution ↓ | ↑ \]](#)

358. Meet Again**Brainteasers** Lvl 1/10 (Easy)*Akuna Capital, IMC Trading*

The clock's minute and hour hands meet at 12:00 PM. After how many hours will they meet again?
Enter your answer rounded to 5 decimal places.

[\[Hint ↓ | Solution ↓ | ↑ \]](#)

359. Non-empty Buckets**Probability** Lvl 3/10 (Easy)

N/A

Given K balls and N buckets, find the expected number of buckets that contain at least one ball. Each ball is randomly placed in a bucket. Consider $N = 10$ and $K = 5$.

Enter your answer rounded to 5 decimal places.

[\[Hint ↓ | Solution ↓ | ↑ \]](#)

360. Maximize Product**Brainteasers** Lvl 5/10 (Medium)

N/A

You have 100 coins and are allowed to arrange them into any number of stacks with any number of coins in each stack. The payout is determined by the product of the number of coins in each stack. What is the maximum payout achievable?

[\[Hint ↓ | Solution ↓ | ↑ \]](#)

361. End With 2018**Probability** Lvl 3/10 (Easy)

N/A

Alice starts with an empty string and randomly appends one of the digits 2, 0, 1, or 8 until the string ends with 2018. What is the probability Alice appends less than 9 digits before stopping?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

362. Balls In Garbage Bag

Probability Lvl 7/10 (Hard)

Hudson River Trading

There are m^2 balls and m buckets. Each ball is of one of m types, with m balls for each type. When a ball is drawn and a bucket is randomly chosen, if the bucket does not contain a ball of the same type, the ball is placed in the bucket. If the bucket already contains a ball of that type, the ball is discarded into a garbage bag. What is the expected number of balls that end up in the garbage bag?

Enter the answer for $m = 10$ rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

363. Coin Coloring Game

Probability Lvl 7/10 (Hard)

N/A

Reimu and Sanae play a game using 4 fair coins. Initially both sides of each coin are white. Starting with Reimu, they take turns to color one of the white sides either red or green. After all sides are colored, the 4 coins are tossed. If there are more red sides showing up, then Reimu wins, and if there are more green sides showing up, then Sanae wins. However, if there is an equal number of red sides and green sides, then neither of them wins. Given that both of them play optimally to maximize the probability of winning, what is the probability that Reimu wins?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

364. Sufficient Sum II

Probability Lvl 9/10 (Very Hard)

Akuna Capital, IMC Trading

Let $X_1, X_2, \dots$ be i.i.d. $\text{Unif}(0, 1)$ random variables, and for $x > 0$ define $N_x = \min\{n : X_1 + \dots + X_n > x\}$. Find $E[N_{\ln(2)}]$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

365. Asking For A Date

Probability Lvl 7/10 (Hard)

N/A

Every day, Heesu talks to Sally with some probability p . One day, after not talking to Sally the previous day, Heesu resolves to ask Sally out on a date. From now on, each day, if Heesu has talked to Sally each of the past four days, then Heesu will ask Sally out on a date. Heesu's friend remarked that at this rate, it would take Heesu an expected 2800 days to finally ask Sally out. Find p .

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

366. Digit Picking

Probability Lvl 5/10 (Medium)

N/A

Bob randomly picks 3 distinct numbers from the set $1, 2, 3, \dots, 7, 8, 9$ and arranges them in descending order to form a 3-digit number. Alice randomly picks 3 distinct numbers from the set $1, 2, 3, \dots, 6, 7, 8$ and also arranges them in descending order to form a 3-digit number. What is the probability that Bob's number is larger than Alice's number?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

367. Equal Heads

Probability Lvl 1/10 (Easy)

N/A

Two players each flip a fair coin four times and record the number of heads they get. What is the probability that both players flip the same number of heads?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

368. Car Crash

Probability Lvl 2/10 (Easy)

Five Rings, Jump Trading, Optiver, Susquehanna International Group

At a certain busy intersection, the likelihood of having at least one car accident during a one-hour period is $\frac{8}{9}$. Car crashes are assumed to happen at a steady rate, and each incident occurs independently of others. Determine the probability of at least one car accident occurring within a 30-minute period.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

369. Interesting Average II

Brainteasers Lvl 5/10 (Medium)

N/A

Let $S = 1, 2, \dots, 2008$. For any nonempty subset $A \subseteq S$, define $m(A)$ to be the median of A (when A has an even number of elements, $m(A)$ is the average of the middle two elements). Determine the average of $m(A)$, when A is taken over all nonempty subsets of S .

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

370. Dice Permutation

Probability Lvl 6/10 (Medium)

N/A

What is the expected number of rolls of a 6-sided die until all six faces appear in some order within six consecutive rolls?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

371. Is It A Perfect Square?

Brainteasers Lvl 4/10 (Medium)

Two Sigma

Is it possible for a number composed of one hundred 0's, one hundred 1's, and one hundred 2's (in some order) to be a perfect square?

Enter 1 if it is possible and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

372. Die Roll LCM

Probability Lvl 7/10 (Hard)

N/A

A fair 10-sided die (faces numbered 1 through 10) is rolled repeatedly until the least common multiple (LCM) of all the outcomes observed so far first exceeds 2000. What is the expected number of rolls needed?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

373. Higher Roll II

Probability Lvl 4/10 (Medium)

N/A

Alice rolls a 20-sided die once. Bob rolls a 15-sided die r times and keeps the maximum value from his rolls. What is $\lim_{r \rightarrow \infty} \mathbb{P}(\text{Alice's roll is strictly greater than Bob's maximum})$?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

374. N Point Variance

Probability Lvl 4/10 (Medium)

Susquehanna International Group

You randomly select N points independently and uniformly within the unit square $[0, 1] \times [0, 1]$. Let I_i be the indicator variable equal to 1 if the i -th point lies inside the quarter-circle defined by $x^2 + y^2 = 1$ in the first quadrant, and equal to 0 otherwise. What is the variance of the random variable $\frac{I_1 + I_2 + \dots + I_N}{N}$ in terms of N ?

Enter the answer for $n = 10$ rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

375. Game Of Cuts

Brainteasers Lvl 5/10 (Medium)

Susquehanna International Group

In a certain two-player game, we start with a rectangular 1985×2015 grid of squares. On each turn, a player either makes a horizontal cut and takes away the portion of the rectangle above the cut, or makes a vertical cut and takes away the portion to the right. Whichever player takes the last square (in the bottom left corner) loses. If both players play perfectly, who has a winning strategy—the first or the second player?

Enter 1 for Alice and 2 for Bob.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

376. Next Date I

Brainteasers Lvl 4/10 (Medium)

N/A

Find the next date where all digits in the format MM/DD/YYYY are distinct. For example, 02/19/3456 is a valid date.

Enter the value of MM+DD+YYYY. For example, if the answer is 02/19/3456, you should enter 02+19+3456=3477

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

377. Dice Game I

Probability Lvl 4/10 (Medium)

Akuna Capital, Citadel, DRW, Hudson River Trading, Optiver, Susquehanna International Group

You roll a die and get paid what the die shows. If you prefer, you can choose to roll again and take the value of the second roll instead. What is the expected value of the amount you get if you play optimally?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

378. Poisoned Wine II

Brainteasers Lvl 10/10 (Very Hard)

Five Rings, Jane Street, Susquehanna International Group, Two Sigma

A king has 5 servants who willingly risk their lives to identify a poisonous bottle among n bottles. It is known that exactly one of these bottles contains poison. Any servant who consumes from the poisoned bottle will die exactly 1 month later; those who drink from the safe bottles will remain unharmed. The servants plan to taste the wine over a period of 3 months. What is the maximum number of bottles n such that the king can definitively determine which bottle is poisoned?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

379. Poisson Cards

Probability Lvl 8/10 (Hard)

N/A

Given a box with X cards, where X follows a Poisson distribution with mean 6, find the expected number of ways to select 4 cards from this box. $\binom{n}{k}$ will be considered zero if $n < k$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

380. Min Collinearity

Brainteasers Lvl 4/10 (Medium)

N/A

Define a lattice point to be a point (x, y) whose coordinates x and y are integers. Three points are collinear if there is a line that passes through them. For example, the points $(1, 3)$, $(2, 6)$, and $(3, 9)$ are collinear. What is the minimum number n such that, given n lattice points, with $0 \leq x \leq 4$ and $0 \leq y \leq 4$, there must be 3 of these points collinear?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

381. Losing It All

Probability Lvl 2/10 (Easy)

Jump Trading

Alice is participating in a game where she has a winning chance of 0.1. To play, she pays 10 dollars, and a win earns her 80 dollars. Starting with 30 dollars, what is the probability, that she wins exactly once before losing all her money?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

382. Handy Subsets

Combinatorics Lvl 10/10 (Very Hard)

N/A

Let $T = \{1, 2, 3, \dots, 14, 15\}$. A subset S of T is considered handy if the sum of all the elements of S is a multiple of 5. For instance, the empty set is handy (because its sum is 0) and T itself is handy (because its sum is 120). Compute the number of handy subsets of T .

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

383. Last To Front

Probability Lvl 5/10 (Medium)

Graviton

5 people stand in a line facing one direction. In every round, the person at the front moves randomly to any position in the line, including the front or the end. What is the expected number of rounds needed for the last person of the initial line to appear at the front of the line?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

384. Coin Toss Movement

Probability Lvl 4/10 (Medium)

N/A

Bob starts at 0 on the real number line. He tosses a fair coin 8 times. When he gets heads, he moves 1 unit in the positive direction; when he gets tails, he moves 1 unit in the negative direction. What is the probability that he reaches 4 at some time?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

385. 1s And 9s

Probability Lvl 5/10 (Medium)

N/A

An 8-digit number is generated by first selecting a random integer k uniformly from the set $\{0, 1, 2, \dots, 8\}$. Afterward, an 8-digit number is chosen uniformly at random from the set of numbers with exactly k ones and $8 - k$ nines. Find the expected value of this number.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

386. Low Median

Probability Lvl 1/10 (Easy)

IMC Trading

Consider drawing three independent random numbers, X_1, X_2, X_3 , from the uniform distribution $[0, 1]$ and then sorting them in ascending order. Let $X_{(2)}$ denote the median of these sorted values. What is the probability that this median value $X_{(2)}$ is less than $\frac{2}{3}$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

387. Connected Components I

Probability Lvl 4/10 (Medium)

Jump Trading

You are given 15 squares arranged in a row. Each square is independently colored either black with probability $\frac{1}{3}$ or white with probability $\frac{2}{3}$. A connected component is defined as a maximal sequence of adjacent squares that are all the same color. For example, in the coloring BBWWBBB, there are three connected components: the first two black squares form one component, the next two white squares form another, and the final three black squares form the third. What is the expected number of connected components in such a randomly colored row?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

388. 3 Heads 3 Tails II

Probability Lvl 4/10 (Medium)

N/A

Alice and Bob are engaged in a coin-tossing game. Alice wins if three heads come up, while Bob wins if three tails are tossed. The heads and tails don't need to be in a row. How many tosses, on average, will it take to declare a winner?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

389. Keep The GCD High

Brainteasers Lvl 10/10 (Very Hard)

N/A

In a box, we have cards with all the pairs (a, b) where $1 \leq a < b \leq 2003$ (no cards are repeated and all cards have only one pair). Alice and Bob are going to play by turns to remove cards from the box. At each turn, when they remove a card with the pair (a, b) , they write on a blackboard the product ab of the numbers in the card. The first one to make the greatest common divisor of the numbers on the blackboard to be 1 loses. If Alice plays first, who has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

390. Moth On A Cube

Combinatorics Lvl 4/10 (Medium)

N/A

A moth begins its journey at vertex A of a cube and aims to reach vertex B, which is diagonally opposite to A. The moth must travel along the edges of the cube, and a single movement along an edge is considered a "step." The goal is for the moth to reach vertex B in five steps or fewer, and it will stop immediately upon reaching vertex B. Determine the total number of possible paths the moth can take to achieve this objective.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

391. Expected Arc Length I

Probability Lvl 5/10 (Medium)

N/A

Thirty points $X_1, X_2, \dots, X_{30}$ are independently and uniformly chosen at random on the circumference of a unit circle. What is the expected length of the arc between X_1 and the nearest of the remaining 29 points?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

392. Dice Difference II

Probability Lvl 4/10 (Medium)

N/A

Alice rolls three fair dice. What is the probability that the difference between the highest and lowest numbers rolled is exactly four?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

393. Game On 2 Piles

Brainteasers Lvl 4/10 (Medium)

N/A

Alice and Bob are playing a game with two piles of stones, each containing the same number of stones. The players take turns removing stones from one pile at a time. On each turn, a player may remove any number of stones, even the entire pile, but only from one pile. The game ends when the last stone is removed, and the player who does so wins. Alice takes the first turn. Who has a winning strategy if the first pile contains 1933 stones and the second pile contains 2032 stones?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

394. 5-runs

Probability Lvl 10/10 (Very Hard)

N/A

We have twenty squares in a row. Each of the squares is painted either black or white with equal probability. A 5-run is any set of exactly five black squares in a row, and once you use a black square as part of one 5-run, you don't use it again. For example, in the sequence WWBBBBBW there is exactly one 5-run (the five consecutive B's in the middle). On the other hand, in the sequence WWBBBBBBBBBB, there are two 5-runs. What is the expected number of 5-runs you will see in a randomly colored line of twenty squares?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

395. Dice Difference I

Probability Lvl 5/10 (Medium)

Akuna Capital

On average, how many rolls of a fair 6-sided die are needed before rolling two consecutive numbers that differ by 2?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

396. 7994 Stones

Brainteasers Lvl 8/10 (Hard)

N/A

There are 7994 stones on a table. Alice and Bob play by turns to remove stones from the table. At each turn, they must remove 1, 3, 4, 5, or 7 stones. Whoever removes the last stone wins. If Alice plays first, who has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

397. Measuring Everything II

Brainteasers Lvl 9/10 (Very Hard)

Susquehanna International Group

What is the minimum number of weights needed to measure any integer weight from 1 to 100 grams on a standard balance with two pans, where these weights can be placed on both pans?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

398. Ending At 1

Probability Lvl 4/10 (Medium)

N/A

Every second, Andrea writes down a random digit uniformly chosen from the set $\{1, 2, 3, 4\}$. She stops when the last two numbers she has written sum to a prime number. What is the probability that the last number she writes down is 1?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

399. Points On A Square

Probability Lvl 5/10 (Medium)

Hudson River Trading

Two points are chosen uniformly at random on the sides of a square with side length 1. What is the probability that the distance between them is greater than 1?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

400. Standing Table III

Probability Lvl 8/10 (Hard)

Citadel, Jane Street

Five points are chosen independently and uniformly at random from the area of a circular table (not restricted to the circumference). Legs are attached at those five interior points. What is the probability that the table stands?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

401. Hungry Driver

Brainteasers Lvl 8/10 (Hard)

Akuna Capital, Optiver

A truck driver has been tasked with transporting 3000 apples between towns A and B , which are 1000 miles apart. However, if the truck driver is driving while there are apples in his truck, he will eat them at a rate of 1 apple per mile. The truck has a maximum capacity of 1000 apples at any one time. The driver may deposit apples at any point between A and B . How can he get the maximum possible number of apples from town A to town B ?

Enter the maximum possible number of apples that can be transported as the answer.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

402. Sufficient Sum III

Probability Lvl 9/10 (Very Hard)

Akuna Capital

Let $X_1, X_2, \dots$ be i.i.d. $\text{Unif}(0, 1)$ random variables, and for $x > 0$ define $N_x = \min\{n : X_1 + \dots + X_n > x\}$. Compute the variance $\text{Var}(N_{\ln(2)})$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

403. Conditional One

Probability Lvl 2/10 (Easy)

N/A

You draw five distinct integers one after another from the set $\{1, 2, \dots, 100\}$ without replacement. You are told afterwards that the five drawn numbers happened to appear in increasing order (so each draw was larger than the previous). Given this information, what is the probability that the number 1 was the first number drawn?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

404. Average Drawn

Probability Lvl 3/10 (Easy)

N/A

A box contains 20 balls labeled 1 through 20. Three balls are drawn at random without replacement. What is the probability that the number on one of the drawn balls is exactly the average of the numbers on the other two?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

405. Too Many Hats

Brainteasers Lvl 10/10 (Very Hard)

Susquehanna International Group

N people form a team and devise a strategy for a game before entering a room. Upon entering, each person receives a hat with one of the first N natural numbers written on it, with possible duplicates among the numbers. For instance, if $N = 3$, the hats might display the numbers 2, 0, and 2. Each person can see the numbers on the hats of the others but cannot see their own hat number. They then simultaneously guess the number on their own hats. What strategy can the team employ to ensure that at least one member correctly guesses their hat number?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

406. Watermelon Weight

Brainteasers Lvl 1/10 (Easy)

N/A

A 100 kg watermelon is initially composed of 99% water. After being exposed to the sun for some time, the water content decreases to 98%. What is the current weight of the watermelon?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

407. Intersecting Subsets

Probability Lvl 5/10 (Medium)

N/A

Let $S = \{1, 2, \dots, 25\}$. Two subsets A and B are chosen independently and uniformly at random from the power set of S . Define the random variable $N = |A \cap B|$. Compute the ratio $\frac{\mathbb{E}[N]}{\text{Var}(N)}$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

408. Bernoulli Matrix

Probability Lvl 2/10 (Easy)

Jump Trading

Let A be a random 2×2 matrix where each element is independently drawn from a Bernoulli(0.5) distribution. Compute $\mathbb{E}[\det(A - A^T)]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

409. B Before O II

Combinatorics Lvl 4/10 (Medium)

N/A

Determine how many anagrams of BOOLAHUBBOO have at least two B's appearing before the first O.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

410. Bacteria Population III

Brainteasers Lvl 6/10 (Medium)

N/A

You have an empty, infinitely large petri dish. Each day, you can add any number of bacteria to the dish. Then, at night, every bacterium divides into two—that is, the total number of bacteria doubles overnight. You can repeat this process as many days as you like. Your goal is to end up with exactly 2025 bacteria in the dish at some point. What is the minimum number of bacteria you need to add over all the days (in total), assuming you have unlimited time?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

411. Sorted Rows

Probability Lvl 6/10 (Medium)

N/A

The integers from 1 to 25, inclusive, are randomly placed into a 5 by 5 grid such that in each row, the numbers are increasing from left to right. If the columns from left to right are numbered 1, 2, 3, 4, and 5, then what is the expected column number of the entry 23?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

412. Coin Reflip Game

Probability Lvl 2/10 (Easy)

N/A

Flip four coins. For every head, you get \$1. You may reflip one coin after the four flips. Calculate the expected returns.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

413. Large Language Model

Combinatorics Lvl 7/10 (Hard)

Graviton

A group consists of 69 members. For any two members a and b , there exists a language that a speaks but b does not, and there is a language that b speaks but a does not. What is the minimum number of distinct languages spoken within this group? Assume that each of the 69 members speaks the same number of languages. For example, it is not possible for the first member to speak 4 distinct languages while the second member speaks 9 distinct languages.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

414. Strange Die

Probability Lvl 5/10 (Medium)

N/A

Jim has two fair 6-sided dice, one whose faces are labelled from 1 to 6, and the second whose faces are labelled from 3 to 8. Twice, he randomly picks one of the dice (each die equally likely) and rolls it. Given the sum of the resulting two rolls is 9, what is the probability that he rolled the same die twice?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

415. Cannons

Combinatorics Lvl 2/10 (Easy)

IMC Trading

There are n pirates, each equipped with 2 cannons. Each pirate points their cannons at 2 distinct pirates among the remaining $n - 1$. Determine the minimum value of n such that it is possible for no two pirates to be mutually targeting each other with their cannons.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

416. Field Construction

Probability Lvl 2/10 (Easy)

N/A

A field is being set up with target dimensions of 300 feet in length and 200 feet in width. Due to measurement inaccuracies, the length error X follows a uniform distribution $X \sim \text{Unif}(-15, 15)$, and the width error Y follows a uniform distribution $Y \sim \text{Unif}(-5, 15)$, with X and Y being independent. The actual dimensions of the field are $300 + X$ and $200 + Y$. Determine the expected area of the field.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

417. Chocolate Bar

Brainteasers Lvl 3/10 (Easy)

N/A

There is a 6×8 rectangular chocolate bar made up of small 1×1 bits. We want to break it into the 48 bits. We can break one piece of chocolate horizontally or vertically, but cannot break two pieces together! What is the minimum number of breaks required?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

418. Unfair Rock Paper Scissors

Probability Lvl 5/10 (Medium)

N/A

Alice and Bob are playing rock-paper-scissors. Alice, however, is cheating, so in each round, she has a $\frac{3}{5}$ chance of winning, $\frac{2}{5}$ chance of drawing, and $\frac{2}{5}$ chance of losing. The first person to win 5 more rounds than the other person wins the match. What is the probability Alice wins?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

419. Sum Over One I

Probability Lvl 7/10 (Hard)

N/A

Suppose we have independent random variables $X_1, X_2, \dots, X_n$ that are uniformly distributed on the interval $[0, 1]$. Let $U = \min(X_1, \dots, X_n)$ and $V = \max(X_1, \dots, X_n)$. Find the probability that $U + V > 1$.

Enter the answer for $n = 10$ rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

420. Exactly 2 Consecutive Heads

Probability Lvl 4/10 (Medium)

N/A

You are given a coin for which you can choose the probability p of flipping heads. You will flip the coin exactly four times. What value of p will maximize the likelihood of getting exactly two consecutive heads within the four flips? Sequences like $HHHT$ and $HHHH$ are not permitted, but a sequence like $HTHH$ is acceptable.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

421. Reach Multiple Of 7

Probability Lvl 3/10 (Easy)

N/A

A and B are playing a game in which they alternately roll a fair six-sided die, with A rolling first. Each roll, regardless of who rolls, is added to a shared total score. The first player to reach a multiple of 7 wins. What is the probability that B wins?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

422. Painted Cube I

Probability Lvl 4/10 (Medium) *Five Rings, Hudson River Trading, Jane Street, Susquehanna International Group*

A $10 \times 10 \times 10$ cube is painted red on its surface and then cut into 1000 $1 \times 1 \times 1$ cubes. One of these smaller cubes is selected at random. Determine the expected number of red faces on the selected cube.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

423. Nearest Circular Neighbor

Probability Lvl 5/10 (Medium)

N/A

Imagine selecting 20 independent and identically distributed (IID) points randomly on the edge of a unit circle, labeled $P_1, P_2, \dots, P_{20}$. Determine the expected arc length, in degrees, between P_1 and its closest neighboring point.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

424. Archery I

Probability Lvl 5/10 (Medium)

N/A

Bob is practicing archery. He hits his first target but misses his second target. For $n \geq 3$, the probability of hitting the n th target is equal to the proportion of targets he hit during his first $n - 1$ attempts. How many targets can Bob expect to hit in 100 attempts?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

425. Same Sum Subsets

Brainteasers Lvl 10/10 (Very Hard)

Optiver

You choose 10 different numbers between 1 and 100. Is it possible to select them in a way that no two nonempty, disjoint subsets of those 10 numbers have the same sum?

Enter 1 if it is possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

426. Pistol Shots

Brainteasers Lvl 3/10 (Easy)

N/A

A pistol can fire six shots in succession, with a total time of 60 seconds from the first shot to the sixth. How much time would it take to fire three shots? The answer is NOT 30 seconds.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

427. Buying Tickets

Probability Lvl 8/10 (Hard)

N/A

At a theater box office, there are 12 people waiting in line to purchase tickets. Six of them have only \$5 bills, while the other six possess only \$10 bills. The ticket seller, however, starts without any change. Each individual intends to buy a ticket costing \$5. What is the probability that everyone can successfully purchase their ticket without any person needing to change their position?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

428. First Ace

Probability Lvl 6/10 (Medium)

Akuna Capital, Citadel, Hudson River Trading, IMC Trading, Optiver

In a standard deck of 52 playing cards, how many cards on average must be flipped to reveal the first ace?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

429. Inflating Balloon

Brainteasers Lvl 2/10 (Easy)

N/A

You are trying to inflate a large balloon. One pump can inflate it in 20 minutes, while another pump can do it in 10 minutes. Once the balloon is fully inflated, it takes 15 minutes for air to slowly leak out after a valve is opened. How long does it take (in minutes) to inflate the balloon when both pumps are working simultaneously and the valve is open?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

430. Expected Squared Area

Probability Lvl 8/10 (Hard)

N/A

On a 9×9 grid of points with unit spacing, select three points independently and uniformly (with replacement). So, the same point may be selected multiple times. Denote them (x_1, y_1) , (x_2, y_2) , (x_3, y_3) . What is the expected square of the area of the triangle they form?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

431. Singer's Location

Probability Lvl 2/10 (Easy)

Citadel

An artist has an 80% chance of being in Philadelphia and a 20% chance of being in Baltimore. If he is in Baltimore, there is an 80% probability he will perform, while in Philadelphia, the probability of performing is 10%. What is the probability that, given he performs, he is in Philadelphia?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

432. Probability Of Everything

Probability Lvl 6/10 (Medium)

N/A

We roll a 6-sided die n times. What is the probability that all faces have appeared?

Enter the answer for $n = 10$ rounded to 5 decimal places. You may use a calculator when computing the answer.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

433. Party Groups II

Probability Lvl 7/10 (Hard)

Jane Street

There are 50 guests at a party, and they form groups by placing their names into a hat. Each guest randomly picks a name, joining the group of the person they selected. If a guest picks their own name, they form a group alone. If Guest A picks Guest B's name, Guest B picks Guest C's, and Guest C picks Guest A's, they form a closed group. What is the expected size of a group?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

434. Heads Run II

Probability Lvl 10/10 (Very Hard)

N/A

A fair coin is tossed 11 times. What is the probability that the sequence never contains 4 heads or 4 tails in a row (i.e., there is no occurrence of "HHHH" or "TTTT" anywhere in the sequence)?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

435. Largest 2 Factor

Probability Lvl 6/10 (Medium)

N/A

For a positive integer n , let $v(n)$ denote the largest integer j such that n is divisible by 2^j . Let a and b be chosen uniformly and independently at random from among the integers between 1 and 32, inclusive. What is the probability that $v(a) > v(b)$?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

436. Trailing Zeros II

Brainteasers Lvl 5/10 (Medium)

Susquehanna International Group

How many trailing zeros does the value $300 \cdot 305 \cdot 310 \cdot \dots \cdot 1090 \cdot 1095 \cdot 1100$ have?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

437. Sum Of Products

Brainteasers Lvl 7/10 (Hard)

N/A

For each integer from 1 through 2019, Bob calculated the product of its digits. Compute the sum of all 2019 of Bob's products.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

438. Ants On An Octahedron

Probability Lvl 7/10 (Hard)

N/A

Six ants simultaneously stand on the six vertices of a regular octahedron, with each ant at a different vertex. Simultaneously and independently, each ant moves from its vertex to one of the four adjacent vertices, each with equal probability. What is the probability that no two ants arrive at the same vertex?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

439. Broken Stick II

Probability Lvl 5/10 (Medium)

Jump Trading

You have a stick of length 1 meter. You randomly select two points independently along the stick and break it at those points, creating three pieces. What is the probability that each of the three pieces has length at least $\frac{1}{8}$ meters?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

440. Prime Sum

Probability Lvl 1/10 (Easy)

Jane Street

Two distinct prime numbers are chosen uniformly at random from the set of prime integers between 1 and 20. Determine the probability that the sum of the two selected primes is even.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

441. Double Factorial

Brainteasers Lvl 6/10 (Medium)

N/A

The double-factorial of a positive integer n , denoted $n!!$, is the product of all even positive whole numbers less than or equal to n . For instance, $7!! = 6!! = 2 \cdot 4 \cdot 6$. Which is greater, $(1012!!)!$ or $(1012!)!!$? The double-factorial of a positive integer n , denoted $n!!$, is the product of all even positive whole numbers less than or equal to n . For instance, $7!! = 6!! = 2 \cdot 4 \cdot 6$. Which is greater, $(1012!!)!$ or $(1012!)!!$?

Enter 1 for $(1012!!)!$ and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

442. Jumping Frog II

Probability Lvl 6/10 (Medium)

IMC Trading

A frog is on a circle with 100 numbered lily pads, from 0 to 99. The frog starts on lily pad 0 and hops to the nearest lily pad either to the left or right with equal probability each time. What is the probability that when the frog lands on lily pad 50 for the first time, it has visited all other lily pads except 50?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

443. Graph Travel I

Probability Lvl 5/10 (Medium)

Akuna Capital

You have an undirected graph consisting of 10 nodes. Each node can reach any other node, including itself, with an equal probability of $\frac{1}{10}$. What is the expected number of steps needed to visit every node at least once?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

444. Never Divisible By 3

Probability Lvl 8/10 (Hard)

Graviton

Consider the numbers $1, 2, \dots, 13$ written down one by one in a random order. Determine the probability that at no point during this process does the sum of the numbers written so far become divisible by 3.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

445. Expected Minimum Distance

Probability Lvl 4/10 (Medium)

N/A

We have a unit circle. We pick four points independently so that each point is uniformly distributed over the interior of the disk—that is, the chance of landing in any region is proportional to its area. Let R be the distance from the center to the closest of those four points. What is the expected value of R ?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

446. 800 Marbles

Probability Lvl 2/10 (Easy)

N/A

A bag contains 800 marbles, each in one of 100 colors, with eight marbles of each color. Anna draws marbles from the bag one by one without replacement, and she will stop as soon as she has drawn eight marbles of the same color. Given that Anna has not yet stopped after drawing 699 marbles, what is the probability that she will stop immediately upon drawing the 700th marble?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

447. Set Intersection

Probability Lvl 4/10 (Medium)

N/A

Given two sets A with 5000 elements and B with 8000 elements, their intersection $A \cap B$ contains 500 elements. If we randomly sample, with replacement, 500 elements from A (called A') and 800 elements from B (called B'), what is the expected number of elements in $A' \cap B'$? Note that the sampling is done with replacement and independently.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

448. Remainders

Brainteasers Lvl 4/10 (Medium)

N/A

Find the smallest positive integer N such that when N is divided by 2, 3, 4, 5, or 6, the remainder is always 1, and when N is divided by 11, there is no remainder.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

449. Exactly 6

Probability Lvl 1/10 (Easy)

N/A

What is the probability that each number appears exactly 6 times if a fair die is rolled 36 times?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

450. Jumping Frog VII

Probability Lvl 6/10 (Medium)

Five Rings

A frog starts at point 0 and wants to reach point 10. At each jump, the frog can leap any number of points. What is the probability that the number of jumps taken to reach point 10 is odd if all unique ways of reaching point 10 are equally likely?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

451. Russian Roulette II

Probability Lvl 3/10 (Easy)

Jane Street

You and a friend are playing Russian Roulette with a six-chambered revolver loaded with one bullet. The cylinder is spun once at the start and is not re-spun between turns. You alternate pulling the trigger until someone fires the bullet and loses. If your friend goes first, what is the probability that you win?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

452. Segment Overlap

Probability Lvl 4/10 (Medium)

Akuna Capital, Graviton

Let 4 points P_1, P_2, P_3, P_4 be selected randomly in the interval $[0, 1]$. What is the probability that the segment joining P_1 and P_2 overlaps with the segment joining P_3 and P_4 ?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

453. 4 Die Sum

Probability Lvl 3/10 (Easy)

N/A

Determine the probability of rolling a total sum of 20 with four fair six-sided dice.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

454. Poisoned Chocolates

Brainteasers Lvl 8/10 (Hard)

IMC Trading, Jump Trading

Alice and Bob take turns removing pieces from a 4×5 rectangular chocolate bar divided into unit squares, where $m > 1$ and $n > 1$. Each turn involves selecting a square and removing that square along with all squares

above and/or to its right. For example, a player might transform the configuration as follows:

```

o  o  o  o  o
o  o  o  o  o
o  o  o  o  o
o  o  o  o  o

```

to

```

o  o  o
o  o  o
o  o  o
o  o  o  o  o

```

Each player aims to avoid being left with the poisonous lower-left square. Alice goes first. Which player, if any, has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

455. Green And Yellow Dice

Probability Lvl 3/10 (Easy)

DRW

You have two identical 10-sided dice—one green and one yellow—both numbered from 1 to 10. You roll both dice together. If the two dice display the same number, the game finishes. However, if the yellow die is at least 3 greater than the green die, you earn \$1 and roll again. In any other scenario, you receive nothing but still roll both dice again. The game continues until both dice match. What is your expected total payout from this game?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

456. Count Handshake Pairs

Combinatorics Lvl 6/10 (Medium)

N/A

Eight celebrities meet at a party. It so happens that each celebrity shakes hands with exactly two others. A fan makes a list of all unordered pairs of celebrities who shook hands with each other. If order does not matter, how many different lists are possible?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

457. 2x3 Grid

Probability Lvl 3/10 (Easy)

N/A

Each cell of a 2×3 grid is filled with numbers 0, 1, or 2. What is the probability that the sum of the numbers in every row and in every column is a multiple of 3?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

458. Forming Quadratic Polynomial III

Probability Lvl 8/10 (Hard)

N/A

Let b and c be integers chosen randomly (uniformly and independently) from the set $\{-6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$. (Note that b and c can be equal.) What is the probability that the two roots of the quadratic $x^2 + bx + c$ are consecutive integers?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

459. 2D Jumps

Probability Lvl 6/10 (Medium)

Jump Trading

Starting at the origin, you make 10 jumps. On each jump you choose a random direction (uniform on $[0, 2\pi]$) and a random length (uniform on $[0, 1]$) and then move by that amount in that direction. What is the expected square of your final Euclidean distance from the origin after 10 jumps?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

460. Digit Sum I

Combinatorics Lvl 3/10 (Easy)

N/A

What is the sum of the digits of all numbers from 1 to 1 million, inclusive? For instance, the sum of the digits of 45 is 9.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

461. Conditional Inequality

Probability Lvl 3/10 (Easy)

Five Rings

Let X and Y be independent and identically distributed random variables, each following a uniform distribution on $(0, 1)$. Determine $P(X - Y > \frac{1}{2} \mid X + Y < 1)$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

462. Yet Another Game

Brainteasers Lvl 10/10 (Very Hard)

N/A

All positive integers from 1 to n are arranged in a row in increasing order. Alice and Bob are playing a game in which they take turns erasing two consecutive numbers from the row and replacing them either with their product or their sum. Alice wins if the last remaining number is odd, while Bob wins if it is even. Alice takes the first turn. Which player, if any, has a winning strategy for $n = 2024$?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

463. Same Color Corners

Brainteasers Lvl 5/10 (Medium)

N/A

On a 2D complex plane, all the integer-component points are coloured either White or Black. Is it possible to find a rectangle parallel to the axes which has all corners of the same color?

Enter 1 if it is possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

464. No Chords Intersecting

Probability Lvl 6/10 (Medium)

N/A

Twelve points $k_1, k_2, \dots, k_{12}$ are chosen independently and uniformly at random on the circumference of a circle. These points are then paired into six chords by connecting each point k_i with the point k_{i+6} , where the indices are taken modulo 12. What is the probability that none of the six chords intersect each other in the interior of the circle?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

465. Drunk Passenger III

Probability Lvl 10/10 (Very Hard)

N/A

A line of 100 airline passengers is waiting to board a plane. They each hold a ticket to one of the 100 seats on that flight. For convenience, let's say that the n th passenger in line has a ticket for seat number ' n '. Being drunk, the first 10 people in line pick a random seat (equally likely for each seat). All of the other 90 passengers are sober, and will go to their assigned seats unless it is already occupied; if it is occupied, they will then find a free seat to sit in, at random. What is the expected number of passengers that sit on their own seat?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

466. Last Number Deleted

Brainteasers Lvl 7/10 (Hard)

Citadel

Consider the integers from 1 to 2000 arranged in increasing order along the circumference of a circle, proceeding clockwise. Beginning at 1, you delete this number. Then, you move to the next number, 2, and retain it. You continue this alternating process—removing one number, then keeping the next—until completing a full cycle around the circle. Upon completing each full rotation, the process repeats: starting with the deletion of the first remaining number, followed by keeping the second, and so on. Determine the last number that will be deleted from the circle.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

467. Digit Sum II

Combinatorics Lvl 3/10 (Easy)

N/A

How many integers less than or equal to 100,000 have digits that sum to 9?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

468. Digit Multiply II

Brainteasers Lvl 5/10 (Medium)

N/A

Consider the set S consisting of all integers whose digits multiply to 96. Let x and y represent the smallest and largest integers in S , respectively. Determine the difference $y - x$. Note that the digits cannot include 1, ensuring that S contains a finite number of elements.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

469. 500 Sided Dice

Probability Lvl 4/10 (Medium)

Graviton, Hudson River Trading, Susquehanna International Group

When rolling 100 dice, each with 500 sides, what is the probability that the total sum of the faces is divisible by 500?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

470. Moving The Piece II

Brainteasers Lvl 8/10 (Hard)

Citadel

Given a grid with 183 rows and 181 columns, Alice and Bob take turns moving a piece to an adjacent square (diagonally connected squares are not considered adjacent). Once a square is visited, it cannot be revisited. The piece starts in the second column of the first row, which cannot be revisited either. The loser is the player that does not have a legal move to make. If Alice goes first, who has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

471. Grid Arrangement I

Combinatorics Lvl 5/10 (Medium)

N/A

How many 4×4 arrays whose entries are 0s and 1s are there such that the row sums (the sum of the entries in each row) are 1, 2, 3, and 4, in some order, and the column sums (the sum of the entries in each column) are also 1, 2, 3, and 4, in some order? For example, the array

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

satisfies the condition.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

472. Information Transfer

Probability Lvl 5/10 (Medium)

N/A

Imagine there are n individuals, labeled $A_1, A_2, \dots, A_n$. Person A_1 is given a binary message, either 0 or 1, and passes this message to A_2 . In general, each person A_i passes the message to the next person A_{i+1} . However, A_i passes the same message with probability p and the opposite message with probability $1 - p$, where $0 < p < 1$, independently from others. Define A_i as the event that A_i passes the original message from A_1 to A_{i+1} . Let p_i represent the probability of this occurring. Find $\lim_{n \rightarrow \infty} p_n$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

473. Chain Link

Brainteasers Lvl 4/10 (Medium)

Five Rings

What is the fewest number of cuts needed in a chain of 21 links to be able to give someone any number of links, from 1 to 21?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

474. Pandemic I

Brainteasers Lvl 2/10 (Easy)

N/A

In a hall with 1000 individuals, one person is known to be ill while the remaining 999 are healthy. Each minute, every individual randomly converses with another person in the room. There is no restriction on the frequency

of these interactions. If a sick person talks to a healthy person, the healthy individual contracts the illness and remains sick for the duration of the event. Determine the minimum amount of time (in minutes) required for the entire population in the hall to become sick.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

475. Expected Absolute Difference

Probability Lvl 2/10 (Easy)

DRW, Jane Street

Two fair 30-sided dice, each with faces numbered 1 through 30, are rolled independently. Let X and Y represent the outcomes of these rolls. Find the expected value of the absolute difference between X and Y , i.e., calculate $E[|X - Y|]$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

476. Game On Jar

Probability Lvl 8/10 (Hard)

Akuna Capital, Jane Street

A jar holds the numbers 1 through 100. The game proceeds as follows: each round, Rick draws a number from the jar, records it, and puts it back. The last number recorded determines the dollar amount awarded to Rick. He may play as many rounds as desired, with each round costing \$1. Assuming optimal play, what is the fair value of this game?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

477. 3 Bus Stops

Brainteasers Lvl 3/10 (Easy)

Jane Street

An unknown number of people are on a bus. At the first stop, $\frac{3}{4}$ of the passengers get off, and then 7 new passengers board. This process occurs again at two additional stops. What is the minimum possible number of people that could be on the bus after these stops?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

478. Least Multiple Of 15

Brainteasers Lvl 2/10 (Easy)

Jane Street

Determine the smallest positive multiple of 15 that consists solely of the digits 1 and 0.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

479. Password

Combinatorics Lvl 5/10 (Medium)

N/A

Determine the number of 7-digit codes where each digit is at least as large as the previous one. Digits can be repeated and can start with 0. For example, codes like 2233445 and 2222222 are valid, but 1234564 is not.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

480. Bridge Crossing

Brainteasers Lvl 5/10 (Medium)

N/A

Alice, Bob, Charlie, and Daniel must cross a river using a rickety wooden bridge that can support only two people at once. It is nighttime, and they have a single lantern, which is required to make the crossing safely. The speed of each crossing is determined by the slower person in the pair. The goal is to get all four across in the shortest time possible. Alice takes 10 minutes to cross, Bob takes 5 minutes, Charlie takes 2 minutes, and Daniel takes 1 minute. What is the least amount of time in which they can all reach the other side?

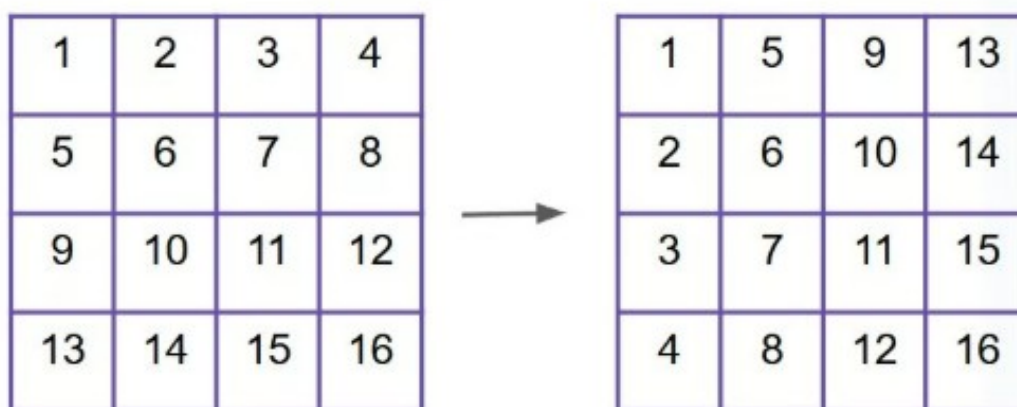
[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

481. Grid Conversion

Brainteasers Lvl 4/10 (Medium)

Graviton

Consider a 4×4 grid filled with integers. In a single operation, you are allowed to increase or decrease all the elements in a specific row or column by 1. The question is: can we transform the left grid below into the right grid using these operations any number of times?



Enter 1 if it is possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

482. Ants On A Circle IV

Probability Lvl 10/10 (Very Hard)

N/A

Ant Alice is one of 24 ants randomly placed on a circular track that is 1 meter long. Each ant faces either clockwise or counterclockwise and moves at a speed of 1 cm/sec. When two ants collide, they reverse directions. What is the probability that, after 100 seconds, Alice is exactly back at her starting point.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

483. Largest 3

Probability Lvl 2/10 (Easy)

Jane Street, Two Sigma

Five fair dice are rolled, and we sum the largest three upfaces. What is the probability that this sum equals 18?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

484. Increasing Game

Brainteasers Lvl 4/10 (Medium)

DRW

Alice and Bob are engaged in a game where Alice starts by choosing a number between 1 and 8. After that, Bob selects a number which is greater than Alice's choice by a value between 1 and 8. They continue taking turns in this manner, each time selecting a number between 1 and 8 larger than the previous one. The winner is the first player to call out the number 200. Which of the players has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

485. Learning To Walk

Probability Lvl 8/10 (Hard)

Two Sigma

A baby is learning to walk with the assistance of its living room couch. The baby starts at the couch and at each given time the baby will make a decision. It will either take a brave step forward, stand in place not knowing what to do, or fearfully take a step back towards the couch with probabilities 0.2, 0.5, and 0.3 respectively. The baby will never go behind the couch (so when at the couch the baby has probability 0.2 of moving forward and probability 0.8 of staying at the couch).

If you were to observe this baby for an extremely long amount of time, what proportion of the time would the baby be at the couch?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

486. Penguin Walk I

Probability Lvl 4/10 (Medium)

N/A

Consider a penguin starting at position 0 on the integer line. Each step, it moves to position $i+1$ with probability $\frac{2}{3}$ and to position $i-1$ with probability $\frac{1}{3}$. Determine the expected number of steps for the penguin to reach position 1.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

487. Circle In 3 Quadrants

Probability Lvl 7/10 (Hard)

N/A

The center of a circle lies on the bisector of quadrant I of the coordinate system. The distance between the center of the circle and the origin of the coordinate system is d , and the radius of the circle is r , where both r and d are independent random variables uniformly distributed in $[0, 1]$. Find the probability that part of the circle lies in quadrants II and IV, but not in quadrant III.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

488. Passing The Parcel

Probability Lvl 4/10 (Medium)

DRW

You and four others are seated in a circle. You start with the parcel, and each person, in turn, can either pass the parcel to the left, pass it to the right, or keep it, with each choice being equally likely. The game continues until someone decides to keep the parcel. What is the probability that you will be the one to end the game by keeping the parcel?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

489. Hands Interchange

Brainteasers Lvl 4/10 (Medium)

Susquehanna International Group

Emma went for a walk between 4:00 and 5:00 PM. When she returned, it was between 7:00 and 8:00 PM, and the positions of the hour and minute hands had swapped. What time did Emma leave?

Enter your answer as the number of minutes after 4:00 PM at which she left (rounded to 1 decimal place).

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

490. Creating Zero

Brainteasers Lvl 4/10 (Medium)

N/A

You are given the set $\{1, 2, 3, \dots, 50\}$. In each step, you select two numbers, a and b , and replace them with $|a - b|$. After each operation, the set reduces by one element. Is it possible to end with the number 0?

Enter 1 if it is possible and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

491. Two Coins Remaining II

Brainteasers Lvl 10/10 (Very Hard)

N/A

You have n coins placed in a row. Initially, all coins are heads. There is one allowable action: you can remove a coin that is not at one of the ends of a row, and flip its neighbors. The removed coin must be heads. For example, $HTHTH$ becomes $HHHH$ if you remove the middle coin. For which n can you reach a state with only two coins remaining?

Enter 1 if it is possible and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

492. Uniform Random Variables I

Probability Lvl 3/10 (Easy)

N/A

Let $P_1, P_2, \dots, P_{20}$ be independent and identically distributed random variables, each following a uniform distribution on the interval $(0, 1)$. Determine $P[P_1 > P_{10} | P_{10} < P_{20}]$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

493. Jumping Frog VIII

Probability Lvl 7/10 (Hard)

Jane Street

A frog starts at 0 on the number line and makes a sequence of jumps to the right. In any one jump, independent of previous jumps, the frog leaps a positive integer distance m with probability $\frac{1}{2^m}$. What is the probability that the frog will eventually land at 10?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

494. Even Or Odd

Probability Lvl 3/10 (Easy)

N/A

A dealer rolls a fair 6-sided die. You have the option to wager on whether the result will be even or odd. If you bet 4 dollars on the outcome being even, you'll earn 6 dollars if you are right. If you bet 6 dollars on the

outcome being odd, you will get 9 dollars if you are correct. What is the optimal amount you should bet in this game?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

495. Additional Info Coin Flips I

Probability Lvl 7/10 (Hard)

Jump Trading

Alice and Bob each toss a fair coin independently until they get heads. Given that Alice achieves her first heads in fewer tosses than Bob does, what is the expected number of tosses Alice makes?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

496. Bulb Toggle II

Brainteasers Lvl 5/10 (Medium)

Graviton

There are 2021 bulbs numbered $1, \dots, 2021$, all initially off. For $k = 1, 2, \dots, 1011$ the k -th person visits and toggles every $(2k - 1)$ -th bulb-i.e., if the bulb is on, he switches it off and if it is off, he switches it on. So person 1 toggles every bulb, person 2 toggles every 3rd bulb (3, 6, 9...), person 3 every 5th (5, 10, 15...), etc. After all 1011 people have gone, how many bulbs are on?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

497. Game On Grid

Brainteasers Lvl 8/10 (Hard)

N/A

Alice and Bob are playing a game on a 4×4 grid. They take turns coloring one square each, starting with Alice, using the same color. The player who completes any 2×2 block of their color loses the game. Who, if anyone, has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

498. Remove Neighbor

Brainteasers Lvl 5/10 (Medium)

N/A

Sofiya and Marquis play a game by taking turns. They form a circle with 2023 other people, and on each turn, Sofiya or Marquis can remove one of their neighbors to the left or to the right from the circle. The person who removes the other player wins. If Sofiya starts, who has the winning strategy?

Enter 1 for Sofiya and 2 for Marquis.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

499. Double Value

Probability Lvl 1/10 (Easy)

N/A

Consider a standard deck of cards where each card is assigned a value based on its number. Also, Ace is worth 1, Jack is worth 11, Queen is worth 12, and King is worth 13. Any card belonging to the diamond suit has its value doubled. For example, the 2 of Diamonds is worth 4. If a card is drawn randomly, what is the expected value of the card?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

500. Coin Toss Game II

Probability Lvl 5/10 (Medium)

Graviton

Alice and Bob are playing a coin-tossing game where they alternate turns. The game ends when the sequence "HT" (heads followed by tails) occurs, with the player who flips the tail being the winner. Alice starts first. What is the probability that Alice will win?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

501. Apples And Oranges

Probability Lvl 5/10 (Medium)

N/A

There are 4 apples and 100 oranges in a basket. They are removed one by one, without replacement, until all 4 apples are extracted. What is the expected number of oranges left in the basket?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

502. King, Queen, Ace II

Probability Lvl 6/10 (Medium)

N/A

A standard deck of 52 cards is shuffled, and cards are drawn one by one without replacement. Find the expected number of cards drawn until we have observed at least one king and either a queen or an ace.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

503. One Colored Face

Probability Lvl 4/10 (Medium)

Citadel, Jane Street

A cube of side length 3 is painted red on all of its exterior faces and then cut into 27 unit cubes. One of these small cubes is chosen at random. Given that the selected cube has at least five unpainted faces, what is the probability that it has exactly one painted face?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

504. Same Arc II

Probability Lvl 8/10 (Hard)

Citadel

Points are dropped one by one uniformly at random on the circumference of a circle. After each drop we check whether the set of points obtained so far all lie inside some semicircular arc. Let X be the number of points drawn up to and including the first time the current set does not lie inside any semicircle. What is $E[X]$?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

505. Count Digits

Brainteasers Lvl 6/10 (Medium)

N/A

Determine the decimal expansions of 2^{1000} and 5^{1000} , and place them side by side to create a new number, denoted as x . Calculate the total number of digits in x .

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

506. Truck Parking

Probability Lvl 7/10 (Hard)

N/A

A row of 16 parking spaces is gradually filled. Twelve cars, each taking one space, park randomly in the lot. Afterward, Bob arrives in his truck, which requires two adjacent spaces. What is the probability that Bob can park?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

507. Unknown Balls

Probability Lvl 6/10 (Medium)

N/A

A box contains 100 balls, each either orange or black. The number of black balls in the box is uniformly distributed between 0 and 100. After drawing a ball at random (without replacement), it is found to be black. What is the probability that the next ball drawn will also be black?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

508. Coin Transactions

Probability Lvl 8/10 (Hard)

N/A

A, B, C, and D play a game where each starts with four coins. The game consists of four rounds. In each round, four balls are placed in an urn: one green, one red, and two white. Each player draws a ball randomly without replacement. The player who draws the green ball gives one coin to the player who draws the red ball. What is the probability that, at the end of the fourth round, all players still have four coins?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

509. Red Blue Game

Probability Lvl 8/10 (Hard)

N/A

A bin contains 3 red and 3 blue balls. You draw them one by one without replacement. Before each draw you must guess "red" or "blue." A correct guess earns \$1, an incorrect one earns \$0. Under optimal play, what is your expected total winnings?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

510. Hard Card Expectation

Probability Lvl 7/10 (Hard)

Citadel, Five Rings, Jump Trading, Optiver

You draw cards from a standard deck of 52 cards one by one without replacement. Determine the expected number of draws to get either 2 kings or at least one king and at least one ace.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

511. Poisoned Wine I

Brainteasers Lvl 7/10 (Hard)

Five Rings, Susquehanna International Group, Two Sigma

A king has 10 servants who are tasked with determining which of the bottles of wine is poisoned. It is known that exactly one of the bottles is poisonous. If a servant drinks from the poisoned bottle, they will die within one month; otherwise, they will remain unharmed. The servants have agreed to participate in this wine tasting process for one month. What is the maximum number of bottles n that can be tested while ensuring the king can accurately identify the poisoned one?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

512. Non-negative Expectation

Probability Lvl 3/10 (Easy)

Jane Street

You roll two fair dice. If you roll double 6s, you win 100 dollars. If you roll a 6 and a non-6, you lose x dollars. For any other outcome, you keep rerolling until you get either double 6s or a 6 and a non-6. What is the maximum value of x such that the expected value of the game is non-negative?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

513. Regular 21-gon

Probability Lvl 5/10 (Medium)

Optiver

Let $\mathcal{S}$ be the set of line segments between any two vertices of a regular 21-gon. If we select two distinct line segments from $\mathcal{S}$ at random, what is the probability they intersect? Note that (for this problem) line segments are considered to intersect if they share a common vertex, as well as when the intersection occurs within the polygon.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

514. Maximum Till Now I

Probability Lvl 6/10 (Medium)

Akuna Capital, Citadel, Five Rings, Hudson River Trading

Consider an infinite sequence of points $x_1, x_2, x_3, \dots$ chosen independently and uniformly from the interval $[0, 1]$. Let k be the first index where x_n is not the maximum of the set $\{x_1, x_2, \dots, x_n\}$. Calculate $E(k)$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

515. Home Run

Probability Lvl 4/10 (Medium)

N/A

Alice and Bob are competing in a home run contest. In each round, a player takes a series of swings, with each swing resulting in either a home run or an out. An out ends the player's turn. The probability that Alice hits a home run on any swing is $\frac{1}{3}$, while Bob's probability of hitting a home run on any swing is $\frac{1}{2}$. Determine the probability that Alice hits more home runs than Bob in one round.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

516. Game Of Placing Kings

Brainteasers Lvl 5/10 (Medium)

Jump Trading

Alice and Bob take turns placing kings on a 9×9 grid. A king cannot be placed on a square that is already occupied, nor can it be placed on a square that is attacked by an existing king. A player loses if they have no

legal move left—meaning there are no safe, unoccupied squares available. Alice moves first. Who has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

517. Same Temperature

Brainteasers Lvl 7/10 (Hard)

N/A

Prove that there exist two diametrically opposite points on the Earth's surface with the same temperature, assuming that the temperature varies continuously across the surface.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

518. 4x4 Array

Combinatorics Lvl 4/10 (Medium)

N/A

In how many ways can we enter numbers from the set $\{1, 2, 3, 4\}$ into a 4×4 array so that all of the following conditions hold? (a) Each row contains all four numbers. (b) Each column contains all four numbers. (c) Each "quadrant" contains all four numbers. (The quadrants are the four corner 2×2 squares.)

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

519. Chess Tournament VII

Probability Lvl 6/10 (Medium)

N/A

In a single-elimination (knockout) chess tournament with 128 players, each competitor has a unique rating. In every match, the higher-rated player wins with probability $\frac{2}{3}$. The tournament is structured in 7 rounds, resulting in a final match. What is the probability that the two highest-rated players meet in the final round and the second-highest rated player wins?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

520. Final Number Possibilities

Brainteasers Lvl 6/10 (Medium)

Graviton

The numbers $1, 2, \dots, 100$ are written on a blackboard. You can repeatedly select two numbers, a and b , erase them, and replace them with the value $ab + a + b$. Continue this process until only one number remains. How many distinct final numbers can result from performing this operation?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

521. Car Clusters

Probability Lvl 8/10 (Hard)

Citadel, Optiver

There are N cars, each traveling at a distinct, randomly assigned speed in the same direction. The cars start at different positions on a straight road with sufficient distance between them. When a faster car catches up to a slower car, it adopts the slower car's speed, effectively forming a cluster. Over time, all cars will settle into clusters, with each cluster moving at a distinct speed. Given that $N = 10$, determine the expected number of clusters.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

522. Strange Party

Brainteasers Lvl 3/10 (Easy)

Jane Street

At a party, $\frac{5}{12}$ of the attendees are adults. A bus carrying 50 people arrives, and now $\frac{11}{25}$ of the attendees are adults. What is the minimum number of adults who could have been present before the bus arrived?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

523. Blackboard

Probability Lvl 9/10 (Very Hard)

Jane Street

Starting from 1000 on a blackboard, you repeatedly replace the current number n with a strictly smaller integer chosen uniformly at random from $\{0, 1, \dots, n-1\}$. You stop when you reach 0. What is the probability that the numbers 15, 30, and 45 all appear on the board at some point during this process?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

524. Phone Connection

Brainteasers Lvl 4/10 (Medium)

N/A

There are 25 phones. A wire always joins exactly two distinct phones (one wire per pair at most, and a wire does not connect more than two phones). Is it possible to wire them so that every phone is connected to exactly 15 other phones?

Enter 1 if it is possible and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

525. Triangle Formation I

Probability Lvl 6/10 (Medium)

Akuna Capital

Let X, Y , and Z be independent random variables uniformly distributed on the interval $[0, 1]$. Determine the probability that these values can form the side lengths of a triangle.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

526. 3 Consecutive Heads II

Probability Lvl 2/10 (Easy)

Jane Street

A fair coin is flipped 6 times. Find the probability of getting exactly 3 consecutive heads at some point in the 6 flips. For example, HHHTTT and HHHTHH are valid, but HHTHHT and HHHHTT are invalid.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

527. Too Many Unknowns

Probability Lvl 10/10 (Very Hard)

N/A

Alice, Bob, Charlie, Diana, Emma, and Fred sit in a circle, in that order, and each roll a fair and normal six-sided die (with values 1 to 6 written on the faces). Each person looks at his or her own roll, and also looks at the roll of either the person to the right or to the left, deciding at random. Then, at the same time, Alice, Bob, Charlie, Diana, Emma, and Fred each state the expected sum of the dice rolls based on the information

they have. All six people say different numbers; in particular, Alice, Bob, Charlie, and Diana say 19, 22, 21, and 23, respectively. Compute the product of the dice rolls.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

528. Dice Triangle

Probability Lvl 3/10 (Easy)

N/A

Three fair six-sided dice, each numbered 1 through 6, are rolled. What is the probability that the three numbers that come up can form the sides of a triangle?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

529. Triangle Sum

Combinatorics Lvl 5/10 (Medium)

Akuna Capital

Consider a triangle where the integers $1, 2, \dots, 9$ are assigned to its vertices and sides. An integer is placed at each vertex, and the remaining 6 integers are distributed along the sides. An integer at a vertex contributes to both adjoining sides. The sum of the integers on each side of the triangle, including the vertex values, is 17. Determine the number of unique ways to distribute the integers on the sides. Triangles that can be obtained by rotation are considered identical, and the order of integers along any side does not matter.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

530. Is It Fibonacci?

Brainteasers Lvl 5/10 (Medium)

N/A

Let $f_1, f_2, f_3, \dots$ be a sequence of numbers such that $f_n = f_{n-1} + f_{n-2}$ for every integer $n \geq 3$. If $f_7 = 83$, what is the sum of the first 10 terms of the sequence?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

531. Dice Triangle II

Probability Lvl 8/10 (Hard)

N/A

A random triangle is produced as follows. A pair of standard dice is rolled independently three times to get three random numbers between 2 and 12, inclusive, by adding the numbers that come up on each pair rolled. Call these three random numbers a , b , and t . The random triangle has two sides of lengths a and b with the angle between them measuring $15(t - 1)$ degrees. What is the probability that the triangle is a right triangle?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

532. Marble Swap III

Probability Lvl 5/10 (Medium)

N/A

You have 3 marbles: red, blue, and red, placed in that order. In each move, you randomly swap the center marble with one of the 2 edge marbles. After 15 moves, what is the probability that the marbles are back in their original order (red, blue, red)?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

533. Jet Trail Intersection

Probability Lvl 7/10 (Hard)

N/A

Two jets start from distinct points and travel in perfectly straight lines on an infinite plane, leaving visible trails behind them. What is the probability that their trails will intersect? Note that the trails are rays extending infinitely in one direction, rather than lines which extend infinitely in both directions.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

534. Increasing Decreasing I

Combinatorics Lvl 4/10 (Medium)

N/A

Determine the number of 9-digit numbers that can be formed using unique digits from 1 to 9, such that the first five digits are in increasing order and the last five digits are in decreasing order.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

535. Game On Circle

Brainteasers Lvl 5/10 (Medium)

N/A

Consider a circle with $n = 2024$ points. Alice and Bob play a game where Alice starts. Each turn, Alice colors an uncolored point red, and Bob colors an uncolored point blue. After all n points are colored, Alice determines the longest sequence of consecutive red points, R , and Bob determines the longest sequence of consecutive blue points, B . Alice wins if $R > B$, Bob wins if $B > R$, and neither wins if $R = B$. Does Alice have a winning strategy? Enter 1 for Yes and 2 for No.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

536. Tetrahedral Die I

Probability Lvl 5/10 (Medium)

N/A

The four faces of a tetrahedral die are labelled 0, 1, 2, and 3, and the die has the property that, when it is rolled, the die promptly vanishes, and a number of copies of itself appear equal to the number on the face the die landed on. For example, if it lands on the face labelled 0, it disappears. If it lands on the face labelled 1, nothing happens. If it lands on the face labelled 2 or 3, there will then be 2 or 3 copies of the die, respectively (including the original). Suppose the die and all its copies are continually rolled, and let p be the probability that they will all eventually disappear. Find p .

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

537. Highest Roll EV

Probability Lvl 5/10 (Medium)

N/A

You roll a fair, six-sided die until you get a 4. What is the expected value of the highest number you roll through this process?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

538. Chords Intersection II

Probability Lvl 8/10 (Hard)

Graviton, Hudson River Trading, Jump Trading

Ten points are placed at random on the circumference of a circle. We choose any one of these points and join it to one of the remaining nine points, selected uniformly at random. From this newly reached point, we draw a chord to one of the remaining eight unused points, again chosen uniformly at random. We continue in this manner, each time connecting the current point to a new unused point, until all points have been connected. In this process, 9 chords will be formed. What is the expected number of pairs of chords that intersect strictly inside the circle?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

539. Increasing Birthdays

Probability Lvl 2/10 (Easy)

Akuna Capital, IMC Trading, Jane Street, Jump Trading

Seven people with distinct ages sit randomly around a circular table. Calculate the probability that they are seated in increasing age order, either in clockwise or counterclockwise direction.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

540. Binary Trailing Zeros

Brainteasers Lvl 5/10 (Medium)

Five Rings

How many trailing zeros are present in the BINARY representation of $142!$?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

541. Vertical Hit

Probability Lvl 3/10 (Easy)

N/A

A frog sitting at the point $(1, 2)$ begins a sequence of jumps, where each jump is parallel to one of the coordinate axes and has length 1, and the direction of each jump (up, down, right, or left) is chosen independently at random. The sequence ends when the frog reaches a side of the square with vertices $(0, 0)$, $(0, 4)$, $(4, 4)$, and $(4, 0)$. What is the probability that the sequence of jumps ends on a vertical side of the square?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

542. Bacteria Population I

Probability Lvl 4/10 (Medium)

N/A

Consider a bacterial colony beginning with a single cell. In each generation, every cell has an independent chance of either dying without leaving offspring or producing a random positive integer number of offspring. The probability of extinction for a colony starting with one cell is $\frac{1}{3}$. Determine the probability of extinction for a colony that starts with three cells.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

543. Count Regions IV

Combinatorics Lvl 10/10 (Very Hard)

N/A

Determine the maximum number of regions into which 3-dimensional space can be divided using n spheres. For example, one sphere divides space into 2 regions: the interior of the sphere and the exterior.

Enter the answer for $n = 10$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

544. Expected Ping-pong Games

Probability Lvl 3/10 (Easy)

N/A

Two players compete in a sequence of independent ping-pong games. In each game, Player A wins with probability 0.3 and Player B wins with probability 0.7. They keep playing until one of them has won 21 games total, at which point the match ends. What is the expected total number of games played in a single match?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

545. Small Chords

Probability Lvl 8/10 (Hard)

N/A

Four points are chosen independently and uniformly at random on the circumference of a circle. Every pair of points is joined by a chord, so there are six chords in total. What is the probability that all six chords are shorter than the radius of the circle?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

546. Large Cycles

Probability Lvl 8/10 (Hard)

N/A

Let C_n denote the number of cycles in a random permutation of $\{1, 2, 3, \dots, 2n\}$ whose lengths are greater than n . Calculate $\lim_{n \rightarrow \infty} \mathbb{E}[C_n]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

547. Cheating Husbands

Brainteasers Lvl 4/10 (Medium)

N/A

In a town of 100 married couples, some husbands are unfaithful, but each wife is unaware of her own husband's behavior while knowing the truth about every other husband. If a wife discovers her husband has cheated, she will expel him from the house at midnight. The husbands remain silent about their actions. One day, the mayor announces that at least one husband is cheating. After the announcement, everyone remains silent, waiting to see if any husbands are expelled. For nine consecutive nights, no husbands are kicked out, but on the 10th night, several husbands are expelled simultaneously. How many husbands were expelled?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

548. Largest Prime Probability

Probability Lvl 6/10 (Medium)

DRW

If we roll a standard die twice and sum, the probability that the sum is prime is $\frac{15}{36} = \frac{5}{12}$. If we renumber the faces of the die, with all faces being different, what is the largest probability of a prime sum that can be achieved?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

549. Jumping Frog VI

Probability Lvl 3/10 (Easy)

DRW, Five Rings

A frog is at point 0 and wants to reach point 10. With each jump, the frog can cover any number of points. In how many ways can the frog reach point 10? For example, one possible way is $0 \rightarrow 4 \rightarrow 5 \rightarrow 8 \rightarrow 10$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

550. Dice Game VIII

Probability Lvl 6/10 (Medium)

N/A

A casino game involves a dealer with a 20-sided die. Over the course of 100 rounds, the player has two options in each round. The player can either roll the die, which replaces the current face value with a new number but yields no monetary gain, or the player can take an amount of dollars from the casino equivalent to the current face value of the die. After opting to take the money, the dealer will remove the die from the table, necessitating a roll in the subsequent turn. What is the expected payout with optimal strategy?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

551. Simulate 1/3

Probability Lvl 2/10 (Easy)

Akuna Capital, Optiver

We design a scheme to simulate an event with a probability of $\frac{1}{3}$ using the following procedure. Flip a fair coin twice. If the result is heads-heads (HH), the event occurs. If the result is heads-tails (HT) or tails-heads (TH), the event does not occur. If the result is tails-tails (TT), repeat the entire process from the beginning. Determine the expected number of coin flips required to simulate this event.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

552. Count Sequences I

Combinatorics Lvl 4/10 (Medium)

DRW

Consider a sequence $s_0, s_1, \dots, s_{10}$, where $s_0 = 0$ and $s_{10} = 4$. For each i between 1 and 10 (inclusive), the difference between consecutive terms satisfies $|s_i - s_{i-1}| = 1$. Determine the number of such sequences.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

553. Diagonal Cut II

Brainteasers Lvl 4/10 (Medium)

Susquehanna International Group

We have a rectangular grid consisting of 234×306 small squares. If a diagonal is drawn from one corner to the opposite corner, how many squares does the diagonal cut through?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

554. Last Ball

Brainteasers Lvl 4/10 (Medium)

N/A

A bag contains 20 blue balls and 13 red balls. Each time you draw two balls at random without replacement, you follow these rules: If the two balls are the same color, you add a blue ball to the bag. If the two balls are of different colors, you add a red ball. With an unlimited supply of blue and red balls, what will be the color of the final remaining ball in the bag after repeatedly applying this process?

Enter 1 for blue and 2 for red.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

555. Treasure Island

Brainteasers Lvl 6/10 (Medium)

N/A

An old parchment provides directions to a treasure chest on an island: "Locate an unmarked grave and two tall oak trees. Walk from the grave to the left tree, counting the steps. Upon reaching the left tree, turn left by 90 degrees and walk the same number of steps. Mark this point with a flag. Return to the grave, then walk towards the right tree, counting the steps. Upon reaching the right tree, turn right by 90 degrees and walk the same number of steps. Mark this point with another flag. The treasure is at the midpoint between the two flags." Sailors on the island find two tall oak trees but cannot locate the unmarked grave. It can be proven that the treasure must lie at one of finitely many discrete locations. How many such locations are there?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

556. Vector Toggle

Probability Lvl 10/10 (Very Hard)

Jump Trading

We have a 10 dimensional vector $(0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$. In one move a coordinate is chosen uniformly at random from the ten coordinates and that coordinate is toggled, so a 0 becomes 1 and a 1 becomes 0. What is the expected number of moves to reach the vector $(1, 0, 0, 0, 0, 0, 0, 0, 0, 0)$?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

557. Prisoner Hats-multicolor

Brainteasers Lvl 10/10 (Very Hard)

IMC Trading, Susquehanna International Group

One hundred prisoners are lined up, each wearing a randomly assigned hat from 10 distinct colors. Each prisoner can see the hats of those in front but not their own or the ones behind. Starting with the last prisoner in the line and proceeding forward, each prisoner must state a single word—specifically a color, such as "red" or "blue." If their guess matches their hat color, they are freed; otherwise, they are executed. All prisoners can hear the previous responses. What strategy will maximize the number of prisoners saved?

Enter the maximum number of prisoners that can be GUARANTEED to be saved under an optimal strategy.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

558. Tennis Odds

Probability Lvl 5/10 (Medium)

Citadel, Optiver, Susquehanna International Group

In a tennis match starting at 30-30, with Alice having a 0.6 probability of winning each serve independently, what is the probability that Alice wins the game, given the standard tennis scoring rules: •The scores progress as 0-15-30-40. •If a player has 40 and wins the next point while the opponent has less than 40, they win the game. •If both players have 40, the next point gives an advantage. The player with advantage must win again to win the game, or else the score resets to 40-40.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

559. Pizza Standard Deviation

Probability Lvl 1/10 (Easy)

Citadel

A pizza shop employs 25 independent workers, each producing pizzas with a standard deviation of 60 pizzas per day. Determine the standard deviation of the total number of pizzas produced by all workers in a day.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

560. Circuit

Probability Lvl 5/10 (Medium)

Citadel, Jane Street

An electrician connects 12 light bulbs, where each pair is linked by a wire with a probability of $\frac{1}{2}$, independently of others. A group of k light bulbs forms a complete k -circuit if every pair within the group has a wire connecting them. Find the expected number of 4-circuits in this setup.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

561. Expected Number Of Rectangles

Probability Lvl 6/10 (Medium)

Jane Street

Across the face of a rectangular table, you idly draw lines that are parallel to its edges. Each time you draw a line, there is a 50% chance it will be in each direction, and you never draw over an existing line or the edge of the table. After a few minutes, you notice that you've drawn 20 lines. What is the expected number of rectangles that the table will be partitioned into?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

562. Triangles Inside Triangle

Combinatorics Lvl 3/10 (Easy)

N/A

A dot is marked at each vertex of a triangle ABC . Then, 2, 3, and 7 more dots are marked on the sides AB , BC , and CA , respectively. How many triangles have their vertices at these dots?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

563. Divisible By 37

Brainteasers Lvl 6/10 (Medium)

N/A

Consider a number of the form $37xyz$, where x, y, z are digits. How many such numbers are there such that $37xyz$, $37yzx$, and $37zxy$ are all divisible by 37?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

564. 3 Heads 3 Tails III

Probability Lvl 2/10 (Easy)

Susquehanna International Group

Alice and Bob are engaged in a coin-tossing game. Alice wins if three heads come up, while Bob wins if three tails are tossed. The heads and tails don't need to be in a row. How many tosses, on average, will it take to declare a winner?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

565. Unlucky Seven

Probability Lvl 5/10 (Medium)

Jane Street

You roll a fair 6-sided die and can either keep the result and receive that value in dollars, or roll again. If you roll again, you must pay the value of your first roll if the sum of both rolls is 7 or more. If the sum is less than 7, you receive the second roll's value in dollars. What is your expected payout under optimal play?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

566. Toss Exceedance

Probability Lvl 4/10 (Medium)

N/A

A fair coin is tossed repeatedly until the number of heads exceeds the number of tails by 4 or until the number of tails exceeds the number of heads by 6. What is the expected number of tosses?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

567. Tennis Tournament

Probability Lvl 3/10 (Easy)

N/A

You are participating in a tennis tournament with Alice and Bob, and the event includes a series of three matches. To claim victory in the tournament, you must win two games in a row. Your likelihood of defeating Bob is higher than that of defeating Alice. If Alice is represented by A and Bob by B, which sequence of matches would be more advantageous: playing Alice, Bob, and then Alice (ABA) or Bob, Alice, and then Bob (BAB)?

Enter 1 for ABA and 2 for BAB

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

568. Chess Tournament VI

Probability Lvl 3/10 (Easy)

DRW

In a chess tournament with 50 players numbered from 1 to 50, a series of 49 matches determines the winner. The tournament begins with two randomly selected players competing in the first match. In each subsequent match, the loser is eliminated, and the winner plays against another randomly selected player who has not yet been eliminated. Suppose that the player with the larger number always defeats the player with the smaller number. What is the probability that player 49 and player 50 play the final match?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

569. Expected Wait Time II

Probability Lvl 4/10 (Medium)

Jane Street, Two Sigma

A bus arrives every 10 minutes on a fixed schedule. The driver independently refills on gas with a 10% probability for each appearance, adding 1 hour to the bus's travel time if it occurs. For instance, the bus can arrive at times 0, 10, 20, 30, etc. In this example, the bus takes 70 minutes to arrive after the fourth arrival time as it refills gas here. If you arrive at a uniformly random time throughout the day, what is the expected time until the next bus appears?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

570. Weighted Die

Probability Lvl 6/10 (Medium)

N/A

Consider a 6-sided die with faces numbered 1 through 6, where the die is weighted such that the probability of rolling each value from 2 to 6 is the same as the probability of rolling these values when rolling the die twice and summing the results. Let p_1 represent the probability of rolling a 1 on this die. p_1 is a positive solution to

the polynomial equation

$$\sum_{k=1}^6 c_k p_1^k = 1$$

where $c_1, \dots, c_6$ are some real constants. Find the sum

$$\sum_{k=1}^6 c_k.$$

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

571. Ordered Pairs In Permutation

Probability Lvl 7/10 (Hard)

N/A

Consider a random permutation of the set $\{1, 2, \dots, 2015\}$. In other words, for each $1 \leq i \leq 2015$, i is sent to the element a_i where $a_i \in \{1, 2, \dots, 2015\}$ and if $i \neq j$, then $a_i \neq a_j$. What is the expected number of ordered pairs (a_i, a_j) with $i - j > 155$ and $a_i - a_j > 266$?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

572. Variance Maximum

Probability Lvl 5/10 (Medium)

Susquehanna International Group

Let X be a random variable defined on the interval $[-1, 1]$. Determine the maximum possible value of the variance $\text{Var}(X)$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

573. Additional Info Die Rolls

Probability Lvl 6/10 (Medium)

N/A

Roll a fair six-sided die repeatedly until a 6 appears. Given that the first 6 occurs before the first 5, find the expected number of rolls.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

574. Word Problem I

Probability Lvl 4/10 (Medium)

IMC Trading

Given a word made from the letters a, e, i, o, and u (with repetition allowed), what is the probability that no two consecutive letters are the same and no letter appears more than twice?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

575. Truth

Brainteasers Lvl 4/10 (Medium)

Jane Street

You have 100 statements written on a sheet of paper. The first statement says, "At most 0 of these 100 statements are true." The second says, "At most 1 of these 100 statements are true." In general, the n -th statement says, "At most $n - 1$ of these 100 statements are true." Determine how many of these statements are true.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

576. Normal Distributions

Probability Lvl 6/10 (Medium)

Akuna Capital, Citadel, Jane Street, Two Sigma

Let $X_1 \sim N(0, 9)$ and $X_2 \sim N(0, 16)$ be independent random variables. Compute $E[X_1 | X_1 + X_2 = 5]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

577. Sports Betting

Probability Lvl 2/10 (Easy)

N/A

Alice and Bob place a 10 dollars wager on opposing outcomes of a baseball game. During the game, Alice raises the bet to 20 dollars. If Bob doesn't match her raise, he forfeits the 10 dollars he has already bet. Let p represent the probability that Bob's team will win at the time Alice increases the wager. Determine the smallest value of p at which Bob should agree to match Alice's bet.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

578. Even Head Stop I

Probability Lvl 4/10 (Medium)

N/A

A coin is tossed repeatedly until two heads occur (not necessarily consecutively). For example, the coin toss will be stopped in both of the following cases: TTTHH and HTTTH. What is the probability that the total number of tosses is even?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

579. Interesting Constraints

Brainteasers Lvl 6/10 (Medium)

N/A

Let $x_1, x_2, \dots, x_{10}$ be 10 numbers. Suppose that $x_i + 2x_{i+1} = 1$ for each i from 1 through 9. What is the value of $x_1 + 512x_{10}$?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

580. Mathematical Expression I

Probability Lvl 7/10 (Hard)

Akuna Capital

An arithmetic expression is created by inserting either a plus sign or a multiplication sign in each of the 11 spaces between consecutive $\sqrt{3}$'s in a row of twelve $\sqrt{3}$'s. The signs are chosen uniformly and independently at random. What is the probability that the resulting expression evaluates to $12\sqrt{3}$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

581. Inside The Range

Combinatorics Lvl 6/10 (Medium)

Five Rings, Jump Trading

Determine the number of non-negative integral solutions to the inequality $6 \leq x_1 + x_2 + x_3 + x_4 + x_5 \leq 10$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

582. Increasing Rolls Expectation

Probability Lvl 5/10 (Medium)

Jump Trading

A fair 10-sided die is rolled once. After that, keep rolling the die as long as each new roll is strictly larger than the previous one. The process stops at the first roll that is not larger than the one before it. What is the expected sum of all rolls except the last roll?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

583. Card Picking II

Probability Lvl 6/10 (Medium)

N/A

A thoroughly shuffled deck is dealt card by card from the top. The 27th card turned over happens to be the first ace. Determine the probability that the next card, the 28th, is the 2 of hearts.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

584. Prime Dice

Probability Lvl 2/10 (Easy)

Jane Street

Two fair 6-sided dice are rolled, with each die having the first six prime numbers on its faces. Determine the probability that the sum of the numbers on the top faces of both dice is also a prime number.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

585. Fish Estimate

Brainteasers Lvl 1/10 (Easy)

Susquehanna International Group

You tag 50 fish in a pond. The following day, you catch 40 fish, of which 10 are tagged. Estimate the total number of fish in the pond.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

586. Proportional Dice

Probability Lvl 3/10 (Easy)

N/A

You have two 6-sided dice: one is fair, and the other has sides that appear in proportion to their values. If you roll both dice, what is the probability that the sum of the outcomes is odd?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

587. Moving Ants II

Probability Lvl 2/10 (Easy)

N/A

Consider the vertices of a square: $A_1 = (0, 0)$, $A_2 = (1, 0)$, $A_3 = (1, 1)$, and $A_4 = (0, 1)$. An ant starts at A_1 at time $t = 0$. In subsequent steps, the ant randomly chooses and moves to a neighboring vertex (for example, if the ant is at A_1 , it can move to A_2 or A_4 , with equal probability). Let T be the minimum number of steps needed to visit all four vertices. What is the probability that $T = 4$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

588. Red Card Prediction**Probability** Lvl 6/10 (Medium)

N/A

You are playing a game using a standard deck of 52 playing cards. Cards are revealed one by one from the top of the deck. At any moment-provided there is at least one card remaining-you may choose to stop. When you stop, the next unrevealed card is turned over. If this card is red (hearts or diamonds), you win; if it is black (spades or clubs), you lose. What is the maximum probability of winning under an optimal stopping strategy?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

589. Infection Spread**Brainteasers** Lvl 6/10 (Medium)

N/A

Consider an $n \times n$ checkerboard where an infection spreads as follows: a square becomes infected if at least two of its four adjacent squares are already infected. Is it possible to eventually infect the entire board if we start with fewer than n initially infected squares? Note that you need to prove your claim rigorously.

Enter 1 if it is possible and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

590. Win-win**Probability** Lvl 5/10 (Medium)

N/A

Alice, Beth, Carla, Dana, and Eden play a game in which each of them simultaneously closes her eyes and randomly chooses two of the others to point at (one with each hand). A participant loses if she points at someone who points back at her; otherwise, she wins. Find the probability that all five girls win.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

591. Parking Lot**Probability** Lvl 9/10 (Very Hard)

DRW, Graviton, Two Sigma

Two cars are parked in positions 1 and 2 of a parking lot, while positions 3 and 4 are empty. Each minute, a car may move forward to the next position, independent of the movement of the other car (so, both cars may move in the same minute), subject to the following conditions: (i) The space directly ahead of the car must be unoccupied before any cars move and (ii) A car can only move if it is not already in position 4. (iii) If a car is eligible to move, it advances to the next position with a probability of $\frac{1}{2}$, independent of the movement of the other car. How many minutes, on average, will it take for the two cars to reach positions 3 and 4?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

592. Head-tail Product**Probability** Lvl 3/10 (Easy)

Graviton, Jump Trading

A fair coin is tossed n times. Let X represent the number of heads and Y the number of tails. Determine the expected value $E[XY]$. Enter the answer for $n = 10$ rounded to 5 decimal places.

Enter the answer for $n = 10$ rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

593. Guess The End

Probability Lvl 10/10 (Very Hard)

N/A

Twenty-five ants are randomly placed on a meter-long rod, with the thirteenth ant from the west end designated as Ant Alice. Each ant has an equal probability of facing east or west. They move forward at a speed of 1 cm/sec, and upon colliding, they reverse directions. Determine the probability that Alice falls off the end of the rod in the direction she was originally facing.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

594. Optimal Die Choice

Probability Lvl 5/10 (Medium)

Jane Street

You have a fair 20-sided die and 100 moves in a game. The die starts with an upface of 1. On each move, you can either roll to re-roll the die or take to cash out the current upface. You can take multiple times without rolling in between. For instance, you could take 100 times at 3 and secure \$300 guaranteed. Your strategy is to take the first time the upface reaches at least a chosen threshold n . What is your expected payout if you select n optimally?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

595. Two Random Points

Probability Lvl 6/10 (Medium)

N/A

On a circle of radius 1, two points are chosen in sequence as follows. First, a point is picked so that its distance R_1 from the center is uniformly distributed on $[0, 1]$. Then, knowing $R_1 = r_1$, a second point is picked uniformly at random from among all points in the disk whose distance from the center exceeds r_1 . What is the probability that the second point's distance R_2 from the center exceeds $\frac{1}{2}$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

596. Passing The Coin I

Probability Lvl 3/10 (Easy)

N/A

Five players, labeled 1 through 5, are playing a game with a biased coin that lands heads with probability $\frac{1}{3}$ and tails with probability $\frac{2}{3}$. The game begins with player 1 holding the coin. On their turn, a player flips the coin. If it lands heads, that player wins immediately. If it lands tails, they pass the coin to the next player in order: player 1 passes to player 2, player 2 to player 3, and so on, with player 5 passing it back to player 1. This cycle continues until some player eventually flips heads and wins the game. What is the probability that player 1 wins the game?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

597. Jumping Frog IX

Probability Lvl 5/10 (Medium)

N/A

Consider lily pads arranged in a row numbered $0, 1, 2, \dots, 11$. There are predators on lily pads 3 and 6, and a morsel of food on lily pad 10. A frog starts on pad 0, and from any given lily pad n , it has a probability of $\frac{1}{2}$ to hop to pad $n + 1$ and a probability of $\frac{1}{2}$ to jump to pad $n + 2$. Determine the probability that the frog reaches pad 10 without ever landing on pads 3 or 6.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

598. Expected Determinant

Probability Lvl 6/10 (Medium)

IMC Trading

Let A be a 5×5 matrix with each entry randomly selected from the interval $[1, 25]$. What is the expected value of the determinant of the matrix?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

599. Least N

Brainteasers Lvl 5/10 (Medium)

N/A

Find the least n such that any subset of $\{1, 2, \dots, 100\}$ with n elements must have two elements with a difference of 9.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

600. Forming Quadratic Polynomial II

Probability Lvl 7/10 (Hard)

N/A

Let b and c be random integers from the set $\{1, 2, \dots, 100\}$, chosen uniformly and independently. What is the probability that the roots of the quadratic $x^2 + bx + c$ are real?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

601. Crazy Weights Arrangement

Combinatorics Lvl 8/10 (Hard)

Jane Street

Alice has a balance and n weights, each weighing $2, 2^2, \dots, 2^n$. She must place these weights on the balance one by one, ensuring that at no point is the right pan heavier than the left. At each step, she can choose an unused weight and place it on either the left or right pan. Determine the number of ways she can complete this task while adhering to the rule, keeping in mind that the sequence of placements matters and different sequences of placements are considered different even if the final arrangement is identical. Enter the answer for $n = 10$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

602. Vending Machine I

Probability Lvl 5/10 (Medium)

Optiver

A vending machine dispenses random stickers from a set of 5 unique stickers. Each sticker is equally likely to appear and independent of previous selections. If you purchase 7 stickers from the machine, how many unique stickers can you expect to collect?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

603. Early Head Probability

Probability Lvl 4/10 (Medium)

N/A

Alice and Bob are both flipping a coin with a probability of $\frac{1}{3}$ of landing heads per flip. What is the probability that Alice gets her first heads in fewer flips than Bob?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

604. Shadow

Brainteasers Lvl 4/10 (Medium)

N/A

Imagine a cube suspended in the air in front of you, with a light source projecting its shadow onto a wall. You are allowed to freely rotate the cube and aim to determine the largest-sided polygon that can appear as the shadow.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

605. Rabbit On A Square Grid

Combinatorics Lvl 5/10 (Medium)

N/A

A rabbit enjoys hopping around an infinite square grid in its backyard. It expends 1 Joule of energy to hop one step north or south, and 1 Joule to hop one step east or west. One morning, the rabbit wakes up on the grid with 100 Joules of energy and continues to hop until it falls asleep with 0 Joules of energy. Determine the number of possible locations where the rabbit could have gone to sleep.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

606. Game Of Moving Rooks

Brainteasers Lvl 4/10 (Medium)

N/A

Consider a standard chessboard with a rook positioned in the bottom-right corner. The rook can only move vertically upwards or horizontally to the left. Alice and Bob are playing a game where they alternate moves with the rook. The objective is to move the rook to the top-left square, and the player who achieves this wins the game. If Alice goes first, who has a winning strategy under optimal play?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

607. Origin In Between

Probability Lvl 1/10 (Easy)

Optiver

2 points are selected randomly in the interval $(-2, 4)$. Determine the probability that the origin lies in between those 2 points.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

608. Increasing Dice Rolls I

Probability Lvl 4/10 (Medium)

IMC Trading

You roll three fair dice sequentially. What is the probability that the outcomes are in strictly increasing order?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

609. Tenth Toss**Probability Lvl 7/10 (Hard)***Graviton*

You flip a coin until you get two consecutive heads. What is the probability that you stop on the 10th toss?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

610. Three In A Line**Brainteasers Lvl 6/10 (Medium)***N/A*

A 9×9 grid is given. Alice and Bob take turns alternately. On each turn, a player may fill any empty cell with either 0 or 1. Both players may choose either number on any turn, independent of their own or the opponent's previous choices. The player who first creates three identical numbers consecutively in a straight line, whether horizontally, vertically, or diagonally, wins. If Alice goes first, who has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

611. Matching Cards Removal**Probability Lvl 6/10 (Medium)***Susquehanna International Group*

Imagine you have 30 cards with values from 1 to 15, each value appearing twice in the deck. You randomly draw 2 cards at a time from the 30 cards. If the drawn cards have the same value, they are removed from the deck; if not, they are returned to the deck. Each pair of draws is considered a turn. Determine the expected number of turns required to remove all the cards.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

612. Blue Marble Remaining**Probability Lvl 6/10 (Medium)***N/A*

Kate has two bags X and Y. Bag X contains 5 red marbles (and nothing else). Bag Y contains 4 red marbles and 1 blue marble (and nothing else). Kate chooses one of her bags at random (each with probability $\frac{1}{2}$) and removes a random marble from that bag (each marble in that bag being equally likely). She repeats the previous step until one of the bags becomes empty. At that point, what is the probability that the blue marble is still in bag Y?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

613. Expected Collisions II**Probability Lvl 10/10 (Very Hard)***N/A*

Twenty-five ants are randomly positioned on a 1-meter rod, with the thirteenth ant from the west being our friend, Ant Alice. Each ant faces either east or west, with equal probability. They all move forward at 1 cm/sec in the direction they are facing, and upon collision with another ant, they reverse their direction. What is the expected number of collisions that Alice herself has?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

614. Minimum Correlation

Probability Lvl 4/10 (Medium)

N/A

Consider 10 random variables, each with a mean of 0 and a variance of 1, and all sharing the same pairwise correlation. What is the minimum possible value for this pairwise correlation?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

615. Complex Circle

Probability Lvl 4/10 (Medium)

N/A

Consider Z as a randomly chosen point on the unit circle in the complex plane, where $|Z| = 1$. Find $E[Z^2]$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

616. Higher Sum

Probability Lvl 2/10 (Easy)

Five Rings

Two standard six-sided dice are rolled. What is the probability that the sum of the rolled numbers is greater than or equal to their product?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

617. 5 Points On A Sphere

Probability Lvl 10/10 (Very Hard)

N/A

Five points are chosen uniformly and independently at random on the surface of a sphere. Next, 2 of these 5 points are randomly picked, with every pair equally likely. What is the probability that the 2 points are separated by the plane containing the other 3 points?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

618. Count Sequences II

Combinatorics Lvl 4/10 (Medium)

N/A

How many sequences of 0s and 1s of length 19 are there that begin with a 0, end with a 0, contain no two consecutive 0s, and contain no three consecutive 1s?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

619. Remove Or Flip

Brainteasers Lvl 5/10 (Medium)

Graviton

On a table there are 4013 tokens which are red on one side and black on the other. Alice and Bob play by turns. In each turn, it is permitted to remove any number of tokens from one color or flip any number of tokens of the same color. The one that removes the last token wins. If Alice plays first, who has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

620. Unit Digit Probability I

Probability Lvl 1/10 (Easy)

N/A

Four numbers are selected from the set $\{1, 2, \dots, 9\}$ with repetition allowed. Determine the probability that the product of the four selected numbers has a units digit of 1, 3, 7, or 9.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

621. Interview III

Probability Lvl 9/10 (Very Hard)

N/A

You are interviewing n candidates for a position, each with a distinct and unknown rating — higher ratings indicate better candidates. The candidates are interviewed one by one in a random order. After each interview, you learn the candidate's relative ranking among all those interviewed so far. Immediately after each interview, you must decide whether to hire that candidate or reject them permanently — you cannot return to a previously rejected candidate. You are allowed to select two candidates in this process. Your goal is to select the best and worst candidates out of all n . What is the optimal strategy to maximize the probability of choosing the best and worst candidates, and what is this maximum probability in the limit as $n \rightarrow \infty$?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

622. Expected Value

Probability Lvl 9/10 (Very Hard)

N/A

Consider the randomly generated base 10 real number $r = 0.p_0p_1p_2\dots$, where each p_i is a digit from 0 to 9, generated as follows: p_0 is generated uniformly at random from 0 to 9, inclusive, and for all $i \geq 0$, p_{i+1} is generated uniformly at random from p_i to 9, inclusive. Compute the expected value of r .

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

623. Forming Quadratic Polynomial I

Probability Lvl 8/10 (Hard)

N/A

Alice has eight slips of paper labeled 1, 2, 4, 8, 16, 32, 64, and 128. He randomly draws three slips with replacement, labeled a, b , and c . What is the probability that a quadratic polynomial with coefficients a, b , and c , in some order, will have two distinct real roots?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

624. Rope Variance

Probability Lvl 4/10 (Medium)

Five Rings

A rope of length 1 meter is cut at a random point along its length. Let the length of the longer piece be denoted as L . Determine the variance of L .

Enter your answer rounded to 5 decimal places.

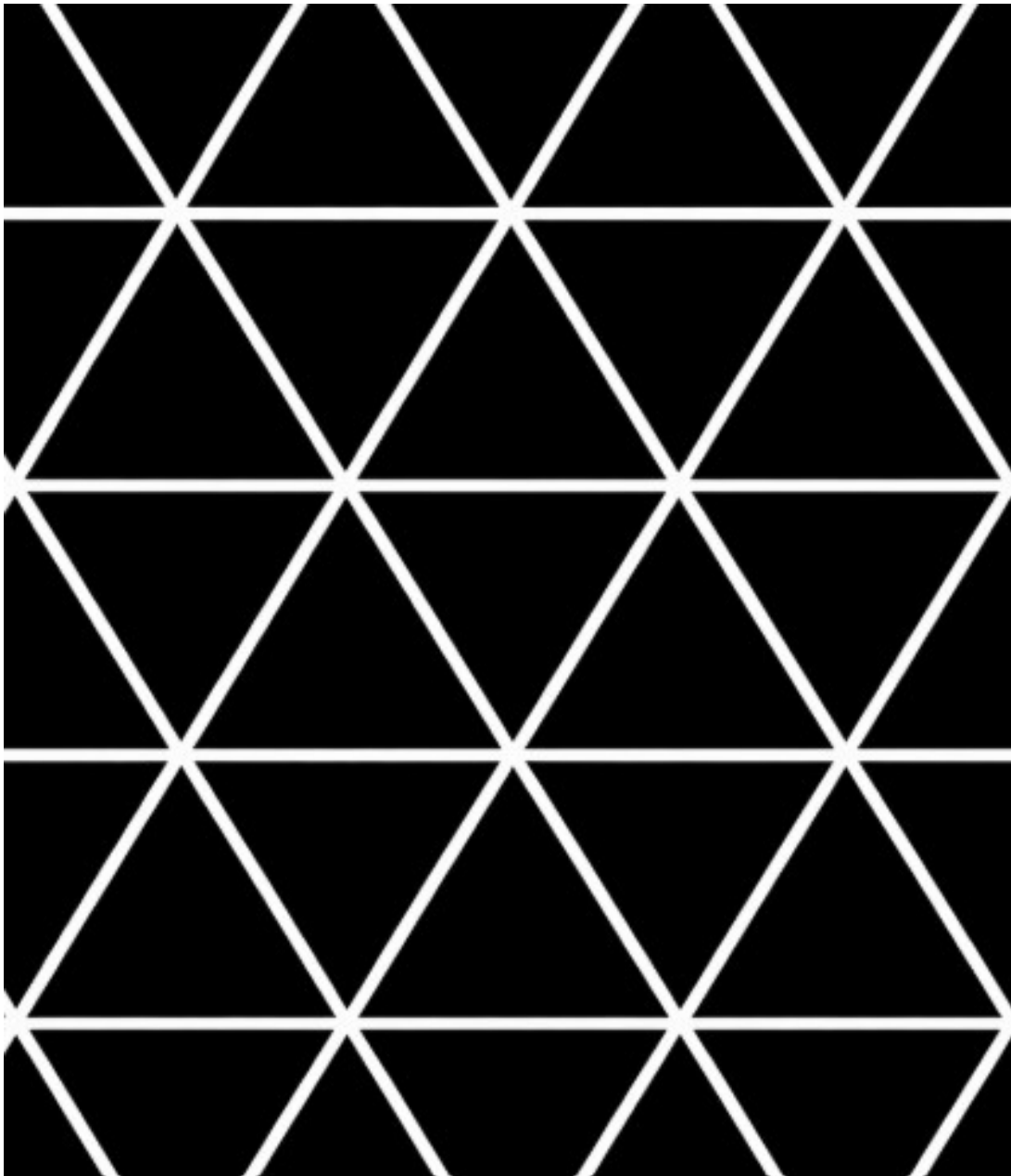
[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

625. Triangular Grid

Probability Lvl 6/10 (Medium)

N/A

Imagine an infinite triangular grid made up of equilateral triangles with sides of 1 unit as shown in the figure.



On each turn, you randomly choose the next grid line to follow. After 5 turns, what is the probability that you have moved beyond 1 unit from your starting point at least once?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

626. Strange Comparison I

Brainteasers Lvl 4/10 (Medium)

N/A

Which number is greater: 31^{11} or 17^{14} ?

Enter 1 for 31^{11} and 2 for 17^{14} .

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

627. Mathematical Expression II

Probability Lvl 5/10 (Medium)

N/A

Martha writes down a random mathematical expression consisting of 3 single-digit positive integers with an addition sign "+" or a multiplication sign "×" between each pair of adjacent digits. (For example, her expression could be $4 + 3 \times 3$, with value 13.) Each positive digit is equally likely, each arithmetic sign ("+" or "×") is equally likely, and all choices are independent. What is the expected value (average value) of her expression?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

628. Boy-girl Adjacent Pairs

Probability Lvl 4/10 (Medium)

N/A

In a class with 15 boys and 10 girls lined up randomly, what is the expected number of times a boy and a girl are adjacent?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

629. Optimal Bid

Probability Lvl 4/10 (Medium)

N/A

You can place a single bid on an item valued randomly between 0 and 1000 by its current owner. If the item's worth to you is 80% higher than to them, you need to determine the optimal bid. If your offer exceeds the owner's valuation, they will accept it. What should your bid be?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

630. 3 Consecutive Heads I

Probability Lvl 2/10 (Easy)

N/A

An unbiased coin is tossed 5 times. Determine the likelihood of getting a sequence of exactly 3 consecutive heads during these 5 tosses.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

631. Two Rolls Reward

Probability Lvl 3/10 (Easy)

N/A

You are playing a game that proceeds in rounds. In each round, you roll an 8-sided die twice. If the two rolls show the same number, the game ends immediately. If the second roll is at least 2 greater than the first, you earn 1 dollar and continue to the next round. In all other cases, you earn nothing, but still proceed to the next round. What is the expected total amount of money you will earn by the time the game ends?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

632. Broken Elevator

Probability Lvl 10/10 (Very Hard)

N/A

Fine Hall has a broken elevator. Every second, it goes up a floor, goes down a floor, or stays still. You enter the elevator on the lowest floor, and after 8 seconds, you are again on the lowest floor. If every possible such path is equally likely to occur, what is the probability you experience no stops?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

633. 3 Lights

Probability Lvl 5/10 (Medium)

N/A

Three lights are placed horizontally on a line on the ceiling. All the lights are initially off. Every second, Neil picks one of the three lights uniformly at random to switch: if it is off, he switches it on; if it is on, he switches it off. When a light is switched, any lights directly to the left or right of that light also get turned on (if they were off) or off (if they were on). What is the expected number of lights that are on after Neil has flipped switches three times?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

634. Dice Game II

Probability Lvl 4/10 (Medium)

Akuna Capital, Citadel, DRW, Hudson River Trading, Optiver

You can roll a die up to three times, earning the value shown on the die. You have the option to stop after each throw and walk away with the money earned. The earnings are not additive. What is the expected payoff of this game?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

635. Checks

Probability Lvl 5/10 (Medium)

N/A

Three knights and three kings are randomly placed on different squares of a 4×4 chessboard. A "check" is counted each time a knight attacks a king. If a single knight threatens multiple kings, each threat counts as a separate check. Similarly, if multiple knights threaten the same king, each of those is also counted individually. What is the expected number of checks?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

636. Arrange Teams

Brainteasers Lvl 5/10 (Medium)

Susquehanna International Group

In a completed round-robin tournament with n teams where each team played every other team once and there were no ties, is it always possible to arrange the teams in a sequence such that each team won against the team listed immediately after it? Prove your claim.

Enter 1 if it is possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

637. Interview IV

Probability Lvl 9/10 (Very Hard)

N/A

You are interviewing 6 candidates for a position, each with a distinct and unknown rating — higher ratings indicate better candidates. The candidates are interviewed one by one in a random order. After each interview, you learn the candidate's relative ranking among all those interviewed so far. Immediately after each interview, you must decide whether to hire that candidate or reject them permanently — you cannot return to a previously rejected candidate. You are allowed to select only one candidate in this process. Your goal is to select the best or the worst candidate out of all six. What is the optimal strategy to maximize the probability of choosing the best or the worst candidate, and what is this maximum probability?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

638. Coin And Die Together

Probability Lvl 2/10 (Easy)

N/A

Consider a fair coin and a fair 6-sided die. The die begins with the number 1 face up. A step starts with a toss of the coin: if the coin comes out heads, we roll the die; otherwise (if the coin comes out tails), we do nothing else in this step. After 5 such steps, what is the probability that the number 1 is face up on the die?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

639. Finding Seven

Probability Lvl 7/10 (Hard)

N/A

Alice rolls three fair six-sided dice and can choose to reroll any subset of the dice, including none or all three. After the reroll, she wins if the sum of the numbers on the three dice is exactly 7. Alice strategizes to maximize her probability of winning. What is the probability that her optimal strategy involves rerolling exactly two of the dice?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

640. Choose Values III

Probability Lvl 5/10 (Medium)

N/A

Alice and Bob are playing a dice game where each uses an identical 6-sided die. Alice designs the die by choosing the face values (each between 1 and 6) and assigning a probability to each value. Once designed, the die is cloned and given to Bob as well. Both players roll their dice once; the player with the higher number wins, and a tie occurs if both roll the same value. How should Alice choose the face values and their probabilities to maximize her chance of winning?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

641. Cherries

Probability Lvl 4/10 (Medium)

N/A

Alice has a bowl with 5 red cherries and 8 purple cherries. She removes cherries one by one until none are left. What is the probability that when the last red cherry is taken, exactly 2 purple cherries remain?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

642. Even Tails Run**Probability** Lvl 6/10 (Medium)

N/A

You flip a fair coin 10 times, generating a sequence of heads (H) and tails (T). In this context, a run of tails refers to a group of one or more consecutive T's, uninterrupted by heads. We are interested in sequences where every run of tails has even length—that is, we never see a single T by itself, nor any group of tails of length 3, 5, etc. For example, sequences like HHHH, HHTT, and TTTT are valid because all runs of tails have even length (0, 2, or 4). In contrast, sequences like HHHT and TTTH are not valid, as they contain a run of tails of length 1 and 3, respectively. What is the probability that a random sequence of 10 fair coin flips contains no run of tails of odd length?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

643. Same Birthday Probability**Probability** Lvl 4/10 (Medium)

N/A

In a class of 23 students, each born in the same year with 365 days, what is the probability that at least two students share the same date of birth?

Enter your answer rounded to 5 decimal places (you may use a calculator for the final numerical computation).

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

644. Matrix Attack I**Probability** Lvl 4/10 (Medium)

N/A

The numbers 1 through 9 are randomly assigned to the 9 cells of a 3×3 matrix, with each number appearing exactly once. What is the probability that the sum of the numbers in each row and each column is even?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

645. Count Of Rational Numbers**Combinatorics** Lvl 3/10 (Easy)

N/A

Determine how many rational numbers can be expressed in the form $\frac{a}{b}$ where a and b are relatively prime positive integers and their product $a \cdot b$ is 25! Note: 25! means 25 factorial.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

646. Shooting Arrows I**Probability** Lvl 1/10 (Easy)

Citadel, DRW

You shoot 3 arrows that land randomly on a circular target divided into three concentric rings with radii 1 meter, 2 meters, and 3 meters. What is the probability that at least one of the arrows lands within the innermost circle (radius 1 meter), assuming all arrows hit the target?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

647. Card Picking I**Probability** Lvl 4/10 (Medium)

N/A

A deck of cards is thoroughly shuffled. Cards are dealt sequentially until the ace of diamonds is revealed. Determine the probability that exactly one king, one queen, and one jack appear before the ace of diamonds.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

648. Game On 753 Coins

Brainteasers Lvl 4/10 (Medium)

N/A

Suppose Alice has 753 coins. She offers to give these coins to her friend Bob if he wins the following game; otherwise, he must pay her the equivalent amount. The game is played as follows: All of Alice's coins start in a single pile. Bob can repeatedly perform the following actions: from any pile containing at least 3 coins, Bob can remove one coin and then divide the remaining coins into two separate piles. Each pile must have at least one coin, but the sizes do not need to be equal. Bob wins if he can divide all piles so that each pile contains exactly three coins. He loses if he cannot achieve this. Should Bob accept Alice's wager? Enter 1 for Yes and 2 for No.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

649. Triangle Inside Circle III

Probability Lvl 7/10 (Hard)

N/A

Three random points are chosen independently and uniformly along the circumference of a unit circle and labeled A , B , and C . Determine the expected perimeter of the triangle formed by these points ABC .

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

650. Coin Toss Game VI

Probability Lvl 3/10 (Easy)

N/A

Alice flips a fair coin until she gets a head. She earns the smaller value between 64 dollars and 2^n dollars, where n is the number of flips. What is her expected reward?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

651. Count Paths I

Combinatorics Lvl 3/10 (Easy)

N/A

How many paths lead from $(0, 0)$ to $(8, 10)$, given that only moves of one unit right or one unit up are allowed?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

652. Game On Divisors II

Brainteasers Lvl 9/10 (Very Hard)

Susquehanna International Group

Alice and Bob are participants in a game governed by the following rules. The game board contains numbers from 1 to 1000000.

Alice goes first. During each turn, a player must remove a number from the board that has not been previously chosen. Additionally, the player is required to remove all divisors (that are still remaining) of that number in the same move. The game continues until one player removes the final number on the board, thereby winning. Who has a winning strategy under optimal play?

Enter 1 for Alice and 2 for Bob.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

653. Polygon Triangle

Probability Lvl 6/10 (Medium)

N/A

Three distinct vertices are chosen at random from the vertices of a regular polygon with 17 sides. Each set of three vertices determines a triangle. What is the probability that the center of the polygon lies inside the triangle determined by these three vertices?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

654. Ants On A Wire

Brainteasers Lvl 3/10 (Easy)

N/A

Alice and Bob each have a group of ants. Alice has 40 ants, and Bob has 80 ants. They stand at opposite ends of a narrow wire and release all of their ants onto it. The ants begin crawling along the wire, and every time two ants bump into each other, they reverse direction and continue moving forward. Determine how many total collisions occur between the ants.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

655. Clock Angle

Brainteasers Lvl 1/10 (Easy)

Akuna Capital, DRW

Determine the angle between the hour and minute hands of a clock at 3:15 PM.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

656. Weighted Coins

Probability Lvl 7/10 (Hard)

N/A

Keith has 10 coins labeled 1 through 10, where the i th coin has weight 2^i . The coins are all fair, so the probability of flipping heads on any of the coins is $\frac{1}{2}$. After flipping all of the coins, Keith takes all of the coins which land heads and measures their total weight, W . What is the probability that $137 \leq W \leq 1061$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

657. 5000 Sets

Combinatorics Lvl 6/10 (Medium)

N/A

You have 5000 distinct finite sets. Their intersection is empty. However, the intersection of any two is nonempty. What is the smallest possible number of elements contained in their union?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

658. Three Steps

Probability Lvl 8/10 (Hard)

N/A

An ant starts on the number line at 0 and crawls in the following manner. For $n = 1, 2, 3$, it chooses a time duration t_n and an increment x_n independently and uniformly at random from the interval $(0, 1)$. During the

n -th step of the process, the ant moves x_n units in the positive direction, using up t_n minutes. If the total elapsed time exceeds 1 minute during the n -th step, it stops at the end of that step; otherwise, it continues with the next step, taking at most 3 steps in total. What is the probability that the ant's position when it stops will be greater than 1?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

659. Just Even Stop

Probability Lvl 4/10 (Medium)

N/A

A fair coin is flipped repeatedly until two consecutive heads or two consecutive tails appear for the first time. What is the probability that the total number of flips is even?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

660. Shooting Stacks

Combinatorics Lvl 3/10 (Easy)

Optiver

A shooter is aiming at a set of targets organized into four vertical stacks. The stacks contain 2, 3, 3, and 4 targets, respectively. The shooter selects a stack for each shot and hits the lowest unbroken target in the chosen stack. Once all targets in a stack are hit, that stack can no longer be selected. Determine the total number of distinct sequences in which the shooter can hit all the targets.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

661. Segments Inside Square I

Probability Lvl 7/10 (Hard)

N/A

Take the unit square with corners at $(0, 0)$, $(1, 0)$, $(0, 1)$, and $(1, 1)$. Call $A = (0, 0)$ and $B = (1, 0)$. Pick a point P uniformly at random in the square and draw the segment AP . Independently pick a point Q uniformly at random in the square and draw the segment BQ . What is the probability that the two line segments AP and BQ intersect?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

662. Dice Game VI

Probability Lvl 5/10 (Medium)

Optiver

You have a fair 6-sided die. After rolling it, you can either receive the observed value or roll again. If you roll again and the total of both rolls is 7 or higher, you pay the amount of your first roll. If the total is less than 7, you receive the sum of both rolls. What is the expected payout?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

663. Fibonacci Error

Brainteasers Lvl 5/10 (Medium)

N/A

Suppose you are computing terms in the Fibonacci sequence and correctly calculate $F_0, F_1, \dots, F_{99}$. However, you mistakenly compute F_{100} as $G_{100} = 1 + F_{100}$. If this error carries over into your calculations of $F_{101}, \dots, F_{110}$, let G_{110} denote the incorrect value you compute for F_{110} . Determine $G_{110} - F_{110}$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

664. Digit Sum V

Brainteasers Lvl 2/10 (Easy)

N/A

Let $S(n)$ denote the sum of the digits of the integer n . If $S(n) = 2018$, what is the smallest possible value $S(n+1)$ can be?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

665. Expected Swaps

Probability Lvl 10/10 (Very Hard)

Two Sigma

An array of length n contains distinct numbers arranged in a random order. The bubble sort algorithm sorts the array by repeatedly swapping adjacent elements that are inverted (i.e., in the wrong relative order) until no inversions remain. What is the expected number of swaps required to sort the array?

Enter the answer for $n = 10$ rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

666. Interesting Average I

Probability Lvl 6/10 (Medium)

N/A

A number is interesting if it is a 6-digit integer that contains no zeros, its first 3 digits are strictly increasing, and its last 3 digits are non-increasing. What is the average of all interesting numbers?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

667. Changing Number

Brainteasers Lvl 7/10 (Hard)

Jane Street

What is the smallest positive integer that ends in the digit 4 such that moving its last digit to the front (for example, $1234 \rightarrow 4123$) results in a number that is four times the original?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

668. Acute Triangle

Probability Lvl 8/10 (Hard)

Jane Street, Susquehanna International Group

Three points are chosen randomly along the circumference of a circle and joined to form a triangle. What is the probability that the resulting triangle is acute-angled?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

669. Segment Intersection

Brainteasers Lvl 4/10 (Medium)

N/A

Is it possible to draw 69 line segments in the plane such that each segment intersects exactly three others?

Enter 1 if it's possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

670. Powerful Three

Brainteasers Lvl 5/10 (Medium)

N/A

Does there exist a positive integral power of 3 whose decimal representation ends with 100 zeros followed by a 1? In other words, is there an exponent $k > 0$ such that the last 101 decimal digits of 3^k are 000...001?

Enter 1 if such a positive integer exists and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

671. Socks Remaining II

Probability Lvl 8/10 (Hard)

Jane Street

You are taking out socks one by one at random from a drawer that contains 10 red socks, 20 blue socks, and 30 green socks. What is the probability that, after all the red socks have been taken out, there is still at least one blue sock and at least one green sock left in the drawer?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

672. Tetrahedron Coloring I

Combinatorics Lvl 2/10 (Easy)

N/A

In how many different ways can you paint the four faces of a tetrahedron using exactly four colors so that each face gets a different color, counting two colorings as the same whenever one can be rotated to match the other?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

673. Domino Tiling I

Combinatorics Lvl 4/10 (Medium)

N/A

How many ways are there to tile a 2×10 rectangular grid with 2×1 dominos, when each domino may be placed either vertically or horizontally and two tilings are considered different if they differ in which unit squares each domino covers?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

674. Domino Tiling II

Combinatorics Lvl 8/10 (Hard)

N/A

How many ways are there to tile a 3×8 rectangular grid with 2×1 dominos, when each domino may be placed either vertically or horizontally and two tilings are considered different if they differ in which unit squares each domino covers?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

675. Largest Mean Median Difference

Brainteasers Lvl 2/10 (Easy)

N/A

Five students take a test where any integer score from 0 to 100 (inclusive) is allowed. The mean of the scores is defined as the average of all five scores, and the median is the middle score when the five scores are arranged in increasing order. What is the largest possible value of the (absolute) difference between the mean and the median?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

676. New Language

Combinatorics Lvl 1/10 (Easy)

N/A

Arin speaks a made-up language called Lumese, whose alphabet contains exactly five “letters”: vak, vek, vik, vok, vuk. A Lumese word may have length 1, 2, 3, 4, or 5 letters. In any word, Arin may choose to add emphasis to any letter independently (an emphasized letter is treated as different from its unemphasized version). How many distinct Lumese words are possible in total?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

677. Constrained Subsets III

Combinatorics Lvl 4/10 (Medium)

N/A

Find the number of subsets of $\{1, 2, 3, \dots, 64\}$ whose elements sum to 2072.

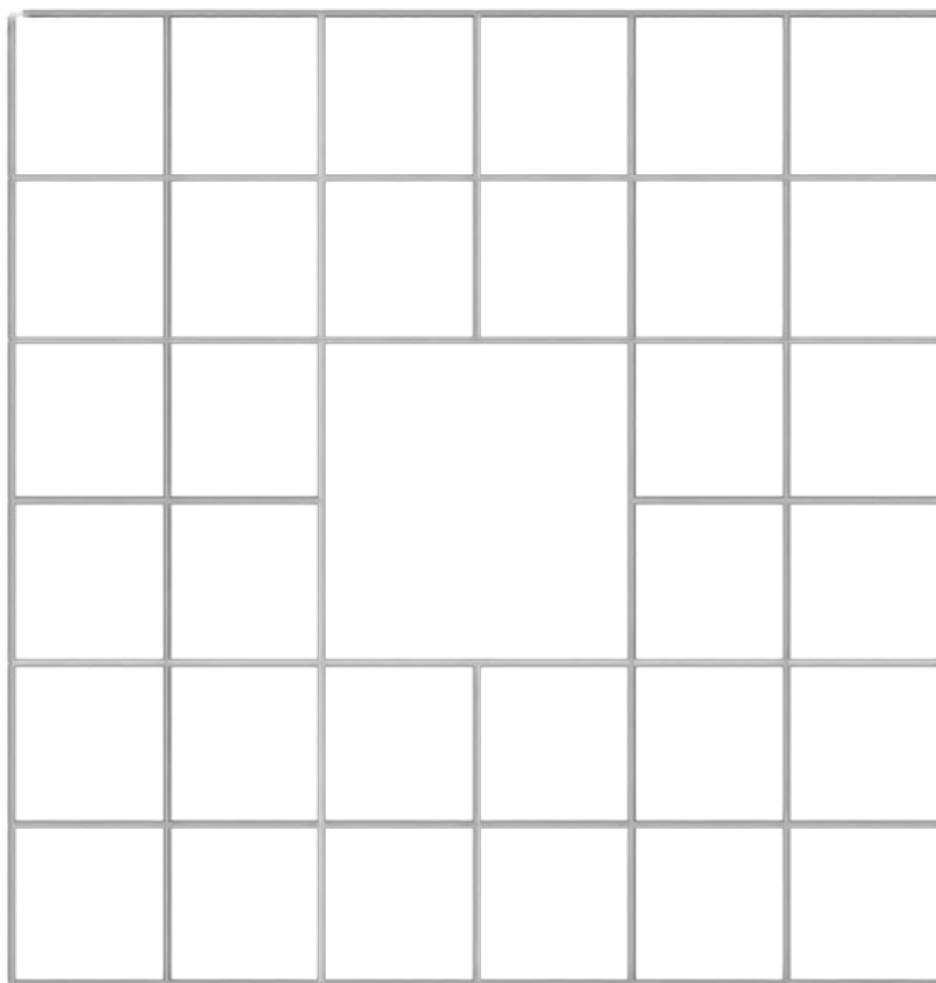
[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

678. Count Rectangles II

Combinatorics Lvl 4/10 (Medium)

N/A

A square grid (shown in the figure) is formed by seven equally spaced vertical lines and seven equally spaced horizontal lines, but a central portion has been removed so that the visible grid has a hollow box in the middle. How many non-degenerate rectangles are there whose four edges lie entirely on the visible grid lines of this figure?



[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

679. Tricky Palindrome I

Combinatorics Lvl 2/10 (Easy)

N/A

A palindrome is a positive integer n not divisible by 10 whose decimal digits read the same forward and backward (for example, 6 and 39793). Determine how many palindromes are less than 1000.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

680. Tricky Palindrome II

Combinatorics Lvl 4/10 (Medium)

N/A

A palindrome is a positive integer whose decimal digits read the same forwards and backwards (and which is not divisible by 10). Determine how many four-digit integers n have the property that both n and $2n$ are palindromes.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

681. Tricky Palindrome III

Combinatorics Lvl 7/10 (Hard)

N/A

A palindrome is a positive integer whose decimal digits read the same forwards and backwards (and which is not divisible by 10). Suppose a positive integer n has the property that each of $n, 2n, 3n, \dots, 9n$ is a palindrome. Prove that every decimal digit of n is either 0 or 1.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

682. Skipping 2s

Combinatorics Lvl 6/10 (Medium)

Citadel

Ava types the positive integers in order starting from 1. Her keyboard's 2 key is broken, so she skips every number that contains the digit 2 anywhere in its decimal representation. What is the 2026th number that Ava types?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

683. Arrow On Dice

Combinatorics Lvl 5/10 (Medium)

N/A

A standard six-sided die has faces labeled 1 through 6. On each face you draw exactly one arrow that points to one of the four faces that share an edge with it (you may not point to the opposite face). The arrows must form a single directed 6-step cycle: starting at any face and following the arrows you visit every face exactly once and return to your starting face. How many different ways are there to draw the arrows so that this happens?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

684. Grid Arrangement III

Combinatorics Lvl 6/10 (Medium)

N/A

The squares of a 3×3 grid are filled with positive integers so that the top-left square contains 1, the bottom-right square contains 171, and every entry divides the number immediately to its right and the number immediately below it (whenever those neighbors exist). How many such fillings are possible?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

685. Tetrahedron Coloring II

Combinatorics Lvl 3/10 (Easy)

N/A

In how many distinct ways can the four vertices of a tetrahedron be colored red, blue, or green so that no face has all three vertices the same color? Two colorings are considered the same if one can be rotated in space to match the other. Also, you do not need to use all three colors.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

686. Logarithm Comparison

Brainteasers Lvl 4/10 (Medium)

Hudson River Trading

Which is larger: $\log_{1111}(1112)$ or $\log_{1112}(1113)$?

Enter 1 for $\log_{1111}(1112)$ and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

687. Circular Frog

Probability Lvl 7/10 (Hard)

Citadel

Ten points are marked equally spaced on a circle. A frog starts at one of the points. Every second the frog jumps to one of the two adjacent points (clockwise or anticlockwise), each with probability $1/2$. What is the expected time (in seconds) until the frog has visited all ten points?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

688. Square Factor

Probability Lvl 3/10 (Easy)

N/A

A random positive divisor of $2^3 \cdot 3^5 \cdot 5^7 \cdot 7^9$ is chosen uniformly at random. What is the probability that the divisor is a perfect square?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

689. Sum Sum Sum

Brainteasers Lvl 7/10 (Hard)

Citadel

Take 19^{100} . Add its digits. Keep adding the digits of each result until only one digit remains. What is that final digit?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

690. Constrained Combinatorics

Combinatorics Lvl 4/10 (Medium)

N/A

How many ordered quadruples (w, x, y, z) of four distinct numbers chosen from $1, 2, \dots, 9$ satisfy $x < w$, $x < y$, and $z < y$?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

691. Evenbin I

Brainteasers Lvl 4/10 (Medium)

N/A

A positive integer is called evenbin if the number of 1's in its binary representation is even. For example, $12_{10} = 1100_2$ is an evenbin number. What is the 2050th smallest evenbin number?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

692. Prime Set

Brainteasers Lvl 3/10 (Easy)

N/A

The average of a set of distinct primes is 27. What is the largest prime that can belong to this set?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

693. Catch Me If You Can I

Brainteasers Lvl 4/10 (Medium)

N/A

A police officer tries to catch a thief running along the positive x -axis. Every second the officer can aim at any real point on the road; if the thief is at that point at that second the thief is caught. The thief starts at the origin and runs with an unknown positive integer speed (constant). A firing sequence is a list $a_1, a_2, a_3, \dots$ meaning the officer aims at a_1 after 1 second, at a_2 after 2 seconds, and so on. Does there exist a firing sequence that guarantees the officer will catch the thief eventually? If yes, give such a sequence. If no, prove it.

Enter 1 if such a sequence exists and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

694. Catch Me If You Can II

Brainteasers Lvl 8/10 (Hard)

N/A

A police officer tries to catch a thief running along the non-negative x -axis. The thief starts at an unknown integer position $a \geq 0$ and runs to the right with an unknown positive integer speed b . Each second the officer may check (aim at) any real point on the road; if the thief is at that point at that second, the thief is caught. A firing sequence is a list $x_1, x_2, x_3, \dots$ where x_n is the point checked at time n . Does there exist a sequence that guarantees the officer will eventually catch the thief? If yes, give such a sequence. If no, prove it.

Enter 1 if such a sequence exists and 2 otherwise.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

695. Game On Divisors IV

Brainteasers Lvl 7/10 (Hard)

N/A

Alice and Bob play a game with a positive integer N . Alice goes first, and the players alternate turns. On a player's turn, if the current number is N , the player must choose a proper divisor d of N (that is, $1 < d < N$), replace N by $N - d$, and pass the resulting number to the opponent. If a player is given a number that is either prime or equal to 1, that player immediately loses. For $N \in \{1, 2, \dots, 1000\}$, how many values of N are winning positions for Alice, assuming both players play optimally?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

696. Evenbin II

Brainteasers Lvl 4/10 (Medium)

N/A

A positive integer is called evenbin if the number of 1's in its binary representation is even. For example, $12_{10} = 1100_2$ is an evenbin number. What is the 2050th smallest evenbin number? What is the sum of all evenbin numbers between 0 and 511 inclusive?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

697. Hexagon Inside Hexagon

Probability Lvl 5/10 (Medium)

Citadel

We have a regular hexagon of side length 1. On each side we pick a point uniformly at random (independently) and then join these six chosen points in order to form a new hexagon. What is the expected area of this inner hexagon?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

698. Subset Chain

Combinatorics Lvl 4/10 (Medium)

N/A

Determine the number of sequences of sets $S_1, S_2, \dots, S_{999}$ such that

$$S_1 \subseteq S_2 \subseteq \dots \subseteq S_{999} \subseteq \{1, 2, \dots, 999\}$$

Here $A \subseteq B$ means that all elements of A are also elements of B .

If the answer is of the form 10^x , enter the value of x .

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

699. Triangle Tiling

Combinatorics Lvl 5/10 (Medium)

N/A

An isosceles right triangle has legs of length 5. In how many different ways can it be tiled (partitioned) into isosceles right unit triangles whose legs have length 1?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

700. Constrained Subsets IV

Brainteasers Lvl 4/10 (Medium)

N/A

Let S be a subset of $\{1, 2, 3, \dots, 2000\}$ with the property that for any two distinct $x, y \in S$ the difference $x - y$ does not divide the sum $x + y$. What is the largest possible size of S ?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

701. Rook Moves

Probability Lvl 5/10 (Medium)

N/A

Consider an 8×8 grid. A rook starts at the bottom-left square. Each minute, it moves to a different square in the same row or the same column, choosing uniformly at random among all such squares. The rook is required to move at every step and cannot remain in place. What is the expected time, in minutes, for the rook to reach the top-right square?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

702. Real Addition I

Brainteasers Lvl 3/10 (Easy)

N/A

Alice and Bob play a game on a blackboard. The number 0 is written at the start. Alice goes first and they alternate moves. On a turn a player replaces the current number x by any real number y satisfying $0 < y - x < 1$, i.e. each move increases the board by some positive real less than 1. The first player who writes a number ≥ 1000 wins. Who has a winning strategy?

Enter 1 for Alice and 2 for Bob.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

703. Real Addition II

Brainteasers Lvl 3/10 (Easy)

N/A

Alice and Bob play a game on a blackboard. The number 0 is written at the start. Alice goes first and they alternate moves. On a turn a player replaces the current number x by any real y with $0 < y - x < 1$, so each move increases the board by some positive amount less than 1. The player who first writes a number ≥ 1000 on their 1001st turn or later wins, but if a player writes a number ≥ 1000 on their 1000th turn or earlier, they lose immediately. Determine who has a winning strategy.

Enter 1 for Alice and 2 for Bob.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

704. Strange Comparison II

Brainteasers Lvl 4/10 (Medium)

N/A

Which is larger: 7^{92} or 8^{91} ?

Enter 1 for 7^{92} and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

705. GCD Expectation

Probability Lvl 5/10 (Medium)

Jump Trading

A divisor d is chosen uniformly at random from all positive divisors of 2025^{2025} . Find the expected value $E[\gcd(d, 2025)]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

706. Expected Good Points

Probability Lvl 6/10 (Medium)

N/A

Fifty points are placed independently and uniformly at random on the interval $[0, 1]$. For each point, its neighbor is the closest other point. A point is called good when it is the neighbor of its neighbor (i.e. mutual nearest neighbors). Find the expected number of good points.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

707. Count Triangular Regions I

Combinatorics Lvl 3/10 (Easy)

N/A

In triangle ABC , mark 10 distinct points on side BC and join each to vertex A , mark 10 distinct points on side AC and join each to vertex B . Assuming no three drawn segments meet at a single interior point, into how many regions is the triangle divided?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

708. Count Triangular Regions II

Combinatorics Lvl 4/10 (Medium)

N/A

In a triangle, from each vertex draw 10 distinct straight segments to ten distinct points on the opposite side (so each vertex sends 10 segments to the opposite side). Assume no three drawn segments meet at the same interior point. How many smaller regions does the triangle get divided into?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

709. Repeated Erase

Probability Lvl 9/10 (Very Hard)

N/A

Two distinct integers are chosen uniformly at random from $\{1, 2, \dots, 1000\}$ and encircled, call the smaller one a and the larger b . Afterwards the following procedure runs: repeatedly pick two (distinct) numbers remaining on the board uniformly at random and erase the smaller of the two. The procedure continues until a single number remains. What is the probability that the smaller encircled number a is erased before the larger encircled number b ?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

710. Everything Odd

Probability Lvl 5/10 (Medium)

N/A

A fair three-sided die (faces 1, 2, 3) is rolled repeatedly until each face has appeared an odd number of times. What is the expected number of rolls?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

711. Expectation Lower Bound

Probability Lvl 5/10 (Medium)

N/A

Let X be a nonnegative continuous random variable with probability density $f(x)$ satisfying $0 \leq f(x) \leq 1$ for all $x \geq 0$. What is the smallest possible value of $E[X]$? Note that an intuitive answer doesn't count, you will have to prove it mathematically.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

712. Expected Stock Price

Probability Lvl 3/10 (Easy)

Citadel

A stock starts at price 100. Each day its price either rises by 5% (multiplier 1.05) with probability $\frac{3}{5}$, or falls by 3% (multiplier 0.97) with probability $\frac{2}{5}$. What is the expected stock price after 10 days?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

713. Even Lattice Points

Probability Lvl 5/10 (Medium)

N/A

A lattice point is a point whose coordinates are both integers. Alex starts at $(0, 2004)$ and walks in a straight line to a uniformly random lattice point chosen from the interior of the square with vertices $(0,0), (0,99), (99,99), (99,0)$. (Interior lattice points have integer coordinates $1 \leq a, b \leq 98$.) What is the probability that the line segment from $(0, 2004)$ to the chosen point contains an even number of lattice points, counting both endpoints?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

714. Polygon Constraints

Brainteasers Lvl 4/10 (Medium)

N/A

Each vertex of a convex n -gon is labeled with a number from the set $\{1, 2, \dots, 10\}$; labels may repeat. It is required that for every unordered pair of distinct labels i, j there is at least one side of the polygon whose endpoint labels are i and j . What is the smallest possible n ? Note that you must also prove that a configuration achieving your minimum value of n actually exists.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

715. Divisible Permutation

Combinatorics Lvl 5/10 (Medium)

N/A

Arrange the numbers $\{1, 2, \dots, 25\}$ in a sequence $\{a_1, a_2, \dots, a_{25}\}$ with the property that whenever the index m divides the index n (so $m \mid n$), the entry in position n is a multiple of the entry in position m ; i.e. $a_m \mid a_n$ whenever $m \mid n$. How many permutations $\{a_1, \dots, a_{25}\}$ of $\{1, \dots, 25\}$ satisfy this condition?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

716. Lottery

Probability Lvl 4/10 (Medium)

N/A

A lottery has tickets labeled $1, 2, \dots, 1000$. The lottery is run as follows: draw a random ticket; if it shows an odd number stop and award prizes to everyone whose ticket was drawn (including that odd ticket); if it is even, draw again and repeat until an odd number appears. You hold ticket number 1000. What is the probability you win a prize?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

717. Central Cities

Combinatorics Lvl 3/10 (Easy)

N/A

Ten cities in a state are joined by some undirected road, altogether there are 40 roads. Call a city central if it has a direct road to every other city. What is the largest possible number of central cities?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

718. Expected Wait Time III

Probability Lvl 4/10 (Medium)

N/A

Two buses arrive at a bus stop sometime between 10:00 and 11:00 - Bus A's first arrival time after 10:00 is chosen uniformly in $[0, 60]$ minutes and independently Bus B's first arrival time is chosen uniformly in $[0, 60]$. After their first visits Bus A repeats every 3 minutes and Bus B repeats every 5 minutes. You arrive at a time chosen uniformly at random between 11:00 and 2:00. What is the expected waiting time (in minutes) until the first of the two buses arrives?

Enter your answer in minutes rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

719. Expected Wait Time IV

Probability Lvl 5/10 (Medium)

N/A

Two buses both stop at a location together at 10:00 AM. After that, Bus A repeats every 3 minutes and Bus B repeats every 5 minutes (perfectly periodic). You arrive at a time chosen uniformly at random between 11:00 AM and 2:00 PM. What is your expected waiting time until the first of the two buses arrives?

Enter your answer in minutes rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

720. Expected Occupied Seats

Probability Lvl 6/10 (Medium)

N/A

Six seats are placed in a row. People arrive one at a time, each arriving person chooses uniformly at random one of the currently available (unoccupied and unblocked) seats and sits there. Whenever someone sits in a seat, the one or two seats immediately adjacent on each side become blocked and may never be occupied. The arrivals continue until no available seats remain. What is the expected number of occupied seats at the end?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

721. N-Palindrome

Brainteasers Lvl 3/10 (Easy)

N/A

An n -string is a string of digits formed by writing the numbers $1, 2, \dots, n$ in some order (in base ten). For example, one possible 10-string is 71053246189. What is the smallest $n > 1$ such that there exists a palindromic n -string?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

722. Maximum Fraction Sum

Brainteasers Lvl 3/10 (Easy)

N/A

Two positive rational numbers x and y , when written in lowest terms, have the property that the sum of their numerators is 9 and the sum of their denominators is 10. What is the largest possible value of $x + y$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

723. Circular Seating

Probability Lvl 4/10 (Medium)

N/A

Twenty guests take seats around a round table by the following random procedure. At each step a guest is chosen uniformly at random from those not yet seated, and that guest then chooses an empty seat uniformly at random, continue until all guests are seated. Alice is one of the guests. What is the probability that, at the moment Alice sits down, both neighboring seats (the one immediately to her left and the one immediately to her right) are already occupied?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

724. 2, 4 and 6

Brainteasers Lvl 6/10 (Medium)

N/A

The numbers 2, 4 and 6 are written on a board. Repeatedly do the following: choose (with repetition allowed) three numbers x, y, z already on the board and compute

$$F(x, y, z) = xyz + xy + yz + zx + x + y + z.$$

If $F(x, y, z)$ is not already on the board and $F(x, y, z) \leq 2000$, write it on the board. Keep repeating until no new numbers can be added. How many numbers are on the board at the end?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

725. Half Permutation

Probability Lvl 4/10 (Medium)

N/A

A permutation of the first n positive integers is called "half" if, for all $i > 1$, i comes after $\lfloor i/2 \rfloor$ in the permutation. What is the probability that a random permutation of the first 14 integers is half?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

726. Proportional Coordinates

Probability Lvl 5/10 (Medium)

N/A

A person assigns to each lattice point (x, y) with $1 \leq x, y < 100$ an integer from 1 to 100, inclusive. For every fixed row $y = k$ the 99 assigned numbers in that row are all distinct and none equals k . Likewise, for every fixed column $x = k$ the 99 numbers in that column are all distinct and none equals k . A point (x, y) is then chosen at random with probability proportional to the integer assigned to it. Find the expected value of $x + y$ for the chosen point.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

727. Grid Frog II

Combinatorics Lvl 8/10 (Hard)

Citadel

A 3×8 grid has one frog in each of its 24 cells. Simultaneously, every frog must jump to one of its neighboring cells (only cells sharing a side count; diagonal moves are not allowed). After all jumps, exactly one frog must occupy each cell again. In how many different ways can all the frogs move so that no two frogs land in the same square?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

728. Max Min Ratio

Probability Lvl 5/10 (Medium)

N/A

Let $X_1, \dots, X_{10}$ be independent random variables, each uniformly distributed on $[0, 1]$. Write

$$m = \min(X_1, \dots, X_{10}), \quad M = \max(X_1, \dots, X_{10})$$

Compute the expected value $E\left[\frac{m}{M}\right]$. Note that this may not be the same as $E[m]/E[M]$ as for two random variables X and Y , $E[X/Y] \neq E[X]/E[Y]$ in general.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

729. Increasing Random Variables

Probability Lvl 6/10 (Medium)

Jane Street

Three numbers are chosen independently at random: a_1 uniformly from $[0, 1]$, a_2 uniformly from $[0, 2]$, and a_3 uniformly from $[0, 3]$. What is the probability that they appear in strictly increasing order: $a_1 < a_2 < a_3$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

730. Largest In Smallest

Probability Lvl 4/10 (Medium)

N/A

Four independent random variables are drawn: $A_1 \sim \text{Unif}(0, 2)$, $A_2 \sim \text{Unif}(0, 4)$, $A_3 \sim \text{Unif}(0, 6)$, and $A_4 \sim \text{Unif}(0, 8)$. What is the probability that A_1 is the largest of the four, i.e.

$$\Pr(A_1 > \max(A_2, A_3, A_4))?$$

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

731. Alice Rolled The 6

Probability Lvl 4/10 (Medium)

N/A

Alice and Bob take turns rolling a fair six-sided die, with Alice rolling first. They stop as soon as someone rolls a 6. Given that Alice is the player who rolls the first 6, what is the expected total number of die rolls (counting both Alice's and Bob's rolls) made in the game?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

732. Given LCM

Combinatorics Lvl 3/10 (Easy)

N/A

How many ways are there to choose positive integers x and y such that the lowest common multiple of x and y is 216?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

733. Conditional Normal I

Probability Lvl 3/10 (Easy)

N/A

Let $X \sim N(0, 1)$. Find the conditional expectation $E[X \mid X^4]$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

734. Conditional Normal II

Probability Lvl 4/10 (Medium)

N/A

Let $X, Y \sim N(0, 1)$ IID. Find the conditional expectation $E[X \mid X^4 + Y^4]$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

735. Conditional Normal III

Probability Lvl 4/10 (Medium)

N/A

Let $X, Y \sim N(0, 1)$ IID. Find the conditional expectation $E[X \mid X^4 + Y]$.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

736. Triangle Formation III

Probability Lvl 5/10 (Medium)

N/A

Let X and Y be independent $\text{Unif}(0, 1)$ random variables. Using the three positive numbers $X, Y, 1$ as side-lengths, what is the probability that they form a triangle which is obtuse?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

737. Same Part

Probability Lvl 5/10 (Medium)

N/A

A unit stick is marked at two random independent points (uniform on the stick). Then four independent random points are chosen (uniform on the stick) and the stick is cut at those four points, producing five pieces. What is the probability that the two originally marked points lie in the same resulting piece?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

738. Elevator I

Probability Lvl 3/10 (Easy)

N/A

An elevator runs continuously between floors 1 and 50 at a constant speed. It reverses direction only at the end floors (1 and 50). The elevator does not linger - it only stops at a floor if somebody wants to board, and any boarding/alighting takes negligible time. Also, the time it takes for the elevator to stop is negligible. You live on floor 40 and arrive there at a time chosen uniformly at random. When the elevator next reaches your floor to pick you up, what is the probability that it is moving down?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

739. Unfair Advantage

Probability Lvl 7/10 (Hard)

N/A

A well-shuffled standard 52-card deck is dealt so that Player 1 receives the top 13 cards (in order) and Player 2 receives the remaining 39 cards (also keeping order). The two players then reveal cards simultaneously one round at a time: on round k (for $k = 1, \dots, 13$) Player 1 shows the k -th card from his 13 and Player 2 shows the k -th card from his 39 (which come from original deck positions 1..13 and 14..26 respectively). If no one has yet revealed an ace after 13 simultaneous rounds, Player 2 continues alone through his remaining cards (rounds 14 to 39 of his deck). The first player to reveal an ace wins; if both reveal their first ace in the same simultaneous round the game is a tie. What is the probability that Player 1 wins?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

740. Monotonic Subsequences

Combinatorics Lvl 3/10 (Easy)

N/A

Arrange the numbers 1 through 12 in a row so that the subsequence formed by the six even numbers is monotonic (either entirely increasing or entirely decreasing) and the subsequence formed by the six odd numbers is also monotonic. How many such permutations are there?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

741. Possible Cover

Brainteasers Lvl 5/10 (Medium)

N/A

An 8×8 chessboard has its top-left and bottom-right squares removed. Can the remaining 62 squares be exactly covered by thirty-one 1×2 dominos, each placed either horizontally or vertically?

Enter 1 if it possible and 2 otherwise.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

742. Alternating Sum Sum

Brainteasers Lvl 8/10 (Hard)

N/A

Let $I = 1, 2, \dots, 30$. For any nonempty subset $M \subseteq I$ write its elements in decreasing order $i_1 > i_2 > \dots > i_k$. Define the alternating sum of M to be $i_1 - i_2 + i_3 - \dots + (-1)^{k-1} i_k$. (For the empty set we define the alternating sum to be 0).

What is the sum of the alternating sums of all subsets of I (including the empty set)?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

743. 3 Eggs

Brainteasers Lvl 7/10 (Hard)

Optiver, Jane Street

You have three identical eggs and a building with an unknown threshold level X : an egg breaks if dropped from any floor $\geq X$ and survives if dropped from any floor $< X$. You may perform at most 10 drops. What is the largest number of floors for which you can always determine X in the worst case?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

744. Expected Minimum

Probability Lvl 4/10 (Medium)

N/A

Five integers are chosen independently and uniformly at random from the set $1, 2, 3, 4, 5$ (the same integer may be selected multiple times). What is the expected value of the minimum of these 5 values?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

745. Grid Arrangement IV

Probability Lvl 4/10 (Medium)

Citadel, Jane Street, Graviton

Place the numbers 1 through 9 at random into the cells of a 3×3 grid, each number used exactly once. What is the probability that the sum of the three entries in every row and every column is odd?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

746. Subset Length Sum

Combinatorics Lvl 7/10 (Hard)

N/A

For each integer n , let $s(n)$ denote the number of subsets of $\{1, 2, \dots, 13\}$ whose elements sum to exactly n .

Compute $\sum_{n=46}^{86} s(n)$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

747. Extended Rock-Paper-Scissors

Probability Lvl 6/10 (Medium)

N/A

Five players repeatedly play independent rounds of rock–paper–scissors. In each round every remaining player chooses uniformly at random one of Rock, Paper, Scissors, independently of others and of previous rounds. If exactly two distinct gestures appear in a round, then the players who chose the losing gesture are eliminated and the players who chose the winning gesture survive to the next round. If exactly one gesture or all three gestures appear, no one is eliminated that round. Starting with five players, find the expected number of rounds until only a single player remains.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

748. King Before Ace

Probability Lvl 8/10 (Hard)

Citadel

A standard 52-card deck is shuffled and cards are turned over one at a time without replacement. Given that the first king appears earlier in the deck than the first ace, find the expected number of cards that lie strictly between these two cards (i.e., the number of cards between the first king and the first ace, not counting the king or ace themselves).

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

749. To Infinity And Beyond II

Probability Lvl 5/10 (Medium)

N/A

Let X_1 be chosen uniformly on $(0, \frac{1}{2})$. For every $n \geq 2$, given $X_{n-1} = x$, choose X_n uniformly on $(x, x + 2^{-n})$. Let $X = \lim_{n \rightarrow \infty} X_n$. Find the variance $\text{Var}(X)$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

750. Everything Contained

Probability Lvl 5/10 (Medium)

N/A

We form 8 random intervals in $(0, 1)$ by repeating the following independent experiment 8 times: pick two points uniformly at random in $(0, 1)$ and take the interval with those points as endpoints. What is the probability that one of these 8 intervals contains every other interval?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

751. Disjoint Arcs

Probability Lvl 5/10 (Medium)

Citadel

Pick four distinct points on a circle in this order: first choose point A , then choose B . Draw the red arc by moving counterclockwise from A to B . Next pick C and then D , and draw the green arc by moving counterclockwise from C to D . What is the probability that the red arc and the green arc are disjoint (they have no common points)?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

752. Maximize RPS Wins

Brainteasers Lvl 4/10 (Medium)

Citadel

Alice and Bob play 10 rounds of Rock–Paper–Scissors. Alice's moves are: 3 Rocks, 5 Scissors, 2 Papers. Bob's moves are: 4 Rocks, 2 Scissors, 4 Papers. You may choose which of Alice's moves is paired against which of Bob's moves (one-to-one). No round may end in a tie. What is the maximum number of rounds Alice can win?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

753. Maximum Within An Interval

Probability Lvl 4/10 (Medium)

Citadel

Five independent points are chosen uniformly at random from the interval $[0, 5]$. What is the probability that the largest of the five points lies in $[3, 4]$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

754. Bulb Toggle I

Brainteasers Lvl 4/10 (Medium)

Citadel

You have 1000 light bulbs in a row, initially all off and numbered 1 through 1000. Person k (for $k = 1, 2, \dots, 1000$) visits and toggles every k -th bulb (on $\rightarrow$ off, off $\rightarrow$ on). After all 1000 people have gone, how many bulbs are finally on?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

755. Ten Cars Meet

Probability Lvl 7/10 (Hard)

Jump Trading

Ten cars arrive independently and uniformly at random between 10:00 and 11:00 (so arrival times are iid Uniform $[0, 60]$ minutes). Each car stays exactly 15 minutes after it arrives. What is the probability that there exists a time during the hour when all ten cars are present simultaneously?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

756. Perfect Rank Guess

Probability Lvl 9/10 (Very Hard)

Citadel , Two Sigma

Four numbers $X_1, \dots, X_4$ are drawn iid Uniform $[0, 1]$. They are revealed one at a time (in the order drawn). Immediately after each reveal you must guess the final rank of that revealed number among the four (rank 1 = smallest, rank 4 = largest). You win only if all four guesses are correct. What strategy maximizes the winning probability, and what is the maximum probability?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

757. Interesting Average III

Probability Lvl 4/10 (Medium)

N/A

Determine the average of all ten-digit positive integers $d_9d_8\dots d_1d_0$ that satisfy $|d_i - i| \leq 1$ for each $i \in \{0, 1, \dots, 9\}$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

758. Seven Colors II

Probability Lvl 7/10 (Hard)

N/A

You have 700 balls placed uniformly at random in a row: 100 of each colour - violet, indigo, blue, green, yellow, orange, red. You are told that among the first 350 positions there are exactly 15 more red balls than yellow balls. Given this information, find the probability that the ball in position 351 is violet.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

759. Seven Colors III

Probability Lvl 8/10 (Hard)

N/A

You have 700 balls placed uniformly at random in a row: 100 of each colour - violet, indigo, blue, green, yellow, orange, red. You are told that among the first 350 positions there are exactly 15 more red balls than yellow balls. Given this information, find the probability that the ball in position 351 is red.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

760. Maximum Till Now II

Probability Lvl 5/10 (Medium)

Hudson River Trading

You observe an infinite sequence $X_1, X_2, X_3, \dots$ of independent, identically distributed real random variables drawn from a continuous distribution with an unknown cumulative distribution function F . After seeing X_1 you keep sampling $X_2, X_3, \dots$ and stop at the first index $i > 1$ for which $X_i < X_1$. Call that stopping index τ . Compute $\mathbb{E}[\tau]$.

Enter your answer rounded to 5 decimal places. Enter -1 for infinity. Enter -2 if we need the cumulative distribution function F to calculate the answer.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

761. Partial Exponential I

Probability Lvl 4/10 (Medium)

N/A

Let $X_1, \dots, X_{10}$ be independent identically distributed exponential random variables with rate 1. For $k = 1, \dots, 10$ set $S_k = \sum_{i=1}^k X_i$. Given $S_{10} = 15$ compute $\mathbb{E}[S_4 | S_{10}]$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

762. Partial Exponential II

Probability Lvl 4/10 (Medium)

N/A

Let $X_1, \dots, X_{10}$ be independent identically distributed exponential random variables with rate 1. For $k = 1, \dots, 10$ set $S_k = \sum_{i=1}^k X_i$. Given $S_4 = 5$, find $\mathbb{E}[S_{10} | S_4]$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

763. Signal To Noise Ratio

Probability Lvl 4/10 (Medium)

N/A

Let X_1 and X_2 be independent random variables with $\frac{\mathbb{E}[X_1]}{\sqrt{\text{Var}(X_1)}} = 4$, $\frac{\mathbb{E}[X_2]}{\sqrt{\text{Var}(X_2)}} = 1$, and $\text{Var}(X_1) = \text{Var}(X_2) = \sigma^2$. For $0 \leq p \leq 1$ define $Z_p = pX_1 + (1-p)X_2$. Find the value of p that maximizes $\frac{\mathbb{E}[Z_p]}{\sqrt{\text{Var}(Z_p)}}$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

764. Your Choice

Probability Lvl 6/10 (Medium)

N/A

You may write any positive integers from 1 to 100 (repetitions allowed) on 100 cards, then show the completed deck to a friend who chooses a number $n \in 1, \dots, 100$ to maximize his expected payout. After you shuffle the deck and reveal the top card, if it equals the friend's chosen number n you pay him n , otherwise you pay

nothing. Design the multiset of 100 numbers on the cards to minimize the friend's maximum expected payout, and determine that minimal expected payout under optimal play by both players.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

765. Monochromatic Removal

Probability Lvl 6/10 (Medium)

Jane Street

You have 30 red marbles and 60 green marbles arranged uniformly at random in a row. Repeatedly do the following: reveal and remove marbles from the left end while they are the same color as the leftmost marble; when you first see a marble of the other color, put that one back among the remaining marbles, reshuffle the remaining pile uniformly at random, and repeat. Stop when only marbles of one color remain and those last marbles are drawn. What is the probability that the final marble drawn is green?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

766. Exponential Inequality I

Probability Lvl 3/10 (Easy)

N/A

Let X_1 and X_2 be independent exponential random variables with rates 4 and 6 respectively, i.e. $X_1 \sim \text{Exp}(4)$ and $X_2 \sim \text{Exp}(6)$. What is the probability that $X_1 < X_2$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

767. Tired Rabbit

Probability Lvl 5/10 (Medium)

N/A

A rabbit starts at 0. At each step it jumps +1 with probability $\frac{1}{3}$ and -1 with probability $\frac{2}{3}$, independently. The process ends as soon as the rabbit reaches +2 or -2 , or when 8 steps have been taken, whichever comes first. What is the expected number of steps taken?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

768. Exponential Inequality II

Probability Lvl 7/10 (Hard)

N/A

Let $X_1, \dots, X_5$ be independent $\text{Exp}(1)$ random variables and let X be independent $\text{Exp}(5)$. Find $\mathbb{P}(X > X_1 + \dots + X_5)$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

769. Give And Take

Probability Lvl 7/10 (Hard)

Jane Street

You play 100 turns with two boxes. Each turn can be a Give or a Take. On a Give, the game sponsor (a third party) adds \$1 into one of the two boxes chosen uniformly at random (and you do not know which box the money was added to). On a Take you collect the entire content of one box chosen uniformly at random and that box is then emptied. You never observe intermediate amounts you get, so your sequence of Gives and Takes

must be fixed in advance. Choose a schedule to maximize your expected total collected after the 100 turns and find the maximum expected total.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

770. Maxima Sum I

Probability Lvl 6/10 (Medium)

N/A

A random permutation of the numbers $1, 2, \dots, 7$ is chosen uniformly. Consider the numbers from the second position onwards that are greater than the number immediately preceding them. What is the probability that the sum of these numbers is odd?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

771. Maxima Sum II

Probability Lvl 6/10 (Medium)

N/A

A random permutation of the numbers $1, 2, \dots, 7$ is chosen uniformly. Consider the numbers from the second position onwards that are greater than all the numbers preceding them. What is the probability that the sum of these numbers is odd?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

772. Nearest Diagonal II

Probability Lvl 4/10 (Medium)

N/A

A point is chosen uniformly at random from the rectangle with vertices $(0, 0), (2, 0), (2, 1), (0, 1)$. What is the expected perpendicular distance from this point to the nearest diagonal?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

773. Fixed Maximum

Probability Lvl 6/10 (Medium)

N/A

You independently draw 6 numbers, each chosen uniformly at random from the set $1, 2, \dots, 20$, and each draw is made with replacement. Given that the largest of the 6 numbers is exactly 15, what is the expected value of the sum of the 6 numbers?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

774. It Took Ten Turns

Probability Lvl 9/10 (Very Hard)

N/A

We keep choosing independent random numbers from the interval $(0, 1)$, each with a uniform distribution. Let S_n be the sum of the first n numbers, and let T be the first time the running sum becomes greater than 1. Given that it takes exactly 10 numbers to cross 1, meaning $T = 10$, find the expected value of S_{10} .

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

775. Increasing Decreasing II

Probability Lvl 8/10 (Hard)

N/A

The numbers $1, 2, \dots, 8$ are arranged uniformly at random in a line. Let a_i be the number in the i -th position. Find the probability that

$$a_1 < a_2 > a_3 < a_4 > a_5 < a_6 > a_7 < a_8.$$

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

776. Swap Swap

Probability Lvl 6/10 (Medium)

N/A

Sixteen numbers $1, 2, \dots, 16$ are arranged in increasing order. At each move, two distinct positions are chosen uniformly at random and the numbers in those positions are swapped. After 5 moves, what is the expected number of numbers that are still in their original positions?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

777. Exponential Pentagon

Probability Lvl 7/10 (Hard)

N/A

Let X_1, X_2, X_3, X_4, X_5 be independent random variables, each having the exponential distribution with rate 1. What is the probability that these five numbers can be taken as the side lengths of a pentagon?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

778. Constrained Maximum

Probability Lvl 6/10 (Medium)

N/A

Let X be a nonnegative random variable with $\mathbb{E}[X] = 5$ and $\text{Var}(X) = 25$. What is the largest possible value of $P(X = 0)$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

779. Strange Product

Probability Lvl 10/10 (Very Hard)

N/A

Let $X_1, X_2, X_3, \dots$ be i.i.d. $\text{Unif}(0, 1)$ random variables. Define

$$P = (1 + X_1)(1 + X_1X_2)(1 + X_1X_2X_3) \cdots$$

Find the expected value of P . Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

780. Keep Multiplying

Probability Lvl 8/10 (Hard)

N/A

We start at 1. Every second, we choose an independent $U \sim \text{Unif}(0, 1)$ and multiply the current number by U . What is the expected time until the number first becomes smaller than $\frac{1}{2}$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

781. Maximum Exponential

Probability Lvl 6/10 (Medium)

N/A

Let $X_1 \sim \text{Exp}(1)$, $X_2 \sim \text{Exp}(2)$, $X_3 \sim \text{Exp}(3)$, $X_4 \sim \text{Exp}(4)$ be independent random variables. Find $\mathbb{E}[\max(X_1, X_2, X_3, X_4)]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

782. Winner Takes All

Probability Lvl 7/10 (Hard)

N/A

Four players start with 1, 4, 6, and 8 dollars, respectively. In each round, two players who are still alive are chosen to play a fair game (each player wins with probability $1/2$). The winner gets 1 dollar from the other, so the total amount of money never changes. Any player who reaches 0 is eliminated. The game stops as soon as only one player is left. What is the expected number of rounds until the game ends?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

783. Spaced Uniforms I

Probability Lvl 9/10 (Very Hard)

N/A

Ten points are chosen independently and uniformly at random from the interval from 0 to 1. What is the probability that no two of the points are within $1/20$ of each other?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

784. Strictly Behind

Probability Lvl 6/10 (Medium)

N/A

A rabbit starts at 0 and a frog starts at 4 on an infinite number line. Each minute, the rabbit and the frog independently move one unit left or one unit right, each with probability $\frac{1}{2}$. What is the probability that after 10 minutes, both of them are back at their starting positions, and the rabbit stays strictly behind the frog at every moment during the 10 minutes?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

785. Spaced Uniforms II

Probability Lvl 9/10 (Very Hard)

N/A

Ten points are chosen independently and uniformly at random on the interval $[0, 1]$. Let M be the minimum distance between any two of the points. Find $\mathbb{E}[M]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

786. Second Last Roll

Probability Lvl 7/10 (Hard)

N/A

A fair six-sided die is rolled repeatedly until the running total first becomes greater than 100. What is the expected value of the roll immediately before the final roll? You need to rigorously prove your answer, intuition alone doesn't count!

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

787. All Suits

Probability Lvl 7/10 (Hard)

N/A

A standard deck of 52 cards is shuffled uniformly at random. The cards are then revealed one by one without replacement. What is the expected number of cards revealed until all four suits have appeared at least once?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

788. Subset Divisibility

Probability Lvl 10/10 (Very Hard)

N/A

From the set $\{1, 2, 3, \dots, 29\}$, we choose 6 distinct numbers uniformly at random. What is the probability that their sum is divisible by 29?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

789. Hit 995

Probability Lvl 8/10 (Hard)

N/A

A rabbit starts at position 0 on the integer number line. Each second, it hops one unit to the left or one unit to the right, each with probability $\frac{1}{2}$. Let N be the number of times the rabbit lands on position 995 before it ever lands on position -5 . Find $\mathbb{E}[N]$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

790. Penguin Walk III

Probability Lvl 4/10 (Medium)

N/A

A penguin starts at position 0 on the integer line. At each move, it goes one step to the right with probability p and one step to the left with probability $1 - p$. Suppose $p = \frac{1}{3}$. Find the probability that the penguin ever reaches position 7.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

791. Penguin Walk IV

Probability Lvl 5/10 (Medium)

N/A

A penguin is moving along the integer line. At each step, it moves one unit to the right with probability p and one unit to the left with probability $1 - p$. Suppose $p = \frac{1}{3}$. The penguin is currently at position 4. Given that it will eventually reach position 7, what is the probability that its next move is to position 5?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

792. Uniform Product

Probability Lvl 7/10 (Hard)

N/A

Let $X_1, X_2, X_3, X_4, X_5, X_6, X_7$ be independent random variables, each uniformly distributed on $(0, 1)$. Define $Y = X_1 X_2 X_3 X_4 X_5 X_6 X_7$. Find the value of the probability density function of Y at $y = \frac{1}{2}$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

793. Last Uniform

Probability Lvl 8/10 (Hard)

N/A

We draw independent random variables $Y_1, Y_2, Y_3, \dots$, each uniformly distributed on $(0, 1)$. We keep drawing until the running total first becomes greater than 1. What is the expected value of the final draw, that is, the draw that makes the sum exceed 1?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

794. Consecutive Stocks

Brainteasers Lvl 10/10 (Very Hard)

N/A

A trader buys a positive integer number of shares on each of 365 consecutive days. Over the full 365 days, he buys exactly 600 shares. Can you always find a consecutive block of days during which he bought exactly 129 shares?

Enter 1 for YES and 2 for NO.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

795. Penguin Walk V

Probability Lvl 4/10 (Medium)

N/A

A penguin starts at position 0 on the integer line and takes a random walk. At each step, it moves from position i to $i + 1$ with probability p , and from i to $i - 1$ with probability $1 - p$. Suppose $p = \frac{1}{3}$. What is the expected number of steps until the penguin first reaches position 1?

Enter your answer rounded to 5 decimal places. If the answer is infinite, enter -1.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

796. Penguin Walk VI

Probability Lvl 4/10 (Medium)

N/A

A penguin starts at 0 on the integer line. On each step, it moves one unit to the right with probability p , and one unit to the left with probability $1 - p$. If $p = \frac{2}{3}$, what is the expected number of steps needed for the penguin to reach position 1 for the first time?

Enter your answer rounded to 5 decimal places. If the answer is infinite, enter -1.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

797. Penguin Walk VII

Probability Lvl 4/10 (Medium)

N/A

A penguin starts at position 0 on the integer line and takes a random walk. At each step, it moves from position i to position $i + 1$ with probability p , and from position i to position $i - 1$ with probability $1 - p$. If $p = 1/2$, find the expected number of steps until the penguin first reaches position 1.

Enter your answer rounded to 5 decimal places. If the answer is infinite, enter -1.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

798. Penguin Walk VIII

Probability Lvl 4/10 (Medium)

N/A

A penguin starts at position 0 on the integer line and takes a random walk. At each step, it moves from position i to position $i + 1$ with probability p , and from position i to position $i - 1$ with probability $1 - p$. If $p = \frac{2}{3}$, find the expected number of steps needed for the penguin to reach position 7 for the first time.

Enter your answer rounded to 5 decimal places. If the answer is infinite, enter -1.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

799. Row Expectation I

Probability Lvl 6/10 (Medium)

N/A

A 3×12 grid is filled by placing the numbers 1 through 36 uniformly at random, with no repeats. What is the expected value of the smallest number in the second row?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

800. Gojo's Six Eyes

Probability Lvl 8/10 (Hard)

N/A

Consider the six points on the unit sphere given by $(0,0,1)$, $(1,0,0)$, $(0,1,0)$, $(-1,0,0)$, $(0,-1,0)$, $(0,0,-1)$. Join every pair of these points to form line segments. Now choose a unit normal vector uniformly at random, and consider the plane through the origin perpendicular to this vector. This plane cuts the sphere into two halves. What is the expected number of these line segments that are intersected by the plane?

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

801. XY To The Power Z

Probability Lvl 7/10 (Hard)

N/A

Let X , Y , and Z be independent random variables, each uniformly distributed on $(0,1)$. Define $W = (XY)^Z$.

Find the probability density function of W , and then evaluate it at $w = \frac{1}{2}$.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

802. Penguin Walk IX

Probability Lvl 5/10 (Medium)

N/A

A penguin starts at 0 on the number line. At each step, it moves right by 1 with probability p and left by 1 with probability $1 - p$. Let T be the number of steps needed for the penguin to reach 1 for the first time. Find $\text{Var}(T)$ when $p = \frac{2}{3}$.

If the variance is infinite, return -1 .

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

803. Penguin Walk X

Probability Lvl 5/10 (Medium)

N/A

A penguin starts at 0 on the integers and performs a random walk. At each step, it moves from position i to position $i + 1$ with probability p , and from i to position $i - 1$ with probability $1 - p$. Suppose $p = \frac{2}{3}$. Find the variance of the total number of steps needed for the penguin to reach 7 for the first time.

If the variance is infinite, return -1 .

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

804. Ten-Dimensional Ball

Probability Lvl 6/10 (Medium)

N/A

A point $X = (X_1, X_2, \dots, X_{10})$ is chosen uniformly at random from the entire 10-dimensional ball centered at the origin with radius 12, including its interior. Find the variance of the first coordinate X_1 .

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

805. Row Expectation II

Probability Lvl 7/10 (Hard)

N/A

A 4×6 grid is filled by placing the numbers $1, 2, \dots, 24$ into the 24 cells uniformly at random, with no repetition. What is the expected value of the third largest number in the first row?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

806. Conditional Normal IV

Probability Lvl 7/10 (Hard)

N/A

Let N_1 and N_2 be independent standard normal random variables. Find the value of $\mathbb{E}[N_1^4 \mid N_1^2 + N_2^2 = 6]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

807. Infinite Sum Expectation

Probability Lvl 8/10 (Hard)

N/A

Let $X_1, X_2, X_3, \dots$ be independent random variables, each uniformly distributed on $(0, 1)$. Define N to be the first index k such that $X_k < X_{k+1}$. If no such index exists, set $N = \infty$. Now consider $S = \sum_{i=1}^N \frac{X_i}{2^i}$. Find $\mathbb{E}[S]$.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

808. Secret Marbles I

Probability Lvl 7/10 (Hard)

Jane Street

Two players, A and B , each independently choose an integer from 1 to 100. Player A secretly puts a marbles in a box, and player B secretly puts b marbles in the same box. Then two marbles are drawn WITH replacement. For EACH draw, if the marble comes from A , then A is paid $100 - a$ from a third party. If the marble comes from B , then B is paid $100 - b$ from a third party. Assume both players choose their numbers optimally. Find the expected total payout of player A .

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

809. Secret Marbles II

Probability Lvl 7/10 (Hard)

N/A

Two players, A and B , each secretly choose an integer from 1 to 100. Player A puts a marbles in a box and player B puts b marbles in the same box. Then one marble is drawn.

If the drawn marble came from A , then A receives $100 - a$ dollars from B . If the drawn marble came from B , then A pays $100 - b$ dollars to B . Both players choose optimally. How many marbles should player A choose?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

810. Second Last I

Probability Lvl 9/10 (Very Hard)

N/A

Let $X_1, X_2, \dots$ be i.i.d. random variables, each uniformly distributed on $(0, 1)$. Define N to be the first index n such that X_n is not larger than all of $X_1, \dots, X_{n-1}$, in other words, N is the first time the sequence fails to set a new record. Find the expected value of the largest number among the first N chosen values, that is, compute $\mathbb{E}[X_{N-1}]$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

811. Second Last II

Probability Lvl 9/10 (Very Hard)

N/A

Let $X_1, X_2, \dots$ be independent random variables, each uniformly distributed on $(0, 1)$. Define N to be the first time that the new observation is not a new minimum, meaning N is the smallest index such that $X_N > \min X_1, \dots, X_{N-1}$.

In other words, we keep recording smaller and smaller values until the first time a draw fails to beat the current record low. Find the expected value of that final record low, that is, $\mathbb{E}[\min X_1, \dots, X_{N-1}]$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

812. Powerful Scaling

Probability Lvl 6/10 (Medium)

N/A

Let $X_0 = 1$. For each $n \geq 0$ define X_{n+1} to be uniformly distributed on the interval $(0, X_n^3)$. Find $\mathbb{E}[X_5]$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

813. One Before Ten

Probability Lvl 5/10 (Medium)

N/A

Ten cards labeled 1 through 10 are shuffled uniformly at random and placed in a row from left to right. Given that card 10 appears somewhere to the right of card 1, what is the expected value of the card in the first position?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

814. Frog Distance I

Probability Lvl 8/10 (Hard)

N/A

Ten frogs are placed independently and uniformly at random on the interval $[0, 1]$. What is the probability that the distance between the leftmost frog and the rightmost frog is between $\frac{1}{4}$ and $\frac{1}{2}$?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

815. Frog Distance II

Probability Lvl 8/10 (Hard)

N/A

Ten frogs are placed independently and uniformly at random on the interval $[0, 1]$. Given that the distance between the leftmost frog and the rightmost frog is more than $\frac{1}{4}$, what is the probability that this distance is less than $\frac{1}{2}$?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

816. Additional Info Coin Flips II

Probability Lvl 7/10 (Hard)

N/A

Three people, A , B , and C , each flip a fair coin repeatedly until they get their first heads. Given that B needed more tosses than A , what is the probability that C needed more tosses than B ?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

817. Power Distance II

Probability Lvl 6/10 (Medium)

N/A

A particle starts at 0. Every second, it moves one step to the left or one step to the right with equal probability, independently of all other moves. What is the expected value of the fourth power of its position after 30 seconds?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

818. Normal Ratio I

Probability Lvl 6/10 (Medium)

N/A

Let X and Y be independent normal random variables with mean 0 and variance 1. What is the probability that $1 < \frac{X}{Y} < \sqrt{3}$?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

819. Normal Ratio II

Probability Lvl 6/10 (Medium)

N/A

Let $X \sim N(0, 2)$ and $Y \sim N(0, 1)$ be independent normal random variables. What is the probability that $\sqrt{2} < \frac{X}{Y} < \sqrt{6}$?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

820. Exponential Ratio

Probability Lvl 4/10 (Medium)

N/A

Let X and Y be independent exponential random variables with rates 1 and 2, respectively. What is the probability that $1 < \frac{X}{Y} < 2$?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

821. Square Intersection II

Probability Lvl 5/10 (Medium)

N/A

We choose two points independently and uniformly at random from the interior of a square. What is the probability that the line through these two points intersects the left side and the bottom side of the square?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

822. Increasing Dice Rolls IV

Probability Lvl 7/10 (Hard)

N/A

We roll a fair six-sided die 10 times. We are told that the results are nondecreasing, meaning each roll is at least as large as the one before it. Given this information, what is the probability that a 3 appears at least once?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

823. Hexagon Intersection

Probability Lvl 6/10 (Medium)

N/A

A regular hexagon has its sides labeled A, B, C, D, E, F in clockwise order. Two points are chosen independently and uniformly at random from inside the hexagon. What is the probability that the line joining these two points intersects side A and side D ?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

824. Sum Over One II

Probability Lvl 7/10 (Hard)

N/A

Ten independent numbers are chosen from the interval $[0, 1]$ uniformly at random. What is the probability that the sum of the smallest number and the SECOND largest number is greater than 1?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

825. Last Card Survivor

Probability Lvl 6/10 (Medium)

N/A

A deck contains the cards $1, 2, \dots, 100$, arranged from top to bottom in increasing order. The deck is not shuffled. Repeatedly, the top two cards are taken out, one of them is chosen at random and discarded, and the other is placed at the bottom of the deck. This continues until only one card remains. Find the expected label of the final card.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

826. Triangle Formation IV

Probability Lvl 6/10 (Medium)

N/A

Three numbers are chosen independently and uniformly at random from the interval $[0, 1]$. Given that these three numbers can be used as the side lengths of a triangle, what is the expected value of the smallest of the three numbers?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

827. Random Tree Construction

Probability Lvl 7/10 (Hard)

N/A

Bob starts with a rooted tree consisting of just one vertex, the root. On each of nine turns, he chooses one existing vertex uniformly at random and adds a new child to it. After nine turns, the tree has 10 vertices in total. Let $S(v)$ be the size of the subtree rooted at vertex v . Alex then chooses one of the 10 vertices uniformly at random, and calls it v_1 . Find the expected value of $S(v_1)$.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

828. Subset Subset

Probability Lvl 7/10 (Hard)

N/A

Consider the set $\{1, 2, \dots, 8\}$. We put every non-empty subset of this set into a bag. So there are $2^8 - 1 = 255$ subsets in total. Each day, we choose one subset uniformly at random that still remains in the bag. After choosing a subset S , we remove S and every subset of S that still remains in the bag. What is the expected number of days until the bag becomes empty?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

829. Socks Remaining III

Probability Lvl 8/10 (Hard)

Jane Street

A drawer has 10 red socks, 20 blue socks, and 30 green socks. The socks are drawn one at a time in random order without replacement. We stop as soon as one color has completely disappeared from the drawer, so that only the other two colors are still present. What is the expected number of socks left in the drawer at that moment?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

830. Expected Head-Tail Ratio

Probability Lvl 7/10 (Hard)

N/A

A fair coin is tossed again and again, and we stop at the first time the number of tails is strictly greater than the number of heads. What is the expected fraction of all tosses that are tails at the stopping time? It may be helpful to think of the summation that appears while solving this problem as Taylor expansion of arcsin.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

831. Too Many Slips

Probability Lvl 9/10 (Very Hard)

N/A

Suppose we have an unlimited supply of four slips with values 0.01, 0.05, 0.10, and 0.25 dollars each. Every second, we choose one of the four slips uniformly at random, add its value to our running total, and keep it. We repeat this process until the total sum becomes an integer. What is the expected value of the total sum at the moment we stop?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

832. Gold Rush

Brainteasers Lvl 7/10 (Hard)

Jane Street

We have a gold bar of length 1 meter, and 3 people place marks on it one after another. The first person places a mark anywhere on the bar. Then the second person places a mark, knowing where the first mark is. Finally, the third person places a mark, knowing the locations of the first two marks. After all three marks are placed, each point on the gold bar belongs to the person whose mark is closest to that point. Each person wants to maximize the amount of gold they receive. If a player has multiple positions that give the same maximum amount of gold, they choose uniformly at random among those positions. What are the possible positions (from the left end) where the first player should place their mark in order to maximize the amount of gold they eventually receive?

If there are multiple possible answers, enter the smallest one (rounded to 5 decimal places).

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

833. Expected Number After Swaps

Probability Lvl 5/10 (Medium)

N/A

We have the number 123456789. Then we repeat the following operation twice: choose two different digits uniformly at random and swap their positions. What is the expected value of the final nine-digit number?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

834. Direction Change III

Probability Lvl 6/10 (Medium)

N/A

Consider a 5×5 grid. Alice starts at the bottom-left corner and wants to reach the top-right corner using only right and upward moves. Assume every such path is equally likely. Given that Alice changes direction at least 6 times during his journey, what is the expected number of direction changes?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

835. Submarine Distance

Probability Lvl 6/10 (Medium)

N/A

A submarine starts at the origin and chooses a direction uniformly at random from the angles between 0° and 90° , measured counterclockwise from the positive x -axis. A second submarine starts at $(10, 0)$ and chooses a direction uniformly at random from the angles between 90° and 180° , also measured counterclockwise from the positive x -axis. Both submarines then move at speed 1 unit per second along their chosen directions. What is the probability that at some point during their motion, the distance between the two submarines becomes less than 4 units?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

836. Nearest Paint I

Probability Lvl 8/10 (Hard)

N/A

Ten people are placed independently and uniformly at random on a 100-meter line segment. Each person paints the straight segment joining themselves to their nearest neighbor among the other nine people. Some parts may be painted more than once. Find the expected length of the part of the segment that is not painted at all.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

837. Nearest Paint II

Probability Lvl 8/10 (Hard)

N/A

Ten people are placed independently and uniformly at random along a circular track of circumference 100 meters. Each person paints the shorter arc from themselves to their nearest neighbor along the circle. Some arcs may be painted more than once. Find the expected total length of the part of the circle that is not painted at all.

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

838. Normal Inequality V

Probability Lvl 5/10 (Medium)

N/A

Let N_1, N_2 , and N_3 be independent standard normal random variables. What is the probability that $N_1^5 + N_2^5 > N_3^5$?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

839. Expected Hits I

Probability Lvl 6/10 (Medium)

N/A

A simple symmetric random walk starts at 0 on the integer line. At each step, it moves to the left or right by 1 unit with equal probability. The walk stops as soon as it reaches either -10 or $+20$. What is the expected number of visits to $+4$ before the walk stops?

Enter your answer rounded to 5 decimal places.

[[Hint ↓](#) | [Solution ↓](#) | [↑](#)]

840. Expected Hits II

Probability Lvl 6/10 (Medium)

N/A

A particle starts at 0 on the integer line. At each second, it moves one step to the right with probability $\frac{1}{3}$ and one step to the left with probability $\frac{2}{3}$. The walk stops as soon as it reaches either -10 or $+20$. What is the expected number of times the particle visits $+4$ before the walk stops?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

841. Jumping Frog X

Probability Lvl 6/10 (Medium)

N/A

A simple random walk starts at point 1 on a circle with 100 labeled points $1, 2, \dots, 100$ in clockwise order. Each second, the walker moves one step clockwise with probability $\frac{1}{3}$ and one step counterclockwise with probability $\frac{2}{3}$. What is the probability that when the walk first reaches point 15, every other point has already been visited at least once?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

842. Elevator II

Probability Lvl 7/10 (Hard)

N/A

Bob is located on the 20th floor of a 50-floor building. There are 4 elevators, and each elevator moves continuously up and down between the 1st floor and the 50th floor at constant speed, instantly reversing direction at the ends. None of the elevators stops unless someone presses a button. Bob decides to leave at a time chosen uniformly at random during the day, and he boards the first elevator that reaches his floor after he decides to leave. What is the probability that this first elevator is moving downward when it arrives?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

843. Unknown Gunmen I

Probability Lvl 8/10 (Hard)

N/A

Two players, A and B , begin at position 0 and move together toward a target at position 1 at the same constant speed. Player A hits the target with probability equal to the current position x whenever he shoots. Player B hits with probability x^2 . Each has exactly one bullet and can choose to fire at any position x between 0 and 1. The players observe each other at all times. If a player hits the target before the other player, that player wins the game. If both fire at precisely the same instant and both hit, the round ends in a tie, both players return to position 0 with fresh bullets, and the game restarts from the beginning. Similarly, if both players miss, the game restarts. Assuming both play optimally, what is the probability that A wins?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

844. Random And Rational

Probability Lvl 6/10 (Medium)

Jane Street

Three players A , B , and C each choose a number from 0 to 100. The player who chooses the largest number must pay each of the other two players an amount equal to the number that player chose. Player C chooses his number uniformly at random from 0 to 100. Players A and B are perfectly rational, and B knows exactly what number A will choose. Player A knows that B has this advantage. What number should A choose in order to maximize his expected payoff?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

845. Line Switch

Probability Lvl 2/10 (Easy)

Jane Street

You are standing in a pizza shop line with 100 people ahead of the counter. A second identical line suddenly opens. If you move to the new line, you are placed uniformly at random into one of the available spots there. Every person acts rationally and wants to end up as early as possible in line. What is the minimum position from the front of the original line at which switching leads to a better expected position?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

846. Unknown Gunmen II

Probability Lvl 8/10 (Hard)

N/A

Two players, A and B, move together at the same constant speed from position 0 toward a target at position 1. At any position $x \in [0, 1]$, player A hits the target with probability x if he shoots, while player B hits the target with probability x^2 if he shoots. Each player observes the other at all times. In each round, A has two bullets and B has one bullet. If one player hits the target before the other player hits, that player wins. If they shoot at exactly the same instant and both hit, the round is a tie and the game restarts from position 0 with fresh bullets. Similarly, if all players miss their shots, the game restarts. Assuming both players play optimally, what is the probability that A wins?

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

847. Match Fixing

Probability Lvl 6/10 (Medium)

Jane Street

Four distinct numbers are chosen from $1, 2, \dots, 50$. These four numbers are then randomly assigned to players A, B, C, D . The players with the largest and smallest numbers form one team, while the two players with the middle numbers form the other team. The team with the larger sum wins money equal to the difference between the two team sums, paid by the other team.

Before the numbers are chosen, you are allowed to force your own number to be any integer x from 1 to 50. The other three numbers are then chosen uniformly at random from the remaining 49 numbers, with no repetitions. What value of x should you choose to maximize your team's expected net payoff?

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

848. Extra Information I

Probability I Lvl 6/10 (Medium)

Citadel

Two players, A and B, each independently receive a number uniformly at random from $[0, 1]$. Each player sees only their own number. Player A moves first and chooses either to keep his number or redraw. If A redraws, he receives a fresh independent uniform $[0, 1]$ number and must keep it. Player B observes only whether A kept or redrew, not A's actual number. Player B then makes the same choice: keep his own original number or redraw once and keep the new number. Both players know all rules and strategies, and A knows that B will observe whether A kept or redrew. The player with the higher final number wins. Under optimal play, what is the probability that B wins? You may use a calculator to solve any polynomial or nonlinear equations that may appear in the solution.

Enter your answer rounded to 5 decimal places.

[[Hint](#) ↓ | [Solution](#) ↓ | ↑]

Hints

1. Non-empty Intersection

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Name the left endpoints $l_1, l_2, \dots, l_5$. Name the right endpoints $r_1, r_2, \dots, r_5$. So, we have 10 entities. By symmetry, each of the permutations of these entities that satisfy $l_i < r_i$ are equally likely, and that forms our sample space. How many such permutations are there?

Hint 2: There are $10!$ total permutations of the 10 entities. Half of them have $l_1 < r_1$, half have $l_2 < r_2$, and so on. So, the total number of permutations in which all $l_i < r_i$ is

$$\frac{10!}{2^5}$$

Which permutations lead to a non-empty intersection?

Hint 3: There will be a non-empty intersection if and only if all the 5 left endpoints lie to the left of all the 5 right endpoints.

2. Standing Table II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "standing table I" and "same arc I" first.

Hint 2: The table will stand exactly when the pentagon formed by joining the five attachment points contains the centre of the circle. This happens precisely when the five points are NOT all contained in any semicircular arc.

3. Game Of Chords

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

Hint 2: Alice can exploit the symmetry by drawing a diameter on her first move.

4. Unequal Neighbors

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Color the grid in an alternating black-and-white pattern, like a chessboard.

5. Find The Units Digit

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $S(n)$ be the sum of "product of nonzero digits" over all n -digit strings (allowing leading zeros). Now to build an $(n+1)$ -digit string, choose its last digit $d \in \{0, 1, \dots, 9\}$ and an n -digit prefix.

- If $d = 0$, the product of nonzeros doesn't change, contributing $S(n)$.
- If $d = k \in \{1, \dots, 9\}$, each n -digit prefix's product gets multiplied by k , contributing $kS(n)$.

Hence,

$$S(n+1) = S(n) + \sum_{k=1}^9 kS(n) = (1 + (1 + 2 + \dots + 9))S(n) = 46S(n).$$

6. Ship Destroyer

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics. Each bomb drop is one of 16 equally likely outcomes, so there are in total 16^4 possible ordered sequences of four bomb-drops. How many sequences result in the destruction of all ships?

Hint 2: To ensure every ship is destroyed, each of the 4 rows must contain at least one bomb OR each of the 4 columns must contain at least one bomb. Equivalently, let E_R be the event "the bombs land in four distinct rows," and E_C be the event "the bombs land in four distinct columns." A ship in row i is wiped out as soon as at least one bomb falls in row i , and a ship in column j is wiped out as soon as at least one bomb falls in column j . Therefore, all ships are destroyed if and only if at least one of the events E_R OR E_C occurs. Use principle of inclusion and exclusion.

7. Moving The Piece I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use proof by coloring.

8. Even Marbles Everywhere

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics. How many ways are there to fill the chessboard such that there are even white marbles in every row and every column?

Hint 2: If you randomly fill the first seven squares of a row, you can always get an even number of white marbles in that row. If the number of white marbles in the row (after filling the first 7 columns) is even, you can put a black marble in the last cell. If the number of white marbles in the row (after filling the first 7 columns) is odd, you can put a white marble in the last cell.

Hint 3: For every arrangement in the top-left 7×7 grid, we can get an arrangement such that every row and every column has even white marbles.

9. Hard Grid Arrangement

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider what happens after you fill the first row.

10. All Odds Before Even

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics.

11. Standing Table I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For the table to remain upright, the center of mass (center of the circle) must lie within the triangle formed by the three legs.

12. 10 Consecutive Heads

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

13. Chords Intersection I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

14. 1 Leftward Jump

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To calculate the number of sequences that result in exactly one leftwards jump, we need to consider how to divide the set $[2, 3, 4, 5, 6, 7]$ into two distinct groups, A and B.

15. Marble Swap II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What happens if the blue marble is in the middle at the end of the 9th move? Which swaps lead to the blue marble remaining in the middle? What happens if the blue marble is in one of the end positions at the end of the 9th move? Which swaps lead to the blue marble to the middle?

Hint 2: Let p be the probability that the blue marble is in the middle at the end of the 9th move. Now use the law of total probability to calculate the probability that the blue marble is in the middle at the end of the 10th move— p will cancel out.

16. Scary Permutation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What are the possible choices for the last number?

Hint 2: The last number must be either 1 or 9. Try to prove it.

Hint 3: From the previous hint, the last number must be either 1 or 9. Suppose it is 1. Then from the second to the 8th position, we have to place the numbers 2 to 9. The 8th position must be either 2 or 9. And so on.

17. Grid Frog I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Color the grid like a chessboard. Every frog on a white cell must jump to a black cell, and every frog on a black cell must jump to a white cell. Thus, the ways in which the white-cell frogs move are independent of the black-cell frogs' moves as frogs from different colored cells cannot land on the same square. We can count the white-cell configurations and then square the result.

Hint 2: Let $f(n)$ be the number of valid jump-patterns for the frogs on the white squares of a $2 \times n$ grid. Assume top-left corner is colored white. Focus on the frog in column 1, row 1 (a white cell). It has two options: - Jump down into column 1, row 2. The remaining white frogs simply arrange themselves in columns 2 through n . This reduces the problem to counting $f(n-1)$ for a $2 \times (n-1)$ grid. - Jump right into column 2, row 1. That move forces the frog in column 2, row 2 (also white) to jump left into column 1, row 2, occupying both columns 1-2 completely. The remaining white frogs must then arrange themselves in the last $n-2$ columns, giving $f(n-2)$. So, we have the following recurrence:

$$f(n) = f(n-1) + f(n-2)$$

18. At Least Six I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: Use the current running sum as the state. Write E_i for the expected number of additional rolls needed when your current total is i . Once your total is already 6 or more, you need no further rolls, so $E_i = 0$ for $i \geq 6$. If your total is $i < 6$, then on the next roll you add $1, 2, \dots, 6$ each with probability $\frac{1}{6}$. In every case you have used one roll and your new total is $i + k$, so

$$E_i = 1 + \frac{1}{6} \sum_{k=1}^6 E_{i+k}.$$

19. Game Of Placing Knights

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

Hint 2: Bob has a winning strategy—he can create a situation where he keeps copying Alice’s moves, so she runs out of moves first.

20. Same Arc I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Label the points in clockwise order as $1, 2, \dots, N$. For each index i let E_i be the event that every point lies inside the clockwise semicircular arc of length π whose left endpoint is point i . What is $P(E_i)$?

Hint 2: $P(E_i)$ is $\frac{1}{2^{N-1}}$.

Hint 3: The E_i s are mutually exclusive.

21. Increasing Dice Rolls III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Simply count the total number of possible outcomes and the number of favorable outcomes.

22. Fill In The Blanks

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Is it possible for the result to be even?

23. Exponent Perfect Square

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We know $\sqrt{p^p} = p^{p/2}$. When is this guaranteed to be an integer?

24. Odd Heads II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the law of total probability. To get odd heads, the fair coin can show heads and then the other 99 coins can show an even number of heads, or the fair coin can show tails and then the other 99 coins can show an odd number of heads.

25. Dice Rolls Product

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The six counts $N_1, \dots, N_6$ are clearly not independent (as they follow the constraint $\sum N_i = 10$), so we cannot write

$$E[N_1 N_2 \cdots N_6] = E[N_1] E[N_2] \cdots E[N_6].$$

Instead, introduce indicator variables for each face on each roll: for $i = 1, \dots, 10$, let

$$a_i = \begin{cases} 1, & \text{roll } i \text{ shows a 1} \\ 0, & \text{otherwise} \end{cases}, \quad b_i = \begin{cases} 1, & \text{roll } i \text{ shows a 2} \\ 0, & \text{otherwise} \end{cases}, \dots, \quad f_i = \begin{cases} 1, & \text{roll } i \text{ shows a 6} \\ 0, & \text{otherwise} \end{cases}$$

What is the product $N_1 N_2 N_3 N_4 N_5 N_6$ in terms of these indicators? Notice that many terms in the expansion of this product consist of products of independent random variables.

Hint 2: We can write $N_1 = \sum_{i=1}^{10} a_i$, $N_2 = \sum_{i=1}^{10} b_i$, $\dots$, $N_6 = \sum_{i=1}^{10} f_i$, so the product is

$$N_1 N_2 \cdots N_6 = \left(\sum_i a_i \right) \left(\sum_j b_j \right) \cdots \left(\sum_k f_k \right).$$

Expanding this product yields a sum of terms of the form

$$a_{i_1} b_{i_2} \cdots f_{i_6}$$

over all choices of indices $i_1, \dots, i_6 \in \{1, \dots, 10\}$. But whenever any two of those indices coincide—say $i_1 = i_2$ —we have

$$a_{i_1} b_{i_2} = a_{i_1} b_{i_1} = 0,$$

because a single roll cannot show both "1" and "2." Hence only the terms with all six indices distinct survive in expectation. These terms (with all six indices distinct) are products of independent random variables, so we can factor them when calculating the expectation.

26. Jumping Frog V

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let d_n represent the number of distinct ways a frog can reach point n . Consider the possible sizes of the first jump, which can range from 2 up to n . If the frog hops k points in its first move, it must find a unique path for the remaining $n - k$ points. Now, treat the k th point as the new starting point, with the top being at $n - k$.

27. Dot Product

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: Let's define p_n as the probability that the dot product of the first n entries of two binary vectors is odd. Express p_{n+1} in terms of p_n to obtain a recurrence.

28. Marble Swap I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Number the marbles as 1, 2, 3 (red, blue, green respectively). In a permutation π , an inversion is a pair of indices $i < j$ such that $\pi(i) > \pi(j)$. Any single swap always changes the total number of inversions by an odd number.

Hint 2: We start with the identity permutation $[1, 2, 3]$, which is completely ordered and therefore has zero inversions—an even number. After performing 125 swaps, since each swap changes the inversion count by an odd number, the total change in inversion count after 125 swaps must be an odd number. So the resulting permutation must have an odd number of inversions.

29. Monster Escape I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use proof by coloring.

Hint 2: If we color the grid in alternating black and white colors like a chessboard, the bottom-left and bottom-right corners will have opposite colors. What does this mean?

30. Chords Forming Quadrilateral

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The total number of possible chords is $\binom{8}{2} = 28$. Thus, the number of ways to select 4 chords is $\binom{28}{4}$. How can we determine the number of selections that result in a convex quadrilateral?

31. Monster Escape II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: They can only meet on the line $y = 4 - x$ (one of the diagonals).

32. Matrix Attack III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Create some matrices of sizes 2×2 , 3×3 , 4×4 , and 5×5 that satisfy the given conditions. Do you notice any pattern?

Hint 2: The pattern is this — for any even-sized $n \times n$ matrix satisfying the two conditions, the numbers 1 through n cannot occur exactly once on the main diagonal. For any odd-sized $n \times n$ matrix satisfying the two conditions, the numbers 1 through n always occur exactly once on the main diagonal.

33. Bulbs In A Grid

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Start with the central light bulb. Analyze the situations when it is turned on and when it is off.

34. Chess Tournament III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Each match eliminates exactly one player, so exactly $32 - 1 = 31$ matches are played.

35. Count Paths II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve count paths I first.

Hint 2: If you are at a point (x, y) , where both x and y are even, and then move one unit to the right, the next move must also be towards right. Similarly, if you are at a point (x, y) , where both x and y are even, and then move one unit up, the next move must also be upward.

Hint 3: Initially, we are at $(0, 0)$. Both the x and y coordinates are even. Each step leads to another step in the same direction. So, the problem can be rephrased as: how many paths go from $(0, 0)$ to $(8, 10)$ such that in one step, you move either 2 units to the right or 2 units upward?

36. Five Light Bulbs

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion with states/Markov chains.

Hint 2: Use the number of bulbs that are currently on to represent the states. So, let E_i be the expected additional time (in seconds) to reach the state "all off" starting from a configuration with exactly i bulbs on.

37. Letter-envelope Pairing

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What are the choices for glove numbered 1? It can either be paired with itself or with glove 2.

Hint 2: The glove numbered 1 can either be paired with itself or with glove 2. If glove 1 is paired with itself, the problem reduces to arranging the remaining 9 gloves. On the other hand, if glove 1 is paired with glove 2, then glove 2 must necessarily be paired with glove 1. In this case, the problem reduces to arranging the remaining 8 gloves.

Hint 3: Use recursion.

38. Frog Return

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the parity (odd/even) of the jump distances.

39. Direction Change I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics. Note that the condition is equivalent to the first and last step being in different directions, as if you switch directions an odd number of times, you must end in a different direction than you started.

40. 3 Dice Minimum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the tail sum formula. For a non-negative discrete random variable X , it holds that

$$E[X] = \sum_{k \geq 0} P(X \geq k).$$

41. Game On Divisors III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve smaller cases first: what happens if we start with 1, 2, 3, or 4 stones?

Hint 2: If it still isn't clear, try solving for 5, 6, 7, up to 10 stones.

Hint 3: Odd numbers are losing states while even numbers are winning states.

42. Jumping Frog IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let x_i be the number of ways to reach point i , starting at point 0. Then, $x_n = x_{n-1} + x_{n-3} + x_{n-5} + \dots + \beta$, where $\beta = 1$ if n is even and $\beta = 0$ if n is odd.

43. Dice Game III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

44. 105 Pebbles

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume the first operation is done. What happens to the gcd of the piles in subsequent operations?

45. Missing Pairwise Product

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: When you have four numbers a, b, c , and d , the six pairwise products can be arranged into three pairs, like this: ab and cd , ac and bd , ad and bc . What's special about this arrangement? The product of the two products in each group is always the same, namely $abcd$.

46. Constrained Subsets II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The sum of the elements of the set $\{1, 2, 3, \dots, 30\}$ is 465. 232 is almost half of the sum.

Hint 2: Use symmetry. For every subset A having sum less than or equal to 232, we have a subset having sum greater than 232—the complement of A .

47. Fifth Card

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Among the five cards, at least two will share the same suit. You can place one of these cards in the first position, allowing your friend to deduce the suit of the hidden card.

Hint 2: Let's number the cards from 1 to 13, with ace numbered as 10, jack as 11, queen as 12, and king as 13. The second, third, and fourth cards can be arranged in $3! = 6$ ways. So, we can encode 6 values using the second, third, and fourth cards. For example, suppose the second, third, and fourth cards are 2, 5, and 12. Then the arrangement (2, 5, 12) can represent 1, (2, 12, 5) can represent 2, (5, 2, 12) can represent 3, and so on.

Hint 3: But the hidden card can have 13 values. We only know its suit (from the first card) and a value between 1 to 6 based on the second, third, and fourth cards. How do we know the VALUE of the hidden card?

48. Expected Area

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The area A of a right triangle with legs of lengths X and Y is given by

$$A = \frac{1}{2}XY.$$

Hint 2: Since X and Y are independent, the expectation factorizes:

$$E[A] = E\left[\frac{1}{2}XY\right] = \frac{1}{2}E[X]E[Y].$$

49. Difficult Die Constraints

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Ignore rolls showing 1, 4, 5 because they do not affect when the first 2 or 3 appears and they do not change the count of 6s relative to the first appearances of 2 and 3. Thus we can think of a three sided die labeled 2, 3, 6 and ask for the probability of obtaining exactly two 6s before both 2 and 3 have appeared and it will not change anything.

Hint 2: Working only with the relevant outcomes 2, 3, 6, the stopping sequence, recording only the first occurrences of 2 and 3 and keeping every 6, must be one of the following six sequences:

2, 6, 6, 3; 3, 6, 6, 2; 6, 2, 6, 3; 6, 3, 6, 2; 6, 6, 2, 3; 6, 6, 3, 2.

Note that in this sequence representation every occurrence of 6 is counted, whereas only the first occurrences of 2 and 3 are counted. For example the actual sequence 2, 2, 2, 6, 6, 2, 3 corresponds to 2, 6, 6, 3 in this representation.

50. Game Of Discarding Cards

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The cards are discarded only when one card is red and one card is black. Hence, discarded cards contain the same number of red and black cards.

51. Normal Inequality IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What is the distribution of $Y - 4X$?

52. Ants On A Circle I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Instead of thinking that ants reverse direction at collisions, imagine they pass through one another without changing direction. Since the ants are indistinguishable, this makes no difference.

53. Ants On A Circle II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If all ants move clockwise or all move anticlockwise, they will each return to their original positions. But can they all return to their own positions in any other case?

Hint 2: They cannot return to their original positions in any other case when the number of ants is odd. Try to prove this.

54. Ants On A Circle III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "ants on a circle I" and "ants on a circle II" first.

Hint 2: If all ants move clockwise or all move anticlockwise, they will each return to their original positions. But can they all return to their own positions in any other case?

Hint 3: If the number of ants is even (as in this problem), then, apart from the cases where all ants move clockwise or all move anticlockwise, the only way for each ant to return to its original position is if an equal number of ants initially move clockwise and anticlockwise. Try to prove this.

55. Game Of Evaluation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Call the maximal sequence of numbers connected by multiplication signs ($\times$) a term. In other words, a term is a portion of the expression that is bounded on both sides by a $+$, $-$, or the start/end of the expression, and contains only multiplication symbols within it. When is the value of a term odd?

Hint 2: A term has an odd value if and only if it consists of a single odd number. In other words, it must be an odd number that is bounded on both sides by a $+$, $-$, or the start or end of the expression.

Hint 3: Suppose Alice, on her first move, places a plus sign after 1. The remaining expression then consists of the numbers 2 through 100. For this part of the expression, Alice's goal is to make its value even, while Bob's goal is to make it odd. Can Alice force this portion of the expression to be even?

56. Ant On A Cube I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Markov chains.

57. Dice Game VII

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Denote by X_1 the value of the first roll and by T the total sum of all rolls through the stopping time. Because each of the six faces on the first roll is equally likely, we have

$$E[T] = \sum_{i=1}^6 E[T|X_1 = i] \cdot P(X_1 = i) = \frac{1}{6} \sum_{i=1}^6 E[T|X_1 = i].$$

Hint 2: Fix i and write $T = i + T'$, where T' is the sum of all subsequent rolls until a face other than i appears. Thus

$$E[T|X_1 = i] = i + E[T'|X_1 = i].$$

$E[T'|X_1 = i]$ can be found using recursion.

58. Duck Escape

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The fox greedily chases the duck. If the duck's angular speed is greater than the fox's, the duck can reach the opposite side of the center from the fox.

59. Count Regions I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: It is clear that n lines will divide the plane into a maximum number of pieces if no two of them are parallel and no three of them are concurrent.

Hint 2: The given problem has an inductive structure. If you know the answer for $n - 1$ lines, how can you determine the answer for n lines? What does adding one more line do?

60. Count Regions III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Count Regions I" first.

Hint 2: n planes will divide 3-dimensional space into a maximum number of parts if any three of them have exactly one point in common, and no four of them have a point in common.

Hint 3: You will find it difficult to determine the number of new regions formed when the $(k + 1)$ th plane is added. Use the result from "Count Regions I" about the maximum number of regions a plane can be divided into by n lines.

61. Square Intersection I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the 2 random points be P and Q . Let L denote the straight line through the random points P and Q . For almost every choice of P and Q the line L meets the boundary of the square in exactly two points and those two intersection points lie on two of the four sides of the square. Let I_{bottom} be the indicator that the line meets the segment AB . For any pair of random points P and Q , the sum of these four indicators equals 2 (the line meets exactly two sides).

Hint 2: $E[I_{bottom}]$ is equal to the probability that L intersects AB .

62. Couple I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

63. Even Ratio

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Because X and Y are continuous and independent on $(0, 1)$, the random ratio Y/X takes values in $(0, \infty)$. Hence $[Y/X]$ is a positive integer, and

$$\{[Y/X] = n\} \iff n-1 < \frac{Y}{X} \leq n \iff (n-1)X < Y \leq nX.$$

Hint 2: We want that chosen integer n to be even. Write $n = 2k$ with $k = 1, 2, 3, \dots$. Then

$$[Y/X] = 2k \iff (2k-1)X < Y \leq 2kX.$$

Because these events are disjoint for different k ,

$$P([Y/X] \text{ is even}) = \sum_{k=1}^{\infty} P((2k-1)X < Y \leq 2kX).$$

Hint 3: Fix any integer $k \geq 1$. We must compute $P((2k-1)X < Y < 2kX)$, given that (X, Y) is uniform over the unit square $\{0 < x < 1, 0 < y < 1\}$.

One convenient way is to integrate in the y -direction: for each fixed $y \in (0, 1)$, the inequalities $(2k-1)X < y \iff X < \frac{y}{2k-1}$, $y \leq 2kX \iff X \geq \frac{y}{2k}$.

Hence, at a given y , the set of x -values satisfying both is $x \in [\frac{y}{2k}, \frac{y}{2k-1})$, provided $\frac{y}{2k} \leq 1$ and $\frac{y}{2k-1} \leq 1$. But for $k \geq 1$ and $y \in (0, 1)$, we always have $0 < \frac{y}{2k} < \frac{y}{2k-1} \leq 1$. Thus, for each $y \in (0, 1)$, the length of the allowable x -interval is

$$\frac{y}{2k-1} - \frac{y}{2k} = y\left(\frac{1}{2k-1} - \frac{1}{2k}\right) = \frac{y}{2k(2k-1)}$$

Since (X, Y) is uniform on the unit square, its joint density is 1. Integrating that interval-length over $y \in (0, 1)$ gives

$$P((2k-1)X < Y < 2kX) = \int_0^1 \frac{y}{2k(2k-1)} dy = \frac{1}{4k(2k-1)}$$

Hint 4: The alternating series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$ is known to converge to $\ln(2)$.

64. 2 Eggs

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Assume we use a strategy with N drops. If the unknown threshold floor falls within the range $[1, N-1]$, how can we determine this threshold? We start by dropping the egg from the N th floor, where it will not break. Next, we continue dropping the egg from the 1st floor, 2nd floor, and so on until we identify the threshold floor. Therefore, the first move in our strategy must be to drop the egg from the N th floor. What should the second move be?

65. Heads Run I

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Use recursion. What are the initial choices?

Hint 2: Let a_n be the number of length- n sequences of H's and T's without ever having three consecutive heads (no "HHH"). To build a valid length- n sequence, look at how it starts: * If it starts with T, the remaining $n-1$ flips can be any valid sequence of length $n-1$. * If it starts with HT, the remaining $n-2$ flips can be any valid sequence of length $n-2$. * If it starts with HHT, the remaining $n-3$ flips can be any

valid sequence of length $n - 3$. These three cases are disjoint and cover all possibilities, so for $n \geq 3$:

$$a_n = a_{n-1} + a_{n-2} + a_{n-3}$$

Hint 3: We need 3 base values to start this recurrence. * An empty sequence ($n = 0$) is valid: $a_0 = 1$ * For $n = 1$, the two sequences "H" and "T" are both valid: $a_1 = 2$ * For $n = 2$, all four sequences of length 2 avoid "HHH": $a_2 = 4$

66. Smallest Set

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write the elements of S in their ternary expansion (i.e. base 3).

67. Variable Game

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: Use the round number as state. Let p_n be the probability that you win the n -th round. We can express p_n using p_{n-1} .

68. Chess Tournament II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In the first round, there are 128 positions. Assume that in the first round, the player at position 1 plays the player at position 2, the player at position 3 plays the player at position 4, and so on. In the second round, the winner of the match between positions 1 and 2 will play the winner of the match between positions 3 and 4 - and so on for subsequent rounds.

Without loss of generality, we can assume that the highest-rated player is occupying position 1. Which of the remaining positions should the second- and third-highest-rated players occupy so that the highest-rated and the third-highest-rated players play in the final?

69. Game Of Smaller Number

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the Law of Total Expectation.

70. Missing Sequence

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since the sequence must remain strictly increasing, choose a and b from the integers between 6 and 29 inclusive. Consider which values should be excluded.

71. Passing The Coin II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Solve "passing the coin I" first.**Hint 2:** If player 3 gets the coin, the problem reduces to "passing the coin I".**72. Remove The Subtree**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use linearity of expectation.**73. Interesting Summation III**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Let S represent the sum in the given equation. We can expand the first few terms of the sum as:

$$S = \sum_{k=0}^{\infty} \frac{F_k}{10^{k+1}} = \frac{F_0}{10} + \frac{F_1}{10^2} + \frac{F_2}{10^3} + \frac{F_3}{10^4} + \dots$$

This can also be written as:

$$S = \frac{F_0}{10} + \frac{F_1}{10^2} + \left(\frac{F_0}{10^3} + \frac{F_1}{10^3} \right) + \left(\frac{F_1}{10^4} + \frac{F_2}{10^4} \right) + \dots$$

Using the definition of the Fibonacci sequence, we can regroup the terms as:

$$S = \frac{F_0}{10} + \frac{F_1}{100} + \left(\frac{F_0}{10^3} + \frac{F_1}{10^4} + \dots \right) + \left(\frac{F_1}{10^3} + \frac{F_2}{10^4} + \dots \right)$$

74. Interesting Summation V[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Let $n[K]$ denote the number of such integers and $E[K]$ denote the expected value of such integers. The required sum is $n[K] \cdot E[K]$.**Hint 2:** To find $n[K]$, use recursion. To find $E[K]$, find the expected value of each digit and then use linearity of expectation.**75. Pair Partition**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Arrange the numbers 1 through 20 in a 2×10 array: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 A pair $\{a, b\}$ with $|a - b| = 1$ corresponds to a horizontal domino (two adjacent cells in the same row). A pair with $|a - b| = 10$ corresponds to a vertical domino (one cell in the top row and the one directly below it).

Hint 2: Every valid pairing is exactly a perfect tiling of the above 2×10 board by dominoes.

76. Last One Off

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume each ant is carrying a flag. Imagine at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem, as the flags simply go past each other. This simplification is commonly used in many collision problems.

77. Strictly Within

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In how many ways can you select a single interval? Note that an interval on the set $\{-n, \dots, n\}$ can either be a single point or a genuine two-endpoint interval.

Hint 2: There are $2n + 1$ single point intervals and $\binom{2n+1}{2}$ two-point intervals. So, in total, there are

$$(2n + 1) + \binom{2n + 1}{2} = (2n + 1)(n + 1)$$

possible intervals. Since we choose two intervals independently (first I_1 , then I_2), the size of our sample space is

$$((2n + 1)(n + 1))^2.$$

Hint 3: We wish to count those ordered pairs (I_1, I_2) for which I_2 lies strictly inside I_1 . Notice immediately that I_1 cannot be a singleton, so I_1 must be a two-endpoint interval. Now, I_2 can be a two-endpoint interval or a singleton. Count the number of favorable cases for each of them.

78. Get King And Ace

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve the problem "first ace" first.

Hint 2: Let the 8 special cards be the 4 kings and 4 aces and the 44 ordinary cards be the rest. Label the draws $1, 2, \dots$ until we have seen at least one king and at least one ace. Split the problem into two stages - (i) Stage 1 (time to the first special) and (ii) Stage 2 (additional draws until the opposite type appears).

79. Max Equals Min

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: $a \leq 0$, as $\min\{0, b\} \leq 0$. Additionally, $b \geq 0$, since $\max\{0, a\} \geq 0$.

80. Uniform Sum Probability

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let p_1, p_2, p_3, p_4, p_5 , and p_6 be the probabilities for one 6-sided die, and q_1, q_2, q_3, q_4, q_5 , and q_6 be the probabilities for another. Suppose that these dice together yield sums with uniform probabilities. That is, suppose $P(\text{sum} = k) = \frac{1}{11}$ for $k = 2, \dots, 12$. We can generate equations in $p_1, p_2, \dots, p_6$ and $q_1, q_2, \dots, q_6$ and try to solve them to prove possibility/impossibility.

81. Stop Or Continue

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: You have a choice: either stop and take your current score S , or continue, in which case you risk losing everything if you roll a 10. The key is to find the smallest score S such that stopping is better than rolling again.

Hint 2: Assume your current score is S . Calculate the expected score if you continue-call this $f(S)$. If you stop, your score is S . Therefore, you should stop when

$$S \geq f(S).$$

82. Greedy Pirates I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Here, the number of pirates is 5. What if the number of pirates was 1, 2, 3...?

83. Coin Toss Game V

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The probability that Alice needs a flips, Bob needs b flips, and Charlie needs c flips to get their first heads is $\left(\frac{1}{2}\right)^a \cdot \left(\frac{1}{2}\right)^b \cdot \left(\frac{1}{2}\right)^c$. The probability that $a < b < c$ is given by

$$\sum_{a=1}^{\infty} \sum_{b=a+1}^{\infty} \sum_{c=b+1}^{\infty} \left(\frac{1}{2}\right)^a \left(\frac{1}{2}\right)^b \left(\frac{1}{2}\right)^c$$

84. Expected Ones In Binary

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $f(n)$ be the answer for a and b selected from $\{0, 1, \dots, 2^n - 1\}$. Use recursion.

85. Points On The Same Face

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since a cube has 8 vertices, each vertex has a $\frac{1}{8}$ probability of being chosen as P . There is a $\frac{1}{8}$ chance that $Q = P$, a $\frac{3}{8}$ chance that Q is connected to P by an edge, a $\frac{3}{8}$ chance that Q is diagonally across a face from P , and a $\frac{1}{8}$ chance that Q is diagonally across the cube from P . Treat these 4 possibilities as separate cases.

86. Product Of Cards

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Try to locate the values 9 and 10 in the fewest possible moves.

87. Gold Auction

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If there are n uniformly distributed random variables in the range $[500, 1000]$, the expected maximum value is given by $1000 - \frac{500}{n+1}$. This is because n variables divide the segment $[500, 1000]$ into $n + 1$ segments of equal length on average. The expected payout when recruiting n people is therefore $1000 - \frac{500}{n+1} - 5n$.

88. Cutting Cube

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The central cube has six faces. At least 6 cuts are necessary to cut through all six faces, regardless of how the blocks are repositioned after each cut.

89. Three Pegs

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think in terms of parities of the x and y coordinates of grid points.

90. Chocolate Problem

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "first ace" first.

Hint 2: We can use the same method as in "first ace," applying it repeatedly for each large chocolate.

91. Fixed Points I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In a random function $f : S_n \rightarrow S_n$, the probability of any given value being a fixed point is $\frac{1}{n}$. Since there are n elements in the set S_n , the number of fixed points, F_n , follows a binomial distribution $F_n \sim \text{Binom}(n, \frac{1}{n})$.

92. Reaching 50

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Work backward from 50. What number closest to 50 guarantees a win if you reach that number?

93. Game On Divisors I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the prime numbers within the set S .

94. At Least Six II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "at least six I" first.

Hint 2: Write E_m for the expected number of additional rolls needed when your current total is m . Clearly $E_m = 0$ for all $m \geq n$. Compute E_{n-1} , E_{n-2} , E_{n-3} , and E_{n-4} . Identify a pattern from these values, then prove the pattern using induction.

95. Painted Cube III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are 2^6 possible colorings of the cube. Consider the color that appears with greater frequency. The property obviously holds true if 5 or 6 of the faces are colored the same, which for each color can happen in $6 + 1 = 7$ ways. What about other cases?

96. Sufficient Sum I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Examine the low-dimensional cases ($n = 1, 2, 3$) by direct geometric reasoning, observe a pattern, and then confirm it in general by induction.

Hint 2: For $n = 1$, the probability is 1. For $n = 2$, it is $\frac{1}{2}$ and for $n = 3$ it is $\frac{1}{6}$. So, the pattern is $\frac{1}{n!}$ for general n .

97. Interesting Summation II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The average value of each digit is 4.5.

98. Random Pairing

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the principle of inclusion-exclusion.

99. Odd Heads I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The number of tosses is irrelevant.

100. Tetrahedral Die II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use tail sum formula.

101. Greedy Pirates II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: How would you approach this problem if it involved only gold and no silver?

102. Horse Racing

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To identify the top three fastest horses, first organize all 25 horses into five groups, each containing five horses. Conduct a race for each group to determine the ranking within each group. After these races, focus on the fastest horses from each group and race them to find the fastest overall. Use this race to narrow down the potential candidates for the second and third fastest horses.

103. 3 Heads 3 Tails I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

104. Going In Circles

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For a simple symmetric random walk starting at 0 with boundaries at $-b$ and a , the expected time to hit either boundary is $a \times b$. Can we transform our "circular" problem to a "linear" problem?

Hint 2: We can transform our "circular" problem to a "linear" problem as follows: (i) Consider point 1 on the circle as point 0 on the line. (ii) A single counterclockwise move from circle-point 1 brings you directly to circle-point 0; on the line this is moving from 0 to -1 , the left absorbing boundary. (iii) Conversely, going

clockwise all the way around the circle (99 steps) also brings you from circle-point 1 back to circle-point 0; on the line this is moving from 0 to +99, the right absorbing boundary.

105. Even Majority

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For each roll of values (x, y, z) where at least two numbers are even, the corresponding roll $(7 - x, 7 - y, 7 - z)$ will have fewer than two even values. Therefore, the number of rolls with two or more even values is equal to those with fewer than two. Since there are $6^3 = 216$ total possible rolls, half of them, or 108, will have at least two even numbers. The desired probability is the number of outcomes where the sum is at most 7 and there are at least two even numbers, divided by 108.

106. Candy Distribution

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the seats be numbered from 1 to 2022. Also, let Bob be seated at seat 1. Let s_i denote the seat number of the i -th candy. What happens to $\sum_{i=1}^{2023} s_i^2$ in one move, if the candies never reach or cross Bob?

107. Roll Till Even

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What is the distribution of the number of rolls till you get an even number.

108. Triangle Inside Circle I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The triangle will be obtuse when all three vertices are within the same half of the circle.

109. Lily Pad

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the 8 lily pads as equivalent to a single lily pad that has been growing for 3 days.

110. Passing The Bottle

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Each of Moor's friends is equally likely to be the last to touch the bottle. Prove it!

111. Jumping Frog I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Get a recurrence relation in p_k and solve it.

112. Penguin Walk II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider what happens on the first step, then use the law of total probability. Note that the probability of reaching position 1 from 0 is the same as that of reaching position 0 from -1.

113. Interview I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: It is clearly never optimal to select a candidate who isn't the best among those interviewed so far. If a candidate is not the best up to this point, they cannot be the best overall.

Hint 2: Suppose you select the i th candidate who is the best among those interviewed so far. Using conditional probability, we can show that if the current candidate is the best among the first i interviewed, then the probability that they are the best overall is $\frac{i}{6}$.

Hint 3: Using Hint 2, we see that it is better to select a "best till now" candidate as late as possible. However, if we delay the selection too long, we risk encountering no "best till now" candidates later in the process. To balance this trade-off, the optimal strategy is to reject the first k candidates outright, then select the first "best till now" candidate that appears afterward. To determine the best value of k , we construct a probability function in terms of k and choose the value that maximizes this probability.

114. Free Throws

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We must hit exactly 65 of the remaining 98 free throws. This can occur in $\binom{98}{65}$ possible sequences. Now, focus on calculating the probability of each such sequence.

115. Starting With Heads

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Define $f(n)$ as the number of sequences of length n with no consecutive heads. Express $f(n)$ in terms of smaller values, such as $f(n-1)$ and $f(n-2)$.

116. Circular Table I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assuming B as boy and G as girl, there must be a sequence BGGGG occurring somewhere in the circle. Let E_i be the event that this sequence occurs at position i (assuming seats are labeled from 1 to 12). These events are disjoint.

117. Gold Thief

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the weights of the coins be $w_1, w_2, \dots, w_5$. Consider the total of all pairwise sums of weights.

118. Hanging Chain

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The entire chain is 80 feet, while the lowest point is at a vertical distance of 40 feet.

119. Coin Sequence Guess

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Choose a non-empty, proper subset A of the sample space. Ask, "Is our sequence s in A ?"

120. Higher Roll I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Condition on Alice's roll.

Hint 2: If Alice's roll is greater than 15, she has a higher value with probability 1. If her roll is less than or equal to 15, then whether it is strictly higher than Bob's depends on the exact value she rolls. Use the law of total probability.

121. Dice Game IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Calculate the probability of winning exactly k dollars in terms of k . Then use the usual formula for expectation,

$$E[X] = \sum xP(X = x).$$

122. Half Right Hats

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: First, solve the "prisoner hats" problem.

123. Pppower

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Clearly, p^p ends with a 1 when p itself ends with 1. Can you identify any other values of p for which p^p ends with 1?

Hint 2: There are no other such values of p .

124. Grid Arrangement II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let a_{ij} represent the cell at i -th row and j -th column. a_{11} can either be $+1$ or -1 (2 ways). Assume it is $+1$. Now, we need to fill one more $+1$ in the first row (3 ways) and one more $+1$ in the first column (3 ways). Assume we fill a_{21} and a_{12} with $+1$. Now, what happens if we place $+1$ or -1 at a_{22} ?

125. Distant Points

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: By rotational symmetry, fix the first point at 0° . To be at least 90° from this, the second point must lie somewhere between 90° and 270° .

Hint 2: Split that 180° region (for the second point) into two equal halves, $[90^\circ, 180^\circ]$ and $[180^\circ, 270^\circ]$, and observe they are symmetric. So, we compute the result for the case where the second point lies in the interval $[90^\circ, 180^\circ]$, then double it.

Hint 3: Write the second point's angle as

$$\phi = 90^\circ + x, \quad 0 \leq x \leq 90^\circ$$

Find the conditional probability (in terms of x) that the third point does not lie within a 90° arc of the first two points, given that $0 \leq x \leq 90$. Then use the law of total probability for the continuous case.

126. Probabilistic Strategy IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The chances that two mathematicians directly across from each other guess right are independent, with each individual probability being $\frac{1}{2}$ because they cannot see or control the colors of their own hats. Thus the answer is at most $(\frac{1}{2})^2 = \frac{1}{4}$. Try to construct a strategy that can achieve this probability.

127. Optimal Card Guessing

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $R_1, R_2, \dots, R_6$ represent the amounts Alice wins on turns 1 through 6. By linearity of expectation, the total expected amount won is $E[R_1] + E[R_2] + \dots + E[R_6]$. The optimal strategy for Alice is to always guess the card type that appears most frequently at each stage.

128. k-inversions

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

Hint 2: Use indicator variables. For a fixed choice $(i_1, \dots, i_6)$, let

$$X_{i_1, \dots, i_6} = \begin{cases} 1, & \text{if the entries at those positions form a strictly decreasing sequence} \\ 0, & \text{otherwise.} \end{cases}$$

129. Five In Numerator

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We know $10! = 2^8 3^4 5^2 7^1$. Because $\gcd(a, b) = 1$ for fractions in lowest terms and $a \cdot b = 10!$, each prime power $2^8, 3^4, 5^2, 7^1$ must go entirely to either a or b , giving $2^4 = 16$ ordered pairs (a, b) and exactly half have $a < b$, so $|S| = 8$.

Hint 2: For the numerator to be divisible by 5 we must assign 5^2 to a . The other prime powers $2^8, 3^4, 7$ can each be assigned independently to a or b , but which ones result in fractions between 0 and 1?

130. W-permutation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Say that $P(2)$ and $P(4)$ are the low values; say that $P(1)$, $P(3)$, and $P(5)$ are the high values. It's easy to see that 1 must be low. It's easy to see that 4 and 5 must be high. So the low values are either $\{1, 2\}$ or $\{1, 3\}$.

131. Prisoner Hats

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The prisoner at the back can count the hats of each color and provide valuable information to the next prisoner.

132. Contiguous THTH

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Conditional on having exactly four heads and seven tails, every arrangement of those eleven flips is equally likely. The total number of such arrangements is $\binom{11}{4} = \frac{11!}{4!7!} = 330$. Thus our sample space has 330 equally likely sequences.

Hint 2: To count how many of those 330 sequences contain the pattern THTH at least once, apply the inclusion-exclusion principle to the eight possible starting positions of a block of length four within an eleven-flip sequence.

133. 50 Wise Men

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Reminder from the first chapter on brainteasers—always begin with a simpler version of the problem whenever possible.

Hint 2: What if there are only 2 wise men? The strategy is simple—one wise man could decide not to interact with the glass at all. The other wise man could decide to flip the glass upside down the first time he sees it in the bottom-down position, then refrain from further interaction. Eventually, the first wise man will observe the glass in an upside-down position, indicating that the second wise man has been summoned. He can then confidently claim that both men have been summoned, securing their freedom.

Hint 3: What if there are 3 wise men? Two of the wise men can flip the glass only once, and only if they observe the glass in a bottom-down position. The remaining wise man can flip the glass whenever he sees the glass upside down. This way, the remaining wise man can count the number of times the glass has been flipped.

134. Rolling Everything I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The time until the first result appears is 1. After that, the random time until a second (different) result appears is geometrically distributed with a success parameter of $\frac{5}{6}$, hence with a mean of $\frac{6}{5}$, and so on.

135. Socks Remaining I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If the color of the last sock is blue (probability 0.4), then it must have been the red socks that were all removed, and the number of socks "left in the drawer" is the number of blue socks encountered starting from the end of the arrangement before any red sock is found.

136. Consecutive Integral Envelopes

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If we decide to always pick up the first envelope, we have no choice but to get the expected value of $x + 2$. Similarly, if we decide to always pick up the third envelope, the expected payout remains $x + 2$. Therefore, to achieve a higher expected payout, we must decide after opening the first two envelopes—whether to retain the second envelope or opt for the third.

137. Trailing Zeros I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Trailing zeros in a factorial result from multiples of 10, which are products of 2 and 5. Since there are more factors of 2 than 5, the number of trailing zeros is determined by the power of 5 in the prime factorization of $100!$.

138. Expected Minimum Till 5

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let X be the random variable representing the minimum number rolled before observing a 5. Note that $\Pr(X = 6) = 0$ since a 5 will always be rolled. Determine $\Pr(X = 1)$, $\Pr(X = 2)$, and so forth.

139. Palindrome Summation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Notice that $10001 + 99999 = 110000$.

140. Matrix Attack II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the total number of possible initial matrices. Additionally, evaluate the total number of matrices that can be obtained using the allowed operations.

141. Consecutive 1s

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Count the number of ways to get at least 3 consecutive ones. Consider cases for exactly 3, 4, or 5 consecutive ones.

142. Ball With Maximum Value

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

143. Digit Multiply I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Examine the prime factorization of 10,000 to determine how to optimally arrange the factors to achieve the smallest integer.

144. Regular Octagon

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use complementary counting. Do casework with the following cases—the diagonals intersect on the octagon, the diagonals never intersect, the diagonals intersect outside of the octagon.

145. Payment[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion or Markov chains. Consider the states as the amount of payment already made.

146. Rock Paper Scissors[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If Alice plays Scissors, Bob must have chosen either Rock or Paper to determine a winner. Thus, out of the 6 times Alice played Scissors, she would have 4 wins from the times Bob played Paper and 2 losses when he played Rock.

147. Party Groups I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To solve this problem, we model the group formation as cycles in a random permutation of the 50 guests. Each guest picking a name corresponds to choosing a random permutation, and each closed cycle in this permutation represents one group. Thus, the expected number of groups equals the expected number of cycles in a random permutation of 50 elements.

148. Min, Median, Max Guess[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Intuitively, if the observed X_1 is very small you should guess "minimum," if it is very large you should guess "maximum," and otherwise you should guess "median." So, the strategy is of the form: If $X_1 \leq \alpha$, guess it is minimum. If $X_1 \geq \beta$, guess it is maximum. If $\alpha \leq X_1 < \beta$, guess it is median. We can use symmetry to simplify this strategy.

Hint 2: Using symmetry, the strategy can be simplified. Fix a cutoff $\alpha \in [0, \frac{1}{2}]$ and use the rule: If $X_1 \leq \alpha$, guess it is the minimum. If $X_1 \geq 1 - \alpha$, guess it is the maximum. If $\alpha < X_1 < 1 - \alpha$, guess it is the median. Why choose $1 - \alpha$ for the upper cutoff? Since $X_1 \sim \text{Unif}(0, 1)$ implies $1 - X_1 \sim \text{Unif}(0, 1)$, the situation " X_1 is the maximum and $X_1 \geq 1 - \alpha$ " corresponds exactly under the map $x \mapsto 1 - x$ to " X_1 is the minimum and $X_1 \leq \alpha$." To treat minimum and maximum symmetrically, the upper threshold must be $1 - \alpha$.

149. Normal Inequality III[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Because N_1, N_2 , and N_3 are independent standard normal variables, the point (N_1, N_2, N_3) has a spherically symmetric distribution in $\mathbb{R}^3$.

150. Permutation Variance[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: $X + Y = 1000 \Rightarrow Y = 1000 - X$ So, $X - Y = 2X - 1000 \Rightarrow \text{Var}(X - Y) = 4 \cdot \text{Var}(X)$ (since variance is unaffected by constants and scales quadratically with coefficients).

Hint 2: $\text{Var}(X) = E[X^2] - (E[X])^2$ Use indicator variables to compute $E[X]$ and $E[X^2]$.

151. Probabilistic Strategy II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Examine the sample space carefully.

152. 3D Tic-tac-toe

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Every straight-line path of five consecutive cells in the $5 \times 5 \times 5$ cube must lie along one of three direction-types (up to reversal):

- Axis-parallel lines, which run along the x , y , or z axes.
- Face-diagonals, which lie in one of the coordinate planes (e.g., the " $+x, +y$ " diagonal in an xy -plane).
- Space-diagonals, which run corner-to-corner through the interior (e.g., the " $+x, +y, +z$ " diagonal).

153. Color And Roll

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There is a solution using the law of total probability. However, a clever approach is to notice that all the faces of the smaller cubes are equally likely to show as the up face.

154. Expected 2s

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Bayes' formula to calculate the probability that we draw from the first and second urns.

155. Drunk Passenger I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Can the last passenger arrive at any other seat than 1 or 100?

156. Broken Calculator

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the binary representation of numbers.

157. Square Fitting II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Solve "square fitting I" first.**Hint 2:** Use the Chebyshev distance

$$M = \max\{|X|, |Y|\}$$

for a random location (X, Y) . Because $|X|$ and $|Y|$ are independent and uniform on $[0, 1]$, the chance that both lie within $[-m, m]$ is m^2 . Equivalently,

$$\Pr(M \leq m) = m^2, \quad 0 \leq m \leq 1,$$

so M has density

$$f_M(m) = 2m, \quad 0 \leq m \leq 1.$$

Label the nine independent distances $M_1, \dots, M_9$. When you order them from smallest to largest, the fourth-smallest value $M_{(4)}$ captures exactly how far out you must go in Chebyshev distance to include the fourth-closest location. Then use the formula for the density of the 4th order statistic for i.i.d. samples:

$$f_{M_{(4)}}(m) = \frac{9!}{3!5!} [F_M(m)]^3 [1 - F_M(m)]^5 f_M(m) = 504m^6 (1 - m^2)^5 (2m).$$

158. Increasing Dice Rolls II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use combinatorics. There are 6^{10} total sequences. How many of those are non-decreasing?**159. Direction Change II**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Denote a rightward movement as R and an upward movement as U . The total number of paths, allowing for any sequence of direction changes, corresponds to the number of ways to arrange 8 R 's and 8 U 's.**Hint 2:** If there are 4 direction changes, the path consists of 5 blocks, with each block being in the same direction (for example, $RRUURRUUUUUURRRR$ is a valid path with 4 direction changes). Each block must have a length of at least one. To determine the number of such arrangements, apply the stars and bars method.**160. Close Factors**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Consider the prime factorization of 1234567890.**161. Non-absorbing Random-walk**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion / Markov chains with the positions $1, \dots, n$ as states.

162. Equal Logarithms

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Take the case that $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor = -1$. In this case, both x and y lie in the interval $[1/2, 1)$. The probability of this is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. Similarly, in the case that $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor = -2$, x and y lie in the interval $[1/4, 1/2)$, and the probability is $\frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$. Can we generalize this?

163. Mathematical Expression III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are 8 gaps between the digits of 123456789. Each of those gaps can either be empty or filled with $+$ or filled with $\times$ (3 choices).

164. Dice Multiple I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the probability that the sum of the first k rolls is a multiple of 6.

165. Self-avoiding Paths

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion. Suppose you have the answer for length $n - 1$. How can you find the answer for length n ?

Hint 2: To build every path of length n , start with a path of length $n - 1$ and append one more step. Notice that:

- Any path of length $n - 1$ can always be extended by a right step without revisiting, because moving right increases your x -coordinate into unvisited territory.
- To extend by up or down, you must avoid backtracking the very last vertical move.

In particular:

- If the $(n - 1)$ -step path ended with a vertical move, you have only two safe extensions (right or continuing in the same vertical direction).
- If it ended with a right move, there is no forbidden vertical backtracking, so you have three choices (right, up, or down).

Hint 3: Exactly a_{n-2} of the length- $(n - 1)$ paths end with a right move: each is formed by taking any self-avoiding path of length $n - 2$ and appending a right step. The remaining $a_{n-1} - a_{n-2}$ paths end vertically. Thus,

$$a_n = \underbrace{2}_{\text{paths ending vertically}} (a_{n-1} - a_{n-2}) + \underbrace{3}_{\text{paths ending with right}} a_{n-2} = 2a_{n-1} + a_{n-2}$$

166. SOS Game[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The only way to secure a win on your next move is to force your opponent into the configuration "S-blank-blank-S," which will be referred to as a "pit."

167. Moving Ants I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What are the total number of ways in which the 3 ants can move? Which ones will lead to the ants not meeting at a vertex?

168. Moving Ants III[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: When two ants collide, we can assume they continue moving in their original direction. Thus, it is as if no collisions occur.

169. Constrained Subsets I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If the smallest number is a , the largest number is $13 - a$. Also, $a < 13 - a \Rightarrow a \leq 6$. So, the smallest-largest pairs are $(1, 12)$, $(2, 11)$, $(3, 10)$, $(4, 9)$, $(5, 8)$, $(6, 7)$. Can we count the number of subsets corresponding to each?

Hint 2: The number of subsets corresponding to the pair $(1, 12)$ are 2^{10} as there are 10 elements between 1 and 12—and we may either include or not include each of those elements in our subset (2 choices).

170. Sufficient Sum IV[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $f(x)$ be the expected number of additional variables needed to make the running total exceed 2, given that the current sum is x , where $0 \leq x \leq 2$. So, we need $f(0)$. After one more draw $U \sim \text{Unif}(0, 1)$, the process either stops immediately or continues from the new sum $x + U$. So the expected remaining time satisfies a recursion in terms of $f(x + U)$. Using the law of total expectation, and then differentiating, we can get a differential equation in x .

171. Numerator I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We observe that $\frac{1}{5} < \frac{179}{720} < \frac{1}{4}$. So, $\frac{1}{5}$ is our initial unit fraction.

172. Predicting King Draws[[Question ↑](#) | [Solution ↓](#) | [↑](#)]*No hints provided.***173. 40-sided And 60-sided Die**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Condition on whether the number on the 60-sided die exceeds 40 or falls within the range 1 to 40. If the 60-sided die shows a value greater than 40, it wins automatically with probability 1, which happens with probability $\frac{1}{3}$. If it shows a number between 1 and 40, the situation resembles rolling two identical dice.

174. Count Bitonic Permutations[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What positions can the number 1 occupy? After putting 1, what positions can the number 2 occupy?

175. Uniform Number Bet I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The optimal strategy is of the threshold form: you keep drawing until you see a value at least as large as some cutoff t , and then you stop (of course, you need to prove why this is true!).

Hint 2: Let V be your expected net payoff (the final random number minus all the fees) when you follow the threshold rule with cutoff t . On your first free draw, you obtain a number $X \sim U(0, 1)$. Two things can happen: - If $X \geq t$, you stop immediately, pay no fee, and your net payoff is simply X . - If $X < t$, you pay a fee c to regenerate, and thereafter your situation (and your expected net payoff) is exactly the same as at the start—namely V . Thus conditioning on the first draw,

$$V = \mathbb{E}[\text{net payoff}] = \int_0^t (V - c)dx + \int_t^1 xdx = t(V - c) + \frac{1 - t^2}{2}.$$

Obtain the expression for V in terms of t .

Hint 3: Differentiate V with respect to t and set it to 0 to find the optimal t . Once you have the optimal t , you can get optimal V using the expression you found in Hint 2.

176. Asset Price[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the price increase for x days and decrease for $10 - x$ days. For what x does the price remain unchanged after 10 days?

177. Prism Count

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider how rectangular prisms can be defined by selecting intersections along the three axes of the cube.

178. Bulb Factory

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the law of total expectation.

179. Moving Checker

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the central 3×3 grid. Start by analyzing the first row. What is the probability that a checker placed at the top-left corner will reach a corner? Calculate these probabilities for the entire top row, then use these values to find the probabilities for the center and last rows of the 3×3 grid.

180. XXYYYYZZ

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The positions of the Ys fully determine the end configuration.

181. Factorial Factor

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The prime factorization of 2000 is $2000 = 2^4 \cdot 5^3$. Therefore, we need to determine the probability that a randomly selected divisor of $20!$ is divisible by both 2^4 and 5^3 .

182. Interesting Division

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If n is prime, it is easy to see that n^2 cannot divide $n!$. Thus, n must be composite.

183. Bulb Toggle III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: A bulb k will be off at the end of the day if it has an odd number of distinct prime factors. This includes all prime numbers, powers of prime numbers, and numbers with exactly three distinct prime factors.

184. Broken Stick I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The first breaking point X is randomly located in between $[0, \frac{1}{2}]$ and the second breaking point Y is located in between $[\frac{1}{2}, 1]$. We want to determine $P[|X - Y| \geq \frac{1}{3}]$.

185. Factorial Vs Exponent[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write $99!$ as the product of three parts: the numbers 1 through 49, the single factor 50, and the numbers 51 through 99. Pair each small factor k ($1 \leq k \leq 49$) with the corresponding large factor $100 - k$ (which runs through 99 down to 51). Then

$$99! = \left(\prod_{k=1}^{49} k \right) \cdot 50 \cdot \left(\prod_{k=1}^{49} (100 - k) \right) = 50 \cdot \prod_{k=1}^{49} (k(100 - k)).$$

Now use the identity $(x - y)(x + y) = x^2 - y^2$.

Hint 2: Put $x = 50$ and $y = 50 - k$. Then for each k between 1 and 49 we have

$$k(100 - k) = (50 - (50 - k)) \cdot (50 + (50 - k)) = 50^2 - (50 - k)^2 \leq 50^2.$$

186. Expected Arc Length III[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "expected arc length II" first.

Hint 2: In what follows, assume all angles are measured relative to the positive x-axis in the counterclockwise direction. Condition on the event that no cut falls in the open interval $(0, \frac{\pi}{2})$, so that the two fixed points indeed share one arc. Under this conditioning, all three cuts are still i.i.d. uniform on $[\frac{\pi}{2}, 2\pi]$. Let $A = \min\{\text{the three cut-angles}\}$, $B = \max\{\text{the three cut-angles}\}$. Then the arc containing angles 0 through $\frac{\pi}{2}$ "wraps" from B around past 0 up to A , so its length is $L = (2\pi - B) + A$.

187. Heavy Group[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To solve the problem, note that we can ignore any uniform offset in the weights, as it doesn't affect the relative comparisons. Thus, we can treat the weights as if they were 1, 2, 3, 4, 5, 6 grams instead of 201, 202, ..., 206.

188. First And Last Flip[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

189. Integral Angles[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Each interior angle of a regular n -sided polygon is $\frac{360}{n}$ degrees.

190. Perfect Ratio[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Because X and Y are continuous and independent on $(0, 1)$, the ratio Y/X ranges over $(0, \infty)$ with a continuous distribution, so

$$\{[Y/X] = k^2\} = \{k^2 - 1 < Y/X \leq k^2\} = \{(k^2 - 1)X < Y \leq k^2 X\}.$$

These events are disjoint for different k . Therefore,

$$P(S \text{ is a square}) = \sum_{k=1}^{\infty} P((k^2 - 1)X < Y \leq k^2 X).$$

Hint 2: If $k = 1$, the condition is $0 < Y \leq 1 \cdot X$, i.e. $Y < X$. By symmetry of two IID Uniform(0, 1) variables,

$$P(Y < X) = \frac{1}{2}.$$

If $k \geq 2$, we need

$$(k^2 - 1)X < Y \leq k^2 X, \quad 0 < X < 1, \quad 0 < Y < 1.$$

Equivalently, at each fixed $y \in (0, 1)$, that range becomes

$$\frac{y}{k^2} \leq X < \frac{y}{k^2 - 1}.$$

Since $\frac{y}{k^2 - 1} < 1$ for all $y < 1$ when $k \geq 2$, the x -interval is

$$\left[\frac{y}{k^2}, \frac{y}{k^2 - 1} \right).$$

Because (X, Y) is uniform on the unit square, its probability is

$$\int_0^1 \left(\frac{y}{k^2 - 1} - \frac{y}{k^2} \right) dy = \int_0^1 y \left(\frac{1}{k^2 - 1} - \frac{1}{k^2} \right) dy = \frac{1}{k^2(k^2 - 1)} \cdot \frac{1}{2} = \frac{1}{2k^2(k^2 - 1)}.$$

Hint 3: Use the summation

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

191. Winning Odds[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To win the game, a player must win two successive rounds.

192. Biased Coin Game I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write the expected value of Bob's returns in terms of p_1 and p_2 .

193. 3 Marbles Remaining

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Compute the probability that Deepali ends with exactly 3 red marbles remaining. Then double that probability to account for her possibly ending with 3 blue marbles instead.

194. Infinite Notes

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Identify the smallest obtainable values that leave remainders of 1, 2, 3, and 4 when divided by 5.

195. Polynomial Coefficient

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The coefficients of $p(x)$ are exactly the coefficients used to represent the integer $p(p(1) + 1)$ in base $p(1) + 1$.

196. Game On Prime Numbers

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: After Bob says 3, Alice has the choice to say either 5 or 7. Is one of them winning?

197. Double Headed Coin I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Bayes' Theorem.

198. Tic Tac Toe

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics.

199. Dominant Flip

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Without loss of generality, assume the first flip is heads. We must get at least 50 heads in the next 99 flips for heads to be dominant at the end of 100 flips. The number of such sequences is

$$\binom{99}{50} + \binom{99}{51} + \cdots + \binom{99}{99}$$

200. Queen Before King

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are 8 cards in total that are kings or queens. Consider only the positions of these cards in the sequence. The first 3 cards are 2 queens and 1 king. Therefore, the remaining 5 cards have 2 queens and 3 kings. The problem asks, what is the probability that the last card among these 5 cards is a king.

201. Expected Arc Length II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $\theta_1, \theta_2, \theta_3$ be i.i.d. Uniform $[0, 2\pi)$, where the angles are measured relative to the x-axis counter-clockwise. Define

$$A = \min\{\theta_1, \theta_2, \theta_3\}, \quad B = \max\{\theta_1, \theta_2, \theta_3\}.$$

The arc containing the fixed point at angle 0 is the "wrap-around" piece from B back to A , whose length is

$$L = (2\pi - B) + A.$$

Hint 2: When n points are placed independently and uniformly at random on a segment of length L :

$$\mathbb{E}[\min] = \frac{L}{n+1}, \quad \mathbb{E}[\max] = \frac{nL}{n+1}.$$

202. Normal Inequality II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The joint distribution of the variables $X_1, X_2, \dots, X_{10}$ is symmetric: the distribution of $(-X_1, -X_2, \dots, -X_{10})$ is the same as the distribution of $(X_1, X_2, \dots, X_{10})$.

203. Draw A Spade

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The desired probability is the relative probability that Jill draws the spade.

204. Median Roll

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry. The probability of getting x on a given roll and the probability of getting $7 - x$ on a given roll are the same ($\frac{1}{6}$).

205. Permutation Return

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In a permutation π , an inversion is a pair of indices $i < j$ such that $\pi(i) > \pi(j)$. Any single swap always changes the total number of inversions by an odd number.

Hint 2: We start with the identity permutation $[1, 2, 3, \dots, 100]$, which is completely ordered and therefore has zero inversions—an even number. After performing 25 swaps, since each swap changes the inversion count by an odd number, the total change in inversion count after 25 swaps must be an odd number. So the resulting permutation must have an odd number of inversions.

206. Getting Singular Matrix

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use law of total expectation based on the first value edited.

207. Chess Tournament V

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Player 49 remains in the tournament if either:

- he does not participate in any match during the first 10 matches, or
- he participates in the first 10 matches but player 50 does not participate in any match during the first 10 matches.

208. Gambler's Ruin II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Markov chains.

209. Sum Of Divisor Ratio

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the relationship between a divisor of 10000 that is a multiple of 100 and a divisor of 100.

210. No Intersections

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider one of the points. To which points can it be joined? For each case, what are the probabilities of no intersections?

211. Expected Area Of Intersection

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let s be the side length of H . There are 20 ways to pick 3 of the 6 vertices of a hexagon. Of these, there are 3 possibilities: (1) there are no pairs of adjacent vertices, which accounts for 2 of the ways, (2) there is a single pair of adjacent vertices, which accounts for 12 of the ways, and (3) the 3 vertices are consecutive, which accounts for 6 of the ways.

212. Blue, Ugly And Special

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Just use $E(x) = \sum x \cdot p(x)$.

213. Card Picking III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The problem is essentially asking - if you arrange 52 cards in a sequence, what is the probability that the 28th card is the 2 of hearts, given that the 27th card is the first ace encountered in the first 27 cards? Therefore, we can use conditional probability.

Hint 2: Using conditional probability, the solution is simply the number of ways to arrange the cards such that the first ace is in the 27th position and the 2 of hearts is in the 28th position, divided by the number of ways to arrange the cards such that the first ace is in the 27th position.

214. Probability Of Subset

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Label the ground set $S = \{1, 2, \dots, 10\}$. For each element $i \in S$, there are four equally likely ways it can lie in the random pair (X, Y) : * in neither X nor Y * in Y only * in X only * in both X and Y What is the probability of each of these cases?

Hint 2: Uniform randomness makes each of the 4 cases in Hint 1 have probability $\frac{1}{4}$.

Hint 3: The condition $X \subseteq Y$ fails when an element is "in X only."

215. Circle Fitting II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let R denote the radius of the smallest concentric circle that contains the two closest points to the center. To compute the expected area, we use LOTUS:

$$E[\pi R^2] = \int_0^1 \pi r^2 f_R(r) dr,$$

where f_R is the probability density of R .

Hint 2: To find the probability density of R , first find its cumulative distribution function and then differentiate it.

216. Triangle Inside Circle II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The total number of ways of choosing 3 points from 1153 points is $\binom{1153}{3}$. Which of these lead to obtuse-angled triangles? The triangle will be obtuse when all three vertices are within the same half of the circle.

217. Game Of Subtracting Digits

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Try analyzing simpler cases first. For example, what happens if the starting number is 11, 12, ..., 20? Which of these are winning or losing states? Can you spot a pattern?

Hint 2: A number is a losing position exactly when it is divisible by 10.

218. Count Rankings

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The competition has 9 games. Each game has 2 possible winners.

219. Count Rectangles I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Every rectangle with sides on the grid is determined by choosing two distinct vertical grid lines (one for the left side and one for the right) and two distinct horizontal grid lines (one for the top and one for the bottom).

220. Line Splits I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Divide the line segments into 3 intervals $(0, 1)$, $(1, 2)$ and $(2, 3)$. There are 3^3 ways to arrange the points in these intervals. Count the number of ways where the middle point lies between 1 and 2 units.

221. Sharing A Secret

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For every pair of individuals, there must be a lock to which neither member of the pair has a key.

222. Falling Ants

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume each ant is carrying a flag. Imagine at each collision, flags are exchanged between the colliding ants. Thus at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem as the flags simply go past each other. Note that this simplification is used in many collision problems.

223. Random Segments

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: It might initially seem like there are countless possibilities for how Leo's path could unfold. However, it actually stops as soon as a very simple condition is met. What is that condition?

Hint 2: The time until the drawing process first "hits" an existing segment is precisely the time until the first occurrence of two identical turns in a row.

224. Even Squares

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Label the squares by their row $i \in \{1, \dots, 9\}$ and column $j \in \{1, \dots, 9\}$. How many of the rectangles include square (i, j) ?

Hint 2: To choose a rectangle covering (i, j) , pick its top boundary among the i horizontal grid lines above or through row i , and its bottom boundary among the $9 - i + 1$ lines at or below row i ; similarly, pick its left boundary among the j vertical lines at or before column j , and its right boundary among the $9 - j + 1$ lines at or after column j . Thus the total number of such rectangles is $i(10 - i) \times j(10 - j)$. When is it odd/even?

225. Triangle Formation II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For the three pieces of a rod to form a triangle, each piece must be less than half the length of the rod. This can be ensured if one breaking point is in the left half and the other in the right half of the rod, with the distance between these points being less than half the rod's length. Consider the probabilities of these conditions being met.

226. Get Heads

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

227. Distinct Faces

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Apply the formula $E[x] = \sum_{i=1}^{\infty} P[x \geq i]$ for a non-negative integer-valued random variable x .

228. Initial Roll

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Condition on the value of the first roll.

229. Digit Sum IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider each number as having six digits, ranging from 000000 to 999999, and find the average sum of digits for such numbers.

230. Baseball Series

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If we imagine that the series always continues to seven games even after one team has won four, this will never change the winner of the series.

231. Collinear Integer Points

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the midpoints of the line segments.

232. Gambler's Ruin I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Markov chains.

233. Local Maxima

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the linearity of expectation.

234. Expected Collisions I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume each ant is carrying a flag. Imagine that at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem, as the flags simply go past each

other. This simplification is commonly used in many collision problems.

235. Only Consecutive Heads

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics. The problem reduces to determining the number of ways that heads can appear only in consecutive pairs when a coin is flipped 10 times.

236. Constrained Digits

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Note that $a > 0$, so that $b > c$, and $e \geq 0$ so that $d \geq c$. Conversely, for each choice of (b, c, d) with $b > c$ and $d \geq c$, there exists a unique pair (a, e) such that $abcde$ is a number having the desired property.

237. Max Fill

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In a 2×2 subgrid, what's the maximum number of cells that can be filled?

Hint 2: The maximum number of filled cells in a 2×2 subgrid is 2. Since there are 16 non-overlapping 2×2 subgrids in an 8×8 grid, the total maximum number of filled cells is 32. Can we achieve this?

238. Up, Right And Diagonal

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Define the random variables x_i, y_i for $1 \leq i \leq 100$ where each x_i equals 1 if on the i th move, Tim makes a contribution to the right and zero otherwise. y_i is equal to 1 if on the i th move we make a contribution upwards and 0 otherwise. Hence, the answer is

$$E \left[\sum_{i=1}^n x_i \cdot \sum_{j=1}^n y_j \right]$$

239. Product Of Solutions

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In some (many?) ways, this problem is similar to choosing 5 objects from 19 objects.

240. Ranged Guess

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Denote by p_n the first player's win probability when the secret lies in $\{1, \dots, n\}$. Solve for small n to find a pattern.

Hint 2: Solving for small n , we have: $n = 1$: you win with probability 1. $n = 2$: you must guess one of $\{1, 2\}$; if wrong you lose, so $p_2 = \frac{1}{2}$. $n = 3$: best first guesses are 1 or 3, giving $p_3 = \frac{2}{3}$. $n = 4$: the maximum probability of winning is $p_4 = \frac{1}{2}$ (calculated using the values for $n = 1, 2, 3$ for each possible initial choice and selecting the maximum). $n = 5$: the highest probability of winning is $\frac{3}{5}$ (calculated using the values for $n = 1, 2, 3, 4$ for each possible initial choice and selecting the maximum). In fact, one can prove by induction that $p_n = \begin{cases} \frac{1}{2}, & n \text{ even} \\ \frac{n+1}{2n}, & n \text{ odd} \end{cases}$.

241. Circular Regions

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What are the possible number of regions. What is the probability of each?

242. Integer Selection

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Bayes' formula. The solution involves dividing the number of selections where only 7 is unchosen by the number of selections where 7 is unchosen.

243. Double Headed Coin II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Bayes' theorem.

244. M,P,F,G

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Even though Alex stops with G, let's imagine that she picks all 4 buttons no matter what.

245. Tetrahedral Dice Game

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let's call b the outcome of the blue die and r the outcome of the red die. First, let's find the $P(b > r)$, $P(b = r)$, and $P(b < r)$ and their outcomes, as they each yield a different result. Total cases: $4 \times 4 = 16$. Of these 16 cases, there are 4 ways to create an outcome such that $b = r$. Thus, $P(b = r) = \frac{4}{16} = \frac{1}{4}$. With the 12 remaining cases, half will result in $b > r$ and the other half will be $r > b$. Both outcomes have a probability of $\frac{12}{16} \times \frac{1}{2} = \frac{3}{8}$. Thus, $P(b > r) = P(b < r) = \frac{3}{8}$.

246. Probabilistic Strategy I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If Alice and Bob guess randomly, the probability of winning is $\frac{1}{4}$. Explore whether there is a more effective strategy by examining the sample space $\{HH, TT, TH, HT\}$ more closely.

247. Coin Toss Game I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion with states or Markov chains.

Hint 2: Use the current suffix of the coin-toss sequence as your states. Let p be the probability Bob wins from the very start (no memory). Observe:

- If you've just seen a single H (state H), the game's prospects are the same as at the very start, so the win-probability from state H is also p .
- Let P_{HH} be the win-probability from state "just saw HH."
- Let P_{HT} be the win-probability from state "just saw HT".

248. Adding 1 Or 2[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

249. Positive Product[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We require that an even number of real numbers are negative (or equivalently, positive), which occurs with probability

$$\frac{1}{3^{2020}} + \binom{2020}{2} \frac{2^2}{3^{2020}} + \cdots + \binom{2020}{2020} \frac{2^{2020}}{3^{2020}}$$

Then use the binomial theorem to evaluate the above sum.

250. Not That Easy[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The required probability is not $\frac{1}{2}$. Assuming the answer is $\frac{1}{2}$ based on the two possibilities — double-headed or one-head-one-tail — is incorrect. Given that a randomly selected coin shows heads, these two possibilities are not equally likely and hence, we can't use the formula for probability for this sample space.

251. Measuring Everything I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think in terms of binary representation.

252. Count Solutions

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To determine the number of non-negative odd integer solutions to $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 96$, we can represent each x_i as an odd integer. Let's use a transformation: let $x_i = 2y_i + 1$, where y_i is a non-negative integer.

253. Dice Pattern Game I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We need to choose the option that requires a higher expected number of rolls. Think in terms of transition graphs.

254. Power Distance I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let x_i represent the particle's movement at time i . The variables $x_1, \dots, x_n$ are IID with probabilities $P[x_i = 1] = P[x_i = -1] = \frac{1}{2}$. The distance d_n after n seconds is given by $d_n = \sum_{i=1}^n x_i$.

We need to compute $E[d_n^2]$:

$$E[d_n^2] = E\left[\left(\sum_{i=1}^n x_i\right)^2\right] = E\left[\sum_{i=1}^n x_i^2 + \sum_{i \neq j} x_i x_j\right] = \sum_{i=1}^n E[x_i^2] + \sum_{i \neq j} E[x_i x_j]$$

255. Power Factor

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the prime factorization of 10^{99} . How many factors does it have? How many of them are divisible by 10^{85} ?

256. Training Sessions

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Number the employees as 1 through 5. Any scheduling of these employees can be seen as a permutation of 111222333444555.

257. Same Cards

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What is the total number of ways to distribute 12 cards among 4 players? The first player can receive any 3 cards from the 12 cards. The second player can receive any 3 cards from the remaining 9 cards, and so on.

Hint 2: The total number of ways to distribute 12 cards among 4 players is given by:

$$\binom{12}{3} \cdot \binom{9}{3} \cdot \binom{6}{3} \cdot \binom{3}{3}$$

Hint 3: How many ways can you distribute the cards such that each player receives exactly 1 king, 1 queen, and 1 jack? The first player can receive any of the 4 kings. The second player can receive any of the remaining 3 kings, and so on.

258. Ant On A Cube II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Markov chains.

259. 2 Or 11

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let E be the expected number of rolls required. Let E_1 be the expected number of rolls required after you roll a 1. Can you generate 2 equations in these variables?

260. Geometric Progression

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are $\binom{100}{3}$ ways to choose 3 distinct integers from $\{1, 2, \dots, 100\}$. How many of these selections form a geometric sequence?

Hint 2: Label the three chosen numbers, in increasing order, as $a < b < c$. They form a geometric progression exactly if $b^2 = ac$. Equivalently, there exists some reduced fraction $r = \frac{m}{n} > 1$ so that $(a, b, c) = (a, ar, ar^2) = (a, a\frac{m}{n}, a\frac{m^2}{n^2})$. Since $b = a\frac{m}{n}$ and $c = a\frac{m^2}{n^2}$ must all be integers ≤ 100 , it follows that a is a multiple of n^2 . Write $a = kn^2$. Then $b = kmn$, $c = km^2$, all in $\{1, \dots, 100\}$. The strongest condition is $km^2 \leq 100$. Hence for each coprime pair (m, n) with $m > n$, k runs from 1 to $\lfloor \frac{100}{m^2} \rfloor$. Every (kn^2, kmn, km^2) is then a valid geometric triple.

261. Sensor Signal

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We analyze the scenario using Poisson thinning. The signals generated by each sensor follow independent Poisson processes. The adjusted rates for these processes are $6 \cdot \frac{1}{6} = 1$ and $10 \cdot \frac{1}{5} = 2$.

262. Uniform Random Variables II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since the random variables are IID, they are exchangeable.

263. Russian Roulette I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

264. Drunk Passenger II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve Drunk Passenger I first.

Hint 2: Use indicator variables and linearity of expectation.

265. Binary Cards Removal

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: On the first turn, Bob should remove the odd numbers. That way, the minimum difference between two numbers left is at least 2.

266. Early 5s

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

267. Bacteria Population

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Given that the probability of extinction for a colony with a single cell is $\frac{1}{3}$, the extinction probability for a colony with three cells can be determined by considering the extinction of each independent lineage.

268. Cube Fitting I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In each direction, only 6 cubes with dimensions $3 \times 3 \times 3$ can fit.

269. Nine Children

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the ages of the nine children be $(a - 4d), (a - 3d), \dots, (a + 3d), (a + 4d)$, and that of the father be m , such that $a > 4d$ and $m > a + 4d$ for $a, d, m \in \mathbb{N}$. According to the condition:

$$(a - 4d)^2 + (a - 3d)^2 + \dots + (a + 4d)^2 = m^2$$

Now, expand the LHS.

270. Unit Digit Probability II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Only the units digit is relevant, so we can consider the first set as $\{1, 3, 5, 7, 9\}$ instead.

271. Grid Paint I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the pigeonhole principle.

Hint 2: The blue squares act as pigeons and rows act as holes.

Hint 3: Note that if your answer is c^* , you need to prove that if we have $c^* - 1$ columns, there exists a configuration such that no row contains at least 15 blues.

272. Dice Game V

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let E_i denote the expected value if Alice accepts a value of i or higher and re-rolls otherwise. Can you determine E_i for i ranging from 1 to 6?

273. No Rocks Allowed

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since paper doesn't defeat scissors or paper, there is no advantage in playing it. Let your probability of playing rock be r and scissors be $1 - r$. Similarly, let your opponent play paper with probability p and scissors with $1 - p$. Construct a function $E(r, p)$ to represent your payout as a function of r and p , and find the equilibrium.

274. Expected Remainder

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the binary representation of M .

275. Coin Inside Square

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think about the area within the square where the coin's center must land for the coin to remain completely inside the square.

276. Coin Toss Game IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If Bob gets first heads in n th turn, Alice must get her first heads at or after $4n$ turns. Use law of total probability.

277. Infectious Ants

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume each ant is carrying a flag. Imagine that at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem, as the flags simply go past each other. This simplification is commonly used in many collision problems.

278. Unique Ids

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: a_5 must be zero.

279. Functional Equation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We can get $f(x)$ explicitly.

Hint 2: $f(1) + f(2) + \dots + f(n) = n^3$ $f(1) + f(2) + \dots + f(n-1) = (n-1)^3$ Subtract the above 2 equations.

280. Expected Wait Time I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If you arrive at a random time, the time until the next bus arrives can be thought of as a uniformly distributed variable over the interval from 0 to 10 minutes. We aim to find the expected value of this waiting time.

281. Weighing Coins I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The optimal first move is to take 6 out of the 8 coins and place 3 coins on each side of the balance scale.

282. Coin Toss Game III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Sorry for repeating the problem, but when would you get HHH before THH?

283. Card Flipping Game

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Alice has a winning strategy if she turns down the 3rd card in her first move. Try to use symmetry and brute force to find this strategy.

284. Divisible By 5

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

Hint 2: Since only divisibility by 5 matters, consider all numbers modulo 5. Among the numbers 1 through 21, there are four numbers congruent to 2 (mod 5), four congruent to 3 (mod 5), four congruent to 4 (mod 5), and five congruent to 1 (mod 5). If Alice removes one of the numbers congruent to 1 (mod 5) on her first turn, can she use a symmetry-based strategy to ensure that the sum of the last two remaining numbers is a multiple of 5?

285. Get More Heads

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

286. To Infinity And Beyond I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Compute expectations by conditioning stepwise. The mean of a $\text{Uniform}(a, b)$ variable is $(a + b)/2$, so

$$\mathbb{E}[X_n \mid X_{n-1} = x] = \frac{x + (x + 2^{-n})}{2} = x + 2^{-(n+1)}$$

Taking unconditional expectation and using the tower property gives the recursion

$$\mathbb{E}[X_n] = \mathbb{E}[X_{n-1}] + 2^{-(n+1)}, \quad n \geq 2$$

287. Tails Removal

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The game concludes when, at some point, all remaining coins either land heads or tails after a round of flips.

288. 1 Dot Removed

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The probability of choosing a face with an even number and the probability of choosing a face with an odd number are both $\frac{1}{2}$. Removing 1 dot changes the parity.

289. Even Head Stop II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion. If the first toss results in tails, the problem becomes finding the probability that the total number of tosses is odd, which is the complement of the original problem.

290. Normal Inequality I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the radial symmetry of the joint distribution of X and Y .

291. Contained Projection

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Denote by X_{AB} the indicator random variable that is 1 if the projection of D onto the line AB falls between A and B , and 0 otherwise. The total number of "good" sides is

$$X = X_{AB} + X_{BC} + X_{CA}$$

so by linearity of expectation,

$$E[X] = E[X_{AB}] + E[X_{BC}] + E[X_{CA}]$$

Hint 2: By symmetry, each side of triangle ABC has the same probability of admitting an interior projection of D . Thus,

$$E[X] = 3 \cdot P(X_{AB} = 1)$$

292. Boat And Steamer

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the river's flow speed be v . Use the relationship between speed, time, and distance for both cases to express the river's length L in terms of v , then solve for v by equating the two expressions for L .

293. King, Queen, Ace I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve the "First Ace" problem first.

Hint 2: Separate the problem into two stages. In the first stage, we draw cards until we see the first card that is either a king, a queen, or an ace. The first relevant card encountered may be a king, queen, or ace. With probability $\frac{1}{3}$, we see a king first, in which case the process stops immediately. With the remaining probability $\frac{2}{3}$, the first divider card is either a queen or an ace, and we must keep drawing until we encounter the other of the two or a king. Use the law of total expectation.

294. Coin Toss Game VII

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion to get the expected value of the game.

295. Interview II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "interview I" first.

Hint 2: Let $\alpha = k/n$. When n is very large, the Riemann sum

$$\frac{1}{n} \sum_{i=k+1}^n \frac{k}{i-1}$$

converges to the integral

$$\int_{\alpha}^1 \frac{\alpha}{x} dx = \alpha [\ln x]_{\alpha}^1 = -\alpha \ln \alpha.$$

Thus, in the limit $n \rightarrow \infty$, the success probability as a function of the skip-fraction α is

$$f(\alpha) = -\alpha \ln \alpha.$$

296. Plus One, Minus One

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Apply the method of proof by coloring, along with the invariance principle.

297. Largest Sum Subset

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: A valid set T can be constructed by selecting all prime numbers from the set S and including the number 1. Can we find a better combination?

298. Careful Soldiers

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Examine the two soldiers who are closest to each other. They are clearly watching one another. If any other soldier observes one of these two, it results in one soldier being watched twice, meaning there must be at least one soldier who is not being observed. Now consider the alternative scenario—if no other soldier watches these two.

299. Dice Difference III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let e_i be the expected number of rolls required to obtain two consecutive rolls with a difference no greater than 1 after rolling the value i .

Hint 2: Note that $e_1 = e_6$, $e_2 = e_5$, and $e_3 = e_4$.

300. Two Coins Remaining I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the invariance principle.

301. Conditional Expected Rolls

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let N be the number of rolls until you stop and B be the condition that the first 6 occurs before the first 5. The probability that you never roll a 5 or 6 for $n - 1$ turns and then roll a 6 on the n -th turn is $\left(\frac{4}{6}\right)^{n-1} \cdot \frac{1}{6}$. So, the expected number of rolls until we obtain the first 6 and never roll a 5 before stopping is:

$$\sum_{n=1}^{\infty} n \cdot \left(\frac{4}{6}\right)^{n-1} \cdot \frac{1}{6}$$

Let's denote this expectation by $E(N = n \cap B)$. The required expectation is $E(N|B) = \frac{E(N = n \cap B)}{P(B)}$.

302. Maximising Heads II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: After the entire process, what is the probability that a given coin shows tails?

303. Green Ball Draw

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics to calculate the number of arrangements with the 4th green ball fixed at the 9th position.

304. Same Values On Dice[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For the initial rolls, there can be three distinct values, two distinct values, or all the same value. Calculate the probability that the second set of rolls matches the initial set for each of these cases, then use the law of total probability.

305. Meeting Probability[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use geometric probability.

306. Knight Return[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Kac's lemma.

307. Rolling Everything II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: Use the number of distinct values seen till now as states. Call E_k the expected number of additional moves needed to see all six faces, given that we have already seen k distinct faces.

308. No Empty Bucket[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Suppose the first head occurs on the k -th toss. Then, each of the k marbles independently lands in any of the three buckets with probability $\frac{1}{3}$, and by inclusion-exclusion the probability that all three buckets are nonempty equals

$$1 - 3 \left(\frac{2}{3}\right)^k + 3 \left(\frac{1}{3}\right)^k.$$

Hint 2: Use the law of total probability by conditioning on k . This results in three terms, each of which is a geometric series.

309. Painted Cube II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Using conditional probability, the answer is the ratio of these two numbers.

310. Random Point On Disc

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For any point (p, q) in the disc, $(-p, q)$ has the same probability density.

311. 14 Marbles

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If a marble of a particular color is placed at some position, what are the possible positions for the other marble of the same color?

312. Heaven Or Hell

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion and symmetry.

313. Largest Area Expectation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $P(i)$ represent the probability that the largest area that is entirely painted red is i . There are 4 possible values of i : 0, 1, 4, and 9. Our answer is $\sum_{i \in \{0,1,4,9\}} P(i)i$

Hint 2: Finding $P(0)$ and $P(9)$ is trivial. Find $P(4)$ using the principle of inclusion and exclusion. Then compute $P(1) = 1 - [P(0) + P(4) + P(9)]$.

314. No Fives

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The probability in question is equivalent to the probability of rolling a 6 before a 5 when rolling a die infinitely many times. What is the complement of this event?

Hint 2: The complement of the above event is rolling a 5 before a 6. By symmetry, both of them are equal.

315. Maximum Rolled Value

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To determine the probability that the maximum value is exactly 4, subtract the probability that the maximum is less than or equal to 3 from the probability that it is less than or equal to 4.

316. Random Knight Movement

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Look at the knight's horizontal motion and vertical motion separately.

Hint 2: Let a_i be the change in the knight's x -coordinate on the i -th move. Then the final x -coordinate is $a_1 + a_2 + \cdots + a_{100}$. Square this expression, expand it, and notice that the cross terms like $a_i a_j$ have expectation $\mathbb{E}[a_i]\mathbb{E}[a_j]$ (since a_i and a_j are independent), which is 0 because $\mathbb{E}[a_i] = 0$.

317. Game On 1-10

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Among all the prime numbers on the board, which one is most advantageous to pick first?

318. Easy Card Expectation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

319. Bot Move

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: It is always better to choose the number $n + 3$ rather than n as it defeats all the numbers that n would have defeated and the number $n + 2$.

320. Rigged Game

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the probability of heads be p . Calculate the expected number of tosses to get 2 consecutive heads in terms of p . Then equate the expression to 30 as this would mean that the player loses on average.

321. Maximising Heads I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Observe that flipping exactly two coins at a time always preserves the parity of the number of heads. That is, the number of heads stays even or odd depending on what it was initially.

Hint 2: So whatever the parity of the number of heads after the initial toss, we can only reach configurations with the same parity. But since we can perform the operation any number of times, we can reach any number of heads with the same parity as the starting number. That means: if we start with 1 or 3 heads (odd), we can reach 3 heads (maximum possible odd value ≤ 4), and if we start with 0, 2, or 4 heads (even), we can reach 4 heads.

322. Distinct Number Count

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

323. Non-linearity Of Expectation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let I_{ij} be the indicator that the ball placed on move i goes into box j . If X_j is the number of balls in box j after move 6, then

$$X_j = \sum_{i=1}^6 I_{ij}, \quad S = \sum_{j=1}^6 X_j^2.$$

Use linearity of expectation and independence to calculate the expectation.

Hint 2: Indicators from different moves are independent. So, using linearity of expectation, for fixed j ,

$$\mathbb{E}[X_j^2] = \mathbb{E}\left[\left(\sum_{i=1}^6 I_{ij}\right)^2\right] = \sum_{i=1}^6 \mathbb{E}[I_{ij}] + 2 \sum_{1 \leq a < b \leq 6} \mathbb{E}[I_{aj}]\mathbb{E}[I_{bj}].$$

324. Noodle Loop

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

325. Dice Sum Parity

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: If the first roll is a 6, we immediately stop, and the sum is guaranteed to be even with a probability of 1.

Hint 3: If we roll a 2 or 4 on the first roll, the problem remains unchanged (since the rolled value is even). Therefore, if the probability of the sum being even at the start is p , it will remain p .

Hint 4: If we roll a 1, 3, or 5 on the first roll, the problem becomes complementary (since the rolled value is odd). From this point onward, we need the sum to remain odd until a 6 is rolled. Therefore, if the probability of the sum being even at the start is p , it will change to $1 - p$.

326. Nearest Diagonal I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the tail integral formula.

327. No Adjacent Standing Pair

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are solutions involving casework, but using recursion leads to easier solutions. Note that there are $2^8 = 256$ cases in total, so it suffices to find the number of satisfying cases. Define h_n as the number of linear sequences of h 's and t 's of length n where no h 's are adjacent and the sequence ends with an h . Similarly, define t_n as the number of linear sequences of h 's and t 's of length n where no h 's are adjacent and the sequence ends with a t (ignore the fact that they are in a circle for now).

328. Odd Heads III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: Let p_n be the probability that we have an odd number of heads after n tosses. In terms of p_n , what is the probability that we have an odd number of heads after $n + 1$ tosses?

329. Handshakes On A Chessboard

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation. Focus on the edges that split squares rather than the squares themselves.

330. Erase

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: You cross out the 1234 from the first set, because you want it to start with a 5 to be as large as possible.

331. Permutation Blocks

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let σ be a permutation on $\{1, \dots, n\}$. Call $m \in \{1, \dots, n\}$ a breakpoint of σ if $\{\sigma(1), \dots, \sigma(m)\} = \{1, \dots, m\}$. Notice that the maximum partition is into k blocks, where k is the number of breakpoints: if our breakpoints are $m_1, \dots, m_k$, then we take $\{1, \dots, m_1\}, \{m_1 + 1, \dots, m_2\}, \dots, \{m_{k-1} + 1, \dots, m_k\}$ as our contiguous blocks.

332. Count Paths III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the reflection principle.

333. Tigers And The Sheep

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider cases with fewer tigers and 1 sheep, such as 1 tiger and 1 sheep, 2 tigers and 1 sheep, etc., to identify the pattern.

334. Unique Birthdays Expectation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation by breaking the number of unique birthdays into indicator variables representing whether the i -th person has a unique birthday different from others.

335. Simpson's Paradox

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: It is possible that team Y ends up with a higher pass-completion percentage.

336. 2014 Stones

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the binary representation of the current number of stones.

337. Chess Tournament IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve chess tournament I first.

338. Interesting Summation I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For $\frac{1}{k}$ to have a terminating decimal, k must be of the form $2^a \cdot 5^b$, where a and b are non-negative integers.

339. Chess Tournament I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In the first round, there are 128 positions. Assume that in the first round, the player at position 1 plays the player at position 2, the player at position 3 plays the player at position 4, and so on. In the second round, the winner of the match between positions 1 and 2 will play the winner of the match between positions 3 and 4 - and so on for subsequent rounds.

Without loss of generality, we can assume that the highest-rated player is occupying position 1. Which of the remaining 127 positions will lead the second-highest-rated player to play the highest-rated player in the

final?

340. How Many 5s

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let n be the roll number on which you first get a 6, and let t be the total number of 5s rolled before that. Using the Law of Total Expectation, we have:

$$E[t] = E[E[t|n]]$$

341. Sphere Variance

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

342. B Before O I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To find that, place B's and O's first and then arrange the remaining letters.

343. Count Regions II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve Count Regions I first.

Hint 2: n circles will divide the plane into a maximum number of pieces if every two of them intersect (that is, if no two of them are tangent and none of them lies entirely within or outside another), and no three of them are concurrent.

344. Square Fitting I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Find the cumulative distribution function (CDF) of the side length, and then differentiate it to obtain the probability density function (PDF).

Hint 2: Once you have the PDF, use LOTUS to find the expected area.

345. Ant On An Icosahedron

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let A be the starting vertex. From A , there are 5 vertices that are distance 1 away, 5 vertices that are distance 2 away, and 1 vertex that is distance 3 away. (By “distance,” we mean the fewest number of edges on a path.) After one step of the random walk, the ant will be distance 1 away from A with probability 1. After two steps, the ant will be distance 0 away (from A) with probability $\frac{1}{5}$, will be distance 1 away with probability $\frac{2}{5}$, and will be distance 2 away with probability $\frac{2}{5}$. After three steps, the ant will be distance 0 away with probability $(\frac{2}{5})(\frac{1}{5}) = \frac{2}{25}$, will be distance 1 away with probability $(\frac{1}{5})(1) + (\frac{2}{5})(\frac{2}{5}) + (\frac{2}{5})(\frac{2}{5}) = \frac{13}{25}$, will be distance 2 away with probability $(\frac{2}{5})(\frac{2}{5}) + (\frac{2}{5})(\frac{2}{5}) = \frac{8}{25}$, and will be distance 3 away with probability $(\frac{2}{5})(\frac{1}{5}) = \frac{2}{25}$.

346. Count 9s

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Count how many times the digit 9 appears in the interval $[1, 99]$. Then, use this result to determine the frequency of 9 in the range $[1, 1000]$.

347. Grid Travel I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Observe that Johnny needs to pass through the points $(0, 0)$, $(1, 0)$, $(2, 0)$, ..., $(5, 0)$ in that order, and he needs to pass through $(0, 1)$, $(1, 1)$, $(2, 1)$, ..., $(5, 1)$ in that order, or else he will intersect his own path.

348. Jumping Frog III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

349. Seven Colors I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the law of total probability.

Hint 2: Let C be the event that among the first eight positions there are exactly six more red balls than yellow balls. Under C only two patterns for the first eight are possible: either there are six red and zero yellow among the first eight (so the other two of the first eight come from the 100 balls that are neither red nor yellow), or there are seven red and one yellow among the first eight (so none of the first eight are from those 100).

350. Triangle Angle I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: $\tan(\theta) = \frac{B}{A}$. If $\theta > \frac{\pi}{4}$, then $\tan(\theta) > 1$, implying $\frac{B}{A} > 1$, or $A > B$.

351. Ant On A Cube III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: There are 8 states corresponding to the 8 vertices. However, many of these states are symmetric.

352. Odd Sum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To make the sum of the 4 integers odd, we will choose either 3 odd and 1 even (case 1) or 1 odd and 3 even (case 2). Use combinatorics to find the number of ways for each case.

353. Coin Equalizer

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We need to count the number of sequences of 5 heads and 5 tails such that they are never equal in number before the 10th flip. This can be done using the reflection principle.

354. Combining Turtles

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Fix a particular turtle. The probability it does not get combined with another turtle when there are n total turtles is $1 - \frac{2}{n}$.

355. Roll Die Till 65

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

356. Assigning Numbers To Dice

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Place the number 1 on a side of the dice and consider it the top. Now, find the number of ways of placing the other numbers.

357. Diagonal Cut I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The width and height have no common factors. What does that mean?

358. Meet Again

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Analyze the relative motion of the hands. The minute hand moves at 360 degrees per hour, while the hour hand moves at 30 degrees per hour.

359. Non-empty Buckets

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

360. Maximize Product

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To maximize the product, arrange the coins into stacks of nearly equal size. This approach ensures that the product of the number of coins in each stack is as large as possible.

361. End With 2018

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The probability that the last 4 digits of any sequence is 2018 is

$$\frac{1}{4^4} = \frac{1}{256}$$

The probability we finish after appending n digits is the probability that the last 4 digits are 2018, and we did not finish before n digits. However, if our last digits are 2018, that implies we did not finish with 2018 after $n - 1$, $n - 2$, or $n - 3$ digits.

362. Balls In Garbage Bag

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the expected number of balls with a particular number put in the garbage. This is equal to the expected total number of buckets which never receive a ball with THAT number.

363. Coin Coloring Game

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: They would always have an equal probability of winning by symmetry.

364. Sufficient Sum II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "summation series I" first.

Hint 2: Use tail sum formula.

365. Asking For A Date

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

366. Digit Picking

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If Bob gets a 9, his number will be larger. If both get the numbers from one to eight, they have equal chance of getting a bigger number due to symmetry.

367. Equal Heads

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Calculate the probability distribution for the numbers of heads in four coin flips and use this to find the probability that both people get the same number of heads.

368. Car Crash

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The number of car accidents in separate time intervals are independent due to their constant rate and the independence of each incident.

369. Interesting Average II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For any subset A , we can define the "reflected subset" $A^0 = \{i \mid 2009 - i \in A\}$. Then $m(A) = 2009 - m(A^0)$.

370. Dice Permutation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

371. Is It A Perfect Square?

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The number will be divisible by 3.

372. Die Roll LCM

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Observe that the LCM of the numbers 1 through 10 is 2520, and no smaller LCM of a subset of these faces exceeds 2000. Therefore the stopping condition "LCM $\leq$ 2000" is equivalent to "the running LCM of our rolls has reached exactly 2520." To reach 2520, our sequence of rolls must collectively supply the prime factors 2^3 , 3^2 , 5^1 , and 7^1 . Among the faces 1–10, the only rolls that contribute 7^1 are rolls of 7; those that contribute 3^2 are rolls of 9; those that contribute 2^3 are rolls of 8; and for the factor 5^1 we may roll either 5 or 10. Thus we have four "coupon-types" to collect:

- A "7" (probability $\frac{1}{10}$)
- An "8" (probability $\frac{1}{10}$)
- A "9" (probability $\frac{1}{10}$)
- A "5-or-10" (probability $\frac{2}{10}$)

Hint 2: Use principle of inclusion and exclusion for expected values. If $p_1, \dots, p_4$ are the probabilities of the four types, then the expected number of rolls is

$$E = \sum_{i=1}^4 \frac{1}{p_i} - \sum_{1 \leq i < j \leq 4} \frac{1}{p_i + p_j} + \sum_{1 \leq i < j < k \leq 4} \frac{1}{p_i + p_j + p_k} - \frac{1}{p_1 + p_2 + p_3 + p_4}$$

373. Higher Roll II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve higher roll I first.

374. N Point Variance

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Variance is given by:

$$\text{Var}(X) = E(X^2) - E(X)^2$$

Hint 2: If X is an indicator variable, $X^2 = X$. Hence,

$$\text{Var}(X) = E(X) - E(X)^2$$

375. Game Of Cuts

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What happens if the starting rectangular grid is actually a square, with the same length and breadth?

376. Next Date I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To find the earliest possible date with distinct digits in the format MM/DD/YYYY, start by minimizing the year, as it has the greatest impact on the date. Note that the first digit of the month (MM) can be either 0 or 1. If it is 0, the day (DD) must be 01, 02, or 31 (30 is not possible as it would repeat the 0). If the first digit of the month is 1, the second digit of the month must be 0 or 2, and the day must include a 0.

377. Dice Game I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Determine the expected value of a single roll and decide when it's better to roll again versus keeping the first roll.

378. Poisoned Wine II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think in terms of base 4 expansion.

379. Poisson Cards

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The required expectation is $E\left[\binom{X}{4}\right]$, where X follows a Poisson distribution with mean 6. Using the law of unconscious statistician, this is equal to:

$$\sum_{n=0}^{\infty} \binom{n}{4} \cdot P(X = n)$$

380. Min Collinearity

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use pigeonhole principle.

Hint 2: If there are 11 points, some row must contain 3 points, which would definitely be collinear. So, 11 is an upper bound for the answer.

381. Losing It All

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To meet the problem's criteria, Alice needs to have played precisely 11 games and secured a win in one of the first 3 rounds.

382. Handy Subsets[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the elements that are included in the set red, and the elements that are not included blue. We will identify each subset using this coloring. Imagine three parts of the set T : $\{1, 2, \dots, 5\}$, $\{6, 7, \dots, 10\}$, $\{11, 12, \dots, 15\}$. If each of these 3 parts is individually monochromatic (consists of the same color), then the resulting subset will be handy. There are 2^3 such subsets. For example, consider $\{1, 2, \dots, 5\}$ and $\{11, 12, \dots, 15\}$ are red and $\{6, 7, \dots, 10\}$ is blue. Then the resulting subset is $\{1, 2, \dots, 5, 11, 12, \dots, 15\}$ which is handy. What about the case when the 3 parts are not individually monochromatic?

383. Last To Front[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: Let the last person in the original line be person 5. For $n = 1, 2, 3, 4, 5$, define E_n as the expected number of rounds needed for person 5 to reach the front when they are currently in position n from the front. Get a recurrence relation for E_n .

384. Coin Toss Movement[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For 6 to 8 heads, we are guaranteed to hit 4. For less than or equal to 3 heads, we will never hit 4. What about the cases where we have 4 or 5 heads?

385. 1s And 9s[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

386. Low Median[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If the median is less than $\frac{2}{3}$, then either exactly one number must be greater than $\frac{2}{3}$ or all three numbers must be less than $\frac{2}{3}$.

387. Connected Components I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation and indicator variables.

Hint 2: Index the squares from 1 to 15 and let X_i be the indicator random variable for the event "a new connected component begins at square i ."

388. 3 Heads 3 Tails II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The probability Alice wins is $\frac{1}{2}$ due to symmetry. Additionally, the expected number of tosses for Alice to win is the same as for Bob, thanks to symmetry. Using the law of total expectation, the answer is simply twice the expected number of tosses for Alice to win.

389. Keep The GCD High[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Note that 997 is prime. By removing the card (1, 997), a player is going to lose if and only if he removes a card that does not have a multiple of 997.

390. Moth On A Cube[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The moth can reach vertex B in exactly 3 steps or in exactly 5 steps.

391. Expected Arc Length I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Fix point X_1 at angle 0. Then the other 29 points are independently and uniformly distributed on the circle. We are interested in the event that no point lies within an arc distance x on either side of X_1 .

Hint 2: For a point not to lie within a distance x on either side of X_1 , it must fall in the complementary region. Since the total circle has circumference 2π and the excluded arc is $2x$ (namely, x on each side), the favorable region has length $2\pi - 2x$. Therefore, for a single point, the probability of falling outside the excluded interval is

$$\frac{2\pi - 2x}{2\pi} = 1 - \frac{x}{\pi}$$

For all 29 points, the probability that each one lies outside the forbidden region is

$$\left(1 - \frac{x}{\pi}\right)^{29}$$

392. Dice Difference II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let e_i be the expected number of rolls required to obtain two consecutive rolls with a difference no greater than 1 after rolling the value i .

Hint 2: Note that $e_1 = e_6$, $e_2 = e_5$, and $e_3 = e_4$.

393. Game On 2 Piles[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What happens if there are equal stones in both piles?

394. 5-runs

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What if overlaps were allowed?

Hint 2: If overlaps were allowed, you could simply use indicator variables for the squares 1 to 16, where the i th indicator would be 1 if the squares from the i th position to the $(i+4)$ th position are all black. Try to count 5-runs using the overlapping runs of different sizes.

Hint 3: Let R_k denote the number of k -runs. This is easy to count. Now, the following is true:

$$\text{5-runs} = (R_5 - R_6) + (R_{10} - R_{11}) + (R_{15} - R_{16}) + R_{20}$$

395. Dice Difference I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let E_i be the expected number of rolls required after rolling an i . We can generate six equations in six variables easily. Use symmetry to reduce the number of variables.

396. 7994 Stones

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Notice that the multiples of 8 are losing positions.

397. Measuring Everything II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Every natural number can be represented as the difference of two numbers whose base 3 representations contain only 0's and 1's.

398. Ending At 1

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let p_n be the probability that the last number she writes down is 1 when the first number she writes down is n . Then use the law of total probability.

399. Points On A Square

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Fix one edge for the first point to lie on. If the second point lies on the opposite edge, the distance would be greater than one. If the second point lies on the same edge, the distance would be less than one. What if the other point lies on an adjacent edge?

400. Standing Table III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "standing table I", "standing table II" and "same arc I" first.

Hint 2: This problem is identical to "standing table II."

401. Hungry Driver

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Instead of one driver taking multiple trips, let's imagine that we have multiple drivers making one trip.

402. Sufficient Sum III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "summation series i" and "summation series ii" first.

Hint 2: From the result in "summation series ii", we have the mean of N_x over the interval $0 \leq x \leq 1$. What remains is to determine $E[N_x^2]$.

Hint 3: Write N_x in terms of its tail indicators: define $I_k = 1\{N_x > k\}$, so that

$$N_x = \sum_{k \geq 0} I_k$$

Then

$$N_x^2 = \left(\sum_{k \geq 0} I_k \right)^2 = \sum_{k \geq 0} I_k + 2 \sum_{0 \leq j < k} I_j I_k$$

But whenever $j < k$, the event $\{N_x > k\}$ implies $\{N_x > j\}$, so $I_j I_k = I_k$. Hence

$$N_x^2 = \sum_{k \geq 0} I_k + 2 \sum_{k \geq 0} k I_k = \sum_{k \geq 0} (2k + 1) I_k$$

and taking expectation gives

$$E[N_x^2] = \sum_{k=0}^{\infty} (2k + 1) P(N_x > k)$$

403. Conditional One

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Knowing the draws are in increasing order is equivalent to first choosing a 5-element subset of $\{1, \dots, 100\}$ uniformly at random and then listing its elements in increasing order. Under this conditioning every 5-subset is equally likely.

404. Average Drawn

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are $\binom{20}{3}$ possible ways to choose any three balls. How many of these selections correspond to a choice where one ball's number is the average of the other two?

Hint 2: Pick any two distinct drawn numbers a and c with $a < c$. For their average $\frac{a+c}{2}$ to be an integer - and so possibly equal to the third drawn number - a and c must have the same parity, both even or both odd. If they do, the average is an integer strictly between a and c , so the three numbers $\{a, \frac{a+c}{2}, c\}$ form a valid triple with one number equal to the average of the other two. So, how many such triplets are possible?

405. Too Many Hats

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "prisoner hats-multicolor" first.

406. Watermelon Weight

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the amount of non-water content in the watermelon, which remains constant before and after drying. Use this to find the new total weight.

407. Intersecting Subsets

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Introduce for each element $i \in S$ the indicator

$$X_i = \begin{cases} 1, & \text{if } i \in A \cap B, \\ 0, & \text{otherwise.} \end{cases}$$

Hint 2: Use linearity of expectation to calculate $\mathbb{E}[N]$.

Hint 3: Each X_i is a Bernoulli random variable, so $\text{Var}(X_i) = p(1-p)$, where p is the Bernoulli parameter representing success. Also, the X_i are independent. So,

$$\text{Var}(N) = \sum_{i=1}^{25} \text{Var}(X_i).$$

408. Bernoulli Matrix

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: A and A^T have the same diagonal elements. Hence, $A - A^T$ will have 0 as their diagonal elements.

409. B Before O II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Place B's and O's first. There are 3 B's and 4 O's. Then arrange the remaining letters.

410. Bacteria Population III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think in terms of the binary representation of the target number of bacteria.

411. Sorted Rows

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: 23 cannot be in columns 1 or 2. Calculate the probabilities that 23 is in columns 3, 4, and 5. Then, use $E(x) = \sum x \cdot p(x)$ to find the expected value.

412. Coin Reflip Game

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are 4 normal flips. Imagine that there is a fifth flip conditioned on there being a tail in the first 4 flips.

413. Large Language Model

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since there are 69 people, we need 69 sets of languages. They must be pairwise distinct, with each set having the same size. We need to achieve this using the minimum number of languages.

Hint 2: The problem can be rephrased as follows: What is the minimum n such that the maximum value of $\binom{n}{r}$ is at least 69?

414. Strange Die

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Note that there is symmetry: the probability of rolling a 9 is the same for both two rolls of the first die and two rolls of the second die.

415. Cannons

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider that there are $\binom{n}{2}$ pairs of pirates, and each pirate points their cannons at 2 distinct pirates. The condition $\binom{n}{2} \geq 2n$ must be satisfied.

416. Field Construction

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The expected area is $E[(300 + X)(200 + Y)] = E[60000 + 300Y + 200X + XY]$. Use the linearity of expectation.

417. Chocolate Bar

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What happens to the number of total pieces after every cut?

418. Unfair Rock Paper Scissors

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Markov chain with the difference in score being the state.

419. Sum Over One I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If $X \sim \text{Unif}(0, 1)$, then $1 - X \sim \text{Unif}(0, 1)$ as well. What if we adjust the problem to use the distribution $1 - X_i$?

420. Exactly 2 Consecutive Heads

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are three scenarios to account for: $HHT-$, $THHT$, and $-THH$, where the dashes $(-)$ can be either H or T . Calculate the probability for each case in terms of p , then find the value of p that maximizes this using calculus.

421. Reach Multiple Of 7

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The probability that B wins on their first turn is $\frac{1}{6}$, as only one of the six possible outcomes will make the total score a multiple of 7. For example, if A rolls a 1 on their first turn, B must roll a 6 on their first turn to win, which has a probability of $\frac{1}{6}$. Similarly, if A rolls a 4 on their first turn, B must roll a 3 on their first turn to win, with the probability again being $\frac{1}{6}$. This pattern holds for all subsequent turns—only one of the six outcomes will make the total score a multiple of 7. Therefore, on the first turn, A cannot win.

For every other turn, the probability of winning in the current turn is $\frac{1}{6}$ for the player whose turn it is.

Hint 2: Use recursion. Let p represent the total probability of the current player winning. B wins on their first turn with probability $\frac{1}{6}$. If B doesn't win initially ($\frac{5}{6}$), A must also fail to win on their turn ($\frac{5}{6}$), returning the dice to B. The probability of B winning is again p .

422. Painted Cube I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There exists a solution using casework. However, a clever approach is to use linearity of expectation. What is the probability that a given face is red?

423. Nearest Circular Neighbor

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Fix the location of P_1 . Let L represent the arc length (in radians) between P_1 and the closest neighboring point. To calculate $E[L]$, we integrate as follows:

$$E[L] = \int_0^\pi P[L \geq x] dx$$

424. Archery I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

425. Same Sum Subsets

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The two subsets being disjoint is redundant. We can simply throw away the common elements of the two sets with equal sums to get disjoint sets with equal sums. Now use the pigeonhole principle.

426. Pistol Shots

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think in terms of intervals between bullets rather than the bullets.

Hint 2: There are 5 intervals between the 1st and 6th shots (1-2, 2-3, 3-4, 4-5, 5-6). There are 2 intervals between the 1st and 3rd bullets (1-2, 2-3).

427. Buying Tickets

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let's solve for $2n$ people. We assign $+1$ to individuals carrying \$5 bills and -1 to those carrying \$10 bills. Let (a, b) denote the state of the walk after a steps, ending at position b . This can be viewed as a random walk, beginning at $(0, 0)$ and concluding at $(2n, 0)$ after $2n$ steps. Out of these $2n$ steps, we must choose n steps as $+1$, resulting in $\binom{2n}{n}$ possible paths. We are concerned with paths where $b \geq 0$ and $0 < a < 2n$.

428. First Ace

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the method of random points on a segment. If you remove the aces, there are 48 cards in total. The four aces would divide these 48 cards into 5 segments.

429. Inflating Balloon

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Calculate the combined rate of inflation (fraction of balloon inflated per unit time) from both pumps and subtract the rate of air leakage. The time to fully inflate the balloon is the reciprocal of the net rate.

430. Expected Squared Area

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Using shoelace formula, the area can be written as,

$$A = \frac{1}{2}(x_1y_2 + x_2y_3 + x_3y_1 - (x_2y_1 + x_3y_2 + x_1y_3)).$$

Hint 2: Squaring gives

$$A^2 = \frac{1}{4}(x_1^2y_2^2 + x_2^2y_3^2 + x_3^2y_1^2 + x_2^2y_1^2 + x_3^2y_2^2 + x_1^2y_3^2 + \text{cross terms}).$$

Each cross term—such as $x_1x_2y_2y_3$ or $x_1^2y_1y_3$ —factors into at least one factor raised to the first power. To ensure these cross-term expectations vanish, center the grid coordinates from -4 to $+4$ so that $E[x_i] = E[y_i] = 0$. By independence each cross-term expectation factors into a product of individual expectations, and since one factor is zero, the term vanishes.

431. Singer's Location

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use Bayes' theorem to find the probability that he came from Philadelphia given that he performed (event E). Calculate the probability of E occurring from each location, then apply Bayes' formula.

432. Probability Of Everything

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Count the number of roll sequences that do not contain all faces instead. Use principle of inclusion and exclusion.

433. Party Groups II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve party groups I first.

434. Heads Run II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve Heads Run I first.

Hint 2: This problem can be reduced to the Heads Run I problem and therefore has the same answer.

435. Largest 2 Factor

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If p is the probability that $v(a) = v(b)$, then the answer is $\frac{1-p}{2}$, by symmetry.

436. Trailing Zeros II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Rewrite the expression in terms of factorials.

437. Sum Of Products

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Group together numbers with the same number of digits to compute this sum.

438. Ants On An Octahedron

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $ACEDFB$ be the octahedron such that points $CDEF$ are coplanar, C/D are opposite each other, and E/F are opposite each other. The set of points that the ants from A/B can travel to is $\{C, D, E, F\}$, the set of points that the ants from C/D can travel to is $\{A, B, E, F\}$, and the set of points that the ants from E/F can travel to is $\{A, B, C, D\}$. So the problem is analogous to finding the number of ways to take 2 elements from each of these sets to make the set $\{A, B, C, D, E, F\}$.

439. Broken Stick II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let E denote the event that each of the three pieces has length at least $\frac{1}{8}$ meters. Let the first random point be x and the second random point be y . The required probability is $P(E) = P(y > x)P(E|y > x) + P(y < x)P(E|y < x)$. Clearly, $P(y > x) = P(y < x) = \frac{1}{2}$. How do we find $P(E|y > x)$ and $P(E|y < x)$?

Hint 2: Use geometric probability to find $P(E|y > x)$ and $P(E|y < x)$.

440. Prime Sum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For the sum to be even, both selected prime numbers must be odd.

441. Double Factorial

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For any even n , we compare terms to find that $(n!!)^2 = (n)(n) \cdot (n-2)(n-2) \cdot (n-4)(n-4) \cdots (2)(2) > n(n-1) \cdot (n-2)(n-3) \cdot (n-4)(n-5) \cdots (2)(1) = n!$

442. Jumping Frog II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: You can assume the frog starts at lily pad 49 (or 51).

443. Graph Travel I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: This problem is analogous to the Coupon Collector's problem.

444. Never Divisible By 3

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since we only care about divisibility by 3, we can consider this list mod 3, i.e., 1, 2, 0, ..., 1.

445. Expected Minimum Distance

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use tail integral formula.

Hint 2: For a single point chosen uniformly in the unit disk, the probability that its distance R_i from the center is at most x equals the ratio of the area of the smaller circle of radius x to the area of the unit circle. Hence

$$P(R_i \leq x) = \frac{\pi x^2}{\pi \cdot 1^2} = x^2, \quad 0 \leq x \leq 1$$

so

$$P(R_i > x) = 1 - x^2$$

Call

$$M = \min\{R_1, R_2, R_3, R_4\}$$

Since the four points are independent,

$$P(M > x) = [P(R_i > x)]^4 = (1 - x^2)^4, \quad 0 \leq x \leq 1$$

446. 800 Marbles

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: In order to not stop after 699 marbles, the last 101 marbles must consist of 2 marbles of one color, and one marble from each other color.

447. Set Intersection

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Use linearity of expectation.

448. Remainders

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Examine $N - 1$ rather than N .

449. Exactly 6

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Using combinatorics, how many sequences can be formed such that each number appears exactly 6 times?

450. Jumping Frog VII

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Each unique path to reaching point 10 can be viewed as a subset of the set $\{1, 2, \dots, 9\}$. Each unique path corresponds to a distinct subset of these points. The goal is to count the number of such subsets that are of odd length (since the final jump to point 10 will result in an even total number of jumps).

451. Russian Roulette II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Number the chambers 1 to 6. Suppose the bullet is in the 6th chamber. At which starting positions would you win?

452. Segment Overlap

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Without loss of generality, assume P1 is the smallest point. What positions of P2 lead to overlap.

453. 4 Die Sum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The total number of outcomes is $6^4 = 1296$. How many of those sum to 20? The number of possible combinations that sum to 20 are few. Use brute force to find the possible combinations.

454. Poisoned Chocolates

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use "stealing the strategy".

455. Green And Yellow Dice

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

456. Count Handshake Pairs

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the celebrities get into one or more circles so that each circle has at least three celebrities, and each celebrity shook hands precisely with their neighbors in the circle. What are the different scenarios now?

457. 2x3 Grid

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics. What happens if you fill any two elements in the same row?

458. Forming Quadratic Polynomial III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What is the form of quadratic equations with consecutive roots?

459. 2D Jumps

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Each jump i contributes $\Delta x_i = r_i \cos \theta_i$, $\Delta y_i = r_i \sin \theta_i$, where $r_i \sim \mathcal{U}(0, 1)$ and $\theta_i \sim \mathcal{U}(0, 2\pi)$, all independent. The final displacement is $X = \sum_{i=1}^{10} \Delta x_i$, $Y = \sum_{i=1}^{10} \Delta y_i$.

Hint 2: The Euclidean distance is $\sqrt{X^2 + Y^2}$, so we seek $\mathbb{E}[(X^2 + Y^2)]$. Expanding $X^2 + Y^2$ produces diagonal terms $\mathbb{E}[r_i^2(\cos^2 \theta_i + \sin^2 \theta_i)]$ and cross-terms of the form $\mathbb{E}[\Delta x_i \Delta x_j]$, $\mathbb{E}[\Delta y_i \Delta y_j]$ for $i \neq j$. Cross terms vanish (how?). The remaining terms are easy to evaluate.

460. Digit Sum I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider each number as having six digits, ranging from 000000 to 999999, and find the average sum of digits for such numbers.

461. Conditional Inequality

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use geometric probability.

462. Yet Another Game

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We can think of the row as a sequence of 0's and 1's because we only care about parity. Use induction.

463. Same Color Corners

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use pigeonhole principle.

464. No Chords Intersecting

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Label the twelve points in clockwise order as $P_1, P_2, \dots, P_{12}$. Forming 6 chords from the 12 points is equivalent to dividing the 12 points into 6 groups, each containing 2 points. The total number of such groupings is:

$$\frac{12!}{2^6 \cdot 6!} = 10,395$$

This forms our sample space. How many of these groupings result in a configuration where none of the six

chords intersect inside the circle?

Hint 2: Use recursion to count the number of configurations where none of the six chords intersect inside the circle.

Hint 3: Define $f(n)$ to be the number of noncrossing pairings of $2n$ labeled points placed in clockwise order on a circle. We seek $f(6)$. To derive a recursion for $f(n)$, imagine the $2n$ points are $P_1, P_2, \dots, P_{2n}$. In any noncrossing matching, P_1 must be paired with some P_{2r} where $2 \leq 2r \leq 2n$. As soon as P_1 is joined to P_{2r} , the straight chord (P_1, P_{2r}) partitions the remaining $2n - 2$ points into two disjoint arcs: * the "inside" arc consisting of $P_2, P_3, \dots, P_{2r-1}$ (a total of $2r - 2$ points), and * the "outside" arc consisting of $P_{2r+1}, \dots, P_{2n}$ (a total of $2n - 2r$ points). For the entire matching to remain noncrossing, each of those arcs must themselves be matched in a noncrossing way. Consequently, once we decide to pair P_1 with P_{2r} , there are exactly $f(r - 1)$ ways to noncrossingly match the inside arc and $f(n - r)$ ways to match the outside arc. Summing over all $r = 1, 2, \dots, n$ yields the recursion:

$$f(n) = \sum_{r=1}^n f(r-1)f(n-r), \text{ with } f(0) = 1.$$

465. Drunk Passenger III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "drunk passenger I" and "drunk passenger II" first.

Hint 2: Use indicator variables and linearity of expectation.

466. Last Number Deleted

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In the first step, you retain all integers n that satisfy $n \equiv 0 \pmod{2}$. What occurs in the second step?

467. Digit Sum II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Reduce the problem to a stars and bars problem.

468. Digit Multiply II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Analyze the prime factorization of 96. Determine how to arrange these factors to achieve the smallest and largest possible numbers.

469. 500 Sided Dice

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve for 2 500-sided dice.

470. Moving The Piece II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use proof by coloring.

471. Grid Arrangement I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Of the sixteen entries in one such array, there are six 0s and ten 1s. The rows and the columns each have zero, one, two, and three 0s, in some order. Once we decide the positions of the 0s, we form one such array.

472. Information Transfer

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Establish a recurrence relation for p_i in terms of p_{i-1} .

473. Chain Link

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider breaking the chain into sections that represent sums of powers of 2, similar to how binary numbers work.

474. Pandemic I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The minimum amount of time will be taken if every infected person talks to a healthy person every minute.

475. Expected Absolute Difference

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Calculate the probability of the difference being equal to k , then use the formula $E(X) = \sum x \cdot p(x)$.

476. Game On Jar

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The strategy is of the form—pick again if you get a value less than x . Stop when you get more than or equal to x .

477. 3 Bus Stops

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume that the initial number of people on the bus is x . What is the expression for the number of people remaining after the 3 bus stops.

478. Least Multiple Of 15

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For a number to be divisible by 15, it must be divisible by both 3 and 5.

479. Password

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: This is equivalent to solving: $x_0 + x_1 + x_2 + \cdots + x_9 = 7$ where x_i represents the number of times digit i appears in the code.

480. Bridge Crossing

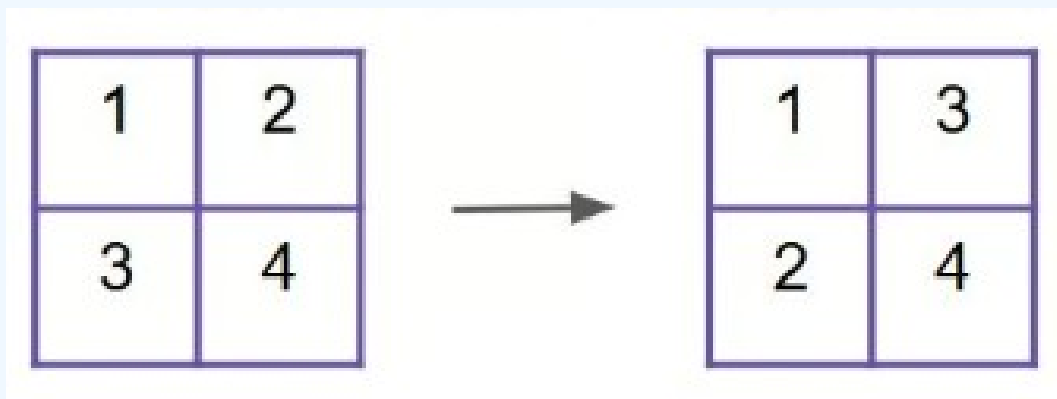
[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Alice and Bob should cross the bridge together, but not during the initial trip. If one of them crosses first, their return trip will add too much time.

481. Grid Conversion

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What if it was a 2×2 grid filled with 1, 2, 3, 4? Could we convert the left grid to the right grid?



Hint 2: Through experimentation, we observe that for a 2×2 grid, it is possible to transform the left grid into the right grid. We can achieve this by increasing all elements in the 2nd column by 1 and then decreasing all elements in the 2nd row by 1. Now extend this approach to a 3×3 grid.

Hint 3: For a 3×3 grid, we can convert the left grid to the right grid as follows: increase the 3rd column 2 times, decrease the 3rd row 2 times, increase the first row 2 times, and decrease the 1st column 2 times. Now, what about a 4×4 grid?

482. Ants On A Circle IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume each ant is carrying a flag. Imagine at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem as the flags simply go past each other. Note that this simplification is used in many collision problems.

483. Largest 3

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The sum of the three upfacs can be 18 only if all the upfacs are 6.

484. Increasing Game

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If we can guarantee that Alice chooses 191, we can guarantee that Alice will win.

485. Learning To Walk

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Suppose we create a state diagram with states $S_0, S_1, \dots, S_n$, where S_n represents the time a baby spends at the state n steps from the couch. Write the transition equations. Now, we need to find S_0 . There is a pattern that allows us to find S_0 easily.

486. Penguin Walk I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

487. Circle In 3 Quadrants

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use geometric probability.

488. Passing The Parcel

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let p_0 be the probability of winning when you have the parcel. Let p_1 be the probability of winning when one of your immediate neighbors has the parcel. Let p_2 be the probability of winning when one of the remaining two players has the parcel. Set up and solve the equations for these probabilities.

489. Hands Interchange

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The minute hand moves 6 degrees per minute, and the hour hand moves 0.5 degrees per minute.

490. Creating Zero

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use invariance principle.

491. Two Coins Remaining II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Any group of 5 consecutive heads can be reduced to a group of 2 heads using the following sequence of steps $HHHHH \rightarrow TTHH \rightarrow THT \rightarrow HH$. So, if the initial number of coins is $3k + 2$ for some integer k , we can reach 2 coins by removing 2 coins repeatedly using the above sequence of operations. If the initial number of coins is $3k$ for some integer k , we can reach 3 coins showing heads by removing 3 coins repeatedly using the above sequence of operations. Once we have 3 coins showing heads, we can remove the middle coin and end up with 2 coins.

492. Uniform Random Variables I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: By the definition of conditional probability, we need to compute:

$$P[P_1 > P_{10}, P_{20} > P_{10}] / P[P_{10} < P_{20}]$$

Since these variables are exchangeable, $P[P_{10} < P_{20}]$ is just $\frac{1}{2}$.

493. Jumping Frog VIII

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In order to find the probability of landing on 10, we must multiply the amount of successful combinations by the probability of those combinations. Notice for any successful combination of steps, the probability must always be $\frac{1}{2^{10}}$.

494. Even Or Odd

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let's say you wager x units on even and y units on odd. What is the expected outcome?

495. Additional Info Coin Flips I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the formula for conditional expectation,

$$E[N|A] = \sum_{n=1}^{\infty} n \cdot P(N = n|A)$$

Hint 2: Let X and Y be the counts of tosses until first heads for Alice and Bob respectively. We seek

$$\mathbb{E}[X \mid X < Y]$$

Since each toss is fair, the chance Alice needs exactly n tosses (fail the first $n - 1$, then succeed) is

$$\Pr(X = n) = \left(\frac{1}{2}\right)^n$$

The chance Bob's first head occurs after n tosses (i.e., he fails the first n) is

$$\Pr(Y > n) = \left(\frac{1}{2}\right)^n$$

Hence

$$\Pr(X = n; X < Y) = \left(\frac{1}{2}\right)^{2n}$$

$$\Pr(X < Y) = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{2n} = \frac{1}{3}$$

Therefore

$$\Pr(X = n \mid X < Y) = 3 \left(\frac{1}{4}\right)^n$$

496. Bulb Toggle II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If light bulb m is left on, it means that m has an odd number of odd factors.

497. Game On Grid

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Analyze the game from Bob's point of view. Consider a strategy where Bob copies Alice's moves with a specific offset.

498. Remove Neighbor

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are an odd number of people in the circle beside Sofiya and Marquis.

499. Double Value

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The probability of drawing a diamond is $\frac{1}{4}$. Given this, what is the expected value of a drawn card if it is a diamond? The probability of not drawing a diamond is $\frac{3}{4}$. Given this, what is the expected value of a drawn card if it is not a diamond? Use the law of total expectation.

500. Coin Toss Game II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If the first toss is tails, the game resets and effectively starts at the second turn, when Bob tosses the coin.

501. Apples And Oranges

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The 4 apples divide the oranges into 5 regions.

502. King, Queen, Ace II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve the "First Ace" problem first.

Hint 2: Think of the twelve cards that matter for the stopping rule, namely the four kings, the four queens and the four aces, as dividers that interrupt runs of ordinary cards. Draw until the first divider appears. The first divider that appears can be a king, a queen or an ace and these three types are equally likely because there are four of each initially. If the first divider is a king we still need a queen or an ace; if the first divider is a queen or an ace we still need a king. Treat these two cases separately and use the same divider reasoning for the remaining deck after the first divider. Use the law of total expectation to find the overall expected value.

503. One Colored Face

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the conditional probability formula

$$P(B_1 | A) = \frac{|A \cap B_1|}{|A|},$$

where B_1 is the event that the selected cube has exactly one painted face, and A is the event that the cube has at least five unpainted faces.

Hint 2: In the $3 \times 3 \times 3$ cube, there is only one small cube that is completely surrounded on all sides — the interior cube — which has zero painted faces. So there is exactly one cube in B_0 , the set of cubes with zero painted faces.

Hint 3: For cubes with exactly one painted face, we look at the center cube of each face of the large cube. Each of the 6 faces has one such center cube, and these cubes are not on edges or corners, so they each have exactly one face painted. Thus, there are 6 such cubes in B_1 .

504. Same Arc II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "same arc I" first.

Hint 2: From "same arc I" we know for $N > 2$ we have the formula $\frac{N}{2^{N-1}}$. For $N = 1$ and $N = 2$ the probability is 1. By the definition of X we therefore have, for integers $N \geq 1$,

$$P(X > N) = \begin{cases} 1, & N = 1, 2, \\ \frac{N}{2^{N-1}}, & N \geq 3. \end{cases}$$

Now, use the tail sum formula.

505. Count Digits

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For an integer k , the number of digits is given by the greatest integer function $\lfloor \log_{10}(k) \rfloor + 1$.

506. Truck Parking

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: It is easier to calculate the probability that Bob cannot park. Bob cannot park if none of the remaining 4 spaces are adjacent. To count the arrangements where this happens, establish a one-to-one correspondence with a more straightforward set.

507. Unknown Balls

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $n = 100$. By symmetry, the probability that the first ball to roll out is black is $\frac{1}{2}$. The probability that there are exactly k black balls and the first ball to roll out is black is $\frac{k}{n(n+1)}$. Thus, the probability that there are k black balls given that the first ball is black is $\frac{2k}{n(n+1)}$.

508. Coin Transactions

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Define a cycle of length n to be a sequence of n transactions, starting and ending with the same person. For example, $A \rightarrow B, B \rightarrow C, C \rightarrow A$ would be a cycle of 3. What are the possible cases that lead to everyone ending up with 4 coins?

509. Red Blue Game

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use dynamic programming.

Hint 2: We let $E(r, b)$ be the maximum expected dollars you can earn when there are r red marbles and b blue marbles remaining, and you must guess a color before each draw. At state (r, b) you compare two choices: Guess red, which yields the expectation:

$$R = \frac{r}{r+b}(1 + E(r-1, b)) + \frac{b}{r+b}(0 + E(r, b-1))$$

Guess blue, which yields:

$$B = \frac{b}{r+b}(1 + E(r, b-1)) + \frac{r}{r+b}(0 + E(r-1, b))$$

You choose whichever is larger, so the recurrence for $r, b \geq 1$ is

$$E(r, b) = \max\{R, B\}$$

Hint 3: The base cases are:

$$E(r, 0) = r, \quad E(0, b) = b$$

since with only one color left you always guess that color and succeed each draw.

510. Hard Card Expectation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve the problem "first ace" first.

511. Poisoned Wine I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: How many binary numbers can you generate using 10 bits?

512. Non-negative Expectation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the method of reduced sample space.

513. Regular 21-gon

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the case where the diagonals we choose do not share vertices. In this case, the probability that they intersect is simply $\frac{1}{3}$.

514. Maximum Till Now I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use tail sum formula.

515. Home Run

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

516. Game Of Placing Kings

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

Hint 2: Alice has a winning strategy—she can create a scenario where she copies Bob's moves, so Bob runs out of moves first.

517. Same Temperature

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the function representing the difference in temperature between two opposite points.

518. 4x4 Array

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Call a filled 4×4 array satisfying the given conditions cool. There are $4!$ possibilities for the first row; without loss of generality, let it be 1 2 3 4. Since each quadrant must contain all four numbers, we have exactly four possibilities for the second row, namely: (i) 3 4 1 2 (ii) 3 4 2 1 (iii) 4 3 1 2 (iv) 4 3 2 1

519. Chess Tournament VII[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Solve "Chess Tournament I" first.**Hint 2:** For the top 2 rated players to meet in the final round, they must be placed in opposite halves of the bracket in the first round (just like in "Chess Tournament I"). Then, both players must reach the final by winning all of their matches in the preceding rounds.**520. Final Number Possibilities**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Whenever there is a sequence of operations, look for what doesn't change.**Hint 2:** Let the numbers written on the board be $a_1, a_2, a_3, \dots, a_n$. The product $(a_1 + 1)(a_2 + 1)(a_3 + 1) \dots (a_n + 1)$ is invariant.**521. Car Clusters**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** When would a given car form a new cluster?**522. Strange Party**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** If p denotes the total number of people present before the bus arrives, then p must satisfy: $p \equiv 0 \pmod{12}$ $p + 50 \equiv 0 \pmod{25}$ **523. Blackboard**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Suppose the starting number is x . What is the probability that the number $x - 1$ appears on the board at any point? The first draw is uniform on $\{0, 1, \dots, x - 1\}$, so the chance you draw $x - 1$ immediately is $\frac{1}{x}$. What if the initial number is $x + 1$? Then your first move lands uniformly in $\{0, 1, \dots, x\}$. With probability $\frac{1}{x+1}$ you hit $x - 1$ immediately; with probability $\frac{1}{x+1}$ you land at x , and from there the chance to hit $x - 1$ is $\frac{1}{x}$. Altogether, $\frac{1}{x+1} + \frac{1}{x+1} \cdot \frac{1}{x} = \frac{1}{x}$.**Hint 2:** Seeing that both "start at x " and "start at $x + 1$ " give probability $\frac{1}{x}$, one conjectures that no matter what integer $m \geq x$ you start from, the probability you ever reach $x - 1$ is always $\frac{1}{x}$.**524. Phone Connection**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: How many wires are needed for such a configuration?

Hint 2: Model the phones as vertices of a graph and each wire as an edge between the two phones it joins. Let d_i be the degree of vertex i , the number of wires incident to phone i . The total number of wire-ends counted over all phones is $\sum_{i=1}^{25} d_i$, and every wire contributes exactly two ends, so the number of wires equals $\frac{1}{2} \sum_{i=1}^{25} d_i$.

525. Triangle Formation I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To determine when a triangle can't be formed, one of the following conditions must be true: $X > Y + Z$, $Y > X + Z$, or $Z > X + Y$. Since these events are mutually exclusive (only the largest side can exceed the sum of the others), we calculate the probability of one of these events and multiply the result by 3 due to symmetry.

526. 3 Consecutive Heads II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The sequences can be of the forms HHH_ _ _ _ HHH_ _ _ _ HHH_, and _ _ _ HHH.

527. Too Many Unknowns

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: It can be proven that no two people look at each other's dice.

528. Dice Triangle

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: It is easier to calculate the probability that they don't form a triangle. Use symmetry!

529. Triangle Sum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The total sum of the numbers from 1 to 9 is 45. Since the sum of the integers on each side of the triangle is 17, the total sum across all three sides is 51. However, the integers at the vertices are counted twice (once for each adjoining side). Therefore, the sum of the numbers at the vertices must be $51 - 45 = 6$.

530. Is It Fibonacci?

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Express everything in terms of f_1 and f_2 .

531. Dice Triangle II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If $t = 7$, then the triangle is a right triangle. This occurs with probability $\frac{1}{6}$. If $t > 7$, it is not possible to be a right triangle because a right triangle cannot have an obtuse angle. If $t < 7$, either the side of length a or the side of length b would have to be the hypotenuse, so that either $a \cos((t-1)15^\circ) = b$ or $b \cos((t-1)15^\circ) = a$, which means that $\cos((t-1)15^\circ)$ must be rational.

532. Marble Swap III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We only need to track whether the blue marble is in the center, because if it is, the order is back to red-blue-red, and if it isn't, it must be on one of the edges (and the two reds are on the other two spots).

Hint 2: Use recursion. Let p_n be the probability that after n moves the blue marble sits in the center. Use the law of total probability to express p_{n+1} in terms of p_n to get the recurrence.

533. Jet Trail Intersection

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If the trails are extended infinitely in both directions, they will intersect with probability 1 (since the lines lie in the same plane, and the probability of them being parallel is negligible). Can this be used to calculate the probability of the trails intersecting?

Hint 2: Let the intersection point of the lines be X . What is the probability that the first ray passes through X ? What is the probability that the second ray also passes through X ?

534. Increasing Decreasing I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The middle digit must be 9.

535. Game On Circle

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the game from Bob's perspective and use symmetry.

536. Tetrahedral Die I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the desired probability be p . Get an equation in p using the law of total probability.

537. Highest Roll EV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since, you stop at 4, your highest roll is 4, 5, or 6. Calculate the probabilities of each of them.

538. Chords Intersection II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "chords intersection I" first.

539. Increasing Birthdays

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: How many circular permutations are there with 7 people? How many of them lead to them sitting in age order, either clockwise or anticlockwise?

540. Binary Trailing Zeros

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The number of trailing zeros in the binary representation of $142!$ corresponds to the highest power of 2 dividing $142!$.

541. Vertical Hit

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If the frog is on one of the diagonals, the probability of hitting a vertical side is $\frac{1}{2}$.

542. Bacteria Population I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Given that the probability of extinction for a colony with a single cell is $\frac{1}{3}$, the extinction probability for a colony with three cells can be determined by considering the extinction of each independent lineage.

543. Count Regions IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Count Regions I", "Count Regions II", and "Count Regions III" first.

544. Expected Ping-pong Games

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The expected value of the ratio of A's to B's score is 3:7. What does that imply?

545. Small Chords

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: A chord is shorter than the radius exactly when the central angle it subtends is less than 60° .

Hint 2: Fix a point i as the starting point and look at the 60° clockwise arc from it. Define E_i as the event that, starting at point i , all the other points lie in that arc. These events are disjoint, so you can add their probabilities.

546. Large Cycles

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use indicator variables and linearity of expectation.

Hint 2: Fix some k with $n < k \leq 2n$. Let Y_k be the number of k -cycles in the permutation. For each element $i \in \{1, \dots, 2n\}$, define the indicator

$$X_i = \begin{cases} 1, & \text{if } i \text{ lies in a cycle of length } k, \\ 0, & \text{otherwise.} \end{cases}$$

Since each k -cycle contains exactly k elements, we have

$$Y_k = \frac{1}{k} \sum_{i=1}^{2n} X_i.$$

547. Cheating Husbands

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What happens if only one husband cheated?

548. Largest Prime Probability

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To get prime sums other than 2, we need to have both even and odd faces. If three faces are even and three are odd, then there will be 18 odd sums out of the 36 possible combinations. If two faces are odd and four are even, there will be 16 odd sums, and if one face is odd and five are even, then there will be 10

odd sums.

549. Jumping Frog VI

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Any valid path looks like:

$$0 \rightarrow x_1 \rightarrow x_2 \rightarrow \cdots \rightarrow x_n \rightarrow 10$$

The first and last points are fixed (the first point is 0, and the last point is 10). The points $x_1, x_2, \dots, x_n$ can be any values in the closed interval $[1, 9]$. They must be distinct ($x_1 \neq x_2 \neq \cdots \neq x_n$) and in increasing order ($x_1 < x_2 < \cdots < x_n$).

Hint 2: The problem is equivalent to finding a (possibly empty) subset of the set:

$$\{1, 2, 3, \dots, 9\}$$

550. Dice Game VIII

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The optimal strategy is as follows: if the number rolled is greater than or equal to x , take the money and re-roll; if it is less than x , simply re-roll.

551. Simulate 1/3

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

552. Count Sequences I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: You start at 0 and end at 4. At each stage, you add or subtract 1 from the previous number to get the current number. Is the number of $+1$ and -1 operations fixed?

Hint 2: The number of $+1$ (and hence -1) is fixed. The number of $+1$ is 7.

Hint 3: The problem is equivalent to finding the number of arranging 7 $+1$ s and 3 -1 s.

553. Diagonal Cut II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve diagonal cut I first.

554. Last Ball[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** What happens to the number of red balls after each operation?**555. Treasure Island**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** It does not matter much where the grave is! There are only two places where the treasure can be.**556. Vector Toggle**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Let $t = (1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$ and $t_0 = (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$. Using Kac's lemma, compute the expected number of moves for the process to return to t (starting from t). Then, use this result to determine the expected number of moves required to reach t starting from t_0 .**Hint 2:** Starting from t the chain makes one move and lands at one of the ten neighbors of t with probability $\frac{1}{10}$ each. If H denotes the expected time to hit t when started from any neighbor of t then the expected return time to t satisfies

$$\text{expected return to } t = 1 + \frac{1}{10} \sum_{\text{neighbors } k} \mathbb{E}_k[T_t] = 1 + H,$$

because the neighbors are equivalent by symmetry and each appears with probability $\frac{1}{10}$ as the first step.**557. Prisoner Hats-multicolor**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Solve the "prisoner hats" problem first.**558. Tennis Odds**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use the method of reduced sample space.**559. Pizza Standard Deviation**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Given that the workers are independent, the variances add up.

560. Circuit

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation by breaking the problem into subsets of 4 bulbs.

561. Expected Number Of Rectangles

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The expected number of rectangles added to the initial rectangle is the sum of the expected number of new rectangles obtained with each new line that is drawn for each of 20 new lines.

562. Triangles Inside Triangle

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Which combinations of 3 dots do not yield a triangle?

563. Divisible By 37

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: $37xyz$ can be written as:

$$37000 + 100x + 10y + z$$

So, $37xyz$ is divisible by 37 if and only if xyz is divisible by 37.

Hint 2: 148 is divisible by 37. 481 is also divisible by 37. 814 is also divisible by 37.

Hint 3: Consider $A = xyz$ and $B = yzx$, where x, y, z are digits. If A is divisible by 37, then B is also divisible by 37.

564. 3 Heads 3 Tails III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The probability Alice wins is $\frac{1}{2}$ due to symmetry. Additionally, the expected number of tosses for Alice to win is the same as for Bob, thanks to symmetry. Using the law of total expectation, the answer is simply twice the expected number of tosses for Alice to win.

565. Unlucky Seven

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There is no point in re-rolling if the first turn yields 3, 4, 5, or 6.

566. Toss Exceedance

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: This can be modelled as a simple symmetric random walk.

567. Tennis Tournament

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let x and y represent your chances of beating Alice and Bob, respectively. Calculate the winning probabilities for both game orders (ABA and BAB) and determine which one is greater, using the fact that $x < y$.

568. Chess Tournament VI

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The given condition can occur only if either player 50 does not play till the last match OR player 49 does not play till the last match.

569. Expected Wait Time II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the law of total expectation. What is the probability of arriving at a time when the bus refills on gas? What is the probability of arriving at a time when the bus follows the normal schedule? What is the expected waiting time in each case?

570. Weighted Die

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the probability of getting the value i on the die be p_i . We know that $\sum_{i=1}^6 p_i = 1$. Express each p_i in terms of p_1 using the condition given in the problem: the probability of rolling each value from 2 to 6 is the same as the probability of rolling these values when rolling the die twice and summing the results.

571. Ordered Pairs In Permutation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: First, observe that the total number of ordered pairs a_i, a_j satisfying $i - j > 155$ is equal to

$$(2015 - 156) + (2015 - 157) + \cdots + 1 = \binom{2015 - 155}{2} = 1728870,$$

where we count by casework on $j = 1, 2, \dots, 1859$.

572. Variance Maximum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In the interval $[-1, 1]$, $X^2 \leq 1$. Thus the maximum value of $E[X^2]$ is 1. Recall that $\text{Var}(X) = E[X^2] - E[X]^2$.

573. Additional Info Die Rolls

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let A be the event that the first 6 appears before the first 5. By definition,

$$E[N|A] = \sum_{n=1}^{\infty} n \cdot P(N = n|A)$$

Hint 2: On any roll, the chance of a "special" face (5 or 6) is $\frac{1}{3}$, so

$$P(\text{first 5 or 6 at } n) = \left(\frac{2}{3}\right)^{n-1} \cdot \left(\frac{1}{3}\right)$$

Given that a special face appears, the chance it's a 6 is $\frac{1}{2}$, hence

$$P(N = n, A) = \left(\frac{2}{3}\right)^{n-1} \cdot \left(\frac{1}{6}\right)$$

$$P(N = n|A) = \frac{P(N = n, A)}{P(A)} = \left(\frac{2}{3}\right)^{n-1} \cdot \left(\frac{1}{3}\right)$$

574. Word Problem I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics. Find the number of desired words.

575. Truth

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assuming a statements are true, which of the written statements are true?

576. Normal Distributions

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: $f_{X_1|X_1+X_2=5}(x) = \frac{f_{X_1, X_1+X_2}(x, 5)}{f_{X_1+X_2}(5)}$

Hint 2: Note that if $X_1 = x$, then $X_2 = 5 - x$ to satisfy the condition $X_1 + X_2 = 5$.

577. Sports Betting

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If Bob doesn't match the bet, his expected payoff is -10. Bob should match Alice's bet if the expected payoff after matching the bet is more than -10.

578. Even Head Stop I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

No hints provided.

579. Interesting Constraints

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Change the variables to y_i where $y_i = x_i - \frac{1}{3}$.

580. Mathematical Expression I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: After the plus signs and multiplication signs are inserted, we obtain a sum of powers of $\sqrt{3}$, which has the form $a + b\sqrt{3}$, where a and b are nonnegative integers. If any power is an even power of $\sqrt{3}$, we would have $a > 0$, which cannot occur.

581. Inside The Range

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Note that the number of non-negative integer solutions to $x_1 + x_2 + \cdots + x_n \leq k$ is equivalent to the number of solutions to $x_1 + x_2 + \cdots + x_n + x_{n+1} = k$, where x_{n+1} is an additional variable representing the "slack" in the inequality.

582. Increasing Rolls Expectation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: Let E_k be the expected sum of the rolls from the current roll onward, including the current roll itself, given that the current roll is k . Find a recurrence relation in k .

Hint 3: For $k < 10$, the next roll must be one of $k+1, k+2, \dots, 10$ in order for the process to continue. So

$$E_k = k + \frac{1}{10}(E_{k+1} + E_{k+2} + \cdots + E_{10}).$$

583. Card Picking II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The problem is essentially asking-if you arrange 52 cards in a sequence, what is the probability that the 28th card is the 2 of hearts, given that the 27th card is the first ace encountered in the first 27 cards? Therefore, we can use conditional probability.

Hint 2: Using conditional probability, the solution is simply the number of ways to arrange the cards such that the first ace is in the 27th position and the 2 of hearts is in the 28th position, divided by the number of ways to arrange the cards such that the first ace is in the 27th position.

584. Prime Dice

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For the sum to be prime, it must be odd.

585. Fish Estimate

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: One-fourth of the captured fish were tagged, implying that the tagged fish represent approximately one-fourth of the total fish population in the pond.

586. Proportional Dice

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let die A be the fair die and die B be the weighted die. Denote the probability of die B rolling an even number as x (so the probability of it rolling odd is $1 - x$). Now, apply the law of total probability.

587. Moving Ants II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If the ant takes 4 steps, how many possible paths are there? Out of these, how many paths cover all 4 vertices?

588. Red Card Prediction

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use martingales.

Hint 2: Label the deck's cards $1, 2, \dots, 52$ in the order they will be revealed, and let $\mathcal{F}_n$ be the information available after the first n cards have been turned face up. At that time there remain $52 - n$ cards, of which r_n are red. Define the stochastic process:

$$X_n = \frac{r_n}{52 - n}$$

$\{X_n\}$ is a martingale.

589. Infection Spread[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** What happens to the perimeter of the infected region?**590. Win-win**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use combinatorics. To calculate the number of ways such that everyone wins, consider a directed graph whose vertices are the five participants. Draw an edge from one participant to another if the first points to the second.**591. Parking Lot**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** For $1 \leq i < j \leq 4$, let (i, j) be the situation where the cars are in spaces i and j . During the process described in the problem, we will definitely be in the states $(1, 2)$, $(1, 3)$, $(2, 4)$, and $(3, 4)$. We may or may not be in one (and only one) of the states $(1, 4)$ and $(2, 3)$. On average, it takes two minutes to get from the initial state $(1, 2)$ to $(1, 3)$. From $(1, 3)$, in any given minute, there is a $\frac{3}{4}$ probability of leaving that state. So, on average, it takes $\frac{4}{3}$ minutes to leave state $(1, 3)$.**592. Head-tail Product**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** The number of heads X follows a binomial distribution with parameter $p = \frac{1}{2}$. Hence, the mean $E[X] = np = \frac{n}{2}$ and the variance $E[X^2] - E[X]^2 = np(1-p) = \frac{n}{4}$, which implies $E[X^2] = \frac{n}{4} + E[X]^2 = \frac{n}{4} + \left(\frac{n}{2}\right)^2$.**593. Guess The End**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** The number of ants falling off the east end of the rod is the same as the number of ants facing east at the start, since the number of ants facing east never changes.**594. Optimal Die Choice**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Maximize this by taking the derivative and setting it to 0.**595. Two Random Points**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Because R_1 is uniform on $[0, 1]$, the chance it lies in the tiny interval $[r, r + dr]$ is proportional to that interval's length. Since the total length of $[0, 1]$ is 1,

$$P\{R_1 \in [r, r + dr]\} = 1 \cdot dr = dr,$$

so the probability density of R_1 is

$$f_{R_1}(r) = 1, \quad 0 \leq r \leq 1.$$

Hint 2: Once the first point is at distance r_1 , the second must lie in the annulus between radii r_1 and 1, which has area $\pi(1 - r_1^2)$. A thin ring at radius r_2 of width dr_2 in that annulus has area $2\pi r_2 dr_2$. Uniform selection over the annulus gives

$$P\{R_2 \in [r_2, r_2 + dr_2] \mid R_1 = r_1\} = \frac{2\pi r_2 dr_2}{\pi(1 - r_1^2)} = \frac{2r_2}{1 - r_1^2} dr_2, \quad r_1 \leq r_2 \leq 1,$$

so the conditional density is

$$f_{R_2|R_1}(r_2|r_1) = \frac{2r_2}{1 - r_1^2}.$$

Now, use the law of total probability for the continuous case.

596. Passing The Coin I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: What are the initial possibilities? Either Player 1 tosses a heads and wins immediately, or he gets a tails and passes the turn to Player 2. How can Player 1 still win in the case where Player 1 tosses a tails?

Hint 3: Use the law of total probability and Hint 2 to set up the recursion.

597. Jumping Frog IX

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion. Let p_n be the probability of landing on lily pad n . If there are no restrictions,

$$p_n = \frac{1}{2} \cdot p_{n-1} + \frac{1}{2} \cdot p_{n-2}.$$

598. Expected Determinant

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What happens to the determinant if we swap the first 2 rows?

599. Least N

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $A = \{18a + 1, 18a + 2, \dots, 18a + 9 : a = 0, 1, 2, 3, 4, 5\}$. Then, notice that A has 54 elements, and no two elements have a difference of 9. So, we know $n \geq 55$.

Hint 2: Use the pigeonhole principle to prove that any subset with 55 elements must have two elements with a difference of 9.

600. Forming Quadratic Polynomial II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The roots of a quadratic with integer coefficients are real when its discriminant is nonnegative. The discriminant of the quadratic $x^2 + bx + c$ is $b^2 - 4c$. We therefore need to determine the number of lattice points (b, c) with $1 \leq b, c \leq 100$ such that $b^2 - 4c \geq 0$, or $c \leq \frac{b^2}{4}$. If $b \geq 20$, then $\frac{b^2}{4} \geq 100$, so all the lattice points (b, c) for which $b \geq 20$ and $1 \leq c \leq 100$ contribute, and these total 8100. If $b = 1$, then $\frac{b^2}{4} = \frac{1}{4}$, so there is no lattice point of the form $(1, c)$ that contributes to our count. What about $1 < b < 20$?

601. Crazy Weights Arrangement

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

602. Vending Machine I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Define an indicator variable I_i for each of the 5 unique stickers, where $I_i = 1$ if the i -th sticker is among the 7 stickers purchased and $I_i = 0$ otherwise.

603. Early Head Probability

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry. The probability that Alice gets her first heads in fewer flips than Bob is given by $\frac{1-p}{2}$, where p is the probability that they obtain their first heads on the same toss.

604. Shadow

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Visualize the cube from above and consider which edges are visible as you rotate the cube.

605. Rabbit On A Square Grid

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Suppose the rabbit starts at $(0, 0)$ and goes to sleep at (x, y) . Then $-100 \leq y \leq 100$. For each of the values of y , what are the possible values of x ?

606. Game Of Moving Rooks

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider a 2×2 chessboard.

607. Origin In Between

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: One point must be to the left of the origin and one point must be to the right of it. Use the law of total probability.

608. Increasing Dice Rolls I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To obtain rolls in strictly increasing order, all values must be distinct AND they must appear in increasing order. How to calculate this probability?

Hint 2: Use multiplication rule.

609. Tenth Toss

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The last two tosses must be heads, and the 8th toss must be a tail. Therefore, the first seven tosses must be such that no two consecutive heads appear.

Hint 2: How to count the number of sequences of first seven tosses such that no two consecutive heads appear? Recursion!

610. Three In A Line

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

Hint 2: Suppose Alice places a 1 in the center square. What would happen if, on every subsequent turn, she mirrors Bob's move across the center of the grid?

Hint 3: If Alice simply mirrors Bob's moves across the center, then Bob can simply complete a line first. But what if Alice continues to mirror Bob's moves except when Bob's next move could create three identical numbers in a line?

611. Matching Cards Removal[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Define D_i as the number of turns needed to remove the i th pair of cards after $i - 1$ pairs have been removed. The total number of turns is $D = \sum_{i=1}^{15} D_i$. Apply the linearity of expectation to find the expected number of turns.

612. Blue Marble Remaining[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Find the expected number of marbles remaining in bag Y. The blue marble is 1 of the 5 marbles in bag Y. So the probability that the blue marble remains in bag Y is $\frac{1}{5}$ of the expected number of marbles remaining in bag Y.

613. Expected Collisions II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume each ant is carrying a flag. Imagine that at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem, as the flags simply go past each other. This simplification is commonly used in many collision problems.

614. Minimum Correlation[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the random variables be $Y_1, Y_2, \dots, Y_{10}$, and let the minimum pairwise correlation be m . For 10 random variables, each with mean 0 and variance 1, let $Y = \frac{1}{10}(Y_1 + Y_2 + \dots + Y_{10})$. Hence, we have:

$$\text{Var}(Y) = \frac{1}{100} \left(\sum_{i=1}^{10} \text{Var}(Y_i) + \sum_{i \neq j} \text{Cov}(Y_i, Y_j) \right)$$

Note that the variance must be non-negative.

615. Complex Circle[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider $Z = e^{i\theta}$.

616. Higher Sum[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let x and y be the values rolled. Then, $x + y \geq xy$ implies that $(x - 1)(y - 1) \leq 1$.

617. 5 Points On A Sphere

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The convex hull of the 5 points is a triangular bipyramid.

618. Count Sequences II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Suppose the number of 0s is n . We can construct the sequence in two steps: Step 1: Place $n - 1$ of 1s between the 0s, Step 2: Place the remaining $19 - n - (n - 1) = 20 - 2n$ of 1s into the $n - 1$ spots where there is a 1.

619. Remove Or Flip

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: 4013 is odd. So, in the beginning the number of red and black tokens is different.

620. Unit Digit Probability I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Which numbers must be excluded to ensure the product ends in 1, 3, 7, or 9?

621. Interview III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "interview I" and "interview II" first.

622. Expected Value

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: By linearity of expectation, we have that $E[r] = E[\frac{p_0}{10}] + E[\frac{p_1}{10^2}] + \dots = \sum_{i=0}^{\infty} \frac{E[p_i]}{10^{i+1}}$, so it suffices to compute the expected value of each digit.

623. Forming Quadratic Polynomial I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Compute the complement: namely, we compute the probability that regardless of the ordering of a , b , and c , no quadratic equation Alice creates will have 2 distinct real roots. For this to be the case, it is sufficient that the largest possible discriminant is nonpositive.

624. Rope Variance

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What is the distribution of the length of the longer piece?

625. Triangular Grid

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion. Calculating the probability that you never go beyond 1 unit from your starting point in the 5 turns is easier.

626. Strange Comparison I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Compare each number to a nearby power of 2.

Hint 2: Since $31 < 32$, $31^{11} < 32^{11} = (2^5)^{11} = 2^{55}$. Can we apply a similar approach to 17^{14} ?

627. Mathematical Expression II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Martha can produce four different expression types, each equally likely: $a + b + c$, $a + bc$, $ab + c$, and abc . Therefore, to find the expected value, we can take the average of the expected values for each of these expression types.

628. Boy-girl Adjacent Pairs

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

629. Optimal Bid

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

No hints provided.

630. 3 Consecutive Heads I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The sequence should appear as HHH-, -HHH- or -HHH. Count the possibilities for each pattern.

631. Two Rolls Reward[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use recursion.**632. Broken Elevator**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Suppose there are u ups, d downs, and s seconds at which the elevator stays still. Since the elevator returns to its original height, $u = d$. Since 8 seconds elapse, $u + d + s = 2u + s = 8$. It is clear that s is even, so $s \in \{0, 2, 4, 6, 8\}$, and $u = d = \frac{8-s}{2}$. We do casework based on the value of s .

633. 3 Lights[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Due to symmetry, the probability of the rightmost light being on is equal to the probability of the leftmost light being on. To find the expected number of lights on after three seconds, calculate the probability for each light and then use $E[x] = \sum x \cdot p(x)$.

634. Dice Game II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Identify when to stop after each roll based on the expected value of subsequent rolls.**635. Checks**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use linearity of expectation.

Hint 2: Let K_1, K_2, K_3 denote the three knights and k_1, k_2, k_3 the three kings. For each pair (i, j) with $i, j \in \{1, 2, 3\}$, define the indicator variable

$$I_{ij} = \begin{cases} 1, & \text{if knight } K_i \text{ attacks king } k_j, \\ 0, & \text{otherwise.} \end{cases}$$

The total number of checks is then

$$T = \sum_{i=1}^3 \sum_{j=1}^3 I_{ij}.$$

636. Arrange Teams[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider a recursive approach.

637. Interview IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Interview I" first.

Hint 2: A similar strategy to that of "Interview I" would work: skip some candidates, then select the first "best or worst" candidate that appears afterward.

638. Coin And Die Together

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If we get at least one heads, what is the probability that the number 1 is face up on the dice?

639. Finding Seven

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If the sum of the dice is 7, we reroll no dice. If the sum of any two dice is ≤ 6 , we reroll 1 die because rolling 1 die will give Alice a $\frac{1}{6}$ chance of winning. Thus, we consider the scenario where the sum of any two dice is 7 or greater.

640. Choose Values III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Because Alice and Bob end up using exactly the same die (same face values and probabilities), symmetry forces $P(\text{Alice wins}) = P(\text{Bob wins})$.

Hint 2: Since one of "Alice wins," "Bob wins," or "tie" must happen, we have $2P(\text{Alice wins}) + P(\text{tie}) = 1$, so Alice's win probability is $P(\text{Alice wins}) = \frac{1}{2}(1 - P(\text{tie}))$. Thus maximizing her chance of victory is equivalent to minimizing the probability of a tie.

641. Cherries

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To have k cherries left in the bowl, these must be all purple with the $k + 1$ -th cherry from the end being red. This simplifies to finding how many ways we can arrange m red cherries and n purple cherries with the last k being purple and the $k + 1$ -th from the end being red.

642. Even Tails Run

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion. What are the initial choices?

Hint 2: Let a_n be the number of length- n sequences with every run of tails even. Think about how such a sequence starts: - If the first flip is H, then the remaining $n - 1$ flips can be any valid sequence of length $n - 1$, giving a_{n-1} possibilities. - If the first flip is T, you must immediately flip T again (to make that first run of tails have length 2), and after those two Ts the remaining $n - 2$ flips form any valid sequence of length $n - 2$. That contributes a_{n-2} more possibilities. Hence every valid sequence of length n comes from either an H-start or a TT-start, and:

$$a_n = a_{n-1} + a_{n-2}$$

643. Same Birthday Probability

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: It is easier to calculate the probability that all students have unique birthdays.

Hint 2: The probability that all of them have unique birthdays is $\frac{365}{365} \times \frac{364}{365} \times \cdots \times \frac{343}{365}$.

644. Matrix Attack I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Try to construct such a matrix.

645. Count Of Rational Numbers

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: From $\gcd(a, b) = 1$, we note that for each prime p dividing $25!$, either $p \mid a$ or $p \mid b$.

646. Shooting Arrows I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To find the probability that at least one arrow lands within the innermost circle (radius 1 meter), first calculate the probability that none of the arrows land within this circle. Then, subtract this value from 1.

647. Card Picking I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider only the king (all 4 suits), queen (all 4 suits), jack (all 4 suits), and ace of diamonds. Visualize the possible arrangements of these 13 cards.

648. Game On 753 Coins

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume there is a winning strategy that takes n operations. Can you create an equation in n ?

649. Triangle Inside Circle III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The triangle is formed by connecting each pair of points with a chord. Since the points are IID, the side lengths are symmetrical. Therefore, the expected perimeter is three times the expected length of one side. If P is the perimeter and S_{AB} is the length of the chord between points A and B , we have $E[P] = 3E[S_{AB}]$.

650. Coin Toss Game VI

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the case when $n \leq 6$ and when $n > 6$ separately.

651. Count Paths I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Label each upward step as U and each rightward step as R . Then, establish a one-to-one correspondence with a simpler set that is easier to count.

Hint 2: There is a one-to-one correspondence between each valid path and a string consisting of 8 R 's and 10 U 's.

652. Game On Divisors II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use "stealing the strategy".

Hint 2: Suppose Alice doesn't have a winning strategy. What if she just removes 1 in this case?

653. Polygon Triangle

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider a regular 17-sided polygon and think about selecting three vertices to form a triangle. Analyze how the position of the triangle relative to the center of the polygon affects whether it contains the center. Focus on counting the triangles based on the number of edges in the minor arc formed by these selected vertices.

654. Ants On A Wire

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Rather than considering that the ants reverse their direction after colliding, we can treat them as passing through each other.

655. Clock Angle

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The hour hand moves 0.5 degrees every minute.

656. Weighted Coins

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: These weights form binary numbers, except the "1" is omitted. Thus the numbers that are generated are exactly the even numbers between 2 and 2046, inclusive.

657. 5000 Sets

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let U be the union of the finite sets. A 12-element set has $2^{12} = 4096$ subsets, therefore, there must be at least 13 elements in U .

658. Three Steps

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use geometric probability. What is the probability that the ant takes exactly 2 steps and moves past point 1? What is the probability that the ant takes exactly 3 steps and moves past point 1?

659. Just Even Stop

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion. On the first toss, you can get either heads or tails. On the second toss, there is a $\frac{1}{2}$ probability of getting the same outcome as the first toss, resulting in an even number of tosses. What happens if the second toss differs from the first?

660. Shooting Stacks

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We can create a one-to-one correspondence between the given problem and some permutation.

Hint 2: Consider each target in a stack as labeled by the stack it belongs to: 1 for the first stack, 2 for the second stack, 3 for the third stack, and 4 for the fourth stack. There is a one-to-one correspondence between the given problem and permutations consisting of 2 1s, 3 2s, 3 3s and 4 4s.

661. Segments Inside Square I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Denote by L the infinite straight line passing through the random points P and Q . Assume L intersects AB . What is the probability that the segments AP and BQ intersect in this case?

Hint 2: Assume L does not intersect AB . What is the probability that the segments AP and BQ intersect in this case?

Hint 3: If L intersects AB , the probability that AP and BQ intersect is 0. If L does not intersect AB , the probability that AP and BQ intersect is $\frac{1}{2}$.

662. Dice Game VI

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Determine when to keep the first roll versus when to roll again based on expected values.

663. Fibonnaci Error

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: You would compute G_{101} as $G_{100} + F_{99} = 1 + (F_{100} + F_{99}) = 1 + F_{101}$. Next, you would find G_{102} as $G_{101} + G_{100} = 2 + (F_{101} + F_{100}) = 2 + F_{102}$. Observe how the error accumulates across calculations.

664. Digit Sum V

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To minimize $S(n+1)$, when we add 1 to n , we should change as many digits as possible from 9 to 0 due to the carry.

665. Expected Swaps

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: An inversion is defined as a pair of indices (i, j) such that $i < j$ and $a[i] > a[j]$. The number of swaps required by the bubble sort algorithm is equal to the number of inversions.

666. Interesting Average I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We calculate the expected value of each digit, then use linearity of expectation to find the total expected value.

667. Changing Number[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the number be N . Since it ends in 4, we can write $N = 10A + 4$ where A is the integer formed by all but the last digit of N . If N has $n + 1$ digits, then A has n . Moving the final 4 to the front produces the number $M = 4 \cdot 10^n + A$.

Hint 2: Our requirement $M = 4N$ becomes $4 \cdot 10^n + A = 4(10A + 4) = 40A + 16$. How to use the above equation to find N ?

668. Acute Triangle[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Thales' theorem states that if one side of an inscribed triangle is a diameter of the circle, then the angle opposite that side is a right angle. So, when is the triangle acute-angled?

Hint 2: The triangle is acute-angled if and only if the center of the circle lies within the triangle.

Hint 3: To calculate the probability that the center of the circle lies within the triangle, use the law of total probability for the continuous case.

669. Segment Intersection[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Model each line segment as a vertex in an undirected graph, and draw an edge between two vertices if the corresponding segments intersect. What is the number of edges if there are 69 line segments and each segment intersects exactly three others?

Hint 2: In any undirected graph, each edge contributes 2 to the total degree count, so the number of edges must be half the sum of the degrees (over all vertices).

670. Powerful Three[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Note that $3^{4n} = (3^2)^{2n} = 9^{2n} = (-1 + 10)^{2n}$. Now use binomial theorem to expand this.

Hint 2: Expanding the expression using the binomial theorem gives,

$$(-1 + 10)^{2n} = \binom{2n}{0}(-1)^{2n} + \binom{2n}{1}(-1)^{2n-1}10 + \binom{2n}{2}(-1)^{2n-2}10^2 + \cdots + \binom{2n}{2n}10^{2n}.$$

The first term is exactly 1. Every term from the 102nd term onward contains a factor of 10^{101} or higher, so such terms automatically end with at least 101 zeros. What about other terms? Can we choose n so that each of these terms has at least 101 trailing zeros?

671. Socks Remaining II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Condition on the color of the last sock drawn. Compute the probability of the required event separately in the case where the last sock is green and in the case where the last sock is blue, then combine these using the law of total probability.

Hint 2: If the last sock is green (probability $30/60$), then the remaining 59 positions form a random arrangement of the other 29 green socks together with the 10 red and 20 blue socks. Among the 30 red-or-blue socks in these positions, each is equally likely to be the last among red and blue, so the probability that blue is last is $20/30$. Similarly, do it for the case where the last sock is blue.

672. Tetrahedron Coloring I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume we have 4 colors - 1, 2, 3, and 4. There are $4! = 24$ ways to assign four distinct colors to the four faces if we ignore symmetry, but many of these assignments are equivalent under rotations of the tetrahedron. One quick way to account for rotations is to fix the bottom as color 1.

673. Domino Tiling I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: For $n \geq 3$ consider the leftmost column: either it is covered by a single vertical domino (leaving a $2 \times (n-1)$ region, which can be tiled in a_{n-1} ways), or it is covered by two horizontal dominos that occupy the first two columns (leaving a $2 \times (n-2)$ region, which can be tiled in a_{n-2} ways). These two cases are disjoint and exhaust all possibilities.

674. Domino Tiling II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Domino Tiling I" first.

675. Largest Mean Median Difference

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the five scores, arranged in nondecreasing order, be $a \leq b \leq c \leq d \leq e$, where each score is an integer between 0 and 100. Since there are five scores, the median is the middle one, namely c , and the mean is $\frac{a+b+c+d+e}{5}$. We want to make the absolute difference between the mean and the median as large as possible. To achieve this, we can either make the mean much larger than the median, or make the median much larger than the mean.

676. New Language

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are five basic letters, and each letter can independently be either emphasized or not. Therefore, each position in a word has $5 \times 2 = 10$ possible choices. Hence, the number of distinct words of length k is 10^k .

677. Constrained Subsets III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The sum of all numbers 1 through 64 is $\frac{64 \cdot 65}{2} = 2080$, which is very close to 2072.

Hint 2: A subset of $\{1, \dots, 64\}$ has sum 2072 exactly when its complement has sum $2080 - 2072 = 8$. So there is a one-to-one correspondence between subsets summing to 2072 and subsets summing to 8.

678. Count Rectangles II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Count Rectangles I" first.

Hint 2: First, count the total number of rectangles assuming no central section is removed. Next, count the rectangles that should be excluded because of the missing central section. Finally, subtract this count from the total to obtain the required number of rectangles.

679. Tricky Palindrome I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Break the palindromes under 1000 into cases by their number of digits.

680. Tricky Palindrome II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write a generic four-digit palindrome in the form $\overline{abba}$, where a is the thousands (and units) digit and b is the hundreds (and tens) digit. Is it possible for $a \geq 5$?

Hint 2: One can show that necessarily $a \leq 4$. Under this restriction on a , examine whether b can be ≥ 5 .

Hint 3: It can also be shown that $b \leq 4$.

681. Tricky Palindrome III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write the decimal representation of n . Because n itself is a palindrome, its first (most significant) digit equals its last (units) digit; call that common digit a . First show that $a = 1$.

Hint 2: Suppose we have already shown that the first (most significant) digit and the last (units) digit are both equal to 1. Now let b be the most significant digit that is neither 0 nor 1. All digits to the left of b are therefore 0s or 1s. Since n is a palindrome, these same digits appear in reverse order at the rightmost end of the number. Now consider what happens when this number is multiplied by 9.

682. Skipping 2s

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think in terms of base-9 representation.

683. Arrow On Dice

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are four possible faces to move to from face 1, namely the four faces adjacent to it. Before reaching the face opposite to 1, we may visit exactly 1, 2, or 3 of these adjacent faces. Analyze these possibilities case by case.

684. Grid Arrangement III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Factor 171 as $3^2 \cdot 19$ and place the 19's and the 3's in the squares independently. The two primes act independently, so the total number of fillings is

$$(\text{number of valid patterns for } 19) \times (\text{number of valid patterns for } 3).$$

Hint 2: Let a_i be the number of 19's in the i th column. The number of ways to fill the grid 19's so that the with divisibility property is satisfied is equal to the number of nondecreasing sequences a_1, a_2, a_3 where each $a_i \in 0, 1, 2, 3$ and the sequence is not $(0, 0, 0)$ and not $(1, 1, 1)$.

Hint 3: For arrangements of divisibility by 3, consider 3 cases-(i) 9 divides the middle square, (ii) 3 doesn't divide the middle square, and (iii) 3 divides the middle square but 9 doesn't.

685. Tetrahedron Coloring II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: No face may be monochromatic, so no color can appear three or four times among the four vertices. Thus the only possible color-count patterns are $2 + 2$ (two colors used twice) or $2 + 1 + 1$ (one color used twice, the other two used once). Count each case separately.

686. Logarithm Comparison

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider the function $f(x) = \log_x(x+1)$. Check if it is increasing or decreasing in the interval $[1111, 1112]$.

687. Circular Frog

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let E_k be the expected number of steps needed to increase the size of the visited contiguous arc from $k-1$ points to k points, given the frog starts at one of the arc's end points. By linearity of expectation, the total expected time to visit all 10 points equals $\sum_{k=2}^{10} E_k$. Now use linearity of expectation.

Hint 2: To compute E_k use the standard formula for the expected hitting time of either boundary in a simple symmetric random walk on a finite interval: if the walk runs on $1, 2, \dots, n$ and starts at i , the expected time to hit 1 or n is $(i-1)(n-i)$. How can we model our situation as a linear simple symmetric random walk?

688. Square Factor

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, then the number of positive divisors of n is $(a_1+1)(a_2+1) \cdots (a_k+1)$.

Hint 2: A divisor is a perfect square exactly when each prime's exponent in the divisor is an even integer between 0 and the prime's exponent in the original number (inclusive).

689. Sum Sum Sum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Find an invariant.

Hint 2: The sum of digits is invariant under modulo 9.

690. Constrained Combinatorics

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Choose any set of four distinct numbers and write them in increasing order $t_1 < t_2 < t_3 < t_4$. For a fixed set t_1, t_2, t_3, t_4 , count how many permutations (w, x, y, z) of these values meet the three inequalities.

691. Evenbin I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For the numbers from 1 to 20, determine which ones are evenbin. Do you notice any pattern?

Hint 2: Observe that for every $k \geq 1$ the numbers $2k$ and $2k + 1$ have binary representations that differ only in the last digit, so their counts of 1's have opposite parity. Therefore exactly one number in each pair $(2k, 2k + 1)$ is evenbin.

692. Prime Set

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: To make the largest prime as large as possible, the remaining primes in the set should be as small as possible. However, one can show that 2 cannot belong to the set. Hence, we take the smallest possible primes starting from 3. With this choice, try to find an upper bound for the largest prime that can belong to our set.

Hint 2: The primes lesser than 27 (and ≥ 3) are 3,5,7,11,13,17,19 and 23. It is best to include all of these numbers in the set (why).

693. Catch Me If You Can I

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Such a sequence does exist. Try to find it.

694. Catch Me If You Can II

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Solve "Catch Me If You Can I" first.

Hint 2: Such a sequence does exist. Try to find it.

695. Game On Divisors IV

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Solve "Game on Divisors III" first. However, note that in this problem we only consider proper divisors (that is, the divisor must be greater than 1).

696. Evenbin II

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Look at each bit position separately. Fix a bit position i (where $i = 0, 1, \dots, 8$). Count how many of the even-parity numbers have a 1 in that position.

697. Hexagon Inside Hexagon

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Label the original regular hexagon H . The area of the inner hexagon equals the area of H minus the total area of the six outer corner regions that get cut away. Each of those six cut-away regions is a triangle formed at a vertex of H by the two chosen points on the two sides meeting at that vertex. So, we need to calculate the expected area of this triangle.

698. Subset Chain

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Look at each element individually, rather than each subset.

699. Triangle Tiling

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: $\sqrt{2}$ is irrational. If a, b, c, d are rational, then $a + b\sqrt{2} = c + d\sqrt{2}$ implies $a = c$ and $b = d$. Therefore, the hypotenuse of the tiling area must be lined with 5 hypotenuses of tiles.

Hint 2: Removal of these 5 tiles on the hypotenuse leaves a shape consisting of 10 unit squares in a staircase shape. These squares must each be filled with a pair of tiles joined along their hypotenuses (why?).

700. Constrained Subsets IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Find an upper bound for the answer and show that this upper bound is achievable.

Hint 2: Divide the numbers $1, 2, \dots, 2000$ into consecutive blocks of three: $\{1, 2, 3\}$, $\{4, 5, 6\}$, $\{7, 8, 9\}$, $\dots$. Each block can contribute at most one element to S .

701. Rook Moves

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion/markov chains.

Hint 2: Using symmetry, we can reduce the number of states to just 2. Let the expected number of minutes it will take the rook to reach the upper right corner from the top or right edges be E_e , and let the expected number of minutes it will take the rook to reach the upper right corner from any other square be E_c . Note that this is justified because the expected time from any square on the top or right edges is the same, as is the expected time from any other square (this is because swapping any two rows or columns does not affect the movement of the rook).

702. Real Addition I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: What if, instead of 1000, the winning condition was to write a number at least 1? Who would win then? And what if the target were 2, 3, and so on?

703. Real Addition II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Real Addition I" first.

704. Strange Comparison II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the binomial expansion for $(7 + 1)^{91}$.

705. GCD Expectation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write 2025 as a product of prime powers: $2025 = 3^4 \cdot 5^2$. Hence $2025^{2025} = 3^{8100} \cdot 5^{4050}$, and any divisor d can be written uniquely as $d = 3^a 5^b$ with integers $0 \leq a \leq 8100, 0 \leq b \leq 4050$. Choosing d uniformly is equivalent to choosing a uniformly from $\{0, \dots, 8100\}$ and b uniformly from $\{0, \dots, 4050\}$, independently.

Hint 2: The greatest common divisor with 2025 is $\gcd(d, 2025) = 3^{\min(a, 4)} \cdot 5^{\min(b, 2)}$. Because a and b are independent, the expectation splits: $E[\gcd(d, 2025)] = E[3^{\min(a, 4)}] \cdot E[5^{\min(b, 2)}]$.

706. Expected Good Points

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Label the ordered sample by $X_{(1)} < X_{(2)} < \dots < X_{(50)}$ and write the consecutive gaps $G_i = X_{(i+1)} - X_{(i)}$ for $i = 1, \dots, 49$. Under what condition, expressed in terms of the gaps, do two consecutive points $X_{(i)}$ and $X_{(i+1)}$ become mutual nearest neighbors?

Hint 2: The nearest neighbor of any interior point $X_{(i)}$ must be one of its two adjacent points, so two adjacent points $X_{(i)}$ and $X_{(i+1)}$ form a mutual nearest-neighbor pair exactly when their gap G_i is smaller than both neighboring gaps G_{i-1} and G_{i+1} . For the first and last gaps the condition is similar: the pair $(X_{(1)}, X_{(2)})$ is mutual iff $G_1 < G_2$, and $(X_{(49)}, X_{(50)})$ is mutual iff $G_{49} < G_{48}$.

Hint 3: Because the spacings G_i are exchangeable and continuous, the probability that an interior gap is the smallest among its triple (G_{i-1}, G_i, G_{i+1}) equals $1/3$, while the probability an endpoint gap is smaller than its single neighbor equals $1/2$.

707. Count Triangular Regions I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $m = 10$ be the number of segments from A and $n = 10$ the number from B . Draw all m segments from A first: these radiate from A to the opposite side and split the triangle into $m + 1$ regions. Now, what happens when segments from B are drawn?

708. Count Triangular Regions II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Count Triangular Regions I" first.

709. Repeated Erase

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion. Condition on the first number K that is erased, noting that this reduces the problem to the same situation with one fewer number.

Hint 2: To solve the recursion, compute smaller values and guess a pattern. Then verify the pattern by substitution.

710. Everything Odd

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

Hint 2: Use the number of faces that have appeared an odd number of times as the state. For example, if the roll sequence is 112133, then the state equals 2 because two faces, 1 and 2, have appeared an odd number of times.

711. Expectation Lower Bound

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the tail-integral formula $E[X] = \int_0^\infty \Pr(X > t)dt$.

712. Expected Stock Price

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let M_i be the daily multiplicative factor on day i (so $M_i \in 1.05, 0.97$). The price after 10 days is

$$S_{10} = 100 \prod_{i=1}^{10} M_i$$

Hint 2: $E[XY] = E[X]E[Y]$ if X and Y are independent.

713. Even Lattice Points[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For two lattice points (x_1, y_1) and (x_2, y_2) , how many lattice points lie on the line segment joining them (including the endpoints)?

Hint 2: For two lattice points (x_1, y_1) and (x_2, y_2) the number of lattice points on the line segment joining them equals $\gcd(|x_2 - x_1|, |y_2 - y_1|) + 1$.

714. Polygon Constraints[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Fix a particular label, say $x \in \{1, \dots, 10\}$. Every occurrence of x at a vertex can contribute at most two distinct neighbor labels (one on each side). To ensure that x appears adjacent to each of the other nine labels at least once, the label x must occur on at least $\lceil 9/2 \rceil = 5$ different vertices.

715. Divisible Permutation[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The number of pairs (n, m) with $n \mid m$ must equal the number of pairs with $a_n \mid a_m$, but since each pair of the former type is also of the latter type, the converse must be true as well. Thus, $n \mid m$ if and only if $a_n \mid a_m$.

Hint 2: For each $n = 1, 2, \dots, 6$, the number of values divisible by n uniquely determines n , so $n = a_n$. What about other numbers?

716. Lottery[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Notice that your chances of winning remain the same if we imagine the lottery instead drawing all tickets in random order, and awarding prizes to the holder of the earliest odd-numbered ticket drawn and to the holders of all even-numbered tickets drawn before it.

717. Central Cities[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If there are h central cities, every pair of central cities must be joined by a road, and every central city must connect to each of the $10 - h$ non-central cities. Thus even if we put no roads among the non-central cities, the roads forced by the central cities already number

$$\binom{h}{2} + h(10 - h) = \frac{h(19 - h)}{2}$$

718. Expected Wait Time III[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Measure time from 10:00. Let $T_A \sim \text{Unif}(0, 60)$ and $T_B \sim \text{Unif}(0, 60)$ be the two independent start offsets, and let your arrival U be uniform on $[60, 240]$ (11:00–14:00). Because the arrival-window length $240 - 60 = 180$ is a multiple of both periods 3 and 5, the remainder of U modulo 3 is uniform on $[0, 3)$ and the remainder modulo 5 is uniform on $[0, 5)$. Moreover these remainders are independent of T_A and T_B . Therefore the residual time until the next Bus A at your arrival is distributed as $X \sim \text{Unif}(0, 3)$ and the residual until the next Bus B is $Y \sim \text{Unif}(0, 5)$, with X and Y independent. Your wait is $W = \min(X, Y)$.

Hint 2: To compute $E[W]$, use the tail integral formula.

719. Expected Wait Time IV[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Because both schedules start together at 10:00 and are periodic, their combined pattern repeats every $\text{lcm}(3, 5) = 15$ minutes. The arrival window from 11:00 to 14:00 is 3 hours = 180 minutes, which is an integer multiple of 15. Thus it suffices to compute the expected wait when your arrival is uniform in a single 15-minute cycle.

Hint 2: Inside one 15-minute cycle (measuring minutes after the cycle start) Bus A arrives at 0, 3, 6, 9, 12 and Bus B at 0, 5, 10. The ordered event times are 0, 3, 5, 6, 9, 10, 12, 15, so the open gaps between consecutive arrivals have lengths 3, 2, 1, 3, 1, 2, 3.

Hint 3: If you land uniformly in a gap of length g , your waiting time is uniform on $[0, g]$ so its conditional mean is $g/2$. Just use law of total expectation.

720. Expected Occupied Seats[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion.

721. N-Palindrome[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: In a palindrome, every digit must occur an even number of times except possibly one digit, which may occur an odd number of times if the total length is odd.

722. Maximum Fraction Sum[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For fixed denominators $a < b$ (with sum 10), we maximize the sum of the fractions by giving the smaller denominator as large a numerator as possible: $8/a + 1/b$.

723. Circular Seating[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry. Only Alice and the guests seated immediately to her left and right matter.

724. 2, 4 and 6[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The polynomial simplifies in a convenient way:

$$F(x, y, z) = (x + 1)(y + 1)(z + 1) - 1.$$

Thus when a number n appears on the board we automatically have $n + 1$ appearing in the multiplicative set generated by the numbers $2 + 1, 4 + 1, 6 + 1 = 3, 5, 7$. Conversely, whenever a product of three already-available “ $n + 1$ ” values appears, the corresponding product minus one is produced by the rule. It is therefore natural to work with the shifted set

$$S = \{m = n + 1 : n \text{ appears on the board}\}.$$

Hint 2: Initially S contains 3, 5, 7. The rule says that if $a, b, c \in S$ then $abc \in S$ (because the corresponding $(abc) - 1$ will be written on the board). So S is closed under multiplying any three of its elements.

725. Half Permutation[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: 1 must be the first number in the permutation, this happens with probability $\frac{1}{14}$. 2 must come before 4, 5, 8, 9, 10, and 11, this happens with probability $\frac{1}{7}$. Continue in the same manner for 3, 4, 5, 6, 7.

726. Proportional Coordinates[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: By linearity of expectation and symmetry it suffices to compute $E[y]$ and then double it, since $E[x] = E[y]$ and $E[x + y] = 2E[y]$.

727. Grid Frog II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve “Grid Frog I” and “Domino Tiling II” first.

728. Max Min Ratio[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the law of total expectation conditioning on the maximum:

$$E\left[\frac{m}{M}\right] = E\left[E\left[\frac{m}{M} \mid M\right]\right].$$

Hint 2: Conditioning on $M = b$ fixes one sample at b and leaves the other $n - 1$ samples independent and uniformly distributed on $[0, b]$. Thus, given $M = b$, the minimum m is the minimum of $n - 1$ iid $\text{Unif}(0, b)$ variables.

729. Increasing Random Variables

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use geometric probability.

Hint 2: The total region is a cuboid with side lengths 1, 2 and 3. The favorable region can be computed using a triple integral.

730. Largest In Smallest

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the law of total probability by conditioning on the value of $A_1 = x$.

Hint 2: Since the variables are independent,

$$\Pr(A_1 > \max(A_2, A_3, A_4)) = \int_0^2 f_{A_1}(x) \Pr(A_2 < x) \Pr(A_3 < x) \Pr(A_4 < x) dx.$$

Here $f_{A_1}(x)$ is the density of a $\text{Unif}(0, 2)$ random variable.

731. Alice Rolled The 6

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the game end on Alice's k -th roll, so the total number of rolls is $2(k - 1) + 1 = 2k - 1$. First compute the conditional probability that the game ends on Alice's k -th roll (given that the game ends on Alice's turn), and then use the standard expectation formula $\sum_x xP(X = x)$.

732. Given LCM

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If we have prime factorizations for x and y , then the LCM is obtained by taking, for each prime, the largest exponent appearing in the two numbers. Use this fact to determine what exponents are allowed in x and y .

733. Conditional Normal I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: $-X$ has the same distribution as X (since $X \sim N(0, 1)$).

734. Conditional Normal II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: $-X$ has the same distribution as X (since $X \sim N(0, 1)$).

735. Conditional Normal III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: $-X$ has the same distribution as X (since $X \sim N(0, 1)$).

736. Triangle Formation III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use geometric probability.

Hint 2: We require the triangle inequality to be satisfied. In addition, for the triangle to be obtuse, we must have $1^2 > X^2 + Y^2$.

737. Same Part

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: All orderings of the six distinct points along the stick are equally likely.

738. Elevator I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If the elevator is above your floor when you arrive, it will be moving downward when it next reaches your floor. If it is below your floor at your arrival time, it will be moving upward when it reaches your floor.

739. Unfair Advantage

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: A brute-force solution is possible, but a much simpler solution exists using symmetry. Can you identify any symmetry?

Hint 2: Partition outcomes into the four mutually exclusive, exhaustive events:

Event A: Player 1 has at least one ace among his 13 cards and gets his first ace before Player 2 gets any ace. Call its probability p . Event B: Player 2 has at least one ace among his first 13 cards and gets his first ace before Player 1 gets any ace. By symmetry $P(B) = p$. Event T (tie): Both players have at least one ace among their first 13 cards and their first aces occur in the same simultaneous round. Call its probability t . Event N: Neither player has any ace among the first 13 cards (all 4 aces lie in the final 26 cards). Its

probability is $\frac{\binom{26}{4}}{\binom{52}{4}}$.

These four probabilities sum to 1. So, we only need to calculate the tie probability t . Calculate that using brute force.

740. Monotonic Subsequences

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: First select the positions where the even numbers will be placed. Once these positions are fixed, how many different permutations are possible?

741. Possible Cover

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use proof by coloring.

742. Alternating Sum Sum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the subsets of I be divided into two classes A and B such that each subset in A contains the number 30 and each subset in B does not contain the number 30, and define the alternating sum of the empty set to be zero. Let the set $a_1 < a_2 < \dots < a_k < 30$ in A and the set $a_1 < a_2 < \dots < a_k$ in B be paired. What is the total of the alternating sums of such a pair?

743. 3 Eggs

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "2 Eggs" first.

744. Expected Minimum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use tail sum formula.

745. Grid Arrangement IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since only the parity of the sums matters, we can reduce all numbers modulo 2. Among $1, \dots, 9$, there are five odd numbers and four even numbers. Thus we can assume that we are placing five 1s and four 0s into the nine cells of the grid.

746. Subset Length Sum[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use symmetry.

Hint 2: The sum of all numbers from 1 to 13 is $1 + 2 + \cdots + 13 = 91$. Take any subset of $\{1, \dots, 13\}$ whose elements add up to some value k . The elements not chosen form its complement, and their sum is $91 - k$. Thus every subset with total k naturally pairs with another subset whose total is $91 - k$. In other words, the number of subsets giving sum k is exactly the same as the number giving sum $91 - k$.

747. Extended Rock-Paper-Scissors[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use recursion.**748. King Before Ace**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Solve the “first ace” problem first.

Hint 2: Just as in the “first ace” problem, think of the 4 kings and 4 aces as divider cards. These 8 special cards split the deck into 9 gaps containing the remaining 44 non-special cards. By symmetry, the expected number of ordinary cards in each gap is $44/9$.

It is given that the first special card is a king. The number of cards between the first king and the first ace depends on the position at which the first ace appears among the remaining 7 special cards. We can therefore condition on the location of the first ace within these 7 cards and use the law of total expectation.

749. To Infinity And Beyond II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write the increments $U_n := X_n - X_{n-1}$ for $n \geq 2$. By construction, conditional on $X_{n-1} = x$ the increment U_n is $\text{Uniform}(0, 2^{-n})$; in particular its law does not depend on x . Hence each U_n (for $n \geq 2$) is independent of the past and of X_1 , and we have

$$X = X_1 + \sum_{n=2}^{\infty} U_n$$

Hint 2: Because the U_n are independent, variances add.

750. Everything Contained[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Label the intervals $[x_i, y_i]$ for $i = 1, \dots, 8$. After sampling we have 16 labeled endpoints. Count the number of permutations of the 16 endpoints that produce valid intervals (the sample space), and then count the number of permutations in which one interval contains all the others.

751. Disjoint Arcs[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Consider all circular permutations of A, B, C , and D . By symmetry, each permutation is equally likely. Which permutations result in disjoint arcs?

752. Maximize RPS Wins[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Show that achieving n points is possible, but achieving $n + 1$ points is impossible.

753. Maximum Within An Interval[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Note that $\Pr(\max X_i \leq t) = \left(\frac{t}{5}\right)^5$.
Can you express the required probability in terms of this?

754. Bulb Toggle I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: A bulb numbered n is toggled once for every positive divisor of n . (Person d toggles bulb n exactly when $d \mid n$.) Since bulbs start off, the final state of bulb n is:

- on if n has an odd number of divisors,
- off if n has an even number of divisors.

755. Ten Cars Meet[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: All ten cars are present at some common time iff the earliest arrival and the latest arrival are within 15 minutes, i.e.

$$\max_i X_i - \min_i X_i \leq 15,$$

where $X_1, \dots, X_{10}$ are iid Uniform $[0, 60]$.

Hint 2: Just derive the joint density of the minimum m and maximum M of n iid Uniform $[0, 60]$ and integrate.

Hint 3: Here is a way to derive the joint density of the minimum m and maximum M of n i.i.d. Uniform $[0, 60]$ random variables.

Let the minimum be u and the maximum be v , with $0 \leq u \leq v \leq 60$. For this to occur, one of the n values must fall in $[u, u + du]$, one of the remaining $n - 1$ values must fall in $[v, v + dv]$, and the other $n - 2$ values must lie in $[u, v]$, which has length $v - u$.

The minimum can be chosen in n ways and then the maximum in $n - 1$ ways. Thus the probability of this

event is

$$n(n-1) \frac{du}{60} \frac{dv}{60} \left(\frac{v-u}{60} \right)^{n-2}.$$

Hence the joint density of m and M is

$$f_{m,M}(u, v) = \frac{n(n-1)}{60^n} (v-u)^{n-2}, \quad 0 \leq u \leq v \leq 60.$$

For $n = 10$, this becomes

$$f_{m,M}(u, v) = \frac{90}{60^{10}} (v-u)^8.$$

756. Perfect Rank Guess

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use recursion/dynamic programming. Assume there are n values in total (instead of 4). The first rank can take any value from 1 to n . Evaluate all n possibilities and select the one that maximizes the overall probability of winning.

Hint 2: Note that the optimal probability of winning does not depend on the length of the interval, it depends only on the number of points. Thus, even if the numbers were chosen from the interval $[a, b]$, the optimal probability of winning would remain the same. Let P_n denote the optimal probability of winning when there are n points in the interval under the rules of the problem.

Suppose the first observed value is t . Also, suppose that you choose some rank r right now. Let $a = r - 1$ and $b = n - r$. The probability your choice is correct and you then succeed on the remaining groups can be at most

$$\Pr(\text{true rank} = r \mid t) \times (\text{best possible continuation success}).$$

But the best possible continuation success - no matter how clever you are later - is exactly $P_a P_b$ (the optimal success probabilities on the left-interval group and right-interval group). Therefore for any strategy that chooses r at this point,

$$\text{conditional success} \leq \Pr(\text{rank} = r \mid t) \cdot P_a P_b$$

Denote $\Pr(\text{rank} = r \mid t) \cdot P_a P_b$ by $S_r(t)$. Therefore,

$$\text{conditional success} \leq S_r(t)$$

That is true pointwise for every r . So whatever your strategy does at this moment, its conditional success $v(t)$ cannot exceed $\max_r S_r(t)$. Now define the strategy that, for this t , chooses

$$r^*(t) = \arg \max_r S_r(t)$$

and thereafter plays optimally on the two subproblems (using the same strategy). This is the optimal strategy.

757. Interesting Average III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We are interested in finding the expected value of $\sum_{i=0}^9 d_i \cdot 10^i$ where each digit d_i is chosen independently with a uniform distribution over its possible values.

758. Seven Colors II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use symmetry.

Hint 2: We are given additional information about the first half of the sequence (positions 1 to 350). However, a specific violet ball is still equally likely to lie in the first half (1–350) or the second half (351–700). Prove this using symmetry.

759. Seven Colors III[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Solve "Seven Colors II" first.**760. Maximum Till Now II**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use tail sum formula.**761. Partial Exponential I**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Because the X_i are iid the information S_{10} treats every coordinate the same, so the conditional expectations $\mathbb{E}[X_i | S_{10}]$ are all equal.

762. Partial Exponential II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write $S_{10} = S_4 + T$ with $T = \sum_{i=5}^{10} X_i$. Independence of the X_i implies the block $X_5, \dots, X_{10}$ (hence T) is independent of S_4 , so $\mathbb{E}[T | S_4] = \mathbb{E}[T]$.

763. Signal To Noise Ratio[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Compute $\frac{\mathbb{E}[Z_p]}{\sqrt{\text{Var}(Z_p)}}$ as a function of p .

764. Your Choice[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let k_n be the number of cards showing n . If the friend picks n his expected payout equals $\frac{n \cdot k_n}{100}$. To minimize the friend's best choice we must minimize

$$M = \max_{1 \leq n \leq 100} n \cdot k_n$$

subject to the integer constraints $k_n \geq 0$ and $\sum_{n=1}^{100} k_n = 100$.

Hint 2: For a candidate value M we have the constraint $k_n \leq \lfloor M/n \rfloor$ for every n , so a necessary and sufficient feasibility condition is

$$S(M) = \sum_{n=1}^{100} \left\lfloor \frac{M}{n} \right\rfloor \geq 100.$$

765. Monochromatic Removal

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Use symmetry!

Hint 2: Think about the moment two turns before the process ends, when there are still marbles of both colors but the game will finish in exactly two more rounds. Call the remaining counts r reds and g greens. If the game were to end in exactly 2 rounds, what are the possible permutations of these marbles? Which permutation is more likely to occur?

766. Exponential Inequality I

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Use the law of total probability (for continuous random variables).

767. Tired Rabbit

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Use tail sum formula.

768. Exponential Inequality II

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Solve "Exponential Inequality I" first. Then recall the memoryless property of exponential distributions.

769. Give And Take

[[Question](#) ↑ | [Solution](#) ↓ | ↑]

Hint 1: Suppose the total number of Give and Take is fixed. Among all sequences containing those turns, which ordering maximizes the expected total amount collected? Think about small values of takes, like 1, 2, or 3.

Hint 2: It can be proven that any adjacent pair “Take then Give” can be swapped to “Give then Take” with a strict gain. Thus, it is optimal to perform all Take actions at the end.

770. Maxima Sum I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let N be the number of selected numbers among 3, 5, 7 that are greater than the number immediately to their left. The sum is odd exactly when N is odd, because only odd selected numbers affect parity.

Hint 2: For any set $x_1 < x_2 < \dots < x_k$ chosen from 3, 5, 7, the probability that all of them are greater than the number immediately to their left is

$$\frac{x_1 - 1}{7} \cdot \frac{x_2 - 2}{6} \cdots \frac{x_k - k}{8 - k}.$$

Hint 3: Use the principle of inclusion and exclusion.

771. Maxima Sum II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let N be the number of selected numbers among 3, 5, 7 that are greater than the number immediately to their left. The sum is odd exactly when N is odd, because only odd selected numbers affect parity.

Hint 2: For any set $x_1 < x_2 < \dots < x_k$ chosen from 3, 5, 7, the probability that all of them are selected is

$$\frac{x_1 - 1}{7} \cdot \frac{1}{8 - x_1} \cdot \frac{1}{8 - x_2} \cdots \frac{1}{8 - x_k}.$$

Hint 3: Use the principle of inclusion and exclusion.

772. Nearest Diagonal II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the tail integral formula.

773. Fixed Maximum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let S be the sum of the 6 numbers and let M be their maximum. We want $\mathbb{E}[S \mid M = 15]$. It is very easy to calculate expectations of the form $\mathbb{E}[S \mid M \leq m]$, for some fixed m . How can we use this?

Hint 2: The event $M = 15$ means $M \leq 15$ but not $M \leq 14$. So, we can write

$$\mathbb{E}[S \mid M = 15] = \frac{\mathbb{P}(M \leq 15)\mathbb{E}[S \mid M \leq 15] - \mathbb{P}(M \leq 14)\mathbb{E}[S \mid M \leq 14]}{\mathbb{P}(M = 15)}.$$

774. It Took Ten Turns

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Given $S_9 = x$ and $T = 10$, the 10th draw is uniform on $(1 - x, 1)$, so

$$\mathbb{E}[X_{10} \mid S_9 = x, T = 10] = \frac{(1 - x) + 1}{2} = 1 - \frac{x}{2}.$$

Hence

$$\mathbb{E}[S_{10} \mid T = 10] = \mathbb{E}[S_9 + X_{10} \mid T = 10] = \mathbb{E}\left[S_9 + 1 - \frac{S_9}{2} \mid T = 10\right] = 1 + \frac{1}{2}\mathbb{E}[S_9 \mid T = 10].$$

So, we only need to find $\mathbb{E}[S_9 \mid T = 10]$.

Hint 2: $T = 10$ means that after 9 draws the sum is still at most 1, but after the 10th draw it becomes greater than 1.

It can be proven that the conditional density of S_9 given $T = 10$ is proportional to

$$f_{S_9|T=10}(x) \propto x^9.$$

Then we can find the conditional density of S_9 by normalizing. Once we have the conditional density, we can easily find $\mathbb{E}[S_9 \mid T = 10]$.

775. Increasing Decreasing II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics. To find the number of permutations that satisfy the given pattern of inequalities, use recursion (but how:).

Hint 2: Place the largest number 8. Since the peaks occur at the even positions, 8 must lie in one of those positions. If 8 is in position $2k$, then the numbers to its left and right form two smaller problems of the same type.

776. Swap Swap

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation.

Hint 2: Let I_j be the indicator variable for the event that the number j is in its original position after 5 swaps. Then the required expected number is

$$\mathbb{E}\left[\sum_{j=1}^{16} I_j\right] = \sum_{j=1}^{16} \mathbb{E}[I_j].$$

By symmetry, all 16 numbers have the same probability of being fixed, so if p_5 denotes the probability that number 1 is in its original position after 5 swaps, then the answer is $16p_5$.

Hint 3: p_5 can be found using recursion.

Hint 4: Define p_n to be the probability that after n swaps, number 1 is in its correct position. Then after one more swap, there are two cases.

If number 1 is already correct, it stays correct provided it is not selected for swapping. The number of pairs of positions not involving position 1 is $\binom{15}{2}$, out of the total $\binom{16}{2}$, so this happens with probability $\frac{\binom{15}{2}}{\binom{16}{2}} = \frac{7}{8}$.

If number 1 is not currently correct, then it becomes correct only if the swap chooses exactly the two positions consisting of its current position and the correct position. There is exactly one such pair, so this happens with probability $\frac{1}{\binom{16}{2}} = \frac{1}{120}$.

777. Exponential Pentagon

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Five positive numbers can form a pentagon exactly when no one of them is at least the sum of the other four. So the only way the five lengths fail to form a pentagon is if one side is too long, meaning that for some i we have $X_i > \sum_{j \neq i} X_j$.

These five bad events are disjoint, because two different sides cannot both be larger than the sum of the other four. Therefore, by symmetry,

$$P(\text{pentagon}) = 1 - 5P(X_1 > X_2 + X_3 + X_4 + X_5).$$

Hint 2: $P(X_1 > X_2 + X_3 + X_4 + X_5)$ can be found using the result from the problem "Exponential Inequality II". Solve it first!

778. Constrained Maximum

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: There are many ways to do this. One approach is to let $p = P(X = 0)$. Let Y denote the random variable describing the distribution of X given that X is nonzero. Find variance of Y in terms of p and use the fact that variance is always non-negative to get an inequality in terms of p . Then show that the maximum value allowed by this inequality can be attained.

779. Strange Product

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

No hints provided.

780. Keep Multiplying

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $f(x)$ be the expected additional time needed to go below $\frac{1}{2}$, given that the current value is x , where $\frac{1}{2} \leq x \leq 1$. Use the law of total expectation, then differentiate the resulting equation to get a differential equation.

781. Maximum Exponential

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the tail integral formula.

782. Winner Takes All

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let N be the total number of rounds played, and let M_i be the number of rounds in which player i takes part. Since every round involves exactly two players, we have

$$2N = M_1 + M_2 + M_3 + M_4.$$

Taking expectations gives

$$2\mathbb{E}[N] = \mathbb{E}[M_1] + \mathbb{E}[M_2] + \mathbb{E}[M_3] + \mathbb{E}[M_4].$$

Hint 2: To compute $\mathbb{E}[M_i]$ for each player, you can model each player's money as a simple symmetric random walk.

783. Spaced Uniforms I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think of placing the points from left to right. After the first point is chosen, reserve a segment of length $1/20$ immediately to its right (no point can lie in this reserved segment if the condition of the problem is to be satisfied). After the second point is chosen, reserve another segment of length $1/20$ to its right (again, no point can be placed in this reserved segment), and continue this way. The total length of the interval is 1. What is the length of the region in which the points can be placed so that the condition of the problem is satisfied?

784. Strictly Behind

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use combinatorics. The total number of possible paths over 10 minutes is $2^{10} \cdot 2^{10} = 2^{20}$, since each of the two independently makes one of two moves every minute. Now, try to count how many of these paths satisfy the required conditions.

Hint 2: Let D_t = frog position – rabbit position after t minutes. Then $D_0 = 4$, and each minute D_t changes by $+2$, 0 , or -2 . The value stays positive exactly when the rabbit is strictly behind the frog. We want two things to happen by time 10: both the rabbit and the frog return to where they started, and D_t never drops to 0 or below. The first condition is straightforward to handle. For the second condition, use the reflection principle.

785. Spaced Uniforms II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Solve "Spaced Uniforms I" first.**Hint 2:** Use tail integral formula.**786. Second Last Roll**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** The first roll and the second last roll have the same probability distribution (but how?).**787. All Suits**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Use linearity of expectation.**Hint 2:** Let T be the number of cards needed. If we look at the cards one at a time, then $T = \sum_x I_x$, where $I_x = 1$ if the card x is revealed before all four suits have appeared, and $I_x = 0$ otherwise. By symmetry, every card has the same probability of having this property, so using linearity of expectation,

$$\mathbb{E}[T] = 52p,$$

where p is the probability that a fixed card is revealed before all four suits have been seen.**Hint 3:** Fix one particular card, and suppose it is a heart. Let A_S be the event that no spade has appeared before this card, and define A_H, A_C, A_D, A_S similarly for the four suits. Then the chosen card is revealed before all four suits have been seen exactly when at least one of these events occurs. So, $p = \Pr(A_H \cup A_C \cup A_D \cup A_S)$. This can be calculated using principle of inclusion-exclusion.**788. Subset Divisibility**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]**Hint 1:** Consider any 6-element subset of $\{1, 2, 3, \dots, 29\}$. Now create a new subset by adding 1 to each element and wrapping around cyclically so that 29 goes back to 1. Repeat this process 28 more times to get 29 subsets in all. If the original subset has sum congruent to $r \pmod{29}$, then after adding k to every element, the sum becomes congruent to $r + 6k \pmod{29}$. Among these 29 sums, how many are congruent to 0 $\pmod{29}$?**Hint 2:** Among the 29 subsets generated in the first hint, exactly one has sum congruent to 0 $\pmod{29}$.**789. Hit 995**[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For a simple symmetric random walk on the integers, if the walk starts at x with $-a < x < b$, then the probability of hitting b before $-a$ is

$$\frac{x+a}{a+b}.$$

790. Penguin Walk III

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Penguin Walk II" first.

Hint 2: Use the strong markov property.

791. Penguin Walk IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve Penguin Walk II and Penguin Walk III first.

Hint 2: Use conditional probability. The answer is simply the probability that the penguin first steps to the right from position 4 and then eventually reaches position 7, divided by the probability that it eventually reaches position 7 starting from position 4.

792. Uniform Product

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If $X \sim \text{Unif}(0, 1)$, the random variable $-\ln X$ has the $\text{Exp}(1)$ distribution.

Hint 2: Let $S = -\ln Y$. Since $Y = X_1 X_2 X_3 X_4 X_5 X_6 X_7$, we have

$$S = -\ln X_1 - \ln X_2 - \ln X_3 - \ln X_4 - \ln X_5 - \ln X_6 - \ln X_7.$$

Now for each $X_i \sim \text{Unif}(0, 1)$, the random variable $-\ln X_i$ has the $\text{Exp}(1)$ distribution.

Hint 3: Because the X_i are independent, the seven terms are independent $\text{Exp}(1)$ random variables, so S has a Gamma distribution with shape 7 and rate 1.

793. Last Uniform

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For $0 \leq x \leq 1$, let $f(x)$ be the expected value of the last draw if we use the stopping rule "keep drawing until the sum exceeds x " instead of 1. We want $f(1)$.

Use the law of total expectation, then differentiate to get a differential equation.

794. Consecutive Stocks

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the pigeonhole principle.

Hint 2: Let $S_0 = 0$, and for each $k = 1, 2, \dots, 365$, let S_k be the total number of shares bought in the first k days. Then

$$0 = S_0 < S_1 < \dots < S_{365} = 600,$$

because the trader buys at least one share each day.

Now consider the two sets of numbers

$$S_0, S_1, \dots, S_{365}$$

and

$$S_0 + 129, S_1 + 129, \dots, S_{365} + 129.$$

In total, set A contains 366 values and set B contains 366 values, giving $366 \times 2 = 732$ values altogether. How many distinct values are there across the two sets? If this number is less than 732, then at least two of the values must be equal.

795. Penguin Walk V

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Penguin Walk I" first.

796. Penguin Walk VI

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use law of total expectation by conditioning on the first step and use the strong markov property.

797. Penguin Walk VII

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use law of total expectation by conditioning on the first step and use the strong markov property.

798. Penguin Walk VIII

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Penguin Walk VI" first.

799. Row Expectation I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since the grid is filled by a random permutation of $1, 2, \dots, 36$, the twelve numbers in the second row are just a uniformly random set of 12 distinct numbers chosen from $1, 2, \dots, 36$. Use the tail sum formula.

Hint 2: Let X be the minimum entry in the second row. To have $X \geq k$, every number in the second row must come from the set $k, k+1, \dots, 36$, which has $37-k$ numbers. So

$$\Pr(X \geq k) = \frac{\binom{37-k}{12}}{\binom{36}{12}}.$$

Therefore,

$$\mathbb{E}[X] = \frac{1}{\binom{36}{12}} \sum_{k=1}^{36} \binom{37-k}{12}.$$

Now let $j = 37 - k$. As k runs from 1 to 36, j runs from 36 down to 1, so

$$\mathbb{E}[X] = \frac{1}{\binom{36}{12}} \sum_{j=1}^{36} \binom{j}{12}.$$

Use the hockey stick identity to evaluate the sum.

800. Gojo's Six Eyes

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation by defining indicator variables for each edge, where each indicator equals 1 if the edge is intersected by the plane and 0 otherwise.

801. XY To The Power Z

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $U = XY$. First find the density of U on $(0, 1)$. A clean way is to first find the CDF $F_U(u) = \Pr(XY \leq u)$ by conditioning on X and then averaging over X .

Hint 2: After that, condition on $U = u$,

$$W = u^Z.$$

For fixed $u \in (0, 1)$, figure out the range of W , then compute the conditional density (again by computing the cumulative distribution first and then differentiating) and average over u .

802. Penguin Walk IX

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We know that $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. We have already calculated $\mathbb{E}[X]$ in Penguin Walk I.

Hint 2: We can calculate $\mathbb{E}[X^2]$ using recursion and the Markov property, similar to how we calculated $\mathbb{E}[X]$ in Penguin Walk I.

803. Penguin Walk X

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Penguin Walk IX" first.

804. Ten-Dimensional Ball

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: We know that $\text{Var}(X_1) = \mathbb{E}[X_1^2] - \mathbb{E}[X_1]^2$. By symmetry of the ball about the origin, $\mathbb{E}[X_1] = 0$, so the problem reduces to finding $\mathbb{E}[X_1^2]$.

Hint 2: Let $R = \sqrt{X_1^2 + \cdots + X_{10}^2}$. For a uniform point in a 10-ball, $\Pr(R \leq r)$ is proportional to the volume of a 10-ball of radius r , so it grows like r^{10} . Use this to find the distribution of R , compute $\mathbb{E}[R^2]$, and then use symmetry to relate $\mathbb{E}[R^2]$ to $\mathbb{E}[X_1^2]$.

805. Row Expectation II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think of the 6 numbers in the first row as 6 bars placed among 24 positions, with the remaining 18 positions as stars. The third largest number is the same as the location of the fourth bar from the left.

Hint 2: The 18 stars are split into 7 blocks, so each block has expected size $\frac{18}{7}$ by symmetry.

806. Conditional Normal IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let

$$R^2 = N_1^2 + N_2^2.$$

Because (N_1, N_2) is a standard bivariate normal vector, its distribution is rotationally symmetric in the plane.

Hint 2: Conditioning on $R^2 = 6$ means we are looking at points on the circle of radius $\sqrt{6}$. By rotational symmetry, the direction on this circle is uniform, so we may write

$$(N_1, N_2) \mid (R^2 = 6) = (\sqrt{6} \cos \theta, \sqrt{6} \sin \theta),$$

where $\theta \sim \text{Unif}(0, 2\pi)$.

807. Infinite Sum Expectation

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Rewrite S as an infinite sum with an indicator:

$$S = \sum_{i=1}^{\infty} \frac{X_i}{2^i} \mathbf{1}_{N \geq i}.$$

Then use linearity of expectation, and focus on computing $\mathbb{E}[X_i \mathbf{1}_{N \geq i}]$ for a fixed i .

808. Secret Marbles I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Focus on one draw first. If A puts a marbles and B puts b marbles, then the probability the drawn marble belongs to A is $\frac{a}{a+b}$. So the expected payout to A from one draw is $\frac{a}{a+b}(100 - a)$.

Since there are two draws with replacement, the total expected payout is just twice the one-draw expected payout.

Hint 2: Fix b and maximize the one-draw expected payout as a function of a . Differentiate with respect to a , set the derivative equal to 0, and solve for the best response a^* . Then use symmetry to set $a = b = x$.

809. Secret Marbles II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write the expected payoff to A by separating the two cases: the marble comes from A , or it comes from B . That gives a function of a and b . Then focus on one player at a time by fixing b and optimizing only over a .

Hint 2: Differentiate the expected payoff with respect to a , set the derivative equal to 0, and solve for the best response a^* in terms of b . Then use symmetry at equilibrium by setting $a = b = x$. This should give a simple equation for x , and that value is the number of marbles player A should choose.

810. Second Last I[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Define $f(x)$ as the expected final record value when the current maximum is x . Use the law of total expectation and differentiate it to get a differential equation for $f(x)$. Split into two cases: the next sample is below x , in which case the process stops, or it is above x , in which case the new state becomes that larger value.

811. Second Last II[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Define $f(x)$ as the expected final record low given that the current minimum is x . Use the law of total expectation by splitting into two cases: the next sample exceeds x , in which case the process stops and the value remains x , or it is less than x , in which case the new state becomes that smaller value (uniform on $(0, x)$). This leads to an integral equation for $f(x)$. Differentiate it to obtain a differential equation for $f(x)$.

812. Powerful Scaling[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Condition on X_n and rewrite the next variable as $X_{n+1} = X_n^3 U$, where $U \sim \text{Unif}(0, 1)$. Then compute $\mathbb{E}[X_{n+1}^p \mid X_n]$ by pulling the X_n -term outside the conditional expectation.

Hint 2: Use the fact that $\mathbb{E}[U^p] = \frac{1}{p+1}$ for $U \sim \text{Unif}(0,1)$. Then work backward from $\mathbb{E}[X_5]$, repeatedly applying the same step to reduce the power each time until you reach X_1 .

813. One Before Ten

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: To find the expected first card, it is enough to determine the probability that each possible card appears first, conditioned on the event that 10 appears somewhere to the right of 1.

814. Frog Distance I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let m and M be the minimum and maximum frog positions. First find the joint density $f_{m,M}(u,v)$ by looking at the event that one frog falls near u , one frog falls near v , and the remaining frogs lie between them.

Hint 2: Once you have $f_{m,M}(u,v)$, the required probability is obtained by integrating over all pairs (u,v) with $\frac{1}{4} \leq v - u \leq \frac{1}{2}$.

815. Frog Distance II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Frog Distance I" first.

816. Additional Info Coin Flips II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Using conditional probability,

$$P(C > B \mid B > A) = \frac{P(A < B < C)}{P(A < B)}$$

First find $p = P(A < B)$ by conditioning on the first tosses of A and B . If both get tails, the situation restarts and gives $\frac{1}{4}P(A < B)$. If A gets tails and B gets heads, then $A < B$ already happens. The other cases do not lead to $A < B$. Set up a recursion for p .

Hint 2: Now let $q = P(A < B < C)$ and condition on the first tosses of all three players. If all three get tails, the situation restarts and gives $\frac{1}{8}q$. If A gets heads while B and C both get tails, then the remaining probability is exactly $P(B < C)$, which is the same as $P(A < B)$. The other cases do not lead to $A < B < C$. Set up a recursion for q .

817. Power Distance II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let $S_{30} = X_1 + X_2 + \cdots + X_{30}$, where each X_i is $+1$ or -1 with probability $\frac{1}{2}$. Expand S_{30}^4 . After taking expectation, any term with some X_i appearing an odd number of times disappears, because $\mathbb{E}[X_i] = 0$ and the variables are independent.

Hint 2: So only two kinds of terms matter: one variable appearing four times, and two different variables each appearing twice. The first contributes 30. For the second, the coefficient of $X_i^2 X_j^2$ is 6 for each pair $i < j$, and since $X_i^2 = 1$, this gives $6\binom{30}{2}$. Add the two contributions.

818. Normal Ratio I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Think of the pair (X, Y) as a random point in the plane. For independent standard normal variables, its distribution is rotationally symmetric, so the only thing that matters is the angle the point makes with the axes.

Hint 2: Rewrite $\frac{X}{Y}$ in terms of the polar angle. Since $\frac{X}{Y} = \cot \Theta$, the inequality

$$1 < \frac{X}{Y} < \sqrt{3}$$

turns into an angular interval. Then use the fact that the angle is uniform on $[0, 2\pi]$ to get the probability from the arc length.

819. Normal Ratio II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Normal Ratio I" first.

Hint 2: Write $X = \sqrt{2}Z$, where $Z \sim N(0, 1)$. Since X and Y are independent, Z and Y are independent standard normal random variables. Then

$$\begin{aligned} \sqrt{2} < \frac{X}{Y} < \sqrt{6} \\ \implies \sqrt{2} < \frac{\sqrt{2}Z}{Y} < \sqrt{6}. \\ \implies 1 < \frac{Z}{Y} < \sqrt{3}. \end{aligned}$$

Note that Z and Y are independent random variables with mean 0 and variance 1. So, this problem is equivalent to "Normal Ratio I".

820. Exponential Ratio

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Write the event $1 < \frac{X}{Y} < 2$ as an inequality in the (x, y) -plane. Since both variables are positive, this becomes a region between the lines $x = y$ and $x = 2y$. Use the joint density of two independent exponentials, $f_{X,Y}(x, y) = 2e^{-x-2y}$ for $x > 0$ and $y > 0$, and integrate it over that region.

821. Square Intersection II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the square be the unit square positioned in the first quadrant, with its bottom-left vertex at the origin and its sides parallel to the coordinate axes. For the line to meet the left side and the bottom side, both chosen points must lie in the triangular region near the lower-left corner bounded by the two sides and the diagonal $x + y = 1$.

Hint 2: First find the probability that both points fall inside that triangle. Then, condition on that event and use symmetry inside the triangle: the line through two random points there is equally likely to exit through any of the three side pairs (why?).

822. Increasing Dice Rolls IV

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Count all nondecreasing sequences of length 10, and then count how many of those contain at least one 3.

Hint 2: A nondecreasing sequence of die rolls is determined only by how many 1s, 2s, 3s, 4s, 5s, and 6s appear. We can use stars and bars here.

Hint 3: To count the number of sequences that include a 3, first count the number of nondecreasing sequences that avoid 3 completely by allowing only 1, 2, 4, 5, 6. Then subtract this from the total number of nondecreasing sequences.

823. Hexagon Intersection

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Square Intersection II" first.

Hint 2: Look at the rectangle inside the hexagon for which sides A and D are opposite sides. A line can hit both A and D only if both chosen points lie inside this rectangle.

824. Sum Over One II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let X be the smallest number and Y be the second largest number. Just derive the joint density of (X, Y) and calculate the required probability as a double integral.

825. Last Card Survivor

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Since each card has the same chance of surviving, the expected value of the survivor is just the average of the two labels. Track how many averaging steps each card goes through.

826. Triangle Formation IV[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use geometric probability, and compute the required probabilities by evaluating volumes of regions using triple integrals.

827. Random Tree Construction[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Instead of trying to track the whole tree, look at a single fixed vertex v . We can calculate the final answer by averaging over all vertices.

Hint 2: Pick a vertex created at time i . For each later vertex $j > i$, first calculate the probability that j lies in the subtree of i . Then use linearity of expectation:

$$\mathbb{E}[S(i)] = 1 + \sum_{j>i} \Pr(j \text{ belongs to the subtree of } i).$$

So the whole problem becomes finding the above probability for each later vertex.

828. Subset Subset[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use linearity of expectation using indicator variables. For each non-empty subset S , what is the probability that S is the one that actually gets chosen before any larger set that contains it?

Hint 2: If $|S| = k$, count how many supersets of S there are inside $1, 2, \dots, 8$. Note that among all those supersets, each is equally likely to be the first one chosen.

829. Socks Remaining III[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let R, B , and G be the positions of the last red sock, last blue sock, and last green sock in a random ordering of all 60 socks.

The process stops when the first color runs out, so the stopping time is $\min(R, B, G)$. Therefore, the number of socks remaining is $60 - \min(R, B, G)$. So we only need $\mathbb{E}[\min(R, B, G)]$.

Hint 2: Use the identity for three numbers:

$$\min(R, B, G) = R + B + G - \max(R, B) - \max(R, G) - \max(B, G) + \max(R, B, G).$$

Taking expectations gives

$$\mathbb{E}[\min(R, B, G)] = \mathbb{E}[R] + \mathbb{E}[B] + \mathbb{E}[G] - \mathbb{E}[\max(R, B)] - \mathbb{E}[\max(R, G)] - \mathbb{E}[\max(B, G)] + \mathbb{E}[\max(R, B, G)].$$

Each of the terms above is easy to compute.

830. Expected Head-Tail Ratio[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let τ be the first time tails are strictly ahead of heads. Think about what must be true for τ to equal $2k + 1$: the first $2k$ tosses must be perfectly balanced, with tails never exceeding heads, and the last toss must be a tail. We can use the reflection principle to count the balanced length- $2k$ paths in which tails never exceed heads before the end, and use this to find $\Pr(\tau = 2k + 1)$.

Hint 2: After you get $\Pr(\tau = 2k + 1)$, sum the expected tail proportion over all k and recognize the resulting series as the Taylor expansion of $\arcsin x$.

831. Too Many Slips[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Work in cents and reduce the problem modulo 100. The process then becomes a finite Markov chain on the residues $0, 1, \dots, 99$, and the stopping time is exactly the first return time to state 0. We can calculate this return time using Kac's lemma discussed in the chapter on return times in Markov chains section.

Hint 2: Once we have the expected draws needed to reach an integer, we can use Wald's equality to determine the expected sum of all the values drawn.

832. Gold Rush[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: If $x < y$, then the third player has three possible moves: place the mark just to the left of x (and get x), place it just to the right of y (and get $1 - y$), or place it anywhere between x and y (all "in between" points lead to the same amount of gold- $\frac{y-x}{2}$). Comparing the gold obtained from these three choices determines the third player's optimal move. Equivalent conditions can be written down for the case $y < x$.

Hint 2: Knowing what the third player will do at the end (Hint 1), what should the second player do? Then, knowing what the second player will do, what should the first player do?

833. Expected Number After Swaps[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use the linearity of expectation. What is the expected value of the digit in the i th position?

834. Direction Change III[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Direction Change II" first.

835. Submarine Distance

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the first submarine choose angle $\alpha \in [0, \pi/2]$, and let the second choose angle $\pi - \beta$ where $\beta \in [0, \pi/2]$. Then their positions after t seconds are

$$A(t) = (t \cos \alpha, t \sin \alpha)$$

and

$$B(t) = (10 - t \cos \beta, t \sin \beta).$$

So the squared distance between them is

$$d^2(t) = (10 - t(\cos \alpha + \cos \beta))^2 + t^2(\sin \beta - \sin \alpha)^2.$$

This is an upward opening quadratic in t and hence, its minimum is attained at its vertex $(\frac{-B}{2A})$.

Hint 2: Using trigonometric identities, the minimum distance simplifies to

$$d_{\min} = 10 \left| \sin \left(\frac{\beta - \alpha}{2} \right) \right|$$

Now, we can convert the condition $d_{\min} < 4$ into a condition on $|\beta - \alpha|$. Then interpret (α, β) as a uniform point in the square $[0, \pi/2] \times [0, \pi/2]$, so the probability is just the area of the diagonal band $|\beta - \alpha| < c$ divided by the area of the square.

836. Nearest Paint I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Let the people be arranged from left to right, and let the 11 gaps be $G_0, G_1, \dots, G_{10}$, where G_0 and G_{10} are the two end gaps, and $G_1, \dots, G_9$ are the gaps between consecutive people.

The two end gaps are never painted, because no one paints beyond the nearest neighbor. Also, G_1 and G_9 are always painted. So the only gaps that can sometimes remain unpainted are the seven middle gaps $G_2, \dots, G_8$. It is easy to see that a middle gap is unpainted precisely when it is larger than the two gaps next to it.

Hint 2: Now the expected length of each of the 11 gaps is $\frac{100}{11}$. So for any middle gap, the expected total of the three consecutive gaps around it is $\frac{300}{11}$. Therefore the expected unpainted length contributed by each of the seven middle gaps is $\frac{300}{11} \cdot E$, where E is the average value of the middle variable y over the region $x + y + z = 1$, subject to $y > x$ and $y > z$.

837. Nearest Paint II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Nearest Paint I" first.

838. Normal Inequality V

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Use symmetry.

839. Expected Hits I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For a simple symmetric random walk on $[a, b]$ with absorbing boundaries at a and b , the probability of hitting b before a , starting from x , is

$$P_x(T_b < T_a) = \frac{x - a}{b - a}.$$

Use this first to find the chance of ever reaching $+4$ from 0 , and then again to find the chance of returning to $+4$ after one visit there.

Hint 2: Once the walk reaches $+4$, look at what happens after the next step. From $+4$, it moves to $+5$ or $+3$ with probability $\frac{1}{2}$ each. Compute the probability of coming back to $+4$ from each of those positions before absorption, average them, and then treat repeated returns as a geometric series.

840. Expected Hits II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: For a biased random walk with right-step probability p and left-step probability q , let $r = \frac{q}{p}$. Then the absorbing-probability formula on $[a, b]$ is

$$P_x(T_b < T_a) = \frac{1 - r^{x-a}}{1 - r^{b-a}}.$$

Use this first on $[-10, 4]$ to find the probability that the walk ever reaches $+4$ starting from 0 .

Hint 2: After the walk reaches $+4$, look at the first step away from $+4$. It goes to $+5$ with probability $\frac{1}{3}$ and to $+3$ with probability $\frac{2}{3}$. Use the same absorbing-probability formula to find the chance of returning to $+4$ from each of those positions before absorption, average those two probabilities, and then convert repeated returns into a geometric series.

841. Jumping Frog X

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Before worrying about all 100 points, notice that to have visited every point by the time you first reach 15, the walk must have reached both neighboring points 14 and 16 earlier.

Hint 2: Use the absorbing probability formula for biased random walk:

$$P_i(\text{hit } N \text{ before } 0) = \frac{1 - (q/p)^i}{1 - (q/p)^N}.$$

842. Elevator II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Look at one elevator by itself first. A complete trip from the 1st floor to the 50th floor and back takes 98 time units, so a random departure time means the elevator is equally likely to be at any point in its cycle. Figure out how the 20th floor splits that cycle into two parts: one corresponding to the next arrival being upward, and the other corresponding to the next arrival being downward.

Hint 2: Once you know the waiting-time distribution for a single elevator, treat the 4 elevators as independent. The first elevator to arrive is downward exactly when one elevator arrives at time t going down and the other 3 all arrive later than t . That leads to an integral of the form

$$4 \int f_d(t) S(t)^3 dt$$

where $f_d(t)$ is the density for a downward arrival and $S(t)$ is the survival function for one elevator.

843. Unknown Gunmen I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: The hit probabilities increase with position, so both players want to wait as long as possible but must guard against the other shooting first.

Hint 2: Consider the balancing point where $x = 1 - x^2$, let γ be the root of this equation in $(0, 1)$, so $\gamma = \frac{\sqrt{5}-1}{2}$. It can be proven that the optimal strategy is for A to shoot exactly at γ and for B to wait completely.

844. Random And Rational

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Fix A's choice as a , and think from B's point of view. Since B knows a , compare two possible strategies for B : choosing just below a , which gives him almost a , or choosing some $b > a$, which might make him the largest number and force him to pay.

Hint 2: Knowing what B does (hint 1), what should A do?

845. Line Switch

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Suppose it is worth switching when your current position is N , but not when your position is any smaller than that. Then exactly $100 - N + 1$ people, namely those in positions $N, N + 1, \dots, 100$, will switch.

846. Unknown Gunmen II

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Solve "Unknown Gunmen I" first.

847. Match Fixing

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Fix your chosen number as x . The only cases that matter are when x is one of the two middle numbers. If x is the smallest or largest of the four numbers, then your team is the extreme team, and the expected payoff in those cases cancels out by symmetry (why?).

Hint 2: Split into two cases: exactly one of the other three numbers is below x , or exactly two of them are below x . Find the expected payoff in each case, multiply by the probability of that case, and add. This gives an expression $E(x)$, which you can then maximize by differentiating and checking the nearest integers.

848. Extra Information I

[[Question ↑](#) | [Solution ↓](#) | [↑](#)]

Hint 1: Assume A uses a cutoff strategy: he keeps his number if it is at least t , and redraws otherwise.

Hint 2: If B sees that A redraw, A's final number is just uniform on $[0, 1]$, so B should compare keeping value b with redrawing, which gives win probability $\frac{1}{2}$. If B sees that A kept, then B knows A's value is uniform on $[t, 1]$. Find B's new keep cutoff k in terms of t . Then determine t by making A indifferent at the boundary: when A's value is exactly t , keeping and redrawing must give him the same winning probability.

Solutions

1. Non-empty Intersection

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Full video solution [here](#).

Name the left endpoints $l_1, l_2, \dots, l_5$, and the right endpoints $r_1, r_2, \dots, r_5$. So, we have 10 entities. By symmetry, each of the permutations of these entities that satisfy $l_i < r_i$ are equally likely, and that forms our sample space. There are $10!$ total permutations of the 10 entities. Half of them satisfy $l_1 < r_1$, half satisfy $l_2 < r_2$, and so on. So the total number of valid permutations (with all $l_i < r_i$) is

$$\frac{10!}{2^5}$$

There will be a non-empty intersection if and only if all the 5 left endpoints lie to the left of all the 5 right endpoints. The number of such permutations is

$$5! \cdot 5!$$

-arrange the l_i 's in the first 5 positions, and the r_i 's in the last 5.

So, the required probability is

$$\frac{5! \cdot 5!}{10!/2^5} = \frac{8}{63}$$

Answer:

2. Standing Table II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Full video solution [here](#).

The table will stand exactly when the pentagon formed by joining the five attachment points contains the centre of the circle. This happens precisely when the five points are NOT all contained in any semicircular arc. From "same arc I", we know that when five points are chosen independently along the circumference, the probability that all of them lie within some semicircular arc is

$$\frac{5}{16}$$

Hence, the probability that they do not all lie within a semicircular arc is

$$1 - \frac{5}{16} = \frac{11}{16}$$

Answer:

3. Game Of Chords

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Alice wins by using a symmetry pairing strategy. On her first move, Alice draws the diameter that joins a chosen pair of opposite points. This choice makes the current configuration symmetric with respect to reflection about that diameter.

After each move of Bob, Alice replies by drawing the reflection of Bob's chord across the diameter. Two facts make this reply always legal and available: 1. Before Bob moves, the position is symmetric, so any chord present has its reflected counterpart already present. When Bob adds a chord, he breaks the symmetry by

creating a chord whose reflection is not present, so that reflected chord is free to be drawn. 2. Reflecting preserves the noncrossing property inside the circle, so if Bob's chord did not intersect any existing chords, then its reflection also does not.

Therefore, Alice can always mirror Bob's move, restoring symmetry after her turn. Because Alice always has a legal reply whenever Bob does, Bob will be the first to run out of moves, and thus Alice wins under optimal play.

Answer: 1

4. Unequal Neighbors

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Color the grid in an alternating black-and-white pattern, like a chessboard. Using the given operation, adjust the values to make all black cells even and all white cells odd: 1. For black cells: - If a black cell is already even, no change is needed. - If it is odd, increase it by 1. 2. For white cells: - If a white cell is already odd, no change is needed. - If it is even, increase it by 1. After these adjustments: - Every black cell (even) is surrounded by white cells (odd). - Every white cell (odd) is surrounded by black cells (even). This ensures that all neighboring cells have differing values, satisfying the condition. Hence, it is always possible to modify the grid accordingly.

Answer: 1

5. Find The Units Digit

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $S(n)$ be the sum of "product of nonzero digits" over all n -digit strings (allowing leading zeros). For $n = 1$, the ten strings are "0" through "9," whose nonzero-digit products are $1, 1, 2, 3, \dots, 9$. Summing these gives

$$S(1) = 1 + 1 + 2 + 3 + \dots + 9 = 46.$$

To build an $(n + 1)$ -digit string, choose its last digit d and an n -digit prefix:

- If $d = 0$, contribute $S(n)$.
- If $d = k \in \{1, \dots, 9\}$, contribute $kS(n)$.

Thus

$$S(n + 1) = S(n) + \sum_{k=1}^9 kS(n) = 46S(n).$$

Since $S(1) = 46$, by induction

$$S(n) = 46^n \implies S(100) = 46^{100}.$$

The final task is to extract the last decimal digit of 46^{100} . Any power of an integer ending in 6 also ends in 6, so the required digit is 6.

Answer: 6

6. Ship Destroyer

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Observe that each bomb drop is one of 16 equally likely outcomes, so there are in total 16^4 possible ordered sequences of four bomb-drops. To ensure every ship is destroyed, each of the 4 rows must contain at least one bomb OR each of the 4 columns must contain at least one bomb. Equivalently, let E_R be the event "the bombs land in four distinct rows," and E_C be the event "the bombs land in four distinct columns." A ship

in row i is wiped out as soon as at least one bomb falls in row i , and a ship in column j is wiped out as soon as at least one bomb falls in column j . Therefore, all ships are destroyed if and only if at least one of the events E_R OR E_C occurs. To count $|E_R|$, note that we require the four bombs to occupy four different rows (but columns may repeat). Since bombs are ordered, there are $4! = 24$ ways to choose a distinct row for each bomb, and for each bomb, 4 possible columns, giving $4^4 = 256$ choices in total for the column positions. Hence

$$|E_R| = 4! \times 4^4 = 24 \times 256 = 6,144.$$

By exactly the same reasoning, $|E_C| = 6,144$, because selecting four distinct columns for the bombs also gives $4! \times 4^4$ sequences. Next, $|E_R \cap E_C|$ is the number of sequences in which the bombs occupy four distinct rows and four distinct columns simultaneously — that is, no two bombs share a row or a column. We first choose a permutation of rows for the four bombs ($4!$ ways) and independently choose a permutation of columns for those bombs ($4!$ ways), giving

$$|E_R \cap E_C| = 4! \times 4! = 24 \times 24 = 576.$$

Finally, by the principle of inclusion-exclusion, the number of bomb-drop sequences that achieve "all ships destroyed" is

$$|E_R \cup E_C| = |E_R| + |E_C| - |E_R \cap E_C| = 6,144 + 6,144 - 576 = 11,712.$$

Therefore, the desired probability is

$$\frac{11,712}{16^4} \approx 0.1787.$$

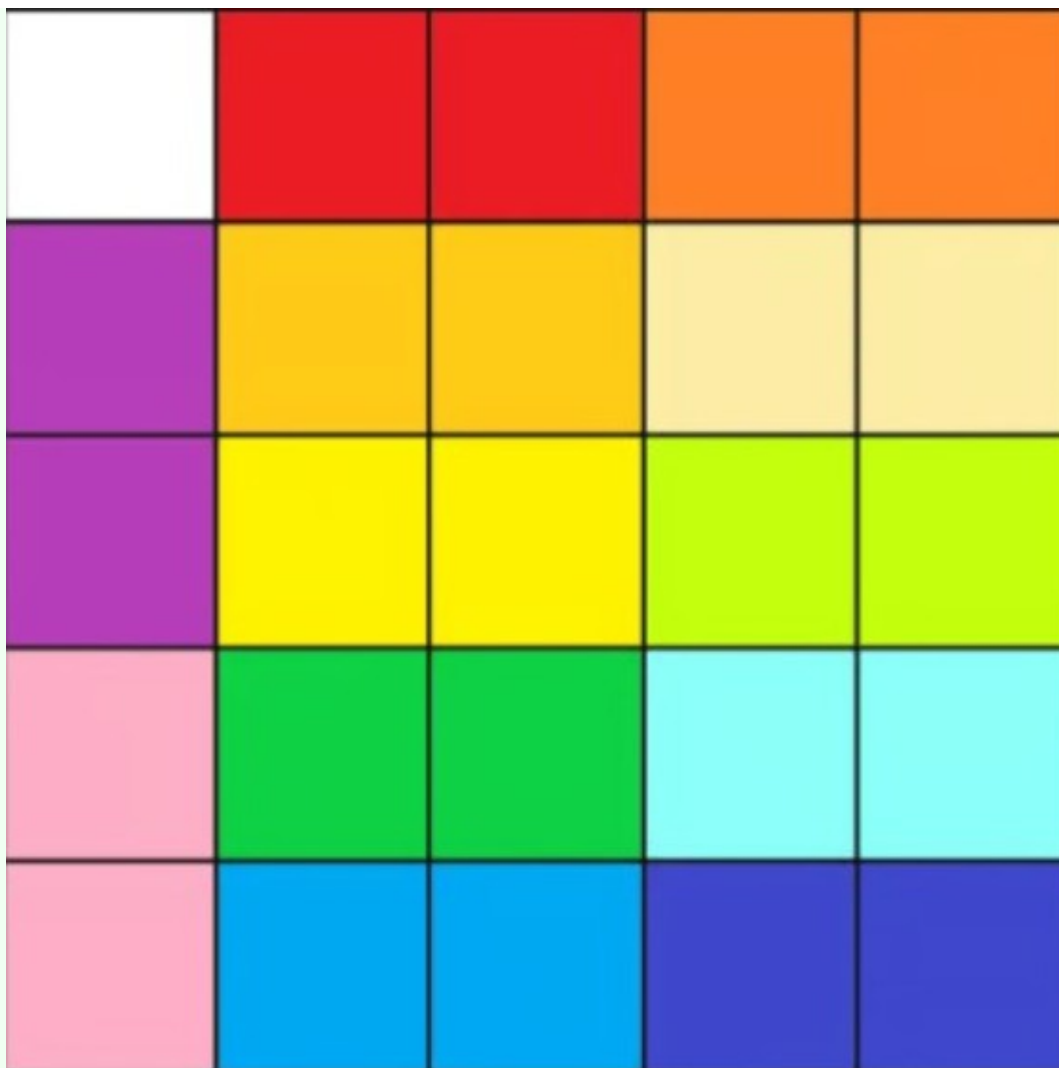
Answer: 0.17871

7. Moving The Piece I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let m be the number of rows and n be the number of columns. Bob has a winning strategy whenever $m \times n$ is odd. We will solve the problem for a 5×5 grid, and the same reasoning can be applied to the 183×181 grid.

Since $m \times n$ is odd, if we ignore the starting square, we can divide the grid into pairs of two (as shown in the image).



The first player always moves on the first square of a pair, while the second player can always move on the second square of the pair. Hence, the second player will always be able to make the last move by moving to the same colored square the piece currently stands on (after Alice's move).

Therefore, Bob has a winning strategy for this case.

Answer: 2

8. Even Marbles Everywhere

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total number of configurations is 2^{64} , since there are 64 squares and each of them can be filled with a white or black marble (2 ways). Now, consider the top-left 7×7 grid. Irrespective of how it is filled, there is a unique way to fill the rightmost column and bottom row to ensure every row and every column has an even number of white marbles: 1. Place marbles in the first 7 cells of the last column such that the first 7 rows have even white marbles. 2. Similarly, place marbles in the first 7 cells of the last row such that the first 7 columns have even white marbles. At this stage, the entire grid is filled except for the bottom-right cell. To ensure the last row and last column both have an even number of white marbles:

- If both are odd, place a white marble in the bottom-right cell.
- If both are even, place a black marble in the bottom-right cell.

What if the 8th row has an even number of white marbles while the 8th column has an odd number (or vice

versa)? This case cannot occur - prove it yourself! Thus, there are 2^{49} ways to fill the grid such that every row and every column has an even number of white marbles. The required probability is:

$$\frac{2^{49}}{2^{64}} = \frac{1}{2^{15}}$$

Answer:

9. Hard Grid Arrangement

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider the arrangement of the first row of the grid. There are 2^n possible ways to fill the first row, as each cell can either be 0 or 1. However, if the first row is not alternating (i.e., it contains two consecutive 0's or 1's), the entire grid becomes predetermined. This is because if we have a 00 (or 11) in any adjacent columns, the next row in those columns must be 11 (or 00) to satisfy the condition that every 2×2 sub-grid sums to 2. As a result, the arrangement of the entire grid becomes determined once the first row is fixed.

On the other hand, if the first row is alternating, such as 01010...01010..., then the second row can be either 10101...10101... or 01010...01010..., as both will maintain the required sum of 2 for all 2×2 sub-grids. Therefore, for an alternating first row, there are two possibilities for each subsequent row. This gives 2^{n-1} possible arrangements when the first row is alternating.

Thus, if the first row is not alternating, each arrangement of the first row would give exactly one arrangement of the whole grid. There are $2^n - 2$ such arrangements. If the first row is alternating like 010101...010101..., we have 2^{n-1} arrangements of the entire grid. If the first row is alternating like 101010...101010..., we have 2^{n-1} arrangements of the entire grid. Hence, our answer is

$$2^n - 2 + 2^{n-1} + 2^{n-1} = 2^{n+1} - 2.$$

Answer:

10. All Odds Before Even

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Look only at the FIRST time each face appears among the rolls. The relative order in which the six faces FIRST show up is equally likely to be any permutation of the numbers 1 through 6 by symmetry. So we can replace the repeated rolling process by a random ordering of the six faces. Instead of tracking every roll, we only track the order in which the six faces appear for the first time, which forms a random permutation of 1, 2, 3, 4, 5, 6.

In this permutation, the first three faces to appear must be exactly the odd faces 1, 3, and 5, in any order, and the last three faces must be the even faces 2, 4, and 6, in any order. The number of favorable orderings is therefore $3! \cdot 3!$, since the odd faces can be arranged among themselves in $3!$ ways and the even faces can be arranged among themselves in $3!$ ways.

The total number of possible orderings of all six faces is $6!$. Hence the probability is

$$\frac{3! \cdot 3!}{6!} = \frac{36}{720} = \frac{1}{20}.$$

So the required probability is .

Answer:

11. Standing Table I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For the table to stand, the center of mass (center of the circle) must be within the triangle created by the three legs. By fixing the position of the first leg arbitrarily, the second leg can be placed anywhere. However, to ensure the triangle contains the center, the third leg must be positioned in a region equivalent to the length of the arc between the first two legs, reflected over the center. The possible location for the second leg can range from 0 to π . On average, this region measures $\frac{\pi}{2}$. Consequently, the probability is

$$\frac{\frac{\pi}{2}}{2\pi} = \frac{1}{4}.$$

Answer:

12. 10 Consecutive Heads

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E_n represent the expected number of additional tosses needed to get 10 consecutive heads, given that you already have n consecutive heads. The goal is to find E_0 , the expected number of tosses starting from 0 heads in a row.

The recurrence relations are as follows:

- $E_{10} = 0$, since no more tosses are needed once we have 10 heads.
- For $n < 10$, if the next toss is heads (which occurs with probability $\frac{1}{2}$), we move to state $n + 1$, and if the next toss is tails (also with probability $\frac{1}{2}$), we go back to state E_0 . Thus:

$$E_n = 1 + \frac{1}{2}E_{n+1} + \frac{1}{2}E_0$$

Solving this recurrence gives $E_0 = 2046$. Thus, the expected number of tosses to get 10 consecutive heads is 2046.

Answer:

13. Chords Intersection I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p_{ij} be the probability that chord i and chord j intersect. Consider the 4 endpoints of these two chords as A, B, C , and D in clockwise order. Point A can be connected to point B, C , or D (3 cases). Out of these 3 cases, only 1 case (point A connected to point C) results in intersecting chords. Thus, the probability that chord i and chord j intersect is $\frac{1}{3}$. Let x_{ij} be an indicator variable where $x_{ij} = 1$ if chords i and j intersect, and $x_{ij} = 0$ otherwise. The expected number of intersections when 10 chords are drawn is:

$$\sum_{i < j} x_{ij} \cdot P[x_{ij} = 1]$$

Since $\frac{1}{3}$ is the probability of any two specific chords intersecting, and there are $\binom{10}{2}$ possible pairs of chords, we have: Expected number of intersections = $\binom{10}{2} \cdot \frac{1}{3} = \frac{10 \cdot 9}{2} \cdot \frac{1}{3} = 15$

Answer:

14. 1 Leftward Jump

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We start by determining the total number of possible sequences the frog can follow, which is given by $6! = 720$.

Next, to find the sequences that involve exactly one leftwards jump, we look at how to split the numbers $[2, 3, 4, 5, 6, 7]$ into two subsets, A and B. The frog will first jump from stone 1 to the stones in group A in ascending order, followed by jumping to the stones in group B, also in ascending order.

However, if group B is either empty or consists of a suffix of $[2, 3, 4, 5, 6, 7]$, the frog does not make any leftwards jump, so we disregard those cases. There are $2^6 = 64$ possible ways to select group B, but after removing the cases where no leftwards jumps occur, we obtain the final count, which leads to the answer $2^6 - 7 = 57$, yielding the probability $\frac{57}{720}$.

Answer:

15. Marble Swap II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p be the probability that after 9 moves the blue marble sits in the center. Then with probability $1 - p$ it is in one of the two end positions. Now consider the 10th move: If the blue marble is already in the center (with probability p), the only way to keep it there is to swap the two reds. Since there are 3 equally likely swaps (swap red-blue, swap blue-red, or swap red-red), the chance of choosing the red-red swap is

$$\frac{1}{3}$$

If the blue marble is not in the center (with probability $1 - p$), say it is in one of the ends, the only swap that moves it into the center is the swap between that end and the center. Again, that is exactly 1 of the 3 equally likely swaps, so the chance is

$$\frac{1}{3}$$

Putting these together, the probability that the blue marble lands in the center after 10 moves is

$$p \cdot \frac{1}{3} + (1 - p) \cdot \frac{1}{3} = \frac{1}{3}(p + 1 - p) = \frac{1}{3}$$

Therefore, the probability of ending up in the original order (red-blue-red) after 10 moves is

$$\frac{1}{3}$$

Answer:

16. Scary Permutation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The last number must be either 1 or 9. Here's why—suppose the last number is something other than 1 or 9. Let's call it n , where $2 \leq n \leq 8$. Now, assume that the first number is greater than n . Since n is the last number, $n - 1$ must be somewhere to its left. But $n - 1$ can't be the first number, because the first number is bigger than n . Then for $n - 1$ to satisfy the condition given in the problem, $n - 2$ or n has to be somewhere to its left (as $n - 1$ cannot be the first number in the permutation). But we already know n is the last number, so it cannot be to the left of $n - 1$. That means $n - 2$ must be to the left of $n - 1$. Then $n - 3$ must be to the left of $n - 2$, and so on. This keeps going indefinitely. But 1 is the smallest number, so this process cannot go on forever — contradiction. Now suppose the first number is smaller than n . A similar argument works with $n + 1, n + 2, \dots$. But the numbers only go up to 9, so this also leads to a contradiction. So, the last number must be either 1 or 9. Let's say the last number is 1. Then the second last number must be either 2 or 9 — that is, the smallest or largest remaining number — just like how we narrowed down the

last number. The same logic applies at every step: at each of the positions from 2 through 8, we always have two valid choices — the minimum or the maximum of the unused numbers. So the total number of valid permutations is

$$2^8 = 256$$

Answer: 256

17. Grid Frog I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Color the grid like a chessboard with the top-left cell white. Every frog on a white cell must jump to a black cell, and every frog on a black cell must jump to a white cell. Thus, the ways in which the white-cell frogs move are independent of the black-cell frogs' moves. We can count the white-cell configurations and then square the result. Let $f(n)$ be the number of valid jump-patterns for the frogs on the white squares of a $2 \times n$ grid. Focus on the frog in column 1, row 1 (a white cell). It has two options: - Jump down into column 1, row 2. The remaining white frogs simply arrange themselves in columns 2 through n . This reduces the problem to counting $f(n-1)$ for a $2 \times (n-1)$ grid. - Jump right into column 2, row 1. That move forces the frog in column 2, row 2 (also white) to jump left into column 1, row 2, occupying both columns 1-2 completely. The remaining white frogs must then arrange themselves in the last $n-2$ columns, giving $f(n-2)$. Hence, we have the recurrence:

$$f(n) = f(n-1) + f(n-2)$$

The base cases are: $f(1) = 1$ (only one frog, must jump down) $f(2) = 2$ (either both frogs move vertically, or both move horizontally) This is the Fibonacci recurrence, so: $f(8) = 34$ Since the eight black-cell frogs also have 34 independent ways to jump, the total number of collision-free arrangements is: $34 \times 34 = 1156$

Answer: 1156

18. At Least Six I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write E_i for the expected number of additional rolls needed when your current total is i . Once your total is already 6 or more, you need no further rolls, so $E_i = 0$ for $i \geq 6$. If your total is $i < 6$, then on the next roll you add $1, 2, \dots, 6$ each with probability $\frac{1}{6}$. In every case you have used one roll and your new total is $i+k$, so

$$E_i = 1 + \frac{1}{6} \sum_{k=1}^6 E_{i+k}.$$

For i in the range 0 to 5, all future states $i+k \geq 6$ contribute $E_{i+k} = 0$, so the sum effectively runs only over those k with $i+k < 6$. Working backwards: At $i = 5$: any roll of 1-6 reaches at least 6, so

$$E_5 = 1.$$

At $i = 4$: only rolling a 1 leaves you at total 5; larger rolls finish the game. Hence

$$E_4 = 1 + \frac{1}{6}E_5 = 1 + \frac{1}{6} \times 1 = \frac{7}{6}.$$

At $i = 3$: rolling a 1 or 2 leads to totals 4 or 5, so

$$E_3 = 1 + \frac{1}{6}(E_4 + E_5) = 1 + \frac{1}{6} \left(\frac{7}{6} + 1 \right) = \frac{49}{36}.$$

At $i = 2$: only rolls 1–3 keep you below 6, giving

$$E_2 = 1 + \frac{1}{6}(E_3 + E_4 + E_5) = \frac{343}{216}.$$

At $i = 1$: rolls 1–4 keep you below 6, so

$$E_1 = 1 + \frac{1}{6}(E_2 + E_3 + E_4 + E_5) = \frac{2401}{1296}.$$

At $i = 0$: rolls 1–5 lead to totals 1–5, so

$$E_0 = 1 + \frac{1}{6}(E_1 + E_2 + E_3 + E_4 + E_5) = \frac{16807}{7776} \approx 2.16139.$$

Thus, the expected number of rolls needed to reach a summation series of at least 6 is

$$\mathbb{E}[T] = E_0 = \frac{16807}{7776} \approx 2.16139.$$

Answer: 2.16139

19. Game Of Placing Knights

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider a horizontal line passing through the middle of the board, dividing it into a top half and a bottom half. Bob can mirror Alice's moves across this line. For example, if Alice places a knight in the top left corner, Bob places his knight in the bottom left corner.

By doing this, Bob ensures that after every pair of moves, the board remains symmetric about the central line. So, whenever it's Alice's turn and she has a legal move, the square symmetric to that move is also legal for Bob. As a result, Bob will always have a move whenever Alice does. Since Alice starts first, she will eventually run out of legal moves before Bob, and thus, Bob wins.

Answer: 2

20. Same Arc I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the points in clockwise order as $1, 2, \dots, N$. For each index i let E_i be the event that every point lies inside the clockwise semicircular arc of length π whose left endpoint is point i . First compute $P(E_i)$. Fix the location of the i -th point. The clockwise semicircle that starts at that point covers exactly one half of the circle, and each of the remaining $N - 1$ points falls into that semicircle independently with probability $\frac{1}{2}$. Therefore

$$P(E_i) = \left(\frac{1}{2}\right)^{N-1}$$

Next show that two different events E_i and E_j cannot both occur (i.e., the E_i s are mutually exclusive), except in the degenerate case where some points lie exactly on semicircle endpoints. Suppose E_i holds, so every point is contained in the semicircle S_i that begins at point i . If $j \neq i$ then point j lies strictly inside S_i rather than at its left endpoint. The semicircle S_j that begins at j therefore starts strictly to the right of point i , so S_j cannot contain the point i . Hence S_j does not contain all points, and E_j fails. The only way E_i and E_j could both hold is if some point lies exactly on the boundary between semicircles, which is a probability-zero event; in all other cases the events E_i are pairwise disjoint. Because the events E_i are disjoint and each has the same probability $\frac{1}{2^{N-1}}$, the probability that the N points all lie in some semicircle

equals the sum of the $P(E_i)$ over $i = 1, \dots, N$. Therefore the desired probability is

$$\frac{N}{2^{N-1}}$$

Answer:

21. Increasing Dice Rolls III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Each roll is independent and equally likely, so the ordered pair (X_1, X_2) is uniformly distributed over the $6 \times 6 = 36$ possible pairs. If the first roll is 1, the second can be $2, \dots, 6$ giving 5 favorable outcomes. If the first is 2, the second can be $3, \dots, 6$ giving 4, and so on. Thus the total number of favorable pairs is

$$5 + 4 + 3 + 2 + 1 = 15$$

Since all 36 outcomes are equally likely, the required probability is

$$\frac{15}{36} = \frac{5}{12}$$

Answer:

22. Fill In The Blanks

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Given that there are 5 odd numbers and 4 even numbers, the result will always be an odd number, regardless of how the $+$ and $-$ signs are assigned. The sum of an odd number of odd numbers is always odd, and adding or subtracting even numbers does not change this property. Therefore, the result can never be 0, which is an even number. Hence, the probability of obtaining a final result of 0 is 0.

Answer:

23. Exponent Perfect Square

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Given $\sqrt{p^p} = p^{p/2}$, this expression is an integer if $p/2$ is an integer, which happens when p is even. Thus, there are 151 even integers satisfying this. Additionally, if p is a perfect square, say $p = n^2$ for some integer n , then $\sqrt{p^p} = \sqrt{(n^2)^{(n^2)}} = \sqrt{n^{2n^2}} = n^{n^2}$, which is an integer. Therefore, we need to include odd perfect squares as well, which are $11^2, 13^2, 15^2, 17^2$, and 19^2 . In total, the number of valid integers is $151 + 5 = 156$.

Answer:

24. Odd Heads II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Use the law of total probability. To get odd heads, the fair coin can show heads and then the other 99 coins can show an even number of heads, or the fair coin can show tails and then the other 99 coins can show an odd number of heads. Let the probability that the other 99 coins show an even number of heads be e . Then the probability that they show an odd number of heads is $1 - e$. Now, using the law of total probability, we

have:

$$P(\text{odd heads}) = \frac{1}{2} \cdot e + \frac{1}{2} \cdot (1 - e) = \frac{1}{2}$$

Answer: 0.50000

25. Dice Rolls Product

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The six counts $N_1, \dots, N_6$ are clearly not independent (they must sum to 10), so we cannot write

$$E[N_1 N_2 \cdots N_6] = E[N_1] E[N_2] \cdots E[N_6].$$

Instead, introduce indicator variables for each face on each roll: for $i = 1, \dots, 10$, let

$$a_i = \begin{cases} 1, & \text{roll } i \text{ shows a 1} \\ 0, & \text{otherwise} \end{cases}, \quad b_i = \begin{cases} 1, & \text{roll } i \text{ shows a 2} \\ 0, & \text{otherwise} \end{cases}, \dots, \quad f_i = \begin{cases} 1, & \text{roll } i \text{ shows a 6} \\ 0, & \text{otherwise} \end{cases}$$

Then

$$N_1 = \sum_{i=1}^{10} a_i, \quad N_2 = \sum_{i=1}^{10} b_i, \quad \dots, \quad N_6 = \sum_{i=1}^{10} f_i$$

so the product is

$$N_1 N_2 \cdots N_6 = \left(\sum_i a_i \right) \left(\sum_j b_j \right) \cdots \left(\sum_k f_k \right).$$

Expanding this product yields a sum of terms of the form

$$a_{i_1} b_{i_2} \cdots f_{i_6}$$

over all choices of indices $i_1, \dots, i_6 \in \{1, \dots, 10\}$. But whenever any two of those indices coincide—say $i_1 = i_2$ —we have

$$a_{i_1} b_{i_2} = a_{i_1} b_{i_1} = 0,$$

because a single roll cannot show both "1" and "2." Hence only the terms with all six indices distinct survive in expectation. There are

$$(10)_6 = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5$$

ways to choose an ordered 6-tuple of distinct rolls. Moreover, for a fixed choice $(i_1, \dots, i_6)$, the indicators $a_{i_1}, b_{i_2}, \dots, f_{i_6}$ involve disjoint rolls and are therefore independent. Since each indicator equals 1 with probability $\frac{1}{6}$,

$$E[a_{i_1} b_{i_2} \cdots f_{i_6}] = E[a_{i_1}] E[b_{i_2}] \cdots E[f_{i_6}] = \left(\frac{1}{6} \right)^6.$$

Putting this together,

$$E[N_1 \cdots N_6] = \sum_{(i_1, \dots, i_6) \text{ distinct}} E[a_{i_1} b_{i_2} \cdots f_{i_6}] = (10)_6 \cdot \left(\frac{1}{6} \right)^6 = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{6^6} = \frac{175}{54}.$$

Answer: 3.24074

26. Jumping Frog V

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let d_n denote the number of distinct paths to reach point n . The frog can jump any number of points from 2 to n , inclusive, on the first move. After jumping k points, the problem becomes finding a path through the remaining $n - k$ points. Hence, we get:

$$d_n = d_{n-2} + d_{n-3} + d_{n-4} + \cdots + d_1 + d_0.$$

Now, observe the tail term $d_{n-3} + d_{n-4} + \cdots + d_1 + d_0$. This is essentially the same as starting from point 1 and jumping strictly more than one point at each step. In this case, point 3 is the first available jump, meaning the frog must find a path through the remaining $n - 1$ points, which reduces to d_{n-1} . Thus, the recurrence becomes:

$$d_n = d_{n-1} + d_{n-2}.$$

The initial conditions are $d_1 = 0$ and $d_2 = 1$, which can be counted directly. This is exactly the Fibonacci sequence shifted by 1 index, where $F_0 = 0$ and $F_1 = 1$. Therefore, we conclude:

$$d_n = F_{n-1}.$$

In particular, for point 10:

$$d_{10} = F_9 = 34.$$

Answer: 34

27. Dot Product

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Full video solution [here](#).

Let p_n be the probability that the dot product of the first n entries of two binary vectors is odd. Now consider the $(n+1)$ -th entry: The dot product of this entry will be 1 only if both entries are 1, which occurs with probability:

$$\frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$$

So, the probability that the dot product of the $(n+1)$ -th entries is 0 is:

$$1 - \frac{3}{8} = \frac{5}{8}$$

Based on this, we can write the recurrence relation:

$$p_{n+1} = p_n \cdot \frac{5}{8} + (1 - p_n) \cdot \frac{3}{8}$$

- The first term accounts for the case where the dot product of the first n entries is already odd, and the $(n+1)$ -th product is 0, so the total remains odd.
- The second term accounts for the case where the dot product of the first n entries is even, and the $(n+1)$ -th product is 1, flipping the parity to odd.

Base case ($n = 1$): The dot product is odd only if both entries are 1, which happens with probability:

$$p_1 = \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$$

This recurrence is linear and can be solved explicitly. The closed-form solution is:

$$p_n = \frac{1}{2} - \frac{1}{2^{2n+1}}$$

So, the probability that the dot product of the first 10 entries is odd is:

$$p_{10} = \frac{1}{2} - \frac{1}{2^{21}}$$

Answer:

28. Marble Swap I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In a permutation π , an inversion is a pair of indices $i < j$ such that $\pi(i) > \pi(j)$. Any single swap always changes the total number of inversions by an odd number. That is, the parity of the number of inversions—whether it's odd or even—flips with each swap (the proof is simple and is left to the reader).

Number the marbles as 1, 2, 3 (red, blue, green respectively). We start with the identity permutation $[1, 2, 3]$, which has zero inversions—an even number. After 125 swaps, since each swap flips the parity of the inversion count, the total number of inversions becomes odd.

The original permutation has zero inversions. We want to know the probability that the permutation returns to that same original order after exactly 125 swaps. But since 125 swaps lead to a permutation with an odd number of inversions, and the original has an even number (zero), it's impossible to return to the original permutation in an odd number of swaps.

The probability is 0.

Answer:

29. Monster Escape I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If we color the grid in alternating black and white colors like a chessboard, the bottom-left and bottom-right corners will have opposite colors. After each move, both Mario and the monster switch to a square of the opposite color from where they started—this is true for both of them. Since they began on squares of opposite colors, they will always be on squares of different colors after every move. Therefore, they can never land on the same square, no matter how they move.

So, the probability that Mario survives is 1.

Answer:

30. Chords Forming Quadrilateral

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total number of possible chords is $\binom{8}{2} = 28$. Thus, the number of ways to choose 4 chords is $\binom{28}{4}$. Now, each set of 4 points on the circle forms exactly one convex quadrilateral. Therefore, the number of ways to select 4 chords that form a convex quadrilateral is $\binom{8}{4}$. Hence, the required probability is:

$$\frac{\binom{8}{4}}{\binom{28}{4}} = \frac{70}{20475} = \frac{2}{585}$$

Thus, the probability that the selected 4 chords form a convex quadrilateral is $\frac{2}{585}$.

Answer:

31. Monster Escape II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that Mario's coordinate sum increases by 1 every second, while the monster's coordinate sum decreases by 1. Starting with sums of 0 for Mario and 8 for the monster, if they meet, they must do so after 4 seconds on the diagonal $y = 4 - x$. In the first 4 seconds, each can move freely in their permitted directions, resulting in

$$2^4 = 16$$

possible paths to the diagonal for each. Thus, there are

$$16 \times 16 = 256$$

ways for both to reach the diagonal. The number of ways in which Mario and the monster can meet is equal to the number of ways of traveling from one corner to the other, as each way of traveling from one corner to the other can be viewed as a way of Mario and the monster meeting. There are

$$\binom{8}{4} = 70$$

ways for them to meet out of the 256 possible paths, giving a probability of

$$\frac{70}{256}$$

Answer:

32. Matrix Attack III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will show that under these two conditions, every number $k \in \{1, \dots, 7\}$ must appear exactly once on the main diagonal — so the probability is 1.

Total count per symbol: Since each of the 7 columns is a permutation of $\{1, \dots, 7\}$, the number k appears exactly once in each column. Hence, overall in the entire matrix, k appears exactly 7 times.

Off-diagonal occurrences come in pairs: By symmetry, whenever k appears in an off-diagonal entry $a_{ij} = k$ with $i \neq j$, it must also appear at $a_{ji} = k$. Thus, the total number of times k appears off the main diagonal is an even number.

Diagonal count is odd: Let d_k be the number of times k appears on the diagonal. Then:

$$(\text{off-diagonal appearances}) + d_k = 7 \Rightarrow d_k \equiv 7 \pmod{2}$$

Since the off-diagonal appearances are even, d_k must be odd.

Diagonal length forces $d_k = 1$: There are exactly 7 diagonal positions total. If some k appeared 3 or more times on the diagonal, that would leave too few slots for the other six symbols to each appear an odd number of times. Concretely, the diagonal has 7 entries and there are 7 different symbols k . Each symbol must occupy an odd positive number of those 7 spots, and the only way to fit 7 symbols into 7 spots with each appearing oddly often is for each to appear exactly once.

Therefore, every symbol 1 through 7 appears exactly once on the main diagonal. There is no choice or random variation left: under the given constraints the diagonal must be a permutation of $\{1, \dots, 7\}$. Therefore, the required probability is 1.

Answer:

33. Bulbs In A Grid

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The light bulbs are arranged as follows:

1	2	3
4	5	6
7	8	9

There are $2^9 = 512$ possible configurations of the light bulbs being on or off. To find the configurations where no two adjacent bulbs are on, consider the following cases:

Case 1: Bulb 5 is on. In this scenario, bulbs 2, 4, 6, and 8 must be off to avoid adjacency. However, bulbs 1, 3, 7, and 9 are not restricted and can be either on or off. Therefore, there are $2^4 = 16$ configurations in this case.

Case 2: Bulb 5 is off. Now, consider three sub-cases for bulbs 4 and 6:

Sub-case 1: Both 4 and 6 are on. Here, bulbs 1, 3, 7, and 9 must all be off to avoid adjacency, while bulbs 2 and 8 are free to be either on or off. This gives $2^2 = 4$ configurations.

Sub-case 2: One of 4 or 6 is on. Let's assume bulb 4 is on and bulb 6 is off (we multiply by 2 later for the opposite case). Since 4 is on, bulbs 1 and 7 must be off, while bulbs 2 and 3 either have exactly one on or neither is on, resulting in 3 possible configurations. Similarly, bulbs 8 and 9 have 3 possibilities, leading to $2 \times 3 \times 3 = 18$ configurations.

Sub-case 3: Both 4 and 6 are off. In this situation, bulbs 1 through 3 can either all be off (1 way), have exactly one on (3 ways), or have 1 and 3 on with 2 off (1 way). This gives 5 configurations for bulbs 1-3. The same logic applies to bulbs 7 through 9, yielding $5^2 = 25$ configurations.

Summing all cases, we get:

$$16 + 4 + 18 + 25 = 63$$

Therefore, the probability is:

$$\frac{63}{512}$$

Answer:

34. Chess Tournament III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Each match eliminates exactly one player, so exactly $32 - 1 = 31$ matches are played, each of which consists of a different pair of players. Among the $\binom{32}{2} = \frac{32 \cdot 31}{2} = 496$ pairs of players, each pair is equally likely to play each other at some point during the tournament. Therefore, the probability that Ava and Tiffany form one of the 31 pairs of players that play each other is $\frac{31}{496} = \frac{1}{16}$.

Answer:

35. Count Paths II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If you are at a point (x, y) , where both x and y are even, and then move one unit to the right, the next move must also be towards right. Similarly, if you are at a point (x, y) , where both x and y are even, and then move one unit up, the next move must also be upward. Initially, we are at $(0, 0)$. Both the x and y coordinates are even. Each step leads to another step in the same direction. So, the problem can be rephrased as: how many paths go from $(0, 0)$ to $(8, 10)$ such that in one step, you move either 2 units to the right or 2 units upward? This can be simplified further to: how many paths go from $(0, 0)$ to $(4, 5)$ such that in one step, you move either 1 unit to the right or 1 unit upward? Using the method described in count paths I, the required number of paths is:

$$\frac{(4+5)!}{4!5!} = 126$$

Answer: 126

36. Five Light Bulbs

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E_i be the expected additional time (in seconds) to reach the state "all off" starting from a configuration with exactly i bulbs on. We start in state 0 (all off), but one flip immediately moves us to state 1, so the total expected return time is

$$1 + E_1.$$

By definition, $E_0 = 0$. For $1 \leq i \leq 4$, in state i we flip an on-bulb (moving to $i - 1$) with probability $\frac{i}{5}$, or an off-bulb (moving to $i + 1$) with probability $\frac{5-i}{5}$. Thus for each such i , we have the recursion

$$E_i = 1 + \frac{i}{5}E_{i-1} + \frac{5-i}{5}E_{i+1}.$$

Finally, in state 5, every flip turns an on-bulb off, so

$$E_5 = 1 + E_4.$$

Writing these out explicitly gives the system

$$\begin{cases} E_0 = 0 \\ E_1 = 1 + \frac{1}{5}E_0 + \frac{4}{5}E_2 \\ E_2 = 1 + \frac{2}{5}E_1 + \frac{3}{5}E_3 \\ E_3 = 1 + \frac{3}{5}E_2 + \frac{2}{5}E_4 \\ E_4 = 1 + \frac{4}{5}E_3 + \frac{1}{5}E_5 \\ E_5 = 1 + E_4 \end{cases}$$

Solving this linear system yields

$$E_1 = 31.$$

Therefore the expected return time to all-off starting from all-off is

$$1 + E_1 = 32.$$

Alternate solution: An alternate approach uses a fundamental theorem of finite Markov chains: in an irreducible, aperiodic chain, the expected return time to state s is the reciprocal of its stationary probability π_s . Here the chain on the $2^5 = 32$ bulb-configurations is symmetric under flipping any single bulb, so the stationary distribution is uniform ($\pi_s = \frac{1}{32}$ for every state s). Hence the expected return time to the "all off" state is

$$\frac{1}{\pi_{\text{all off}}} = 32.$$

Answer: 32

37. Letter-envelope Pairing

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total number of pairings is $10!$. We need to count the number of pairings such that the absolute difference between the numbers of the left and right gloves is at most 1. This can be done using recursion. The glove numbered 1 can either be paired with itself or with glove 2.

- If glove 1 is paired with itself, the problem reduces to arranging the remaining 9 gloves.
- If glove 1 is paired with glove 2, then glove 2 must necessarily be paired with glove 1, reducing the problem to arranging the remaining 8 gloves.

Let $f(n)$ be the number of ways of arranging n consecutive gloves such that the absolute difference between the numbers of the left and right gloves is at most 1. Based on the above argument:

$$f(n) = f(n-1) + f(n-2)$$

With base cases:

$$f(1) = 1, f(2) = 2$$

By applying the recurrence relation:

- $f(3) = f(1) + f(2) = 1 + 2 = 3$
- $f(4) = f(3) + f(2) = 3 + 2 = 5$
- $f(5) = f(4) + f(3) = 5 + 3 = 8$
- $f(6) = f(5) + f(4) = 8 + 5 = 13$
- $f(7) = f(6) + f(5) = 13 + 8 = 21$
- $f(8) = f(7) + f(6) = 21 + 13 = 34$
- $f(9) = f(8) + f(7) = 34 + 21 = 55$
- $f(10) = f(9) + f(8) = 55 + 34 = 89$

Thus, $f(10) = 89$. The required probability is:

$$\frac{f(10)}{10!} = \frac{89}{3628800}$$

Answer:

38. Frog Return

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The jumps range from 1 to 25, with 13 odd numbers and 12 even numbers. The sum of any combination of these numbers must be odd. Therefore, the grasshopper cannot return to its starting position after 25 jumps. Thus, the probability is 0.

Answer:

39. Direction Change I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total number of ways to reach $(16, 16)$ is $\binom{32}{16}$.

Now, let's calculate the number of paths such that Bob changes direction an odd number of times. Note that the condition is equivalent to the first and last step being in different directions. If the first step is in the x direction and the last step is in the y direction, it suffices to count the number of paths from $(1, 0)$ to $(16, 15)$, of which there are $\binom{30}{15}$. Similarly, in the other case, it suffices to count the number of paths from $(0, 1)$ to $(15, 16)$, of which there are also $\binom{30}{15}$. Therefore, the total number of paths is $2 \cdot \binom{30}{15}$.

So, the required probability is $\frac{2 \cdot \binom{30}{15}}{\binom{32}{16}} = \frac{16}{31}$.

Answer:

40. 3 Dice Minimum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will use the tail sum formula. For a non-negative discrete random variable X , it holds that

$$E[X] = \sum_{k \geq 0} P(X \geq k).$$

In this case, let X_1, X_2, X_3 be the values of the three rolls, and let $X = \min\{X_1, X_2, X_3\}$. Note that $X \geq t \Leftrightarrow X_1 \geq t, X_2 \geq t, X_3 \geq t$. By independence, we thus have

$$E[X] = \sum_{k=0}^{100} P(X_1 \geq k)P(X_2 \geq k)P(X_3 \geq k).$$

This simplifies to

$$E[X] = \frac{1}{100^3} \sum_{k=0}^{100} (100 - k)^3.$$

Finally, we compute the sum and simplify:

$$E[X] = \frac{1}{100^3} \sum_{k=1}^{100} k^3 = \frac{1}{100^3} \left(\frac{100(100+1)}{2} \right)^2 = 25.5025.$$

Answer: 25.50250

41. Game On Divisors III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Check small sizes to see the pattern. With 1 stone the mover must remove 1, thereby taking the last stone and losing, so 1 is losing. With 2 stones the mover can remove 1 (a valid divisor) leaving 1 for the opponent who then loses, so 2 is winning. With 3 the mover can only remove 1 or 3; removing 3 loses immediately, removing 1 leaves 2 which is winning for the next player, so 3 is losing. With 4 the mover can remove 1 and leave 3, which is losing for the opponent, so 4 is winning. If we continue this process up to (say) 10 stones, we find that odd numbers are losing states, while even numbers are winning states. Here's a simple proof for the above claim. Suppose the pile has an odd number of stones. All the factors of an odd number are also odd, so any move will remove an odd number of stones, leaving an even number. If the pile has an even number of stones, the player can remove one stone to make it odd, forcing the opponent into an odd state. This means the opponent will eventually be forced to take the last stone and lose. Since odd states are losing, Alice will lose as we start with an odd number of stones and hence Bob has the winning strategy.

Answer: 2

42. Jumping Frog IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let x_i be the number of ways to reach point i , starting at point 0. Then, $x_n = x_{n-1} + x_{n-3} + x_{n-5} + \cdots + \beta$, where $\beta = 1$ if n is even and $\beta = 0$ if n is odd.

Now, notice that $x_{n-2} = x_{n-3} + x_{n-5} + \cdots + \beta$, where $\beta = 1$ if n is even and $\beta = 0$ if n is odd.

Hence, $x_n = x_{n-1} + x_{n-2}$. $x_0 = 1, x_1 = 1, x_2 = 1$.

So, x_{10} is simply the 10th Fibonacci number, which is 55.

Answer: 55

43. Dice Game III[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p be the probability that Player 1 ultimately wins. If Player 1 doesn't win on their first toss, the remaining players each have an equal chance of being "next" to win, so they all have an equal probability of winning, which is $\frac{1-p}{5}$. On the first toss, Player 1 either rolls a 1 and wins instantly, or rolls a different number and effectively becomes one of the "other" players. Therefore:

$$p = \frac{1}{6} + \frac{5}{6} \times \frac{1-p}{5}$$

Solving the above equation, we get $p = \frac{2}{7}$.

Answer: **44. 105 Pebbles**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since all three piles are odd, the first operation must be merging two existing piles, resulting in an even-sized pile and an odd-sized pile. Let g denote the greatest common divisor (gcd) of these two pile sizes. Since one pile is odd-sized and the other pile is even-sized, g must be odd. For an odd g , no matter what sequence of operations is carried out (merger of two existing piles or division of an even pile into two equal piles), every pile size will continue to be a multiple of g . If $g > 1$, then we can never reach a configuration where all piles are size 1. Starting with $\{5, 49, 51\}$, three possible states are $\{54, 51\}$, $\{5, 100\}$, and $\{49, 56\}$. The gcd is 3, 5, and 7, respectively, in these three states. Therefore, we can never achieve a configuration with 105 piles, each of size 1.

Answer: **45. Missing Pairwise Product**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

When you have four numbers a, b, c , and d , the six pairwise products can be arranged into three pairs, like this: ab and cd , ac and bd , ad and bc . What's special about this arrangement? The product of the two products in each group is always the same, namely $abcd$. Now, we are given five of the products: $\{9, 10, 12, 27, 30\}$, and we must identify two pairs that give the same product. Looking at these numbers, their pairwise products are:

$$9 \times 30 = 270$$

$$10 \times 27 = 270$$

$$12 \times x$$

Thus,

$$12 \times x = 270$$

which gives

$$x = \frac{270}{12} = 22.5$$

Therefore, the missing product is $x = 22.5$.

Answer: **46. Constrained Subsets II**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The sum of the elements of the set $\{1, 2, 3, \dots, 30\}$ is 465. Hence, for every subset A having sum less than or equal to 232, we have a subset having sum greater than 232—the complement of A . Hence, the number of subsets having sum less than or equal to 232 is equal to the number of subsets having sum greater than 232. Now, the number of subsets having sum less than or equal to 232, plus the number of subsets having sum greater than 232, must equal the total number of subsets of $\{1, 2, 3, \dots, 30\}$ —as every subset of $\{1, 2, \dots, 30\}$ has sum either less than or equal to 232, or greater than 232. So, the required number of subsets is:

$$\frac{2^{30}}{2} = 2^{29}$$

Answer: 536870912

47. Fifth Card

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Among the five cards, at least two will share the same suit. You can place one of these cards in the first position, allowing your friend to deduce the suit of the hidden card. Let's number the cards from 1 to 13, with ace numbered as 10, jack as 11, queen as 12, and king as 13. The second, third, and fourth cards can be arranged in $3! = 6$ ways. So, we can encode 6 values using the second, third, and fourth cards. For example, suppose the second, third, and fourth cards are 2, 5, and 12. Then the arrangement (2, 5, 12) can represent 1, (2, 12, 5) can represent 2, (5, 2, 12) can represent 3, and so on. To find the value of the hidden card, our friend can add this value to the first card and take it modulo 13. The difference in values taken modulo 13 between any 2 cards cannot exceed 6. So, we can always arrange the first and fifth cards such that $(\text{first card} + \text{value from 2nd, 3rd, and 4th card}) \pmod{13} = \text{fifth card}$. For example, if the first and fifth cards are 1 and 13, we can place the fifth card as the first card and the first card as the fifth (hidden card). Then we can arrange the second, third, and fourth cards for value 1. Our friend would then guess the fifth card as $(13 + 1) \pmod{13} = 1$, which is correct.

48. Expected Area

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The area A of a right triangle with legs of lengths X and Y is given by

$$A = \frac{1}{2}XY.$$

Since X and Y are independent and identically distributed, the expectation factorizes:

$$E[A] = E\left[\frac{1}{2}XY\right] = \frac{1}{2}E[X]E[Y].$$

Each of X and Y is uniform on $\{1, 2, 3, 4, 5\}$, so

$$E[X] = E[Y] = \frac{1 + 2 + 3 + 4 + 5}{5} = \frac{15}{5} = 3.$$

Plugging in,

$$E[A] = \frac{1}{2} \times 3 \times 3 = \frac{9}{2} = 4.5.$$

Therefore, the expected area of the triangle is 4.5 square units.

Answer: 4.50000

49. Difficult Die Constraints

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Ignore rolls showing 1, 4, 5 because they do not affect when the first 2 or 3 appears and they do not change the count of 6s relative to the first appearances of 2 and 3. Thus we can think of a three sided die labeled 2, 3, 6 and ask for the probability of obtaining exactly two 6s before both 2 and 3 have appeared and it will not change anything. Working only with the relevant outcomes 2, 3, 6, the stopping sequence, recording only the first occurrences of 2 and 3 and keeping every 6, must be one of the following six sequences:

$$2, 6, 6, 3; \quad 3, 6, 6, 2; \quad 6, 2, 6, 3; \quad 6, 3, 6, 2; \quad 6, 6, 2, 3; \quad 6, 6, 3, 2.$$

Note that in this sequence representation every occurrence of 6 is counted, whereas only the first occurrences of 2 and 3 are counted. For example the actual sequence 2, 2, 2, 6, 6, 2, 3 corresponds to 2, 6, 6, 3 in this representation.

We compute the probability of each admissible sequence and sum them. For example, consider the sequence 6, 6, 2, 3. First, we require two 6s to occur before any 2 or 3, which has probability $(1/3)^2$. Next, a 2 must appear before either a 6 or a 3, otherwise another 6 would give more than two 6s, or a 3 would appear before 2, contradicting the desired sequence. This occurs with probability $1/3$. Finally, a 3 must appear before a 6, otherwise we would again exceed two 6s. This last step has probability $1/2$. Hence,

$$P(6, 6, 2, 3) = \left(\frac{1}{3}\right)^2 \cdot \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{54}$$

Proceeding in the same way, we obtain,

$$P(6, 6, 3, 2) = \frac{1}{54}, \quad P(6, 3, 6, 2) = \frac{1}{36}, \quad P(6, 2, 6, 3) = \frac{1}{36}$$

$$P(2, 6, 6, 3) = \frac{1}{24}, \quad P(3, 6, 6, 2) = \frac{1}{24}$$

Adding these probabilities yields

$$2 \cdot \frac{1}{54} + 2 \cdot \frac{1}{36} + 2 \cdot \frac{1}{24} = \frac{19}{108}$$

Answer:

50. Game Of Discarding Cards

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The cards are discarded only when one card is red and one card is black. Hence, discarded cards contain the same number of red and black cards. Therefore, the number of red and black cards that are NOT discarded are equal. So, you and the dealer always have the same number of cards. As a result, you can never win.

Answer:

51. Normal Inequality IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that $Y - 4X$ is a linear combination of independent standard normal variables. Since both X and Y are $N(0, 1)$, their means are zero. Thus, the mean of $Y - 4X$ is also zero because it is given by

$$E[Y - 4X] = E[Y] - 4E[X] = 0 - 4(0) = 0.$$

Since $Y - 4X$ is a linear combination of normal variables, it also follows a normal distribution with mean 0. By the symmetry of the normal distribution, the probability that $Y - 4X > 0$ is $\frac{1}{2}$. Therefore, the

probability that $Y > 4X$ is $\frac{1}{2}$.

Answer:

52. Ants On A Circle I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Instead of thinking that ants reverse direction at collisions, imagine they pass through one another without changing direction. Since the ants are indistinguishable, this makes no difference. Under this pass-through picture each ant follows a continuous path at the same constant speed and therefore makes exactly one full circuit in one minute, so every ant ends up at the same geometric point where it started. Note that this does not necessarily mean each individual ant returns to its original position; it only means that the set of points occupied by the ants initially and finally is the same. Therefore the required probability is 1.

Answer:

53. Ants On A Circle II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If all ants move clockwise or all move anticlockwise, they will each return to their original positions. We now show that they cannot return to their original positions in any other case when there are 9 ants. Note that an ant can return to its original position only if it moves a full 360° clockwise, a full 360° anticlockwise, or has a net movement of 0° . Moreover, if the ants are not all moving in the same direction, each ant will collide with at least one other and reverse its direction. Consequently, every ant must have a net movement of 0° for all of them to return to their original positions. Imagine the net angular displacement of ant i after one minute as θ_i , measured positive for clockwise and negative for counterclockwise. For every ant to be at its original location we must have $\theta_i = 0$ for every i , hence

$$\sum_{i=1}^9 \theta_i = 0.$$

Let x be the number of ants that initially chose clockwise and $9 - x$ the number that initially chose counterclockwise. The quantity x is invariant in time because at each collision two ants reverse direction, so one clockwise ant becomes counterclockwise and one counterclockwise becomes clockwise. Thus the sum of the θ_i would be

$$\sum_{i=1}^9 \theta_i = x \cdot 360^\circ + (9 - x) \cdot (-360^\circ) = (2x - 9) \cdot 360^\circ.$$

So we must have $(2x - 9) \cdot 360^\circ = 0$, which is impossible for $x \in \{1, 2, \dots, 8\}$. Since there are 2^9 equally likely initial direction assignments and exactly two favorable assignments, the required probability is

$$\frac{2}{2^9} = \frac{1}{256}.$$

Hence the probability that every distinguishable ant is exactly at its own starting point after one minute is $\frac{1}{256}$.

Answer:

54. Ants On A Circle III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If all ants move clockwise or all move anticlockwise, they will each return to their original positions. We now show that if the number of ants is even (as in this problem), then, apart from the cases where all ants move clockwise or all move anticlockwise, the only way for each ant to return to its original position is if an equal number of ants initially move clockwise and anticlockwise. Note that an ant can return to its original position only if it moves a full 360° clockwise, a full 360° anticlockwise, or has a net movement of 0° . Moreover, if the ants are not all moving in the same direction, each ant will collide with at least one other and reverse its direction. Consequently, every ant must have a net movement of 0° for all of them to return to their original positions. Imagine the net angular displacement of ant i after one minute as θ_i , measured positive for clockwise and negative for counterclockwise. For every ant to be at its original location we must have $\theta_i = 0$ for every i , hence

$$\sum_{i=1}^{10} \theta_i = 0.$$

Let x be the number of ants that initially chose clockwise and $10 - x$ the number that initially chose counterclockwise. The quantity x is invariant in time because at each collision two ants reverse direction, so one clockwise ant becomes counterclockwise and one counterclockwise becomes clockwise. Thus the sum of the θ_i would be

$$\sum_{i=1}^{10} \theta_i = x \cdot 360^\circ + (10 - x) \cdot (-360^\circ) = (2x - 10) \cdot 360^\circ.$$

So we must have $(2x - 10) \cdot 360^\circ = 0$, which gives $x = 5$. This shows that, for all ants to return to their original positions (assuming not all move clockwise or all move anticlockwise), it must be that $x = 5$, i.e., five ants move clockwise and five move anticlockwise. However, we still need to show that if five ants move clockwise and five move anticlockwise, then all will indeed return to their original positions. So far, we've only established that this condition is necessary, not that it is sufficient. To prove sufficiency, note first that the initial and final sets of positions are identical (from Ants on a Circle I). Next, the circular order of the ants remains unchanged, since whenever two ants meet, they simply reverse directions rather than passing through each other, preserving their relative order around the circle. Finally, when five ants move clockwise and five anticlockwise, we have $\sum_{i=1}^{10} \theta_i = 0$. Together, these facts imply that each ant must return to its original position when an equal number of ants move clockwise and anticlockwise. Hence, the total number of configurations in which all ants return to their original positions is

$$1 + 1 + \binom{10}{5},$$

the first 1 corresponding to all ants moving clockwise, the second to all moving anticlockwise, and $\binom{10}{5}$ representing the cases where five ants move clockwise and five move anticlockwise. So, the required probability is

$$\frac{1 + 1 + \binom{10}{5}}{2^{10}} = \frac{127}{512}.$$

Answer: 0.24805

55. Game Of Evaluation

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Call the maximal sequence of numbers connected by multiplication signs ($\times$) a term. In other words, a term is a portion of the expression that is bounded on both sides by a $+$, $-$, or the start/end of the expression, and contains only multiplication symbols within it. A term has an odd value if and only if it consists of a single odd number. In other words, it must be an odd number that is bounded on both sides by a $+$, $-$, or the start or end of the expression.

Suppose Alice places a $+$ between 1 and 2 on her first move. She can then force the value of the remaining expression, from 2 through 100, to be even using the following strategy.

Label the gap between i and $i + 1$ as g_i . For instance, the gap between 2 and 3 is g_2 , between 3 and 4 is g_3 ,

and so on. Pair the gaps as

$$(g_2, g_3), (g_4, g_5), \dots, (g_{98}, g_{99}).$$

Whenever Bob places a symbol in one gap of a pair, Alice places a multiplication sign in the other gap of that same pair. This ensures that no term from 2 to 100 consists of a single odd number surrounded by + or − on both sides. Consequently, every term in this portion of the expression is even, and all are separated by + or −. Therefore the value of this part of the expression is even, making $1 + (\text{even})$ odd, and Alice wins.

Answer: 1

56. Ant On A Cube I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The cube has eight vertices: one is the starting vertex, v_0 , three vertices are adjacent to it (v_1), three vertices are adjacent to the opposite vertex (v_2), and the last one is the opposite vertex v_3 . We need to find the expected number of steps, $E[v_i]$, for the ant to reach v_3 starting from v_i . By setting this up as a Markov chain, we have the following system of equations:

$$E[V_0] = 1 + \frac{1}{3}E[V_1] + \frac{1}{3}E[V_1] + \frac{1}{3}E[V_1]$$

$$E[V_1] = 1 + \frac{1}{3}E[V_0] + \frac{1}{3}E[V_2] + \frac{1}{3}E[V_2]$$

$$E[V_2] = 1 + \frac{1}{3}E[V_1] + \frac{1}{3}E[V_1] + \frac{1}{3}E[V_3]$$

$$E[V_3] = 0$$

After solving the system of equations, we find that $E[V_0] = 10$.

Answer: 10

57. Dice Game VII

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We denote by X_1 the value of the first roll and by T the total sum of all rolls through the stopping time. Because each of the six faces on the first roll is equally likely, we have

$$E[T] = \sum_{i=1}^6 E[T|X_1 = i] \cdot P(X_1 = i) = \frac{1}{6} \sum_{i=1}^6 E[T|X_1 = i].$$

Fix i and write $T = i + T'$, where T' is the sum of all subsequent rolls until a face other than i appears. Thus

$$E[T|X_1 = i] = i + E[T'|X_1 = i].$$

On each roll after the first, the probability of rolling the same face i (and continuing) is $\frac{1}{6}$, while the probability of rolling a different face (and stopping) is $\frac{5}{6}$. When the process stops, the final roll is equally likely to be any of the five numbers in $\{1, 2, 3, 4, 5, 6\} \setminus \{i\}$, whose average value is

$$\frac{1 + 2 + \dots + 6 - i}{5} = \frac{21 - i}{5}.$$

Let $\mu_i = E[T'|X_1 = i]$. Conditioning on the next roll yields

$$\mu_i = \frac{1}{6}(i + \mu_i) + \frac{5}{6} \cdot \frac{21 - i}{5}.$$

$$\Rightarrow \mu_i = \frac{21}{5}.$$

Therefore, for every i ,

$$E[T|X_1 = i] = i + \frac{21}{5} = \frac{5i + 21}{5}.$$

Finally, averaging over $i = 1, \dots, 6$,

$$E[T] = \frac{1}{6} \sum_{i=1}^6 \frac{5i + 21}{5}.$$

$$E[T] = \frac{231}{30}.$$

Answer:

58. Duck Escape

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The duck swims in a circle of radius $r < \frac{R}{4}$, where R is the radius of the lake. The center of the duck's circle coincides with the lake's center. The fox is four times faster than the duck. By swimming in this smaller circle, the duck gradually increases its angular separation from the fox. The fox constantly runs around the lake's shore, trying to stay closest to the duck. However, since the duck is moving slightly faster in terms of angular velocity, the fox begins to lag behind. Eventually, the duck and fox end up on opposite sides of the lake. Once directly across from the fox, the duck swims straight to the shore. The duck travels a distance of $\frac{3R}{4}$, while the fox must run πR . Given the speed ratio, it will take $3R$ time units for the duck and $3.14R$ for the fox. Thus, the duck gains a slight head start to escape.

Answer:

59. Count Regions I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

It is clear that n lines will divide the plane into a maximum number of pieces if no two of them are parallel and no three of them are concurrent. Consequently, we have only to determine into how many pieces n mutually non-parallel lines, no three of which are concurrent, divide the plane. Suppose that k of the lines have already been drawn in the plane; let us draw the $(k+1)$ st line and see by how much it increases the number of pieces into which the plane is divided. The $(k+1)$ st line meets each of the k lines which have already been drawn; the k points of intersection divide it into $k+1$ parts. Consequently, the $(k+1)$ st line cuts exactly $k+1$ of the parts into which the plane has already been divided. Since it splits each of these parts into two pieces, drawing the $(k+1)$ st line increases the number of pieces by $k+1$. But if only one line is drawn, it will divide the plane into two pieces. It follows from this that after n lines have been drawn the plane will have been divided into

$$2 + 2 + 3 + 4 + \dots + n$$

parts (drawing the second line increases the number of parts by 2, drawing the third line increases it by 3 more, drawing the fourth increases it by 4 more, etc.). Consequently, the greatest number of parts into which n straight lines can divide the plane is

$$2 + 2 + 3 + \dots + n = (1 + 2 + 3 + \dots + n) + 1 = \frac{n(n+1)}{2} + 1 = \frac{n^2 + n + 2}{2}$$

Answer:

60. Count Regions III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

n planes will divide 3-dimensional space into a maximum number of parts if any three of them have exactly one point in common, and no four of them have a point in common. Suppose that k planes have already been drawn. Let us determine how much the $(k+1)$ st plane increases the number of regions in space. This plane intersects each of the first k planes in a line. Furthermore, any two of these lines intersect in exactly one point (since any three planes intersect at exactly one point), and no three lines are concurrent (since that would require four or more planes to intersect at a single point, which contradicts our assumption). Consequently, these k lines divide the $(k+1)$ st plane into

$$\frac{k^2 + k + 2}{2}$$

regions (this result comes from "Count Regions I"). Each of these regions is the surface where the $(k+1)$ st plane intersects a volume formed by the first k planes. So the $(k+1)$ st plane cuts exactly $\frac{k^2+k+2}{2}$ of the existing regions in half, thus increasing the number of regions by that amount. Since a single plane divides space into 2 parts, it follows that the total number of regions formed by n planes is:

$$2 + \frac{1^2 + 1 + 2}{2} + \frac{2^2 + 2 + 2}{2} + \cdots + \frac{(n-1)^2 + (n-1) + 2}{2}$$

This simplifies to:

$$\sum_{k=0}^{n-1} \frac{k^2 + k + 2}{2}$$

Evaluating the sum:

$$\begin{aligned} \sum_{k=0}^{n-1} \frac{k^2 + k + 2}{2} &= \frac{1}{2} \left(\sum_{k=0}^{n-1} k^2 + \sum_{k=0}^{n-1} k + \sum_{k=0}^{n-1} 2 \right) \\ &= \frac{1}{2} \left(\frac{(n-1)n(2n-1)}{6} + \frac{(n-1)n}{2} + 2n \right) \\ &= \frac{n^3 + 5n + 6}{6} \end{aligned}$$

So, the maximum number of regions that n planes can divide 3D space into is:

$$\frac{n^3 + 5n + 6}{6}$$

Answer:

61. Square Intersection I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the 2 random points be P and Q . Let L denote the straight line through the random points P and Q . For almost every choice of P and Q the line L meets the boundary of the square in exactly two points and those two intersection points lie on two of the four sides of the square. Let I_{bottom} be the indicator that the line meets the segment AB . For any pair of random points P and Q , the sum of these four indicators equals 2 (the line meets exactly two sides). By rotational symmetry all four sides are equally likely to be hit, so the expected value of each indicator is the same. Taking expectations,

$$4 \cdot E[I_{bottom}] = E[\text{sum of indicators}] = 2,$$

hence

$$E[I_{bottom}] = \frac{1}{2},$$

which is the probability that the line L intersects a given side. So the probability that L intersects AB is $\frac{1}{2}$.

Answer: 0.50000

62. Couple I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let I_i be the indicator function that takes value 1 when couple i sits together and 0 otherwise. Then the number of couples that sit together is given by $\sum_{i=1}^{10} I_i$. By linearity of expectation,

$$E \left[\sum_{i=1}^{10} I_i \right] = \sum_{i=1}^{10} E[I_i].$$

We need to find $E[I_i]$, the probability that couple i sits together. Fix the husband from couple i in an arbitrary seat. The two people adjacent to him must be women due to the alternating seating arrangement. Therefore, out of the 10 possible seats for his wife, 2 are next to him. Thus, the probability that the wife of this man is seated next to him is $\frac{2}{10}$. Hence, $E[I_i] = \frac{2}{10}$, and so

$$E \left[\sum_{i=1}^{10} I_i \right] = \sum_{i=1}^{10} \frac{2}{10} = 2.$$

Answer: 2

63. Even Ratio

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Because X and Y are continuous and independent on $(0, 1)$, the random ratio $\frac{Y}{X}$ takes values in $(0, \infty)$. Hence $\lfloor Y/X \rfloor$ is a positive integer, and

$$\{\lfloor Y/X \rfloor = n\} \iff n-1 < \frac{Y}{X} \leq n \iff (n-1)X < Y \leq nX$$

We want that chosen integer n be even. Write $n = 2k$ with $k = 1, 2, 3, \dots$. Then

$$\lfloor Y/X \rfloor = 2k \iff (2k-1)X < Y \leq 2kX$$

Because these events are disjoint for different k ,

$$P(\lfloor Y/X \rfloor \text{ is even}) = \sum_{k=1}^{\infty} P((2k-1)X < Y \leq 2kX)$$

Fix any integer $k \geq 1$. We must compute $P((2k-1)X < Y < 2kX)$, given that (X, Y) is uniform over the unit square $\{0 < x < 1, 0 < y < 1\}$. One convenient way is to integrate in the y -direction: for each fixed $y \in (0, 1)$, the inequalities $(2k-1)X < y \iff X < \frac{y}{2k-1}$, $y \leq 2kX \iff X \geq \frac{y}{2k}$. Hence, at a given y , the set of x -values satisfying both is $x \in [\frac{y}{2k}, \frac{y}{2k-1})$, provided $\frac{y}{2k} \leq 1$ and $\frac{y}{2k-1} \leq 1$. But for $k \geq 1$ and $y \in (0, 1)$, we always have $0 < \frac{y}{2k} < \frac{y}{2k-1} \leq 1$. Thus, for each $y \in (0, 1)$, the length of the allowable x -interval is

$$\frac{y}{2k-1} - \frac{y}{2k} = y \left(\frac{1}{2k-1} - \frac{1}{2k} \right) = \frac{y}{2k(2k-1)}$$

Since (X, Y) is uniform on the unit square, its joint density is 1. Integrating that interval-length over $y \in (0, 1)$ gives

$$P((2k-1)X < Y < 2kX) = \int_0^1 \frac{y}{2k(2k-1)} dy = \frac{1}{4k(2k-1)}$$

Therefore,

$$P(\lfloor Y/X \rfloor \text{ is even}) = \sum_{k=1}^{\infty} \frac{1}{4k(2k-1)} = \frac{1}{4} \sum_{k=1}^{\infty} \frac{1}{k(2k-1)}$$

Notice that for each $k \geq 1$,

$$\frac{1}{k(2k-1)} = 2\left(\frac{1}{2k-1} - \frac{1}{2k}\right)$$

Hence, $\sum_{k=1}^{\infty} \frac{1}{k(2k-1)} = 2 \sum_{k=1}^{\infty} \left(\frac{1}{2k-1} - \frac{1}{2k}\right) = 2\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots\right) = 2\ln(2)$. Therefore,

$$P(\lfloor Y/X \rfloor \text{ is even}) = \frac{1}{4}(2\ln(2)) = \frac{1}{2}\ln(2)$$

Answer: 0.34657

64. 2 Eggs

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider a strategy involving N drops. If the unknown threshold floor is between 1 and $N-1$, we can drop the egg from the N -th floor, which will not break. We then proceed to drop the egg sequentially from the 1st floor, 2nd floor, etc., until the threshold is found. Thus, the first move should be dropping the egg from the N -th floor. For the second move, we should drop the egg from the $(N + (N-1))$ -th floor. Since we've already used one drop at the N -th floor, we have $N-1$ drops remaining. This pattern continues. Consequently, the maximum number of floors that can be covered with N drops is:

$$N + (N-1) + (N-2) + \dots + 1 = \frac{N(N+1)}{2}$$

This sum must be at least 100. Therefore, $N \geq 14$. Thus, the minimum number of drops required is 14.

Answer: 14

65. Heads Run I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let a_n be the number of length- n sequences of H's and T's without ever having three consecutive heads. Based on the analysis in hints: * $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$ * $a_0 = 1$, $a_1 = 2$, $a_2 = 4$ We compute the values up to $n = 10$:

$$a_3 = a_2 + a_1 + a_0 = 4 + 2 + 1 = 7$$

$$a_4 = a_3 + a_2 + a_1 = 7 + 4 + 2 = 13$$

$$a_5 = a_4 + a_3 + a_2 = 13 + 7 + 4 = 24$$

$$a_6 = a_5 + a_4 + a_3 = 24 + 13 + 7 = 44$$

$$a_7 = a_6 + a_5 + a_4 = 44 + 24 + 13 = 81$$

$$a_8 = a_7 + a_6 + a_5 = 81 + 44 + 24 = 149$$

$$a_9 = a_8 + a_7 + a_6 = 149 + 81 + 44 = 274$$

$$a_{10} = a_9 + a_8 + a_7 = 274 + 149 + 81 = 504$$

There are $2^{10} = 1024$ total sequences of 10 coin flips, all equally likely. Hence, the probability of no run of three heads is:

$$\frac{a_{10}}{2^{10}} = \frac{504}{1024} = \frac{63}{128}$$

Answer: 0.49219

66. Smallest Set

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write the elements of S in their ternary expansion (i.e. base 3). Then the second condition translates into, if $d_1 d_2 \dots d_k \in S$, then $d_1 d_2 \dots d_k 0$ and $d_1 d_2 \dots d_k 1$ are also in S . It follows that S is the set of nonnegative integers whose tertiary representation contains only the digits 0 and 1. Since $2 \cdot 3^6 < 2008 < 3^7$, there are $2^7 = 128$ such elements less than 2008. Therefore, there are 128 such non-negative elements.

Answer:

67. Variable Game

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Denote by p_n the probability that you win the n -th round. By the problem's rules, your chance of winning round n depends only on whether you won or lost round $n - 1$. Concretely:

* If you lost round $n - 1$, then $P(\text{win } n \mid \text{lose } n - 1) = \frac{1}{2}$. * If you won round $n - 1$, then $P(\text{win } n \mid \text{win } n - 1) = \frac{1}{3}$.

By the law of total probability,

$$p_n = P(\text{win } n) = P(\text{win } n \mid \text{win } n - 1)p_{n-1} + P(\text{win } n \mid \text{lose } n - 1)(1 - p_{n-1}).$$

Substituting the conditional probabilities gives the recursion

$$p_n = \frac{1}{3}p_{n-1} + \frac{1}{2}(1 - p_{n-1}) = \frac{1}{2} - \frac{p_{n-1}}{6},$$

with the initial condition $p_1 = \frac{1}{2}$.

We now iteratively compute up to p_5 :

$$\begin{aligned} * p_1 &= \frac{1}{2} \quad * p_2 = \frac{1}{2} - \frac{1}{6} \times \frac{1}{2} = \frac{1}{2} - \frac{1}{12} = \frac{5}{12} \quad * p_3 = \frac{1}{2} - \frac{1}{6} \times \frac{5}{12} = \frac{1}{2} - \frac{5}{72} = \frac{31}{72} \quad * p_4 = \frac{1}{2} - \frac{1}{6} \times \frac{31}{72} = \frac{1}{2} - \frac{31}{432} = \frac{185}{432} \\ * p_5 &= \frac{1}{2} - \frac{1}{6} \times \frac{185}{432} = \frac{1}{2} - \frac{185}{2592} = \frac{1111}{2592} \end{aligned}$$

Hence the probability you win the 5th round is

$$p_5 = \frac{1111}{2592} \approx 0.42863$$

Answer:

68. Chess Tournament II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In the first round, there are 128 positions. Assume that in the first round, the player at position 1 plays the player at position 2, the player at position 3 plays the player at position 4, and so on. In the second round, the winner of the match between position 1 and position 2 will play the winner of the match between position 3 and position 4 - and so on for subsequent rounds. Without loss of generality, we can assume that the highest-rated player is occupying position 1.

The second-highest-rated player must occupy one of the positions 2-64 (63 favorable positions) so that he gets eliminated by the highest-rated player before the final. The third-highest-rated player must occupy one of the positions 65-128 (64 favorable positions) so that he plays the highest-rated player in the final. Hence, the required probability is:

$$\frac{63}{127} \times \frac{64}{126} = \frac{32}{127}$$

Answer:

69. Game Of Smaller Number

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let P_b denote Bob's payout when he selects the value b . Let A represent the integer chosen by Alice. The case where $A = b$ is irrelevant, as the payout remains 0 for both players. According to the Law of Total Expectation, we have:

$$E[P_b] = E[P_b \mid b > A] \cdot P[b > A] + E[P_b \mid b < A] \cdot P[b < A].$$

The probability $P[b > A]$ is given by

$$P[b > A] = \frac{b-1}{100}$$

and

$$P[b < A] = \frac{100-b}{100}$$

since there are $b-1$ integers in $\{1, \dots, b-1\}$ and $100-b$ integers in $\{b+1, \dots, 100\}$. To find $E[P_b \mid b > A]$, if $A < b$, the values are uniformly distributed over $\{1, \dots, b-1\}$, yielding an expected payout of

$$E[P_b \mid b > A] = \frac{b}{2}.$$

For $E[P_b \mid b < A]$, Bob simply receives his selected amount b . Combining these components, we arrive at:

$$E[P_b] = \left(\frac{b-1}{100}\right) \left(\frac{b}{2}\right) + \left(\frac{100-b}{100}\right) (b) = \frac{201b - 3b^2}{200}.$$

Taking the derivative and setting it to zero, the optimal value for b is 33.5. Hence, the maximum profit occurs at $b = 33$ or $b = 34$. The required maximum profit is

$$E[P_{33}] = E[P_{34}] = 16.83.$$

Answer:

70. Missing Sequence

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We assign a and b from the integers between 6 and 29 while ensuring the sequence is strictly increasing. First, exclude 6 and 20, as they would create an arithmetic progression with 4 terms. This leaves $29 - 6 + 1 - 2 = 22$ integers to choose from, giving $\binom{22}{2} = 231$ possible pairs of a and b . However, we must exclude additional pairs that form arithmetic sequences of 4 numbers. These are $(a, b) \neq (7, 9), (12, 21), (16, 28)$. Therefore, the final count is $231 - 3 = 228$.

Answer:

71. Passing The Coin II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Suppose Player 1 and then Player 2 both toss tails. In that case, Player 3 takes the coin. According to the result from "Passing the Coin I," Player 3's probability of winning is $\frac{81}{211}$. So, the probability that player 3 wins is:

$$\frac{2}{3} \times \frac{2}{3} \times \frac{81}{211} = \frac{36}{211}$$

Answer:

72. Remove The Subtree

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let X_i be an indicator variable that takes the value 1 when node i is chosen by Alice and 0 otherwise. The total number of steps is given by $\sum_{i=1}^7 X_i$. To find the expected number of steps, we first need the probability that node i was chosen given that it was removed. The probability that node i was chosen is $\frac{1}{\text{size of tree}}$. The probability that node i was removed is $\frac{\text{depth of node } i}{\text{size of tree}}$, since choosing any ancestor of node i will remove it. Thus, the probability that node i was chosen given that it was removed is:

$$\frac{\text{probability that node } i \text{ was chosen and removed}}{\text{probability that node } i \text{ was removed}} = \frac{\frac{1}{\text{size of tree}}}{\frac{\text{depth of node } i}{\text{size of tree}}} = \frac{1}{\text{depth of node } i}$$

Therefore, the expected number of steps required to remove the entire tree is:

$$E\left(\sum_{i=1}^7 X_i\right) = \sum_{i=1}^7 E(X_i) = \frac{1}{1} + \frac{1}{2} + \frac{1}{2} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{10}{3}$$

Answer:

73. Interesting Summation III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We start with the sum S and expand the terms as:

$$S = \sum_{k=0}^{\infty} \frac{F_k}{10^{k+1}} = \frac{F_0}{10} + \frac{F_1}{10^2} + \frac{F_2}{10^3} + \frac{F_3}{10^4} + \dots$$

This can be rewritten as:

$$S = \frac{F_0}{10} + \frac{F_1}{10^2} + \left(\frac{F_0}{10^3} + \frac{F_1}{10^3}\right) + \left(\frac{F_1}{10^4} + \frac{F_2}{10^4}\right) + \dots$$

Using the Fibonacci sequence definition, we regroup the terms:

$$S = \frac{F_0}{10} + \frac{F_1}{100} + \left(\frac{F_0}{10^3} + \frac{F_1}{10^4} + \dots\right) + \left(\frac{F_1}{10^3} + \frac{F_2}{10^4} + \dots\right)$$

Since $F_0 = 0$, we can add a $\frac{F_0}{10^2}$ term to the second parenthesized group without changing the value of the sum. Notably, the first parenthesized expression equals $\frac{S}{100}$, and the second parenthesized expression equals $\frac{S}{10}$. Therefore, we can rewrite the equation as:

$$S = \frac{0}{10} + \frac{1}{100} + \frac{1}{100}S + \frac{1}{10}S$$

This simplifies to:

$$\frac{89}{100}S = \frac{1}{100} \implies S = \frac{1}{89}$$

Answer:

74. Interesting Summation V

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let f_n be the number of length- n digit sequences over $\{0, 1, \dots, 9\}$ with no two adjacent 1's. For $n = 1$, any digit $0, 1, \dots, 9$ is allowed, so $f_1 = 10$. For $n = 2$, there are $10^2 = 100$ total pairs and only "11" is forbidden, so $f_2 = 100 - 1 = 99$. For $n \geq 3$, consider the first digit.

- If it is non-1 (9 choices), the remaining $n - 1$ digits form any valid sequence of length $n - 1$, contributing $9f_{n-1}$.
- If the first digit is 1 (1 choice), the second must be non-1 (9 choices), and the remaining $n - 2$ digits form any valid sequence of length $n - 2$, contributing $9f_{n-2}$.

Hence the recursion:

$$f_n = 9f_{n-1} + 9f_{n-2}$$

Compute up to $n = 5$:

$$f_3 = 9 \times 99 + 9 \times 10 = 981$$

$$f_4 = 9 \times 981 + 9 \times 99 = 9720$$

$$f_5 = 9 \times 9720 + 9 \times 981 = 96309$$

Thus there are $n[K] = f_5 = 96309$ valid five-digit strings. Let d_i be the i -th digit from the left.

To find the probability p_i that the i -th digit $d_i = 1$, count sequences with $d_i = 1$:

- For $i = 1$:

Place a 1 in the first position (1 way), force $d_2 \neq 1$ (9 ways), and the remaining 3 digits form any valid length-3 sequence ($f_3 = 981$ ways), giving

$$\#\{d_1 = 1\} = 1 \times 9 \times 981 = 8829, \quad p_1 = \frac{8829}{96309}$$

- For $i = 2$: Choose $d_1 \neq 1$ (9 ways), set $d_2 = 1$ (1 way), force $d_3 \neq 1$ (9 ways), and the last two any valid length-2 sequence ($f_2 = 99$ ways), giving

$$\#\{d_2 = 1\} = 9 \times 1 \times 9 \times 99 = 8019, \quad p_2 = \frac{8019}{96309}$$

- For $i = 3$: Any valid length-2 prefix ending in non-1 ($9 \times 10 = 90$ such), then $d_3 = 1$, $d_4 \neq 1$ (9 ways), d_5 any of 10 digits, giving

$$\#\{d_3 = 1\} = 90 \times 1 \times 9 \times 10 = 8100, \quad p_3 = \frac{8100}{96309}$$

By symmetry:

$$p_5 = p_1 = \frac{8829}{96309}, \quad p_4 = p_2 = \frac{8019}{96309}$$

When a digit is not 1, it is one of the 9 other digits $\{0, 2, \dots, 9\}$, whose average is $\frac{44}{9}$. Hence, for each position i ,

$$E[d_i] = p_i \times 1 + (1 - p_i) \times \frac{44}{9}$$

A short calculation shows:

$$E[d_1] = E[d_5] = \frac{436509}{96309}, \quad E[d_2] = E[d_4] = \frac{439659}{96309}, \quad E[d_3] = \frac{439344}{96309}$$

Finally, writing

$$K = d_1 \times 10^4 + d_2 \times 10^3 + d_3 \times 10^2 + d_4 \times 10 + d_5$$

Linearity of expectation gives

$$E[K] = 10^4 E[d_1] + 10^3 E[d_2] + 10^2 E[d_3] + 10 E[d_4] + E[d_5]$$

Multiplying by the total count $f_5 = 96309$ gives the exact integer sum

$$S = E[K] \times 96309 = 4,853,516,499$$

Therefore, the sum of all integers between 0 and 99999 with no two consecutive 1's is 4,853,516,499.

Answer:

75. Pair Partition

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Arrange the numbers 1 through 20 in a 2×10 array: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20. A pair $\{a, b\}$ with $|a - b| = 1$ corresponds to a horizontal domino (two adjacent cells in the same row). A pair with $|a - b| = 10$ corresponds to a vertical domino (one cell in the top row and the one directly below it). Hence every valid pairing is exactly a perfect tiling of this 2×10 board by dominoes. Let $T(n)$ be the number of domino-tilings of a $2 \times n$ rectangle. By looking at the left end of the board one sees the recursion:

- If the first column is covered by a vertical domino, the rest is a $2 \times (n - 1)$ tiling: $T(n - 1)$ ways.
- If the first two columns are covered by two horizontal dominoes, the rest is a $2 \times (n - 2)$ tiling: $T(n - 2)$ ways.

Thus,

$$T(n) = T(n - 1) + T(n - 2),$$

with initial conditions $T(1) = 1$, $T(2) = 2$. Therefore, the number of ways to partition $\{1, \dots, 20\}$ into 10 pairs each differing by 1 or 10 is $T(10) = 89$.

An important note: Now, consider the set $1, 2, 3 \dots 18$. Suppose we want to divide this set into 9 pairs such that the difference between the numbers in each pair is either 1 or 9. If we follow the same domino tiling approach as above, we find that the number of valid pairings is 55. However, there's one additional pairing configuration that the tiling method misses: pairing 1 with 2, 3 with 4, and so on up to 17 with 18. In this configuration, all pairs differ by 1, and it forms a valid complete pairing. Therefore, the total number of valid pairings is 56.

Answer:

76. Last One Off

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume each ant is carrying a flag. Imagine at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem, as the flags simply go past each other. This simplification is commonly used in many collision problems. Without loss of generality, we may assume that Alice departs by the east end of the rod, which means that the 12 ants to her east do the same. If she is last off, it must be that the 12 ants west of Alice drop off the west end; it follows that initially, exactly 12 flags, thus 12 ants, faced west. This happens with probability $\frac{\binom{25}{12}}{2^{24}}$, about 31%. However, Alice is not necessarily the last ant off in these cases; in some cases, her western neighbor would be the last ant off. The time each flag is fated to spend on the rod is independent and uniformly random between 0 and 100 seconds. Thus, the probability that the longest-lived flag is one of the 13 east-facing flags is $\frac{13}{25}$. It follows that the correct value is $\frac{13}{25} \times \frac{\binom{25}{12}}{2^{24}}$, about 0.16118.

Answer:

77. Strictly Within

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

An interval on the set $\{-n, \dots, n\}$ can either be a single point or a genuine two-endpoint interval. There

are $2n + 1$ single-point intervals, and

$$\binom{2n+1}{2} = \frac{(2n+1) \cdot 2n}{2}$$

two-endpoint intervals, so in total

$$(2n+1) + \binom{2n+1}{2} = (2n+1)(n+1)$$

possible intervals. Since we choose two intervals independently (first I_1 , then I_2), the size of our sample space is

$$((2n+1)(n+1))^2.$$

Let's count the favorable intervals now. We wish to count those ordered pairs (I_1, I_2) for which I_2 lies strictly inside I_1 . Notice immediately that I_1 cannot be a singleton, so I_1 must be a two-endpoint interval. Now, I_2 can be a two-endpoint interval or a singleton. We split according to whether I_2 is also two-endpoint or is a singleton:

Case 1: both intervals two-endpoint Writing $I_2 = [p, q]$ with $p < q$ and $I_1 = [r, s]$ with $r < s$, we need $p < r < s < q$. Equivalently, we choose any four distinct points out of the $2n + 1$, and let the outer two be p, q and the inner two be r, s . There are $\binom{2n+1}{4}$ ways to do this.

Case 2: I_1 two-endpoint but I_2 a singleton Let $I_2 = i$, where i must lie between $-n + 1$ and $n - 1$ so that it can be strictly inside some two-endpoint interval. To contain i , a two-endpoint $I_1 = [a, b]$ must satisfy $a < i < b$. There are $(i + n)$ choices for $a \in -n, \dots, i - 1$ and $(n - i)$ choices for $b \in i + 1, \dots, n$. Hence for each i there are $(i + n)(n - i)$ valid I_1 . Summing over all $i = -n + 1, \dots, n - 1$ gives

$$\sum_{i=-n+1}^{n-1} (i+n)(n-i).$$

To see that this equals $\binom{2n+1}{3}$, set $j = i + n$, so j runs from 1 to $2n - 1$ and $n - i = 2n - j$. Then

$$\sum_{i=-n+1}^{n-1} (i+n)(n-i) = \sum_{j=1}^{2n-1} j(2n-j) = 2n \sum_{j=1}^{2n-1} j - \sum_{j=1}^{2n-1} j^2.$$

We know $\sum_{j=1}^m j = \frac{m(m+1)}{2}$ and $\sum_{j=1}^m j^2 = \frac{m(m+1)(2m+1)}{6}$. Here $m = 2n - 1$, so

$$\sum_{j=1}^{2n-1} j = n(2n-1), \quad \sum_{j=1}^{2n-1} j^2 = \frac{(2n-1)(2n)(4n-1)}{6}.$$

Therefore

$$\sum_{j=1}^{2n-1} j(2n-j) = 2n \cdot n(2n-1) - \frac{(2n-1)(2n)(4n-1)}{6} = \frac{2n(2n-1)(2n+1)}{6} = \binom{2n+1}{3}.$$

Adding both cases, the total number of favorable pairs is

$$\binom{2n+1}{4} + \binom{2n+1}{3} = \binom{2n+2}{4},$$

where we've used the identity $\binom{N}{3} + \binom{N}{4} = \binom{N+1}{4}$. Dividing by the total $((2n+1)(n+1))^2$ gives

$$p(n) = \frac{\binom{2n+2}{4}}{((2n+1)(n+1))^2} = \frac{n(2n-1)}{6(n+1)(2n+1)}.$$

Answer: 0.13709

78. Get King And Ace

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the 8 special cards be the 4 kings and 4 aces and the 44 ordinary cards be the rest. Label the draws $1, 2, \dots$ until we have seen at least one king and at least one ace. Split the argument into two stages.

Stage 1 (time to the first special) Let G be the number of ordinary cards observed before the first special; then the time of the first special is $T_1 = G + 1$. The 8 specials split the 44 ordinary cards into 9 gaps; by symmetry the first special falls into a uniformly random one of those 9 gaps, so

$$\mathbb{E}[G] = \frac{44}{9}, \quad \mathbb{E}[T_1] = \frac{44}{9} + 1.$$

Stage 2 (additional draws until the opposite type appears) Condition on the event $G = g$ and on which type (king or ace) appeared first. After seeing g ordinary cards and one special, the remainder of the deck contains $44 - g$ ordinary cards and 7 specials, of which exactly 4 are of the opposite type we still need. The 4 opposite-type specials are uniformly distributed among the remaining cards, so they cut the remaining $44 - g$ ordinary cards into 5 exchangeable gaps. Thus the expected number of ordinary cards before the first opposite-type special, given $G = g$, equals $\frac{44-g}{5}$. Adding the opposite special itself gives the conditional expected additional draws

$$\mathbb{E}[\text{extra draws} \mid G = g] = \frac{44 - g}{5} + 1.$$

Take expectation over G . By linearity,

$$\mathbb{E}[\text{extra draws}] = \mathbb{E}\left[\frac{44 - G}{5} + 1\right] = \frac{44 - \mathbb{E}[G]}{5} + 1 = \frac{44 - \frac{44}{9}}{5} + 1.$$

Add the two stages:

$$\mathbb{E}[N] = \mathbb{E}[T_1] + \mathbb{E}[\text{extra draws}] = \frac{44}{9} + 1 + \frac{44 - \frac{44}{9}}{5} + 1 = \frac{689}{45} \approx 15.31111.$$

Alternate Solution: Let's first compute the expected number of cards until Bob flips over an ace. We can view this as computing the expected position of the first ace in a random permutation of 52 cards. Each permutation includes 4 aces among 52 cards, so there are 48 cards that are not aces. For each of the 48 non-ace cards, it has a $\frac{1}{5}$ probability of being before the first ace. By the Linearity of Expectation, the expected number of cards up to and including the first ace is:

$$\frac{52 - 4}{5} + 1 = \frac{53}{5}$$

Similarly, the expected number of cards to flip over the first king is also $\frac{53}{5}$. Now, define A as the number of cards needed to flip over until Bob sees the first ace, and K as the number of cards until the first king. We need to compute $E[\max(A, K)]$. Using the identity:

$$\max(a, b) = a + b - \min(a, b),$$

we find that:

$$E[\max(A, K)] = E[A] + E[K] - E[\min(A, K)].$$

We already have $E[A] = E[K] = \frac{53}{5}$. To calculate $E[\min(A, K)]$, we use the same reasoning but with 8 dividers (4 aces and 4 kings), yielding:

$$\frac{52 - 8}{9} + 1 = \frac{53}{9}$$

Thus, the final answer is:

$$E[\max(A, K)] = \frac{53}{5} + \frac{53}{5} - \frac{53}{9} = \frac{689}{45}.$$

Answer: 15.31111

79. Max Equals Min[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We know $a \leq 0$, because $\min\{0, b\} \leq 0$. Similarly, $b \geq 0$, as $\max\{0, a\} \geq 0$. Hence, we need to determine the probability that $a \leq 0$ and $b \geq 0$. The total number of ways to select a and b is $\binom{21}{2} = 441$. Among these, there are 11 possible values for both $a \leq 0$ and $b \geq 0$, so the probability is:

$$\frac{11}{21} \times \frac{11}{21} = \frac{121}{441}$$

Answer: **80. Uniform Sum Probability**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p_1, p_2, p_3, p_4, p_5 , and p_6 be the probabilities for one 6-sided die, and q_1, q_2, q_3, q_4, q_5 , and q_6 be the probabilities for another. Suppose that these dice together yield sums with uniform probabilities. That is, suppose $P(\text{sum} = k) = \frac{1}{11}$ for $k = 2, \dots, 12$. Then:

$$p_1 q_1 = \frac{1}{11} \quad \text{and} \quad p_6 q_6 = \frac{1}{11}$$

Also:

$$\frac{1}{11} = P(\text{sum} = 7) \geq p_1 q_6 + p_6 q_1,$$

so:

$$p_1 \frac{1}{11 p_6} + p_6 \frac{1}{11 p_1} \leq \frac{1}{11}$$

i.e.,

$$\frac{p_1}{p_6} + \frac{p_6}{p_1} \leq 1.$$

Now, if we let $x = \frac{p_1}{p_6}$, then we have:

$$x + \frac{1}{x} \leq 1,$$

which is impossible, since for positive real x , $x + \frac{1}{x} \geq 2$. Thus, no such dice are possible.

Answer: **81. Stop Or Continue**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

You have a choice: either stop and take your current score S , or continue, in which case you risk losing everything if you roll a 10. The key is to find the smallest score S such that stopping is better than rolling again.

Let's compute the expected gain from continuing, assuming your current score is S . When you roll again:

- With probability $\frac{9}{10}$, you roll a number from 1 to 9 and add it to your score.
- With probability $\frac{1}{10}$, you roll a 10 and your score becomes 0.

So, the expected value if you choose to continue is:

$$\text{EV}(\text{continue}) = \frac{1}{10} \cdot 0 + \frac{9}{10} \left(\frac{1 + 2 + \dots + 9}{9} + S \right)$$

$$= \frac{9}{10}(S + 5)$$

Now compare this with the option to stop and take your current score S . You should stop when:

$$S \geq \frac{9}{10}(S + 5)$$

$$\Rightarrow S \geq 45$$

So, the optimal strategy is to keep rolling until your score reaches 45 or more, and then stop. Therefore, the minimum score at which you should stop is 45.

Answer: 45

82. Greedy Pirates I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the pirates be called P_5, P_4, P_3, P_2, P_1 . P_5 is the leader. Let's study a simpler case with only 2 pirates. P_2 can propose to take all the 100 gold coins. Even if P_1 votes against this, P_2 will vote in favor, and the proposal will still be accepted (50% acceptance), leaving P_1 with zero coins. The distribution is (100, 0) for levels (2, 1) respectively. Now let's look at the case with 3 pirates. P_3 knows that if this proposal is not accepted, then P_2 will get all the gold and P_1 will get nothing. So P_3 can bribe P_1 with one gold coin. P_1 knows that one gold coin is better than zero, and hence votes in favor of this proposal. Therefore, P_3 can propose the following distribution:

Level-3 pirate: 99, Level-2: 0, Level-1: 1

Since pirate 1 and 3 will vote in favor, this proposal will be accepted. The distribution is (99, 0, 1) for levels (3, 2, 1) respectively. If there are 4 pirates, P_4 can extend this pattern by bribing those who would get nothing in the 3-pirates case, to ensure their votes. So the distribution will be (99, 0, 1, 0) for levels (4, 3, 2, 1) respectively. This will ensure that P_2 will vote in favor, and with the help of self-voting (P_4), we have 50% votes to accept this proposal. Similarly, P_5 can bribe P_3 and P_1 because they get nothing if the proposal is rejected, as P_4 was giving them nothing. Hence, at level-5, the proposal is:

(98, 0, 1, 0, 1)

This proposal will get accepted and provide the maximum amount of gold to the leader.

Answer: 98

83. Coin Toss Game V

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We know that A , B , and C follow independent geometric distributions with parameter $1/2$, representing the number of coin flips needed for each to get heads. The probability mass function (PMF) of each is given by

$$P(A = a, B = b, C = c) = \left(\frac{1}{2}\right)^a \left(\frac{1}{2}\right)^b \left(\frac{1}{2}\right)^c$$

for $a, b, c = 1, 2, 3, \dots$. We want to find the probability that $a < b < c$. First, sum over all possible values of a , starting from 1 to infinity. Next, sum b from $a + 1$ to infinity (since $b > a$), and finally sum c from $b + 1$ to infinity (since $c > b$). This gives the triple sum

$$\sum_{a=1}^{\infty} \sum_{b=a+1}^{\infty} \sum_{c=b+1}^{\infty} \left(\frac{1}{2}\right)^a \left(\frac{1}{2}\right)^b \left(\frac{1}{2}\right)^c$$

Now, we simplify the innermost sum over c :

$$\sum_{c=b+1}^{\infty} \left(\frac{1}{2}\right)^c = \frac{1}{2^b}$$

This reduces the original expression to

$$\sum_{a=1}^{\infty} \sum_{b=a+1}^{\infty} \left(\frac{1}{2}\right)^a \cdot \frac{1}{2^{b+1}}$$

Next, simplify the sum over b :

$$\sum_{b=a+1}^{\infty} \frac{1}{2^{b+1}} = \frac{1}{4^a} \cdot \frac{4}{3}$$

Finally, summing over a , we get

$$\frac{1}{3} \sum_{a=1}^{\infty} \frac{1}{2^a} \cdot 4^a = \frac{1}{3} \sum_{a=1}^{\infty} \frac{1}{8^a} = \frac{1}{21}$$

Thus, the required probability is $\frac{1}{21}$.

Answer: 0.04762

84. Expected Ones In Binary

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $f(n)$ be the answer for a and b selected from $\{0, 1, \dots, 2^n - 1\}$. Note that $f(1) = \frac{3}{4}$. Now, consider $f(n+1)$. We can think of a and b as being selected from $\{0, 1, \dots, 2^n - 1\}$ and then with probability $\frac{1}{2}$ adding 2^n to their sum and with probability $\frac{1}{2}$ adding 2^{n+1} . If nothing is added, the number of ones is just the number of ones in $a+b$. If 2^{n+1} is added, it is that number plus another one in the 2^{n+1} 's place. If 2^n is added, that's an extra 1 if and only if $a+b$ has a 0 in its 2^n 's place, i.e. $a+b < 2^n$. This happens with probability $\frac{2^n(2^n+1)}{2 \cdot 2^n \cdot 2^n} = \frac{2^n+1}{2^{n+1}}$. Thus, we have

$$f(n+1) = f(n) + \frac{1}{4} + \frac{1}{2} \cdot \frac{2^n+1}{2^{n+1}} = f(n) + \frac{1}{2} \left(1 + \frac{1}{2^{n+1}}\right).$$

From here, it is easy to find the explicit formula:

$$f(n) = \frac{n+1}{2} - \frac{1}{2^{n+1}},$$

so

$$f(9) = 5 - \frac{1}{1024} = \frac{5119}{1024}.$$

Answer: 4.99902

85. Points On The Same Face

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since a cube has 8 vertices, each vertex has a $\frac{1}{8}$ probability of being chosen as P . There is a $\frac{1}{8}$ chance that $Q = P$, a $\frac{3}{8}$ chance that Q is connected to P by an edge, a $\frac{3}{8}$ chance that Q is diagonally across a face from P , and a $\frac{1}{8}$ chance that Q is diagonally across the cube from P . We treat these 4 possibilities as separate cases.

If $Q = P$, then the only vertex R that is not contained in a common face with Q and P is the vertex diagonally across the cube from $Q = P$. So there is a $\frac{7}{8}$ chance that P, Q , and R are contained in a common face.

If Q is connected to P by an edge, then the only vertices R that are not contained in a common face with Q and P are one of the two vertices that are the endpoints of the edge that is opposite the edge PQ in the cube. So there is a $\frac{6}{8}$ chance that P, Q , and R are contained in a common face.

If Q is diagonally across a face from P , then there is a unique face that contains both P and Q , so there is a $\frac{4}{8}$ chance that P, Q , and R are contained in a common face.

If Q is diagonally across the cube from P , there is no face that contains P and Q and hence, there is no chance that P, Q , and R are contained in a common face.

Thus, the desired probability is

$$8 \times \frac{1}{8} \left(\frac{1}{8} \cdot \frac{7}{8} + \frac{3}{8} \cdot \frac{6}{8} + \frac{3}{8} \cdot \frac{4}{8} + \frac{1}{8} \cdot 0 \right) = \frac{37}{64}.$$

Answer: 0.57813

86. Product Of Cards

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We can locate the values 9 and 10 within 11 moves. Hence, the payout would be $9 \times 10 - 11 = 79$. The strategy is to fix one card and find its difference with 11 of the other 12 cards. Suppose the fixed card is a face card, which has a value of 0. Its difference with the remaining 11 cards would either contain: (i) Two zeros and a 9 or a 10 — in this case, when we get two zeros, we know the fixed card has a value of 0 (as two zeros cannot occur otherwise). We can also identify the position of 9 or 10 based on the difference, and the last card, which wasn't used for comparison, must be either 9 or 10. (ii) Both 9 and 10 — in this case, we know the fixed card has a value of 0 (as both 9 and 10 cannot appear as differences otherwise). In this case, we directly locate the positions of cards 9 and 10. Now, suppose the fixed card does not have a value of 0. Then at least two of the 11 differences would have the same value (these would be differences with the face cards). These two differences would equal the value of the fixed card. Since we know the value of the fixed card, we can identify the positions of cards 9 and 10. Hence, the maximum guaranteed profit is 79.

Answer: 79

87. Gold Auction

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If there are n uniformly distributed random variables in the range $[500, 1000]$, the expected maximum value is given by $1000 - \frac{500}{n+1}$. This is because n variables divide the segment $[500, 1000]$ into $n + 1$ segments of equal length on average. The expected payout when recruiting n people is therefore $1000 - \frac{500}{n+1} - 5n$. To maximize the expected payout, we define the expected payout function as:

$$E[P(n)] = 1000 - \frac{500}{n+1} - 5n$$

Taking the derivative and setting it equal to 0:

$$E'[P(n)] = \frac{500}{(n+1)^2} - 5 = 0$$

Rearranging gives:

$$\frac{500}{(n+1)^2} = 5 \implies (n+1)^2 = \frac{500}{5} = 100$$

Taking the square root yields:

$$n+1 = 10 \implies n = 9$$

Substituting $n = 9$ back into the expected payout formula:

$$E[P(9)] = 1000 - \frac{500}{9+1} - 5 \times 9 = 905$$

Thus, the maximum expected payout is 905.

Answer: 905

88. Cutting Cube

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The central cube has six faces. At least 6 cuts are necessary to cut through all six faces, regardless of how the blocks are repositioned after each cut. We have a simple strategy that requires 6 cuts. Start by making 4 cuts on the top surface (2 vertical and 2 horizontal), and then make 2 more cuts on the sides to complete the process. Thus, 6 cuts is the minimum required.

Answer: 6

89. Three Pegs

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Initially, the 3 pegs are at $(1, 0)$, $(0, 1)$, and $(1, 1)$. So, initially, the coordinate parities are (odd, even), (even, odd), and (odd, odd). The parity of $(0, 0)$ is (even, even). Note that whenever a peg jumps, the parity of both the x and y coordinates remains unchanged since the jump in any direction (x or y) is always a multiple of 2. So, the jump always looks like $(x, y) \rightarrow (x + 2p, y + 2q)$ for some integers p and q . Hence, we cannot place a peg on $(0, 0)$ since it has a different parity than the other 3 pegs' initial coordinates.

Answer: 2

90. Chocolate Problem

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The bag contains 3 large chocolates (40 g each) and 6 small chocolates (20 g each). Every time you draw, you either eat a small chocolate directly or, if you draw a large one, you split it into two small pieces, eat one (20 g) and return the other. Notice that on every draw you consume exactly one 20 g piece, so the total chocolate eaten equals 20 g times the number of draws until all three large chocolates have been drawn (and thus converted). We model this as three consecutive "stages," where each stage ends the moment a large chocolate is drawn. If at the start of stage i there are M_i small and N_i large chocolates, then by the classic "first ace" result in drawing without replacement the expected draws in that stage is

$$E_i = \frac{M_i + N_i + 1}{N_i + 1}$$

After eating the $E_i - 1$ small pieces that precede the large draw, splitting the large adds one small piece back, so the counts update by

$$M_{i+1} = M_i - \frac{M_i}{N_i + 1} + 1, \quad N_{i+1} = N_i - 1.$$

Starting with $M_1 = 6, N_1 = 3$: Stage 1:

$$E_1 = \frac{6 + 3 + 1}{3 + 1} = \frac{10}{4} = 2.5.$$

Afterward,

$$M_2 = 6 - \frac{6}{4} + 1 = 5.5, \quad N_2 = 2.$$

Stage 2:

$$E_2 = \frac{5.5 + 2 + 1}{2 + 1} = \frac{8.5}{3} \approx 2.8333.$$

Afterward,

$$M_3 = 5.5 - \frac{5.5}{3} + 1 \approx 4.6667, \quad N_3 = 1.$$

Stage 3:

$$E_3 = \frac{4.6667 + 1 + 1}{1 + 1} = \frac{6.6667}{2} = 3.3333.$$

Summing these gives an expected total of

$$E[\text{draws}] = 2.5 + 2.8333 + 3.3333 = 8.6667 \text{ draws.}$$

Since each draw eats one 20 g piece, the expected amount of chocolate consumed is

$$20 \text{ g} \times 8.6667 \approx 173.33 \text{ grams.}$$

Answer:

91. Fixed Points I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In a random function $f : S_n \rightarrow S_n$, the probability of any given value being a fixed point is $\frac{1}{n}$. Since there are n elements in the set S_n , the number of fixed points, F_n , follows a binomial distribution $F_n \sim \text{Binom}(n, \frac{1}{n})$. Using the Poisson limit theorem, as n becomes large, F_n approaches a Poisson distribution with parameter 1, i.e., $F_n \sim \text{Poisson}(1)$. This implies that the probability of having exactly 5 fixed points is given by:

$$P(F_n = 5) = \frac{1^5 e^{-1}}{5!} = \frac{e^{-1}}{5!}$$

Thus, the probability is:

$$P(F_n = 5) = \frac{e^{-1}}{120}$$

Answer:

92. Reaching 50

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If Alice says 39, she can always ensure victory because she can respond to whatever Bob adds by making the total 50. For example, if Bob says 1, making the total 40, Alice can add 10 to win. In every case, Alice can adjust her response so that both turns add up to 11. Working backward from 39, if Alice says 28, she can again control the game to ensure reaching 39. Similarly, if she says 17, she can force the total to 28 on her next turn. Finally, starting with 6 lets her continue this strategy, consistently adding 11 after each of Bob's turns until reaching 50. Therefore, Alice should start by saying 6.

Answer:

93. Game On Divisors I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The prime numbers in $S = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29$ are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29$, totaling 10 integers. A critical observation is that if a player selects the integer 6, they eliminate the primes 2 and 3. No other pairwise products of the remaining prime integers are less than or equal to 30. This leaves 8 primes, and the player who selects first in this scenario will lose, as they can eliminate only one prime factor per turn.

However, this situation is essentially the second turn of the game after eliminating 2 and 3. Thus, Alice should choose to go first and select the integer 6 (alternatively, 12, 18, or 24, which also eliminate both 2 and 3). Following this, Alice and Bob will alternate selections from the remaining primes, forcing Bob to choose a non-prime integer with a prime factor among those already eliminated, leading to his loss. Therefore, Alice should go first.

Additionally, there is another effective initial choice: 5. Regardless of Bob's selection after Alice chooses 5, she can force him to eliminate one prime factor at a time on her subsequent turns. Therefore, both 5 and $5^2 = 25$ are also valid first moves.

In conclusion, Alice should go first, and her possible first moves are 6, 12, 18, 24, 5, or 25.

Answer: 1

94. At Least Six II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write E_m for the expected number of additional rolls needed when your current total is m . Clearly $E_m = 0$ for all $m \geq n$. If $m < n$, then on the next roll you add one of $\{1, \dots, n\}$ uniformly, so you always spend one roll and then your new total is $m + j$. Hence

$$E_m = 1 + \frac{1}{n} \sum_{j=1}^n E_{\min(m+j, n)} = 1 + \frac{1}{n} \sum_{j=1}^{n-m} E_{m+j}$$

since any $m + j \geq n$ contributes zero. It is convenient to re-index by setting $k = n - m$, so

$$D_k := E_{n-k}$$

measures the expected rolls when you are k below the target. Then $D_0 = 0$ and for $k \geq 1$,

$$D_k = 1 + \frac{1}{n} \sum_{j=1}^k D_{k-j} = 1 + \frac{1}{n} \sum_{j=0}^{k-1} D_j$$

We compute the first few: $D_1 = 1$ (the sum is empty). $D_2 = 1 + \frac{1}{n} D_1 = 1 + \frac{1}{n}$.

$$D_3 = 1 + \frac{1}{n} (D_2 + D_1) = 1 + \frac{1}{n} \left(1 + \frac{1}{n} + 1 \right) = 1 + \frac{2}{n} + \frac{1}{n^2}$$

$$D_4 = 1 + \frac{1}{n} (D_3 + D_2 + D_1 + D_0) = 1 + \frac{1}{n} \left(\left(1 + \frac{2}{n} + \frac{1}{n^2} \right) + \left(1 + \frac{1}{n} \right) + 1 + 0 \right) = 1 + \frac{3}{n} + \frac{3}{n^2} + \frac{1}{n^3}$$

From these one sees the pattern

$$D_k = \sum_{i=0}^{k-1} \binom{k-1}{i} \frac{1}{n^i}$$

To prove this, note the base case

$$D_1 = 1 = \sum_{i=0}^0 \binom{0}{i} n^{-i}$$

If for all $j \leq k$ we assume

$$D_j = \sum_{i=0}^{j-1} \binom{j-1}{i} n^{-i}$$

then

$$\sum_{j=0}^{k-1} D_j = \sum_{j=0}^{k-1} \sum_{i=0}^{j-1} \binom{j-1}{i} n^{-i} = \sum_{i=0}^{k-1} n^{-i} \sum_{j=i+1}^{k-1} \binom{j-1}{i}$$

using the identity

$$\sum_{j=i+1}^{k-1} \binom{j-1}{i} = \binom{k-1}{i+1}$$

Hence

$$D_{k+1} = 1 + \frac{1}{n} \sum_{j=0}^k D_j = 1 + \frac{1}{n} \sum_{i=0}^{k-1} \binom{k-1}{i+1} n^{-i} = \sum_{i=0}^k \binom{k}{i} n^{-i}$$

which is exactly the same form with $k \mapsto k+1$. This completes the proof by induction. Finally,

$$E[T] = E_0 = D_n$$

so

$$E[T] = D_n = \sum_{i=0}^{n-1} \binom{n-1}{i} \frac{1}{n^i} = \left(1 + \frac{1}{n}\right)^{n-1}$$

Thus the expected number of rolls to reach a total of at least n is

$$\left(1 + \frac{1}{n}\right)^{n-1}$$

Answer:

95. Painted Cube III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

There are 2^6 possible colorings of the cube. Consider the color that appears with greater frequency. The property obviously holds true if 5 or 6 of the faces are colored the same, which for each color can happen in $6 + 1 = 7$ ways. If 4 of the faces are colored the same, there are 3 possible cubes (corresponding to the 3 possible ways to pick pairs of opposite faces for the other color). If 3 of the faces are colored the same, the property obviously cannot be satisfied. Thus, there are a total of $2(7 + 3) = 20$ ways for this to occur, and the desired probability is:

$$\frac{20}{2^6} = \frac{5}{16}$$

Answer:

96. Sufficient Sum I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We shall first examine the low-dimensional cases by direct geometric reasoning, observe a pattern, and then confirm it in general by induction. When $n = 1$, the event $X_1 \leq 1$ clearly has probability

$$P(X_1 \leq 1) = 1 = \frac{1}{1!}$$

For $n = 2$, one samples (X_1, X_2) uniformly in the unit square $[0, 1]^2$. The region

$$\{x_1 + x_2 \leq 1\}$$

is a right triangle of area

$$\frac{1 \cdot 1}{2} = \frac{1}{2} = \frac{1}{2!}$$

For $n = 3$, (X_1, X_2, X_3) lies in the unit cube $[0, 1]^3$, and

$$\{x_1 + x_2 + x_3 \leq 1\}$$

is a tetrahedron of volume

$$\frac{1}{3!} = \frac{1}{6}$$

These three computations all yield $\frac{1}{n!}$ for $n = 1, 2, 3$, suggesting that in general

$$P(X_1 + \cdots + X_n \leq 1) = \frac{1}{n!}$$

Note that since $(X_1, \dots, X_n)$ is chosen uniformly from the unit n -cube, the total "volume" of outcomes is

$$\text{Vol}([0, 1]^n) = 1$$

By our pattern, the volume of the subset $\{x_1 + \cdots + x_n \leq 1\}$ must therefore be $\frac{1}{n!}$. We prove for all $n \geq 1$ that

$$V_n = \int_{[0,1]^n} 1(x_1 + \cdots + x_n \leq 1) dx_1 \dots dx_n = \frac{1}{n!}$$

Base case ($n = 1$):

$$V_1 = \int_0^1 1(x_1 \leq 1) dx_1 = 1 = \frac{1}{1!}$$

Inductive step. Assume $V_n = \frac{1}{n!}$. Then

$$V_{n+1} = \int_0^1 \left[\int_{[0,1]^n} 1(x_1 + \cdots + x_n \leq 1 - x_{n+1}) dx_1 \dots dx_n \right] dx_{n+1}$$

By the induction hypothesis, the inner integral equals

$$V_n(\text{scaled to side } 1 - x_{n+1}) = \frac{(1 - x_{n+1})^n}{n!}$$

Hence

$$V_{n+1} = \frac{1}{n!} \int_0^1 (1 - x)^n dx = \frac{1}{n!} \cdot \frac{1}{n+1} = \frac{1}{(n+1)!}$$

completing the induction. Therefore, for $n = 5$,

$$P(X_1 + X_2 + \cdots + X_5 \leq 1) = \frac{1}{5!} = \frac{1}{120}$$

Answer: 0.00833

97. Interesting Summation II

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

We can form $10 \times 9 \times 8 = 720$ distinct numbers where each digit a , b , and c is chosen from 0 to 9 and are distinct. Each digit a , b , and c has an average value of 4.5. To find the average value of the repeating decimal $0.\overline{abc}$, consider the contribution of each digit to the overall value. The average value of each digit being 4.5 means the average number formed is:

$$\frac{4.5}{10} + \frac{4.5}{100} + \frac{4.5}{1000} + \dots$$

This is a geometric series with the first term $\frac{4.5}{10}$ and ratio $\frac{1}{10}$. Summing this series gives:

$$\frac{4.5}{10} \times \frac{1}{1 - \frac{1}{10}} = \frac{4.5}{10} \times \frac{10}{9} = \frac{4.5}{9} = 0.5$$

Therefore, the average value of each number is 0.5. With 720 numbers, the total sum is: $0.5 \times 720 = 360$

Answer: 360

98. Random Pairing

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let s_i represent the event that couple i is paired together. We are tasked with finding $P[S_1 \cup S_2 \cup S_3]$. By the inclusion-exclusion principle, this can be written as:

$$P[S_1 \cup S_2 \cup S_3] = 3P[S_1] - 3P[S_1 \cap S_2] + P[S_1 \cap S_2 \cap S_3]$$

To compute $P[S_1]$, we fix one person from couple 1. Since only one of the five remaining people is their partner:

$$P[S_1] = \frac{1}{5}$$

Next, to find $P[S_1 \cap S_2]$, note that if couple 1 is already paired, there are three people left for couple 2. Thus:

$$P[S_1 \cap S_2] = \frac{1}{15}$$

Finally, if two couples are already paired, the third couple must also be paired. So:

$$P[S_1 \cap S_2 \cap S_3] = \frac{1}{15}$$

Plugging these values into the inclusion-exclusion formula:

$$P[S_1 \cup S_2 \cup S_3] = 3 \times \frac{1}{5} - 3 \times \frac{1}{15} + \frac{1}{15} = \frac{7}{15}$$

Answer: 0.46667

99. Odd Heads I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p be the probability that the number of heads in the first 149 tosses is odd. Then, the probability that the number of heads in the first 149 tosses is even is $1 - p$.

Assume the number of heads in the first 149 tosses is odd. To get an odd number of heads in total, the last toss must be a tail. The probability of the last toss being a tail is $\frac{1}{2}$.

Now, assume the number of heads in the first 149 tosses is even. To get an odd number of heads in total, the last toss must be a head. The probability of the last toss being a head is $\frac{1}{2}$.

Hence, the probability that the total number of heads in 150 tosses is odd is:

$$P(\text{odd heads}) = p \cdot \frac{1}{2} + (1 - p) \cdot \frac{1}{2} = \frac{1}{2}$$

Another way to think about this is that only the last coin toss determines whether the number of heads is odd or not. Thus, the probability of having an odd number of heads in 150 tosses is $\frac{1}{2}$.

Answer: 0.50000

100. Tetrahedral Die II[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let X be the random variable representing the number of steps Maryam takes. Then

$$\mathbb{E}(X) = \sum_{k=0}^{\infty} kP(X = k) = \sum_{k=0}^{\infty} P(X \geq k).$$

Now, $P(X \geq k)$ is the probability that Maryam's game lasts at least k rolls. Since the game lasts more than 1 roll, we know $P(X \geq 0) = 1$, so we can assume $k > 0$.

For the game to last at least k rolls, Maryam's first $k - 1$ rolls must be a strictly decreasing sequence. For every subset of $k - 1$ distinct die outcomes, there is a unique way to arrange them in decreasing order. Thus, there are

$$\binom{4}{k-1}$$

ways to get $k - 1$ decreasing outcomes in a row. Since there are 4^{k-1} possible ways to roll the tetrahedral die, we have

$$P(X \geq k) = \frac{\binom{4}{k-1}}{4^{k-1}}.$$

Therefore,

$$\mathbb{E}(X) = \sum_{k=0}^{\infty} \binom{4}{k-1} \left(\frac{1}{4}\right)^{k-1} = \left(1 + \frac{1}{4}\right)^4 = \left(\frac{5}{4}\right)^4 = \frac{625}{256},$$

where we have used the convention that $\binom{n}{k} = 0$ if $k < 0$ or $k > n$.

Answer: 2.44141

101. Greedy Pirates II[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If the problem involved only gold and no silver, the optimum number to say would be 1005 because the amount of gold in each chest might be equal—hence, you need to say at least 1005. And 1005 is sufficient as well as you can take the 1005 chests that have the highest amount of gold. Now, let us return to the original problem. Since the captain wants to keep at least half of the gold, he needs again at least 1005 chests. It turns out that 1005 is also enough in this case. To see this, let us order again the chests according to their amount of gold $a_1 \geq a_2 \geq \dots \geq a_{2009}$. The choice of the captain is as follows. First, he splits the chests into the following groups: $\{a_1\}$, $\{a_2, a_3\}$, $\{a_4, a_5\}$, $\dots$, $\{a_{2008}, a_{2009}\}$. From each group, he picks the one that has more silver. By doing this, he guarantees he keeps half of the total amount of silver. However, in each group, the chest he did not choose has at most the same amount of gold as the chest he chose from the previous group. Thus, he also has at least half of the total amount of gold.

Answer: 1005

102. Horse Racing[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Begin by dividing the 25 horses into five groups: 1-5, 6-10, 11-15, 16-20, and 21-25. Conduct one race per group, resulting in five races. This step provides the relative speeds within each group. Assume that the horses are ranked in numerical order within each group (e.g., in group 6-10, horse 6 is the fastest and horse 10 is the slowest). From these results, select the fastest horse from each group: horses 1, 6, 11, 16, and 21. Eliminate the last two horses from each group since they cannot be among the top three. Race the top five horses to determine the fastest horse overall. This race also helps identify potential candidates for the

second and third places. Specifically, horses 2 and 3 from group 1-5, horses 6 and 7 from group 6-10, and horse 11 from group 11-15 are contenders for the remaining top spots. Finally, race these candidates: horses 2, 3, 6, 7, and 11. Therefore, the total number of races required to determine the top three fastest horses is seven.

Answer: 7

103. 3 Heads 3 Tails I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E_t be the expected number of tosses right after we get a tails. Let E_{tt} be the expected number of tosses right after we get 2 tails in a row. Let E_h be the expected number of tosses right after we get a heads. Let E_{hh} be the expected number of tosses right after we get 2 heads in a row.

We get the following equations: $E_h = 1 + \frac{1}{2}E_t + \frac{1}{2}E_{hh}$ $E_t = 1 + \frac{1}{2}E_h + \frac{1}{2}E_{tt}$ $E_{tt} = 1 + \frac{1}{2}E_h$ $E_{hh} = 1 + \frac{1}{2}E_t$

We can solve the above system of equations easily by noticing that $E_{tt} = E_{hh}$ and $E_t = E_h$ by symmetry.

Solving the above system of equations, we get $E_t = 6$ and $E_h = 6$. So, using the law of total expectation, the expected number of tosses is:

$$\frac{1}{2}(1 + 6) + \frac{1}{2}(1 + 6) = 7$$

Answer: 7

104. Going In Circles

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We transform this problem into a straight-line walk with two absorbing boundaries: (i) Consider point 1 on the circle as point 0 on the line. (ii) A single counterclockwise move from circle-point 1 brings you directly to circle-point 0; on the line this is moving from 0 to -1 , the left absorbing boundary. (iii) Conversely, going clockwise all the way around the circle (99 steps) also brings you from circle-point 1 back to circle-point 0; on the line this is moving from 0 to $+99$, the right absorbing boundary. Thus on the line you start at 0, with absorbing boundaries at -1 and $+99$. For a simple symmetric random walk starting at 0 with boundaries at $-b$ and a , the expected time to hit either boundary is $a \times b$. Here $b = 1$ (distance from 0 to -1) and $a = 99$ (distance from 0 to $+99$), so the expected number of steps until you first absorb-and thus first reach circle-point 0-is $99 \times 1 = 99$.

Answer: 99

105. Even Majority

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For each roll of values (x, y, z) where at least two numbers are even, the corresponding roll $(7 - x, 7 - y, 7 - z)$ will have fewer than two even values. Therefore, the number of rolls with two or more even values is equal to those with fewer than two. Since there are $6^3 = 216$ total possible rolls, half of them, or 108, will have at least two even numbers. We need to count the ways to roll at least two even numbers and have the sum of the dice be no more than 7. Valid outcomes include permutations of $(1, 2, 4)$, $(2, 2, 3)$, $(2, 2, 1)$, and $(2, 2, 2)$. These correspond to the following number of distinct permutations:

- $(1, 2, 4)$ has $3! = 6$
- $(2, 2, 3)$ has 3 permutations
- $(2, 2, 1)$ also has 3
- $(2, 2, 2)$ has 1.

Adding them up gives $6 + 3 + 3 + 1 = 13$ favorable outcomes. Therefore, the probability is $\frac{13}{108}$.

Answer:

106. Candy Distribution

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the seats be numbered from 1 to 2022. Also, let Bob be seated at seat 1. Let s_i denote the seat number of the i -th candy. If the candies never reach or cross Bob, in one move, $\sum_{i=1}^{2023} s_i^2$ increases by 2. Now, if we assume that the candies never reach Bob, this sum will increase indefinitely, which is impossible. Hence, Bob will receive a candy and moves will terminate.

Answer:

107. Roll Till Even

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability of rolling an even number on any roll is $\frac{1}{2}$. Hence, the number of rolls till you get an even number follows a geometric distribution with probability parameter $\frac{1}{2}$. Therefore, the expected number of rolls is $\frac{1}{(1/2)} = 2$.

Answer:

108. Triangle Inside Circle I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total number of ways of choosing 3 points from 1153 points is $\binom{1153}{3}$. The triangle will be obtuse when all three vertices are within the same half of the circle. We can always designate one vertex as the "start" of a half-circle, moving clockwise. Choose one point to be the vertex at the start of this half-circle; there are $\binom{1152}{1}$ ways to choose this point. There are 1152 points remaining, half of which are eligible for the other two points on the same half as the first point in the clockwise direction; so there are $\binom{576}{2}$ ways to choose the other two vertices. So the answer is $\frac{\binom{1152}{1} \times \binom{576}{2}}{\binom{1153}{3}} = \frac{1,725}{2,302}$.

Answer:

109. Lily Pad

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The 8 lily pads are equivalent to one lily pad that has matured for 3 days, because $2^3 = 8$. Therefore, it will take $10 - 3 = 7$ days to completely cover the pond.

Answer:

110. Passing The Bottle

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider some person X that is not Moor, and let L and R be the people to X 's left and right, respectively. For X to be the last person, the bottle must have first gone to L and then all the way around to R before reaching X , or it could have gone to R first and then come around to L before reaching X . We note that, regardless of whether L or R was visited first, the probability of the bottle going around before reaching X is the same. Therefore, since at least one of L or R must have been visited before X , and it does not matter which one was visited first, each of Moor's friends is equally likely to be the last to touch the bottle. The expected distance is then:

$$\frac{1 + 2 + 3 + 4 + 5 + 4 + 3 + 2 + 1}{9} = \frac{25}{9}$$

Answer: 2.77778

111. Jumping Frog I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The frog can reach position k from the positions $k - 1$ and $k - 2$. Hence,

$$p_k = \frac{1}{2}p_{k-1} + \frac{1}{2}p_{k-2}$$

Rearranging gives:

$$p_k - \frac{1}{2}p_{k-1} - \frac{1}{2}p_{k-2} = 0$$

This is a homogeneous second-order recurrence relation with constant coefficients. The characteristic polynomial is:

$$x^2 - \frac{1}{2}x - \frac{1}{2} = 0$$

Solving for x :

$$x = 1, -\frac{1}{2}$$

The general solution must be of the form:

$$p_k = a + b \left(-\frac{1}{2}\right)^k$$

where a and b are constants. Clearly, $p_k = a + b(1)^k$ is not valid as it would imply a constant p_k , which contradicts the fact that $p_0 = 1$ and $p_1 = \frac{1}{2}$. Thus:

$$p_k = a + b \left(-\frac{1}{2}\right)^k$$

Using the initial conditions $p_0 = 1$ and $p_1 = \frac{1}{2}$, we find:

$$p_k = \frac{2}{3} + \frac{1}{3} \left(-\frac{1}{2}\right)^k$$

The maximum probability occurs at $k = 2$, giving:

$$p_{max} = \frac{2}{3} + \frac{1}{3} \left(-\frac{1}{2}\right)^2 = \frac{3}{4}$$

Answer: 0.75000

112. Penguin Walk II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p represent the probability that the penguin moves towards the right at a given step (which is $\frac{1}{3}$ for the given problem). Let s represent the probability that the penguin eventually reaches position 1. First, if the penguin moves right (to position 1) on the first step, which happens with probability p , it has already reached its goal. If it moves left (to position -1) with probability $1 - p$, it must first return to position 0 (probability s) and then proceed to position 1 (also probability s). Hence, the probability of reaching 1 from position -1 is s^2 . Thus, we can set up the equation:

$$s = p + (1 - p)s^2$$

Solving for s , we get $s = 1$ or $s = \frac{p}{1-p}$. Since s represents a probability, $0 \leq s \leq 1$. So, if $p \geq \frac{1}{2}$, $s = 1$ because $\frac{p}{1-p}$ is greater than 1 for $p \geq \frac{1}{2}$. For $p < \frac{1}{2}$, the random walk is biased towards the left, hence $s \neq 1$. For $p < \frac{1}{2}$, $s = \frac{p}{1-p}$. For $p = \frac{1}{3}$, $s = \frac{1/3}{1-1/3} = \frac{1/3}{2/3} = \frac{1}{2}$.

Answer: 0.50000

113. Interview I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

It is clearly never optimal to select a candidate who isn't the best among those interviewed so far. If a candidate is not the best up to this point, they cannot be the best overall. Suppose you select the i -th candidate who is the best among those interviewed so far. Using conditional probability, we can show that if the current candidate is the best among the first i interviewed, then the probability that they are the best overall is

$$\frac{i}{6}$$

Let A_i be the event that the i -th candidate is the best among the first i interviewed, and let A be the event that this same candidate is the best of all six. We wish to compute:

$$P(A|A_i)$$

By the definition of conditional probability:

$$P(A|A_i) = \frac{P(A \cap A_i)}{P(A_i)}$$

Since being the best overall (event A) automatically implies being the best among the first i (event A_i), we have $A \subset A_i$, and hence:

$$P(A \cap A_i) = P(A)$$

Because all six candidates are equally likely to occupy any position in the random interview order, the probability that a particular person is the very best is

$$\frac{1}{6}$$

Likewise, among the first i candidates, each is equally likely to be the best so far, so:

$$P(A_i) = \frac{1}{i}$$

Putting these together gives:

$$P(A|A_i) = \frac{P(A)}{P(A_i)} = \frac{1/6}{1/i} = \frac{i}{6}$$

as claimed. So, we see that it is better to select a "best till now" candidate as late as possible. However, if we delay the selection too long, we risk encountering no "best till now" candidates later in the process. To balance this trade-off, the optimal strategy is to reject the first k candidates outright, then select the first "best till now" candidate that appears afterward. Let's find the probability of selecting the best candidate

in terms of k . Under the rule "skip the first k candidates, then hire the next best till now," we only become eligible at position $k + 1$. If the true best candidate appears in position i , that event has probability

$$\frac{1}{6}$$

Given they appear at $i \geq k + 1$, we will indeed select them if no earlier record-breaker after position k has occurred. Equivalently, the maximum among the first $i - 1$ candidates must lie in the initial block of size k , which has probability

$$\frac{k}{i - 1}$$

Hence for each $k = 0, 1, \dots, 5$ the success probability is:

$$P(k) = \sum_{i=k+1}^6 \left(\frac{1}{6} \times \frac{k}{i-1} \right)$$

Below we evaluate $P(k)$ for each k : $k = 0$ (hire candidate 1 immediately): By convention you must take whoever appears first, so

$$P(0) = \frac{1}{6} \approx 0.167$$

$k = 1$ (skip one, then first "best till now"):

$$P(1) = \frac{1}{6} \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} \right) = \frac{137}{360} \approx 0.381$$

$k = 2$ (skip two):

$$P(2) = \frac{1}{6} \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} \right) = \frac{77}{180} \approx 0.428$$

$k = 3$ (skip three):

$$P(3) = \frac{1}{6} \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} \right) = \frac{47}{120} \approx 0.392$$

$k = 4$ (skip four):

$$P(4) = \frac{1}{6} \left(\frac{1}{4} + \frac{1}{5} \right) = \frac{3}{10} = 0.300$$

$k = 5$ (skip five, must take sixth):

$$P(5) = \frac{1}{6} \cdot \frac{1}{5} = \frac{1}{6} \approx 0.167$$

Observing these values shows the peak occurs at $k = 2$, giving the highest probability:

$$P(2) = \frac{77}{180} \approx 0.428$$

Answer: 0.42778

114. Free Throws

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We must hit exactly 65 of the remaining 98 free throws. This can occur in $\binom{98}{65}$ possible sequences. Now, let's focus on calculating the probability of each such sequence. The probability of each sequence (not just the required one with 65 successful throws) is of the form:

$$\frac{1}{2} \times \frac{a_1}{3} \times \frac{a_2}{4} \times \dots \times \frac{a_{97}}{99}$$

where $a_1, a_2, \dots, a_{97}$ are determined by whether the corresponding free throw is successful or not. Specifically:

* If the third free throw is successful, $a_1 = 2$. * If the third free throw is not successful, $a_1 = 1$. For exactly 65 successful free throws, the product $a_1 \times a_2 \times \cdots \times a_{97}$ should equal $65! \times 33!$. Hence, the probability of each successful sequence is:

$$\frac{65! \times 33!}{99!}$$

Thus, the probability of exactly 65 successful free throws in the remaining 98 throws is:

$$\binom{98}{65} \times \frac{65! \times 33!}{99!} = \frac{1}{99}$$

Answer:

115. Starting With Heads

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $f(n)$ represent the number of sequences of length n where no two heads appear consecutively. For $n = 1$, the number of valid sequences is $f(1) = 2$ (the sequences are T and H). For $n = 2$, the number of valid sequences is $f(2) = 3$ (the sequences are TT, TH, and HT). To avoid consecutive heads, a sequence of length n must start either with T or HT. If it begins with T, the remaining $n - 1$ positions must be a valid sequence of length $n - 1$, which is $f(n - 1)$. If it starts with HT, the remaining $n - 2$ positions must be a valid sequence of length $n - 2$, which is $f(n - 2)$. Therefore, the recurrence relation is:

$$f(n) = f(n - 1) + f(n - 2)$$

with initial values $f(1) = 2$ and $f(2) = 3$. Using this relation, we compute $f(8) = 55$ and $f(10) = 144$. All possible sequences are equally likely. The probability that a sequence of length 10 starts with HT (and thus H) is:

$$P = \frac{f(8)}{f(10)} = \frac{55}{144}$$

Answer:

116. Circular Table I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let's consider the sequence BGGGG where B represents a boy and G represents a girl. This sequence must appear somewhere in the circle. Let E_i be the event that this sequence occurs at position i , with seats numbered from 1 to 12. These events are disjoint because only one such sequence can occur at a time without overlapping. The probability we are interested in is the sum of the probabilities of each E_i , which is $\sum_{i=1}^{12} P[E_i]$. Since all $P[E_i]$ are equal, this summation simplifies to $12 \times P[E_i]$. To find $P[E_i]$, we need to determine the number of ways the sequence BGGGG can occur. The sequence can be arranged in $6 \times 6 \times 5 \times 4 \times 3$ ways. The total number of ways to assign people to those 5 spots is $12 \times 11 \times 10 \times 9 \times 8$. Therefore,

$$P[E_i] = \frac{6 \times 6 \times 5 \times 4 \times 3}{12 \times 11 \times 10 \times 9 \times 8} = \frac{1}{44}$$

Thus, the probability we are looking for is:

$$12 \times \frac{1}{44} = \frac{12}{44} = \frac{3}{11}$$

So, the required probability is $\frac{3}{11}$.

Answer:

117. Gold Thief[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the weights of the coins be $w_1, w_2, \dots, w_5$. When we sum all the pairwise weights, each individual weight appears four times. Thus, we have:

$$4(w_1 + w_2 + w_3 + w_4 + w_5) = 21 + 22 + \dots + 30 = 255$$

Therefore,

$$w_1 + w_2 + w_3 + w_4 + w_5 = \frac{255}{4}$$

Answer:

118. Hanging Chain[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The distance between the tacks must be zero because the length of the entire chain is 80 feet, while the lowest point is at a vertical distance of 40 feet. Hence, the chain must be vertical.

Answer:

119. Coin Sequence Guess[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Select a non-empty proper subset A of the sample space. Ask whether the sequence s is in A . Let $|A|$ denote the number of sequences in A . The probability of guessing correctly is:

$$\frac{|A|}{2^{100}} \cdot \frac{1}{|A|} + \left(1 - \frac{|A|}{2^{100}}\right) \cdot \frac{1}{2^{100} - |A|} = 2^{-99}$$

Thus, the maximum probability of correctly guessing the sequence is 2^{-99} .

Answer:

120. Higher Roll I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Condition on Alice's roll X . If $X > 15$ (i.e. $X \in \{16, 17, 18, 19, 20\}$), then Bob's maximum M can be at most 15, so Alice wins with probability 1 in these cases. Since

$$P(X > 15) = \frac{5}{20} = \frac{1}{4}$$

this contributes $\frac{1}{4}$ to the overall probability. If $X \leq 15$, say $X = r \in \{1, \dots, 15\}$, Alice wins only if Bob's two independent 15-sided rolls both fall below r . The chance that one roll is at most $r - 1$ is $\frac{r-1}{15}$, so the chance both are below r is

$$\left(\frac{r-1}{15}\right)^2$$

Since $P(X = r) = \frac{1}{20}$, the total contribution from $X \leq 15$ is

$$\sum_{r=1}^{15} \frac{1}{20} \left(\frac{r-1}{15} \right)^2 = \frac{1}{20 \cdot 225} \sum_{k=0}^{14} k^2 = \frac{1}{4500} \cdot \frac{14 \cdot 15 \cdot 29}{6} = \frac{1015}{4500} = \frac{203}{900}$$

Adding both parts gives the overall probability Alice's roll exceeds Bob's maximum:

$$\frac{203}{900} + \frac{1}{4} = \frac{428}{900} = \frac{107}{225}$$

Answer: 0.47556

121. Dice Game IV

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

To end the game with exactly k dollars in hand, you must roll a "gain" outcome (1, 2, or 3) exactly k times and then roll a "stop" outcome (4 or 5) on the next roll. Since the probability of a gain on any given toss is

$$\frac{3}{6} = \frac{1}{2}$$

and the probability of stopping with a payout is

$$\frac{2}{6} = \frac{1}{3},$$

the probability of earning exactly k dollars is

$$P(k) = \left(\frac{1}{2} \right)^k \cdot \frac{1}{3}.$$

The expected payout is therefore

$$E = \sum_{k=1}^{\infty} k \cdot P(k) = \sum_{k=1}^{\infty} k \left(\frac{1}{2} \right)^k \cdot \frac{1}{3}.$$

Recalling the standard series

$$\sum_{k=1}^{\infty} k r^k = \frac{r}{(1-r)^2} \quad \text{for } |r| < 1,$$

with $r = \frac{1}{2}$, we get

$$\sum_{k=1}^{\infty} k \left(\frac{1}{2} \right)^k = \frac{\frac{1}{2}}{\left(1 - \frac{1}{2} \right)^2} = \frac{\frac{1}{2}}{\left(\frac{1}{2} \right)^2} = 2.$$

Multiplying by the stopping-probability factor $\frac{1}{3}$ gives

$$E[K] = 2 \times \frac{1}{3} = \frac{2}{3}.$$

Thus, the fair cost to play this game is $\frac{2}{3}$ dollar.

Answer: 0.66667

122. Half Right Hats

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

The prisoners can develop a strategy where 50 of them assume that the total number of red hats is even, while the other 50 assume it is odd. Each prisoner will count the red hats on the other prisoners and make their guess based on this. For instance, if a prisoner believes the total number of red hats is even but sees an odd number of red hats among the others, they would guess "red." This approach guarantees that at least one of the two groups will guess correctly, ensuring that at least half of the prisoners get their hat color right.

123. Pppower

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We claim that this is only possible if p has a units digit of 1. Clearly, p^p ends with a 1 when p itself ends with 1. Additionally, p^p cannot have a units digit of 1 when p has a units digit of 2, 4, 5, 6, or 8. If p has a units digit of 3 or 7, then p^p has a units digit of 1 if and only if $4|p$, a contradiction. Similarly, if p has a units digit of 9, then p^p has a units digit of 1 if and only if $2|p$, also a contradiction. Since there are 200 integers between 1 and 2000, inclusive, with a units digit of 1, there are 200 such p which fulfill our criterion.

Answer: 200

124. Grid Arrangement II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let a_{ij} represent the cell at i -th row and j -th column. a_{11} can either be $+1$ or -1 (2 ways). Assume it is $+1$. Now, we need to fill one more $+1$ in the first row (3 ways) and one more $+1$ in the first column (3 ways). Assume we fill a_{21} and a_{12} with $+1$.

- If we fill a_{22} with $+1$, the entire grid is determined (1 way).
- If we fill a_{22} with -1 , we need to fill one more $+1$ in the second row and second column (4 ways). The rest of the grid is determined.

Hence, the total number of ways is: $2 \times 3 \times 3 \times 1 + 2 \times 3 \times 3 \times 4 = 90$.

Answer: 90

125. Distant Points

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

By rotational symmetry, we place the first point at 0° . The second point must then lie somewhere between 90° and 270° (a span of 180°) to be at least 90° from the first. We split that 180° region into two equal halves, $[90^\circ, 180^\circ]$ and $[180^\circ, 270^\circ]$, and observe they are symmetric. So, we compute the result for the case where the second point lies in the interval $[90^\circ, 180^\circ]$, then double it. Focus on the half $[90^\circ, 180^\circ]$. Write the second point's angle as

$$\phi = 90^\circ + x, \quad 0 \leq x \leq 90^\circ$$

Because its unconditional density is $\frac{1}{360}$, the chance the second point falls in the infinitesimal interval $[\phi, \phi + d\phi]$ is

$$\frac{1}{360} \cdot d\phi = \frac{1}{360} \cdot dx$$

Around the first point we forbid the arc $[-90^\circ, 90^\circ]$ and around the second we forbid $[\phi - 90^\circ, \phi + 90^\circ]$. On $[90^\circ, 180^\circ]$, these two forbidden arcs overlap by exactly $(90^\circ - x)$, so their union has length

$$180^\circ + 180^\circ - (90^\circ - x) = 360^\circ - (90^\circ - x)$$

Hence the allowed portion for the third point is the complement, of length

$$360^\circ - [360^\circ - (90^\circ - x)] = 90^\circ - x$$

and its probability—since the third is uniform over 360° —is

$$\frac{90 - x}{360}$$

We now integrate over $x \in [0, 90]$ and then double for the symmetric half $[180^\circ, 270^\circ]$. By the continuous law of total probability, the overall probability is

$$P = 2 \int_0^{90} \frac{1}{360} \cdot \frac{90 - x}{360} dx = \frac{2}{360^2} \int_0^{90} (90 - x) dx = \frac{2}{129600} \cdot \frac{90^2}{2} = \frac{8100}{129600} = \frac{1}{16}$$

Therefore, the probability that no two of the three points lie within 90° of each other is $\frac{1}{16}$.

Answer:

126. Probabilistic Strategy IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The chances that two mathematicians directly across from each other guess right are independent, with each individual probability being $\frac{1}{2}$ because they cannot see or control the colors of their own hats. Thus the answer is at most $(\frac{1}{2})^2 = \frac{1}{4}$. We show that this is possible. Split the mathematicians, based on position, into two equilateral triangles. Their strategy will be to guess based on the assumption that both triangles have an odd number of red hats and deduce the color of their own hat from those in their triangle. The chance they win is then the probability that this assumption is true, which occurs with probability $(\frac{1}{2})^2 = \frac{1}{4}$.

Answer:

127. Optimal Card Guessing

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Define $R_1, R_2, \dots, R_6$ as the amounts Alice wins in each of the six turns. Using the linearity of expectation, the expected total winnings are $E[R_1] + E[R_2] + \dots + E[R_6]$. Initially, $E[R_1] = \frac{1}{2}$ since Alice has an equal probability of guessing correctly. In the second round, regardless of Alice's first guess, she will know the majority among the remaining cards. This gives $E[R_2] = \frac{3}{5}$. For the third round, Alice will either face a situation where the remaining pile is split 3-1 or 2-2. Based on the observed distribution, the expected value is $E[R_3] = \frac{3}{5}$. In the fourth round, the remaining cards will either be split 2-1 or 3-0. With probability $\frac{1}{10}$, the split will be 3-0, and with probability $\frac{9}{10}$, it will be 2-1. Therefore, $E[R_4] = \frac{7}{10}$. For round five, Alice will either be left with a 2-0 split (probability $\frac{2}{5}$) or a 1-1 split (probability $\frac{3}{5}$). Thus, $E[R_5] = \frac{7}{10}$. Finally, in the sixth round, only one card remains, so $E[R_6] = 1$. Summing everything up:

$$E[R_1] + E[R_2] + E[R_3] + E[R_4] + E[R_5] + E[R_6] = \frac{1}{2} + \frac{3}{5} + \frac{3}{5} + \frac{7}{10} + \frac{7}{10} + 1 = \frac{41}{10}$$

Therefore, the total expected amount Alice wins is $\frac{41}{10}$.

Answer:

128. k-inversions

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For a fixed choice $(i_1, \dots, i_6)$, let

$$X_{i_1, \dots, i_6} = \begin{cases} 1, & \text{if the entries at those positions form a strictly decreasing sequence} \\ 0, & \text{otherwise.} \end{cases}$$

There are $\binom{10}{6}$ such variables. What is the probability that $X_{i_1, \dots, i_6} = 1$? Since our permutation is uniformly random, the six values that land in these slots are equally likely to appear in any of the $6!$ possible orders. Exactly one of those orders is strictly decreasing, so

$$P(X_{i_1, \dots, i_6} = 1) = \frac{1}{6!} = \frac{1}{720}.$$

So, the total number of 6-inversions is

$$X = \sum_{1 \leq i_1 < \dots < i_6 \leq 10} X_{i_1, \dots, i_6}.$$

Taking expectations and using linearity of expectation,

$$E[X] = \sum_{1 \leq i_1 < \dots < i_6 \leq 10} E[X_{i_1, \dots, i_6}] = \binom{10}{6} \times \frac{1}{6!} = \frac{210}{720} = \frac{7}{24}.$$

Therefore, the expected number of 6-inversions in a random permutation of $\{1, 2, \dots, 10\}$ is $\frac{7}{24}$.

Answer: 0.29167

129. Five In Numerator

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We start with

$$10! = 2^8 3^4 5^2 7^1.$$

For any fraction $\frac{a}{b}$ in lowest terms satisfying $a \cdot b = 10!$, each prime power of $10!$ must be assigned completely to either a or b . There are four prime-power blocks $\{2^8, 3^4, 5^2, 7\}$, so there are $2^4 = 16$ possible ordered pairs (a, b) . Because $10!$ is not a perfect square, none of these pairs have $a = b$, so exactly half of the ordered pairs satisfy $a < b$. Hence $|S| = 8$. Now place 5^2 in the numerator. The other prime powers $2^8, 3^4, 7$ can each be assigned independently to a or b , but which ones result in fractions between 0 and 1? The only values of a that produce fractions lying between 0 and 1 are 5^2 and $5^2 \times 7$ (2 possibilities). So, the required probability is $2/8 = 1/4$.

Answer: 0.25000

130. W-permutation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Say that $P(2)$ and $P(4)$ are the low values; say that $P(1)$, $P(3)$, and $P(5)$ are the high values. It's easy to see that 1 must be low. It's easy to see that 4 and 5 must be high. So the low values are either $\{1, 2\}$ or $\{1, 3\}$. Let's look at the first case: the low values are $\{1, 2\}$. There are $2! = 2$ ways to assign them to $P(2)$ and $P(4)$. There are $3! = 6$ ways to assign $\{3, 4, 5\}$ to $P(1)$, $P(3)$, and $P(5)$. So there are $2 \cdot 6 = 12$ W-permutations in this case. Let's now look at the second case: the low values are $\{1, 3\}$. There are $2! = 2$ ways to assign them to $P(2)$ and $P(4)$. For concreteness, suppose that $P(2) = 1$. Then $P(1)$ must be 2. There are $2! = 2$ ways to assign $\{4, 5\}$ to $P(3)$ and $P(5)$. So there are $2 \cdot 2 = 4$ W-permutations in this case.

Altogether, there are $12 + 4 = 16$ W-permutations. So the probability of being a W-permutation is

$$\frac{16}{5!} = \frac{16}{120}$$

which simplifies to

$$\frac{2}{15}$$

Answer: 0.13333

131. Prisoner Hats

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We can label the prisoners from 100 to 1, with prisoner 100 at the back of the line. They agree that prisoner 100 will say "red" if they see an even number of red hats among prisoners 1 to 99, and "blue" if the number is odd. This allows prisoner 99 to look ahead and count the red hats. If prisoner 99 sees an even number and hears prisoner 100 say "red," they deduce they must be wearing a blue hat. Conversely, if they see an even number and prisoner 100 says "blue," indicating an odd number of red hats, then prisoner 99 must be wearing a red hat. Prisoner 98 can use the information from prisoner 99, along with what they can see among the remaining hats, to determine their own hat color. This reasoning continues for all prisoners up to prisoner 1. Using this strategy, 99 prisoners will definitely be freed. However, prisoner 100 must guess, giving them only a 50% chance of being correct.

132. Contiguous THTH

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First note that, conditional on having exactly four heads and seven tails, every arrangement of those eleven flips is equally likely. The total number of such arrangements is the binomial coefficient

$$\binom{11}{4} = \frac{11!}{4!7!} = 330.$$

Thus our sample space has 330 equally likely sequences. Next we must count how many of those 330 sequences contain the pattern THTH at least once. To do so we apply the inclusion-exclusion principle to the eight possible starting positions of a block of length four within an eleven-flip sequence. Label by A_i the set of sequences in which THTH begins at position i , for $i = 1, 2, \dots, 8$. Whenever THTH is fixed in positions i through $i + 3$, those four flips contribute exactly two tails and two heads, leaving five tails and two heads to be arranged in the remaining seven positions. The number of ways to do that is

$$\binom{7}{2} = 21.$$

Since there are eight choices for i , the sum of the sizes of all single events is

$$\sum_{i=1}^8 |A_i| = 8 \times 21 = 168.$$

We must subtract off the overcounting from pairs $A_i \cap A_j$. There are two types of overlaps: 1. Overlapping placements when $j - i = 2$. In this case the two occurrences of THTH overlap to form the string THTHTH of length six, which uses three tails and three heads. That leaves four tails and one head to place in the remaining five positions, giving

$$\binom{5}{1} = 5$$

arrangements for each choice of i . Since i can range from 1 up to 6, there are $6 \times 5 = 30$ sequences counted twice among the single-pattern counts. 2. Non-overlapping placements when $j - i \geq 4$. Here the two patterns occupy disjoint blocks of four. Together they consume four heads and four tails, leaving three tails and no heads to place in the remaining three positions, which can be done in exactly one way each. The number of index-pairs (i, j) with $1 \leq i < j \leq 8$ and $j - i \geq 4$ is $4 + 3 + 2 + 1 = 10$, so there are 10 such disjoint pairs, each contributing one sequence to the double-count. Altogether the sum of the sizes of all pairwise intersections is

$$\sum_{1 \leq i < j \leq 8} |A_i \cap A_j| = 30 + 10 = 40.$$

Finally we must add back the sequences that were subtracted twice when three patterns all overlap. The only way to have three overlapping occurrences is to start at positions $i, i+2, i+4$, yielding the eight-letter string THTHTHTH, which uses all four heads and four of the seven tails. Three tails remain to fill the other three slots in exactly one way. Since i may be 1 through 4, there are 4 such sequences. By inclusion-exclusion, the total number of sequences containing THTH at least once is

$$\left(\sum_i |A_i| \right) - \left(\sum_{i < j} |A_i \cap A_j| \right) + \left(\sum_{i < j < k} |A_i \cap A_j \cap A_k| \right) = 168 - 40 + 4 = 132.$$

Therefore the desired probability is

$$\frac{132}{330} = \frac{2}{5}.$$

Answer: 0.40000

133. 50 Wise Men

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

What if there are only 2 wise men? The strategy is simple—one wise man could decide not to interact with the glass at all. The other wise man could decide to flip the glass upside down the first time he sees it in the bottom-down position, then refrain from further interaction. Eventually, the first wise man will observe the glass in an upside-down position, indicating that the second wise man has been summoned. He can then confidently claim that both men have been summoned, securing their freedom. What if there are 3 wise men? Two of the wise men can flip the glass only once, and only if they observe the glass in a bottom-down position. The remaining wise man can flip the glass whenever he sees the glass upside down. This way, the remaining wise man can count the number of times the glass has been flipped—and once the glass has been flipped twice from the bottom-down position, he can deduce that the other 2 men have been called. This same strategy works for 50 wise men.

134. Rolling Everything I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The time until the first result appears is 1. After that, the random time until a second (different) result appears is geometrically distributed with a success parameter of $\frac{5}{6}$, hence with a mean of $\frac{6}{5}$ (recall that the mean of a geometrically distributed random variable is the inverse of its parameter). After that, the random time until a third (different) result appears is geometrically distributed with a success parameter of $\frac{4}{6}$, hence with a mean of $\frac{6}{4}$. This continues until the random time of appearance of the last and sixth result, which is geometrically distributed with a success parameter of $\frac{1}{6}$, hence with a mean of $\frac{6}{1}$. This shows that the mean total time to get all six results is:

$$\sum_{k=1}^6 \frac{6}{k} = 6 \cdot \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} \right) = 14.7.$$

Answer: 14.70000

135. Socks Remaining I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If the color of the last sock is blue (probability 0.4), then it must have been the red socks that were all removed, and the number of socks "left in the drawer" is the number of blue socks encountered starting from the end of the arrangement before any red sock is found. If indeed the last sock is blue, there are 39 other blue socks distributed in 61 slots (before the first red sock, between the first and second red socks, and so forth, finally after the 60th red sock). Thus, we expect $\frac{39}{61}$ blue socks in that last slot; adding the final blue sock gives an average of $\frac{100}{61}$ blue socks left in the drawer. Similarly, if the last sock is red (probability 0.6), we have 59 other red socks in 41 slots, so on average $\frac{59}{41}$ in the last slot; that leaves $1 + \frac{59}{41} = \frac{100}{41}$ red socks behind. Putting these together, we have that the expected number of socks left is

$$0.4 \times \frac{100}{61} + 0.6 \times \frac{100}{41} \approx 2.12.$$

Answer:

136. Consecutive Integral Envelopes

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If we decide to always pick up the first envelope, we have no choice but to get the expected value of $x + 2$. Similarly, if we decide to always pick up the third envelope, the expected payout remains $x + 2$. Therefore, to achieve a higher expected payout, we must decide after opening the first two envelopes—whether to retain the second envelope or opt for the third. Let us denote the first envelope as e_1 , the second as e_2 , and the third as e_3 . Consider the following strategy: (i) If $e_1 - e_2 = +2$, then we know e_2 is the smallest valued envelope and hence, we pick the third envelope. This case can occur in 1 way. (ii) If $e_1 - e_2 = -2$, then we know e_2 is the largest valued envelope and hence, we pick the second envelope. This case can occur in 1 way. (iii) If $e_1 - e_2 = 1$, then either $e_1 = x + 2$, $e_2 = x + 1$, $e_3 = x + 3$ OR $e_1 = x + 3$, $e_2 = x + 2$, $e_3 = x + 1$. The expected value of the second envelope is $x + 1.5$ while that of the third envelope is $x + 2$. Hence, we choose the third envelope in this case. This case can occur in 2 ways. (iv) If $e_1 - e_2 = -1$, then either $e_1 = x + 1$, $e_2 = x + 2$, $e_3 = x + 3$ OR $e_1 = x + 2$, $e_2 = x + 3$, $e_3 = x + 1$. The expected value of the second envelope is $x + 2.5$ while that of the third envelope is $x + 2$. Hence, we choose the second envelope in this case. This case can occur in 2 ways. The probability of the first case is $\frac{1}{6}$. The expected payout is $x + 3$. The probability of the second case is $\frac{1}{6}$. The expected payout is $x + 2$. The probability of the third case is $\frac{2}{6}$. The expected payout is $x + 2$. The probability of the fourth case is $\frac{2}{6}$. The expected payout is $x + 2.5$. Hence, the overall expected payout is

$$\frac{1}{6}(x + 3) + \frac{1}{6}(x + 2) + \frac{2}{6}(x + 2) + \frac{2}{6}(x + 2.5) = x + 2.33$$

which is better than $x + 2$. Hence, the optimal payout is $x + 2.33$.

Answer:

137. Trailing Zeros I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We obtain a trailing zero for every factor of 10, and 10 is the product of 2 and 5. Since the number of factors of 2 exceeds the number of factors of 5 in the numbers from 1 to 100, we need to count how many times 5 appears as a factor in the prime factorization of $100!$. To determine this, we sum the highest powers of 5

that divide the numbers between 1 and 100. The formula for finding the number of trailing zeros in $n!$ is:

$$\left\lfloor \frac{n}{5} \right\rfloor + \left\lfloor \frac{n}{25} \right\rfloor + \left\lfloor \frac{n}{125} \right\rfloor + \dots$$

For $n = 100$, we calculate:

$$\left\lfloor \frac{100}{5} \right\rfloor = 20$$

$$\left\lfloor \frac{100}{25} \right\rfloor = 4$$

There are no higher powers of 5 less than 100, so the total number of trailing zeros is:

$$20 + 4 = 24$$

Thus, the number of trailing zeros in $100!$ is 24.

Answer: 24

138. Expected Minimum Till 5

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let X denote the minimum number rolled until a 5 is observed. Since the minimum value of X cannot be 6, we compute the expected value $E[X]$ as follows:

$$E[X] = \sum_{x=1}^5 x \cdot \Pr(X = x)$$

To find $\Pr(X = 1)$, calculate the probability that a 1 appears before a 5. By symmetry and the law of total probability, we have:

$$\Pr(X = 1) = \frac{1}{2}$$

To find $\Pr(X = 2)$, we need the probability that a 2 appears before a 5, and a 1 does not appear before a 5. The orderings of $\{1, 2, 5\}$ are: 1. 1, 2, 5 2. 2, 5, 1 3. 2, 1, 5 Only the ordering $\{2, 5, 1\}$ gives $X = 2$. Therefore:

$$\Pr(X = 2) = \frac{1}{6}$$

For $\Pr(X = 3)$, the sequence must include a 3 before a 5, with 1 and 2 appearing in any order after the 3. Thus:

$$\Pr(X = 3) = \frac{2!}{4!} = \frac{1}{12}$$

Similarly:

$$\Pr(X = 4) = \frac{3!}{5!} = \frac{1}{20}$$

$$\Pr(X = 5) = \frac{4!}{5!} = \frac{1}{5}$$

Combining these, we compute the expected value:

$$E[X] = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{12} + 4 \cdot \frac{1}{20} + 5 \cdot \frac{1}{5} = \frac{137}{60}$$

Answer: 2.28333

139. Palindrome Summation

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Notice that $10001 + 99999 = 110000$. In fact, ordering the palindromes in ascending order, we find that the sum of the n -th palindrome and the n -th to last palindrome is 110000. We have $9 \cdot 10 \cdot 10$ palindromes, or 450 pairs of palindromes summing to 110000. Performing the multiplication gives 49500000.

Answer:

140. Matrix Attack II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Given a collection of submatrices and an initial configuration of the 8×8 matrix, the final configuration of the matrix remains the same regardless of the order in which the submatrices are applied. Specifically, applying the same submatrix more than once has no additional effect, allowing us to reduce the collection to a set of unique submatrices.

There are 36 possible 3×3 submatrices and 25 possible 4×4 submatrices. Thus, the total number of unique sets of submatrices is 2^{61} . Therefore, starting from a matrix of all zeros, we can achieve at most 2^{61} distinct configurations out of the 2^{64} possible configurations of the matrix. Consequently, it may not be possible to always eliminate all the ones from the original matrix.

Answer:

141. Consecutive 1s

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Determine the ways to have at least 3 consecutive ones. Break it down into cases: For exactly 3 consecutive ones: Patterns: 111xx, 111x1, x111x, x111x, xx111, 1x111, 1x111 For exactly 4 consecutive ones: Patterns: 1111x, 1111x, x1111, x1111 For exactly 5 consecutive ones: Pattern: 11111 Calculate the probability for each case: Probability of exactly 3 consecutive ones:

$$3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{5}{6}\right)^2$$

Probability of exactly 4 consecutive ones:

$$4 \times \left(\frac{1}{6}\right)^4 \times \left(\frac{5}{6}\right)$$

Probability of exactly 5 consecutive ones:

$$\left(\frac{1}{6}\right)^5$$

Add these probabilities:

$$3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{5}{6}\right)^2 + 4 \times \left(\frac{1}{6}\right)^4 \times \left(\frac{5}{6}\right) + \left(\frac{1}{6}\right)^5$$

Thus, the combined probability is: $\frac{1}{81}$

Answer:

142. Ball With Maximum Value

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We can generalize this problem. Suppose we select n balls from a total of N balls, and let k represent the maximum value of the selected balls. The other $n - 1$ balls must be chosen from the first $k - 1$ balls. Using linearity of expectation and summing over all possible values of k from n to N , we derive the expected maximum value:

$$\mathbb{E} = \sum_{k=n}^N k \cdot \frac{\binom{k-1}{n-1}}{\binom{N}{n}} = \frac{n}{\binom{N}{n}} \sum_{k=n}^N \binom{k}{n}$$

By applying the hockey stick identity, the summation simplifies to:

$$\frac{n}{\binom{N}{n}} \cdot \binom{N+1}{n+1} = \frac{n \cdot (N+1)}{n+1}$$

Substituting $n = 10$ and $N = 20$, we obtain the final result:

$$\mathbb{E} = \frac{10 \cdot 21}{11} = \frac{210}{11}$$

Answer:

143. Digit Multiply I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The prime factorization of 10,000 is

$$10,000 = 10^4 = 2^4 \cdot 5^4$$

Therefore, our integer must contain four 2's and four 5's. Since $2^3 = 8$, three 2's can be combined into a single digit 8. Consequently, we need four 5's, one 8, and one 2. To minimize the integer, arrange these digits in ascending order, as the digits positioned further to the left significantly impact the number's magnitude. Hence, the smallest integer formed is 255558.

Answer:

144. Regular Octagon

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We use complementary counting. Split into three cases in which the diagonals don't intersect strictly within the octagon.

Case 1: The diagonals intersect on the octagon. For each vertex, there are 5 possible diagonals from which we must choose 2 in this case. This gives us $8 \cdot \binom{5}{2} = 80$ total ways.

Case 2: The diagonals never intersect. Take a diagonal separated by 1 vertex. There are 2 other diagonals parallel to this one. There are 4 distinct possible rotations, giving a total of $4 \cdot \binom{3}{2} = 12$ possibilities. The only other way for the diagonals to not intersect is to take a diagonal separated by 2 vertices, in which case there is only 1 parallel choice and 4 rotations. Thus, we have a total of $12 + 4 = 16$ ways for this case.

Case 3: The diagonals intersect outside of the octagon. There are two subcases here. First, we can have two non-parallel diagonals that are each separated by 1 vertex. By rotation, there are 8 ways for this. Second, we can have one diagonal separated by 1 vertex and one separated by 2. For each possible diagonal separated by 1 vertex, there are 2 possibilities for the other diagonal. Thus, we have $8 \times 2 = 16$ possibilities here. That gives us a total of $8 + 16 = 24$ ways for this case.

The total number of ways to select diagonals that do not intersect strictly within the octagon is $80 + 16 + 24 = 120$. There are $8 \times \frac{5}{2} = 20$ diagonals, giving $\binom{20}{2} = 190$ total choices. Thus, by complementary counting, we arrive at a final answer of $1 - \frac{120}{190} = \frac{7}{19}$.

Answer:

145. Payment[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will use recursion. Consider the states as the amount of payment already made. Let E_i be the expected number of notes required to complete the payment if i has already been transferred. We have the following equations:

$$E_0 = \frac{1}{2}E_{10} + \frac{1}{2}E_0 + 1$$

$$E_{10} = \frac{1}{2}E_{20} + \frac{1}{2}E_0 + 1$$

$$E_{20} = \frac{1}{2}E_{30} + \frac{1}{2}E_0 + 1$$

$$E_{30} = \frac{1}{2}E_{40} + \frac{1}{2}E_0 + 1$$

$$E_{40} = \frac{1}{2}E_{50} + \frac{1}{2}E_0 + 1$$

$$E_{50} = 0$$

Solving the above set of equations, we get:

$$E_0 = 62$$

Answer: 62

146. Rock Paper Scissors[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

When Alice played Scissors, Bob had to play either Rock or Paper for a distinct outcome. Out of the 6 times Alice played Scissors, she won 4 rounds when Bob used Paper and lost 2 rounds when he used Rock. In the remaining 4 rounds, Bob played Scissors each time. This resulted in Alice losing 3 times when she played Rock and winning once when she played Paper. In total, Alice won 7 rounds.

Answer: 7

147. Party Groups I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $E(n)$ represent the expected number of cycles in a permutation of n people. We want to find $E(50)$. We approach this using the idea that the first person can either form a cycle with themselves or with someone else.

Case 1: The first person forms a cycle with themselves.

This happens with probability $\frac{1}{n}$ (since they randomly pick their own name). In this case, they form a single cycle, and the remaining $n - 1$ people are left to form cycles. So the expected number of cycles in this case is $1 + E(n - 1)$.

Case 2: The first person forms a cycle with someone else.

This happens with probability $\frac{n-1}{n}$ (since they pick someone else's name). In this case, they and the person they picked are part of a larger cycle, and now we are left with the remaining $n - 1$ people, which gives us the same expected number of cycles, $E(n - 1)$.

So, the recurrence relation is:

$$E(n) = \frac{1}{n}(1 + E(n - 1)) + \frac{n - 1}{n}E(n - 1)$$

Simplifying this:

$$E(n) = E(n-1) + \frac{1}{n}$$

This shows that the expected number of cycles increases by $\frac{1}{n}$ each time we add another person. We can sum this recurrence from $E(1) = 1$, because with 1 person, there is exactly 1 cycle.

Thus, the expected number of cycles in a permutation of 50 people is:

$$E(50) = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{50} \approx 4.49921$$

Answer: 4.49921

148. Min, Median, Max Guess

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Intuitively, if the observed X_1 is very small you should guess "minimum," if it is very large you should guess "maximum," and otherwise you should guess "median." To capture this, fix a cutoff $\alpha \in [0, \frac{1}{2}]$ and use the rule: If $X_1 \leq \alpha$, guess it is the minimum. If $X_1 \geq 1 - \alpha$, guess it is the maximum. If $\alpha < X_1 < 1 - \alpha$, guess it is the median. Why choose $1 - \alpha$ for the upper cutoff? Since $X_1 \sim \text{Unif}(0, 1)$ implies $1 - X_1 \sim \text{Unif}(0, 1)$, the situation " X_1 is the maximum and $X_1 \geq 1 - \alpha$ " corresponds exactly under the map $x \mapsto 1 - x$ to " X_1 is the minimum and $X_1 \leq \alpha$." To treat minimum and maximum symmetrically, the upper threshold must be $1 - \alpha$. Let $p(\alpha)$ be the overall success probability of this cutoff rule. It splits into three contributions: When $X_1 \leq \alpha$ we guess minimum and are correct if both $X_2, X_3 > x$, giving

$$\int_0^\alpha (1-x)^2 dx.$$

When $\alpha < X_1 < 1 - \alpha$ we guess median and succeed if one of X_2, X_3 lies below x and the other above, giving

$$\int_\alpha^{1-\alpha} 2x(1-x) dx.$$

When $X_1 \geq 1 - \alpha$ we guess maximum and are correct if both $X_2, X_3 < x$, giving

$$\int_{1-\alpha}^1 x^2 dx.$$

By the symmetry just noted, the first and third integrals are equal, so

$$p(\alpha) = 2 \int_0^\alpha (1-x)^2 dx + \int_\alpha^{1-\alpha} 2x(1-x) dx.$$

In closed form one finds

$$p(\alpha) = \frac{1 + 6\alpha - 12\alpha^2 + 6\alpha^3}{3}.$$

Maximizing this over $\alpha \in [0, \frac{1}{2}]$ by solving $p'(\alpha) = 0$ yields $\alpha = \frac{1}{3}$. Substituting back gives

$$p\left(\frac{1}{3}\right) = \frac{17}{27} \approx 0.63.$$

Answer: 0.62963

149. Normal Inequality III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Because N_1, N_2 , and N_3 are independent standard normal variables, the point (N_1, N_2, N_3) has a spherically symmetric distribution in $\mathbb{R}^3$.

Write

$$R = \sqrt{N_1^2 + N_2^2 + N_3^2} \quad \text{and} \quad U_3 = \frac{N_3}{R}.$$

Then U_3 is the z -coordinate of a random point uniformly distributed on the unit sphere, and for such a point, U_3 is uniform on $[-1, 1]$.

Now the event $N_1^2 + N_2^2 > N_3^2$ becomes $R^2 - N_3^2 > N_3^2$, which is the same as $1 - U_3^2 > U_3^2$. So

$$1 > 2U_3^2 \quad \implies \quad |U_3| < \frac{1}{\sqrt{2}}.$$

Since U_3 is uniform on $[-1, 1]$, the probability is just the length of the interval $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ divided by 2:

$$\Pr\left(|U_3| < \frac{1}{\sqrt{2}}\right) = \frac{2/\sqrt{2}}{2} = \frac{1}{\sqrt{2}}.$$

So the required probability is $\boxed{\frac{1}{\sqrt{2}}}$.

Answer: 0.70711

150. Permutation Variance

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Since every number is either fixed or not, we always have $X + Y = 1000$. Rearranging this gives $Y = 1000 - X$, so we can write $X - Y = 2X - 1000$. Using properties of variance, this implies:

$$\text{Var}(X - Y) = \text{Var}(2X - 1000) = 4 \cdot \text{Var}(X)$$

(since variance is unaffected by constants and scales quadratically with coefficients). To compute $\text{Var}(X)$, we need to evaluate $E[X^2] - (E[X])^2$. Let's begin by analyzing X . We can express X as the sum of indicator random variables:

$$X = X_1 + X_2 + \cdots + X_{1000}$$

where $X_i = 1$ if position i is a fixed point (i.e., the number in position i is i), and $X_i = 0$ otherwise. Since the permutation is random, each position has a $\frac{1}{1000}$ chance of being a fixed point. That is,

$$E[X_i] = \Pr(\text{element } i \text{ is at position } i) = \frac{1}{1000}$$

Then by linearity of expectation:

$$E[X] = \sum_{i=1}^{1000} E[X_i] = 1000 \cdot \frac{1}{1000} = 1.$$

Next, we compute $E[X^2]$. Since $X = \sum X_i$, we expand:

$$X^2 = \left(\sum X_i\right)^2 = \sum X_i^2 + \sum_{i \neq j} X_i X_j$$

Taking expectations:

$$E[X^2] = \sum E[X_i^2] + \sum_{i \neq j} E[X_i X_j]$$

Because each X_i is 0 or 1, we have $X_i^2 = X_i$, so:

$$\sum E[X_i^2] = \sum E[X_i] = 1000 \cdot \frac{1}{1000} = 1.$$

Now for the cross terms, we need $E[X_i X_j]$ for $i \neq j$. Since the permutation is uniformly random, the distribution of each pair (X_i, X_j) is the same for any distinct i, j , so all $E[X_i X_j] = E[X_1 X_2]$. Let's compute that. The probability that both positions 1 and 2 are fixed is the probability that 1 is in position 1 (which is $\frac{1}{1000}$), and given that, 2 is in position 2 (which is then $\frac{1}{999}$, since one position is already taken). Therefore:

$$E[X_1 X_2] = \Pr(1 \text{ is in position 1 and } 2 \text{ is in position 2}) = \frac{1}{1000} \cdot \frac{1}{999}$$

There are $2 \cdot \binom{1000}{2} = 1000 \cdot 999$ such pairs (i, j) , so the total contribution of cross terms is:

$$\sum_{i \neq j} E[X_i X_j] = 1000 \cdot 999 \cdot \frac{1}{1000} \cdot \frac{1}{999} = 1.$$

Thus:

$$E[X^2] = 1 + 1 = 2 \text{ and } \text{Var}(X) = E[X^2] - (E[X])^2 = 2 - 1 = 1.$$

Finally:

$$\text{Var}(X - Y) = 4 \cdot \text{Var}(X) = 4 \cdot 1 = 4.$$

Answer: 4

151. Probabilistic Strategy II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To maximize their chances of winning, the wizards can use the following strategy: Each wizard should remain silent if they see one black and one white hat. If a wizard observes two black hats, they should announce "white," and if they see two white hats, they should announce "black." With this strategy, the probability of winning the game is $\frac{3}{4}$. This is because the game is lost only if all the hats are either white or black, both of which occur with a probability of $\frac{1}{8}$ each.

Answer: 0.75000

152. 3D Tic-tac-toe

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Every straight-line path of five consecutive cells in a $5 \times 5 \times 5$ cube must fall into one of the following categories: 1. Axis-parallel paths These run along one of the coordinate axes (x , y , or z).

- For x -axis lines: there are 5 choices for y , and 5 for $z \rightarrow 5 \times 5 = 25$ lines.
- Similarly, there are 25 lines along the y -axis and 25 along the z -axis.

Total: $3 \times 25 = 75$ axis-parallel paths 2. Face-diagonal paths These lie entirely within one of the coordinate planes (xy , yz , or xz).

- Each of the 3 planes has 5 layers.
- In each layer, there are 2 diagonals of length 5.

Total: $3 \times (5 \times 2) = 30$ face-diagonal paths 3. Space-diagonal paths These are the long diagonals that go through the cube from one corner to the opposite.

- There are exactly 4 such diagonals in a cube.

Total: 4 space-diagonal paths

Final count: $75 + 30 + 4 = 109$ There are 109 straight-line paths of five consecutive cells in a $5 \times 5 \times 5$ cube.

Answer: 109

153. Color And Roll

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Notice that all the faces of the smaller cubes are equally likely to show as the up face. There are $6 \cdot 5^3$ faces in total. There are $6 \cdot 5^2$ colored faces. Hence, the probability that the colored face shows as the up face is:

$$\frac{6 \cdot 5^2}{6 \cdot 5^3} = \frac{1}{5}$$

Answer: 0.20000

154. Expected 2s

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We start by calculating the probability that we initially drew from each urn given that the drawn paper was a 2. The probability of drawing a 2 from the first urn (Urn A) is $\frac{1}{8}$, and from the second urn (Urn B) is $\frac{1}{2}$. Using Bayes' Theorem, we calculate that the probability we initially chose Urn B given that a 2 was drawn is:

$$P(\text{Urn B} \mid \text{draw is 2}) = \frac{P(\text{draw is 2} \mid \text{Urn B}) \times P(\text{Urn B})}{P(\text{draw is 2})}$$

Here,

$$P(\text{draw is 2}) = \left(\frac{1}{8} \times \frac{1}{2}\right) + \left(\frac{1}{2} \times \frac{1}{2}\right) = \frac{1}{16} + \frac{4}{16} = \frac{5}{16}$$

Thus,

$$P(\text{Urn B} \mid \text{draw is 2}) = \frac{\frac{1}{2} \times \frac{1}{2}}{\frac{5}{16}} = \frac{\frac{1}{4}}{\frac{5}{16}} = \frac{4}{5}$$

Similarly,

$$P(\text{Urn A} \mid \text{draw is 2}) = \frac{\frac{1}{8} \times \frac{1}{2}}{\frac{5}{16}} = \frac{\frac{1}{16}}{\frac{5}{16}} = \frac{1}{5}$$

Given these probabilities, the expected frequency of drawing a 2 in the subsequent 40 draws is a weighted average based on which urn was likely chosen. From Urn A, the probability of drawing a 2 is $\frac{1}{8}$, and from Urn B, it is $\frac{1}{2}$. Thus, the expected number of 2s in 40 draws is:

$$E = 40 \left(\frac{1}{5} \times \frac{1}{8} + \frac{4}{5} \times \frac{1}{2} \right) = 40 \times \left(\frac{1}{40} + \frac{16}{40} \right) = 40 \times \frac{17}{40} = 17$$

Therefore, the expected number of times 2 appears in the next 40 draws is 17.

Answer: 17

155. Drunk Passenger I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that the last passenger can only take seat 1 or 100.

* If any passenger takes seat 1, the cycle stops, and all the subsequent passengers take their own seats (including the last). * Otherwise, if seat 100 is taken before 1, the cycle is paused, i.e., the subsequent

passengers do take their own seats, but the last passenger would take seat 1.

Now, any passenger from 1st to 99th who is picking a random vacant seat will choose between 1, 100 or any other seat equally likely. Thus, by symmetry, 1 or 100, anyone will be taken first - with equal probability.

Hence, the last person ends up in their own seat with a probability of 0.5.

Answer: 0.50000

156. Broken Calculator

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We consider the integers from this process written in binary. The first operation truncates the rightmost digit, while the second operation appends 01 to the right. We cannot have a number with a substring 11. For simplicity, call a string valid if it has no consecutive 1's. Note that any number generated by this process is valid, as truncating the rightmost digit and appending 01 to the right of the digits clearly preserve validity. Since we can effectively append a zero by applying the second operation and then the first operation, we see that we can achieve all valid strings.

Note that 2014 has eleven digits when written in binary, and any valid binary string with eleven digits is at most $10111111111 = 1535$. Therefore, our problem reduces to finding the number of eleven-digit valid strings. Let F_n denote the number of valid strings of length n . For any valid string of length n , we can create a valid string of length $n + 1$ by appending a 0, or we can create a valid string of length $n + 2$ by appending 01. This process is clearly reversible, so our recursion is given by $F_n = F_{n-1} + F_{n-2}$, with $F_1 = 2$ and $F_2 = 3$. This yields a sequence of Fibonacci numbers starting from 2, and some computation shows that our answer is $F_{11} = 233$.

Answer: 233

157. Square Fitting II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Use the Chebyshev distance

$$M = \max\{|X|, |Y|\}$$

for a random location (X, Y) . Because $|X|$ and $|Y|$ are independent and uniform on $[0, 1]$, the chance that both lie within $[-m, m]$ is m^2 . Equivalently,

$$\Pr(M \leq m) = m^2, \quad 0 \leq m \leq 1,$$

so M has density

$$f_M(m) = 2m, \quad 0 \leq m \leq 1.$$

Label the nine independent distances $M_1, \dots, M_9$. When you order them from smallest to largest, the fourth-smallest value $M_{(4)}$ captures exactly how far out you must go in Chebyshev distance to include the fourth-closest location. The density of this fourth order statistic follows the standard formula for i.i.d. samples:

$$f_{M_{(4)}}(m) = \frac{9!}{3!5!} [F_M(m)]^3 [1 - F_M(m)]^5 f_M(m) = 504m^6(1 - m^2)^5(2m).$$

Once you have $M_{(4)}$, the side length of the smallest centered square that reaches Chebyshev distance m is $S = 2m$. Its area is $S^2 = 4m^2$, so the random area covering the four nearest locations is

$$A = 4(M_{(4)})^2.$$

Taking expectations,

$$E[A] = 4 \int_0^1 m^2 f_{M_{(4)}}(m) dm = 4 \int_0^1 m^2 [504m^6(1 - m^2)^5(2m)] dm = \frac{8}{5},$$

a Beta-integral calculation. Therefore, the expected area of the smallest centered square containing the four closest locations is

$$\frac{8}{5}.$$

Answer:

158. Increasing Dice Rolls II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Use combinatorics. There are 6^{10} total sequences. How many of those are non-decreasing?

An excellent observation is that every non-decreasing sequence is equivalent to a “histogram” or vector which gives the number of times each face appears.

For example, the sequence 1, 2, 2, 2, 3, 4, 5, 5, 5, 6 is equivalent to the vector (1, 3, 1, 1, 3, 1).

By equivalent, I mean that there is a one-to-one correspondence between the sequences and vectors. So, counting one is equivalent to counting the other.

Thus, we wish to count how many ways 10 indistinguishable things can be placed into 6 bins, where we allow for zero items to be placed in some bins.

To count that, we observe that this is equivalent to the number of ways to place 16 indistinguishable things into 6 bins, where each bin must contain at least one item. Subtracting one from each bin will give us a vector of the previous sort.

To count this, we can use the stars-and-bars method. Putting 16 things into 6 bins is equivalent to putting 5 bars among 16 stars, such that there is at most one bar between any two stars. For instance, this choice of bars:

|**|**|*|*****|*

represents the vector (3, 5, 2, 1, 4, 1), which, if we subtract one from each component, yields the vector (2, 4, 1, 0, 3, 0), which corresponds to the rolled sequence 1, 1, 2, 2, 2, 3, 5, 5, 5.

Since there are 16 stars, there are 15 places for bars, and hence the number of such sequences is $\binom{15}{5} = 3003$.

Thus, the probability of rolling such a sequence is $\frac{3003}{6^{10}}$.

Answer:

159. Direction Change II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Denote a rightward movement as R and an upward movement as U . The total number of paths, allowing for any sequence of direction changes, corresponds to the number of ways to arrange 8 R 's and 8 U 's.

There are $\frac{16!}{8!8!} = 12870$ such paths.

If there are 4 direction changes, the path consists of 5 blocks, with each block being in the same direction (for example, $RRUURRUUUUUURRRR$ is a valid path with 4 direction changes). Each block must have a length of at least one. We can use stars and bars to determine the number of such paths.

First, we need to determine the first step (R or U) - 2 ways.

Then, we need to split 8 moves into 3 blocks and the other 8 moves into 2 blocks, with each block having size at least 1. Using stars and bars, there are $\binom{7}{2} \cdot \binom{7}{1} = 147$ ways. Hence, the total number of paths with exactly 4 direction changes is $2 \cdot 147 = 294$.

So, the required probability is $\frac{294}{12870}$.

Answer:

160. Close Factors[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We begin with the prime factorization of 1234567890, which is $2 \cdot 3^2 \cdot 5 \cdot 3607 \cdot 3803$. Our goal is to find two factors x_0 and y_0 such that $x_0 y_0 = 1234567890$ and $|x_0 - y_0|$ is minimized. This is achieved by distributing the prime factors as evenly as possible between x_0 and y_0 while closing the numerical gap between them. The key is to multiply the slightly smaller prime, 3607, by a larger combination of the remaining smaller primes. Specifically, we calculate $x_0 = 3607 \cdot 2 \cdot 5 = 36070$ and $y_0 = 3803 \cdot 3^2 = 34227$.

Answer: **161. Non-absorbing Random-walk**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Model the positions $1, \dots, n$ as states of a Markov chain with absorbing state n . Let h_i be the expected number of steps to hit n starting from state i . Then

$$\begin{cases} h_n = 0, \\ h_1 = 1 + h_2 \quad (\text{from 1 you must go to 2}), \\ h_i = 1 + \frac{1}{2}h_{i-1} + \frac{1}{2}h_{i+1}, \quad 1 < i < n. \end{cases}$$

Rewriting the interior equation gives the discrete second difference

$$h_{i+1} - 2h_i + h_{i-1} = -2 \quad (1 < i < n).$$

The general solution of this constant second-difference equation is quadratic. Write

$$h_i = ai^2 + bi + c,$$

and substitute to obtain $2a = -2$, so $a = -1$. Using $h_1 = 1 + h_2$ yields $b = 2$. Finally $h_n = 0$ gives $c = n^2 - 2n$. Thus

$$h_i = -i^2 + 2i + n^2 - 2n = (n - i)(n + i - 2).$$

In particular, starting from 1,

$$h_1 = (n - 1)^2.$$

Answer: **162. Equal Logarithms**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Take the case that $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor = -1$. In this case, both x and y lie in the interval $[1/2, 1)$. The probability of this is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. Similarly, in the case that $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor = -2$, x and y lie in the interval $[1/4, 1/2)$, and the probability is $\frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$.

The probability we are looking for is the sum of the probabilities for each of the cases $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor = -1, -2, -3, \dots$. It is easy to see that the probabilities for $\lfloor \log_2 x \rfloor = \lfloor \log_2 y \rfloor = n$ for $-\infty < n < 0$ form an infinite geometric series that starts at $\frac{1}{4}$ with a common ratio of $\frac{1}{4}$. Using the formula for the sum of an infinite geometric series, we calculate:

$$\text{Sum} = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Thus, the probability is $\frac{1}{3}$.

Answer:

163. Mathematical Expression III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

There are 8 gaps between the digits of 123456789. Each of those gaps can either be empty or filled with + or filled with $\times$ (3 choices). Hence, the number of such expressions are $3^8 = 6561$.

Answer:

164. Dice Multiple I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability that the sum of the first k rolls is a multiple of 6 is $\frac{1}{6}$. This is because, among the six possible outcomes of the k -th roll, only one will result in the sum being divisible by 6. Therefore, the number of rolls required follows a geometric distribution with parameter $\frac{1}{6}$. Consequently, the expected number of rolls is 6.

Answer:

165. Self-avoiding Paths

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We have shown that for $n \geq 2$,

$$a_n = 2a_{n-1} + a_{n-2},$$

with the base cases:

$$a_0 = 1 \quad (\text{the empty path}), \quad a_1 = 3 \quad (\text{one step: right, up, or down}).$$

The characteristic equation

$$r^2 - 2r - 1 = 0$$

has roots $1 \pm \sqrt{2}$, and matching $a_0 = 1, a_1 = 3$ gives the closed-form:

$$a_n = \frac{(1 + \sqrt{2})^{n+1} + (1 - \sqrt{2})^{n+1}}{2}.$$

166. SOS Game

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To win for $n = 11$, Alice can employ a winning strategy. She begins by placing an "S" in the 6th position. If Bob then plays in positions 1 to 5, Alice places another "S" in the 9th position. If Bob plays in positions 7 to 11, Alice will place an "S" in the 3rd position. At this point, six positions remain outside the pit, and it is Bob's turn. If Bob chooses to play outside the pit, Alice will always have an available position outside the pit to fill. Consequently, Bob is ultimately forced to play into the pit, ensuring that Alice maintains her winning strategy.

Answer:

167. Moving Ants I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider 3 ants standing at the vertices of a triangle. Each ant can either move to the right (R) or to the left (L). Therefore, the total number of ways in which the 3 ants can move is $2^3 = 8$. Some examples of these arrangements are RRR, RLR, and LLR. The ants will not meet at a vertex if all of them move in the same direction, either all to the right (RRR) or all to the left (LLL). Hence, there are just 2 such arrangements. Thus, the probability that the ants will not meet at a vertex is $\frac{2}{8} = \frac{1}{4}$. So, the probability that at least two ants will meet at a vertex is $1 - \frac{1}{4} = \frac{3}{4}$.

Answer: 0.75000**168. Moving Ants III**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

When two ants collide, we can assume they continue moving in their original direction. Thus, it is as if no collisions occur.

Think of each ant as carrying a nametag. When two ants collide and reverse, it is exactly the same (for the purpose of who falls when) as if they passed through one another and kept going while the nametags swapped. Thus, for determining the times when ants fall off the stick we may imagine every ant simply walks straight in its initial direction until it falls.

Let an ant start at position $x \in [0, 1]$. If its initial direction is left its fall time is x ; if right its fall time is $1 - x$. The starting position x is uniform on $[0, 1]$ and the initial direction is independent and equally likely to be left or right, so the fall time Y of a single ant has distribution

$$P(Y \leq t) = \frac{1}{2}P(x \leq t) + \frac{1}{2}P(x \geq 1 - t) = \frac{1}{2}t + \frac{1}{2}t = t; 0 \leq t \leq 1$$

Thus $Y \sim \text{Unif}(0, 1)$.

With $n = 1000$ ants the (pass-through) fall times are iid $\text{Unif}(0, 1)$. If M_n denotes their maximum then for $0 \leq t \leq 1$,

$$P(M_n \leq t) = t^n$$

Therefore

$$\mathbb{E}[M_n] = \int_0^1 P(M_n > t) dt = \int_0^1 (1 - t^n) dt = 1 - \frac{1}{n+1} = \frac{n}{n+1}$$

For $n = 1000$ this gives

$$\mathbb{E}[M_{1000}] = \frac{1000}{1001} \approx 0.99900 \text{ seconds.}$$

Answer: 0.99900**169. Constrained Subsets I**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If the smallest number is a , the largest number is $13 - a$. Also, $a < 13 - a \Rightarrow a \leq 6$. So, the smallest-largest pairs are $(1, 12)$, $(2, 11)$, $(3, 10)$, $(4, 9)$, $(5, 8)$, $(6, 7)$. The number of subsets corresponding to the pair $(1, 12)$ are 2^{10} as there are 10 elements between 1 and 12—and we may either include or not include each of those elements in our subset (2 choices). Similarly, for the other smallest-largest pairs, the number of subsets are 2^8 , 2^6 , 2^4 , 2^2 , and 2^0 . So, the total number of desired subsets is:

$$2^{10} + 2^8 + 2^6 + 2^4 + 2^2 + 2^0 = 1365$$

Answer: 1365

170. Sufficient Sum IV[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $f(x)$ be the expected number of additional variables needed to make the running total exceed 2, given that the current sum is x , where $0 \leq x \leq 2$. So, we need $f(0)$.

After one more draw $U \sim \text{Unif}(0, 1)$, the process either stops immediately or continues from the new sum $x + U$. So the expected remaining time satisfies a recursion in terms of $f(x + U)$.

For $1 \leq x \leq 2$, one draw can already finish the process, so

$$f(x) = 1 + \int_0^{2-x} f(x+u)du.$$

Equivalently,

$$f(x) = 1 + \int_x^2 f(t)dt.$$

Differentiating, $f'(x) = -f(x)$.

Also, when $x = 2$, one more draw is always enough, so $f(2) = 1$.

Hence

$$f(x) = e^{2-x}, \quad 1 \leq x \leq 2.$$

Now for $0 \leq x \leq 1$, the next draw cannot yet stop the process, so

$$f(x) = 1 + \int_0^1 f(x+u)du = 1 + \int_x^{x+1} f(t)dt.$$

Differentiating gives

$$f'(x) = f(x+1) - f(x).$$

Since $x+1 \in [1, 2]$, we already know $f(x+1) = e^{2-(x+1)} = e^{1-x}$. Therefore

$$f'(x) + f(x) = e^{1-x}, \quad 0 \leq x \leq 1.$$

Multiplying by e^x , we get, $(e^x f(x))' = e$. So

$$e^x f(x) = ex + C,$$

$$\implies f(x) = e^{-x}(ex + C).$$

Using continuity at $x = 1$, $f(1) = e$. Thus

$$e^{-1}(e + C) = e \implies C = e^2 - e.$$

So, $f(0) = e^2 - e$.

Therefore, the expected number of variables is $e^2 - e$.

Answer: 4.67077

171. Numerator I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Given that $\frac{1}{5} < \frac{179}{720} < \frac{1}{4}$, the first unit fraction is $\frac{1}{5}$. Subtracting $\frac{1}{5}$ from $\frac{179}{720}$ yields: $\frac{179}{720} - \frac{1}{5} = \frac{179}{720} - \frac{144}{720} = \frac{35}{720} = \frac{7}{144}$. Next, $\frac{1}{20} < \frac{7}{144} < \frac{1}{21}$, so we add $\frac{1}{21}$ to our sum. Subtracting $\frac{1}{21}$ from $\frac{7}{144}$ gives: $\frac{7}{144} - \frac{1}{21} = \frac{144}{3024} - \frac{3}{3024} = \frac{1}{1008}$. Therefore, the largest denominator in the unit fraction representation is 1008.

Answer: 1008

172. Predicting King Draws[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To determine $P[X = k]$, where the 5th king appears on the k -th draw, we need exactly 4 kings among the first $k - 1$ cards and the 5th king on the k -th card. 1. Choose 4 positions from $k - 1$: $\binom{k-1}{4}$ 2. Permutations of 5 kings: $P(40, 5)$ 3. Select $k - 5$ non-kings from 480 cards: $P(480, k - 5)$ 4. Total ways to arrange k cards from 520: $P(520, k)$ Thus:

$$P[X = k] = \frac{\binom{k-1}{4} \cdot P(40, 5) \cdot P(480, k - 5)}{P(520, k)}$$

The valid range for k is from 5 to 485. To maximize the probability, find the smallest k for which:

$$\frac{P[X = k]}{P[X = k + 1]} \leq 1$$

This simplifies to:

$$\frac{\binom{k-1}{4} \cdot P(480, k - 5)}{\binom{k}{4} \cdot P(480, k - 4)} \leq 1$$

Further reducing:

$$\frac{(k - 4)(520 - k)}{k(485 - k)} \geq 1$$

Solving this:

$$520k - k^2 - 2080 + 4k \geq 485k - k^2$$

Simplifies to:

$$k \geq \frac{160}{3} \approx 53.3$$

Thus, the optimal k to maximize the probability is 54, as the ratio exceeds 1 at $k = 53$.

Answer: 54

173. 40-sided And 60-sided Die[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To solve this, first consider if the 60-sided die rolls a value above 40, in which case it wins with certainty. This occurs with a probability of $\frac{1}{3}$. If the value is within the range 1 to 40, it's like rolling two identical dice, and we need to compute the probability that the 60-sided die rolls strictly higher than the 40-sided one. In the case where both dice roll within the 1 to 40 range, the probability they are equal is $\frac{1}{40}$, because for any number the 40-sided die shows, the chance of the 60-sided die matching it is 1 out of 40. Hence, the probability that they are not equal is $\frac{39}{40}$. Since either die is equally likely to roll higher, the chance that the 60-sided die rolls the larger number is $\frac{39}{80}$. Combining the two cases, the total probability is:

$$\frac{1}{3} \times 1 + \frac{2}{3} \times \frac{39}{80} = \frac{79}{120}$$

Answer: 0.65833

174. Count Bitonic Permutations[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The number 1 must occupy either the first or the last position. The number 2 must occupy either the first unoccupied or the last unoccupied position, and so on. Each number has 2 options, except the last number, which has only one option. Hence, the number of such permutations is $2^8 = 256$.

Answer: 256

175. Uniform Number Bet I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The optimal strategy is of threshold form: you keep drawing until you see a value at least as large as some cutoff t , and then you stop. Whenever you face the choice "stop now and take your current draw x , or pay a fixed cost c for another independent draw," the optimal decision is always of threshold form because the value of continuing does not depend on x ; it is a constant. Here's why in more detail: denote by V your maximum expected net gain if you start fresh (with no current draw). Now imagine you're in the middle of play and your best draw so far is x . You have two options: * Stop and lock in payoff x . * Continue, pay c , discard x , and effectively "start over," so your expected net payoff from that point on is $V - c$. Since option (2) always yields the same expectation $V - c$ no matter what x is, you compare x directly with the constant $V - c$. If $x \geq V - c$, stopping gives you at least as much as continuing, so you should stop. If $x < V - c$, then the guaranteed x is less than the expected $V - c$ of another try, so you should continue. Thus the optimal rule is: Stop as soon as your current draw x satisfies

$$x \geq V - c$$

and continue otherwise. That single cutoff $t = V - c$ is exactly the threshold. It emerges because the continuation value $V - c$ is a fixed number, independent of the particular x you're holding or the number of rounds you have already played. Let V be your expected net payoff (the final random number minus all the fees) when you follow the threshold rule with cutoff t . On your first free draw, you obtain a number $X \sim U(0, 1)$. Two things can happen: * If $X \geq t$, you stop immediately, pay no fee, and your net payoff is simply X . * If $X < t$, you pay a fee c to regenerate, and thereafter your situation (and your expected net payoff) is exactly the same as at the start—namely V . Thus conditioning on the first draw,

$$V = \mathbb{E}[\text{net payoff}] = \int_0^t (V - c)dx + \int_t^1 xdx = t(V - c) + \frac{1 - t^2}{2}$$

Solving for V gives

$$V = \frac{1 - t^2}{2(1 - t)} - \frac{tc}{1 - t} = \frac{1 + t}{2} - \frac{tc}{1 - t}$$

To find the best cutoff t , we differentiate V with respect to t and set to zero:

$$\frac{dV}{dt} = \frac{1}{2} - \frac{c}{(1 - t)^2} = 0 \Rightarrow (1 - t)^2 = 2c \Rightarrow t = 1 - \sqrt{2c}$$

Substituting $c = 0.045$ gives $\sqrt{2c} = \sqrt{0.09} = 0.3$, so the optimal threshold is

$$t^* = 0.7$$

Finally, plugging $t = 0.7$ back into our formula for V ,

$$V = \frac{1 + 0.7}{2} - \frac{0.7 \times 0.045}{1 - 0.7} = 0.85 - \frac{0.0315}{0.3} = 0.85 - 0.105 = 0.745$$

Therefore, under optimal play - redrawing until you see at least 0.7 - you can expect a net payout of 0.745.

Answer: 0.74500

176. Asset Price

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the asset's price increase for x days and decrease for $10 - x$ days. Starting with an initial price p , the

price after 10 days is given by:

$$p(1.2)^x(0.8)^{10-x}$$

For the price to remain unchanged, we require:

$$p(1.2)^x(0.8)^{10-x} = p$$

Canceling p from both sides, this simplifies to:

$$1.2^x(0.8)^{10-x} = 1$$

$$\Rightarrow 1.2^x = 0.8^{10-x}$$

This equation does not hold for any integer x . Hence, the required probability is zero.

Answer:

177. Prism Count

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In a $5 \times 5 \times 5$ cube, a rectangular prism is defined by selecting two distinct points on each of the three axes to determine its length, width, and height. There are 6 points (including the boundaries) along each axis. Selecting 2 points from these 6 points can be done in: $\binom{6}{2} = 15$ ways. Since this selection process is independent for each of the three axes, the total number of rectangular prisms is: $\left(\binom{6}{2}\right)^3 = 15^3 = 3375$.

Answer:

178. Bulb Factory

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E_b be the expected number of bulbs till we see the next bad bulb if we start with a bad bulb. Let E_g be the expected number of bulbs till we see a bad bulb if we start with a good bulb.

Using the law of total expectation:

$$E_b = \frac{3}{5} \cdot 1 + \frac{2}{5} \cdot (1 + E_g)$$

Now, the number of bulbs till we see a bad bulb after starting with a good bulb follows a geometric distribution with parameter $\frac{3}{5}$. Hence,

$$E_g = \frac{5}{3}$$

Thus, we can calculate:

$$E_b = \frac{3}{5} \cdot 1 + \frac{2}{5} \cdot \left(1 + \frac{5}{3}\right) = \frac{5}{3}$$

Answer:

179. Moving Checker

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We are interested in the likelihood of reaching a corner from various positions in the 3×3 grid. There are five squares where reaching a corner is possible. Using the law of total probability, first observe that each of the nine squares has an equal chance of being selected. From there, we calculate the chances of reaching a corner from each square, proceeding recursively from top to bottom. Starting with the top-left square, there's a $\frac{1}{2}$ chance of directly reaching the left corner and a $\frac{1}{2}$ chance of moving to the top-center, which offers no

chance of reaching a corner. Thus, the overall probability for the top-left square is $\frac{1}{2}$. By symmetry, the top-right square has the same probability, while the top-center square has zero chance of reaching a corner. Moving downwards, the probabilities for the lower squares are calculated using the values from the upper squares. Summing up, we find:

$$\frac{1}{9} \times \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \right) = \frac{5}{18}$$

Answer:

180. XXYYYYZZ

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The key claim is that the positions of the Ys fully determine the end configuration. Indeed, since all Xs are initially left of all Zs, the only successful swaps that can occur will involve Ys. So, picking $\binom{8}{4} = 70$ spots for Ys and then filling in the remaining 4 spots with Xs first and then Zs gives all possible arrangements. It is not hard to show that all of these arrangements are also achievable, just greedily move Ys to their target positions.

Answer:

181. Factorial Factor

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The prime factorization of $20!$ is $20! = 2^{18} \cdot 3^8 \cdot 5^4 \cdot \dots$. A divisor of $20!$ can be expressed in the form $2^a \cdot 5^b \cdot k$, where k is an integer that does not include factors of 2 or 5. The exponents a and b can be any values such that $0 \leq a \leq 18$ and $0 \leq b \leq 4$. To be divisible by 2000, a divisor must have at least 2^4 and 5^3 , which means $a \geq 4$ and $b \geq 3$. There are 15 valid choices for a (from 4 to 18) out of 19 possible choices, and 2 valid choices for b (from 3 to 4) out of 5 possible choices. Thus, the probability is given by:

$$\frac{15}{19} \cdot \frac{2}{5} = \frac{6}{19}$$

Answer:

182. Interesting Division

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First, if n is prime, it is easy to see that n^2 cannot divide $n!$. Therefore, n must be composite. If n is not a perfect square, it is equal to ab with $a, b < n$ and $a \neq b$. Since n, a , and b all occur in the expansion of $n!$, n satisfies the statement. If $n = m^2$ is a perfect square, when $m > 2$, we have $2m < n$. Thus, all perfect squares except 4 are also valid. There are 25 primes less than 100, so the answer is $99 - 25 - 1 = 73$.

Answer:

183. Bulb Toggle III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

A bulb k will be off at the end of the day if it has an odd number of distinct prime factors. This includes all prime numbers, powers of prime numbers, and numbers with exactly three distinct prime factors. To determine the number of bulbs that are on, we analyze the integers from 1 to 100 based on their prime factorization.

Prime Numbers: The primes less than or equal to 100 are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97. This gives us 25 integers.

Powers of Prime Numbers: The powers of primes that are less than or equal to 100 are: $4(2^2)$, $8(2^3)$, $16(2^4)$, $32(2^5)$, $64(2^6)$, $9(3^2)$, $27(3^3)$, $25(5^2)$, $49(7^2)$, $81(3^4)$. This totals 10 integers.

Integers with Exactly Three Distinct Prime Factors: These integers, which have exactly three distinct prime factors and are less than or equal to 100, are: $30 = 2 \cdot 3 \cdot 5$, $42 = 2 \cdot 3 \cdot 7$, $66 = 2 \cdot 3 \cdot 11$, $78 = 2 \cdot 3 \cdot 13$, $70 = 2 \cdot 5 \cdot 7$, $60 = 2^2 \cdot 3 \cdot 5$, $84 = 2^2 \cdot 3 \cdot 7$, $90 = 2 \cdot 3^2 \cdot 5$. This gives us 8 integers.

Adding these counts together, we find: Total bulbs that are off: $25 + 10 + 8 = 43$ Thus, the number of bulbs that are on at the end of the day is: $100 - 43 = 57$ Therefore, 57 bulbs remain on.

Answer: 57

184. Broken Stick I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Given that X is located between $[0, \frac{1}{2}]$ and Y is located between $[\frac{1}{2}, 1]$, the goal is to compute $P[|X - Y| \geq \frac{1}{3}]$. Since the joint probability distribution over this region is uniform, the probability can be determined by comparing the areas. For the condition $|X - Y| \geq \frac{1}{3}$, the complement forms a right triangle with side lengths $\frac{1}{3}$. The ratio of areas for the complement is $\frac{1}{2} / \frac{18}{4} = \frac{2}{9}$, representing the probability of the complement. Thus, the desired probability is:

$$P[|X - Y| \geq \frac{1}{3}] = 1 - \frac{2}{9} = \frac{7}{9}$$

Answer: 0.77778

185. Factorial Vs Exponent

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write $99!$ as the product of three parts: the numbers 1 through 49, the single factor 50, and the numbers 51 through 99. Pair each small factor k ($1 \leq k \leq 49$) with the corresponding large factor $100 - k$. Then

$$99! = \left(\prod_{k=1}^{49} k \right) \cdot 50 \cdot \left(\prod_{k=1}^{49} (100 - k) \right) = 50 \cdot \prod_{k=1}^{49} (k(100 - k)).$$

Now use the identity $(x - y)(x + y) = x^2 - y^2$ with $x = 50$ and $y = 50 - k$. Then for each k between 1 and 49,

$$k(100 - k) = (50 - (50 - k))(50 + (50 - k)) = 50^2 - (50 - k)^2 < 50^2,$$

because $(50 - k)^2 > 0$ when $k \neq 50$. Therefore, every paired product $k(100 - k)$ is strictly less than 50^2 . Multiplying those inequalities for $k = 1$ to 49 gives

$$\prod_{k=1}^{49} k(100 - k) < (50^2)^{49} = 50^{98}.$$

Multiply both sides by the remaining factor 50 to recover $99!$:

$$99! = 50 \cdot \prod_{k=1}^{49} k(100 - k) < 50 \cdot 50^{98} = 50^{99}.$$

Thus

$$99! < 50^{99},$$

so 50^{99} is the greater number.

Answer:

186. Expected Arc Length III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In what follows, assume all angles are measured relative to the positive x-axis in the counterclockwise direction. Condition on the event that no cut falls in the open interval $(0, \frac{\pi}{2})$, so that the two fixed points indeed share one arc. Under this conditioning, all three cuts are still i.i.d. uniform on $[\frac{\pi}{2}, 2\pi]$. Let $A = \min\{\text{the three cut-angles}\}$, $B = \max\{\text{the three cut-angles}\}$. Then the arc containing angles 0 through $\frac{\pi}{2}$ "wraps" from B around past 0 up to A , so its length is $L = (2\pi - B) + A$. For three i.i.d. Uniform $[\frac{\pi}{2}, 2\pi]$ variables (an interval of length $L_0 = 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$), we have $E[A] = \frac{\pi}{2} + \frac{L_0}{4} = \frac{\pi}{2} + \frac{3\pi}{8} = \frac{7\pi}{8}$, $E[B] = \frac{\pi}{2} + \frac{3L_0}{4} = \frac{\pi}{2} + \frac{9\pi}{8} = \frac{13\pi}{8}$. By linearity of expectation, $E[L] = 2\pi - E[B] + E[A] = 2\pi - \frac{13\pi}{8} + \frac{7\pi}{8} = \frac{10\pi}{8} = \frac{5\pi}{4}$. Therefore, the expected length of the arc containing $(1, 0)$ and $(0, 1)$ is $\frac{5\pi}{4}$.

Answer:

187. Heavy Group

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To solve the problem, note that we can ignore any uniform offset in the weights, as it doesn't affect the relative comparisons. Thus, we can treat the weights as if they were 1, 2, 3, 4, 5, 6 grams instead of 201, 202, ..., 206. Fix the weight of 6-grams on one side of the scale. The total weight of all six weights is 21 grams, so the side containing the 6-gram weight will be heavier if its total weight is at least 11 grams. To complete the group with the 6-gram weight, we must select two additional weights from the remaining five (1, 2, 3, 4, 5). There are $\binom{5}{2} = 10$ ways to choose these two weights. Out of these combinations, the following pairs do not reach the required sum of 11: $1 + 2 = 3$, $1 + 3 = 4$. Thus, there are $10 - 2 = 8$ combinations where the side containing the 6-gram weight is heavier. Therefore, the probability that the side with the 6-gram weight is heavier is:

$$\frac{8}{10} = \frac{4}{5}$$

Answer:

188. First And Last Flip

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Call this probability x . We can solve for x using the relation

$$x = \frac{1}{2^{19}} + \frac{(1-x)(2^{19}-1)}{2^{20}}$$

The first term describes the chance that the first 20 flips are the same, and the added term describes the chance of getting the next 20 flips the same, given that the previous flip was not part of such a sequence of 20 flips. Solving for x yields

$$x = \frac{2^{19}-1}{2^{20}-1}$$

Answer:

189. Integral Angles

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The prime factorization of 360 is $360 = 2^3 \cdot 3^2 \cdot 5$. Therefore, the total number of factors of 360 can be calculated as:

$$(3+1)(2+1)(1+1) = 24$$

However, we need to exclude 1 and 2 as factors, since they do not correspond to valid polygons. Thus, there are:

$$24 - 2 = 22$$

values of n for which a regular n -sided polygon has integer-valued interior angles.

Answer: 22

190. Perfect Ratio

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Because X and Y are continuous and independent on $(0,1)$, the ratio Y/X ranges over $(0,\infty)$ with a continuous distribution, so

$$\{[Y/X] = k^2\} = \{k^2 - 1 < Y/X \leq k^2\} = \{(k^2 - 1)X < Y \leq k^2 X\}.$$

These events are disjoint for different k . Therefore,

$$P(S \text{ is a square}) = \sum_{k=1}^{\infty} P((k^2 - 1)X < Y \leq k^2 X).$$

Case $k = 1$: When $k = 1$, the condition is $0 < Y \leq X$, i.e. $Y < X$. By symmetry of two IID Uniform $(0,1)$ variables,

$$P(Y < X) = \frac{1}{2}.$$

Case $k \geq 2$: For fixed $k \geq 2$, we need

$$(k^2 - 1)X < Y \leq k^2 X, \quad 0 < X < 1, \quad 0 < Y < 1.$$

Equivalently, for each fixed $y \in (0,1)$, that range becomes

$$\frac{y}{k^2} \leq X < \frac{y}{k^2 - 1}.$$

Since $\frac{y}{k^2 - 1} < 1$ for all $y < 1$ whenever $k \geq 2$, the x -interval is

$$\left[\frac{y}{k^2}, \frac{y}{k^2 - 1} \right).$$

Because (X, Y) is uniform on the unit square, its probability is

$$\int_0^1 \left(\frac{y}{k^2 - 1} - \frac{y}{k^2} \right) dy = \int_0^1 y \left(\frac{1}{k^2 - 1} - \frac{1}{k^2} \right) dy = \frac{1}{k^2(k^2 - 1)} \cdot \frac{1}{2} = \frac{1}{2k^2(k^2 - 1)}.$$

Putting these together,

$$P(S \text{ is a square}) = \frac{1}{2} + \sum_{k=2}^{\infty} \frac{1}{2k^2(k^2 - 1)} = \frac{1}{2} + \frac{1}{2} \sum_{k=2}^{\infty} \frac{1}{k^2(k^2 - 1)}.$$

It remains to evaluate

$$\sum_{k=2}^{\infty} \frac{1}{k^2(k^2 - 1)}.$$

Observe the telescoping identity for $k \geq 2$:

$$\frac{1}{k^2(k^2 - 1)} = \frac{1}{k^2 - 1} - \frac{1}{k^2},$$

because

$$\frac{1}{k^2 - 1} - \frac{1}{k^2} = \frac{k^2 - (k^2 - 1)}{k^2(k^2 - 1)} = \frac{1}{k^2(k^2 - 1)}.$$

Hence,

$$\sum_{k=2}^{\infty} \frac{1}{k^2(k^2 - 1)} = \sum_{k=2}^{\infty} \left(\frac{1}{k^2 - 1} - \frac{1}{k^2} \right) = \sum_{k=2}^{\infty} \frac{1}{k^2 - 1} - \sum_{k=2}^{\infty} \frac{1}{k^2}.$$

First,

$$\sum_{k=2}^{\infty} \frac{1}{k^2} = \sum_{k=1}^{\infty} \frac{1}{k^2} - 1 = \frac{\pi^2}{6} - 1.$$

Next,

$$\sum_{k=2}^{\infty} \frac{1}{k^2 - 1}.$$

Note that

$$\frac{1}{k^2 - 1} = \frac{1}{2} \left(\frac{1}{k - 1} - \frac{1}{k + 1} \right).$$

Therefore,

$$\sum_{k=2}^{\infty} \frac{1}{k^2 - 1} = \frac{1}{2} \sum_{k=2}^{\infty} \left(\frac{1}{k - 1} - \frac{1}{k + 1} \right).$$

Writing out the first few terms shows telescoping:

$$\left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots = 1 + \frac{1}{2} - \lim_{N \rightarrow \infty} \left(\frac{1}{N+1} + \frac{1}{N+2} \right) = 1 + \frac{1}{2} = \frac{3}{2}.$$

Hence,

$$\sum_{k=2}^{\infty} \frac{1}{k^2 - 1} = \frac{3}{4}.$$

Putting these two results together,

$$\sum_{k=2}^{\infty} \frac{1}{k^2(k^2 - 1)} = \frac{3}{4} - \left(\frac{\pi^2}{6} - 1 \right) = \frac{3}{4} - \frac{\pi^2}{6} + 1 = \frac{7}{4} - \frac{\pi^2}{6}.$$

Therefore,

$$P(S \text{ is a square}) = \frac{1}{2} + \frac{1}{2} \left(\frac{7}{4} - \frac{\pi^2}{6} \right) = \frac{1}{2} + \frac{7}{8} - \frac{\pi^2}{12} = \frac{4+7}{8} - \frac{\pi^2}{12} = \frac{11}{8} - \frac{\pi^2}{12}.$$

Answer: 0.55253

191. Winning Odds

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

To win the game, a player must win two successive rounds. The probability that Alice wins two consecutive rounds is

$$\left(\frac{3}{10} \right)^2$$

and the probability that Bob wins two consecutive rounds is

$$\left(\frac{7}{10}\right)^2$$

Using the method of reduced sample space, the probability that Alice wins the game is the ratio of her winning probability to the total probability (the sum of Alice's and Bob's probabilities), which is:

$$\frac{\left(\frac{3}{10}\right)^2}{\left(\frac{3}{10}\right)^2 + \left(\frac{7}{10}\right)^2} = \frac{9}{9 + 49} = \frac{9}{58}$$

Answer: 0.15517

192. Biased Coin Game I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Determine the expected value of Bob's earnings based on p_1 and p_2 . The probabilities for each outcome are:
 * Both heads (HH): $p_1 p_2$ * Both tails (TT): $(1 - p_1)(1 - p_2)$ * Mixed outcomes (TH or HT): $p_1(1 - p_2) + p_2(1 - p_1) = p_1 + p_2 - 2p_1 p_2$ Bob's gains are:
 * For HH: +\$6 * For TT: +\$4 * For TH or HT: -\$5 The expected value is:

$$6p_1 p_2 + 4(1 - p_1)(1 - p_2) - 5(p_1 + p_2 - 2p_1 p_2)$$

Simplify to:

$$\begin{aligned} 6p_1 p_2 + 4 - 4p_1 - 4p_2 + 4p_1 p_2 - 5p_1 - 5p_2 + 10p_1 p_2 \\ = 20p_1 p_2 - 9p_1 - 9p_2 + 4 \end{aligned}$$

Now, assuming that $p_1 \neq \frac{9}{20}$ (such that we can rearrange and divide), we see that

$$p_2 > \frac{9p_1 - 4}{20p_1 - 9}$$

The RHS is 0 when $p_1 = \frac{4}{9}$, as the numerator vanishes. In addition, let's find when the RHS is at least 1. This implies:

$$9p_1 - 4 \geq 20p_1 - 9$$

which means that $p_1 \leq \frac{5}{11}$. One can verify that the expected value in this case (by plugging into the expected value equation previously) is $p_2 - 1$. However, as a probability is at most 1, unless $p_2 = 1$, Bob has a strictly negative expected value. Similarly, by plugging in $p_1 = \frac{4}{9}$, one can find the expected value to be $-p_2$. Therefore, unless $p_2 = 0$, Bob once again has a strictly negative expected value. Therefore, $a = \frac{4}{9}$ and $b = \frac{5}{11}$, so

$$b - a = \frac{1}{99}$$

Answer: 0.01010

193. 3 Marbles Remaining

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We'll first compute the probability that Deepali ends with exactly 3 red marbles remaining. We'll then double that probability to account for her possibly ending with 3 blue marbles instead. Even though Deepali is supposed to stop once all the remaining marbles have the same color, let's imagine that she continues until she has removed all 20 marbles. For Deepali to end with exactly 3 red marbles in the original game, her final four marbles in the extended game have to be BRRR (blue, red, red, red). Let's analyze the probability of finishing with BRRR, starting from the end. The probability that the final marble is red is $\frac{10}{20}$, or $\frac{1}{2}$. Given that the final marble is red, the probability that the second-to-last marble is red is $\frac{9}{19}$. Given that the final

two marbles are red, the probability that the third-to-last marble is red is $\frac{8}{18}$, or $\frac{4}{9}$. Given that the final three marbles are red, the probability that the fourth-to-last marble is blue is $\frac{10}{17}$. So the probability that the final four marbles are BRRR is

$$\frac{1}{2} \cdot \frac{9}{19} \cdot \frac{4}{9} \cdot \frac{10}{17} = \frac{20}{19 \cdot 17} = \frac{20}{323}$$

That's the probability of ending with exactly 3 red marbles in the original game. So the probability of ending with 3 marbles (of either color) in the original game is twice that, or

$$\frac{40}{323}$$

Answer:

194. Infinite Notes

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The smallest obtainable values that leave specific remainders when divided by 5 are:

- The smallest value that leaves a remainder of 1 when divided by 5 is 21.
- The smallest value that leaves a remainder of 2 when divided by 5 is 42.
- The smallest value that leaves a remainder of 3 when divided by 5 is 63.
- The smallest value that leaves a remainder of 4 when divided by 5 is 84.

Afterwards, by adding multiples of 5 to these values, we can obtain stamps of any number equivalent to these remainders modulo 5. Therefore, the largest value we cannot create would be one that is 4 modulo 5, as that is the highest starting point. Specifically, the largest integer that cannot be created is 79, since any number that is 4 modulo 5 and at least 84 can be created. Thus, 79 is the answer.

Note: For two relatively prime integers a and b , the largest value that cannot be expressed as $ax + by$ (where x and y are non-negative integers) is given by $ab - a - b$.

Answer:

195. Polynomial Coefficient

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Two evaluations of the polynomial are sufficient. The key is that the coefficients of $p(x)$ are the coefficients used to represent the integer $p(p(1) + 1)$ in base $p(1) + 1$. Therefore, Bob should first evaluate $p(1)$. He then evaluates $p(p(1) + 1)$ and converts this value into base $p(1) + 1$ to determine all the coefficients of $p(x)$.

Answer:

196. Game On Prime Numbers

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Alice has a winning strategy. After Bob says 3, Alice has the choice to say either 5 or 7. If saying 7 were a winning strategy, she would say it. But if saying 7 is losing, she can force Bob to say it by saying 5.

Answer:

197. Double Headed Coin I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To determine the probability that the selected coin is the double-headed one given that it shows heads, we apply Bayes' Theorem. Let C denote the event that the chosen coin is the double-headed coin, and R denote the event that the coin shows heads. Bayes' Theorem states:

$$P(C|R) = \frac{P(R|C)P(C)}{P(R)}$$

The probability of selecting any coin is $\frac{1}{100}$, so $P(C) = \frac{1}{100}$. For the double-headed coin, $P(R|C) = 1$, since it always shows heads. For a fair coin, $P(R|\text{fair}) = \frac{1}{2}$, and for the double-tailed coin, $P(R|\text{double-tailed}) = 0$. The total probability of observing heads, $P(R)$, is calculated by summing over all cases:

$$P(R) = P(R|C)P(C) + P(R|\text{fair})P(\text{fair}) + P(R|\text{double-tailed})P(\text{double-tailed})$$

Substituting values:

$$P(R) = (1) \left(\frac{1}{100} \right) + \left(\frac{1}{2} \right) \left(\frac{98}{100} \right) + (0) \left(\frac{1}{100} \right) = \frac{1}{100} + \frac{49}{100} = \frac{50}{100}$$

Finally, we compute $P(C|R)$:

$$P(C|R) = \frac{P(R|C)P(C)}{P(R)} = \frac{1 \cdot \frac{1}{100}}{\frac{50}{100}} = \frac{1}{50}$$

Thus, the probability that the coin is the double-headed one is $\frac{1}{50}$.

Answer:

198. Tic Tac Toe[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total number of ways in which marbles can land are 9^3 , since each marble has 9 possibilities. There are 8 winning lines (3 rows, 3 columns, and 2 diagonals). For each winning line, there are $3!$ ways of arranging the marbles. Hence, the total number of favorable cases is $8 \times 3!$. Hence, the required probability is $\frac{3! \times 8}{9^3} = \frac{16}{243}$.

Answer:

199. Dominant Flip[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Without loss of generality, assume the first flip is heads. For heads to be dominant at the end of 100 flips, we must get at least 50 heads in the next 99 flips. The number of such sequences with at least 50 heads in 99 flips is given by the sum of the binomial coefficients:

$$\binom{99}{50} + \binom{99}{51} + \cdots + \binom{99}{99}$$

Using the identity $\binom{99}{i} = \binom{99}{99-i}$, we observe that this sum is equal to:

$$\binom{99}{0} + \binom{99}{1} + \cdots + \binom{99}{49}$$

Hence, the number of sequences with at least 50 heads in 99 flips is equal to the number of sequences with

fewer than 50 heads in 99 flips. Hence, the required probability is $\frac{1}{2}$.

Answer:

200. Queen Before King

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

There are 8 cards in total that are king or queen. Consider only the positions of these cards in the sequence. The first 3 cards are 2 queens and 1 king. Hence, the remaining 5 cards have 2 queens and 3 kings. The problem asks, what is the probability that the last card among these 5 cards is a king. This probability is simply $\frac{3}{5}$. Therefore, the expected earnings are $\frac{3}{5} \times 100 = 60$ dollars.

Answer:

201. Expected Arc Length II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $\theta_1, \theta_2, \theta_3$ be i.i.d. Uniform $[0, 2\pi)$, where the angles are measured relative to the positive x-axis counter-clockwise. Define

$$A = \min\{\theta_1, \theta_2, \theta_3\}, \quad B = \max\{\theta_1, \theta_2, \theta_3\}.$$

The arc containing the fixed point at angle 0 is the "wrap-around" piece from B back to A , whose length is

$$L = (2\pi - B) + A.$$

When $n = 3$ points are placed uniformly at random on a segment of length 2π , we have:

$$\mathbb{E}[A] = \mathbb{E}[\min] = \frac{2\pi}{4} = \frac{\pi}{2}, \quad \mathbb{E}[B] = \mathbb{E}[\max] = \frac{3 \cdot 2\pi}{4} = \frac{3\pi}{2}.$$

By linearity of expectation:

$$\mathbb{E}[L] = 2\pi - \mathbb{E}[B] + \mathbb{E}[A] = 2\pi - \frac{3\pi}{2} + \frac{\pi}{2} = \pi.$$

Hence, the expected length of the arc containing $(1, 0)$ is π .

Answer:

202. Normal Inequality II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability $P(X_1 > X_2 \cdot X_3 \cdots X_{10})$ is equal to $\frac{1}{2}$. This is because the joint distribution of the variables $X_1, X_2, \dots, X_{10}$ is symmetric: the distribution of $(-X_1, -X_2, \dots, -X_{10})$ is the same as the distribution of $(X_1, X_2, \dots, X_{10})$. This symmetry implies that the probabilities $P(X_1 > X_2 \cdot X_3 \cdots X_{10})$ and $P(X_1 < X_2 \cdot X_3 \cdots X_{10})$ are equal. Additionally, since the probability $P(X_1 = X_2 \cdot X_3 \cdots X_{10})$ is zero for continuous random variables, we conclude that

$$P(X_1 > X_2 \cdot X_3 \cdots X_{10}) = \frac{1}{2}.$$

Answer:

203. Draw A Spade[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The desired probability is the relative probability that Jill draws the spade. In the first round, Jack, Jill, and John draw a spade with probabilities $\frac{1}{4}$, $\frac{3}{4} \cdot \frac{1}{4}$, and $(\frac{3}{4})^2 \cdot \frac{1}{4}$, respectively. Thus, the probability that Jill draws the spade is:

$$\frac{\frac{3}{4} \cdot \frac{1}{4}}{\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + (\frac{3}{4})^2 \cdot \frac{1}{4}} = \frac{12}{37}$$

Answer: **204. Median Roll**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Suppose you roll the values x, y , and z where $x \leq y \leq z$. Hence, the median is y . Now, the probability of getting the values $7 - x, 7 - y$, and $7 - z$ is the same, and $7 - y$ is the median in this case. Hence, the probabilities of getting a median y and $7 - y$ are the same. Let the probability of getting a median i be p_i . Now, $p_1 = p_6, p_2 = p_5$, and $p_3 = p_4$. Also, $p_1 + p_2 + p_3 + p_4 + p_5 + p_6 = 1 \Rightarrow p_1 + p_2 + p_3 = \frac{1}{2}$. So, the required expected value of the median is:

$$\begin{aligned} & p_1 \cdot 1 + p_2 \cdot 2 + p_3 \cdot 3 + p_4 \cdot 4 + p_5 \cdot 5 + p_6 \cdot 6 \\ &= p_1 \cdot 1 + p_2 \cdot 2 + p_3 \cdot 3 + p_3 \cdot 4 + p_2 \cdot 5 + p_1 \cdot 6 \\ &= 7(p_1 + p_2 + p_3) = 7 \cdot \frac{1}{2} = 3.5 \end{aligned}$$

Answer: **205. Permutation Return**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In a permutation π , an inversion is a pair of indices $i < j$ such that $\pi(i) > \pi(j)$. Any single swap always changes the total number of inversions by an odd number. That is, the parity of the number of inversions—whether it's odd or even—flips with each swap (the proof is simple and is left to the reader). We start with the identity permutation $[1, 2, 3, \dots, 100]$, which has zero inversions—an even number. After 25 swaps, since each swap flips the parity of the inversion count, the total number of inversions becomes odd. The original permutation has zero inversions. We want to know the probability that the permutation returns to that same original order after exactly 25 swaps. But since 25 swaps lead to a permutation with an odd number of inversions, and the original has an even number (zero), it's impossible to return to the original permutation in an odd number of swaps. The probability is 0.

Answer: **206. Getting Singular Matrix**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let s represent the number of steps required to reach a singular matrix starting from the identity matrix. Since this depends on which entry is changed first, we apply the Law of Total Expectation to the first edited value. If a diagonal element is selected, the matrix becomes singular immediately. Otherwise, if an

off-diagonal element is edited, the resulting matrix is either:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

or

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Note that for any square matrix A , the determinant satisfies $\det(A) = \det(A^T)$. Given that the elements are chosen randomly, both matrices will take the same expected number of steps to become singular. Let O denote the expected steps to reach a singular matrix from

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Using the Law of Total Expectation, we have:

$$E[T] = \frac{1}{2} \cdot 1 + \frac{1}{2}(1 + E[O]) = 1 + \frac{1}{2}E[O].$$

Now, let's calculate $E[O]$. If any element other than the top right corner is selected, the matrix becomes singular in one step with probability $\frac{3}{4}$. However, selecting the top right corner brings us back to the identity matrix. Therefore:

$$E[O] = \frac{3}{4} \cdot 1 + \frac{1}{4}(1 + E[s]) = 1 + \frac{1}{4}E[s].$$

Substituting $E[O]$ into the equation for $E[s]$ and simplifying gives:

$$E[s] = \frac{12}{7}.$$

Answer: 1.71429

207. Chess Tournament V

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Player 49 remains in the tournament if either he does not play any match in the first 10 matches or he plays in the first 10 matches but player 50 does not play in any match during this period. Each possible sequence of the tournament can be represented as a permutation of the 50 players, denoted $a_1, a_2, \dots, a_{50}$. The first match occurs between a_1 and a_2 , the second match occurs between the winner of the first match and a_3 , and so forth.

Player 49 will not play any match in the first 10 matches if he occupies one of the positions $a_{12}, a_{13}, \dots, a_{50}$. The probability of this case is $\frac{39}{50}$, since there are 39 such positions out of 50.

If player 49 participates in the first 10 matches, he can occupy one of the positions $a_1, a_2, \dots, a_{11}$, while player 50 must occupy one of the positions $a_{12}, a_{13}, \dots, a_{50}$. The number of such arrangements is given by $\binom{11}{1} \binom{39}{1} \cdot 48!$, and the total number of permutations is $50!$. Thus, the probability of this case is:

$$\frac{\binom{11}{1} \binom{39}{1} \cdot 48!}{50!} = \frac{429}{2450}$$

The total probability is the sum of these two cases:

$$\frac{39}{50} + \frac{429}{2450} = \frac{234}{245}$$

Answer: 0.95510

208. Gambler's Ruin II[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let P_i be the probability that the gambler's fortune will reach N instead of 0, starting from an initial state i , where $0 < i < N$. The next state can either be $i + 1$ with probability p or $i - 1$ with probability $q = 1 - p$. This leads to the equation:

$$P_i = p, P_{i+1} + q, P_{i-1}.$$

Rearranging this:

$$P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1}),$$

$$P_{i+1} - P_i = \left(\frac{q}{p}\right)^i (P_1 - P_0).$$

We also have the boundary conditions:

$$P(0) = 0 \quad \text{and} \quad P(N) = 1.$$

Using the recurrence relation, we can express P_i in terms of P_1 :

$$P_1 = p, P_2 + q, P_0,$$

$$P_2 = \frac{1}{p}P_1 = \left(1 + \frac{q}{p}\right)P_1,$$

$$P_3 = \left(1 + \frac{q}{p} + \left(\frac{q}{p}\right)^2\right)P_1,$$

$$\vdots$$

$$P_i = \left(1 + \frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1}\right)P_1.$$

Now, using the boundary condition $P_N = 1$, we get:

$$P_N = 1 = \left(1 + \frac{q}{p} + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{N-1}\right)P_1.$$

This simplifies to:

$$P_N = \frac{1 - \left(\frac{q}{p}\right)^N}{1 - \frac{q}{p}} \cdot P_1 \quad \text{if } \frac{q}{p} \neq 1,$$

or

$$P_N = N \cdot P_1 \quad \text{if } \frac{q}{p} = 1.$$

Solving these equations for P_1 , we can now determine P_i :

$$P_i = \frac{1 - \left(\frac{q}{p}\right)^i}{1 - \left(\frac{q}{p}\right)^N} \quad \text{if } \frac{q}{p} \neq 1,$$

or

$$P_i = \frac{i}{N} \quad \text{if } \frac{q}{p} = 1.$$

Answer: 0.12414

209. Sum Of Divisor Ratio[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let D be the sum of the positive divisors of 100. A divisor of $10000 = 100^2$ that is a multiple of 100 can be written as $100d$ for some positive integer d satisfying $100d|100^2 \Rightarrow d|100$. So, this divisor is 100 times a positive divisor of 100. Thus, the sum of the positive divisors of 10000 that are multiples of 100 can be written as $100D$. The answer is thus 100.

Answer: **210. No Intersections**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume the pairs of points are selected sequentially. Start with one point on the circle. For this point, it can either be paired with its left or right neighbor, or with the point directly across the circle. Any other choice would result in an intersection.

Pairing with a Neighbor: The probability that the first point is paired with a neighbor (left or right) is $\frac{2}{5}$. For the remaining four points, the probability that these pairs do not intersect is $\frac{2}{3}$. Thus, the total probability for this case is

$$\frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$$

Pairing with the Point Across: The probability that the first point is paired with the point directly across is $\frac{1}{5}$. In this scenario, the remaining four points must be paired such that no intersection occurs, which can only happen if the points on either side of the chord are paired correctly. The probability for this case is $\frac{1}{3}$. Thus, the total probability for this case is:

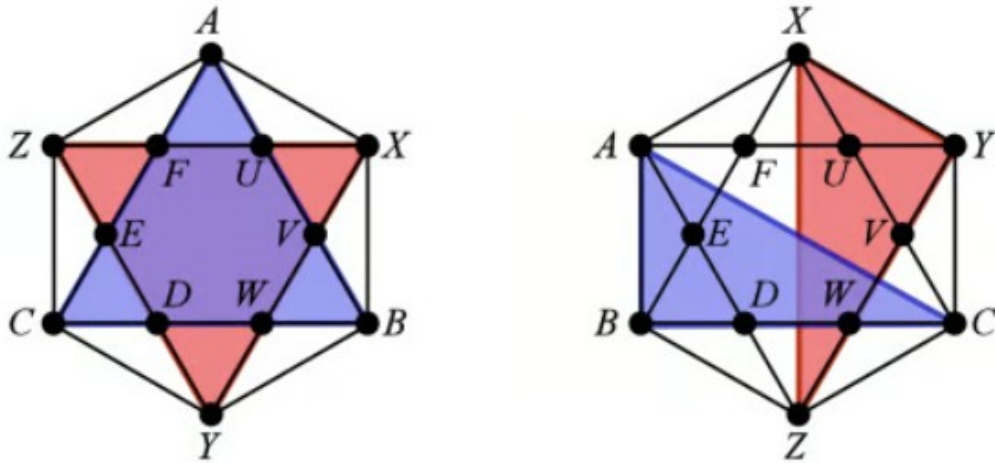
$$\frac{1}{5} \times \frac{1}{3} = \frac{1}{15}$$

Combining both scenarios, the total probability of no intersections is:

$$\frac{4}{15} + \frac{1}{15} = \frac{5}{15} = \frac{1}{3}$$

Answer: **211. Expected Area Of Intersection**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let s be the side length of H . There are 20 ways to pick 3 of the 6 vertices of a hexagon. Of these, there are 3 possibilities: (1) there are no pairs of adjacent vertices, which accounts for 2 of the ways, (2) there is a single pair of adjacent vertices, which accounts for 12 of the ways, and (3) the 3 vertices are consecutive, which accounts for 6 of the ways.



In the figure above, H is the larger hexagon in each picture.

In Case 1, the two triangles $\triangle XYZ$ and $\triangle ABC$ intersect in regular hexagon $UVWDEF$. Note that the distance between UF and DW is equal to s . The corresponding length in H is the distance between AX and YC , which is $\sqrt{3}s$. Therefore, the area of hexagon $UVWDEF$ is

$$\left(\frac{\sqrt{3}s}{3s}\right)^2 \cdot 360 = 120$$

In Case 2, the two triangles $\triangle XYZ$ and $\triangle ABC$ intersect in a region obtained by slicing out a piece of hexagon $UVWDEF$ by perpendicular bisectors to adjacent sides, and so the intersection has an area one-sixth that of hexagon $UVWDEF$, or 20.

In Case 3, triangles $\triangle XYZ$ and $\triangle ABC$ do not intersect.

The expected area is therefore

$$\left(\frac{2}{20}\right) 120 + \left(\frac{12}{20}\right) 20 = 24$$

Answer: 24

212. Blue, Ugly And Special

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

The probability of drawing k marbles is the probability of drawing $k - 1$ blue marbles and then the special marble, which is $p_k = \frac{9}{20} \cdot \frac{1}{20}$. The probability of drawing no ugly marbles is therefore $\sum_{k=1}^{\infty} p_k = \frac{1}{11}$. Then given that no ugly marbles were drawn, the probability that k marbles were drawn is $11p_k$. The expected number of marbles Ryan drew is

$$\sum_{k=1}^{\infty} k(11p_k) = \frac{11}{20} \sum_{k=1}^{\infty} k \left(\frac{9}{20}\right)^{k-1} = \frac{11}{20} \times 400/121 = \frac{20}{11}.$$

(To compute the sum in the last step, let $S = \sum_{k=1}^{\infty} k \left(\frac{9}{20}\right)^{k-1}$ and note that $\frac{9}{20}S = S - \sum_{k=1}^{\infty} \left(\frac{9}{20}\right)^{k-1} = S - \frac{20}{11}$.)

Answer: 1.81818

213. Card Picking III

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Using conditional probability, the solution is simply the number of ways to arrange the cards such that the first ace is at the 27th position and the 2 of hearts is at the 28th position divided by the number of ways to arrange the cards such that the first ace is at the 27th position. The denominator represents the total number of ways to position the first ace as the 27th card dealt. First, we need to place 26 of the 48 non-aces in the first 26 spots. This can be done in ${}^{48}P_{26}$ ways. Next, we place one of the 4 aces in the 27th position, leaving 25 remaining cards to arrange in any order, which gives us ${}^{48}P_{26} \times 4 \times 25!$ possible configurations where the first ace is the 27th card. For the numerator, we must satisfy the additional condition that the 28th card is the 2 of hearts. Therefore, the first 26 cards must exclude both aces and the 2 of hearts. We place 26 of the remaining 47 cards in the first 26 spots, which can be done in ${}^{47}P_{26}$ ways. Then, one of the 4 aces is placed in the 27th position, and the 2 of hearts must be the 28th card. Finally, the remaining 24 cards can be arranged in any order, giving us ${}^{47}P_{26} \times 4 \times 24!$. The probability is the ratio of these values:

$$P = \frac{4 \times {}^{47}P_{26} \times 24!}{4 \times {}^{48}P_{26} \times 25!} = \frac{11}{600}$$

Answer:

214. Probability Of Subset

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the ground set $S = \{1, 2, \dots, 10\}$. For each element $i \in S$, there are four equally likely ways it can lie in the random pair (X, Y) :

- in neither X nor Y
- in Y only
- in X only
- in both X and Y

Uniform randomness makes each of these four cases have probability $\frac{1}{4}$. The condition $X \subseteq Y$ fails exactly when an element is "in X only." Thus, for each i , the probability it does not violate $X \subseteq Y$ is

$$1 - \frac{1}{4} = \frac{3}{4}$$

Since the choices for different i are independent, the chance all ten elements obey $X \subseteq Y$ is

$$\left(\frac{3}{4}\right)^{10}$$

Alternate solution: There are 2^{10} subsets total, and hence 2^{20} equally likely ordered pairs (X, Y) . To count those with $X \subseteq Y$ note that for any fixed Y of size k , there are 2^k choices of $X \subseteq Y$. Summing over $k = 0$ to 10:

$$\sum_{k=0}^{10} \binom{10}{k} 2^k = (1 + 2)^{10} = 3^{10}$$

Hence, the probability is

$$\frac{3^{10}}{2^{20}} = \left(\frac{3}{4}\right)^{10}$$

Answer:

215. Circle Fitting II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let R denote the radius of the smallest concentric circle that contains the two closest points to the center. To compute the expected area, we use LOTUS:

$$E[\pi R^2] = \int_0^1 \pi r^2 f_R(r) dr,$$

where f_R is the density of R . For a single random point in the unit disk, the probability that its distance from the center is at most r is r^2 . Therefore, for the five independent points, the event $\{R \leq r\}$ occurs exactly when at least two of the five points lie within distance r of the center. The probability that zero points lie within this distance is $(1 - r^2)^5$. The probability that exactly one point lies within this distance is $5r^2(1 - r^2)^4$. Hence the cumulative distribution function of R is

$$F_R(r) = 1 - (1 - r^2)^5 - 5r^2(1 - r^2)^4, \quad 0 \leq r \leq 1.$$

Differentiating gives the density

$$f_R(r) = \frac{d}{dr} F_R(r) = 40r^3(1 - r^2)^3, \quad 0 \leq r \leq 1.$$

Applying LOTUS,

$$E[\pi R^2] = \int_0^1 \pi r^2 \cdot 40r^3(1 - r^2)^3 dr = \frac{\pi}{3}.$$

Hence the expected area of the smallest concentric circle containing the two points closest to the center is $\frac{\pi}{3}$.

Answer: 1.04720

216. Triangle Inside Circle II

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

The total number of ways of choosing 3 points from 1153 points is $\binom{1153}{3}$. The triangle will be obtuse when all three vertices are within the same half of the circle. We can always designate one vertex as the "start" of a half-circle, moving clockwise. Choose one point to be the vertex at the start of this half-circle; there are $\binom{1152}{1}$ ways to choose this point. There are 1152 points remaining, half of which are eligible for the other two points on the same half as the first point in the clockwise direction; so there are $\binom{576}{2}$ ways to choose the other two vertices. So the answer is

$$\frac{\binom{1152}{1} \times \binom{576}{2}}{\binom{1153}{3}} = \frac{1,725}{2,302}$$

Answer: 0.74935

217. Game Of Subtracting Digits

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

A number is a losing position exactly when it is divisible by 10. We prove this claim as follows: 1. If n is divisible by 10 then its units digit is 0. Any legal move subtracts some nonzero digit d (coming from a higher place) so

$$n - d \equiv -d \pmod{10}.$$

Since $d \in \{1, \dots, 9\}$ we have $n - d \not\equiv 0 \pmod{10}$. Thus every move from a multiple of 10 lands at a number not divisible by 10. 2. If n is not divisible by 10 then its units digit u is one of $1, 2, \dots, 9$. Subtracting that units digit is a legal move and yields $n - u$, which is divisible by 10. Hence from any non-multiple-of-10 there exists a move to a multiple of 10. Combining the two facts shows that the multiples of 10 are exactly the losing positions: from a multiple of 10 every move goes to a winning position, and from every non-multiple-

of-10 there is a move to a losing position. So, Alice begins by subtracting 4 from 1234, leaving Bob with a multiple of 10, which is a losing position. So, Alice has the winning strategy.

Answer:

218. Count Rankings

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The competition has 9 games. Each game has 2 possible winners. So the tournament has 2^9 possible sequences of winners. Each sequence of winners leads to a unique ranking. So the number of possible rankings is 2^9 , which equals 512.

Answer:

219. Count Rectangles I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Every rectangle with sides on the grid is determined by choosing two distinct vertical grid lines (one for the left side and one for the right) and two distinct horizontal grid lines (one for the top and one for the bottom). Since there are 7 vertical lines and 7 horizontal lines, the number of ways to choose a pair of vertical lines is $\binom{7}{2}$ and similarly $\binom{7}{2}$ for horizontal lines. Multiplying these choices gives the total number of rectangles:

$$\binom{7}{2} \cdot \binom{7}{2} = 21 \cdot 21 = 441.$$

So there are 441 non-degenerate rectangles.

Answer:

220. Line Splits I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Divide the line segments into 3 intervals $(0, 1)$, $(1, 2)$ and $(2, 3)$. There are 3^3 ways to arrange the points in these intervals. There are 4 cases if the middle point lies between 1 and 2 units: 1. All points lie in the interval $(1, 2)$ 2. One point lies in interval $(0, 1)$, and two points lie in the interval $(1, 2)$ 3. One point lies in interval $(0, 1)$, one point lies in interval $(1, 2)$, and one point lies in interval $(2, 3)$ 4. Two points lie in the interval $(1, 2)$ and one point lies in the interval $(2, 3)$ For the first case, there is only one way to get the required arrangement. For the second case, there are 3 ways to get the required arrangement (choose 1 point out of the 3 which goes to interval $(0, 1)$). For the third case, there are $3!$ ways to get the required arrangement. For the fourth case, there are 3 ways to get the required arrangement (similar to case 2). Hence, the total number of favorable arrangements is $1 + 3 + 3! + 3 = 13$. Hence, the required probability is $\frac{13}{3^3} = \frac{13}{27}$.

Answer:

221. Sharing A Secret

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To ensure that every pair of individuals does not both possess a key to at least one lock, each lock must be inaccessible to at least one pair. The key for such a lock will be distributed to the remaining 3 individuals in that pair. Therefore, at least $\binom{5}{2} = 10$ locks are needed. Each lock will have 3 keys, distributed among a unique group of 3 people. Consequently, each individual will hold $\frac{10 \times 3}{5} = 6$ keys. Thus, a minimum of 10 locks is required.

Answer: 10

222. Falling Ants

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume each ant is carrying a flag. Imagine at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem, as the flags simply go past each other. For all the flags to fall off, it would take 100 seconds in the worst case, as an ant can be placed at the westmost point of the rod and might be going east initially. Hence, the flag carried by this ant (initially) will take 100 seconds to fall from the eastmost point of the rod. Additionally, we can arrange the ants such that this last flag to fall off is being carried by Alice at the end. One such arrangement is Alice going towards the west initially, with all other ants to the east of Alice going east initially. This means 100 seconds is the least time for us to be sure that Alice has fallen.

Answer: 100

223. Random Segments

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Leo's drawing process is equivalent to a random walk on the infinite triangular lattice, where each step is a unit segment turned either 120° left or 120° right with equal probability. Notice two key facts:

* Intersection happens exactly when you turn the same way twice in a row. If Leo ever makes two left-turns in a row (or two right-turns in a row), those three unit segments form an equilateral triangle and the third segment necessarily meets one of the earlier ones. * If he alternates left and right forever, he never intersects.

Hence, the time until the drawing process first “hits” an existing segment is precisely the time until the first occurrence of two identical turns in a row. Each turn (left or right) is like a fair coin toss, and we're asking: how many tosses until we see heads-heads or tails-tails?

Let N be the number of coin tosses needed to get two consecutive identical results. Define f_H to be the expected number of additional tosses to finish once you've just tossed an H, and f_T similarly after a T. On the next toss you either:

* With probability $\frac{1}{2}$, repeat the same letter and stop immediately (no further tosses needed), or * With probability $\frac{1}{2}$, flip to the other letter and now need f_T or f_H more tosses.

Thus:

$$f_H = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2} f_T, \quad \text{and} \quad f_T = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2} f_H$$

Solving these gives:

$$f_H = f_T = 2$$

Since no toss has occurred before the first turn, the total expected tosses until two in a row is:

$$N = 1 + f_H = 1 + 2 = 3$$

Finally, Leo starts by drawing one segment along the x-axis (no turn yet), and then makes one segment per toss. Thus, the expected total number of segments drawn—counting the very first one before any turn—is:

$$1 + N = 1 + 3 = 4$$

Answer:

224. Even Squares

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the squares by their row $i \in \{1, \dots, 9\}$ and column $j \in \{1, \dots, 9\}$. How many of the rectangles include square (i, j) ? To choose a rectangle covering (i, j) , pick its top boundary among the i horizontal grid lines above or through row i , and its bottom boundary among the $9 - i + 1$ lines at or below row i ; similarly, pick its left boundary among the j vertical lines at or before column j , and its right boundary among the $9 - j + 1$ lines at or after column j . Thus the total number of such rectangles is $i(10 - i) \times j(10 - j)$. That product is odd exactly when each factor i , $(10 - i)$, j , and $(10 - j)$ is odd. But among $\{1, \dots, 9\}$, i is odd if and only if $10 - i$ is odd—so the condition reduces to “ i is odd and j is odd.” There are 5 odd choices for the row and 5 for the column, giving $5 \times 5 = 25$ squares that lie in an odd number of rectangles. Since all other $81 - 25 = 56$ squares lie in an even number of rectangles, the answer is 56.

Answer:

225. Triangle Formation II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If three pieces of a rod can form a triangle, each piece must be less than half the length of the rod. Therefore, the rod can be divided into three segments that can form a triangle if and only if: 1. One of the breaking points is in the left half of the rod, and the other is in the right half. 2. The distance between the two breaking points is less than half the length of the rod. The probability that the first condition is met is 0.5. Given that the first condition is satisfied, the probability that the second condition is also met is 0.5. Therefore, the combined probability that both conditions are satisfied is $0.5 \times 0.5 = 0.25$. Thus, the probability that the three pieces can form a triangle is 0.25.

Answer:

226. Get Heads

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability that Alice and Bob both flip heads on their first try is $\frac{1}{4}$. The probability that Alice and Bob both flip heads on their second try is $\frac{1}{16}$ (since they each had to flip tails first, then they each had to flip heads). The probability that Alice and Bob both flip heads on their third try is $\frac{1}{64}$, and so on.

So we have this infinite series:

$$\frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \dots$$

This is a geometric series with the first term $\frac{1}{4}$ and a common ratio $\frac{1}{4}$. The sum of this series is:

$$\frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Let p be the probability that Alice needs more coin tosses to get the first head compared to Bob. By symmetry, the probability that Bob needs more coin tosses to get the first head is also p . The total probability must sum to 1:

$$p + p + \frac{1}{3} = 1$$

Thus, $p = \frac{1}{3}$.

Answer:

227. Distinct Faces[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To determine $E[x]$, use the formula $E[x] = \sum_{i=1}^{\infty} P[x \geq i]$ for a non-negative integer-valued random variable x . First, $P[x \geq 1] = 1$ since the first roll always results in a new face. Next, $P[x \geq 2] = \frac{5}{6}$ because you need to roll one of the remaining 5 faces to get a new distinct face. $P[x \geq 3] = \frac{5}{6} \times \frac{4}{6} = \frac{20}{36} = \frac{5}{9}$ as you need to roll one of the remaining 4 faces. $P[x \geq 4] = \frac{5}{9} \times \frac{3}{6} = \frac{15}{54} = \frac{5}{18}$ because you need to roll one of the remaining 3 faces. $P[x \geq 5] = \frac{5}{18} \times \frac{2}{6} = \frac{10}{108} = \frac{5}{54}$ since you need to roll one of the remaining 2 faces. $P[x \geq 6] = \frac{5}{54} \times \frac{1}{6} = \frac{5}{324}$ as you need to roll the last remaining face. Finally, $P[x \geq 7] = 0$ since there are only 6 distinct faces. Summing these values gives the expected number of distinct faces as $\frac{899}{324}$.

Answer: 2.77469**228. Initial Roll**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability of rolling a value that meets or exceeds the initial roll depends on the outcome of that first roll. Hence, we condition on R_1 , the result of the first roll. This implies

$$E[N] = E[E[N|R_1]]$$

Given R_1 , there are $7 - R_1$ values on the die that are at least as large as R_1 , so the chance of rolling such a number in any subsequent roll is $\frac{7-R_1}{6}$. Consequently, the expected number of additional rolls required is $\frac{6}{7-R_1}$. Applying the Law of Total Expectation, we have:

$$E[N] = E\left[\frac{6}{7-R_1}\right] = \sum_{i=1}^6 \frac{6}{7-i} \cdot \frac{1}{6} = \frac{49}{20}$$

Answer: 2.45000**229. Digit Sum IV**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Treat each number as a six-digit number, including leading zeros, ranging from 000000 to 999999 (note that the sum of the digits for 1,000,000 is 1, which will be added separately). Each digit from 0 to 9 appears with equal frequency in each decimal place. Thus, the average value of each digit is $\frac{9}{2}$. Therefore, the average sum of the digits for a six-digit number is: $6 \times \frac{9}{2} = 27$. With one million numbers from 000000 to 999999, the total sum of all digits is: $27 \times 1,000,000 = 27,000,000$. Including the sum of the digits for 1,000,000, which is 1, the final answer is: $27,000,000 + 1 = 27,000,001$.

Answer: 27000001**230. Baseball Series**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that if we imagine the series always continues to seven games even after one team has won four, the winner remains unchanged. Furthermore, the probability that the Red Sox win Game n is exactly the same as the probability that the Yankees win Game $8 - n$. Thus, the probability that the Yankees win at least four games is the same as the probability that the Red Sox win at least four games, which is $\frac{1}{2}$.

Answer: 0.50000

231. Collinear Integer Points[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Alice cannot find such 5 points. There are four possible parity combinations for integer points: Odd, Odd, Even, Odd, Even, Even, Odd, Even. By the Pigeonhole Principle, at least two of the points must share the same parity. When two points of identical parity are selected, the midpoint of the line segment connecting them must also be an integer point.

Answer: 2**232. Gambler's Ruin I**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In this game, player M and player N play until one of them is left without any money. Player M wins 2 out of every 3 games played. At the start, M has \$1, and N has \$2. The state space can be defined as the combination of the money that player M has (m) and the money that player N has (n). Since neither m nor n can be negative, the state space is $\{m\} = \{0, 1, 2, 3\}$. The transition matrix and graph can be used to calculate the probability of M winning. The initial state is $X = 1$ (i.e., M starts with \$1). At state 1, the next state can be state 0 (if M loses a game) with probability $\frac{1}{3}$ or state 2 (if M wins a game) with probability $\frac{2}{3}$. Similarly, we can compute the probabilities for other states. Both state 3 (where M wins the game) and state 0 (where M loses the game) are absorbing states. Using absorption probability equations, we calculate the probability that M reaches the absorbing state of 3. Plugging in the transition probabilities, we find that $a = \frac{4}{7}$. Therefore, starting with \$1, player M has a $\frac{4}{7}$ probability of winning.

Answer: 0.57143**233. Local Maxima**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let I_j be an indicator variable that equals 1 if the number at the j -th position is a local maximum, where $1 \leq j \leq 14$. The total number of local maxima, T , is given by the sum:

$$T = I_1 + I_2 + \cdots + I_{14}.$$

By the principle of linearity of expectation:

$$E[T] = E[I_1] + E[I_2] + \cdots + E[I_{14}].$$

The expected value of an indicator variable is simply the probability that the corresponding event occurs, so:

$$E[I_j] = P(I_j = 1).$$

For positions $2 \leq j \leq 13$, there are neighbors on both sides of j . The number at position j is a local maximum if it is larger than both its neighbors. Out of the $3!$ possible arrangements of the numbers at $j-1$, j , and $j+1$, only 2 result in a local maximum at position j . For example, if the three numbers are 123, then the valid arrangements for a local maximum at the middle are 213 and 312. Hence,

$$E[I_j] = \frac{2}{3!} = \frac{1}{3}.$$

At the endpoints, positions $j = 1$ and $j = 14$, each has only one adjacent number. Out of the $2!$ arrangements of these two numbers, only one forms a local maximum. Thus,

$$E[I_1] = E[I_{14}] = \frac{1}{2}.$$

Finally, summing the expected values, we have:

$$E[T] = \frac{1}{2} \cdot 2 + \frac{1}{3} \cdot 12 = 5.$$

Answer:

234. Expected Collisions I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume each ant is carrying a flag. Imagine that at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past each other. This simplifies the problem, as the flags simply go past each other. What is the probability that two flags cross? No matter where they are, this happens if and only if they face one another, thus with probability $\frac{1}{4}$. By linearity of expectation, then, the expected number of flag crossings is $\binom{25}{2} \times \frac{1}{4} = \frac{25 \times 24}{8} = 75$.

Answer:

235. Only Consecutive Heads

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Total number of outcomes when a coin is flipped 10 times is $2^{10} = 1024$. Since heads are allowed to occur only in pairs, we can have 0, 2, 4, 6, 8, or 10 heads. The number of ways to achieve 0 heads (10 tails) is 1. The number of ways to achieve 2 heads (1 pair) and 8 tails is $\frac{9!}{1!8!}$ since the consecutive head pair can be treated as a single entity. Similarly, the number of ways to achieve 4 heads (2 pairs) and 6 tails is $\frac{8!}{2!6!}$. The number of ways to achieve 6 heads (3 pairs) and 4 tails is $\frac{7!}{3!4!}$. The number of ways to achieve 8 heads (4 pairs) and 2 tails is $\frac{6!}{4!2!}$. The number of ways to achieve 10 heads (5 pairs) and 0 tails is $\frac{5!}{5!0!}$. Hence, the total number of ways in which heads can appear only in consecutive pairs when a coin is flipped 10 times is: $1 + \frac{9!}{1!8!} + \frac{8!}{2!6!} + \frac{7!}{3!4!} + \frac{6!}{4!2!} + \frac{5!}{5!0!} = 1 + 9 + 28 + 35 + 15 + 1 = 89$. Hence, the required probability is $\frac{89}{1024}$.

Answer:

236. Constrained Digits

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that $a > 0$, so that $b > c$, and $e \geq 0$ so that $d \geq c$. Conversely, for each choice of (b, c, d) with $b > c$ and $d \geq c$, there exists a unique pair (a, e) such that $abcde$ is a number having the desired property. Thus, we compute

$$\sum_{c=0}^9 (9-c)(10-c) = \sum_{c=0}^9 c^2 - 19c + 90 = 330.$$

Answer:

237. Max Fill

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First, let's find the maximum number of cells that can be filled in a 2×2 subgrid. In any 2×2 block every cell is a neighbor of every other cell in that block because sharing a corner or an edge suffices. If three or four of the cells in a 2×2 block were filled, take the largest of the filled values in that block. The largest entry is strictly greater than the other two or three filled entries in the block, and hence it is greater than at least two neighboring filled cells, violating the rule. Therefore any 2×2 block can contain at most two filled cells. Since there are 16 non-overlapping 2×2 subgrids in an 8×8 grid, the total maximum number of filled cells is 32. We can achieve this. Fill every cell in rows 1, 3, 5, 7 and leave rows 2, 4, 6, 8 empty. That fills exactly 4 rows $\times$ 8 cells per row = 32 cells. Assign distinct integers across each filled row in strictly increasing order from left to right. Because the empty rows separate the filled rows, any filled cell has filled neighbors only to its immediate left and right in the same row. With a strictly increasing row the value in a filled cell is greater than its left neighbor but less than its right neighbor, so it is greater than at most one neighboring filled cell. So, the maximum achievable number of filled cells is 32.

Answer: 32

238. Up, Right And Diagonal

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Define the random variables x_i, y_i for $1 \leq i \leq 100$ where each x_i equals 1 if on the i th move, Tim makes a contribution to the right and zero otherwise. y_i is equal to 1 if on the i th move we make a contribution upwards and 0 otherwise. Hence, the answer is

$$E \left[\sum_{i=1}^n x_i \cdot \sum_{j=1}^n y_j \right] = \sum_{i,j} E[x_i y_j]$$

If $i \neq j$, then $x_i y_j = 1$ when, on the i -th move, Tim moves to the right or diagonally up-right (probability $1/3 + 1/3 = 2/3$), and on the j -th move, Tim moves up or diagonally up-right (probability $1/9 + 1/3 = 4/9$). Hence, for $i \neq j$, $x_i y_j = 1$ with probability $(2/3)(4/9) = 8/27$.

If $i = j$, then $x_i y_j$ is 1 if and only if Tim jumps up and right, which happens with probability $1/3$.

So, the required expectation is $(100^2 - 100) \cdot 8/27 + 100 \cdot 1/3 = 8900/3$.

Answer: 2966.66667

239. Product Of Solutions

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We view choosing five objects from a row of 19 objects in an unusual way. First, remove two of the chosen objects, the second and fourth, which are not adjacent nor at either end, forming three nonempty groups of consecutive objects. We then have i, j , and k choices for the first, third, and fifth objects. Because this is a reversible process taking a triple (i, j, k) to ijk choices, the answer is:

$$\binom{19}{5} = 11628$$

Answer: 11628

240. Ranged Guess

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Denote by p_n the first player's win probability when the secret lies in $\{1, \dots, n\}$. Solving for small n , we have: $n = 1$: you win with probability 1. $n = 2$: you must guess one of $\{1, 2\}$; if wrong you lose, so $p_2 = \frac{1}{2}$. $n = 3$: best first guesses are 1 or 3, giving $p_3 = \frac{2}{3}$. $n = 4$: the maximum probability of winning is $p_4 = \frac{1}{2}$ (calculated using the values for $n = 1, 2, 3$ for each possible initial choice and selecting the maximum). $n = 5$: the highest probability of winning is $\frac{3}{5}$ (calculated using the values for $n = 1, 2, 3, 4$ for each possible initial choice and selecting the maximum). In fact, one can prove by induction that

$$p_n = \begin{cases} \frac{1}{2}, & n \text{ even} \\ \frac{n+1}{2n}, & n \text{ odd} \end{cases}$$

For $n = 1$, $p_1 = 1 = \frac{1+1}{2 \cdot 1}$. For $n = 2$, you win only by guessing correctly first, so $p_2 = \frac{1}{2}$. These match the formula. Now observe that for the set $\{1, 2, 3, \dots, n\}$, if the first player begins by choosing k , he succeeds immediately with chance $\frac{1}{n}$; otherwise the game reduces to size $k - 1$ or $n - k$ with the opponent moving first, giving

$$p_{n,k} = \frac{1}{n} + \frac{k-1}{n}(1 - p_{k-1}) + \frac{n-k}{n}(1 - p_{n-k}) \quad (i)$$

where $p_{n,k}$ is the probability that the first player wins if he chooses k in the first turn. Assume the formula holds for all $m < n$. When n is odd, say $n = 2m + 1$, consider your first choice $k = 1$. You win immediately with probability $\frac{1}{n}$. If wrong (probability $\frac{2m}{n}$), the secret lies in $\{2, \dots, n\}$, an even-sized interval of length $2m$, where by induction the first-player win rate is $p_{2m} = \frac{1}{2}$. Thus

$$p_n = \frac{1}{n} + \frac{2m}{n} \left(1 - \frac{1}{2}\right) = \frac{1}{n} + \frac{2m}{n} \cdot \frac{1}{2} = \frac{1}{n} + \frac{m}{n} = \frac{1+m}{n} = \frac{1 + \frac{n-1}{2}}{n} = \frac{\frac{n+1}{2}}{n} = \frac{n+1}{2n}.$$

For any other interior first choice (i.e., $1 < k < n$), one of $k - 1$ or $n - k$ is even and the other odd. If k is odd, $k - 1$ is even so $p_{k-1} = \frac{1}{2}$, and $n - k$ is odd and

$$p_{n-k} = \frac{(n-k)+1}{2(n-k)}.$$

Substituting into (i) shows

$$\frac{1}{n} + \frac{k-1}{n} \left(1 - \frac{1}{2}\right) + \frac{n-k}{n} \left(1 - \frac{(n-k)+1}{2(n-k)}\right) = \frac{1}{n} + \frac{k-1}{2n} + \frac{n-k}{n} \left(\frac{2(n-k) - (n-k) - 1}{2(n-k)}\right) = \frac{1}{n} + \frac{k-1}{2n} + \frac{n-k-1}{2n} = \frac{2}{2n} = \frac{1}{n}.$$

If k is even, a similar check yields exactly $\frac{1}{2}$. Thus no interior k (i.e., $1 < k < n$) beats $\frac{n+1}{2n}$, so $p_n = \frac{n+1}{2n}$. When n is even, say $n = 2m$, any guess k splits into two subgames of sizes $k - 1$ and $2m - k$, one odd and one even. By the induction hypothesis, the win chances there are $\frac{k}{2(k-1)}$ or $\frac{1}{2}$ and $\frac{(2m-k)+1}{2(2m-k)}$ or $\frac{1}{2}$ respectively. Substituting those values into equation (i) shows in every case the total equals exactly $\frac{1}{2}$, so $p_{2m} = \frac{1}{2}$. By induction, the formula holds for all n , and in particular for $n = 100$ (even), we have

$$p_{100} = \frac{1}{2}.$$

Answer: 0.50000

241. Circular Regions

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

There are 4 regions when the chords intersect and 3 regions otherwise. The possible arrangements of the 4 points in clockwise direction are as follows: 1. $P_1P_2P_3P_4$ 2. $P_1P_2P_4P_3$ 3. $P_1P_3P_2P_4$ 4. $P_1P_3P_4P_2$ 5. $P_1P_4P_2P_3$ 6. $P_1P_4P_3P_2$ Cases 3 and 5 will lead to intersecting chords while others won't. Hence, the probability of intersection is $\frac{2}{6}$, while that of non-intersection is $\frac{4}{6}$. So, the required expected value is $(\frac{2}{6}) \times 4 + (\frac{4}{6}) \times 3 = \frac{10}{3}$

Answer: 3.33333

242. Integer Selection[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To find the probability, we first count the number of ways where the number 7 is not selected by any of the 7 people. Each person picks a number from the set $\{1, 2, 3, 4, 5, 6\}$, so the total number of possible selections is 6^7 .

Next, to find the cases where only 7 is unselected and all other numbers are chosen, we need exactly two people to pick the same value, and the remaining five people to pick one of the other five values. There are $\binom{7}{2}$ ways to choose the two people who will pick the same number. Then, there are 6 possible values that the pair can select, and the remaining 5 values can be assigned to the other 5 individuals in $5!$ ways.

Thus, the probability is

$$\frac{\binom{7}{2} \cdot 6 \cdot 5!}{6^7} = \frac{35}{648}$$

Answer:

243. Double Headed Coin II[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We start by applying Bayes' Theorem to compute the probability that the coin is double-headed, given k consecutive heads:

$$P(\text{double-headed} \mid k \text{ heads}) = \frac{P(k \text{ heads} \mid \text{double-headed})P(\text{double-headed})}{P(k \text{ heads})}$$

Given:

$$P(\text{double-headed}) = \frac{1}{20}$$

$$P(\text{fair}) = \frac{19}{20}$$

$$P(k \text{ heads} \mid \text{double-headed}) = 1 \quad (\text{since a double-headed coin always lands head})$$

$$P(k \text{ heads} \mid \text{fair}) = \left(\frac{1}{2}\right)^k \quad (\text{since a fair coin lands heads with probability } \frac{1}{2})$$

The total probability of k heads is:

$$P(k \text{ heads}) = P(k \text{ heads} \mid \text{double-headed})P(\text{double-headed}) + P(k \text{ heads} \mid \text{fair})P(\text{fair})$$

Substitute the values:

$$P(k \text{ heads}) = (1) \left(\frac{1}{20}\right) + \left(\left(\frac{1}{2}\right)^k\right) \left(\frac{19}{20}\right)$$

Now, we want the posterior probability to be at least 0.95:

$$\begin{aligned} \frac{\frac{1}{20}}{\frac{1}{20} + \left(\frac{1}{2}\right)^k \frac{19}{20}} &> \frac{19}{20} \\ \Rightarrow \frac{1}{1 + \frac{19}{2^k}} &> \frac{19}{20} \end{aligned}$$

$$\Rightarrow 2^k > 361$$

$$\Rightarrow k \geq 9$$

Hence, our answer is 9.

Answer:

244. M,P,F,G

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Even though Alex stops with G, let's imagine that she picks all 4 buttons no matter what. We'll call this scenario the extended game. The extended game is just a random permutation of the 4 buttons. For the original game to have at least three picks with the third pick being M is equivalent to the extended game having the third pick be M and the fourth pick be G. So in the extended game, the only successful permutations are PFMG and FPMG. Hence the probability of success is

$$\frac{2}{4!} = \frac{2}{24}$$

which simplifies to

$$\frac{1}{12}$$

Answer:

245. Tetrahedral Dice Game

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let's call b the outcome of the blue die and r the outcome of the red die. First, let's find the $P(b > r)$, $P(b = r)$, and $P(b < r)$ and their outcomes, as they each yield a different result. Total cases: $4 \times 4 = 16$. Of these 16 cases, there are 4 ways to create an outcome such that $b = r$. Thus, $P(b = r) = \frac{4}{16} = \frac{1}{4}$. With the 12 remaining cases, half will result in $b > r$ and the other half will be $r > b$. Both outcomes have a probability of $\frac{12}{16} \times \frac{1}{2} = \frac{3}{8}$. Thus, $P(b > r) = P(b < r) = \frac{3}{8}$. Now let's find our earnings given each of these three scenarios. Let's call x our earnings. In the situation $b = r$, the game stops and we earn nothing, so our conditional expectation is $E[x | b = r] = 0$. In the situation $b < r$, we earn nothing but the game doesn't stop, so x remains unchanged. Therefore, $E[x | b < r] = E[x]$. In the final situation where $b > r$, we are paid the difference and are able to continue playing. Of the six total cases where $b > r$, one will result in a difference of 3, two will result in a difference of 2, and three will result in a difference of 1. The average collected difference will be:

$$\frac{1}{6} \cdot 3 + \frac{2}{6} \cdot 2 + \frac{3}{6} \cdot 1 = \frac{5}{3}$$

This will be in addition to your earnings, so $E[x | b > r] = E[x] + \frac{5}{3}$. Putting everything together we get:

$$E[x] = \frac{1}{4}(0) + \frac{3}{8}(E[x]) + \frac{3}{8}\left(E[x] + \frac{5}{3}\right)$$

Solving for $E[x]$, we are left with:

$$\Rightarrow E[x] = 2.5$$

Answer:

246. Probabilistic Strategy I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Alice and Bob can improve their chances by using the following strategy: each guesses that their coin toss result matches the result of the other person's coin toss. Specifically:

- Alice will guess that Bob's result is the same as her own.
- Bob will guess that Alice's result is the same as his own.

With this strategy, they will win if both outcomes are either HH or TT . Hence, they win in two out of the four possible outcomes. Consequently, the probability of winning with this strategy is:

$$\text{Probability} = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{2}{4} = \frac{1}{2}$$

Thus, by employing this optimal strategy, Alice and Bob increase their probability of winning to $\frac{1}{2}$.

Answer:

247. Coin Toss Game I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We label each "memory" of the last one or two flips as a state and write a recursion for the probability that Bob's pattern HHT appears before Alice's pattern HTH.

1. State HH : The next flip is

- H with probability $\frac{1}{2}$, in which case you remain in state HH (win-probability P_{HH}),
- T with probability $\frac{1}{2}$, which completes HHT and Bob wins (win-probability 1).

Hence

$$P_{HH} = \frac{1}{2}P_{HH} + \frac{1}{2} \cdot 1 \implies P_{HH} = 1.$$

2. State HT : The next flip is

- H with probability $\frac{1}{2}$, which completes HTH and Bob loses (win-probability 0),
- T with probability $\frac{1}{2}$, which "forgets" the last two flips and returns you to the start (win-probability p).

Thus

$$P_{HT} = \frac{1}{2} \cdot 0 + \frac{1}{2}p = \frac{1}{2}p.$$

3. State (start or H) - denote this probability by p : From the start, a head moves you to HH and a tail moves you to HT , so

$$p = \frac{1}{2}P_{HH} + \frac{1}{2}P_{HT} = \frac{1}{2} \cdot 1 + \frac{1}{2} \left(\frac{1}{2}p \right) = \frac{1}{2} + \frac{1}{4}p.$$

Solving,

$$p - \frac{1}{4}p = \frac{1}{2} \implies p = \frac{2}{3}.$$

Therefore, Bob wins with probability $\frac{2}{3}$.

Answer:

248. Adding 1 Or 2

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We note that the only way n does not appear in the sequence is if $n - 1$ and then $n + 1$ appears. Hence, we

have $p_0 = 1$, and $p_n = 1 - \frac{1}{2}p_{n-1}$ for $n > 0$. This gives

$$p_n - \frac{2}{3} = -\frac{1}{2} \left(p_{n-1} - \frac{2}{3} \right),$$

so that

$$p_n = \frac{2}{3} + \frac{1}{3} \left(-\frac{1}{2} \right)^n.$$

Thus,

$$p_{20} - p_{15} = [1 - (-2)^5] \cdot \frac{3}{2^{20}}.$$

Answer:

249. Positive Product

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We require that an even number of real numbers are negative (or equivalently, positive), which occurs with probability

$$\frac{1}{3^{2020}} + \binom{2020}{2} \frac{2^2}{3^{2020}} + \cdots + \binom{2020}{2020} \frac{2^{2020}}{3^{2020}}$$

Observe that the binomial expansion of

$$\left(\frac{1}{3} + \frac{2}{3} \right)^{2020}$$

contains this sum. By symmetry, adding

$$\left(\frac{1}{3} - \frac{2}{3} \right)^{2020}$$

produces twice the desired sum. Simplifying, we get that

$$P = \frac{1}{2} + \frac{1}{2 \cdot 3^{2020}}$$

Answer:

250. Not That Easy

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The required probability is not $\frac{1}{2}$. Assuming the answer is $\frac{1}{2}$ based on the two possibilities—double-headed or one-head-one-tail—is incorrect. Given that a randomly selected coin shows heads, these two possibilities are not equally likely and hence, we can't use the formula for probability for this sample space. Thus, the required probability equals the probability of selecting the double-headed coin and obtaining a head, divided by the total probability of obtaining a head - whether from the double-headed coin or from the fair coin.

So, the required probability is

$$\frac{\frac{1}{3} \times 1}{\frac{1}{3} \times 1 + \frac{1}{3} \times \frac{1}{2}} = \frac{2}{3}$$

Answer:

251. Measuring Everything I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Every natural number can be written in the binary system. Thus, to weigh any number of grams of gold from 1 through 100, it is sufficient to have seven weights: 1, 2, 4, 8, 16, 32, 64. These correspond to the powers of 2. On the other hand, six weights are insufficient because we can obtain no more than $2^6 - 1 = 63$ different weights with them (each weight can either be placed or not placed on the left pan). Therefore, seven weights are needed to cover all numbers from 1 to 100.

Answer:

252. Count Solutions

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Substitute $x_i = 2y_i + 1$ for each x_i :

$$(2y_1 + 1) + (2y_2 + 1) + \cdots + (2y_6 + 1) = 96$$

Simplifying, this becomes:

$$2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6) + 6 = 96$$

so

$$y_1 + y_2 + y_3 + y_4 + y_5 + y_6 = 45$$

Now, we want the number of non-negative integer solutions to $y_1 + y_2 + y_3 + y_4 + y_5 + y_6 = 45$. By the "stars and bars" method, the number of solutions is:

$$\binom{50}{5} = 2118760$$

Thus, the number of non-negative odd integer solutions to $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 96$ is 2,118,760.

Answer:

253. Dice Pattern Game I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For both options, consider 3 states: (i) None of the numbers in the current pattern (ii) First number in the current pattern (iii) First and second numbers in the current pattern. For the first option, there is a $\frac{2}{3}$ chance of moving from state (ii) to state (i), while for the second option, there is a $\frac{5}{6}$ chance of moving from state (ii) to state (i). Hence, the second option will require more rolls on average, and therefore, we should select the second option.

Answer:

254. Power Distance I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let x_i represent the particle's movement at time i . Every second, the particle's movement x_i is either -1 or 1 with equal probabilities. Therefore, x_i is IID with $P[x_i = 1] = P[x_i = -1] = \frac{1}{2}$. The distance after n seconds is $d_n = \sum_{i=1}^n x_i$. We compute $E[d_n^2]$ as follows:

$$E[d_n^2] = E\left[\left(\sum_{i=1}^n x_i\right)^2\right] = E\left[\sum_{i=1}^n x_i^2 + \sum_{i \neq j} x_i x_j\right]$$

Since x_i and x_j are IID, $E[x_i x_j] = E[x_i]E[x_j] = 0$. Thus:

$$E[d_n^2] = \sum_{i=1}^n E[x_i^2] = \sum_{i=1}^n 1 = n$$

For $n = 30$, the result is 30.

Answer:

255. Power Factor

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The prime factorization of 10^{99} is $2^{99} \cdot 5^{99}$. The total number of factors of 10^{99} is $(99 + 1) \cdot (99 + 1) = 10,000$. The prime factorization of 10^{99} can be written as $(2^{85} \cdot 5^{85}) \cdot (2^{14} \cdot 5^{14})$. Hence, the total number of factors of 10^{99} that are divisible by 10^{85} is $(14 + 1) \cdot (14 + 1) = 225$. Thus, the required probability is $\frac{225}{10,000}$.

Answer:

256. Training Sessions

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Number the employees as 1 through 5. Any scheduling of these employees can be seen as a permutation of 111222333444555. The total number of such permutations is given by

$$\frac{15!}{3!5}$$

Now, to exclude those permutations where 1 is the first employee, fix 1 at the start, leaving us with the sequence 11222333444555. The number of such permutations is

$$\frac{14!}{2! \times 3!4}$$

Subtracting these from the total, the number of valid permutations (those not starting with 1) is:

$$\frac{15!}{3!5} - \frac{14!}{2! \times 3!4} = 134,534,400$$

Answer:

257. Same Cards

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To calculate the total number of ways to distribute the 12 cards among 4 players, the first player can receive any 3 cards from the 12 cards. The second player can receive any 3 cards from the remaining 9 cards. The third player can receive any 3 cards from the remaining 6 cards. The last player receives the remaining 3 cards. Thus, the total number of ways is:

$$\binom{12}{3} \cdot \binom{9}{3} \cdot \binom{6}{3} \cdot \binom{3}{3}$$

Now, calculate the number of favorable distributions where each player gets exactly 1 king, 1 queen, and 1 jack. The first player can receive any of the 4 kings. The second player can receive any of the remaining 3 kings. The third player can receive any of the remaining 2 kings. The last player receives the remaining

king. Thus, there are $4!$ ways to distribute the kings. Similarly, there are $4!$ ways to distribute the queens and $4!$ ways to distribute the jacks. The total number of favorable distributions is:

$$4! \cdot 4! \cdot 4!$$

Finally, the required probability is:

$$\frac{4! \cdot 4! \cdot 4!}{\binom{12}{3} \cdot \binom{9}{3} \cdot \binom{6}{3} \cdot \binom{3}{3}} = \frac{72}{1925}$$

Answer:

258. Ant On A Cube II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the initial state be E_{00} , where the ant starts at a vertex. Due to symmetry, the expected moves to return are based on the ant being 1 edge away, denoted E_1 , so $E_{00} = E_1 + 1$. Similarly, let E_2 represent the expected moves to return when the ant is 2 edges away and E_3 be the expected moves to return when the ant is at the opposite vertex. The equations for these states are:

$$E_{00} = E_1 + 1$$

$$E_1 = \frac{2}{3}E_2 + \frac{1}{3}E_{00} + 1$$

$$E_2 = \frac{2}{3}E_1 + \frac{1}{3}E_3 + 1$$

$$E_3 = E_2 + 1$$

We know $E_{00} = 0$. Solving this system gives $E_{00} = 8$, meaning the ant takes 8 steps on average to return to its starting point.

Answer:

259. 2 Or 11

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E denote the expected number of rolls needed. We'll apply the law of total expectation to determine E based on the outcome of our first roll. If the first roll is a 2, the game ends immediately. If the first roll is a 1, we then need one additional 1 to complete the game, and we define the expected number of rolls from this point as E_1 . If the roll is neither, we return to our starting condition. Therefore:

$$E = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot (1 + E_1) + \frac{2}{3} \cdot (1 + E)$$

To calculate E_1 , observe that rolling either a 1 or 2 will end the game in one turn. If neither is rolled, we revert to the original state. Thus:

$$E_1 = \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot (1 + E) = 1 + \frac{2}{3}E$$

Solving the above two equations, we get $E = \frac{21}{4}$.

Answer:

260. Geometric Progression

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the three chosen numbers, sorted, as $a < b < c$. They form a geometric progression exactly if $b^2 = ac$. Equivalently, there exists a positive rational $r = \frac{m}{n} > 1$ (in lowest terms) so that $(a, b, c) = (a, ar, ar^2) = (a, a\frac{m}{n}, a\frac{m^2}{n^2})$. Since a, b, c must be integers in $\{1, \dots, 100\}$ with $b = a\frac{m}{n}$ and $c = a\frac{m^2}{n^2}$, it follows that a must be a multiple of n^2 . Write $a = kn^2$ for some positive integer k . Then $b = kn^2 \cdot \frac{m}{n} = kmn$, $c = kn^2 \cdot \frac{m^2}{n^2} = km^2$. All three are integers, distinct because $m > n$, and lie in $\{1, \dots, 100\}$ if and only if $a = kn^2 \leq 100$, $b = kmn \leq 100$, $c = km^2 \leq 100$. The strictest of these is $c = km^2 \leq 100$. Hence for each pair (m, n) with $\gcd(m, n) = 1$ and $m > n$, k can run from 1 up to $\lfloor \frac{100}{m^2} \rfloor$. Every such triple (kn^2, kmn, km^2) is a valid geometric progression inside $\{1, \dots, 100\}$. We now need to count all pairs (m, n) satisfying $2 \leq m \leq 10$, $1 \leq n < m$, $\gcd(m, n) = 1$, and for each, record how many values of k are allowed, namely $\lfloor \frac{100}{m^2} \rfloor$. We list the valid (m, n) :

- $m = 2$. Coprime n is $n = 1$. Since $\lfloor \frac{100}{4} \rfloor = 25$, there are $1 \times 25 = 25$ triples.
- $m = 3$. Coprime n are $n = 1, 2$. $\lfloor \frac{100}{9} \rfloor = 11$, giving $2 \times 11 = 22$.
- $m = 4$. Coprime n are $n = 1, 3$. $\lfloor \frac{100}{16} \rfloor = 6$, giving $2 \times 6 = 12$.
- $m = 5$. Coprime n are $n = 1, 2, 3, 4$. $\lfloor \frac{100}{25} \rfloor = 4$, giving $4 \times 4 = 16$.
- $m = 6$. Coprime n are $n = 1, 5$. $\lfloor \frac{100}{36} \rfloor = 2$, giving $2 \times 2 = 4$.
- $m = 7$. Coprime n are $n = 1, 2, 3, 4, 5, 6$. $\lfloor \frac{100}{49} \rfloor = 2$, giving $6 \times 2 = 12$.
- $m = 8$. Coprime n are $n = 1, 3, 5, 7$. $\lfloor \frac{100}{64} \rfloor = 1$, giving $4 \times 1 = 4$.
- $m = 9$. Coprime n are $n = 1, 2, 4, 5, 7, 8$. $\lfloor \frac{100}{81} \rfloor = 1$, giving $6 \times 1 = 6$.
- $m = 10$. Coprime n are $n = 1, 3, 7, 9$. $\lfloor \frac{100}{100} \rfloor = 1$, giving $4 \times 1 = 4$.

Adding these totals yields $25 + 22 + 12 + 16 + 4 + 12 + 4 + 6 + 4 = 105$ distinct ordered triples (a, b, c) in $\{1, \dots, 100\}$ that satisfy $b^2 = ac$ and $a < b < c$. Since any choice of three distinct numbers from 1 to 100 is equally likely, the probability in question is $\frac{105}{\binom{100}{3}}$. Thus, the answer is $\frac{105}{\binom{100}{3}}$.

Answer: 0.00065

261. Sensor Signal

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We analyze the scenario using Poisson thinning. The signals generated by each sensor follow independent Poisson processes. The adjusted rates for these processes are:

$$\lambda_1 = 6 \cdot \frac{1}{6} = 1 \quad \text{and} \quad \lambda_2 = 10 \cdot \frac{1}{5} = 2.$$

Consequently, the time between signals for each sensor follows exponential distributions: $\text{Exp}(\lambda_1)$ and $\text{Exp}(\lambda_2)$, respectively. The overall time between signals from either sensor is determined by the minimum of the two time intervals, $\min\{X_1, X_2\}$, where:

$$X_1 \sim \text{Exp}(1) \quad \text{and} \quad X_2 \sim \text{Exp}(2).$$

For any $t > 0$, the cumulative behavior of this minimum can be derived as:

$$P[\min\{X_1, X_2\} > t] = P[X_1 > t] \cdot P[X_2 > t] = e^{-t} \cdot e^{-2t} = e^{-3t}.$$

This result demonstrates that $\min\{X_1, X_2\} \sim \text{Exp}(3)$. Since the mean of an $\text{Exp}(\lambda)$ distribution is $\frac{1}{\lambda}$, the expected time between signals is:

$$\frac{1}{3}.$$

Answer: 0.33333

262. Uniform Random Variables II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We need to find the probability that P_1 is the maximum among P_1, P_2, P_{10} , and P_{15} . Due to the exchangeability of the IID random variables, each has an equal chance of being the maximum. Thus, the probability that P_1 is the largest is $\frac{1}{4}$.

Answer:

263. Russian Roulette I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p be the probability that the first player loses. Hence, the probability that you (the second player) lose is $1 - p$. Now, there is a $\frac{1}{6}$ chance that the first person loses on the first shot. If he doesn't lose on the first shot, he effectively becomes the second player, and hence, his probability of losing is now $1 - p$. Hence, using the law of total probability,

$$p = \frac{1}{6} \cdot 1 + \frac{5}{6}(1 - p) \implies p = \frac{6}{11}$$

Answer:

264. Drunk Passenger II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We label the passengers 1 through 100 in the order they board, and we label the seats 1 through 100 so that passenger n "belongs" in seat n . Define the indicator random variable

$$X_i = \begin{cases} 1, & \text{if passenger } i \text{ ends up in seat } i, \\ 0, & \text{otherwise.} \end{cases}$$

Then the total number of passengers in their own seats is

$$X = \sum_{i=1}^{100} X_i,$$

and by linearity of expectation,

$$E[X] = \sum_{i=1}^{100} E[X_i] = \sum_{i=1}^{100} P(X_i = 1).$$

Passenger 1 picks a seat uniformly at random from 1 to 100:

$$P(X_1 = 1) = \frac{1}{100}.$$

From Drunk Passenger I, we know:

$$P(X_{100} = 1) = \frac{1}{2}.$$

Now consider passenger 99. He can end up sitting in one of three seats: seat 1, seat 99, or seat 100. Whenever someone before passenger 99 is forced to choose at random, they choose uniformly from among the remaining seats — which also include these three. The first time someone in this chain of displacements chooses among these three, the chain ends:

* If they choose seat 99, then passenger 99 loses his seat. * If they choose seat 1 or 100, then seat 99 remains untouched.

By symmetry, each of these three outcomes is equally likely, so:

$$P(X_{99} = 1) = \frac{2}{3}.$$

Similarly,

$$P(X_{98} = 1) = \frac{3}{4}, \quad P(X_{97} = 1) = \frac{4}{5}, \quad \dots, \quad P(X_2 = 1) = \frac{99}{100}.$$

So the total expectation is:

$$E[X] = P(X_1 = 1) + \sum_{k=2}^{100} P(X_k = 1) = \frac{1}{100} + \frac{99}{100} + \frac{98}{99} + \dots + \frac{3}{4} + \frac{2}{3} + \frac{1}{2}.$$

Let's simplify the sum. For $t = 2, \dots, 100$, note:

$$\frac{t-1}{t} = 1 - \frac{1}{t}.$$

So,

$$\sum_{k=2}^{100} \frac{k-1}{k} = \sum_{t=2}^{100} \left(1 - \frac{1}{t}\right) = 99 - (H_{100} - 1) = 100 - H_{100},$$

where

$$H_{100} = \sum_{i=1}^{100} \frac{1}{i}$$

is the 100th harmonic number. Adding the $\frac{1}{100}$ from passenger 1 gives:

$$E[X] = 100 - H_{100} + \frac{1}{100}.$$

Therefore, the expected number of passengers who end up in their own seats is

$$100 - H_{100} + \frac{1}{100} \approx 94.82262.$$

Answer: 94.82262

265. Binary Cards Removal

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First, we design a strategy for Bob. In his first turn, he removes the odd numbers, ensuring the minimum difference between two numbers left is at least 2. In his second turn, there are only numbers of the type $4k$ and $4k+2$. As Bob has to remove half of the numbers, he can guarantee that he can remove all the ones of one of these two types. Thus, the minimum difference between two numbers is now at least 4. Bob can keep on doing this and doubling the minimum difference at each turn. The strategy for Alice is similar. If the two numbers with the biggest difference at some moment are x and y , between x and $\frac{x+y}{2}$, or between $\frac{x+y}{2}$ and y , we have half or less than half of all the numbers. Thus, Alice can remove all of these, and the maximum difference between any two numbers left is reduced by at least half. Using this strategy, Bob can guarantee he wins at least 32 dollars, and Alice can guarantee she loses at most 32 dollars. Thus, 32 is the number we wanted.

Answer: 32

266. Early 5s

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Define success as "having two fives before having both 4 and 6," and denote - p_{00} - probability of success before we have any 4, 5, or 6 - p_{10} - probability of success after having one 5, but no 4 or 6 - p_{01} - probability of success after having one from 4 or 6, but no 5s (it doesn't matter if we have 4 or 6) - p_{11} - probability of success after having one from 4 or 6, and one 5. From symmetry, $p_{11} = \frac{1}{2}$. Now, using the law of total probabilities,

$$p_{00} = \frac{3}{6} \cdot p_{00} + \frac{2}{6} \cdot p_{01} + \frac{1}{6} \cdot p_{10}$$

$$p_{10} = \frac{3}{6} \cdot p_{10} + \frac{2}{6} \cdot p_{11} + \frac{1}{6} \cdot 1$$

$$p_{01} = \frac{4}{6} \cdot p_{01} + \frac{1}{6} \cdot p_{11} + \frac{1}{6} \cdot 0$$

From this, we can find $p_{01} = \frac{1}{4}$, $p_{10} = \frac{2}{3}$, and $p_{00} = \frac{7}{18}$. Hence the required probability is $\frac{7}{18}$.

Answer:

267. Bacteria Population

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For a colony of three cells to become extinct, each of the three independent lineages must individually go extinct. Since each cell has a $\frac{1}{3}$ chance of extinction, the probability that all three lineages go extinct is:

$$\left(\frac{1}{3}\right)^3 = \frac{1}{27}$$

Answer:

268. Cube Fitting I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since the length of the cube is 20, fitting cubes of size $3 \times 3 \times 3$, we can place 6 along each edge. A 7th cube would exceed the cube's boundary. Thus, the total number of cubes that can fit is $6^3 = 216$.

Answer:

269. Nine Children

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the ages of the nine children be $(a - 4d), (a - 3d), \dots, (a + 3d), (a + 4d)$, and that of the father be m , such that $a > 4d$ and $m > a + 4d$ for $a, d, m \in \mathbb{N}$. According to the condition:

$$(a - 4d)^2 + (a - 3d)^2 + \dots + (a + 4d)^2 = m^2$$

On expanding the LHS and grouping the terms, we get:

$$3(3a^2 + 20d^2) = m^2$$

Since the left-hand side is a perfect square, it must be a multiple of 9. Hence, d must be a multiple of 3. Let $d = 3k, k \in \mathbb{N}$. Substituting $d = 3k$, we get:

$$9(a^2 + 60k^2) = m^2$$

If $k = 1$, trying values of $a > 12k$, we get $a = 14$, so that the LHS becomes a perfect square. With these

values of k and a , we get $m = 48$. Bigger values of a or k will yield m greater than 90. Hence, the required age is 48.

Answer: 48

270. Unit Digit Probability II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since only the units digit is relevant, we can consider the set $\{1, 3, 5, 7, 9\}$. Note that $x^4 \equiv 1 \pmod{10}$ for all odd digits x , except for 5. Looking at the second set, which consists of integers from 2004 to 2023, there are 20 members of this set. Thus, modulo 4, this set includes 5 values for each residue class $\{0, 1, 2, 3\}$, making the probability equal for all cases.

Next, determine which values result in $m^n \equiv 1 \pmod{10}$. The counts are $4 + 1 + 0 + 1 + 2 = 8$, where $5^4 \equiv 1 \pmod{10}$ only if $n = 0$, which is not applicable here. Out of 20 values, simplifying $\frac{8}{20}$ yields $\frac{2}{5}$.

Answer: 0.40000

271. Grid Paint I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Across all c columns, there are exactly $10c$ blue squares. If we try to keep every row to at most 14 blues, then the total number of blues would be at most: $50 \times 14 = 700$. But when $c = 71$, the actual total is: $10c = 710 > 700$. This contradiction shows that more than 700 blue squares cannot be distributed across 50 rows without forcing one row to exceed 14 blues. So, at least one row must contain 15 blues. This is a direct pigeonhole principle argument: putting 710 blue squares into 50 rows forces one row to contain at least

$$\left\lceil \frac{710}{50} \right\rceil = 15$$

Now, let's construct an example for 70 columns that avoids this issue and keeps every row at exactly 14 blues:

- Number the rows $1, 2, \dots, 50$ and the columns $1, 2, \dots, 70$.
- For each column j , paint blue the 10 rows i satisfying:

$$i \equiv j \pmod{5}$$

Explanation:

- The residues mod 5 are 0, 1, 2, 3, 4, and each residue class contains $50/5 = 10$ rows.
- So each column paints exactly 10 rows blue.

Now fix any row i :

- As j ranges from 1 to 70, the value $j \pmod{5}$ takes each residue exactly $70/5 = 14$ times.
- Thus, each row is painted blue in exactly 14 columns, and no row exceeds 14 blues.

Final conclusion:

- If $c = 71$, some row must have at least 15 blues.
- If $c = 70$, it's possible to keep all rows at exactly 14 blues, so no row reaches 15.

Answer: 71

272. Dice Game V

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Define E_i as the expected value when Alice chooses to accept a roll of i or more, otherwise she re-rolls. To find the optimal strategy, evaluate the expected values for different choices of i :

$$E_1 = \frac{1}{6} \times (1 + 2 + 3 + 4 + 5 + 6) \Rightarrow E_1 = \frac{7}{2}$$

$$E_2 = \frac{1}{6} \times (E_2 - 1) + \frac{1}{6} \times (2 + 3 + 4 + 5 + 6) \Rightarrow E_2 = \frac{19}{5}$$

$$E_3 = \frac{2}{6} \times (E_3 - 1) + \frac{1}{6} \times (3 + 4 + 5 + 6) \Rightarrow E_3 = 4$$

$$E_4 = \frac{3}{6} \times (E_4 - 1) + \frac{1}{6} \times (4 + 5 + 6) \Rightarrow E_4 = 4$$

$$E_5 = \frac{4}{6} \times (E_5 - 1) + \frac{1}{6} \times (5 + 6) \Rightarrow E_5 = \frac{7}{2}$$

$$E_6 = \frac{5}{6} \times (E_6 - 1) + \frac{1}{6} \times 6 \Rightarrow E_6 = 1$$

By comparing these values, we find that the optimal choice yields a result of 4.

Answer:

273. No Rocks Allowed

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Given that playing paper offers no advantage, assume you play rock with probability r and scissors with probability $1 - r$. Let your opponent play paper with probability p and scissors with probability $1 - p$. The expected profit function is:

$$E(r, p) = r \cdot p(-1) + r \cdot (1 - p)(1) + (1 - r) \cdot p(1) + (1 - r) \cdot (1 - p)(0) = r + p - 3rp.$$

To find the equilibrium, take the partial derivatives of $E(r, p)$ with respect to r and p :

$$\frac{\partial E}{\partial r} = 1 - 3p, \quad \frac{\partial E}{\partial p} = 1 - 3r.$$

Setting these equal to zero gives the equilibrium at $r = p = \frac{1}{3}$. Therefore, the expected profit per round is:

$$E\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{1}{3} + \frac{1}{3} - \frac{1}{3} = \frac{1}{3}.$$

Answer:

274. Expected Remainder

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider M in binary. Assume we start with $M = 0$, then add a to M , then add $2b$ to M , then add $4c$ to M , and so on. After the first addition, the first bit (defined as the rightmost bit) of M is toggled with probability $\frac{1}{2}$. After the second addition, the second bit of M is toggled with probability $\frac{1}{2}$. After the third addition, the third bit is toggled with probability $\frac{1}{2}$, and so on for the remaining three additions. As such, the six bits of M are each toggled with probability $\frac{1}{2}$ —specifically, the k -th bit is toggled with probability $\frac{1}{2}$ at the k -th addition and is never toggled afterward. Therefore, each residue from 0 to 63 has probability $\frac{1}{64}$ of occurring, so they are all equally likely. The expected value is then just $\frac{63}{2}$.

Answer:

275. Coin Inside Square[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To determine the probability that the coin is entirely within a square, we need to calculate the region where the coin's center can land while keeping the coin inside the square. Given the coin's radius is 0.5 cm, the center of the coin must be at least 0.5 cm away from any edge. This leaves a 1 cm by 1 cm area for the center to land in. The area of this region is 1 square cm. The total area of the square is 4 square cm, so the probability is $\frac{1}{4}$.

Answer: **276. Coin Toss Game IV**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If Bob gets his first heads in the n th turn, Alice must get her first heads at or after $4n$ turns - which simply means Alice must get tails in her first $4n - 1$ turns. The probability that Bob gets his first heads in the n th turn is $(\frac{1}{2})^n$. The probability that Alice gets tails in her first $4n - 1$ turns is $(\frac{1}{2})^{4n-1}$. Using the law of total probability, the probability that Alice will need at least four times as many flips as Bob to get hers is:

$$\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n \times \left(\frac{1}{2}\right)^{4n-1} = 2 \sum_{n=1}^{\infty} \left(\frac{1}{2^5}\right)^n = \frac{2 \times \frac{1}{2^5}}{1 - \frac{1}{2^5}} = \frac{2}{31}$$

Answer: **277. Infectious Ants**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume each ant is carrying a flag. Imagine that at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem, as the flags simply go past each other. The easiest way to make the required calculation is to think of the flags being infected instead of the ants. Assume that Alice faces east. All west-facing flags ahead of her will cross hers and become "infected," while east-facing flags ahead of her will escape uninfected. Meanwhile, the west-facing flags, after crossing Alice's flag, will infect all east-facing flags behind Alice, while the west-facing flags behind Alice get away unaffected. Since there are an average of 6 west-facing flags ahead of Alice and 6 east-facing flags behind her, this gives an expected total of 13 infected flags (including Alice's), thus leading to 13 infected ants. However, there is a slight issue: if there are no west-facing ants ahead of Alice, then there is no flag to cross Alice's and infect the east-facing flags behind her. This situation happens with a probability of $\frac{1}{2^{12}}$, reducing the expected number of infected ants from 7 (Alice plus an average of 6 east-facing flags behind her) to just 1 (Alice alone). So the correct answer is not 13 but $13 - \frac{6}{2^{12}} \approx 12.9985352$ sick ants on average.

Answer: **278. Unique Ids**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that a_5 must be the smallest digit in the sequence since:

$$a_1 > a_2 > a_3 > a_4 > a_5 < a_6 < a_7 < a_8 < a_9 < a_{10}$$

Thus, $a_5 = 0$. The remaining nine digits must then be divided into two groups: four digits for a_1, a_2, a_3, a_4

(arranged in decreasing order) and five digits for $a_6, a_7, a_8, a_9, a_{10}$ (arranged in increasing order). Each choice of four digits for the first group uniquely determines the arrangement of both groups. The number of ways to choose four digits from the remaining nine is:

$$\binom{9}{4}$$

giving a total of 126 unique IDs.

Answer: 126

279. Functional Equation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

$f(1) + f(2) + \cdots + f(n) = n^3$ $f(1) + f(2) + \cdots + f(n-1) = (n-1)^3$ Subtracting the above equations, we get

$$f(n) = n^3 - (n-1)^3 = 3n^2 - 3n + 1$$

Since f is a polynomial and the identity holds for all positive integers n , it must hold for all x :

$$f(x) = 3x^2 - 3x + 1$$

Therefore

$$f\left(\frac{3}{2}\right) = 3\left(\frac{3}{2}\right)^2 - 3\left(\frac{3}{2}\right) + 1 = \frac{13}{4}$$

Answer: 3.25

280. Expected Wait Time I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let X represent the time (in minutes) until the next bus arrives. Since you arrive at a random time, X is uniformly distributed on the interval $[0, 10]$. For a uniform distribution over $[0, b]$, the expected value $E(X)$ is given by:

$$E(X) = \frac{b}{2}$$

Here, $b = 10$. So:

$$E(X) = \frac{10}{2} = 5$$

Therefore, the expected waiting time until the next bus arrives is 5 minutes.

Answer: 5

281. Weighing Coins I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Start by placing 3 coins on each side of the balance scale. If the scale balances, the heavier coin must be one of the remaining 2 coins, so simply weigh them against each other to identify the heavier one. If the scale does not balance, the heavier coin is among the 3 coins on the heavier side. To find the heavier coin, weigh 2 of the 3 coins against each other. If they are equal, the third coin is the heaviest. If one side is heavier, the heavier coin is on that side. This process requires 2 weighings.

Answer: 2

282. Coin Toss Game III[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Once a tail appears, the sequence THH will inevitably occur before HHH. Therefore, to get HHH before THH, the sequence must start with HHH. This can only happen if HHH appears within the first 3 flips, which occurs with a probability $\frac{1}{8}$.

Answer: **283. Card Flipping Game**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let O denote a card turned up, and X denote a card turned down. Alice turns down the 3rd from either end. This leaves: OOXOOOOO Whatever happens next, Alice can always leave one of the following: OOOXOOO OOXOOXOXO OXOOXOOO In the first case, Alice copies in one side what Bob does in the other side until he gets the last card. In the second case, Alice similarly copies Bob until he gets the last card. In the third case, whatever Bob does, Alice can leave: OXO OXOXOXO OOXOO All these 3 cases lead to Alice's win.

Answer: **284. Divisible By 5**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Alice can guarantee a win using a symmetry-based strategy. Since only divisibility by 5 matters, consider all numbers modulo 5. Among the numbers 1 through 21, there are four numbers congruent to 2 (mod 5), four congruent to 3 (mod 5), four congruent to 4 (mod 5), and five congruent to 1 (mod 5). On her first turn, Alice removes a number congruent to 1 (mod 5). After that, whenever Bob removes a number congruent to k (mod 5), Alice removes a number congruent to $(5 - k)$ (mod 5). This ensures that after each pair of turns, the sum of the remaining numbers is divisible by 5. Consequently, when only two numbers remain, their sum will also be divisible by 5, and Alice wins.

Answer: **285. Get More Heads**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The extra coin in A's set differentiates it from B's set. If we remove one coin from A, the comparison between A and B becomes symmetric. This symmetry provides valuable insights into the problem. Let's remove the last coin from A and compare the number of heads in A's first n coins with B's n coins. There are three possible outcomes:

- E_1 : A's n coins have more heads than B's n coins.
- E_2 : A's n coins have the same number of heads as B's n coins.
- E_3 : A's n coins have fewer heads than B's n coins.

By symmetry, the probability that A has more heads than B is the same as the probability that B has more heads than A. Therefore, $P(E_1) = P(E_3)$. Let $P(E_1) = P(E_3) = x$ and $P(E_2) = y$. Hence, we have the equation:

$$2x + y = 1.$$

For event E_1 , A will always have more heads than B, regardless of the result of A's $(n + 1)$ th coin. For event E_2 , the outcome of A's $(n + 1)$ th coin becomes significant. If it's a head (which occurs with probability 0.5),

A will end up with more heads than B. Thus, the $(n+1)$ th coin increases the probability that A has more heads than B by $0.5y$. Consequently, the total probability that A has more heads than B becomes:

$$x + 0.5y = x + 0.5(1 - 2x) = 0.5.$$

Thus, when A has $n+1$ coins, the probability that A has more heads than B is 0.5.

Answer:

286. To Infinity And Beyond I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Compute expectations by conditioning stepwise. The mean of a $\text{Uniform}(a, b)$ variable is $(a+b)/2$, so

$$\mathbb{E}[X_n \mid X_{n-1} = x] = \frac{x + (x + 2^{-n})}{2} = x + 2^{-(n+1)}$$

Taking unconditional expectation and using the tower property gives the recursion

$$\mathbb{E}[X_n] = \mathbb{E}[X_{n-1}] + 2^{-(n+1)}, \quad n \geq 2$$

Since $\mathbb{E}[X_1] = \text{mean of Uniform}(0, \frac{1}{2}) = \frac{1}{4}$, we sum the increments to obtain for $n \geq 2$

$$\mathbb{E}[X_n] = \frac{1}{4} + \sum_{k=2}^n 2^{-(k+1)}$$

The second term above is a finite geometric sum. Evaluating it and substituting, we get

$$\mathbb{E}[X_n] = \frac{1}{4} + \left(\frac{1}{4} - 2^{-(n+1)} \right) = \frac{1}{2} - 2^{-(n+1)}$$

Because the sequence X_n is nondecreasing and nonnegative, the Monotone Convergence Theorem allows passing the limit inside the expectation:

$$\mathbb{E}[X] = \mathbb{E} \left[\lim_{n \rightarrow \infty} X_n \right] = \lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \lim_{n \rightarrow \infty} \left(\frac{1}{2} - 2^{-(n+1)} \right) = \frac{1}{2}$$

Thus the expected value of the limit is $\mathbb{E}[X] = \frac{1}{2}$.

Answer:

287. Tails Removal

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The game will ultimately end when all remaining coins either show heads or tails after a round of flips. Since both outcomes are equally probable, the probability of the game ending with all heads is $\frac{1}{2}$.

Answer:

288. 1 Dot Removed

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

There are 3 faces with even numbers and 3 faces with odd numbers on the die. The probability of choosing a face with an even number is $\frac{1}{2}$, and the probability of choosing a face with an odd number is also $\frac{1}{2}$. If a dot is removed from an even-numbered face, there would be 4 odd and 2 even numbers remaining (so, the

probability of rolling an even number is $\frac{2}{6}$). If a dot is removed from an odd-numbered face, there would be 2 odd and 4 even numbers remaining (so, the probability of rolling an even number is $\frac{4}{6}$). Hence, using the law of total probability, the probability that an even value is rolled on this die is:

$$\frac{1}{2} \cdot \frac{2}{6} + \frac{1}{2} \cdot \frac{4}{6} = \frac{1}{2}$$

Answer:

289. Even Head Stop II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the required probability be p . Note that if the first toss results in tails, the problem becomes finding the probability that the total number of tosses is odd, which is the complement of the original problem. If the first two tosses are heads, we are finished, and the total number of tosses will be even. If the first toss is heads and the second toss is tails, we are essentially back to the original problem of determining the probability that the total number of tosses is even. Hence, we get the following equation:

$$p = \frac{1}{2}(1 - p) + \frac{1}{4} + \frac{1}{4}p$$

Solving for p , we get:

$$p = \frac{3}{5}$$

Hence, the required probability is $\frac{3}{5}$.

Answer:

290. Normal Inequality I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We use the radial symmetry of the joint distribution of X and Y , which are IID standard normal variables. Radial symmetry implies that probabilities depend only on the angle formed by a line segment from the origin, as the distribution depends only on distance and not direction. The condition $Y > 0$ restricts us to the upper half-plane ($0 \leq \theta < \pi$), while $Y > \sqrt{3}X$ corresponds to the region above the line $Y = \sqrt{3}X$, which makes an angle of $\theta = \arctan(\sqrt{3}) = \pi/3$ with the positive X -axis. Hence, the region of interest lies between $\pi/3$ and π , covering an angular span of:

$$\pi - \pi/3 = 2\pi/3$$

Since the total span of the upper half-plane is π , the probability is the ratio of these spans:

$$P(Y > \sqrt{3}X \mid Y > 0) = \frac{2\pi/3}{\pi} = \frac{2}{3}$$

Thus, the required probability is $\frac{2}{3}$.

Answer:

291. Contained Projection

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

By symmetry, each side of triangle ABC has the same chance of admitting an interior projection of D . Let X_{AB} be the indicator variable for the event that the projection of D onto the line AB lies between A and B . Then the total number of "good" sides is

$$X = X_{AB} + X_{BC} + X_{CA}$$

and by linearity of expectation,

$$E[X] = E[X_{AB}] + E[X_{BC}] + E[X_{CA}] = 3 \cdot P(X_{AB} = 1)$$

To compute $P(X_{AB} = 1)$, note that this occurs if and only if either $\angle DAB$ or $\angle DBA$ is obtuse. This happens precisely when points A and B lie on the same side of the diameter through D , which occurs with probability $\frac{1}{2}$. Hence,

$$E[X] = 3 \cdot \frac{1}{2} = \frac{3}{2}$$

Answer: 1.50000

292. Boat And Steamer

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the length of the river between points A and B be L meters, and the river's flow velocity be v m/s. In the first case, the boat drifts with the river at v m/s, so:

$$L = 6v$$

In the second case, the steamer moves at $v + 1$ m/s downstream, so:

$$L = 4(v + 1)$$

Equating the two expressions for L :

$$6v = 4(v + 1)$$

Simplifying:

$$6v = 4v + 4 \implies 2v = 4 \implies v = 2$$

Substituting $v = 2$ into $L = 6v$:

$$L = 6 \times 2 = 12$$

Thus, the length of the river is 12 meters.

Answer: 12

293. King, Queen, Ace I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

This problem can be viewed as a variant of the "First Ace" Problem. Separate the problem into 2 stages. In the first stage, we draw cards until we see the first card that is either a king, a queen, or an ace. There are $4 + 4 + 4 = 12$ such "divider" cards. The 12 divider cards partition the remaining $52 - 12 = 40$ non-special cards into 13 segments in expectation. Hence, each segment has $\frac{40}{13}$ cards on average, so the expected position of the first divider card is

$$\frac{40}{13} + 1 = \frac{53}{13}$$

Now, the first divider card can be a king, a queen, or an ace, each equally likely since there are four of each initially. With probability $\frac{1}{3}$, we see a king first, in which case the process stops immediately. With the remaining probability $\frac{2}{3}$, the first divider card is either a queen or an ace, and we must keep drawing until we encounter the other of the two or a king. In this latter case, after the first divider card appears, there

are

$$52 - \frac{53}{13} = \frac{623}{13}$$

cards expected to remain. Among these, 8 are "dividers": 4 of the unobserved rank (either queens or aces) and 4 kings. These 8 dividers split the remaining deck into 9 regions, with $\frac{519}{13}$ non-dividers distributed among them. Thus, the expected number of additional cards until the next divider is

$$\frac{519}{117} + 1 = \frac{636}{117}$$

By the law of total expectation, combining both cases gives

$$E = \frac{53}{13} + \frac{2}{3} \times \frac{636}{117} = \frac{901}{117} \approx 7.70085$$

Hence, the expected number of cards drawn until we have either a king or both a queen and an ace is approximately 7.7.

Answer: 7.70085

294. Coin Toss Game VII

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the expected value if you go first be denoted by x . There is a $\frac{1}{2}$ chance of winning 30 dollars, a $\frac{1}{4}$ chance of losing 30 dollars (when you flip tails and your opponent flips heads), and a $\frac{1}{4}$ chance that the game continues, leaving you with the same expected gain x dollars. This gives us the equation:

$$x = 0.5(30) + 0.25(-30) + 0.25x$$

Solving this equation yields $x = 10$. Therefore, the value of going first is at most 10 dollars.

Answer: 10

295. Interview II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

As the number of candidates n grows large, we still follow the rule "skip the first k candidates, then hire the next person who is best among all seen so far." The probability of success can be written exactly as

$$P_n(k) = \sum_{i=k+1}^n \frac{1}{n} \cdot \frac{k}{i-1}$$

because each position i is equally likely to contain the overall best (probability $1/n$), and given it does, we succeed only if the maximum of the first $i-1$ candidates fell in the first k (probability $k/(i-1)$). Let $\alpha = k/n$. When n is very large, the Riemann sum

$$\frac{1}{n} \sum_{i=k+1}^n \frac{k}{i-1}$$

converges to the integral

$$\int_{\alpha}^1 \frac{\alpha}{x} dx = \alpha [\ln x]_{\alpha}^1 = -\alpha \ln \alpha.$$

Thus, in the limit $n \rightarrow \infty$, the success probability as a function of the skip-fraction α is

$$f(\alpha) = -\alpha \ln \alpha.$$

To maximize $f(\alpha)$ on $(0, 1)$, we differentiate:

$$f'(\alpha) = -\ln \alpha - 1,$$

set $f'(\alpha) = 0$, giving $\ln \alpha = -1$ and hence $\alpha = e^{-1}$. Substituting back yields

$$f(1/e) = -\frac{1}{e} \ln \left(\frac{1}{e} \right) = \frac{1}{e}.$$

Therefore, as $n \rightarrow \infty$, the optimal strategy is to skip roughly the first n/e candidates, and the probability of selecting the very best candidate approaches

Answer: 0.36788

296. Plus One, Minus One

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We begin by coloring the grid in an alternating pattern of black and white cells, similar to a checkerboard. The key observation is that the difference between the sum of the values in the black cells and the sum of the values in the white cells remains unchanged (invariant) after each operation. This is because each operation modifies both a black and a white cell by adding or subtracting the same value, leaving the difference intact.

If we aim to make all cells in the grid equal to zero, the difference between the sums of the black and white cells must also be zero at the end. Consequently, for this to be possible, the sum of the values in the black cells must equal the sum of the values in the white cells at the outset. This is a necessary condition.

Moreover, it can be demonstrated that this condition is also sufficient. Starting from the bottom-right cell, we can reduce the value of $a_{n,n}$ to zero by performing operations on adjacent cells $a_{n,n}$ and $a_{n,n-1}$. Then, we proceed to make $a_{n,n-1}$ zero by operating on $a_{n,n-1}$ and $a_{n,n-2}$, and so forth, until the entire last row is filled with zeros. This process is repeated for the remaining rows.

Eventually, only two non-zero cells will remain. Since the sum of the values in the black cells is equal to the sum of the values in the white cells, the remaining two values must be identical, allowing us to simultaneously reduce them to zero, completing the process.

297. Largest Sum Subset

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To ensure that all elements are coprime, a straightforward candidate set would consist of all prime numbers up to 30, along with the number 1, since 1 is coprime to every positive integer. This initial candidate set is

$$T = \{1, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$$

The sum of this set is 130. However, we can improve this by replacing some prime integers with combinations of their products that yield a higher sum. For instance, we can remove 3 and 5 and replace them with 15, which increases the sum to 137. Next, we consider removing 2 from the set since it is the smallest integer besides 1. The highest product we can achieve with the remaining integers is 28, as $2 \times 7 = 28$. We note that 29 is already included in our set, but 30 cannot be added because it is divisible by 3 and 5. Thus, we replace 2 and 7 with 28, which raises our sum by 19, bringing it to 149. The remaining set becomes

$$T = \{1, 3, 5, 11, 13, 17, 19, 23, 28, 29\}$$

Next, we see that 3 can be replaced with 27, which remains coprime with the other elements in the set. Similarly, 5 can be replaced with 25, which also preserves coprimality. This leads to our final set

$$T = \{1, 11, 13, 17, 19, 23, 25, 27, 28, 29\}$$

The sum of the elements in this set is 193.

Answer: 193

298. Careful Soldiers

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Focus on the two soldiers with the shortest distance between them. They are observing each other. If another soldier is watching one of these two, it leads to one soldier being watched twice, indicating that at least one soldier is not being watched.

In the alternative case, if no other soldier watches these two, they can be removed without affecting the others. Since the total number of soldiers is odd, this process can continue until only one soldier remains, who does not observe anyone, leading to a contradiction.

299. Dice Difference III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let e_i be the expected number of rolls required to obtain two consecutive rolls with a difference no greater than 1 after rolling the value i . Note that $e_1 = e_6$, $e_2 = e_5$, and $e_3 = e_4$. Let e be the expected number of rolls required to obtain two consecutive rolls with a difference no greater than 1.

$$e = \frac{1}{6}(1 + e_1) + \frac{1}{6}(1 + e_2) + \frac{1}{6}(1 + e_3) + \frac{1}{6}(1 + e_4) + \frac{1}{6}(1 + e_5) + \frac{1}{6}(1 + e_6)$$

Since $e_4 = e_3$, $e_5 = e_2$, and $e_6 = e_1$:

$$e = 1 + \frac{1}{6}(e_1 + e_2 + e_3 + e_3 + e_2 + e_1) = 1 + \frac{1}{3}(e_1 + e_2 + e_3)$$

Now,

$$e_1 = \frac{1}{6}(1) + \frac{1}{6}(1) + \frac{1}{6}(1 + e_3) + \frac{1}{6}(1 + e_4) + \frac{1}{6}(1 + e_5) + \frac{1}{6}(1 + e_6)$$

Substituting $e_4 = e_3$, $e_5 = e_2$, and $e_6 = e_1$:

$$e_1 = 1 + \frac{1}{6}e_1 + \frac{1}{6}e_2 + \frac{1}{3}e_3$$

Similarly,

$$e_2 = 1 + \frac{1}{6}e_1 + \frac{1}{6}e_2 + \frac{1}{6}e_3$$

$$e_3 = 1 + \frac{1}{3}e_1 + \frac{1}{6}e_2$$

Solving these equations simultaneously:

$$e_1 = \frac{288}{115}, \quad e_2 = \frac{246}{115}, \quad e_3 = \frac{252}{115}$$

Finally,

$$e = 1 + \frac{1}{3}(e_1 + e_2 + e_3) = 1 + \frac{1}{3} \left(\frac{288}{115} + \frac{246}{115} + \frac{252}{115} \right) = \frac{377}{115}$$

Answer: 3.27826

300. Two Coins Remaining I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider what happens to the number of tails after each operation:

- If you remove a coin that is surrounded by tails on both sides, both of its neighbors will turn into heads, and the number of tails decreases by 2.
- If you remove a coin that is surrounded by heads on both sides, both neighbors will turn into tails, increasing the number of tails by 2.
- If you remove a coin that is surrounded by a head on one side and a tail on the other, the number of tails remains unchanged.

Hence, the parity of the number of tails remains invariant, and initially, the number of tails is 0 (even). Therefore, it is impossible to reach the state "HT".

Answer: 2

301. Conditional Expected Rolls

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let N be the number of rolls until you stop and B be the condition that the first 6 occurs before the first 5. The probability that you never roll a 5 or 6 for $n - 1$ turns and then roll a 6 on the n -th turn is $\left(\frac{4}{6}\right)^{n-1} \cdot \frac{1}{6}$. So, the expected number of rolls until we obtain the first 6 and never roll a 5 before stopping is:

$$\sum_{n=1}^{\infty} n \cdot \left(\frac{4}{6}\right)^{n-1} \cdot \frac{1}{6}$$

Let's denote this expectation by $E(N = n \cap B)$. The required expectation is:

$$E(N|B) = \frac{E(N \cap B)}{P(B)} = \frac{\sum_{n=1}^{\infty} n \cdot \left(\frac{4}{6}\right)^{n-1} \cdot \frac{1}{6}}{\frac{1}{2}} = 3$$

Answer: 3

302. Maximising Heads II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will reflip the coins that show tails after the first flip. Hence, the probability that a given coin shows tails after the entire process is $\frac{1}{2} \times \frac{1}{2}$ (since it must show tails after the first flip and again show tails after the second flip). So, the probability that a given coin shows heads is $\frac{3}{4}$. There are 8 coins, so the expected number of heads (and hence, the expected payout) is

$$\frac{3}{4} \times 8 = 6.$$

Answer: 6

303. Green Ball Draw

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Place the 4th green ball at the 9th position. This leaves 3 green balls to be arranged in the first 8 positions. The number of ways to choose these positions is $\binom{8}{3}$. The total number of ways to arrange all 4 green balls

among 15 positions is $\binom{15}{4}$. Therefore, the probability is given by

$$\frac{\binom{8}{3}}{\binom{15}{4}} = \frac{8}{195}$$

Answer:

304. Same Values On Dice

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

When rolling three dice, the possible scenarios for the initial values are having all dice the same, having two distinct values, or having all three distinct values. We analyze each scenario to find the probability that the second roll matches the first.

All Dice Show the Same Value: The probability of this happening is $\frac{1}{36}$, as the first die determines the value and the other two must match it, which occurs with probability $\frac{1}{6}$ per die.

All Dice Show Distinct Values: The probability for this is $\frac{5}{36}$, as there are $6 \times 5 \times 4$ possible distinct outcomes, divided by 6^3 total outcomes.

Two Distinct Values: The probability here is $\frac{5}{12}$, calculated as $1 - \frac{1}{36} - \frac{5}{36}$.

Next, we calculate the probability that the second roll matches the initial roll for each scenario:

All Dice Same: The second roll will match the first if all three dice show the same value, which happens with probability $\frac{1}{216}$ (since there are 6^3 total outcomes).

Two Distinct Values: There are 3 favorable outcomes for matching the initial roll, resulting in a probability of $\frac{1}{72}$ (as there are 6^3 total outcomes).

Three Distinct Values: There are 6 ways to match the values, giving a probability of $\frac{1}{36}$ (with 6^3 total outcomes).

Combining these probabilities using the law of total probability, we get:

$$\frac{1}{36} \cdot \frac{1}{216} + \frac{5}{12} \cdot \frac{1}{72} + \frac{5}{9} \cdot \frac{1}{36} = \frac{83}{3888}$$

Answer:

305. Meeting Probability

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To determine the probability of two individuals meeting at the station, we must calculate the likelihood of an overlap in their arrival times. Assume one individual arrives X minutes after 5:00 AM, and the other arrives Y minutes after 5:00 AM. Both X and Y are independently uniformly distributed between 0 and 60 minutes. We can model this situation on a coordinate plane where the sample space is a square with vertices at $(0, 0)$ and $(60, 60)$. The individuals will meet if $|X - Y| \leq 5$. To find the probability of this event, we need to calculate the area where $|X - Y| \leq 5$ within the square. To do this, we shade the region within the square where $|X - Y| \leq 5$. This shaded region forms a band around the line $X = Y$ with a width of 10 minutes. The area of this band can be found by subtracting the areas of the two triangles outside the band from the area of the square. The area of the shaded region can be calculated as follows:

$$(60 + 55) \times (60 - 55) = 575$$

The area of the entire square is:

$$60 \times 60 = 3600$$

Thus, the probability of the two individuals meeting is the ratio of the area of the shaded region to the area

of the square:

$$\frac{575}{3600} = \frac{23}{144}$$

Answer: 0.15972

306. Knight Return

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

View each of the 64 squares as a state of a Markov chain and let the transition probability from square i to a reachable square j be

$$P_{ij} = \frac{1}{d_i}$$

when a knight can move from i to j , where d_i is the number of legal knight moves from square i . The chain is finite and irreducible, so Kac's lemma applies and the expected return time to state i equals $1/\pi_i$, where π is the stationary distribution. Now, note that the given Markov chain satisfies the following properties: (i) whenever $P_{ij} > 0$ we also have $P_{ji} > 0$, (ii) whenever $P_{ij} > 0$ then $P_{ij} = 1/d_i$. So, using the theorem described in the chapter "Expected First Passage and Return Times", we have for every square j

$$\pi_j = \frac{d_j}{\sum_k d_k}$$

Therefore the expected return time to square i is

$$E[\text{return to } i] = \frac{1}{\pi_i} = \frac{\sum_k d_k}{d_i}$$

Let's compute the degrees on the 8×8 board. There are 4 corner squares with degree 2, 8 squares with degree 3, 20 squares with degree 4, 16 squares with degree 6, and 16 squares with degree 8. Thus

$$\sum_k d_k = 4 \cdot 2 + 8 \cdot 3 + 20 \cdot 4 + 16 \cdot 6 + 16 \cdot 8 = 336.$$

The bottom-left corner is a corner square so its degree is $d_{\text{corner}} = 2$. Hence the expected number of moves to return to the bottom-left corner is

$$\frac{\sum_k d_k}{d_{\text{corner}}} = \frac{336}{2} = 168.$$

Hence, the expected number of moves for the knight to return to its starting square is 168.

Answer: 168

307. Rolling Everything II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will use recursion to solve this problem. Let E_k be the expected number of additional moves needed to see all six faces, given that we have already seen k distinct faces. When k faces are already seen, there remain $6 - k$ unseen faces. In one move of two dice rolls, we have the following possibilities:

No new face: Let p_0 be the probability of this case: both dice land on already-seen faces.

$$p_0 = \left(\frac{k}{6}\right)^2$$

Exactly one new face: Let p_1 be the probability of this case: either one die shows an unseen face and the

other a seen face ($2k(6-k)$ outcomes), or both dice show the same unseen face ($6-k$ outcomes).

$$p_1 = \frac{2k(6-k) + (6-k)}{36} = \frac{(6-k)(2k+1)}{36}$$

Two new faces: Let p_2 be the probability of this case: both dice show two different unseen faces.

$$p_2 = \frac{(6-k)(5-k)}{36}$$

The recurrence is

$$E_k = 1 + p_0 E_k + p_1 E_{k+1} + p_2 E_{k+2}, \quad E_6 = 0$$

so

$$E_k = \frac{1 + p_1 E_{k+1} + p_2 E_{k+2}}{1 - p_0}$$

Now compute E_5 through E_0 by substituting: $k = 5 : p_0 = \frac{25}{36}, p_1 = \frac{11}{36}, p_2 = 0 \implies E_5 = \frac{1}{1 - \frac{25}{36}} = \frac{36}{11}$
 $k = 4 : p_0 = \frac{16}{36}, p_1 = \frac{18}{36}, p_2 = \frac{2}{36} \implies E_4 = \frac{1 + \frac{18}{36} E_5}{1 - \frac{16}{36}} = \frac{261}{55}$ $k = 3 : p_0 = \frac{9}{36}, p_1 = \frac{21}{36}, p_2 = \frac{6}{36} \implies E_3 = \frac{1 + \frac{21}{36} E_4 + \frac{6}{36} E_5}{1 - \frac{9}{36}} = \frac{949}{165}$ $k = 2 : p_0 = \frac{4}{36}, p_1 = \frac{20}{36}, p_2 = \frac{12}{36} \implies E_2 = \frac{1 + \frac{20}{36} E_3 + \frac{12}{36} E_4}{1 - \frac{4}{36}} = \frac{8579}{1320}$ $k = 1 : p_0 = \frac{1}{36}, p_1 = \frac{15}{36}, p_2 = \frac{20}{36} \implies E_1 = \frac{1 + \frac{15}{36} E_2 + \frac{20}{36} E_3}{1 - \frac{1}{36}} = \frac{65609}{9240}$ $k = 0 : p_0 = 0, p_1 = \frac{6}{36}, p_2 = \frac{30}{36} \implies E_0 = 1 + \frac{6}{36} E_1 + \frac{30}{36} E_2 = \frac{70219}{9240}$ Therefore, the expected number of rolls needed to see all six faces at least once is

$$\frac{70219}{9240}$$

Answer: 7.60000

308. No Empty Bucket

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let K be the toss count for the first heads, so $P(K = k) = 2^{-k}$ for $k \geq 1$. Conditional on $K = k$, each of the k marbles independently lands in any of the three buckets with probability $\frac{1}{3}$, and by inclusion-exclusion the probability that all three buckets are nonempty equals

$$1 - 3 \left(\frac{2}{3} \right)^k + 3 \left(\frac{1}{3} \right)^k.$$

Using the law of total probability, the required probability is

$$\sum_{k=1}^{\infty} 2^{-k} \left[1 - 3 \left(\frac{2}{3} \right)^k + 3 \left(\frac{1}{3} \right)^k \right].$$

Split this into three geometric series. First term

$$\sum_{k=1}^{\infty} 2^{-k} = 1.$$

Second term with the multiple inside the sum

$$\sum_{k=1}^{\infty} 2^{-k} \cdot 3 \cdot \left(\frac{2}{3} \right)^k = 3 \sum_{k=1}^{\infty} \left(\frac{1}{3} \right)^k = 3 \cdot \frac{1/3}{1 - 1/3} = 3 \cdot \frac{1}{2}.$$

Third term with the multiple inside the sum

$$\sum_{k=1}^{\infty} 2^{-k} \cdot 3 \cdot \left(\frac{1}{3}\right)^k = 3 \sum_{k=1}^{\infty} \left(\frac{1}{6}\right)^k = 3 \cdot \frac{1/6}{1 - 1/6} = 3 \cdot \frac{1}{5}.$$

Combine the three results

$$1 - \frac{3}{2} + \frac{3}{5} = \frac{1}{10}.$$

Thus the probability that no bucket is empty after the random process is $\frac{1}{10}$

Answer:

309. Painted Cube II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The number of faces that are red and they belong to a corner cube is $8 \times 3 = 24$. The number of faces that are red is $6 \times 3^2 = 54$. Hence, using conditional probability, the required answer is

$$\frac{24}{54} = \frac{4}{9}$$

Answer:

310. Random Point On Disc

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For any point (p, q) in the disc, $(-p, q)$ shares the same probability distribution since the point selection is uniform over the disc. Additionally, the disc is symmetric with respect to reflections along the axes. The transformation $(p, q) \rightarrow (-p, q)$ reflects the unit disc across the q -axis. Consequently, the joint distribution of (P, Q) is identical to that of $(-P, Q)$. This implies that:

$$E[PQ] = E[(-P)Q] = -E[PQ]$$

leading to:

$$E[PQ] = 0$$

Answer:

311. 14 Marbles

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If a red marble occupies an odd-numbered slot, the second red marble must be in an even-numbered slot. We can place the 14 marbles in $7! \cdot 7!$ ways to ensure that there is an even gap between pairs of the same color. The total number of ways to arrange the marbles is $\frac{14!}{2^7}$. Thus, the probability is

$$\frac{7! \cdot 7! \cdot 2^7}{14!} = \frac{16}{429}$$

Answer:

312. Heaven Or Hell

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $P(m, n)$ be the probability that she ascends to heaven from point (m, n) . Then $P(6m, 6n) = 1$ and $P(6m + 3, 6n + 3) = 0$ for all integers $m, n \in \mathbb{Z}$. At all other points,

$$4P(m, n) = P(m - 1, n) + P(m + 1, n) + P(m, n - 1) + P(m, n + 1) \quad (1)$$

This gives an infinite system of equations. However, we can apply symmetry arguments to cut down the number of variables to something more manageable. We have $P(m, n) = P(m + 6a, n + 6b)$ for $a, b \in \mathbb{Z}$, and $P(m, n) = P(n, m)$, and $P(m, n) = P(-m, n)$, and $P(m, n) = 1 - P(m + 3, n + 3)$ (since any path from the latter point to heaven corresponds with a path from the former point to hell, and vice versa). Thus, for example, we have

$$P(1, 2) = P(-1, -2) = 1 - P(2, 1) = 1 - P(1, 2),$$

so $P(1, 2) = \frac{1}{2}$. Applying Equation (1) to points $(1, 1)$, $(0, 1)$, and $(0, 2)$, and using the above symmetries, we get the equations:

$$4P(1, 1) = 2P(0, 1) + 1,$$

$$4P(0, 1) = P(0, 2) + 2P(1, 1) + 1,$$

$$4P(0, 2) = P(0, 1) + \frac{3}{2}.$$

Solving yields $P(1, 1) = \frac{13}{22}$.

Answer: 0.59091

313. Largest Area Expectation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $P(i)$ represent the probability that the largest red square has area i , where $i \in \{0, 1, 4, 9\}$. The expected value is

$$E(A) = \sum_{i \in \{0, 1, 4, 9\}} P(i)i.$$

We begin with the trivial values:

$$P(0) = \left(\frac{1}{2}\right)^9 = \frac{1}{512} \text{ (all squares blue)}, \quad P(9) = \left(\frac{1}{2}\right)^9 = \frac{1}{512} \text{ (all squares red)}.$$

To compute $P(4)$, let E be the event "at least one all-red 2×2 block." There are four overlapping 2×2 blocks in the 3×3 grid:

- a_1 = event that the top-left 2×2 block is all red
- a_2 = event that the top-right block is all red
- a_3 = event that the bottom-left block is all red
- a_4 = event that the bottom-right block is all red

By inclusion-exclusion:

$$P(E) = \sum_i P(a_i) - \sum_{i < j} P(a_i \cap a_j) + \sum_{i < j < k} P(a_i \cap a_j \cap a_k) - P(a_1 \cap a_2 \cap a_3 \cap a_4).$$

Each single block has probability $\left(\frac{1}{2}\right)^4 = \frac{1}{16}$, so

$$\sum_i P(a_i) = 4 \cdot \frac{1}{16} = \frac{1}{4}.$$

There are 4 edge-adjacent pairs covering 6 cells each, and 2 diagonal-adjacent pairs covering 7 cells each, so

$$\sum_{i < j} P(a_i \cap a_j) = 4 \cdot \frac{1}{64} + 2 \cdot \frac{1}{128} = \frac{1}{16} + \frac{1}{64} = \frac{5}{64}.$$

Each triple covers 8 cells, so

$$\sum_{i < j < k} P(a_i \cap a_j \cap a_k) = 4 \cdot \frac{1}{256} = \frac{1}{64}.$$

All four cover 9 cells, so

$$P(a_1 \cap a_2 \cap a_3 \cap a_4) = \left(\frac{1}{2}\right)^9 = \frac{1}{512}.$$

Hence

$$P(E) = \frac{1}{4} - \frac{5}{64} + \frac{1}{64} - \frac{1}{512} = \frac{95}{512}.$$

Since $P(9) = \frac{1}{512}$,

$$P(4) = P(E) - P(9) = \frac{94}{512}.$$

Then

$$P(1) = 1 - (P(0) + P(4) + P(9)) = 1 - \left(\frac{1}{512} + \frac{94}{512} + \frac{1}{512}\right) = \frac{416}{512}.$$

Finally,

$$E(A) = \frac{416}{512} \cdot 1 + \frac{94}{512} \cdot 4 + \frac{1}{512} \cdot 9 = \frac{801}{512}.$$

Answer: 1.56445

314. No Fives

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability in question is equivalent to the probability of rolling a 6 before a 5 when rolling a die infinitely many times. The complement of this event is rolling a 5 before a 6. By symmetry, both of them are equal. Let the required probability be p . Hence,

$$p + p = 1 \Rightarrow p = \frac{1}{2}.$$

So, the required probability is $\frac{1}{2}$.

Answer: 0.50000

315. Maximum Rolled Value

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To find the probability that the maximum value is precisely 4, we need to consider cases where the maximum value is at most 4 but not at most 3. Hence:

$$P(\max = 4) = P(\max \leq 4) - P(\max \leq 3)$$

This can be computed as:

$$P(\max = 4) = \left(\frac{4}{6} \times \frac{4}{6} \times \frac{4}{6}\right) - \left(\frac{3}{6} \times \frac{3}{6} \times \frac{3}{6}\right) = \frac{37}{216}$$

Answer: 0.17130

316. Random Knight Movement

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Solution 1:

Let the knight's horizontal position after 100 moves be X and its vertical position be Y . Then the square of its distance from the origin is $X^2 + Y^2$. So we just need to find $\mathbb{E}[X^2]$ and $\mathbb{E}[Y^2]$.

We first look at the x -coordinate. Let the change in x from the first move be a_1 , from the second move be a_2 , and so on up to a_{100} . Then

$$X = a_1 + a_2 + \cdots + a_{100}.$$

For a single move, the knight's x -change is $-2, -1, 1$, or 2 , each with probability $\frac{1}{4}$. So

$$\mathbb{E}[a_i] = \frac{1}{4}(-2) + \frac{1}{4}(-1) + \frac{1}{4}(1) + \frac{1}{4}(2) = 0$$

and

$$\mathbb{E}[a_i^2] = \frac{1}{4}(2^2) + \frac{1}{4}(1^2) + \frac{1}{4}(1^2) + \frac{1}{4}(2^2) = \frac{5}{2}.$$

Now expand X^2 :

$$X^2 = (a_1 + a_2 + \cdots + a_{100})^2.$$

This gives

$$X^2 = a_1^2 + a_2^2 + \cdots + a_{100}^2 + 2 \sum_{1 \leq i < j \leq 100} a_i a_j.$$

Taking expectations,

$$\mathbb{E}[X^2] = \sum_{i=1}^{100} \mathbb{E}[a_i^2] + 2 \sum_{1 \leq i < j \leq 100} \mathbb{E}[a_i a_j].$$

Since the moves are independent,

$$\mathbb{E}[a_i a_j] = \mathbb{E}[a_i] \mathbb{E}[a_j].$$

But $\mathbb{E}[a_i] = 0$, so every cross term is 0. Therefore only the square terms remain:

$$\mathbb{E}[X^2] = 100 \cdot \frac{5}{2} = 250.$$

The same argument applies to the y -coordinate, so $\mathbb{E}[Y^2] = 250$.

Therefore,

$$\mathbb{E}[X^2 + Y^2] = \mathbb{E}[X^2] + \mathbb{E}[Y^2] = 250 + 250 = 500.$$

So the expected value of the square of the knight's distance from the origin after 100 seconds is 500.

Solution 2:

Suppose the knight is at (a, b) before a turn. Consider the two possible moves $(-2, -1)$ and $(+2, +1)$. We have that

$$\frac{(a-2)^2 + (b-1)^2 + (a+2)^2 + (b+1)^2}{2} = a^2 + b^2 + 5.$$

The six other possible moves can be paired up similarly. Summing all possibilities, the expected value of $a^2 + b^2$ increases by 5 each turn, so after 100 such turns, the answer is

$$100 \times 5 = 500.$$

Answer: 500

317. Game On 1-10[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since each prime number is only divisible by 1 and itself, you can only claim one prime from the board. Therefore, it is optimal to start by choosing 7, the largest prime available. After selecting 7, the dealer claims 1, and the remaining numbers on the board are 2, 3, 4, 5, 6, 8, 9, and 10. You cannot select 2, 3, or 5, as they are primes and no longer have any unclaimed factors. Avoid picking 4 because it prevents you from taking 8 later on. The optimal next move is to take 9, as it is a larger number with only one factor left, which is 3. This reduces the board to 2, 4, 5, 6, 8, and 10. At this point, 6 is the only number with a single remaining factor, so you take it. The dealer now claims 2, leaving the board with 4, 5, 8, and 10. You can now take 8 and 10 in any order, and the dealer will claim 4 and 5. The numbers you claim are 6, 7, 8, 9, and 10, which total a score of 40. The dealer takes 1, 2, 3, 4, and 5, with a total score of 15.

Answer: 40**318. Easy Card Expectation**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To calculate the expected total payout, we use the principle of linearity of expectation, which applies even when the random variables are dependent. Let X_i represent the value of the i -th card drawn. The total payout is:

$$5(X_1 + X_2 + X_3 + X_4)$$

By linearity of expectation, the expected payout becomes:

$$5(E[X_1] + E[X_2] + E[X_3] + E[X_4])$$

The expected value of a single card, $E[X_i]$, is the average of the numbers from 1 to 35, which is:

$$\frac{1 + 35}{2} = 18$$

Thus, $E[X_1] = E[X_2] = E[X_3] = E[X_4] = 18$. Substituting, the expected total payout is:

$$5(18 + 18 + 18 + 18) = 5 \times 72 = 360$$

Answer: 360**319. Bot Move**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

It is always better to choose the number $n + 3$ rather than n as it defeats all the numbers that n would have defeated and the number $n + 2$. Hence, you must pick the numbers from the range $[98, 99, 100]$ with some optimal probability distribution. By symmetry, the opponent would also pick the numbers from the range $[98, 99, 100]$ with the same optimal probability distribution. Also, the expected payoff for each player under optimal play would be 0. Now, suppose that you select the numbers with a non-uniform distribution. Then the opponent will have a strategy that will lead to a positive payoff for them (and hence, a negative payoff for you). For example, suppose you select the number 98 with probability 0.5, 99 with probability 0.4, and 100 with probability 0.1. In this case, the opponent, after looking at your bot's strategy, can decide to always choose the number 99. Then, their expected payoff would be:

$$0.5 \cdot 1 + 0.4 \cdot 0 + 0.1 \cdot (-1) = 0.4$$

Hence, your expected payoff is -0.4 . Therefore, the optimal strategy is to choose the numbers 98, 99, and

100 each with probability $\frac{1}{3}$. So, the required variance is:

$$\frac{1}{3}((98 - 99)^2 + (99 - 99)^2 + (100 - 99)^2) = \frac{1}{3} \cdot 2 = \frac{2}{3}$$

Answer:

320. Rigged Game

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E represent the expected number of flips needed to get two consecutive heads, and E_1 be the expected number of additional flips required given that a head was just flipped. Let p be the probability of flipping heads. We establish the following equations:

$$E = pE_1 + (1 - p)E + 1$$

$$E_1 = (1 - p)E + 1$$

The first equation accounts for the scenarios where either a head or a tail is flipped. If a head is flipped, E_1 flips are needed; if a tail is flipped, E flips are needed, plus one for the current flip. The second equation covers the situation where a head was just flipped: a head requires no additional flips, while a tail necessitates E flips. Solving these equations yields:

$$E = \frac{1}{p} + \frac{1}{p^2}$$

To ensure the player loses on average, we set $E = 30$:

$$\frac{1}{p} + \frac{1}{p^2} = 30$$

Solving for p , the positive root is:

$$p = \frac{1}{5}$$

Answer:

321. Maximising Heads I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First, observe that flipping exactly two coins at a time always preserves the parity of the number of heads. That is, the number of heads stays even or odd depending on what it was initially. For example:

* Flipping two heads turns them into two tails (decreases heads by 2). * Flipping one head and one tail swaps their statuses (net number of heads unchanged). * Flipping two tails turns them into two heads (increases heads by 2).

So whatever the parity of the number of heads after the initial toss, we can only reach configurations with the same parity. But since we can perform the operation any number of times, we can reach any number of heads with the same parity as the starting number.

That means: if we start with 1 or 3 heads (odd), we can reach 3 heads (maximum possible odd value ≤ 4), and if we start with 0, 2, or 4 heads (even), we can reach 4 heads.

So we want the expected value of the maximum number of heads reachable, which depends on the initial number of heads.

Let's compute the probabilities for each possible number of heads after tossing 4 fair coins. Since each coin is independent, the distribution of heads follows a binomial distribution $B(4, \frac{1}{2})$:

* Probability of 0 heads:

$$\binom{4}{0} \cdot \frac{1}{16} = \frac{1}{16}$$

* Probability of 1 head:

$$\binom{4}{1} \cdot \frac{1}{16} = \frac{4}{16}$$

* Probability of 2 heads:

$$\binom{4}{2} \cdot \frac{1}{16} = \frac{6}{16}$$

* Probability of 3 heads:

$$\binom{4}{3} \cdot \frac{1}{16} = \frac{4}{16}$$

* Probability of 4 heads:

$$\binom{4}{4} \cdot \frac{1}{16} = \frac{1}{16}$$

Now, for each case, compute the optimal number of heads reachable:

* If initial heads = 0 $\rightarrow$ max = 4 * If initial heads = 1 $\rightarrow$ max = 3 * If initial heads = 2 $\rightarrow$ max = 4 * If initial heads = 3 $\rightarrow$ max = 3 * If initial heads = 4 $\rightarrow$ max = 4

Now compute the expected number of heads under the optimal strategy:

$$E = \frac{1}{16}(4) + \frac{4}{16}(3) + \frac{6}{16}(4) + \frac{4}{16}(3) + \frac{1}{16}(4) = 3.5$$

So, under the optimal strategy, the expected number of heads you can achieve is 3.5.

Answer: 3.5

322. Distinct Number Count

[[Question](#) $\uparrow$ | [Hint](#) $\uparrow$ | $\uparrow$]

To determine the expected number of distinct points in a set of size n drawn from $\{1, \dots, n\}$, we use the principle of linearity of expectation. Define an indicator variable S_i for each integer i where $S_i = 1$ if integer i is included in the resulting set, and $S_i = 0$ otherwise. Let S denote the total number of distinct points selected. Thus S can be expressed as:

$$S = S_1 + S_2 + \dots + S_n$$

By linearity of expectation, the expected value of S is:

$$E[S] = E[S_1] + E[S_2] + \dots + E[S_n]$$

The expectation $E[S_i]$ is the probability that integer i is included in the selected set. This is given by

$$E[S_i] = 1 - P(i \text{ is not chosen})$$

If each element is drawn independently, the probability that a specific integer i is not chosen in a single draw is $\frac{n-1}{n}$. Therefore, if the draws are made independently, the probability that i is not chosen at all can be given by:

$$\left(\frac{n-1}{n}\right)^n$$

Thus:

$$E[S_i] = 1 - \left(\frac{n-1}{n}\right)^n$$

Summing up for all n integers:

$$E[S] = n \left[1 - \left(\frac{n-1}{n} \right)^n \right]$$

Hence, for $n = 4$, the answer is:

$$E[S] = 4 \left[1 - \left(\frac{3}{4} \right)^4 \right] = 4 \left[1 - \frac{81}{256} \right] = 4 \times \frac{175}{256} = \frac{700}{256} = \frac{175}{64}$$

Answer: 2.73438

323. Non-linearity Of Expectation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let I_{ij} be the indicator that the ball placed on move i goes into box j . If X_j is the number of balls in box j after move 6, then

$$X_j = \sum_{i=j}^6 I_{ij}, \quad S = \sum_{j=1}^6 X_j^2.$$

For each i , the ball on move i is equally likely to go into any of the first i boxes, so for $j \leq i$,

$$\mathbb{E}[I_{ij}] = \frac{1}{i}.$$

Also, indicators from different moves are independent. So for fixed j ,

$$\mathbb{E}[X_j^2] = \mathbb{E} \left[\left(\sum_{i=j}^6 I_{ij} \right)^2 \right] = \sum_{i=j}^6 \mathbb{E}[I_{ij}] + 2 \sum_{j \leq a < b \leq 6} \mathbb{E}[I_{aj}] \mathbb{E}[I_{bj}].$$

This can be rewritten as

$$\mathbb{E}[X_j^2] = \left(\sum_{i=j}^6 \frac{1}{i} \right)^2 + \sum_{i=j}^6 \frac{1}{i} \left(1 - \frac{1}{i} \right).$$

Now sum over $j = 1, 2, \dots, 6$:

$$\mathbb{E}[S] = \sum_{j=1}^6 \left(\sum_{i=j}^6 \frac{1}{i} \right)^2 + \sum_{j=1}^6 \sum_{i=j}^6 \frac{1}{i} \left(1 - \frac{1}{i} \right).$$

The first part is

$$\left(\frac{49}{20} \right)^2 + \left(\frac{29}{20} \right)^2 + \left(\frac{19}{20} \right)^2 + \left(\frac{37}{60} \right)^2 + \left(\frac{11}{30} \right)^2 + \left(\frac{1}{6} \right)^2 = \frac{191}{20}.$$

For the second part, each move i appears in exactly i of the sums over j , so

$$\sum_{j=1}^6 \sum_{i=j}^6 \frac{1}{i} \left(1 - \frac{1}{i} \right) = \sum_{i=1}^6 i \cdot \frac{1}{i} \left(1 - \frac{1}{i} \right) = \sum_{i=1}^6 \left(1 - \frac{1}{i} \right) = \frac{71}{20}.$$

Therefore,

$$\mathbb{E}[S] = \frac{191}{20} + \frac{71}{20} = 13.1.$$

Alternate Solution: Let S_i be the score after the i -th move, with $S_0 = 0$. Note that the final score can be written as the sum of the score increments over the 6 moves, that is, $S_6 = \sum_{i=1}^6 (S_i - S_{i-1})$. Therefore, it suffices to compute the expected increase in score at each move (and then sum them at the end).

When a ball is added to a box that currently has c balls, the contribution of that box changes from c^2 to $(c+1)^2$. So the score increases by

$$(c+1)^2 - c^2 = 2c + 1.$$

So, on move i , if the chosen box has X_i balls before the move, then

$$S_i - S_{i-1} = 2X_i + 1.$$

Just before the i -th move, let there be c_1 balls in box 1, c_2 balls in box 2, $\dots$, c_i balls in box i . Since the chosen box is uniformly random among these i boxes,

$$\mathbb{E}[X_i] = \frac{1}{i}c_1 + \frac{1}{i}c_2 + \dots + \frac{1}{i}c_i.$$

Each term above represents the probability of selecting a box multiplied by the number of balls in that box. Note that just before move i , the total number of balls is $i-1$, so

$$c_1 + c_2 + \dots + c_i = i-1.$$

Hence

$$\mathbb{E}[X_i] = \frac{i-1}{i}.$$

Therefore,

$$\mathbb{E}[S_i - S_{i-1}] = 2\mathbb{E}[X_i] + 1 = 2 \cdot \frac{i-1}{i} + 1 = 3 - \frac{2}{i}.$$

So by linearity of expectation,

$$\mathbb{E}[S_6] = \sum_{i=1}^6 \mathbb{E}[S_i - S_{i-1}] = \sum_{i=1}^6 \left(3 - \frac{2}{i}\right) = 13.1$$

Answer: 13.10000

324. Noodle Loop

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $E[n]$ be the expected number of loops if you start with n noodles. In the first move, you will take out 2 ends and connect them. If the 2 selected ends belong to the same noodle, then we would end up with 1 loop plus $n-1$ noodles. The probability of this case is $\frac{1}{2n-1}$. If the 2 selected ends belong to different noodles, then we would end up with $n-1$ noodles. The probability of this case is $\frac{2n-2}{2n-1}$. Hence,

$$E[n] = \frac{1}{2n-1}(1 + E[n-1]) + \frac{2n-2}{2n-1}E[n-1]$$

$$\Rightarrow E[n] = \frac{1}{2n-1} + E[n-1]$$

Now, $E[1] = 1$, as if you have one noodle to start with, you will always end up with one loop. Hence,

$$E[n] = 1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1}$$

$$\Rightarrow E[100] = 1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{199} \approx 3.28434$$

Answer: 3.28434

325. Dice Sum Parity

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will use recursion. * If the first roll is a 6, we immediately stop, and the sum is guaranteed to be even with a probability of 1. * If we roll a 2 or 4 on the first roll, the problem remains unchanged (since the rolled value is even). Therefore, if the probability of the sum being even at the start is p , it will remain p . * If we roll a 1, 3, or 5 on the first roll, the problem becomes complementary (since the rolled value is odd). From this point onward, we need the sum to remain odd until a 6 is rolled. Therefore, if the probability of the sum being even at the start is p , it will change to $1 - p$. Thus, we get the equation:

$$p = \frac{1}{6} \cdot 1 + \frac{2}{6} \cdot p + \frac{3}{6} \cdot (1 - p) \implies p = \frac{4}{7}$$

Thus, the probability of the sum being even is $\frac{4}{7}$.

Answer: 0.57143

326. Nearest Diagonal I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let D be the distance from the random point (X, Y) to the line through $(0, 0)$ and $(2, 1)$. The line has equation $x - 2y = 0$, so

$$D = \frac{|X - 2Y|}{\sqrt{5}}.$$

Using the tail integral formula,

$$\mathbb{E}[D] = \int_0^\infty \mathbb{P}(D > t) dt.$$

Since the farthest point in the rectangle is at distance $\frac{2}{\sqrt{5}}$ from the line, this becomes

$$\mathbb{E}[D] = \int_0^{2/\sqrt{5}} \mathbb{P}(D > t) dt.$$

Now fix t , and write $s = t\sqrt{5}$. Then $D > t$ is the same as

$$|X - 2Y| > s.$$

So $\mathbb{P}(D > t)$ is the area of the part of the rectangle lying outside the strip

$$|x - 2y| \leq s,$$

divided by the total area 2. A slice calculation shows that, for $0 \leq s \leq 2$, the area of this strip is

$$A(s) = \frac{s(4 - s)}{2}.$$

Therefore,

$$\mathbb{P}(D > t) = 1 - \frac{A(s)}{2} = 1 - \frac{s(4 - s)}{4} = 1 - s + \frac{s^2}{4}.$$

Substituting $s = t\sqrt{5}$,

$$\mathbb{P}(D > t) = 1 - t\sqrt{5} + \frac{5t^2}{4}.$$

Hence

$$\mathbb{E}[D] = \int_0^{2/\sqrt{5}} \left(1 - t\sqrt{5} + \frac{5t^2}{4} \right) dt = \frac{2}{3\sqrt{5}}$$

Answer: 0.29814

327. No Adjacent Standing Pair[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that there are $2^8 = 256$ cases in total, so it suffices to find the number of satisfying cases. Define h_n as the number of linear sequences of h 's and t 's of length n where no h 's are adjacent and the sequence ends with an h . Similarly, define t_n as the number of linear sequences of h 's and t 's of length n where no h 's are adjacent and the sequence ends with a t (ignore the fact that they are in a circle for now). In order for a sequence to end in h , the previous term has to be a t ; a sequence with a length of n ending in h is equivalent to adding h to the end of a sequence of length $n - 1$ that ends with t . This means $h_n = t_{n-1}$. In order for a sequence to end in t , the previous term can be h or t , so $t_n = t_{n-1} + h_{n-1}$. Now, we have to take account of the circle. Notice that t_n does not depend on if the people are in a circle or not, but h_n does. To guarantee that some sequence that ends with h does not start with h , we have to add a t in the beginning of the sequence. Since there are $n - 1$ terms left, there are h_{n-1} sequences which end with h that satisfy the circle condition. The length of our sequence is 8, so we need to find $t_8 + h_7$. Manipulating our recursions, we can find that $t_n = t_{n-1} + t_{n-2}$. Now, we proceed to find our terms.

$$t_1 = 1$$

$$t_2 = 2$$

$$t_3 = 3$$

$$t_4 = 5$$

$$t_5 = 8$$

$$t_6 = 13$$

$$t_7 = 21$$

$$t_8 = 34$$

Since $h_7 = t_6$ due to the fact that $h_n = t_{n-1}$, our answer is

$$\frac{34 + 13}{256} = \frac{47}{256}$$

Answer: 0.18359

328. Odd Heads III[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will use a recursive approach. Define p_n as the probability that after n coin tosses, the total number of heads is odd. The recursion is set up as follows:

$$p_{n+1} = p_n \cdot \frac{2}{3} + (1 - p_n) \cdot \frac{1}{3}$$

The term $p_n \cdot \frac{2}{3}$ represents the situation where we already have an odd number of heads and the $(n + 1)$ -th coin is tails, so the parity remains odd. The term $(1 - p_n) \cdot \frac{1}{3}$ accounts for the case where we had an even number of heads and the $(n + 1)$ -th coin is heads, flipping the parity to odd. Additionally, if we were to flip only a single coin, the probability of getting an odd total number of heads would be $\frac{1}{3}$. This occurs only when the single coin shows heads. Therefore, we have $p_1 = \frac{1}{3}$. The above recursion is linear, so we can solve it explicitly:

$$p_n = \frac{1}{2} - \frac{1}{2} \left(\frac{1}{3} \right)^n$$

Therefore, after 10 tosses, the probability of getting an odd number of heads is:

$$p_{10} = \frac{1}{2} \left(1 - \left(\frac{1}{3} \right)^{10} \right) \approx 0.4999915328$$

Answer:

329. Handshakes On A Chessboard

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Focus on the edges that split squares rather than the squares themselves. Call a square an n -square if the person on that square has n neighbors. There are 8 edges that split a 3-square and a 2-square. The probability that there is a handshake at this edge is $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$. Similarly, there are 20 edges that separate a 3-square and a 3-square, 24 edges that separate a 3-square and a 4-square, and 60 edges that separate a 4-square and a 4-square. Hence, by linearity of expectation, the expected number of handshakes is:

$$8 \times \frac{1}{6} + 20 \times \frac{1}{9} + 24 \times \frac{1}{12} + 60 \times \frac{1}{16} = \frac{335}{36}$$

Answer:

330. Erase

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

You cross out the 1234 from the first set, because you want it to start with a 5 to be as large as possible. You cross out the 1234 from the 2nd set, so your second digit will be as large as possible. Then you cross out the 12 from the 3rd set, so your 3rd digit will be as large as possible. That's 10 digits, leaving you with 553451234512345.

Answer:

331. Permutation Blocks

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let σ be a permutation on $\{1, \dots, n\}$. Call $m \in \{1, \dots, n\}$ a breakpoint of σ if $\{\sigma(1), \dots, \sigma(m)\} = \{1, \dots, m\}$. Notice that the maximum partition is into k blocks, where k is the number of breakpoints: if our breakpoints are $m_1, \dots, m_k$, then we take $\{1, \dots, m_1\}, \{m_1 + 1, \dots, m_2\}, \dots, \{m_{k-1} + 1, \dots, m_k\}$ as our contiguous blocks. Now we just want to find

$$E[k] = E[X_1 + \dots + X_n],$$

where $X_i = 1$ if i is a breakpoint, and $X_i = 0$ otherwise. We use linearity of expectation and notice that

$$E[X_i] = \frac{i!(n-i)!}{n!},$$

since this is the probability that the first i numbers are just $\{1, \dots, i\}$ in some order. Thus,

$$E[k] = \sum_{i=1}^n \frac{i!(n-i)!}{n!} = \sum_{i=1}^n \binom{n}{i}^{-1}.$$

We can compute for $n = 7$ that the answer is $\frac{151}{105}$.

Answer:

332. Count Paths III[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To find the number of valid paths from $(0, 0)$ to $(5, 3)$ that do not cross the line $y = x$, we begin by calculating the total number of paths without any restrictions. This is given by $\binom{8}{3} = 56$, where we take 3 upward moves and 5 rightward moves. Next, we eliminate the paths that cross the line $y = x$. If a path crosses this line, it must first intersect the diagonal from $(-1, 0)$ to $(2, 3)$. For each path that crosses, we can reflect the portion of the path before crossing across the diagonal. Paths that cross correspond to paths from $(-1, 1)$ to $(5, 3)$, and the number of these paths is $\binom{8}{2} = 28$. Therefore, the number of valid paths that do not cross $y = x$ is $56 - 28 = 28$.

Answer: 28**333. Tigers And The Sheep**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

With 1 tiger, it will eat the sheep since there's no threat to itself. With 2 tigers, neither will eat the sheep, as each would risk being eaten by the other. With 3 tigers, each realizes that if one eats the sheep, the remaining 2 tigers would ensure the sheep's survival, so all will try to eat the sheep, resulting in its demise. With 4 tigers, the sheep survives as the tigers anticipate the outcome from the case with 3 tigers. Thus, if there is an even number of tigers, the sheep survives. If there is an odd number, the sheep does not survive. Hence, with 100 tigers, the sheep will survive.

Answer: 1**334. Unique Birthdays Expectation**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Introduce indicator variables $x_1, x_2, \dots, x_N$, where x_i represents the event that the i -th person has a unique birthday among N people. The total number of unique birthdays is:

$$p(N) = x_1 + x_2 + \dots + x_N$$

The expected value of $p(N)$ is:

$$E[p(N)] = E[x_1] + E[x_2] + \dots + E[x_N]$$

By symmetry, each $E[x_i]$ is equal, so:

$$E[p(N)] = N \cdot E[x_1]$$

To compute $E[x_1]$, note that the first person has a fixed birthday, and the remaining $N - 1$ people must have distinct birthdays among the other 364 days. The probability of this is:

$$E[x_1] = \left(\frac{364}{365}\right)^{N-1}$$

Thus:

$$E[p(N)] = N \cdot \left(\frac{364}{365}\right)^{N-1}$$

Since $E[p(N)]$ increases initially and then decreases, the two values of N that yield the same maximum are consecutive. Let these be n^* and $n^* + 1$. Equating their expected values:

$$n^* \cdot \left(\frac{364}{365}\right)^{n^*-1} = (n^* + 1) \cdot \left(\frac{364}{365}\right)^{n^*}$$

Solving this equation gives $n^* = 364$. Hence, the required values are 364 and 365.

Answer:

335. Simpson's Paradox

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Yes, it is possible that team Y ends up with a higher pass-completion percentage. Consider the following example: First half:

- Team X: completes 1 pass out of 1 attempted $\rightarrow 100\%$
- Team Y: completes 2 passes out of 3 attempted $\rightarrow \frac{2}{3} = 66\frac{2}{3}\%$

Second half:

- Team X: completes 1 pass out of 4 attempted $\rightarrow \frac{1}{4} = 25\%$
- Team Y: completes 0 passes out of 1 attempted $\rightarrow 0\%$

In each half, Team X's completion rate exceeds Team Y's (100% vs $66\frac{2}{3}\%$ in the first; 25% vs 0% in the second). Yet overall:

- Team X: $1 + 1 = 2$ successes out of $1 + 4 = 5$ attempts $\Rightarrow \frac{2}{5} = 40\%$
- Team Y: $2 + 0 = 2$ successes out of $3 + 1 = 4$ attempts $\Rightarrow \frac{2}{4} = 50\%$

So Team Y ends up with the higher overall pass-completion percentage ($\frac{1}{2}$ vs $\frac{2}{5}$), even though Team X was better in both halves. This is called Simpson's paradox.

Answer:

336. 2014 Stones

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider the binary representation of the current number of stones. Let's call the 1s in the binary representation at an odd position (counting from the right, corresponding to an odd power of two) as blue and the 1s at an even position red. Alice has a winning strategy. If on her turn there are no red 1s and only one blue 1, then end the game by taking the blue 1. Otherwise, if there are some red 1s and no blue 1s, then choose to take a power of two corresponding to a 0, thereby creating at least one blue 1. Otherwise, take any even power of two.

Answer:

337. Chess Tournament IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

This is the same as putting the teams in a bracket-style tournament at random. The probability that the teams ranked 1 and 2 are not in the same quarter of the draw as the team ranked 3 is the relevant probability, and it is $\frac{24 \cdot 23}{31 \cdot 30} = \frac{92}{155}$.

Answer:

338. Interesting Summation I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To ensure $\frac{1}{k}$ has a finite decimal expansion, k needs to be of the form $2^a \cdot 5^b$, with a and b as non-negative integers. Thus, the sum becomes:

$$\sum_{a=0}^{\infty} \sum_{b=0}^{\infty} \left(\frac{1}{2^a \cdot 5^b} \right)^3$$

This can be rewritten as:

$$\sum_{a=0}^{\infty} \frac{1}{8^a} \cdot \sum_{b=0}^{\infty} \frac{1}{125^b}$$

We now evaluate the sums:

$$\frac{1}{1 - \frac{1}{8}} \cdot \frac{1}{1 - \frac{1}{125}} = \frac{8}{7} \cdot \frac{125}{124}$$

Simplifying gives the result:

$$\frac{1000}{868} = \frac{250}{217}$$

Answer:

339. Chess Tournament I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In the first round, there are 128 positions. Assume that in the first round, player at position 1 plays player at position 2, player at position 3 plays player at position 4, and so on. In the second round, the winner of the match between position 1 and position 2 will play the winner of the match between position 3 and position 4 - and so on for subsequent rounds.

Without loss of generality, we can assume that the highest-rated player is occupying position 1. The second-highest-rated player can play the highest-rated player in the final only if he occupies one of the positions in positions 65 - 128 (64 favorable positions), or else he will be eliminated by the highest-rated player before the final.

Hence, the required probability is:

$$\frac{64}{127}$$

since there are 127 possible positions and 64 favorable positions.

Answer:

340. How Many 5s

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let n be the roll number on which you first get a 6, and let t be the total number of 5s rolled before that. Using the Law of Total Expectation, we have:

$$E[t] = E[E[t|n]]$$

Before getting a 6, we have $n - 1$ rolls where no 6 appears. In these rolls, the numbers 1-5 can occur with equal probability. Thus:

$$E[t|n] = \frac{n-1}{5}$$

Since n follows a geometric distribution with parameter $\frac{1}{6}$:

$$E[n] = 6$$

Therefore:

$$E[t] = E\left[\frac{n-1}{5}\right] = \frac{E[n]-1}{5} = \frac{6-1}{5} = 1$$

Answer:

341. Sphere Variance

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

$$\text{Var}(z) = E[z^2] - (E[z])^2$$

Given the unit sphere with the equation $x^2 + y^2 + z^2 = 1$, and considering the symmetry of the problem, the z^2 term will contribute equally with x^2 and y^2 . Hence:

$$E[z^2] = \frac{1}{3}$$

Since the sphere is symmetric, the expected value of z is zero:

$$E[z] = 0$$

Therefore:

$$\text{Var}(z) = E[z^2] - (E[z])^2 = \frac{1}{3} - 0^2 = \frac{1}{3}$$

Answer:

342. B Before O I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In the word BOOLAHUBBOO, there are 3 B's and 4 O's. To choose positions for these 7 letters, there are $\binom{11}{7}$ ways. Only one specific arrangement places all the B's before the O's (BBB followed by OOOO). The remaining 4 letters (L, A, H, U) can be arranged in the remaining slots in $4!$ different ways. Thus, the total number of anagrams meeting the condition is $4! \times \binom{11}{7} = 7920$.

Answer:

343. Count Regions II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

n circles will divide the plane into a maximum number of pieces if every two of them intersect (i.e., no two are tangent and none lies entirely within or outside another), and no three of them are concurrent. (It can be proven that such sets of circles always exist for any n .) By reasoning similarly to Count Regions I, we observe that the $(k+1)$ st circle increases the number of regions by $2k$, since it intersects each of the first k circles in two points. These $2k$ points divide the $(k+1)$ st circle into $2k$ arcs. Each of these arcs splits an existing region into two. Since one circle divides the plane into two parts, the total number of regions after drawing the n th circle is:

$$\begin{aligned} & 2 + 2 + 4 + 6 + 8 + \cdots + 2(n-1) \\ &= 2 + 2(1 + 2 + 3 + \cdots + (n-1)) \\ &= 2 + 2 \cdot \frac{(n-1)n}{2} = 2 + n(n-1) \end{aligned}$$

$$= n^2 - n + 2$$

Answer: 92.00000

344. Square Fitting I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let S denote the side length of our empty, centered square. Its area is S^2 . A point (X, Y) lies outside that square precisely when

$$\max(|X|, |Y|) > \frac{S}{2},$$

because the square runs from $-\frac{S}{2}$ to $+\frac{S}{2}$ in both coordinates. Define the random variable

$$M = \max(|X|, |Y|),$$

for a single uniform point in $[-1, 1]^2$. Since $|X|$ and $|Y|$ are independent on $[0, 1]$,

$$\Pr(M \leq m) = \Pr(|X| \leq m)\Pr(|Y| \leq m) = m^2, \quad 0 \leq m \leq 1.$$

Thus each point lies outside the centered square of side S with probability

$$1 - \Pr\left(M \leq \frac{S}{2}\right) = 1 - \left(\frac{S}{2}\right)^2$$

With nine independently chosen points, the chance they all lie outside is

$$\Pr(S > s) = \left[1 - \left(\frac{s}{2}\right)^2\right]^9, \quad 0 \leq s \leq 2$$

Differentiating with respect to s yields the density of S :

$$f_S(s) = \frac{d}{ds} \Pr(S \leq s) = 9 \cdot 2 \cdot \frac{s}{4} \left[1 - \left(\frac{s}{2}\right)^2\right]^8 = 9 \frac{s^2}{4} \left(1 - \frac{s^2}{4}\right)^8, \quad 0 \leq s \leq 2$$

The expected area is

$$\mathbb{E}[S^2] = \int_0^2 s^2 f_S(s) ds = \int_0^2 s^2 \left(9 \frac{s^2}{4} \left(1 - \frac{s^2}{4}\right)^8\right) ds = \frac{2}{5},$$

a standard Beta-integral result. Therefore the expected area of the largest empty centered square is

$$\frac{2}{5}.$$

Answer: 0.40000

345. Ant On An Icosahedron

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let A be the starting vertex. From A , there are 5 vertices that are distance 1 away, 5 vertices that are distance 2 away, and 1 vertex that is distance 3 away. (By “distance,” we mean the fewest number of edges on a path.) After one step of the random walk, the ant will be distance 1 away from A with probability 1. After two steps, the ant will be distance 0 away (from A) with probability $\frac{1}{5}$, will be distance 1 away with probability $\frac{2}{5}$, and will be distance 2 away with probability $\frac{2}{5}$. After three steps, the ant will be distance 0 away with probability $\left(\frac{2}{5}\right)\left(\frac{1}{5}\right) = \frac{2}{25}$, will be distance 1 away with probability $\left(\frac{1}{5}\right)(1) + \left(\frac{2}{5}\right)\left(\frac{2}{5}\right) + \left(\frac{2}{5}\right)\left(\frac{2}{5}\right) = \frac{13}{25}$,

will be distance 2 away with probability $\left(\frac{2}{5}\right)\left(\frac{2}{5}\right) + \left(\frac{2}{5}\right)\left(\frac{2}{5}\right) = \frac{8}{25}$, and will be distance 3 away with probability $\left(\frac{2}{5}\right)\left(\frac{1}{5}\right) = \frac{2}{25}$.

We could keep going like this, but let's take a shortcut. After three steps, the ant will be at some vertex, call it B . To end at A after six steps, the ant must go from B to A in the final three steps. We can use our work for the first three steps to figure out the probability of this event. The probability of going from B to A in the final three steps is

$$\left(\frac{2}{25}\right)^2 + \frac{1}{5}\left(\frac{13}{25}\right)^2 + \frac{1}{5}\left(\frac{8}{25}\right)^2 + \left(\frac{2}{25}\right)^2 = \frac{4}{625} + \frac{169}{3125} + \frac{64}{3125} + \frac{4}{625} = \frac{273}{3125}.$$

So the probability of returning to A after six steps is $\frac{273}{3125}$.

Answer: 0.08736

346. Count 9s

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In the interval $[1, 89]$, the digit 9 appears 10 times, specifically in $9, 19, 29, \dots, 89$. In the range $[90, 99]$, 9 appears 11 times. Therefore, 9 appears 20 times in total within $[1, 99]$. Now, consider the range $[1, 1000]$. For the ones and tens places, 9 occurs 20 times in each $[1, 99]$ block as the hundreds place varies from 0 to 9. Hence, 9 appears $10 \times 20 = 200$ times in the ones and tens places. Additionally, 9 appears 100 times in the hundreds position across the interval $[900, 999]$. Thus, the total number of 9s is $200 + 100 = 300$.

Answer: 300

347. Grid Travel I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Observe that Johnny needs to pass through the points $(0, 0), (1, 0), (2, 0), \dots, (5, 0)$ in that order, and he needs to pass through $(0, 1), (1, 1), (2, 1), \dots, (5, 1)$ in that order, or else he will intersect his own path. Thus, the problem is equivalent to interlacing these two sequences together, so that the first term is $(0, 0)$ and the final term is $(5, 1)$. To do this, we need to select 5 positions out of 10 to place the points with x -coordinate 0. Hence, the number of ways is:

$$\binom{10}{5} = 252$$

Answer: 252

348. Jumping Frog III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let d_n denote the number of distinct paths for a frog to reach point n . To determine this, consider the frog's last jump. If the frog's final jump was 1 point, it must have been at point $n - 1$ before this jump. The number of ways to get to point $n - 1$ is d_{n-1} . Similarly, if the last jump was 2 points, the frog must have been at point $n - 2$. The number of ways to reach point $n - 2$ is d_{n-2} . Therefore, the recurrence relation is:

$$d_n = d_{n-1} + d_{n-2}$$

The initial conditions are $d_1 = 1$ and $d_2 = 2$. This recurrence relation corresponds to the Fibonacci sequence shifted by one index. Thus,

$$d_n = F_{n+1}$$

where F_{n+1} is the $(n+1)$ -th Fibonacci number. Specifically, for point 10:

$$d_{10} = F_{11} = 89$$

Answer: 89

349. Seven Colors I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let C be the event that among the first eight positions there are exactly six more red balls than yellow balls. Under C only two patterns for the first eight are possible: either there are six red and zero yellow among the first eight (so the other two of the first eight come from the 100 balls that are neither red nor yellow), or there are seven red and one yellow among the first eight (so none of the first eight are from those 100). Count the number of 8-ball selections that realize each pattern: for the first pattern the number of choices is $\binom{20}{6}\binom{100}{2}$ and for the second it is $\binom{20}{7}\binom{20}{1}$. The probability of the first pattern given C is therefore the number of choices for that pattern divided by the total number of choices consistent with C , namely

$$\Pr(\text{first pattern} \mid C) = \frac{\binom{20}{6}\binom{100}{2}}{\binom{20}{6}\binom{100}{2} + \binom{20}{7}\binom{20}{1}} = \frac{495}{499}$$

and the second pattern thus has probability $\frac{4}{499}$.

Now compute the conditional probability that position 71 is violet in each pattern and use the law of total probability. If the second pattern occurs (seven red and one yellow among the first eight) then all twenty violets remain among the 132 remaining balls, so $\Pr(\text{violet} \mid \text{second pattern}) = \frac{20}{132} = \frac{5}{33}$. If the first pattern occurs (six red, zero yellow, and two drawn from the 100 other balls) then the two “other” draws contain, on average, $2 \cdot \frac{20}{100} = \frac{2}{5}$ violets, so the expected number of violets remaining is $20 - \frac{2}{5} = \frac{98}{5}$, hence

$$\Pr(\text{violet} \mid \text{first pattern}) = \frac{\frac{98}{5}}{132} = \frac{49}{330}$$

Using the law of total probability,

$$\Pr(71\text{st is violet} \mid C) = \frac{495}{499} \cdot \frac{49}{330} + \frac{4}{499} \cdot \frac{5}{33} = \frac{4891}{33934}$$

Answer: 0.14413

350. Triangle Angle I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since $\tan(\theta) = \frac{B}{A}$, and for $\theta > \frac{\pi}{4}$, we know $\tan(\theta) > 1$, meaning $\frac{B}{A} > 1$, or $A < B$. Thus, we seek $P[A < B]$, which equals $\frac{1}{2}$ since A and B are independent and identically distributed (IID), making them symmetric.

Answer: 0.50000

351. Ant On A Cube III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the vertices of the cube be labeled as follows:

- $A = (0, 0, 0)$ — starting point
- $B = (1, 0, 0)$
- $C = (0, 1, 0)$

- $D = (0, 0, 1)$
- $E = (1, 1, 0)$ — trap
- $F = (1, 0, 1)$
- $G = (0, 1, 1)$
- $H = (1, 1, 1)$ — cheese

At first glance, there are 8 states corresponding to the 8 vertices. However, by symmetry, many of these states behave similarly with respect to reaching the cheese or falling into the trap. In fact, vertices B and C are equivalent, and vertices F and G are equivalent. This symmetry allows us to reduce the number of states and simplify the analysis.

Define $P(v)$ as the probability of reaching H (cheese) before E (trap) when starting from vertex v . We have the boundary conditions:

$$P(E) = 0 \quad (\text{trap reached}), \quad P(H) = 1 \quad (\text{cheese reached})$$

By considering the possible moves from each vertex and using symmetry, we can group the vertices into equivalent states. For example, define:

$$x = P(A), \quad y = P(B) = P(C), \quad z = P(D), \quad w = P(F) = P(G)$$

Using this notation, we obtain the following system of equations:

$$\begin{aligned} x &= \frac{2y + z}{3} \\ y &= \frac{x + w}{3} \\ z &= \frac{x + 2w}{3} \\ w &= \frac{y + z + 1}{3} \end{aligned}$$

Solving this system yields:

$$x = \frac{3}{7}$$

Thus, the probability that the mouse reaches the cheese at $H = (1, 1, 1)$ before encountering the trap at $E = (1, 1, 0)$, starting from $A = (0, 0, 0)$, is $\frac{3}{7}$.

Answer: 0.42857

352. Odd Sum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let n : We are choosing 4 integers from 1 to 9. The number of ways of doing so is

$$\binom{9}{4} = \frac{9 \cdot 8 \cdot 7 \cdot 6}{24} = 9 \cdot 7 \cdot 2 = 126.$$

To make the sum of the 4 integers odd, we will choose either 3 odd and 1 even (case 1) or 1 odd and 3 even (case 2). The number of ways to satisfy case 1 is

$$\binom{5}{3} \binom{4}{1} = 10 \cdot 4 = 40.$$

The number of ways to satisfy case 2 is

$$\binom{5}{1} \binom{4}{3} = 5 \cdot 4 = 20.$$

So the total number of successful choices is $40 + 20 = 60$. Hence, the probability of success is

$$\frac{60}{126} = \frac{30}{63} = \frac{10}{21}.$$

Answer: 0.47619

353. Coin Equalizer

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label a head by $+1$ and a tail by -1 . Starting at 0, a return to 0 at time 10 means there are exactly five heads and five tails overall. To make that return the first time at step 10 we must avoid having equal counts at any earlier even time, so we need the number of length-10 sequences with five heads and five tails whose partial sums are nonzero for times $1, 2, \dots, 9$ and equal to zero at time 10. It suffices to count sequences that start with a head, because by symmetry the family of sequences that start with a tail gives the same contribution, and then double the count. This can be done using the reflection principle. First note that the total number of length-10 sequences that start with a head and contain five heads in total is

$$\binom{9}{4} = 126$$

because we must place the remaining four heads among the last nine positions. Among these 126 sequences we call a sequence "bad" if it returns to zero at some even time before 10, and "good" if it does not. To count the bad ones use the reflection principle: for any sequence whose first return to zero occurs at time $2k$ with $1 \leq k \leq 4$, reflect the prefix up to and including that first return (swap heads and tails on that prefix). This reflection produces a distinct sequence with the first symbol a tail and identifies the bad sequences among a fixed family. Summing over the possible first-return times yields a total of 112 bad sequences among those 126 that start with a head. So, we have 14 good sequences. Doubling to include sequences that start with a tail gives the total number of length-10 sequences whose first return to zero is at step 10 as

$$2 \cdot 14 = 28$$

Each of these 28 sequences has probability $p^5(1-p)^5$ because every such sequence contains exactly five heads and five tails. Therefore the desired probability is

$$28p^5(1-p)^5$$

Substituting $p = \frac{1}{3}$ gives

$$28 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^5 = \frac{28 \cdot 2^5}{3^{10}} = \frac{896}{59049} \approx 0.015174$$

So the probability that the first time heads and tails are equal is exactly on the 10th flip is

$$\frac{896}{59049} \approx 0.01517$$

Answer: 0.01517

354. Combining Turtles

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Fix a particular turtle. The probability it does not get combined with another turtle when there are n total turtles is $1 - \frac{2}{n}$. Therefore, the probability it is not combined in 2015 seconds is

$$\prod_{k=0}^{2014} \left(1 - \frac{2}{2017-k}\right) = \prod_{k=0}^{2014} \frac{2015-k}{2017-k} = \frac{2}{(2017)(2016)}$$

Since it is impossible for more than one turtle to never be combined, the desired probability is

$$2017 \left(\frac{2}{(2017)(2016)} \right) = \frac{1}{1008}$$

Answer:

355. Roll Die Till 65

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E be the expected number of rolls to get the sequence "6 followed by 5". Let E_6 be the expected number of additional rolls to get the sequence "6 followed by 5" if we start with a rolled 6. Hence,

$$E_6 = \frac{1}{6}(E_6 + 1) + \frac{4}{6}(E + 1) + \frac{1}{6}(1)$$

$$E = \frac{1}{6}(E_6 + 1) + \frac{5}{6}(E + 1)$$

Solving the above system of equations, we get $E = 36$.

Answer:

356. Assigning Numbers To Dice

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We begin by fixing the number 1 on the top face. There are 5 possible numbers that can be assigned to the bottom face. For the remaining four numbers, there are $4!$ ways to arrange them around the sides. However, since rotations of the dice produce identical arrangements, we need to account for this by dividing by 4. For instance, if the numbers 2, 3, 4, and 5 are arranged in a specific order around the sides, rotating this arrangement gives us four equivalent setups. So, dividing by 4 eliminates these duplicates. Thus, the total number of distinct arrangements is:

$$5 \times \frac{4!}{4} = 5 \times 3! = 30$$

Answer:

357. Diagonal Cut I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The width and height have no common factors. Hence, the diagonal line doesn't go through any intersections, so each time it crosses a horizontal or vertical grid line, it adds 1 to the number of squares touched. And to get from the top left to the bottom right, it must pass through $199 - 1$ vertical grid lines and $991 - 1$ horizontal grid lines. Counting the original square in the top left, this gives a total of $1 + (199 - 1) + (991 - 1) = 1189$ squares cut.

Answer:

358. Meet Again

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Analyze the relative motion of the hands. The minute hand moves at 360 degrees per hour, while the hour hand moves at 30 degrees per hour. The difference in their speeds is therefore 330 degrees per hour, meaning the minute hand gains 330 degrees on the hour hand in one hour. Since a full circle is 360 degrees, it takes the minute hand:

$$\frac{360}{330} \text{ hours, or } \frac{12}{11} \text{ hours,}$$

to catch up to the hour hand. Thus, the hands will meet again exactly $\frac{12}{11}$ hours after 12:00 PM.

Answer: 1.09091

359. Non-empty Buckets

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let I_i be an indicator variable that equals 1 if bucket i contains at least one ball, and 0 otherwise. The total number of buckets with at least one ball is given by $\sum_{i=1}^N I_i$. To find the expected number of buckets with at least one ball, calculate $E\left(\sum_{i=1}^N I_i\right)$. By linearity of expectation, this becomes $\sum_{i=1}^N E(I_i)$, where $E(I_i)$ is the probability that bucket i contains at least one ball. The probability that bucket i is empty can be found as follows: The chance that a particular ball does not go into bucket i is $\frac{N-1}{N}$. Therefore, the probability that none of the K balls end up in bucket i is $\left(\frac{N-1}{N}\right)^K$. Thus, the probability that bucket i contains at least one ball is:

$$1 - \left(\frac{N-1}{N}\right)^K$$

Summing over all buckets, the expected number of buckets with at least one ball is:

$$\sum_{i=1}^N \left(1 - \left(\frac{N-1}{N}\right)^K\right) = N \left(1 - \left(\frac{N-1}{N}\right)^K\right)$$

Considering $N = 10$ and $K = 5$, the expected number of buckets with at least one ball is:

$$10 \left(1 - \left(\frac{10-1}{10}\right)^5\right) = 10 \left(1 - \left(\frac{9}{10}\right)^5\right) \approx 10(1 - 0.59049) = 10 \times 0.40951 = 4.0951$$

Answer: 4.09510

360. Maximize Product

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To achieve the maximum product of the number of coins in each stack, start by dividing the 100 coins into stacks of equal size. Initially, arranging the coins into 10 stacks of 10 coins each results in a product of 10^{10} . By dividing each stack of 10 into stacks of 5 coins, you obtain 5^{20} . For further optimization, consider using stacks of size 3. This configuration yields 33 stacks of 3 coins with one leftover coin. Combining one stack of 3 coins with the leftover coin gives 4×3^{32} . Calculating this, you get $4 \times 1853020188851841 = 7412080755407364$, which is the maximum product achievable.

Answer: 7412080755407364

361. End With 2018

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability that the last 4 digits of any sequence is 2018 is

$$\frac{1}{4^4} = \frac{1}{256}$$

The probability we finish after appending n digits is the probability that the last 4 digits are 2018, and we did not finish before n digits. However, if our last digits are 2018, that implies we did not finish with 2018 after $n-1$, $n-2$, or $n-3$ digits. Therefore, the probability that we finish at the 4th, 5th, 6th, or 7th digits is

$$\frac{1}{256}$$

each. The probability of finishing at the 8th digit is then

$$\frac{1}{256} - \frac{1}{256^2}$$

since we need to avoid overcounting the case of 20182018. Thus, our answer is:

$$4 \cdot \frac{1}{256} + \frac{255}{256^2} = \frac{5 \cdot 256 - 1}{256^2} = \frac{1279}{65536}$$

Answer: 0.01952

362. Balls In Garbage Bag

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider a particular bucket and the m balls with a particular number. The probability that the bucket never receives a ball with that number is

$$\left(\frac{m-1}{m}\right)^m.$$

So the expected total number of buckets which never receive a ball with that number is

$$m \left(\frac{m-1}{m}\right)^m.$$

That is then equal to the expected number of balls with that number put in the garbage, since they did not go into any bucket.

So the total expected number of balls put in the garbage is

$$m^2 \left(\frac{m-1}{m}\right)^m.$$

Answer: 34.86784

363. Coin Coloring Game

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Clearly, Reimu will always color a side red and Sanae will always color a side green, because their situation is never worse off when a side of a coin changes to their own color. Since the number of red-only coins is always equal to the number of green-only coins, no matter how Reimu and Sanae color the coins, they will have an equal probability of winning by symmetry. Therefore, they will cooperate to make sure that the probability of a tie is minimized, which is when all 4 coins have different colors on both sides (which can easily be achieved by Reimu coloring one side of a new coin red and Sanae immediately coloring the opposite

side green). Therefore, the probability of Reimu winning is

$$\frac{\binom{4}{3} + \binom{4}{4}}{2^4} = \frac{5}{16}$$

Answer: 0.31250

364. Sufficient Sum II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

By the tail-sum formula for a nonnegative integer-valued random variable,

$$E[N_x] = \sum_{n=0}^{\infty} P(N_x > n),$$

and the event $N_x > n$ is equivalent to

$$X_1 + \cdots + X_n \leq x.$$

From "summation series I," we know that for i.i.d. $\text{Unif}(0,1)$ variables the probability

$$P(X_1 + \cdots + X_n \leq x) = \frac{x^n}{n!},$$

it follows that

$$E[N_x] = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x.$$

Substituting $x = \ln 2$ gives

$$E[N_{\ln 2}] = e^{\ln 2} = 2.$$

Answer: 2

365. Asking For A Date

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E be the expected number of days it would take Heesu to ask Sally out (in terms of p). Let E_k for $k \in \mathbb{Z}$ denote the expected number of days it would take given that Heesu has talked to Sally each of the last k days but not on the $(k+1)$ -th day before. For all $k \geq 4$, we have $E_k = 0$. Additionally, we have:

$$E_0 = 1 + (1-p)E_0 + pE_1$$

$$E_1 = 1 + (1-p)E_0 + pE_2$$

$$E_2 = 1 + (1-p)E_0 + pE_3$$

$$E_3 = 1 + (1-p)E_0 + pE_4$$

Subtracting the equations, we then have:

$$\frac{1}{p} = E_0 - E_1 = p(E_1 - E_2) = p^2(E_2 - E_3) = p^3(E_3 - E_4) = p^3E_3$$

Then, solving, we obtain:

$$E_3 = \frac{1}{p^4}, \quad E_2 = \frac{1}{p^3} + \frac{1}{p^4}, \quad E_1 = \frac{1}{p^2} + \frac{1}{p^3} + \frac{1}{p^4}, \quad E_0 = \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3} + \frac{1}{p^4}$$

Since $E = E_0$, we have:

$$2800 = 7 + 49 + 343 + 2401 = \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3} + \frac{1}{p^4}$$

So, we find $p = \frac{1}{7}$.

Answer: 0.14286

366. Digit Picking

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If Bob gets a 9, his number will be larger. There are $\binom{8}{2} \cdot \binom{8}{3}$ ways in which this can happen. If both get the numbers from one to eight ($\binom{8}{3} \cdot \binom{8}{3}$ ways), they have equal chance of getting a bigger number due to symmetry. Out of these $\binom{8}{3} \cdot \binom{8}{3}$ ways, there are $\binom{8}{3}$ ways in which they will have the same number. Hence, the number of ways in which Bob will get a bigger number in this case is $\frac{\binom{8}{3} \cdot \binom{8}{3} - \binom{8}{3}}{2}$. The total number of ways in which Bob and Alice can choose a number is $\binom{9}{3} \cdot \binom{8}{3}$. Hence the required probability is

$$\frac{\binom{8}{2} \cdot \binom{8}{3} + \frac{\binom{8}{3} \cdot \binom{8}{3} - \binom{8}{3}}{2}}{\binom{9}{3} \cdot \binom{8}{3}} = \frac{37}{56}$$

Answer: 0.66071

367. Equal Heads

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For each person, the probability of getting exactly k heads in four flips follows a binomial distribution, given by:

$$P(\text{heads} = k) = \binom{4}{k} \left(\frac{1}{2}\right)^4 = \binom{4}{k} \cdot \frac{1}{16}$$

The probabilities for 0, 1, 2, 3, and 4 heads are:

$$P(0 \text{ heads}) = \frac{1}{16}, P(1 \text{ head}) = \frac{4}{16} = \frac{1}{4}, P(2 \text{ heads}) = \frac{6}{16} = \frac{3}{8}$$

$$P(3 \text{ heads}) = \frac{4}{16} = \frac{1}{4}, P(4 \text{ heads}) = \frac{1}{16}$$

Now, the probability that both people flip the same number of heads is the sum of the probabilities that they both get 0, 1, 2, 3, or 4 heads:

$$P(\text{same heads}) = P(0)^2 + P(1)^2 + P(2)^2 + P(3)^2 + P(4)^2$$

Calculating each term:

$$P(0 \text{ heads})^2 = \left(\frac{1}{16}\right)^2 = \frac{1}{256}, P(1 \text{ head})^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

$$P(2 \text{ heads})^2 = \left(\frac{3}{8}\right)^2 = \frac{9}{64}, P(3 \text{ heads})^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

$$P(4 \text{ heads})^2 = \left(\frac{1}{16}\right)^2 = \frac{1}{256}$$

Summing these probabilities:

$$P(\text{same heads}) = \frac{1}{256} + \frac{1}{16} + \frac{9}{64} + \frac{1}{16} + \frac{1}{256} = \frac{35}{128}$$

Therefore, the probability that both people flip the same number of heads is $\frac{35}{128}$.

Answer:

368. Car Crash

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p represent the probability we're looking for. The probability of no accident in a one-hour period is $\frac{1}{9}$, which implies no accident in both 30-minute intervals. Since car crashes are independent across different intervals, the probability of no crash in each 30-minute segment is $1 - p$. Consequently, the probability of no crashes in both segments is $(1 - p)^2$. Therefore, we have:

$$(1 - p)^2 = \frac{1}{9}$$

Solving this gives:

$$p = \frac{2}{3}$$

Answer:

369. Interesting Average II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For any subset A , we can define the “reflected subset” $A^0 = \{i \mid 2009 - i \in A\}$. Then $m(A) = 2009 - m(A^0)$. Note that as A is taken over all nonempty subsets of S , A^0 goes through all the nonempty subsets of S as well. Thus, the average of $m(A)$ is equal to the average of $\frac{m(A) + m(A^0)}{2}$, which is the constant $\frac{2009}{2}$.

Answer:

370. Dice Permutation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let's define E_i as the expected number of rolls needed to get all 6 different faces consecutively when the last i rolls are already different. Our goal is to find E_0 , which represents the expected number of rolls when no rolls have occurred yet. The sequence of equations is as follows:

$$E_0 = 1 + E_1$$

$$E_1 = 1 + \frac{1}{6}E_1 + \frac{5}{6}E_2$$

$$E_2 = 1 + \frac{1}{6}E_1 + \frac{1}{6}E_2 + \frac{4}{6}E_3$$

$$E_3 = 1 + \frac{1}{6}E_1 + \frac{1}{6}E_2 + \frac{1}{6}E_3 + \frac{3}{6}E_4$$

$$E_4 = 1 + \frac{1}{6}E_1 + \frac{1}{6}E_2 + \frac{1}{6}E_3 + \frac{1}{6}E_4 + \frac{2}{6}E_5$$

$$E_5 = 1 + \frac{1}{6}E_1 + \frac{1}{6}E_2 + \frac{1}{6}E_3 + \frac{1}{6}E_4 + \frac{1}{6}E_5 + \frac{1}{6}E_6$$

$$E_6 = 0$$

(since all six faces are already different) By solving this system of equations, we find that $E_0 = 83.2$.

Answer: 83.20000

371. Is It A Perfect Square?

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The number is divisible by 3 (as the sum of digits is divisible by 3). However, it is not divisible by 9 (as the sum of digits is not divisible by 9). Hence, the number formed cannot be a perfect square.

Answer: 2

372. Die Roll LCM

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First observe that the LCM of the numbers 1 through 10 is 2520, and no smaller LCM of a subset of these faces exceeds 2000. Therefore the stopping condition "LCM $\leq$ 2000" is equivalent to "the running LCM of our rolls has reached exactly 2520." To reach 2520, our sequence of rolls must collectively supply the prime factors 2^3 , 3^2 , 5^1 , and 7^1 . Among the faces 1–10, the only rolls that contribute 7^1 are rolls of 7; those that contribute 3^2 are rolls of 9; those that contribute 2^3 are rolls of 8; and for the factor 5^1 we may roll either 5 or 10. Thus we have four "coupon-types" to collect:

- A "7" (probability $\frac{1}{10}$)
- An "8" (probability $\frac{1}{10}$)
- A "9" (probability $\frac{1}{10}$)
- A "5-or-10" (probability $\frac{2}{10}$)

We stop as soon as we have seen at least one of each of these four types. Using the inclusion–exclusion principle for the coupon-collector expected time, if $p_1, \dots, p_4$ are the probabilities of the four types, then the expected number of rolls is

$$E = \sum_{i=1}^4 \frac{1}{p_i} - \sum_{1 \leq i < j \leq 4} \frac{1}{p_i + p_j} + \sum_{1 \leq i < j < k \leq 4} \frac{1}{p_i + p_j + p_k} - \frac{1}{p_1 + p_2 + p_3 + p_4}$$

Here $p_1 = p_2 = p_3 = 0.1$ for the types 7, 8, 9, and $p_4 = 0.2$ for "5 or 10." We compute each term:

Sum over single types:

$$3 \times \frac{1}{0.1} + \frac{1}{0.2} = 30 + 5 = 35$$

Sum over pairs (six pairs total):

- Three pairs among {7, 8, 9}: each has probability 0.2, giving $\frac{1}{0.2} = 5$, so

$$3 \times 5 = 15$$

- Three pairs each combining one of {7, 8, 9} with {5 or 10}: each has probability 0.3, giving $\frac{1}{0.3} = \frac{10}{3}$, so

$$3 \times \frac{10}{3} = 10 \quad \text{Total for pairs is } 15 + 10 = 25$$

Sum over triples (four triples):

- The triple {7, 8, 9} has total probability 0.3 $\Rightarrow \frac{1}{0.3} = \frac{10}{3}$
- Each of the other three triples that include "5 or 10" has probability 0.4 $\Rightarrow \frac{1}{0.4} = 2.5$, so

$3 \times 2.5 = 7.5$ Total for triples is

$$\frac{10}{3} + 7.5 = \frac{10}{3} + \frac{15}{2} = \frac{20 + 45}{6} = \frac{65}{6}$$

All four together:

$$0.1 + 0.1 + 0.1 + 0.2 = 0.5 \Rightarrow \frac{1}{0.5} = 2$$

Putting these into the inclusion-exclusion formula gives:

$$E = 35 - 25 + \frac{65}{6} - 2 = \frac{10 \times 6 + 65 - 12}{6} = \frac{60 + 65 - 12}{6} = \frac{125 - 12}{6} = \frac{113}{6} \approx 18.8333$$

Hence the expected number of rolls needed is $\frac{113}{6}$

Answer: 18.83333

373. Higher Roll II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Condition on Alice's single roll X , which is uniform on $\{1, 2, \dots, 20\}$. There are two cases:

Case 1: $X > 15$. If Alice rolls one of $\{16, 17, 18, 19, 20\}$, then Bob's maximum of any number r of rolls of a 15-sided die can never reach X . Hence in these 5 cases Alice wins with probability 1. Since

$$P(X > 15) = \frac{5}{20} = \frac{1}{4}$$

this contributes $\frac{1}{4}$ to the overall win probability.

Case 2: $X \leq 15$. Suppose $X = R$ for some $R \in \{1, 2, \dots, 15\}$. Bob rolls his 15-sided die r times and keeps the maximum M . Alice wins exactly when $M < R$, i.e., all r of Bob's rolls are drawn from $\{1, 2, \dots, R-1\}$. The chance of one roll landing in $\{1, \dots, R-1\}$ is $\frac{R-1}{15}$, so

$$P(M < R) = \left(\frac{R-1}{15}\right)^r$$

Because $P(X = R) = \frac{1}{20}$, the total contribution from this case is

$$\sum_{R=1}^{15} \frac{1}{20} \left(\frac{R-1}{15}\right)^r = \frac{1}{20} \sum_{k=0}^{14} \left(\frac{k}{15}\right)^r$$

Putting both cases together, the probability that Alice's roll beats Bob's maximum is

$$P_r = \frac{1}{4} + \frac{1}{20} \sum_{k=0}^{14} \left(\frac{k}{15}\right)^r$$

Finally, as $r \rightarrow \infty$, every term $\left(\frac{k}{15}\right)^r$ with $0 \leq k \leq 14$ goes to 0. Therefore,

$$\lim_{r \rightarrow \infty} P_r = \frac{1}{4} + \frac{1}{20} \cdot 0 = \frac{1}{4}$$

Answer: 0.25000

374. N Point Variance

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $X = \frac{I_1 + I_2 + \dots + I_N}{N}$. Then,

$$\text{Var}(X) = \frac{1}{N^2} \text{Var}(I_1 + I_2 + \dots + I_N)$$

Since the points are independently selected, $I_1, I_2, \dots, I_N$ are independent. Hence,

$$\text{Var}(X) = \frac{1}{N^2} (\text{Var}(I_1) + \text{Var}(I_2) + \dots + \text{Var}(I_N)) = \frac{1}{N^2} \cdot N \cdot \text{Var}(I_1) = \frac{\text{Var}(I_1)}{N}$$

Now,

$$\text{Var}(I_1) = E(I_1^2) - E(I_1)^2 = E(I_1) - E(I_1)^2$$

$E(I_1)$ is equal to the probability that the randomly selected point lies within the quarter circle. Using geometric probability, this is equal to $\frac{\pi}{4}$. Hence,

$$E(I_1) = \frac{\pi}{4}$$

So,

$$\text{Var}(I_1) = \frac{\pi}{4} - \left(\frac{\pi}{4}\right)^2$$

Hence,

$$\text{Var}(X) = \frac{1}{N} \left(\frac{\pi}{4} - \left(\frac{\pi}{4}\right)^2 \right)$$

Answer:

375. Game Of Cuts

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The first player wins whenever the starting grid is not a square, i.e., its length and width are different. The strategy is to always leave the other player with a square, since then they will eventually have a 1×1 square and hence, lose. This works because: • Starting from a square, it is impossible to get to a smaller square in 1 turn. • Starting from a non-square, it is always possible to reach a square. Thus, the first player wins whenever the starting grid is not a square, and the second player wins whenever the starting grid is a square. Hence, for the given problem, the first player would win.

Answer:

376. Next Date I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To determine the earliest date with all distinct digits, begin by minimizing the year, which significantly affects the result. Consider that the first digit of the month (MM) is either 0 or 1. If it is 0, then the day (DD) must be either 01, 02, or 31 (since 30 is excluded due to repeating the 0). If the first digit of the month is 1, the second digit of the month must be 0 or 2, and the day should include a 0. Thus, the year should start with 2 to avoid skipping to the year 1000. This limits our format to 0M/1D/2YYYY. To minimize the remaining digits, assign the smallest values possible: $YYY = 345$ (using the smallest available digits). Next, set the month (M) to 6, and choose the smallest day (D) remaining, which is 7. Therefore, the earliest date with all distinct digits is 06/17/2345.

Answer:

377. Dice Game I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First, calculate the expected value for a single roll of a fair six-sided die:

$$\frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$$

With the option to roll again, the optimal strategy is to re-roll if the first roll is 1, 2, or 3, as the expected value of the second roll is 3.5, which is higher. If the first roll is 4, 5, or 6, you should keep it. Thus, the expected value of the game is:

$$\frac{1}{6}(3.5) + \frac{1}{6}(3.5) + \frac{1}{6}(3.5) + \frac{1}{6}(4) + \frac{1}{6}(5) + \frac{1}{6}(6) = 4.25$$

So, under the optimal strategy, the expected payout is \$4.25.

Answer: 4.25000

378. Poisoned Wine II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We can assign some servants to drink at the first month, second month, and third month. With 5 servants, we convert each bottle label $i = b_{0i} + b_{1i} \cdot 4^1 + b_{2i} \cdot 4^2 + b_{3i} \cdot 4^3 + b_{4i} \cdot 4^4$, where each $b_{ki} = 0, 1, 2, 3$ for $0 \leq k \leq 4$. If $b_{ki} = 0$, servant k doesn't drink from bottle i . If $b_{ki} = p$ for $p > 0$, servant k drinks from bottle i in month p . When a servant dies, we record which servant died and what month they died in. From that, we can reconstruct the bottle that was poisoned. For example, if our sequence is 10031, the poisoned bottle was

$$1 + 3 \cdot 4^1 + 0 \cdot 4^2 + 0 \cdot 4^3 + 1 \cdot 4^4 = 278.$$

We have now created a 1-1 correspondence between base 4 expansions of length 5 and the bottle labels, so we can test up to $n = 4^5 = 1024$ bottles.

Answer: 1024

379. Poisson Cards

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The required expectation is $E \left[\binom{X}{4} \right]$, where X follows a Poisson distribution with mean 6. Using the law of unconscious statistician, this is equal to:

$$\sum_{n=0}^{\infty} \binom{n}{4} \cdot P(X = n)$$

This is equal to:

$$\sum_{n=4}^{\infty} \frac{n!}{4! \cdot (n-4)!} \cdot \frac{6^n}{n!} \cdot e^{-6} = e^{-6} \cdot \frac{1}{4!} \sum_{n=4}^{\infty} \frac{6^n}{(n-4)!}$$

Substituting $i = n - 4$, we get:

$$= e^{-6} \cdot \frac{1}{4!} \cdot 6^4 \sum_{i=0}^{\infty} \frac{6^i}{i!}$$

The sum $\sum_{i=0}^{\infty} \frac{6^i}{i!}$ is the Taylor expansion of e^6 . Hence, the required expectation is:

$$E \left[\binom{X}{4} \right] = e^{-6} \cdot \frac{6^4}{4!} \cdot e^6 = \frac{6^4}{4!} = \frac{1296}{24} = 54$$

Answer: 54

380. Min Collinearity[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

It is possible to arrange 10 points such that none of them are collinear. Consider the following arrangement (x denote the chosen points): X-X- -XX- XX— -X-X —XX If there are 11 points, some row must contain 3 points, which would definitely be collinear. So, 11 is the answer.

Answer: **381. Losing It All**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For Alice to win exactly once before losing all her money, she must play exactly 11 games. Additionally, Alice must win within the initial three rounds, which can occur in three different ways. Thus, the probability is calculated as:

$$(0.1) \cdot (0.9)^{10} \cdot 3 \approx 0.10460$$

Answer: **382. Handy Subsets**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider the elements that are included in the set red, and the elements that are not included blue. We will identify each subset using this coloring. Imagine three parts of the set T : $\{1, 2, \dots, 5\}$, $\{6, 7, \dots, 10\}$, $\{11, 12, \dots, 15\}$. If each of these 3 parts is individually monochromatic (consists of the same color), then the resulting subset will be handy. There are 2^3 such subsets. For example, consider $\{1, 2, \dots, 5\}$ and $\{11, 12, \dots, 15\}$ are red and $\{6, 7, \dots, 10\}$ is blue. Then the resulting subset is $\{1, 2, \dots, 5, 11, 12, \dots, 15\}$ which is handy. Next, let's deal with the second case: at least one of the three parts of the coloring is not monochromatic. There are $2^{15} - 2^3 = 32,760$ such colorings. Given such a coloring, consider the first part that is not monochromatic. We can apply an arbitrary cyclic shift of this part of the coloring without affecting which parts are monochromatic. So the set of colorings in this case can be partitioned into classes of 5 colorings each. In every class, we claim that exactly one of the 5 colorings corresponds to a handy subset. (Suppose we have a subset R of 1 to 5, or 6 to 10, or 11 to 15. When we shift it cyclically one position to the right, its sum goes up by $|R| \pmod{5}$. So the 5 possible shifts lead to increases $\pmod{5}$ of 0, $|R|$, $2|R|$, $3|R|$, and $4|R|$. Because $|R|$ is not a multiple of 5, these offsets cover every $\pmod{5}$ residue exactly once.) Hence the number of colorings in this case that correspond to a handy subset is

$$\frac{(2^{15} - 2^3)}{5}$$

which is 6552. Combining the two cases, we see that the number of handy subsets is

$$6552 + 8 = 6560.$$

Answer: **383. Last To Front**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the last person in the original line be person 5. For $n = 1, 2, 3, 4, 5$, define E_n as the expected number of rounds needed for person 5 to reach the front when they are currently in position n from the front.

Clearly, $E_1 = 0$.

Now suppose person 5 is in position $n > 1$. In one round, the front person is taken out and reinserted uniformly into one of the 5 positions of the line.

There are $n - 1$ positions at or before person 5, and $6 - n$ positions after person 5.

If the front person is reinserted into one of the first $n - 1$ positions, then person 5 stays in position n . If the front person is reinserted after person 5, then person 5 moves one step forward, from position n to position $n - 1$.

So we get

$$E_n = 1 + \frac{n-1}{5}E_n + \frac{6-n}{5}E_{n-1}.$$

Rearranging,

$$E_n = E_{n-1} + \frac{5}{6-n}.$$

Now compute the expectations:

$$E_2 = E_1 + \frac{5}{4} = \frac{5}{4},$$

$$E_3 = E_2 + \frac{5}{3} = \frac{5}{4} + \frac{5}{3},$$

$$E_4 = E_3 + \frac{5}{2} = \frac{5}{4} + \frac{5}{3} + \frac{5}{2},$$

$$E_5 = E_4 + 5 = \frac{5}{4} + \frac{5}{3} + \frac{5}{2} + 5.$$

Thus the expected number of rounds is

$$5 \left(\frac{1}{4} + \frac{1}{3} + \frac{1}{2} + 1 \right) = \frac{125}{12}.$$

So the answer is $\frac{125}{12}$ rounds.

Answer:

384. Coin Toss Movement

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For 6 to 8 heads, we are guaranteed to hit 4, so the sum here is

$$\binom{8}{2} + \binom{8}{1} + \binom{8}{0} = 28 + 8 + 1 = 37.$$

For 4 heads, you have to hit the 4 heads at the start, so there's only one way, 1. For 5 heads, we either start off with 4 heads, which gives us $\binom{4}{1} = 4$ ways to arrange the other flips, or we start off with five heads and one tail, which has 6 ways minus the 2 overlapping cases, HHHHHTTT and HHHHTHTT. Total ways: 8. Then we sum to get 46. There are a total of $2^8 = 256$ possible sequences of 8 coin flips, so the probability is

$$\frac{46}{256} = \frac{23}{128}.$$

Answer:

385. 1s And 9s

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let D_i represent the digit in the 10^i position of the number, with D_0 in the ones place. Since k is selected uniformly at random, each digit D_i has an equal probability of being either 1 or 9. This is because the set

of 8-digit integers with k ones is symmetric with the set of integers where the roles of ones and nines are swapped. Therefore, the expected value of each digit D_i is:

$$E[D_i] = \frac{1+9}{2} = 5$$

The number X can be expressed as:

$$X = 10^7 D_7 + 10^6 D_6 + \cdots + 10^1 D_1 + D_0$$

Taking the expectation of both sides, we get:

$$E[X] = 10^7 E[D_7] + 10^6 E[D_6] + \cdots + 10^1 E[D_1] + E[D_0]$$

Substituting $E[D_i] = 5$ for all i , we find:

$$E[X] = 5 \times (10^7 + 10^6 + \cdots + 10^1 + 1) = 5 \times 11111111 = 55555555$$

Answer: 55555555

386. Low Median

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Probability of exactly one number greater than $\frac{2}{3}$: The probability that one number is greater than $\frac{2}{3}$ is $\frac{1}{3}$ and the probability that the other two numbers are less than $\frac{2}{3}$ is $(\frac{2}{3})^2$. Thus, the probability of exactly one number being greater than $\frac{2}{3}$ is:

$$3 \cdot \left(\frac{2}{3}\right)^2 \cdot \left(\frac{1}{3}\right) = 3 \cdot \frac{4}{9} \cdot \frac{1}{3} = \frac{4}{9}$$

Probability of all numbers less than $\frac{2}{3}$: The probability that all three numbers are less than $\frac{2}{3}$ is:

$$\left(\frac{2}{3}\right)^3 = \frac{8}{27}$$

Total Probability: Adding the two probabilities together gives:

$$P\left(X(2) < \frac{2}{3}\right) = \frac{4}{9} + \frac{8}{27}$$

To combine these, find a common denominator:

$$\frac{4}{9} = \frac{12}{27}$$

and

$$P\left(X(2) < \frac{2}{3}\right) = \frac{12}{27} + \frac{8}{27} = \frac{20}{27}$$

Thus, the probability that the median of the three drawn numbers is less than $\frac{2}{3}$ is $\frac{20}{27}$.

Answer: 0.74074

387. Connected Components I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We index the squares from 1 to 15 and let X_i be the indicator random variable for the event "a new connected component begins at square i ." Since every component must start somewhere, the total number

of components is

$$X = \sum_{i=1}^{15} X_i$$

and by linearity of expectation,

$$E[X] = \sum_{i=1}^{15} E[X_i].$$

Now $X_1 = 1$ with probability 1, because the very first square always begins a new component. For each position $i \geq 2$, a new component starts at i exactly when square i has a different color than square $i - 1$. Since each square is black with probability $\frac{1}{3}$ and white with probability $\frac{2}{3}$, these two colors differ with probability

$$\Pr(\text{black then white}) + \Pr(\text{white then black}) = \frac{1}{3} \times \frac{2}{3} + \frac{2}{3} \times \frac{1}{3} = \frac{4}{9}.$$

Thus for $i \geq 2$,

$$E[X_i] = \Pr(X_i = 1) = \frac{4}{9}.$$

Putting these pieces together,

$$E[X] = E[X_1] + \sum_{i=2}^{15} E[X_i] = 1 + 14 \cdot \frac{4}{9} = \frac{65}{9}.$$

Hence the expected number of connected components in the row of 15 randomly colored squares is $\frac{65}{9}$.

Answer: 7.22222

388. 3 Heads 3 Tails II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The chance Alice wins is $\frac{1}{2}$, based on symmetry. The expected tosses for Alice and Bob to win are identical. Applying the law of total expectation, we need to double the expected tosses for Alice. Alice wins after 3 tosses with a probability of $\frac{1}{8}$, after 4 tosses with a probability of $\frac{3}{16}$ (as there are 3 ways to arrange 3 heads and 1 tail, ending in heads), and after 5 tosses with a probability of $\frac{6}{32}$ (6 ways to arrange 3 heads and 2 tails, ending in tails). The expected number of tosses for Alice is calculated as:

$$\frac{1}{8} \cdot 3 + \frac{3}{16} \cdot 4 + \frac{6}{32} \cdot 5 = \frac{66}{32}$$

Multiplying by 2 gives $\frac{33}{8}$.

Answer: 4.12500

389. Keep The GCD High

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that 997 is prime. By removing the card $(1, 997)$, a player is going to lose if and only if he removes a card that does not have a multiple of 997. Note that there are exactly two multiples of 997 between 1 and 2003. There are 2002 cards that contain 997 and 2002 cards that contain 1994, but we are counting twice the card $(997, 1994)$. Thus, there are 4003 cards with a multiple of 997. Since this number is odd, every time Bob removes one of these cards Alice can remove another. Thus Alice has a winning strategy. Note that the only thing needed to do to obtain this strategy is finding a prime p with r multiples between 1 and 2003, where r is 2 or 3 modulo 4. Removing the card $(1, p)$ wins the games.

Answer: 1

390. Moth On A Cube[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider three directions X, Y , and Z , representing the axes along which the moth can initially move. We can represent the moth's path by a sequence of moves in these directions: whenever the moth takes a step in a direction parallel or opposite to X , we note X , and similarly for Y and Z . The moth can reach vertex B in either exactly 3 steps or exactly 5 steps. For a path of 3 steps, the sequence must include one step in each of the directions X, Y , and Z , resulting in XYZ in some order. The number of different possible orders of XYZ is $3! = 6$. For a path of 5 steps, the sequence must be represented by one of the following: $XYZXX$, $XYZYY$, or $XYZZZ$, in some order. The number of possible permutations for each of these sequences is given by the formula for rearranging sequences with repeated elements:

$$\frac{5!}{3!1!1!} = 20 \text{ ways for each sequence.}$$

Since there are three possible sequences ($XYZXX, XYZYY, XYZZZ$), the total number of possibilities is $3 \times 20 = 60$. However, we must subtract the cases where the moth reaches B by the 3rd step, as these were already counted in the 3-step case. There are 3 such sequences for each of the 6 paths, so we subtract $3 \times 6 = 18$. Thus, the total number of valid paths is $60 - 18 + 6 = 48$.

Answer: 48**391. Expected Arc Length I**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Place X_1 at angle 0. For each of the other 29 points, the probability that a given point does not lie within an arc distance x on either side of X_1 is

$$1 - \frac{x}{\pi},$$

since the favorable length is $2\pi - 2x$ out of the total 2π . Thus, the probability that no point lies within the distance x is

$$\left(1 - \frac{x}{\pi}\right)^{29}.$$

This expression represents the probability that the arc distance from X_1 to its nearest neighbor exceeds x . Let A denote the arc distance from X_1 to its nearest neighbor. Then

$$P(A > x) = \left(1 - \frac{x}{\pi}\right)^{29}, \quad 0 \leq x \leq \pi.$$

Using the tail-sum formula for the expected value, we have

$$E(A) = \int_0^\pi P(A > x) dx = \int_0^\pi \left(1 - \frac{x}{\pi}\right)^{29} dx.$$

Let $u = \frac{x}{\pi}$ so that $dx = \pi du$ and u runs from 0 to 1. Then,

$$E(A) = \pi \int_0^1 (1 - u)^{29} du.$$

Evaluating the integral,

$$\int_0^1 (1 - u)^{29} du = \frac{1}{30}.$$

Thus,

$$E(A) = \frac{\pi}{30}.$$

Since we are on a unit circle, this value is the actual arc length. Therefore, the expected arc length between X_1 and its nearest neighbor is

$$\frac{\pi}{30}.$$

Answer: 0.10472

392. Dice Difference II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let e_i be the expected number of rolls required to obtain two consecutive rolls with a difference no greater than 1 after rolling the value i . Note that $e_1 = e_6$, $e_2 = e_5$, and $e_3 = e_4$. Let e be the expected number of rolls required to obtain two consecutive rolls with a difference no greater than 1.

$$e = \frac{1}{6}(1 + e_1) + \frac{1}{6}(1 + e_2) + \frac{1}{6}(1 + e_3) + \frac{1}{6}(1 + e_4) + \frac{1}{6}(1 + e_5) + \frac{1}{6}(1 + e_6)$$

Since $e_4 = e_3$, $e_5 = e_2$, and $e_6 = e_1$:

$$e = 1 + \frac{1}{6}(e_1 + e_2 + e_3 + e_3 + e_2 + e_1) = 1 + \frac{1}{3}(e_1 + e_2 + e_3)$$

Now,

$$e_1 = \frac{1}{6}(1) + \frac{1}{6}(1) + \frac{1}{6}(1 + e_3) + \frac{1}{6}(1 + e_4) + \frac{1}{6}(1 + e_5) + \frac{1}{6}(1 + e_6)$$

Substituting $e_4 = e_3$, $e_5 = e_2$, and $e_6 = e_1$:

$$e_1 = 1 + \frac{1}{6}e_1 + \frac{1}{6}e_2 + \frac{1}{3}e_3$$

Similarly,

$$e_2 = 1 + \frac{1}{6}e_1 + \frac{1}{6}e_2 + \frac{1}{6}e_3$$

$$e_3 = 1 + \frac{1}{3}e_1 + \frac{1}{6}e_2$$

Solving these equations simultaneously:

$$e_1 = \frac{288}{115}, \quad e_2 = \frac{246}{115}, \quad e_3 = \frac{252}{115}$$

Finally,

$$e = 1 + \frac{1}{3}(e_1 + e_2 + e_3) = 1 + \frac{1}{3} \left(\frac{288}{115} + \frac{246}{115} + \frac{252}{115} \right) = \frac{377}{115}$$

Answer: 3.27826

393. Game On 2 Piles

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If the two piles contain the same number of stones, the second player has a winning strategy. On each turn, if the first player removes some stones from one of the piles, the second player should respond by removing the same number of stones from the other pile. This strategy prevents the first player from removing the last stone, leading to a win for the second player. Once the first player removes the last stone from one pile, the second player can then remove all the remaining stones from the other pile. Conversely, if the piles have different numbers of stones, the first player has a winning strategy. On their first move, the first player should remove enough stones from the larger pile to leave both piles with an equal number of stones. By making this move, the first player assumes the advantageous position that the second player had in the

previous exercise, allowing them to employ the same winning strategy to secure victory.

Answer: 1

394. 5-runs

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the squares 1 through 20. For any fixed k , there are $21 - k$ possible starting positions for a block of k consecutive squares, and each such block is entirely black with probability $\frac{1}{2^k}$. Therefore, allowing overlaps,

$$E[R_k] = (21 - k) \frac{1}{2^k}$$

Now, consider 9 consecutive black squares. What would $R_5 - R_6$ be in this case? It would be 1. R_5 will be equal to 5, and R_6 will be 4. So, $R_5 - R_6$ will give us the number of 5-runs in this case. But what if we have 10 consecutive black squares? $R_5 - R_6$ will again be 1, but there are 2 5-runs (since there are 10 consecutive black squares). So, the number of 5-runs is given by $R_5 - R_6 + R_{10}$ in this case. What if there are 11 consecutive black squares? The number of 5-runs would be equal to $R_5 - R_6 + R_{10} - R_{11}$. Proceeding in a similar way, we see

$$\text{5-runs} = R_5 - R_6 + R_{10} - R_{11} + R_{15} - R_{16} + R_{20}$$

Using linearity of expectation,

$$E[\text{5-runs}] = E[R_5] - E[R_6] + E[R_{10}] - E[R_{11}] + E[R_{15}] - E[R_{16}] + E[R_{20}].$$

We compute: $* R_5 = \frac{21-5}{2^5} = \frac{16}{32} = \frac{1}{2}$ $* R_6 = \frac{21-6}{2^6} = \frac{15}{64}$ $* R_{10} = \frac{21-10}{2^{10}} = \frac{11}{1024}$ $* R_{11} = \frac{21-11}{2^{11}} = \frac{10}{2048} = \frac{5}{1024}$
 $* R_{15} = \frac{21-15}{2^{15}} = \frac{6}{32768} = \frac{3}{16384}$ $* R_{16} = \frac{21-16}{2^{16}} = \frac{5}{65536}$ $* R_{20} = \frac{21-20}{2^{20}} = \frac{1}{1048576}$ Hence

$$E[\text{5-runs}] = \frac{284785}{1048576}$$

Answer: 0.27161

395. Dice Difference I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E be the expected number of rolls. Let E_i be the expected number of rolls required after rolling an i . The total expected number of rolls can be written as:

$$E = \frac{1}{6}(1 + E_1) + \frac{1}{6}(1 + E_2) + \frac{1}{6}(1 + E_3) + \frac{1}{6}(1 + E_4) + \frac{1}{6}(1 + E_5) + \frac{1}{6}(1 + E_6)$$

$$E = 1 + \frac{1}{6}(E_1 + E_2 + E_3 + E_4 + E_5 + E_6)$$

Using symmetry, we can simplify this:

$$E_1 = E_6 = E_2 = E_5$$

$$E_3 = E_4$$

So, we have:

$$E = 1 + \frac{1}{6}(4E_1 + 2E_3)$$

Now, let's consider the equations for E_1 and E_3 :

$$E_1 = \frac{1}{6}(1 + E_1) + \frac{1}{6}(1 + E_2) + \frac{1}{6}(1) + \frac{1}{6}(1 + E_4) + \frac{1}{6}(1 + E_5) + \frac{1}{6}(1 + E_6)$$

Using symmetry, this simplifies to:

$$E_1 = 1 + \frac{1}{6}(4E_1 + E_3)$$

Similarly, for E_3 :

$$E_3 = \frac{1}{6}(1) + \frac{1}{6}(1 + E_2) + \frac{1}{6}(1 + E_3) + \frac{1}{6}(1 + E_4) + \frac{1}{6}(1) + \frac{1}{6}(1 + E_6)$$

Using symmetry, this simplifies to:

$$E_3 = 1 + \frac{1}{6}(2E_1 + 2E_3)$$

Solving these equations, we find:

$$E_1 = 5$$

$$E_3 = 4$$

Hence, the expected number of rolls is:

$$E = 1 + \frac{1}{6}(4E_1 + 2E_3)$$

$$E = 1 + \frac{1}{6}(4 \cdot 5 + 2 \cdot 4)$$

$$E = \frac{17}{3}$$

Answer:

396. 7994 Stones

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Notice that the multiples of 8 are losing positions. If a player finds themselves at a multiple of 8, they cannot take all the stones in a single turn. If Alice takes a stones, Bob can respond by taking $8 - a$ stones, ensuring that the remaining number of stones is again a multiple of 8. Initially, 7994 leaves a remainder of 2 when divided by 8. If the first player does not remove exactly 4 stones, the other player can force the game back to a multiple of 8. This pattern continues for subsequent turns. If both players remove precisely 4 stones each turn, the game proceeds systematically until Bob is left with 2 stones on his final turn, securing the win. Thus, Bob has the winning strategy.

Answer:

397. Measuring Everything II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Every natural number can be represented as the difference of two numbers whose base 3 representations contain only 0's and 1's. We can prove this property by writing the original number in base 3 and constructing the required numbers digit by digit from right to left. Now, it is clear that it suffices to have only 5 weights: 1, 3, 9, 27, and 81. Four weights are insufficient because we can only weigh up to

$$3^4 - 1 = 80$$

different weights (each weight can either be placed on the left pan, right pan, or not used at all). Hence, we need 5 weights to weigh any natural number up to 100.

Answer:

398. Ending At 1[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p_n denote the probability that the last number she writes down is 1, given that she starts with n . If she begins by writing 2 or 4, she can continue with either 2 or 4, but she stops the first time she writes 1 or 3. Thus, we have:

$$p_2 = p_4 = \frac{1}{2}$$

If she starts with 1, she stops if she writes 1, 2, or 4, but continues if she writes 3. Therefore,

$$p_1 = \frac{1}{4}(1 + p_3)$$

If she starts with 3, she stops if she writes 2 or 4 and continues otherwise. Hence,

$$p_3 = \frac{1}{4}(p_1 + p_3) = \frac{1}{16}(1 + 5p_3)$$

Solving this gives us $p_3 = \frac{1}{11}$ and $p_1 = \frac{3}{11}$. The probability we want to find is:

$$\frac{1}{4}(p_1 + p_2 + p_3 + p_4) = \frac{1}{4} \left(\frac{3}{11} + \frac{1}{2} + \frac{1}{11} + \frac{1}{2} \right) = \frac{15}{44}$$

Answer: 0.34091**399. Points On A Square**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Fix one edge for the first point to lie on. If the second point lies on the opposite edge, it will be of distance greater than one (with $\frac{1}{4}$ probability), and if it lies on the same edge, then it will be of distance less than one (again $\frac{1}{4}$ probability). Suppose then that the second point lies on one of the other two adjacent edges. If the first point is distance x from this edge, then the other point must lie farther than $\sqrt{1-x^2}$ from the vertex shared by the edges the two points lie on. Therefore, this reduces to a geometric probability problem where we are finding the area of a region outside of a quarter circle in the unit square (we could also view this as the integral $\int_0^1 \sqrt{1-x^2} dx$, although this is a bit more work). Thus, the required probability is:

$$\frac{1}{4} + \frac{1}{2} \left(1 - \frac{\pi}{4} \right) = \frac{6-\pi}{8}$$

Answer: 0.35730**400. Standing Table III**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that the table will stand exactly when the pentagon formed by joining the five attachment points contains the centre of the circle. Write each chosen point in polar coordinates (R, Θ) measured from the circle centre. For a uniformly random point in the disk the angle Θ is uniform on $[0, 2\pi)$ and is independent of the radius R . Therefore for five independent uniform points the five angles $\Theta_1, \dots, \Theta_5$ are independent and each is uniform on $[0, 2\pi)$, and the radii carry no information about direction. The geometric condition that the centre lies outside the convex hull of the five points is equivalent to the existence of a line through the centre that places all five points strictly on one side of that line. Such a line is described by a direction and cuts off a semicircular arc of directions of length π ; a point lies on one given side of the line exactly when its angle lies in that semicircular arc. Hence the property “all five points lie on one side of some line through the centre” depends only on the collection of angles Θ_i and not on the radii. Because those angles are independent uniforms, the probability that all five angles lie in some semicircular region is the

same as the probability that five independent uniform points on the circumference all fall in some semicircle. From "same arc I", this probability equals $\frac{5}{16}$. The event that the convex hull contains the centre is the complement of this event, so the probability the table stands is

$$1 - \frac{5}{16} = \frac{11}{16}.$$

Answer: 0.68750

401. Hungry Driver

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The first thing you should notice about this problem is that if we try to take 1000 apples directly from town A to town B , we will be left with no apples at our destination. This is obvious, as we have to travel a distance of 1000 miles and we lose 1 apple per mile we travel. Instead of one driver taking multiple trips, let's imagine that we have multiple drivers making one trip. There are 3000 apples, so at least 3 drivers must set out from town A , and they are eating 3 apples per mile. After $\frac{1000}{3} \approx 333$ miles, there are 2000 apples left, so we only need 2 drivers to take the remaining apples to town B . After another $\frac{1000}{2} = 500$ miles, there are 1000 apples left, so we kick out another driver. Finally, we drive the remaining 167 miles. The number of apples left when we arrive at B is $1000 - 167 = 833$. To check that this reframing of the problem works, we can examine what happens in the original single-driver setting. The driver sets out from A with 1000 apples, stops when he drives 333 miles, and deposits 667 apples. He then drives back to A , and repeats this process twice more. This means we have gone from having 3000 apples at A to having 2000 apples at a point 333 miles closer to B . He can now repeat this process by driving from A to the point where the apples were deposited, and in 2 trips get 500 miles closer to B (now with 1000 apples left). Finally, he can take these 1000 apples and drive directly to B .

Answer: 833

402. Sufficient Sum III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We begin by writing N_x in terms of its tail indicators: define $I_k = 1\{N_x > k\}$, so that

$$N_x = \sum_{k \geq 0} I_k$$

Then

$$N_x^2 = \left(\sum_{k \geq 0} I_k \right)^2 = \sum_{k \geq 0} I_k + 2 \sum_{0 \leq j < k} I_j I_k$$

Since $I_j I_k = I_k$ for $j < k$, we have

$$N_x^2 = \sum_{k \geq 0} I_k + 2 \sum_{k \geq 0} k I_k = \sum_{k \geq 0} (2k + 1) I_k$$

Taking expectation,

$$E[N_x^2] = \sum_{k=0}^{\infty} (2k + 1) P(N_x > k)$$

Now $P(N_x > k) = P(X_1 + \cdots + X_k \leq x) = \frac{x^k}{k!}$ for $0 \leq x \leq 1$, so

$$E[N_x^2] = \sum_{k=0}^{\infty} (2k + 1) \frac{x^k}{k!} = 2 \sum_{k=0}^{\infty} k \frac{x^k}{k!} + \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Using the identities $\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$, $\sum_{k=0}^{\infty} k \frac{x^k}{k!} = xe^x$, we obtain

$$E[N_x^2] = 2xe^x + e^x = (2x + 1)e^x$$

We already know (from "sufficient sum ii") $E[N_x] = e^x$, so the variance is

$$\text{Var}(N_x) = E[N_x^2] - (E[N_x])^2 = (2x + 1)e^x - e^{2x}$$

Substituting $x = \ln 2$, we have $e^x = 2$, so

$$\text{Var}(N_{\ln 2}) = (2 \ln 2 + 1) \cdot 2 - 4 = 4 \ln 2 - 2$$

Thus the variance of $N_{\ln(2)}$ is $4 \ln 2 - 2$.

Answer: 0.77259

403. Conditional One

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Knowing the draws are in increasing order is equivalent to first choosing a 5-element subset of $\{1, \dots, 100\}$ uniformly at random and then listing its elements in increasing order. Under this conditioning every 5-subset is equally likely, and in the increasing listing the first position is occupied by the smallest element of the chosen subset. Thus the event "the first drawn number is 1" is exactly the event that the chosen 5-subset contains 1. The number of 5-subsets that contain 1 is $\binom{99}{4}$ since we must choose the remaining four elements from the other 99 numbers, while the total number of 5-subsets is $\binom{100}{5}$. Therefore the required probability is

$$\frac{\binom{99}{4}}{\binom{100}{5}} = \frac{1}{20}$$

So the probability that 1 was drawn first, given the observed increasing order, is $\frac{1}{20}$.

Answer: 0.05

404. Average Drawn

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Pick any two distinct drawn numbers a and c with $a < c$. For their average $\frac{a+c}{2}$ to be an integer - and so possibly equal to the third drawn number - a and c must have the same parity, both even or both odd. If they do, the average is an integer strictly between a and c , so the three numbers $\{a, \frac{a+c}{2}, c\}$ form a valid triple with one number equal to the average of the other two. Count the favourable triples: there are 10 even numbers in $\{1, \dots, 20\}$, so $\binom{10}{2} = 45$ unordered pairs of distinct evens, and each pair determines exactly one triple. Similarly, there are $\binom{10}{2} = 45$ unordered pairs of distinct odds, giving another 45 triples. No triple is counted twice because a triple with one number equal to the average has a unique smallest and largest element. Thus the total number of favourable triples is $45 + 45 = 90$. The total number of ways to choose any three balls is $\binom{20}{3} = 1140$. Therefore the required probability is $\frac{90}{1140} = \frac{3}{38}$. So the probability that one drawn number equals the average of the other two is $\frac{3}{38}$.

Answer: 0.07895

405. Too Many Hats

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Using the approach from "prisoner hats-multicolor," assign each of the N colors a value from 0 to $N - 1$. Define v as the weighted sum of the number of balls of each color (with weights equal to the value assigned to each color) modulo N . The value v can range from 0 to $N - 1$. The first prisoner assumes v to be 0 and makes their guess based on the visible hats, calculating what color on their hat will make v equal to 0. The second prisoner assumes the value to be 1 and guesses accordingly, and so on. At least one prisoner will guess correctly, as the true value of v lies in the interval $[0, N - 1]$.

406. Watermelon Weight

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Initially, the watermelon weighs 100 kg, and 99% of it is water. This means the non-water content is 1% of 100 kg, which is 1 kg. After drying, the watermelon is now 98% water, so the non-water content is 2% of the total weight. Let the new weight of the watermelon be x kg. Since the non-water content remains 1 kg, we have:

$$0.02x = 1$$

Solving for x :

$$x = \frac{1}{0.02} = 50 \text{ kg}$$

Thus, the watermelon now weighs 50 kg.

Answer: 50

407. Intersecting Subsets

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Introduce for each element $i \in S$ the indicator

$$X_i = \begin{cases} 1, & \text{if } i \in A \cap B, \\ 0, & \text{otherwise.} \end{cases}$$

Because A and B are each chosen uniformly, the chance that any fixed i lies in A is $\frac{1}{2}$, and independently the same holds for B . Therefore

$$\mathbb{P}(X_i = 1) = \mathbb{P}(i \in A) \cdot \mathbb{P}(i \in B) = \frac{1}{4}.$$

Since $N = \sum_{i=1}^{25} X_i$, linearity of expectation gives

$$\mathbb{E}[N] = \sum_{i=1}^{25} \mathbb{E}[X_i] = 25 \cdot \frac{1}{4} = 6.25.$$

Each X_i is a Bernoulli random variable with parameter $p = \frac{1}{4}$, so its variance is

$$\text{Var}(X_i) = p(1 - p) = \frac{3}{16}.$$

Moreover, because the inclusion of different elements in $A \cap B$ are independent events, the X_i are independent. Hence the variance of their sum is the sum of variances:

$$\text{Var}(N) = \sum_{i=1}^{25} \text{Var}(X_i) = 25 \cdot \frac{3}{16} = 4.6875.$$

Finally, forming the ratio yields

$$\frac{\mathbb{E}[N]}{\text{Var}(N)} = \frac{6.25}{4.6875} = \frac{4}{3} \approx 1.33333.$$

Answer: 1.33333

408. Bernoulli Matrix

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

A and A^T have the same diagonal elements. Hence, $A - A^T$ will have 0 as their diagonal elements. Let a_{ij} represent the element at the i -th row and j -th column of matrix A . Thus:

$$\det(A - A^T) = 0 - (a_{11} - a_{22}) \cdot (a_{22} - a_{11}) = (a_{11} - a_{22})^2.$$

Now, $a_{11} - a_{22} = 0$ if $a_{11} = 0$ and $a_{22} = 0$ OR $a_{11} = 1$ and $a_{22} = 1$. The probability of this is:

$$\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}.$$

Similarly, $a_{11} - a_{22} = 1$ if $a_{11} = 1$ and $a_{22} = 0$. The probability of this is:

$$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Finally, $a_{11} - a_{22} = -1$ if $a_{11} = 0$ and $a_{22} = 1$. The probability of this is:

$$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Hence, the expected value of $(a_{11} - a_{22})^2$ is:

$$0 \cdot \frac{1}{2} + 1^2 \cdot \frac{1}{4} + (-1)^2 \cdot \frac{1}{4} = \frac{1}{2}.$$

Answer: 0.50000

409. B Before O II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To solve this, start by arranging the 3 B's and 4 O's. We need at least 2 B's to appear before the first O. We consider scenarios where 2 or 3 B's come before any O. There are 5 such configurations (BBBOOOO, BBBOOOO, BBOOBOO, BBOOOBO, BBOOOOB). Now, we can place these 7 letters in 11 positions in $\binom{11}{7}$ ways. Hence, the number of ways of arranging B's and O's is $5 \times \binom{11}{7}$. After placing B's and O's, arrange the remaining letters L, A, H, and U, which can be permuted in $4!$ ways. The total number of favorable arrangements is: $5 \times 4! \times \binom{11}{7} = 5 \times 24 \times 330 = 39600$.

Answer: 39600

410. Bacteria Population III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To minimize the total bacteria you ever add, notice that each bacterium you introduce on day d will be doubled every subsequent night, contributing exactly 2^k bacteria to the final count if there are k nights after you add it. Thus, to reach exactly 2025, you should add a single bacterium on each day corresponding to a "1" in the binary expansion of 2025—no more, no less. Write $2025 = (11111101001)_2$. This expansion has

eight 1's (in positions for $2^{10}, 2^9, 2^8, 2^7, 2^6, 2^5, 2^3$, and 2^0). Concretely, you add:

- 1 bacterium ten nights before the end (contributes 2^{10}),
- 1 bacterium nine nights before (contributes 2^9),
- ...
- 1 bacterium three nights before (contributes 2^3),
- 1 bacterium on the final day (contributes $2^0 = 1$).

Because there are exactly eight ones in the binary form, you add only 8 bacteria in total. No scheme can use fewer, since each bacterium must account for one power-of-two summand in the final sum.

Answer: 8

411. Sorted Rows

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We proceed by considering the different cases where 23 could be in columns 3, 4, or 5.

Case 1: 23 is in column 5 If 23 is in column 5, we place it in the rightmost position in a given row. The remaining positions can be filled by choosing 4 elements from the remaining 22. Therefore, the number of ways to fill the row is $\binom{22}{4}$. The probability for this case is calculated as:

$$\text{Probability} = \frac{5 \times \binom{22}{4}}{\binom{25}{5}}$$

Case 2: 23 is in column 4 If 23 is in column 4, the rightmost two elements must be 23 and either 24 or 25. Thus, there are 2 possible ways to choose the rightmost two elements, and the number of ways to fill the remaining positions in the row is $\binom{22}{3}$. The probability for this case is:

$$\text{Probability} = \frac{10 \times \binom{22}{3}}{\binom{25}{5}}$$

Case 3: 23 is in column 3 If 23 is in column 3, the rightmost three elements must be 23, 24, and 25. The remaining 2 elements can be chosen from the other 22, so the number of ways to fill the row is $\binom{22}{2}$. The probability for this case is:

$$\text{Probability} = \frac{5 \times \binom{22}{2}}{\binom{25}{5}}$$

Now, to calculate the expected value $E(x)$, we use the formula:

$$E(x) = 3 \cdot P(\text{column 3}) + 4 \cdot P(\text{column 4}) + 5 \cdot P(\text{column 5})$$

Substituting the values of the probabilities calculated earlier, and simplifying, we find:

$$E(x) = \frac{14}{3}$$

Answer: 4.66667

412. Coin Reflip Game

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For the 4 normal flips, the expected number of heads is 2. The expected number of heads on the fifth flip, conditioned on there being a tail in the first 4 flips, is 0.5. The expected number of heads on the fifth flip, conditioned on there being no tail in the first 4 flips, is 0 (since the fifth flip doesn't happen in this case).

The probability that there is a tail in the first 4 flips is $\frac{15}{16}$ (since there are $2^4 = 16$ possibilities and only one of them has no tails [HHHH]). Hence, the number of possibilities where there is at least one tail is 15. So, the expected number of heads is

$$2 + \frac{15}{16} \cdot 0.5 + \frac{1}{16} \cdot 0 = \frac{79}{32}$$

Answer:

413. Large Language Model

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since there are 69 people, we need 69 sets of languages. They must be pairwise distinct, with each set having the same size. We need to achieve this using the minimum number of languages. The problem can be rephrased as follows: What is the minimum n such that the maximum value of $\binom{n}{r}$ is at least 69? Recall that the maximum value of $\binom{n}{r}$ occurs when $r = \lfloor n/2 \rfloor$ or $r = \lceil n/2 \rceil$. $n = 1$ does not work as the maximum value of $\binom{1}{r}$ occurs when $r = 1$, and the maximum value is $\binom{1}{1} = 1$. $n = 2$ does not work as the maximum value of $\binom{2}{r}$ occurs when $r = 1$, and the maximum value is $\binom{2}{1} = 2$. $n = 3$ does not work as the maximum value of $\binom{3}{r}$ occurs when $r = 1$ or 2, and the maximum value is $\binom{3}{2} = 3$. Continuing in this manner, we find that the smallest value of n that satisfies the condition is 8, as $\binom{n}{r}$ is maximized when $r = 4$, and the maximum value is $\binom{8}{4} = 70$. So, with 8 languages, we can get 70 distinct sets of 4 languages each. So, the required answer is 8.

Answer:

414. Strange Die

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

There is symmetry in the problem. The probability of rolling a 9 is the same whether we roll the first die twice or the second die twice. This symmetry can be observed by recognizing the correspondence between a face f on the first die and the face $9 - f$ on the second die. The probability of rolling a 9 with two rolls of either die is $\frac{4}{36} = \frac{1}{9}$. Listing the possible combinations for the faces of the two dice gives us this value. Next, if both dice are rolled, the probability of rolling a 9 is $\frac{6}{36} = \frac{1}{6}$. Using conditional probability, we calculate the probability that the same die was rolled given that the total sum was 9:

$$P(\text{same die} \mid \text{rolled 9}) = \frac{P(\text{rolled 9} \mid \text{same die})}{P(\text{rolled 9})}$$

The numerator $P(\text{rolled 9} \mid \text{same die})$ is $\frac{1}{9}$, and the denominator is $\frac{1}{2} \cdot \left(\frac{1}{9}\right) + \frac{1}{2} \cdot \left(\frac{1}{6}\right)$. Therefore:

$$P(\text{same die} \mid \text{rolled 9}) = \frac{\frac{1}{9}}{\frac{1}{2} \cdot \frac{1}{9} + \frac{1}{2} \cdot \frac{1}{6}} = \frac{2}{5}$$

Answer:

415. Cannons

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For $n = 2$, the pirates can only point at each other. For $n = 3$, there are 3 pairs of pirates but only 6 cannons, making mutual targeting unavoidable. Thus, $n \geq 4$. For $n = 4$, there are 6 pairs of pirates and 8 cannons, which means by the Pigeonhole Principle, at least 2 pirates will be mutually pointing at each other. For $n = 5$, there are 10 pairs of pirates and 10 cannons. One can arrange the pirates such that pirate i (where $1 \leq i \leq 5$) points at pirates $(i + 1) \pmod{5}$ and $(i + 2) \pmod{5}$. This arrangement avoids mutual

targeting, thus confirming that $n = 5$ satisfies the condition.

Answer: 5

416. Field Construction

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The length measurement is $300 + X$ and the width measurement is $200 + Y$, so the area is $(300 + X)(200 + Y)$. To find the expected area, compute:

$$E[(300 + X)(200 + Y)] = E[60000 + 300Y + 200X + XY]$$

Using the properties of expectation and noting that $E[X] = 0$ and $E[Y] = 5$, we apply linearity of expectation:

$$E[60000 + 300Y + 200X + XY] = E[60000] + 300E[Y] + 200E[X] + E[X]E[Y]$$

Since $E[X] = 0$, the terms involving $E[X]$ vanish. Therefore:

$$E[60000 + 300Y + 200X + XY] = 60000 + 300 \cdot 5 = 61500$$

Thus, the expected area is 61500.

Answer: 61500

417. Chocolate Bar

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For a chocolate of size $m \times n$, we need $mn - 1$ steps. By breaking an existing piece horizontally or vertically, we merely increase the total number of pieces by one. Starting from 1 piece, we need $mn - 1$ steps to get to $m \times n$ pieces. Hence, for the given problem, the required number of cuts is $6 \times 8 - 1 = 47$.

Answer: 47

418. Unfair Rock Paper Scissors

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let a_k represent the probability of Alice winning given that the current score difference is k . We can derive the recurrence relation:

$$a_k = p \cdot a_{k+1} + q \cdot a_{k-1}$$

with boundary conditions $a_5 = 1$ (Alice wins if the score difference reaches 5) and $a_{-5} = 0$ (Bob wins if the score difference reaches -5). The characteristic polynomial for this recurrence relation is:

$$z = p \cdot z^2 - q$$

This has roots $z_1 = 1$ and $z_2 = \frac{q}{p}$. Therefore, the general solution to the recurrence is:

$$a_k = C_1 \cdot 1^k + C_2 \cdot \left(\frac{q}{p}\right)^k$$

Using the boundary conditions $a_{-5} = C_1 + C_2 \left(\frac{q}{p}\right)^{-5} = 0$ and $a_5 = C_1 + C_2 \left(\frac{q}{p}\right)^5 = 1$, we solve for C_1 and

C_2 . From $a_{-5} = 0$, we get:

$$C_2 \cdot \left(\frac{q}{p}\right)^{-5} = -C_1$$

From $a_5 = 1$, we get:

$$C_1 + C_2 \cdot \left(\frac{q}{p}\right)^5 = 1$$

Solving these equations, we find:

$$C_2 = \frac{(q/p)^5}{(q/p)^{10} - 1}, \quad C_1 = \frac{-1}{(q/p)^{10} - 1}$$

Thus, the probability a_0 that Alice wins when the score difference is initially 0 is:

$$a_0 = \frac{(q/p)^5 - 1}{(q/p)^{10} - 1} = \frac{1}{(q/p)^5 + 1}$$

Substituting $p = 2/3$ and $q = 1/3$, we get:

$$a_0 = \frac{1}{(1/3 \div 2/3)^5 + 1} = \frac{1}{(1/2)^5 + 1} = \frac{1}{1/32 + 1} = \frac{1}{(1 + 32)/32} = \frac{32}{33}$$

Wait, the solution in the image has a different calculation for a_0 after substitution. Let's re-evaluate the image's calculation.

Image's calculation:

$$a_0 = \frac{1}{(2/3)^5 + 1} = \frac{243}{243 + 32} = \frac{243}{275}$$

Let's re-check the formula for a_0 . The formula is $\frac{1}{(q/p)^5 + 1}$. If $p = 2/3$ and $q = 1/3$, then $q/p = (1/3)/(2/3) = 1/2$. So, $a_0 = \frac{1}{(1/2)^5 + 1} = \frac{1}{1/32 + 1} = \frac{1}{(1 + 32)/32} = \frac{32}{33}$.

The image seems to have substituted p/q instead of q/p in the denominator, or there's a typo in the formula for a_0 or the substitution. The image shows $(2/3)^5 + 1$ in the denominator, which would imply p/q was used, not q/p . Let's assume the image's final calculation is correct based on its own intermediate steps, even if the formula for a_0 might be miswritten or the substitution is for p/q instead of q/p .

Following the image's final calculation: Substituting $p = 2/3$ and $q = 1/3$, we get:

$$a_0 = \frac{1}{(2/3)^5 + 1} = \frac{1}{32/243 + 1} = \frac{1}{(32 + 243)/243} = \frac{243}{275}$$

This matches the image's final fraction. I will use this value.

Answer: 0.88364

419. Sum Over One I

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let p represent $P[U + V > 1]$ and q denote $P[U + V < 1]$. Since $X \sim \text{Unif}(0, 1)$ implies $1 - X \sim \text{Unif}(0, 1)$, we can use this fact to simplify our analysis. Define U' as $\min\{1 - X_1, \dots, 1 - X_n\}$ and V' as $\max\{1 - X_1, \dots, 1 - X_n\}$. By using these new variables, we find that $P[U' + V' > 1]$ remains p , and $P[U' + V' < 1]$ is still q . Observe that $U' = 1 - V$ because to get the smallest value of $1 - X_i$, you must maximize X_i and subtract that from 1. Similarly, $V' = 1 - U$. Consequently:

$$p = P[U' + V' > 1] = P[(1 - V) + (1 - U) > 1] = P[U + V < 1] = q$$

From this, we conclude that p and q are equal. Since $p + q = 1$, it follows that both p and q are $\frac{1}{2}$.

Answer: 0.50000

420. Exactly 2 Consecutive Heads

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We look at the three cases: $HHT-$, $THHT$, and $-THH$, where the dashes can be either heads or tails. The probability of the first and last cases is $p^2(1-p)$ each, since we get two heads followed by a tail. The second case, $THHT$, has a probability of $p^2(1-p)^2$, as there are two heads and two tails in a specific order. Now, let $f(p)$ be the total probability of these events:

$$f(p) = 2p^2(1-p) + p^2(1-p)^2 = p^4 - 4p^3 + 3p^2.$$

To find the maximum value of $f(p)$, we differentiate it with respect to p and set the derivative to zero:

$$f'(p) = 4p^3 - 12p^2 + 6p = 2p(2p^2 - 6p + 3) = 0 \Rightarrow p = \frac{6 \pm \sqrt{12}}{4}$$

Since $\frac{6+\sqrt{12}}{4} > 1$, we take the negative square root, giving us:

$$p = \frac{3 - \sqrt{3}}{2}.$$

Answer: 0.63397

421. Reach Multiple Of 7

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

A cannot win on their first turn. The probability that B wins on their first turn is $\frac{1}{6}$, as only one of the six possible outcomes will make the total score a multiple of 7. For example, if A rolls a 1 on their first turn, B must roll a 6 on their first turn to win, which has a probability of $\frac{1}{6}$. Similarly, if A rolls a 4 on their first turn, B must roll a 3 on their first turn to win, with the probability again being $\frac{1}{6}$. This pattern holds for all subsequent turns—only one of the six outcomes will make the total score a multiple of 7. Therefore, on the first turn, A cannot win. For every other turn, the probability of winning in the current turn is $\frac{1}{6}$ for the player whose turn it is. We will use recursion. Let p represent the total probability of the current player winning. B wins on their first turn with probability $\frac{1}{6}$. If B doesn't win initially ($\frac{5}{6}$), A must also fail to win on their turn ($\frac{5}{6}$), returning the dice to B. The probability of B winning is again p . So, using the law of total probability,

$$\begin{aligned} p &= \frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot p \\ \Rightarrow p &= \frac{6}{11} \end{aligned}$$

Thus, the probability of B winning is $\frac{6}{11}$.

Answer: 0.54545

422. Painted Cube I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let I_i be the indicator that the i -th face of the cube is red. Hence, the number of red faces on a cube is $I_1 + I_2 + I_3 + I_4 + I_5 + I_6$. By linearity of expectation, the expected number of red faces is

$$E[I_1] + E[I_2] + E[I_3] + E[I_4] + E[I_5] + E[I_6].$$

All the faces are symmetric, and hence, the expected number of red faces is $6E[I_1]$. Now, what is the probability that a given face is red? There are 6×1000 faces in total. There are 6×10^2 red faces. Hence, the probability that a given face is red is

$$\frac{6 \times 10^2}{6 \times 1000} = \frac{1}{10}.$$

Therefore, $E[I_1] = \frac{1}{10}$. Thus, the expected number of red faces is

$$6E[I_1] = 6 \times \frac{1}{10} = \frac{3}{5}.$$

Answer:

423. Nearest Circular Neighbor

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Begin by fixing the position of P_1 . Let L denote the arc length (in radians) between P_1 and its nearest point. We compute the expected value of L as follows:

$$E[L] = \int_0^\pi P[L \geq x] dx$$

The maximum distance between points is π , since we are considering both directions around the circle. The event $\{L \geq x\}$ implies that none of the other 19 points lie within a region spanning x radians in either direction. The valid range for the remaining points is $2(\pi - x)$ radians. Therefore, the probability that all 19 points are outside this region is given by:

$$\left(\frac{2(\pi - x)}{2\pi}\right)^{19} = \left(1 - \frac{x}{\pi}\right)^{19}$$

Thus, the expected distance $E[L]$ is:

$$E[L] = \int_0^\pi \left(1 - \frac{x}{\pi}\right)^{19} dx = \pi \int_0^1 (1 - u)^{19} du = \frac{\pi}{20}$$

Finally, converting to degrees:

$$\frac{180}{\pi} \cdot \frac{\pi}{20} = 9 \text{ degrees.}$$

Answer:

424. Archery I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For the first 2 arrows, one target is hit. For the remaining 98 arrows, the scenario exhibits symmetry - both hit and miss are symmetric. Let the expected number of hits be x and the expected number of misses be y . Hence, $x + y = 98$. Since both hit and miss are symmetric, the expected number of hits and misses must be the same (both equal to x). Therefore, $x + x = 98$, so $x = 49$. Hence, for the remaining 98 arrows, the expected number of hits is 49. So, the total expected number of hits is 50.

Answer:

425. Same Sum Subsets

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The two subsets being disjoint is redundant. We can simply throw away the common elements of the two sets with equal sums to get disjoint sets with equal sums. The number of nonempty subsets of the set of 10 numbers is $2^{10} - 1 = 1023$. The possible sums range only from 1 to $100 + 99 + \cdots + 91 < 1000$, so there are fewer sums than subsets. By the pigeonhole principle, there must be two different subsets with the same sum.

Answer:

426. Pistol Shots

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

There are 5 intervals between the 1st and 6th shots (1-2, 2-3, 3-4, 4-5, 5-6). There are 2 intervals between the 1st and 3rd bullets (1-2, 2-3). 5 intervals took 60 seconds. So, 1 interval would take $\frac{60}{5} = 12$ seconds. Thus, 2 intervals would take $2 \times 12 = 24$ seconds.

Answer:

427. Buying Tickets

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We assign $+1$ for individuals holding \$5 bills and -1 for those with \$10 bills. Let (a, b) denote the state after a steps, where the walk ends at position b . The process is modeled as a random walk, starting from $(0, 0)$ and finishing at $(2n, 0)$ after $2n$ steps. To account for the n steps that are $+1$, there are $\binom{2n}{n}$ possible paths. Our focus is on paths where b is always non-negative, i.e., $b \geq 0$, and $0 < a < 2n$.

To simplify, we first calculate the number of paths that dip below zero, specifically those that reach $b = -1$ at least once within $0 < a < 2n$. Using the reflection principle, we reflect any path that touches $b = -1$ for the first time along the line $y = -1$. These reflected paths will end at $(2n, -2)$ instead of $(2n, 0)$. For such a path, there will be $n - 1$ steps of $+1$ and $n + 1$ steps of -1 , leading to $\binom{2n}{n-1}$ such paths.

Thus, the number of valid paths (where $b \geq 0$) is the total number of paths minus those that dip below zero: $\binom{2n}{n} - \binom{2n}{n-1}$. This simplifies to $\binom{2n}{n} - \frac{n}{n+1} \binom{2n}{n}$, which equals $\frac{1}{n+1} \binom{2n}{n}$.

Therefore, the probability that everyone can buy their ticket without needing to change places is $\frac{1}{n+1}$. For $n = 6$, this gives the final answer as $\frac{1}{7}$.

Answer:

428. First Ace

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If you remove the aces, there are 48 cards in total. The four aces would divide these 48 cards into 5 segments. If two aces are side by side, we say the segment between them is of length 0. If the first card is an ace, the segment before it is of length zero, and similarly for the segment following an ace that is the last card. The 5 segments should average 9.6 cards. The next card is the ace itself, so it is the 10.6th card on average.

Answer:

429. Inflating Balloon

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The first pump inflates the balloon at a rate of $\frac{1}{20}$ per minute, and the second pump inflates it at $\frac{1}{10}$ per minute. Together, they inflate the balloon at a rate of:

$$\frac{1}{20} + \frac{1}{10} = \frac{3}{20} \text{ per minute.}$$

The air leaks out at a rate of $\frac{1}{15}$ per minute. The net inflation rate is:

$$\frac{3}{20} - \frac{1}{15} = \frac{9}{60} - \frac{4}{60} = \frac{5}{60} = \frac{1}{12} \text{ per minute.}$$

Thus, it takes 12 minutes to fully inflate the balloon.

Answer: 12

430. Expected Squared Area

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The area is

$$A = \frac{1}{2} |x_1 y_2 + x_2 y_3 + x_3 y_1 - (x_2 y_1 + x_3 y_2 + x_1 y_3)|.$$

Squaring gives

$$A^2 = \frac{1}{4} (x_1^2 y_2^2 + x_2^2 y_3^2 + x_3^2 y_1^2 + x_2^2 y_1^2 + x_3^2 y_2^2 + x_1^2 y_3^2 + \text{cross terms}).$$

Each cross term—such as $x_1 x_2 y_2 y_3$ or $x_1^2 y_1 y_3$ —factors into at least one factor raised to the first power. To ensure these cross-term expectations vanish, center the grid coordinates from -4 to $+4$ so that $E[x_i] = E[y_i] = 0$. By independence each cross-term expectation factors into a product of individual expectations, and since one factor is zero, the term vanishes. Hence only the squared terms contribute:

$$E[A^2] = \frac{1}{4} \sum_{i \neq j} E[x_i^2 y_j^2] = \frac{1}{4} \sum_{i \neq j} E[x_i^2] E[y_j^2].$$

On $\{-4, -3, \dots, 3, 4\}$,

$$E[x^2] = \frac{(-4)^2 + \dots + 4^2}{9} = \frac{60}{9} = \frac{20}{3}, \quad E[y^2] = \frac{20}{3}.$$

With six such terms,

$$E[A^2] = \frac{1}{4} \cdot 6 \cdot \left(\frac{20}{3}\right)^2 = \frac{200}{3}.$$

Therefore, the expected square of the triangle's area is $\frac{200}{3}$.

Answer: 66.66667

431. Singer's Location

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $P(P) = 0.8$ represent the probability he is in Philadelphia and $P(B) = 0.2$ for Baltimore. The probability he performs given he is in Philadelphia is $P(E|P) = 0.1$, and for Baltimore, it's $P(E|B) = 0.8$. The total probability of performing (event E) is:

$$P(E) = P(E|P) \times P(P) + P(E|B) \times P(B) = (0.1 \times 0.8) + (0.8 \times 0.2) = 0.24$$

Now, the probability he is from Philadelphia given that he performed:

$$P(P|E) = \frac{P(E|P) \times P(P)}{P(E)} = \frac{0.08}{0.24} = \frac{1}{3}$$

Thus, the probability he came from Philadelphia if he performs is $\frac{1}{3}$.

Answer:

432. Probability Of Everything

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Hence, we have to add back in the number of sequences that fail to have three faces. We continue in this way, alternating subtracting and adding numbers of sequences, until we reach the final count: no sequence can fail to have all 6 faces. All together, then, we find that the number of sequences that fail to have all 6 faces is:

$$\binom{6}{1}5^n - \binom{6}{2}4^n + \binom{6}{3}3^n - \binom{6}{4}2^n + \binom{6}{5}1^n$$

Hence, the probability of having all 6 faces appear in n rolls of the die is:

$$\begin{aligned} 1 - \binom{6}{1}\left(\frac{5}{6}\right)^n + \binom{6}{2}\left(\frac{4}{6}\right)^n - \binom{6}{3}\left(\frac{3}{6}\right)^n + \binom{6}{4}\left(\frac{2}{6}\right)^n - \binom{6}{5}\left(\frac{1}{6}\right)^n \\ = 1 - 6\left(\frac{5}{6}\right)^n + 15\left(\frac{2}{3}\right)^n - 20\left(\frac{1}{2}\right)^n + 15\left(\frac{1}{3}\right)^n - 6\left(\frac{1}{6}\right)^n \\ = \frac{6^n - 6 \cdot 5^n + 15 \cdot 4^n - 20 \cdot 3^n + 15 \cdot 2^n - 6}{6^n} \end{aligned}$$

Answer:

433. Party Groups II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We can model the group formation as finding cycles in a random permutation of 50 guests. Each guest picks a guest's name, which can be viewed as a random permutation of the guests. The goal is to compute the expected size of cycles in this permutation. The number of possible cycles of size k in a permutation of length n is $\frac{n!}{k}$. The number of total possible cycles in a permutation of length n is $n! \cdot H_n$, where H_n is the n th harmonic number (since H_n is the average number of cycles and $n!$ is the total number of permutations). Hence, the probability that a particular cycle is of length k is:

$$\frac{n!/k}{n! \cdot H_n} = \frac{1}{kH_n}$$

So, the expected number of cycles is:

$$\sum_{k=1}^n k \cdot \frac{1}{kH_n} = \frac{n}{H_n}$$

For $n = 50$, the answer is approximately 11.11307.

Answer:

434. Heads Run II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We can transform any sequence of 11 coin tosses (each being heads or tails) into a new sequence s of length 10, made up of 1s and 0s. Here's how: For each position i from 1 to 10, define

$$s_i = \begin{cases} 1 & \text{if the } i^{\text{th}} \text{ and } (i+1)^{\text{th}} \text{ tosses are the same (HH or TT)} \\ 0 & \text{if they are different (HT or TH)} \end{cases}$$

Each adjacent pair of tosses has a 50% chance of being the same or different, so:

$$P(s_i = 1) = \frac{1}{2}, \quad P(s_i = 0) = \frac{1}{2}$$

We now want to compute the probability that this sequence s of 0s and 1s contains no three consecutive 1s. This is exactly the same constraint as in the Heads Run I problem (where we were avoiding three consecutive heads in a sequence of coin tosses). Hence, the final probability is:

Answer: 0.49219

435. Largest 2 Factor

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If p is the probability that $v(a) = v(b)$, then the answer is $\frac{1-p}{2}$, by symmetry. We compute p more generally for integers chosen from between 1 and 2^n , inclusive. Note that for a fixed $0 \leq k < n$, the number of integers x between 1 and 2^n , inclusive, such that $v(x) = k$ is equal to 2^{n-k-1} , and that there is only 1 such integer (namely 2^n) such that $v(x) = n$. We compute:

$$\begin{aligned} p &= \frac{1}{2^{2n}} \sum_{k=0}^n (\#\{x \mid v(x) = k\})^2 = \frac{1}{2^{2n}} \left(1 + \sum_{k=0}^{n-1} 2^{2(n-k-1)} \right) \\ &= \frac{1}{2^{2n}} \left(1 + \frac{4^n - 1}{4 - 1} \right), \end{aligned}$$

from which we find the desired probability:

$$\frac{1-p}{2} = \frac{4^n - 1}{3 \cdot 4^n}.$$

Substituting $n = 5$, we find that the answer is $\frac{341}{1024}$.

Answer: 0.33301

436. Trailing Zeros II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Rewrite the product $300 \cdot 305 \cdot 310 \cdots 1090 \cdot 1095 \cdot 1100$ as:

$$\frac{5^{161} \cdot 220!}{59!}$$

The number of factors of 5 in $59!$ is:

$$\left\lfloor \frac{59}{5} \right\rfloor + \left\lfloor \frac{59}{25} \right\rfloor = 11 + 2 = 13$$

The number of factors of 2 in 59! is:

$$\left\lfloor \frac{59}{2} \right\rfloor + \left\lfloor \frac{59}{4} \right\rfloor + \left\lfloor \frac{59}{8} \right\rfloor + \left\lfloor \frac{59}{16} \right\rfloor + \left\lfloor \frac{59}{32} \right\rfloor = 29 + 14 + 7 + 3 + 1 = 54$$

The number of factors of 5 in 220! is:

$$\left\lfloor \frac{220}{5} \right\rfloor + \left\lfloor \frac{220}{25} \right\rfloor + \left\lfloor \frac{220}{125} \right\rfloor = 44 + 8 + 1 = 53$$

The number of factors of 2 in 220! is:

$$\left\lfloor \frac{220}{2} \right\rfloor + \left\lfloor \frac{220}{4} \right\rfloor + \left\lfloor \frac{220}{8} \right\rfloor + \left\lfloor \frac{220}{16} \right\rfloor + \left\lfloor \frac{220}{32} \right\rfloor + \left\lfloor \frac{220}{64} \right\rfloor + \left\lfloor \frac{220}{128} \right\rfloor = 215$$

In the given expression, the number of factors of 5 is:

$$161 + 53 - 13 = 201$$

And the number of factors of 2 is:

$$215 - 54 = 161$$

Since the number of factors of 2 (161) is less than the number of factors of 5 (201), the number of trailing zeros in the product is determined by the number of factors of 2. Thus, the number of trailing zeros is 161.

Answer: 161

437. Sum Of Products

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We group together numbers with the same number of digits to compute this sum. The single-digit numbers from 1 through 9 contribute 45 to the sum. The double-digit numbers from 10 through 99 contribute:

$$\sum_{t=1}^9 \sum_{u=0}^9 tu = \left(\sum_{t=1}^9 t \right) \left(\sum_{u=0}^9 u \right) = 45^2 = 2,025$$

to the sum. The triple-digit numbers from 100 through 999 contribute:

$$\sum_{h=1}^9 \sum_{t=0}^9 \sum_{u=0}^9 htu = \left(\sum_{h=1}^9 h \right) \left(\sum_{t=0}^9 t \right) \left(\sum_{u=0}^9 u \right) = 45^3 = 91,125$$

to the sum. For the four-digit numbers, note that the numbers from 2000 to 2019 all have a zero digit and do not contribute to the sum. However, since the numbers from 1000 to 1999 all have a 1 in the thousands place, these contribute an amount equivalent to the contribution from the three-digit numbers. Thus, the answer is:

$$45 + 2025 + 2(91,125) = 184,320$$

Answer: 184320

438. Ants On An Octahedron

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $ACEDFB$ be the octahedron such that points $CDEF$ are coplanar, C/D are opposite each other, and E/F are opposite each other. The set of points that the ants from A/B can travel to is $\{C, D, E, F\}$, the set of points that the ants from C/D can travel to is $\{A, B, E, F\}$, and the set of points that the ants from E/F can travel to is $\{A, B, C, D\}$. So the problem is analogous to finding the number of ways to take 2 elements

from each of these sets to make the set $\{A, B, C, D, E, F\}$. Case 1: Ants from E/F travel to A/B . If this occurs, then the ants from C/D must travel to E/F , and therefore, the ants from A/B must travel to C/D . So this gives 1 case. Case 2: Ants from E/F travel to A/C . If this happens, then one of the ants from C/D must go to B and one of the ants from A/B must go to D . The remaining ant from C/D can choose to go to E or F , and then the remaining ant from A/B has only 1 choice. So this gives 2 cases. The ants from E/F traveling to A/B is equivalent to them traveling to C/D (both remove 2 elements from another set). The ants from E/F traveling to A/C is equivalent to them traveling to A/D , B/C , or B/D (both remove 1 element from the other 2 sets). For each of the 3 pairs of ants, there are 2 ways the pair of ants can go to 2 points (e.g., $E \rightarrow A$ and $F \rightarrow B$, or $E \leftrightarrow B$ and $F \rightarrow A$). Thus, there are

$$2^3(2 \cdot 1 + 4 \cdot 2) = 80$$

possibilities. There are a total of 2^{12} ways the ants can move, so the answer is

$$\frac{80}{2^{12}} = \frac{5}{256}$$

Answer: 0.01953

439. Broken Stick II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E denote the event that each of the three pieces has length at least $\frac{1}{8}$ meters. Let the first random point be x and the second random point be y . The required probability is $P(E) = P(y > x)P(E|y > x) + P(y < x)P(E|y < x)$. Clearly, $P(y > x) = P(y < x) = \frac{1}{2}$. Now, let's evaluate $P(E|y > x)$. If it is given that $y > x$, each of the three pieces has length at least $\frac{1}{8}$ meters iff: $x \geq \frac{1}{8}$ * $y - x \geq \frac{1}{8}$ * $1 - y \geq \frac{1}{8}$

Let this region be called R_1 . Also, let the region where $y > x$ be called R_2 . Then,

$$P(E|y > x) = \frac{\text{Area}(R_1 \cap R_2)}{\text{Area}(R_2)}$$

If we plot R_1 , we see that it forms a right triangle with base $\frac{5}{8}$ and height $\frac{5}{8}$, fully contained within R_2 . The area of R_2 is $\frac{1}{2}$. Thus,

$$P(E|y > x) = \frac{\frac{1}{2} \times \frac{5}{8} \times \frac{5}{8}}{\frac{1}{2}}$$

Similarly,

$$P(E|y < x) = \frac{\frac{1}{2} \times \frac{5}{8} \times \frac{5}{8}}{\frac{1}{2}}$$

So, the probability that each of the three pieces has length at least $\frac{1}{8}$ meters is

$$P(E) = \frac{1}{2} \times \frac{\frac{1}{2} \times \frac{5}{8} \times \frac{5}{8}}{\frac{1}{2}} + \frac{1}{2} \times \frac{\frac{1}{2} \times \frac{5}{8} \times \frac{5}{8}}{\frac{1}{2}} = \frac{25}{64}$$

Answer: 0.39063

440. Prime Sum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The prime numbers between 1 and 20 are 2, 3, 5, 7, 11, 13, 17, and 19. For the sum of two numbers to be even, either both numbers must be even or both must be odd. Since 2 is the only even prime number, both selected numbers must be odd. The probability that the first selected number is odd is $\frac{7}{8}$. Given that the first selected number is odd, the probability that the second selected number is odd is $\frac{6}{7}$. Hence, using the

multiplication rule, the probability that both selected numbers are odd is:

$$\frac{7}{8} \times \frac{6}{7} = \frac{3}{4}$$

Answer:

441. Double Factorial

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For any even n , we compare terms to find that $(n!)^2 = (n)(n) \cdot (n-2)(n-2) \cdot (n-4)(n-4) \cdots (2)(2) > n(n-1) \cdot (n-2)(n-3) \cdot (n-4)(n-5) \cdots (2)(1) = n!$

A similar comparison yields $(2n)! > (n!)^2$. Putting these together yields

$$\begin{aligned} ((1012!)!!)^2 &> (1012!)! > \left(\left(\frac{1012!}{2} \right)! \right)^2 > ((1012!)!)^2 \\ \implies (1012!)!! &> (1012!)! \end{aligned}$$

Thus, $(1012!)!!$ is greater.

Answer:

442. Jumping Frog II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To jump to lily pad 50, the frog must first hop to lily pad 49 or 51. Due to symmetry, each of these lily pads is equally likely to be the first lily pad the frog reaches. Let's assume the frog hops to lily pad 49 first. We can simplify the problem by treating lily pad 49 as our new starting point, calling it 0. From here, a hop to the right is $+1$, and a hop to the left is -1 . We need to find the probability that the frog, now acting as a "random walker," reaches -98 before reaching 1. Reaching -98 would mean the frog has also visited lily pad 51, the other neighbor of 50. Since -98 is 98 hops away in the opposite direction, the probability of reaching -98 before 1 in this symmetric random walk is $\frac{1}{1+98} = \frac{1}{99}$.

Answer:

443. Graph Travel I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E_i represent the expected number of steps needed to visit a new node after i nodes have already been visited. The number of steps required to reach a new node, given i nodes are already visited, follows a geometric distribution with a probability parameter of $\frac{10-i}{10}$. Thus, the expected number of steps to reach a new node is $\frac{10}{10-i}$.

Therefore, the expected total number of steps to visit all 10 nodes is given by:

$$\sum_{i=0}^9 \frac{10}{10-i} \approx 29.28968$$

Answer:

444. Never Divisible By 3[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since we only care about divisibility by 3, we can consider this list mod 3, i.e., $1, 2, 0, \dots, 1$. With the exception of the first slot, the 0 mod 3 elements can be inserted anywhere without changing the partial sum mod 3. Using this list, it is clear that the only way the other elements can be ordered is $1, 1, 2, 1, 2, 1, 2, 1, 2$. There are $5! \times 4!$ ways to arrange the 5 elements that are 1 mod 3 and the 4 elements that are 2 mod 3 in this manner. Then, there are 9 spots to insert "3" in this list, 10 for "6," and so on. Thus, the probability that any given arrangement is of the desired form is:

$$\frac{5! \times 4! \times 9 \times 10 \times 11 \times 12}{13!} = \frac{1}{182}$$

Answer: 0.00549**445. Expected Minimum Distance**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First, as in Hint 2, we have

$$F_{R_i}(x) = P(R_i \leq x) = x^2$$

because the area of a circle of radius x is πx^2 and the unit disk has area π . Thus

$$P(R_i > x) = 1 - x^2$$

For the minimum M of four independent distances,

$$P(M > x) = (1 - x^2)^4, \quad 0 \leq x \leq 1$$

By the tail-integral formula,

$$\mathbb{E}[M] = \int_0^\infty P(M > x) dx = \int_0^1 (1 - x^2)^4 dx$$

Next, expand by the binomial theorem:

$$(1 - x^2)^4 = 1 - 4x^2 + 6x^4 - 4x^6 + x^8$$

so

$$\begin{aligned} \int_0^1 (1 - x^2)^4 dx &= \int_0^1 (1 - 4x^2 + 6x^4 - 4x^6 + x^8) dx = \left[x - \frac{4}{3}x^3 + \frac{6}{5}x^5 - \frac{4}{7}x^7 + \frac{1}{9}x^9 \right]_0^1 \\ &= 1 - \frac{4}{3} + \frac{6}{5} - \frac{4}{7} + \frac{1}{9} \end{aligned}$$

Combining these fractions gives

$$\int_0^1 (1 - x^2)^4 dx = \frac{128}{315}$$

Therefore, the expected distance from the center to the nearest of the four points is

$$\boxed{\frac{128}{315}}$$

Answer: 0.40635**446. 800 Marbles**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In order to not stop after 699 marbles, the last 101 marbles must consist of 2 marbles of one color, and one marble from each other color. Since each of these marbles is equally likely to be the next to be drawn, and we stop after drawing the next marble as long as it's not one of the two of the same color, the desired probability is simply $\frac{99}{101}$.

Answer:

447. Set Intersection

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assuming sampling is done with replacement and independently, let X_i be an indicator variable for whether an element i in $A \cap B$ is selected in both samples. Specifically, $X_i = 1$ if i is chosen in both samples, and $X_i = 0$ otherwise. Therefore, the size of $A' \cap B'$ can be expressed as:

$$|A' \cap B'| = X = \sum_{i \in A \cap B} X_i$$

To find the expected value $E[|A' \cap B'|]$, compute $E[X]$ as follows:

$$E[X] = \sum_{i \in A \cap B} E[X_i]$$

The probability that i is chosen in both samples is:

$$E[X_i] = \frac{500}{5000} \times \frac{800}{8000} = \frac{1}{100}$$

Since there are 500 elements in $A \cap B$:

$$E[X] = 500 \times \frac{1}{100} = 5$$

Thus, the expected number of elements in $A' \cap B'$ is 5.

Answer:

448. Remainders

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let N represent the driver's number. Given that $N - 1$ must be divisible by 2, 3, 4, 5, and 6, it follows that $N - 1 = k \times \text{lcm}(2, 3, 4, 5, 6) = 60k$, where k is an integer. Testing values of k : - If $k = 0$, $N = 1$, which is not divisible by 11. - If $k = 1$, $N = 61$, which is also not divisible by 11. - For $k = 2$, $N = 121$, which is divisible by 11. Thus, $N = 121$ is the smallest number that satisfies the given conditions.

Answer:

449. Exactly 6

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For an experiment with n trials, with k possible results and $x_1, \dots, x_k$ desired outcomes corresponding to each of the possible results, the number of possible experimental outcomes is given by:

$$\binom{n}{x_1, \dots, x_k} = \frac{n!}{x_1! \cdots x_k!}$$

We know that the probability of any face from the dice showing up is $\frac{1}{6}$, and that we are rolling the dice 36 times. Therefore, we have the solution given below:

$$\binom{36}{6, 6, 6, 6, 6, 6} \cdot \left(\frac{1}{6}\right)^{36} = \frac{36!}{(6!)^6} \cdot \left(\frac{1}{6}\right)^{36} \approx 0.00026$$

Answer:

450. Jumping Frog VII

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Each unique path to point 10 can be represented as a subset of $\{1, 2, \dots, 9\}$. Each subset corresponds to a distinct path, and the task is to find subsets of odd length (since the final jump to point 10 will result in an even total number of jumps). The total number of subsets of $\{1, 2, \dots, 9\}$ is 2^9 . For any subset of odd length L , its complement will have an even length $(9 - L)$. Therefore, the number of subsets with odd length is equal to the number of subsets with even length. Thus, the number of subsets of odd length is $\frac{1}{2} \cdot 2^9 = 2^8$. The required probability is therefore

$$\frac{2^8}{2^9} = \frac{1}{2}$$

Answer:

451. Russian Roulette II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Number the chambers 1 to 6. Suppose the bullet is in the 6th chamber. You win if you start at the 1st, 3rd, or 5th chamber and lose if you start at the 2nd, 4th, or 6th chamber. So, the probability that you win is

$$\frac{3}{6} = \frac{1}{2}$$

Answer:

452. Segment Overlap

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Without loss of generality, assume P1 is the smallest point. If P2 is the second smallest point, there would be no overlap. If P2 is the third or fourth smallest point, there would be overlap. Hence, the required probability is $\frac{2}{3}$.

Answer:

453. 4 Die Sum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total number of outcomes is $6^4 = 1296$. The combinations that sum to 20 are: (i) 6,6,6,2 (ii) 6,6,5,3 (iii) 6,6,4,4 (iv) 6,5,5,4 (v) 5,5,5,5. The number of permutations for each case are: (i) 6,6,6,2: $\frac{4!}{3!} = 4$ (ii) 6,6,5,3: $\frac{4!}{2!} = 12$ (iii) 6,6,4,4: $\frac{4!}{(2! \cdot 2!)} = 6$ (iv) 6,5,5,4: $\frac{4!}{2!} = 12$ (v) 5,5,5,5: 1. So, the total number of permutations that sum to 20 are: $4 + 12 + 6 + 12 + 1 = 35$. Hence, the required probability is $\frac{35}{1296}$.

Answer:

454. Poisoned Chocolates

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Either the first player (Alice) or the second (Bob) must have a winning strategy; suppose it is Bob. Then, in particular, Bob must have a winning answer to Alice's opening move when she merely nibbles off the upper right-hand square. However, whatever Bob's reply is could have been made by Alice as her own opening move, after which she could follow Bob's allegedly winning strategy and chalk up the game. This of course contradicts the assumption that Bob can always win. Hence, it must be Alice that has a winning strategy.

Answer:

455. Green And Yellow Dice

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let e be the expected payout. The probability that the yellow die shows a number at least 3 greater than the green die is $\frac{7}{25}$. You can find this by checking each possible value of the green die and counting how many values on the yellow die satisfy the condition. Summing these up gives 28 out of 100 equally likely outcomes. The probability that both dice show the same number is $\frac{1}{10}$, and thus, the chance of them showing different numbers is $\frac{31}{50}$. Using the Law of Total Expectation based on the outcome of the first roll, we get:

$$e = \frac{7}{25}(e + 1) + \frac{31}{50}e$$

If the yellow die is at least 3 greater than the green, the game effectively restarts with a payout of 1. Solving for e , we arrive at:

$$\frac{1}{10}e = \frac{7}{25}$$

which simplifies to:

$$e = \frac{14}{5}$$

Answer:

456. Count Handshake Pairs

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the celebrities get into one or more circles so that each circle has at least three celebrities, and each celebrity shook hands precisely with their neighbors in the circle. Let's consider the possible circle sizes:

- One big circle with all 8 celebrities: Depending on the ordering of the people in the circle, the fan's list can still vary. Literally speaking, there are $7!$ different circles 8 people can make: fix one of the people, and then there are 7 choices for the person to the right, 6 for the person after that, and so on. But this would be double-counting because, as far as the fan's list goes, it makes no difference if we "reverse" the order of all the people. Thus, there are $\frac{7!}{2} = 2520$ different possible lists here.
- Two circles of sizes 5 and 3: In this case, there are $\binom{8}{5} = 56$ ways to split into the two circles, $4!$ different ways of ordering the 5-circle, and $2!$ ways for the 3-circle, giving a total count of $56 \times 12 \times 1 = 672$.
- Two circles of size 4 each: In this case, there are $\frac{\binom{8}{4}}{2} = 35$ ways to split into the two circles (we divide by 2 because, unlike in the 5 + 3 case, it does not matter which circle is which), and $3! \times 3! = 9$ ways of ordering each, giving a total count of $35 \times 9 = 315$.

Adding them up, we get $2520 + 672 + 315 = 3507$.

Answer: 3507

457. 2x3 Grid

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total number of ways to fill the grid is $3^6 = 729$. Now, to count the grids where the sum of the numbers in every row and in every column is a multiple of 3, note that filling any two elements in the same row fixes the rest of the grid. So the number of such grids is $3 \times 3 = 9$. Hence, the required probability is $\frac{9}{729} = \frac{1}{81}$.

Answer: 0.01235

458. Forming Quadratic Polynomial III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Because b and c are chosen uniformly and independently at random, the answer is the number of pairs (b, c) such that the roots of $x^2 + bx + c$ are consecutive integers divided by the total number of pairs (b, c) . There are a total of $13 \times 13 = 169$ pairs. The condition that a monic quadratic have roots that are consecutive integers is a strong condition, so rather than checking if pairs (b, c) yield monic quadratics with consecutive integer roots, we determine the form of such a quadratic and see which have coefficients belonging to the given set. Such a quadratic has the form $(x - r)(x - (r + 1)) = x^2 - (2r + 1)x + r(r + 1)$, where r is an integer. For the coefficient of x to be in the given set, we need $|2r + 1| \leq 6$, that is $r = -3, -2, -1, 0, 1, 2$. All 6 of these choices satisfy $|r(r + 1)| \leq 6$, hence the answer is $\frac{6}{169}$.

Answer: 0.03550

459. 2D Jumps

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Each jump i contributes $\Delta x_i = r_i \cos \theta_i$, $\Delta y_i = r_i \sin \theta_i$, where $r_i \sim \mathcal{U}(0, 1)$ and $\theta_i \sim \mathcal{U}(0, 2\pi)$, all independent. The final displacement is $X = \sum_{i=1}^{10} \Delta x_i$, $Y = \sum_{i=1}^{10} \Delta y_i$. The Euclidean distance is $\sqrt{X^2 + Y^2}$, so we seek $\mathbb{E}[(X^2 + Y^2)]$. Expanding $X^2 + Y^2$ produces diagonal terms $\mathbb{E}[r_i^2(\cos^2 \theta_i + \sin^2 \theta_i)]$ and cross-terms of the form $\mathbb{E}[\Delta x_i \Delta x_j]$, $\mathbb{E}[\Delta y_i \Delta y_j]$ for $i \neq j$. By independence, each cross-term expectation factors into a product of individual expectations (e.g. $\mathbb{E}[\Delta x_i \Delta x_j] = \mathbb{E}[r_i] \mathbb{E}[\cos \theta_i] \mathbb{E}[r_j] \mathbb{E}[\cos \theta_j]$), and since $\mathbb{E}[\cos \theta_i] = \mathbb{E}[\sin \theta_i] = 0$, these vanish. Only the diagonal terms remain: $\mathbb{E}[X^2 + Y^2] = \sum_{i=1}^{10} \mathbb{E}[r_i^2(\cos^2 \theta_i + \sin^2 \theta_i)] = 10 \cdot \mathbb{E}[r^2]$, and $\mathbb{E}[r^2] = \int_0^1 r^2 dr = \frac{1}{3}$. Thus, $\mathbb{E}[\text{squared distance}] = \frac{10}{3}$.

Answer: 3.33333

460. Digit Sum I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Treat each number as a six-digit number, including leading zeros, ranging from 000000 to 999999 (note that the sum of the digits for 1,000,000 is 1, which will be added separately). Each digit from 0 to 9 appears with equal frequency in each decimal place. Thus, the average value of each digit is $\frac{9}{2}$. Therefore, the average sum of the digits for a six-digit number is:

$$6 \times \frac{9}{2} = 27$$

With one million numbers from 000000 to 999999, the total sum of all digits is:

$$27 \times 1,000,000 = 27,000,000$$

Including the sum of the digits for 1,000,000, which is 1, the final answer is:

$$27,000,000 + 1 = 27,000,001$$

Answer:

461. Conditional Inequality

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The required answer is simply $P(X - Y > \frac{1}{2} \text{ and } X + Y < 1) / P(X + Y < 1)$. The region $X + Y < 1$ is represented by the green area below. The area is $\frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$. The region $X - Y > \frac{1}{2}$ and $X + Y < 1$ is represented by the red area below. The area is $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{16}$. The region $X - Y > \frac{1}{2}$ and $X + Y < 1$ is represented by the red area below. The area is $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{16}$. Hence, the required probability is $\frac{1/16}{1/2} = \frac{1}{8}$.

Answer:

462. Yet Another Game

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let n be the length of the row.

Claim: 1. If n is even and Alice begins, then Alice wins. 2. If n is odd and Bob begins, then Alice wins.

Note that 1 is sufficient for the problem, but we will prove both 1 and 2 simultaneously by induction on n .

The base cases $n = 1, 2, 3$ are trivial. Suppose now the claim holds for $1, 2, \dots, n - 1$, and let's prove it for n .

If n is even, the row is of the form $1, 0, 1, 0, 1, 0, \dots, 1, 0$ and we may take the last pair of $(1, 0)$ and make it a 1. By the induction hypothesis, since now it is Bob's turn, Alice wins.

If n is odd and Bob begins, the row is of the form $1, 0, 1, 0, 1, 0, \dots, 1, 0, 1$.

Bob might pick a pair of the form $(1, 0)$ or a pair of the form $(0, 1)$, and replace this pair with a 0 or a 1. Note that it doesn't matter which pair he chose to replace; there is either a 1 right before it or a 1 right after it: there is a 1 after each pair of the form $(1, 0)$ and there is a 1 before each pair of the form $(0, 1)$. Therefore, the two numbers Bob chose to replace are part of three consecutive numbers in the row, which are $1, 0, 1$ (and Bob chose either the first two or the last two of those three). It doesn't matter what Bob did; we can convert the $1, 0, 1$ that Bob chose his pair from into a 1. For example, if Bob chose the first two and made them a 0, the $1, 0, 1$ became $0, 1$, which Alice can convert into a 1. Therefore, after Alice's second turn, the difference from the original row is that a $1, 0, 1$ was replaced by a 1. Now, since it is Bob's turn, by the induction hypothesis, Alice wins.

Answer:

463. Same Color Corners

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Draw three horizontal lines. We'll find a rectangle with vertices on two of these. The other sides are vertical. At the intersection with the three horizontal lines, every vertical line has three candidate points to serve as vertices of the sought rectangle. Three points may be colored with 2 colors in 8 different ways. So, if you choose 9 vertical lines, there are bound to be triplets of points colored in the same manner. Select any two triplets colored in the same way—this gives you vertical sides. In any triplet, at least two points are of the same color. Select two such—this gives you horizontal sides.

Answer:

464. No Chords Intersecting[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the twelve points in clockwise order as $P_1, P_2, \dots, P_{12}$. Forming 6 chords from the 12 points is equivalent to dividing the 12 points into 6 groups, each containing 2 points. The total number of such groupings is:

$$\frac{12!}{2^6 \cdot 6!} = 10,395$$

This forms our sample space. We now count how many of those matchings are "noncrossing," meaning no two chords intersect inside the circle. Define $f(n)$ to be the number of noncrossing pairings of $2n$ labeled points placed in clockwise order on a circle. We seek $f(6)$. As before, we use the recursion:

$$f(n) = \sum_{r=1}^n f(r-1)f(n-r), \text{ with } f(0) = 1.$$

Starting from $f(0) = 1$, we compute: $* f(1) = f(0)f(0) = 1$ $* f(2) = f(0)f(1) + f(1)f(0) = 1 + 1 = 2$ $* f(3) = f(0)f(2) + f(1)f(1) + f(2)f(0) = 2 + 1 + 2 = 5$ $* f(4) = f(0)f(3) + f(1)f(2) + f(2)f(1) + f(3)f(0) = 5 + 2 + 2 + 5 = 14$ $* f(5) = f(0)f(4) + f(1)f(3) + f(2)f(2) + f(3)f(1) + f(4)f(0) = 14 + 5 + 4 + 5 + 14 = 42$ $* f(6) = f(0)f(5) + f(1)f(4) + f(2)f(3) + f(3)f(2) + f(4)f(1) + f(5)f(0) = 42 + 14 + 10 + 10 + 14 + 42 = 132$
Hence, the number of favorable outcomes is 132. The required probability is:

$$\frac{132}{10,395} = \frac{4}{315}$$

Answer: 0.01270**465. Drunk Passenger III**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We label the passengers 1 through 100 in the order they board, and we label the seats 1 through 100 so that passenger n "belongs" in seat n . Define the indicator random variable:

$$X_i = \begin{cases} 1, & \text{if passenger } i \text{ ends up in seat } i \\ 0, & \text{otherwise} \end{cases}$$

Then the total number of passengers in their own seats is:

$$X = \sum_{i=1}^{100} X_i,$$

and by linearity of expectation,

$$E[X] = \sum_{i=1}^{100} E[X_i] = \sum_{i=1}^{100} P(X_i = 1).$$

The first 10 passengers have a probability $\frac{1}{100}$ each of sitting on their assigned seats. The 100th passenger can only sit on the seats numbered 1, 2, 3, ..., 10, and 100. When he boards the plane, exactly 10 of those seats will be occupied, and all possible configurations are equally likely. Therefore, the probability that the 100th passenger ends up in his assigned seat is $\frac{1}{11}$. Similarly, the probability that the 99th passenger ends up in his assigned seat is $\frac{2}{12}$, and so on. Therefore, the required expected value is:

$$\frac{1}{100} \cdot 10 + \frac{90}{100} + \frac{89}{99} + \dots + \frac{2}{12} + \frac{1}{11} = 67.51591.$$

Answer: 67.51591

466. Last Number Deleted[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Initially, you retain all integers n that satisfy $n \equiv 0 \pmod{2}$. In the subsequent step, you keep integers that satisfy $n \equiv 0 \pmod{4}$. More generally, at step k , you keep all integers n such that $n \equiv 0 \pmod{2^k}$. To determine the last integer, find the largest k such that $2^k \leq 2000$. Here, 2^k will be the final integer kept. For $k = 10$, $2^{10} = 1024$, which is the highest power of 2 less than or equal to 2000.

Answer: **467. Digit Sum II**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To count the number of integers up to 100,000 where the sum of the digits equals 9, we focus on integers with up to 5 digits by padding with leading zeros if necessary. For example, the number 12 becomes 00012. We represent such numbers as $d_1d_2d_3d_4d_5$, where $0 \leq d_i \leq 9$, and need to find the count of such numbers where the digits sum up to 9. We need to find the number of non-negative integer solutions to the equation:

$$d_1 + d_2 + d_3 + d_4 + d_5 = 9$$

This is a classic stars and bars problem. The number of solutions is given by:

$$\binom{9+5-1}{5-1} = \binom{13}{4} = 715$$

Thus, there are 715 integers with up to 5 digits where the sum of the digits equals 9.

Answer: **468. Digit Multiply II**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The prime factorization of 96 is $96 = 3 \times 2^5$. Thus, our number must incorporate five 2s and one 3. To form the largest number, arrange the digits in descending order: the largest possible number is 322222. For the smallest number, combine the digits effectively. Since $2^3 = 8$ and $2 \times 3 = 6$, replace four 2s and the 3 with an 8 and a 6. The smallest number using the digits 8, 6, and 2 is 268. Therefore, the difference between the largest and smallest numbers is $322222 - 268 = 321954$.

Answer: **469. 500 Sided Dice**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Regardless of the sum of the first 99 dice, there is only one value on the last die that will make the total sum divisible by 500. Hence, the probability is $\frac{1}{500}$.

Answer:

470. Moving The Piece II[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Color the entire grid in alternating black and white squares, similar to a chessboard pattern, starting with the top-left square as white. Now, cover the entire grid (except the bottom-right cell) with dominoes. Alice would always move from a black to a white square, and Bob will always move from a white square to a black square. Note that the uncovered bottom-right cell is white, so Bob will never be able to move the piece to this square. Now, Alice can move the piece to the other half of the domino containing the starting square. For all subsequent moves, the second player will move into a new domino, and the first player moves to the other square in that domino. Therefore, Alice will always have a possible move, and hence, she has a winning strategy.

Answer: **471. Grid Arrangement I**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Of the sixteen entries in one such array, there are six 0s and ten 1s. The rows and the columns each have zero, one, two, and three 0s, in some order. Once we decide the positions of the 0s, we form one such array. We perform the following steps:

- There are $\binom{4}{1} = 4$ ways to choose the row with three 0s. After that, there are $\binom{4}{3} = 4$ ways to choose the positions of the three 0s.
- There are $\binom{3}{1} = 3$ ways to choose the column with three 0s. After that, there are $\binom{3}{2} = 3$ ways to choose the positions of the two additional 0s.
- There are $\binom{4}{1} = 4$ ways to choose the position of the last 0: it is the intersection of the column of one 0 in Step 1 and the row of one 0 in Step 2.

Together, the answer is $4 \times 4 \times 3 \times 3 \times 4 = 576$

Answer: **472. Information Transfer**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To find a recurrence relation for p_i , consider the information that person A_i has. If A_i has the original message (with probability p_{i-1}), they pass it correctly to A_{i+1} with probability p . If A_i has the incorrect message (with probability $1 - p_{i-1}$), they pass the incorrect message to A_{i+1} with probability $1 - p$. Thus, the probability p_i is given by:

$$p_i = p \cdot p_{i-1} + (1 - p) \cdot (1 - p_{i-1})$$

$$p_i = (2p - 1) \cdot p_{i-1} + (1 - p)$$

Taking the limit as $i \rightarrow \infty$ (assuming the limit exists), we get:

$$p^* = (2p - 1) \cdot p^* + (1 - p)$$

Solving for p^* :

$$p^* = \frac{1 - p}{2 - 2p} = \frac{1}{2}$$

Answer:

473. Chain Link[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The key idea is to divide the chain into parts where the lengths correspond to powers of 2. This allows you to create any number of links by combining these sections, as any integer can be expressed as a sum of powers of 2. First, observe the powers of 2 relevant to the total length of the chain: $2^0 = 1$, $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$. Now, we can divide the chain into sections with lengths 1, 2, 4, 8, and the remainder, which is 6 (since $1 + 2 + 4 + 8 + 6 = 21$). To achieve this, we need to make four cuts:

- Cut one link to create a section of 1 link.
- Cut another link to create a section of 2 links.
- Cut a third link to create a section of 4 links.
- Cut a fourth link to create a section of 8 links.

The remaining 6-link section stays intact. By combining these pieces, we can form any number of links from 1 to 21. Thus, the least number of links that need to be cut is 4.

Answer: **474. Pandemic I**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The fastest way to spread the illness is for each infected individual to interact with a healthy person every minute. This causes the number of sick people to double each minute. After n minutes, the number of infected individuals will be 2^n . To ensure that all 1000 people are infected, find the smallest n for which 2^n exceeds 1000. This occurs when $n = 10$.

Answer: **475. Expected Absolute Difference**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total number of outcomes when rolling two 30-sided dice is $30^2 = 900$. The number of outcomes where the difference is k is $2(30 - k)$. This occurs because a difference of k can be represented by pairs of the form $(i, i + k)$, where the maximum value of i is $30 - k$. The favorable pairs include $(1, 1 + k)$, $(1 + k, 1)$, $(2, 2 + k)$, $(2 + k, 2)$, ..., $(30 - k, 30)$, $(30, 30 - k)$. Thus, the probability of the difference being k is given by:

$$P(k) = \frac{2(30 - k)}{900}$$

To find the expected value of the absolute difference, we compute:

$$E[|X - Y|] = \sum_{k=0}^{29} k \cdot P(k) = \sum_{k=0}^{29} k \cdot \frac{2(30 - k)}{900} = \sum_{k=0}^{29} \frac{30k - k^2}{450}$$

Calculating this gives an expected value of: $\frac{899}{90}$

Answer: **476. Game On Jar**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The strategy is of the form—pick again if you get a value less than x . Stop when you get more than or equal to x . Let the expected payoff be p . So,

$$\begin{aligned} p &= -1 + \sum_{i=x}^{100} \frac{i}{100} + \sum_{i=1}^{x-1} \frac{p}{100} \\ \Rightarrow p &= -1 + \frac{(100-x+1)(100+x)}{200} + \frac{(x-1)}{100} p \\ \Rightarrow p &= \frac{(101-x)(100+x) - 200}{2(101-x)} \end{aligned}$$

Now,

$$\frac{dp}{dx} = \frac{10001 - 202x + x^2}{4(101-x)^2}$$

Setting $\frac{dp}{dx} = 0$:

$$\begin{aligned} \Rightarrow (10001 - 202x + x^2) &= 0 \\ \Rightarrow x^* &= \frac{202 \pm \sqrt{800}}{2} = 101 \pm 10\sqrt{2} \end{aligned}$$

Since $x \leq 100$,

$$x^* = 101 - 10\sqrt{2} \approx 86.9$$

Hence, the optimal x^* is either 86 or 87. Plugging both values into p , we see $x^* = 87$ maximizes p . The maximum p is $\frac{1209}{14}$.

Answer:

477. 3 Bus Stops

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume that the initial number of people on the bus is x . After the first stop, the number of people remaining are $\frac{1}{4}x + 7$. Similarly, after the second stop, the number of people remaining are $\frac{1}{4}(\frac{1}{4}x + 7) + 7 = \frac{1}{16}x + \frac{35}{4}$. After the third stop, the number of people remaining are $\frac{1}{4}(\frac{1}{16}x + \frac{35}{4}) + 7 = \frac{x+588}{64}$. x should be such that $x + 588$ must be divisible by 64. The least such x is 52. Hence, the initial number of people on the bus is 52, and hence, after 3 stops, there are 10 people remaining.

Answer:

478. Least Multiple Of 15

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For a number to be divisible by 15, it must be divisible by both 3 and 5. If a number is divisible by 5, its unit digit must either be 0 or 5. In our case, it must be 0 since 5 is not allowed. For a number to be divisible by 3, the sum of its digits must be divisible by 3. Hence, we can add three 1s at the beginning, and this would be the smallest number divisible by 15 with digits 0 and 1. So, the answer is 1110.

Answer:

479. Password

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To solve this, we use the "stars and bars" method for counting combinations with repetition. We need to find the number of ways to distribute 7 indistinguishable "stars" (digits) into 10 distinguishable "bins" (possible digit values from 0 to 9) where the order of digits is non-decreasing. We need to find the number of non-negative integral solutions to the equation: $x_0 + x_1 + x_2 + \cdots + x_9 = 7$ where x_i represents the number of times digit i appears. The number of non-negative integral solutions to this equation is given by:

$$\binom{7 + 10 - 1}{10 - 1} = \binom{16}{9}$$

Calculating this:

$$\binom{16}{9} = 11,440$$

Thus, there are 11,440 such non-decreasing 7-digit codes.

Answer:

480. Bridge Crossing

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Instead of sending Alice and Bob first, Charlie and Daniel should make the initial crossing, taking two minutes. Afterward, Daniel returns with the lantern, taking one minute. Then Alice and Bob cross together, which takes ten minutes. Once they are across, Charlie goes back with the lantern, taking two minutes. Finally, Charlie and Daniel cross again, which takes another two minutes. The total time required is 17 minutes.

Answer:

481. Grid Conversion

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

What if it was a 2×2 grid filled with 1, 2, 3, 4, as shown in Hint 1? Could we convert the left grid to the right grid? Through experimentation, we observe that for a 2×2 grid, it is possible to transform the left grid into the right grid. We can achieve this by increasing all elements in the 2nd column by 1 and then decreasing all elements in the 2nd row by 1.

What about a 3×3 grid? For a 3×3 grid, we can convert the left grid to the right grid as follows: increase the 3rd column 2 times, decrease the 3rd row 2 times, increase the first row 2 times, and decrease the 1st column 2 times.

A similar strategy would work for the 4×4 grid. Increase 4th column 9 times, decrease 4th row 9 times, increase 3rd column 6 times, decrease 3rd row 6 times, increase second row 3 times, decrease second row 3 times.

Answer:

482. Ants On A Circle IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume each ant is carrying a flag. Imagine at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. Then, each flag travels exactly once around the track in the given time period, ending where it began. The ants themselves must remain in the same circular order in which they started, so they have experienced (at most) some rotation; every ant must move the same number of positions, say k positions clockwise. In particular, Alice returns to her initial position only if all the ants

do. Note, however, that if m ants are initially facing clockwise, then at any time there are always m ants moving clockwise and $24 - m$ moving counterclockwise. This is because, at each collision, a clockwise ant becomes counterclockwise and vice versa; or, you can think of it as conservation of angular momentum! In any case, the average ant moves $2m - 24$ cm clockwise during the experiment. Thus, we are back to the initial position if and only if $2m - 24$ is a multiple of 24; i.e., if $m = 0, 12$, or 24 . The first two possibilities (where all ants initially faced the same way) have negligible probability, but the last contributes a healthy 16.1180377%. To be precise, there are $2^{24} = 16,777,216$ ways to choose directions for the ants, of which

$$\binom{24}{0} + \binom{24}{12} + \binom{24}{24} = 1 + 1 + 2,704,156$$

bring Alice back to where she started. This gives the probability

$$\frac{2,704,158}{16,777,216} \sim 0.161180377$$

Answer: 0.16118

483. Largest 3

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The sum of the three upfaces can be 18 only if all the upfaces are 6. Hence, the required probability is simply the probability of getting at least 3 sixes in 5 rolls.

- The probability of getting exactly 5 sixes is $\frac{1}{6^5}$.
- The probability of getting exactly 4 sixes is $\binom{5}{1} \cdot \frac{1}{6^4} \cdot \frac{5}{6}$.
- The probability of getting exactly 3 sixes is $\binom{5}{2} \cdot \frac{1}{6^3} \cdot \left(\frac{5}{6}\right)^2$.

Hence, the required probability is

$$\frac{1}{6^5} + \binom{5}{1} \cdot \frac{1}{6^4} \cdot \frac{5}{6} + \binom{5}{2} \cdot \frac{1}{6^3} \cdot \left(\frac{5}{6}\right)^2 = \frac{23}{648}.$$

Answer: 0.03549

484. Increasing Game

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

When Alice begins with 2, Bob is restricted to choosing a number between 3 and 10. No matter what Bob picks, Alice should then choose 11. This creates a consistent strategy where Alice always picks numbers of the form $9p + 2$, where p is an integer. The largest number in this form below 200 is 191. Once Alice says 191, Bob will be forced to choose a number between 192 and 199. Whatever Bob selects, it will be within 8 of 200, allowing Alice to say 200 on her next turn, securing her victory.

Answer: 1

485. Learning To Walk

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Suppose we create a state diagram with states $S_0, S_1, \dots, S_n$, where S_n represents the time a baby spends at the state n steps from the couch. Using our decision matrix

$$\begin{bmatrix} 0.2 & 0.5 & 0.3 \end{bmatrix},$$

we can map the state diagram of the baby. The baby will be at the couch, or S_0 , after being at the S_0 in the previous state and standing still or moving back towards the couch from S_1 . Thus,

$$S_0 = 0.8S_0 + 0.3S_1$$

Similarly, we get:

$$S_1 = 0.2S_0 + 0.5S_1 + 0.3S_2$$

$$S_2 = 0.2S_1 + 0.5S_2 + 0.3S_3$$

$$S_3 = 0.2S_2 + 0.5S_3 + 0.3S_4$$

$$\vdots$$

$$S_n = 0.2S_{n-1} + 0.5S_n + 0.3S_{n+1}$$

By simplifying these equations, we can identify a pattern:

$$S_0 = 0.8S_0 + 0.3S_1 \implies S_0 = 1.5S_1$$

Substituting $S_0 = 1.5S_1$ into our S_1 equation, we get:

$$S_1 = 0.2(1.5S_1) + 0.5S_1 + 0.3S_2 \implies S_1 = 1.5S_2$$

It is evident that all subsequent state equations can be re-written as

$$S_n = 1.5S_{n+1}$$

What that tells us is the baby will spend $\frac{2}{3}$ as much time in S_{n+1} compared to S_n . We know that all these states must sum to one and have a common ratio $\frac{2}{3}$. We can create the infinite geometric sum:

$$S_0 + \frac{2}{3}S_0 + \frac{4}{9}S_0 + \cdots + \left(\frac{2}{3}\right)^n S_0 + \cdots = 1$$

$$\implies S_0 \cdot \frac{1}{1 - \frac{2}{3}} = 1 \implies S_0 = \frac{1}{3}$$

Thus, the baby will be at the couch $\frac{1}{3}$ of the time.

Answer: 0.33333

486. Penguin Walk I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the expected number of steps required to reach position 1 be s . On the first step, the penguin can jump to position 1 with probability $\frac{2}{3}$. In this case, the number of steps required to reach position 1 will be 1. Alternatively, on the first step, the penguin can jump to position -1 with probability $\frac{1}{3}$. The expected number of steps required to reach position 1 in this case will be $2s + 1$ as the expected number of steps required to go from position -1 to 0 is the same as the expected number of steps required to go from position 0 to 1. Hence,

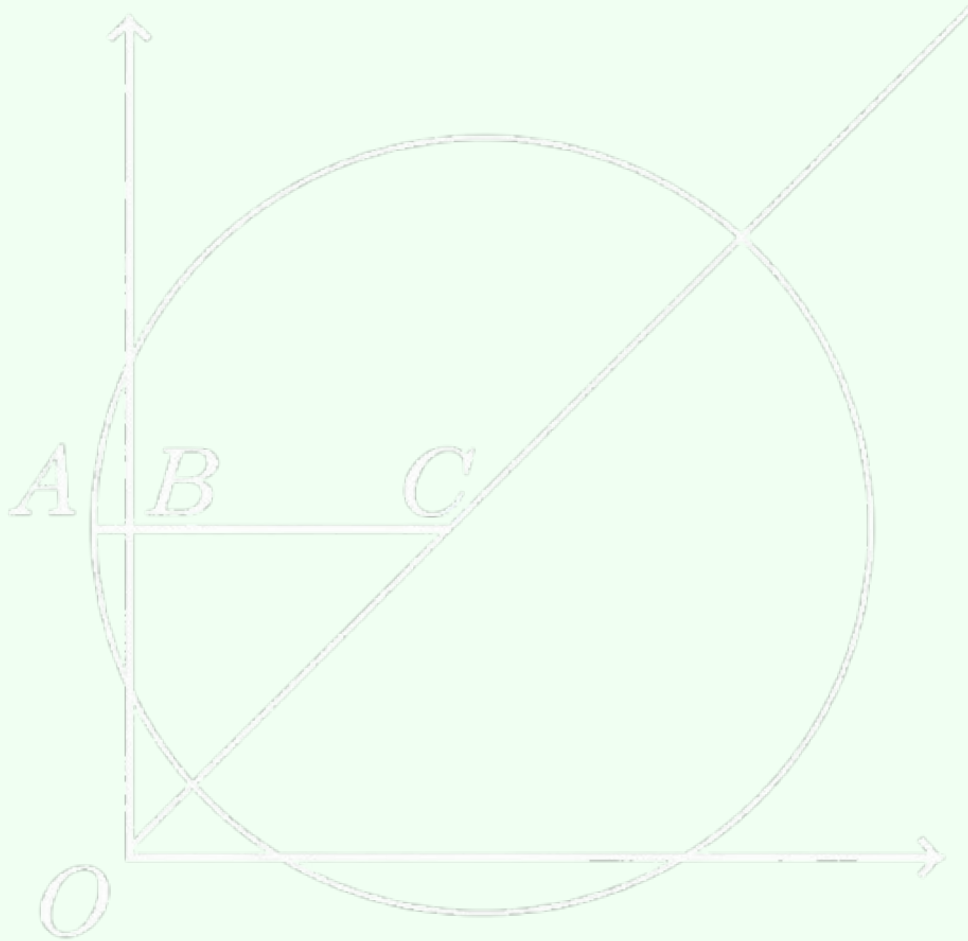
$$\begin{aligned} s &= \frac{2}{3} \cdot 1 + \frac{1}{3} \cdot (2s + 1) \\ \implies s &= 3 \end{aligned}$$

Answer: 3

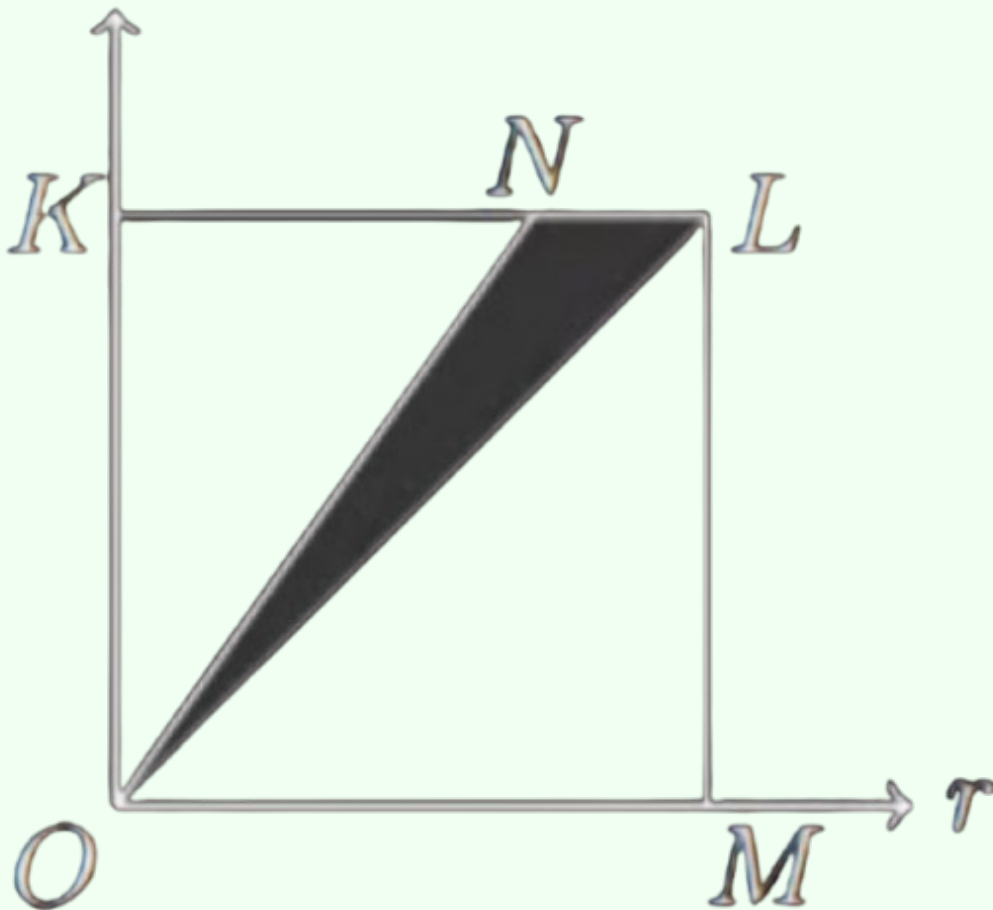
487. Circle In 3 Quadrants

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let C be the center of the circle and let AC be the radius perpendicular to the y -axis. Set $OC = d$ and $AC = r$; it quickly follows that $BC = \frac{d}{\sqrt{2}}$.



Now it follows that our desired condition that some part of the circle lies nontrivially in quadrants II and IV but not III occurs iff $\frac{d}{\sqrt{2}} < r < d$. Since r and d are chosen uniformly and independently randomly over the range $[0, 1]$, we can interpret (r, d) as a random point uniformly chosen in the unit square OKLM, as shown below.



Then the solution to the original two-sided inequality $\frac{\sqrt{d}}{2} < r < d$ is represented by exactly the points inside the gray triangle $\triangle ONL$, which is bounded between the lines $r = d$, $\sqrt{2}r = d$, and $d = 1$. Then $OK = 1$ and $NL = 1 - \frac{\sqrt{1}}{2}$, so it follows that our desired probability is

$$\frac{[ONL]}{[OKLM]} = \frac{NL \cdot KO}{2} = \frac{2 - \sqrt{2}}{4}.$$

Answer:

488. Passing The Parcel

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p_0 be the probability of winning when you have the parcel. Let p_1 be the probability of winning when one of your immediate neighbors has the parcel. Let p_2 be the probability of winning when one of the remaining two players has the parcel. The probabilities satisfy the following equations:

$$p_0 = \frac{1}{3} + \frac{2}{3}p_1$$

$$p_1 = \frac{1}{3}p_0 + \frac{1}{3}p_2$$

$$p_2 = \frac{1}{3}p_1 + \frac{1}{3}p_2$$

Solving these equations, we find:

$$p_0 = \frac{5}{11}$$

Answer:

489. Hands Interchange

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First, we know that the minute hand moves 6 degrees per minute, and the hour hand moves 0.5 degrees per minute. Let's assume Emma left at x minutes past four. At that time, the minute hand will be at $6x$ degrees from the origin, and the hour hand will be at $120 + 0.5x$ degrees, because the hour hand starts at 120 degrees at 4:00 and moves 0.5 degrees for each minute. Similarly, when Emma returned at y minutes past seven, the minute hand would be at $6y$ degrees from the origin, and the hour hand would be at $210 + 0.5y$ degrees, as the hour hand starts at 210 degrees at 7:00 and moves 0.5 degrees per minute. We are given that the positions of the minute and hour hands swap, so we can set up the following system of equations:

$$6x = 210 + 0.5y$$

$$6y = 120 + 0.5x$$

After simplifying and solving for x and y , we get $x \approx 36.9$ and $y \approx 23.03$. Hence, Emma left approximately 36.9 minutes past 4:00, and returned approximately 23.03 minutes past 7:00.

Answer:

490. Creating Zero

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The invariant is the parity of the sum of elements in the set. Initially, the sum is $\frac{50 \times 51}{2} = 1275$, which is odd. After every operation, the sum of elements in the set decreases by $2 \times \min(a, b)$. Hence, the sum of elements in the set remains odd after each operation. Since you cannot achieve an even sum (such as 0) from an odd sum by subtracting even values, it is impossible to get 0 as the last remaining element.

Answer:

491. Two Coins Remaining II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Any group of 5 consecutive heads can be reduced to a group of 2 heads using the following sequence of steps $HHHHH \rightarrow TTHH \rightarrow THT \rightarrow HH$. So, if the initial number of coins is $3k + 2$ for some integer k , we can reach 2 coins by removing 2 coins repeatedly using the above sequence of operations. If the initial number of coins is $3k$ for some integer k , we can reach 3 coins showing heads by removing 3 coins repeatedly using the above sequence of operations. Once we have 3 coins showing heads, we can remove the middle coin and end up with 2 coins. If the initial number of coins is $3k + 1$, we can't end up with 2 coins. Consider the following 2 invariants: 1. The number of tails is always even. Therefore, if we want to end with two coins, they must both be heads or must both be tails. 2. Count each head as $+1$ if it appears to the right of an even number of tails and -1 if it appears to the right of an odd number of tails. Then the total of these $+1$'s and -1 's is constant modulo 3. For example: $HHHHHTHTHTTH = 4 - 2 + 1 - 0 + 1 = 4$. If $n - 1$ is divisible by 3, then invariant 2 above is also congruent to 1 modulo 3. But HH is 2 modulo 3 and TT is 0 modulo 3, so it's impossible.

Answer:

492. Uniform Random Variables I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The conditional probability becomes:

$$\frac{P[P_1 > P_{10}, P_{20} > P_{10}]}{P[P_{10} < P_{20}]}$$

By the symmetry of the variables, $P[P_{10} < P_{20}]$ is $\frac{1}{2}$, meaning either one is equally likely to be larger. For the numerator, we seek the probability that P_{10} is the smallest among P_1, P_{10}, P_{20} . Each variable is equally likely to be the smallest, so the numerator is $\frac{1}{3}$. Thus, the desired probability is:

$$\frac{1}{3} \div \frac{1}{2} = \frac{2}{3}$$

Answer:

493. Jumping Frog VIII[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In order to find the probability of landing on 10, we must multiply the amount of successful combinations by the probability of those combinations. Notice for any successful combination of steps, the probability must always be $\frac{1}{2^{10}}$. Now, we only need to find the amount of possibilities for steps since we know the probability of each combination occurring is the same. This can be done using stars and bars $C_0^9 + C_1^9 + C_2^9 + \dots + C_9^9 = 2^9$. Hence, the final answer is

$$\frac{2^9}{2^{10}} = \frac{1}{2}$$

Answer:

494. Even Or Odd[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume x is the amount bet on even, and y is the amount bet on odd. Your total bet is $4x + 6y$. The expected return can be written as $0.5 \times 6x + 0.5 \times 9y = 3x + 4.5y$. Thus, the expected return on the total bet is $\frac{3}{4} \times (4x + 6y)$. This implies that the expected return is $\frac{3}{4}$ of your wager, meaning the game offers no positive expected value, making it unwise to participate. So, the optimal amount to bet is 0.

Answer:

495. Additional Info Coin Flips I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let X and Y be the counts of tosses until first heads for Alice and Bob respectively. We seek

$$\mathbb{E}[X \mid X < Y].$$

Since each toss is fair, the chance Alice needs exactly n tosses (fail the first $n - 1$, then succeed) is

$$\Pr(X = n) = \left(\frac{1}{2}\right)^n$$

and the chance Bob's first head occurs after n tosses (i.e., he fails the first n) is

$$\Pr(Y > n) = \left(\frac{1}{2}\right)^n.$$

Hence

$$\Pr(X = n; X < Y) = \left(\frac{1}{2}\right)^{2n}$$

$$\Pr(X < Y) = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{2n} = \frac{1}{3}.$$

Therefore

$$\Pr(X = n \mid X < Y) = 3 \left(\frac{1}{4}\right)^n \Rightarrow \mathbb{E}[X \mid X < Y] = 3 \sum_{n=1}^{\infty} n \left(\frac{1}{4}\right)^n.$$

Let

$$S = \sum_{n=1}^{\infty} nr^n \quad \text{where} \quad r = \frac{1}{4}.$$

Then

$$rS = \sum_{n=1}^{\infty} nr^{n+1} = \sum_{m=2}^{\infty} (m-1)r^m.$$

So

$$S - rS = r + \sum_{n=2}^{\infty} r^n = r + \frac{r^2}{1-r} = \frac{r}{1-r}.$$

Therefore

$$S = \frac{r}{(1-r)^2} = \frac{1/4}{(3/4)^2} = \frac{1/4}{9/16} = \frac{4}{9}.$$

Hence, $\mathbb{E}[X \mid X < Y] = 3 \cdot \frac{4}{9} = \frac{4}{3}$.

Answer: 1.33333

496. Bulb Toggle II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If light bulb m is left on, it means that m has an odd number of odd factors. In other words, m is either a perfect square or twice a perfect square. No number is both a perfect square and twice a perfect square, so we can count such m by counting perfect squares and twice perfect squares separately, then adding. The number of perfect squares less than or equal to 2021 is $\lfloor \sqrt{2021} \rfloor = 44$. The number of twice perfect squares less than or equal to 2021 is $\lfloor \sqrt{2021/2} \rfloor = 31$. Therefore, the answer is $44 + 31 = 75$.

Answer: 75

497. Game On Grid

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Whenever Alice picks a square at position (i, j) , Bob should choose the square at $((i+2) \pmod{4}, j)$. By using this strategy, Bob will avoid completing a 2×2 block before Alice, ensuring that he always wins.

Answer: 2

498. Remove Neighbor

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that there are an odd number of people in the circle beside Sofiya and Marquis, so Sofiya and Marquis divide the circle into two arcs, one with an even number of people and the other with an odd number of

people. Sofiya's winning strategy will be to always remove a neighbor from the even side. This would leave Marquis with an odd number of people on either side of him (excluding Sofiya), so continuing this process would also leave Sofiya with an even side to remove a neighbor from. This process always leaves an odd number of people between Marquis and Sofiya whenever it is Marquis' turn, so he will never be able to remove Sofiya from the circle. This secures Sofiya's win.

Answer:

499. Double Value

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability of drawing a diamond is $\frac{1}{4}$. If the card is a diamond, the expected value of the card drawn is:

$$\frac{1}{13} \times (2 + 4 + 6 + \cdots + 26) = 14$$

The probability of not drawing a diamond is $\frac{3}{4}$. If the card is not a diamond, the expected value of the card drawn is:

$$\frac{1}{13} \times (1 + 2 + 3 + \cdots + 13) = 7$$

Using the law of total expectation, the expected value of a random card is:

$$\frac{1}{4} \times 14 + \frac{3}{4} \times 7 = 8.75$$

Answer:

500. Coin Toss Game II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To find the probability that Alice wins, denote this probability as $P(A)$, and note that Bob's winning probability is $P(B) = 1 - P(A)$. Let's analyze the situation based on Alice's first coin toss. If Alice's first toss is a tail (probability $\frac{1}{2}$), the game effectively resets with Bob taking the first toss, making Alice's chance of winning $P(A|T) = P(B) = 1 - P(A)$. If Alice's first toss is a head (also with probability $\frac{1}{2}$), we then consider Bob's toss. If Bob's toss is a tail, Alice loses immediately. But if Bob's toss is another head, the game resets again with Bob as the first player, and Alice's chance of winning is now $P(A|H)$. To summarize, when Alice's first toss is a head, we have:

$$P(A|H) = \frac{1}{2} \times 0 + \frac{1}{2} \times (1 - P(A)) = \frac{1}{2} \times (1 - P(A))$$

This simplifies to:

$$P(A|H) = \frac{1}{3}$$

Now, combining the probabilities for the two possible outcomes of Alice's first toss:

$$P(A) = \frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times (1 - P(A))$$

Solving this equation, we get:

$$P(A) = \frac{4}{9}$$

Answer:

501. Apples And Oranges[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider that the 4 apples partition the 100 oranges into 5 separate sections. On average, these sections are evenly distributed. Therefore, each section is expected to contain 20 oranges. After the final apple is drawn, one such section would remain, containing 20 oranges.

Answer: 20**502. King, Queen, Ace II**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Think of the twelve cards that matter for the stopping rule, namely the four kings, the four queens and the four aces, as dividers that interrupt runs of ordinary cards. Draw until the first divider appears. Just like the “first ace” problem, there are 12 dividers and 40 other cards, so the twelve dividers split those 40 non-dividers into 13 regions in expectation. Hence the expected position of the first divider is $\frac{40}{13} + 1 = \frac{53}{13}$.

The first divider that appears can be a king, a queen or an ace and these three types are equally likely because there are four of each initially. If the first divider is a king we still need a queen or an ace; if the first divider is a queen or an ace we still need a king. Treat these two cases separately and use the same divider reasoning for the remaining deck after the first divider.

Condition on the identity of the first divider. After the first divider the expected number of cards remaining is $52 - \frac{53}{13} = \frac{623}{13}$. If the first divider was a king then there remain 8 relevant dividers (the four queens and the four aces). Those 8 dividers split the remaining cards into 9 regions and the expected number of non-divider cards per region is $\frac{\frac{623}{13} - 8}{9} = \frac{519}{117}$. Therefore the expected number of additional draws until we hit a queen or an ace in this case is $\frac{519}{117} + 1 = \frac{636}{117}$.

If the first divider was a queen or an ace then there remain 4 relevant dividers (the four kings). Those 4 dividers split the remaining deck into 5 regions and the expected number of non-dividers per region is $\frac{\frac{623}{13} - 4}{5} = \frac{571}{65}$. Hence the expected number of additional draws until we see a king in this case is $\frac{571}{65} + 1 = \frac{636}{65}$.

Now, use the law of total expectation. The first divider contributes $\frac{53}{13}$ draws on average, and then with probability $\frac{1}{3}$ the first divider is a king and we add $\frac{636}{117}$ more draws on average, while with probability $\frac{2}{3}$ the first divider is a queen or an ace and we add $\frac{636}{65}$ more draws on average. Thus the total expected draws is $\frac{53}{13} + \frac{1}{3} \cdot \frac{636}{117} + \frac{2}{3} \cdot \frac{636}{65} = \frac{7261}{585} \approx 12.41197$.

So on average about 12.41197 cards must be drawn before we have seen at least one king and at least one card that is either a queen or an ace.

Answer: 12.41197**503. One Colored Face**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We use the conditional probability formula

$$P(B_1 | A) = \frac{|A \cap B_1|}{|A|},$$

where B_1 is the event that the selected cube has exactly one painted face, and A is the event that the cube has at least five unpainted faces. Note that since each small cube has six faces, event A implies the cube has either zero or one painted face.

In the $3 \times 3 \times 3$ cube, there is only one small cube that is completely surrounded on all sides — the interior cube — which has zero painted faces. So there is exactly one cube in B_0 , the set of cubes with zero painted faces. For cubes with exactly one painted face, we look at the center cube of each face of the large cube.

Each of the 6 faces has one such center cube, and these cubes are not on edges or corners, so they each have exactly one face painted. Thus, there are 6 such cubes in B_1 . Altogether, the event A (at least five unpainted faces) includes the one cube from B_0 and the six cubes from B_1 , so

$$|A| = 1 + 6 = 7.$$

Out of these, exactly 6 cubes belong to B_1 , so

$$|A \cap B_1| = 6.$$

Using the formula, we get

$$P(B_1 | A) = \frac{6}{7}.$$

So, the probability that the selected cube has exactly one painted face, given that it has at least five unpainted faces, is $\frac{6}{7}$.

Answer: 0.85714

504. Same Arc II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

From "same arc I" we know for $N > 2$ the probability that N independent uniform points on the circle all lie in some semicircle equals $\frac{N}{2^{N-1}}$. For $N = 1$ and $N = 2$ the probability is 1. By the definition of X we therefore have, for integers $N \geq 1$,

$$P(X > N) = \begin{cases} 1, & N = 1, 2, \\ \frac{N}{2^{N-1}}, & N \geq 3. \end{cases}$$

Use the tail-sum formula for a positive integer-valued random variable:

$$\mathbb{E}[X] = \sum_{n=0}^{\infty} P(X > n).$$

Write the expectation as

$$\mathbb{E}[X] = 1 + 1 + 1 + \sum_{N=3}^{\infty} P(X > N) = 3 + S, \quad S := \sum_{N=3}^{\infty} \frac{N}{2^{N-1}}.$$

So, we have,

$$S = \frac{3}{2^2} + \frac{4}{2^3} + \frac{5}{2^4} + \dots,$$

so

$$\frac{1}{2}S = \frac{3}{2^3} + \frac{4}{2^4} + \frac{5}{2^5} + \dots$$

Subtract the second line from the first, termwise:

$$\frac{1}{2}S = \frac{3}{2^2} + \sum_{k=3}^{\infty} \frac{1}{2^k} = \frac{3}{4} + \sum_{k=3}^{\infty} \frac{1}{2^k}.$$

Now,

$$\sum_{k=3}^{\infty} \frac{1}{2^k} = \frac{1/8}{1 - 1/2} = \frac{1}{4}.$$

Thus

$$\frac{1}{2}S = \frac{3}{4} + \frac{1}{4} = 1,$$

so $S = 2$. Therefore

$$\mathbb{E}[X] = 3 + S = 3 + 2 = 5.$$

Answer: 5

505. Count Digits

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For an integer k , the number of digits is $\lfloor \log_{10}(k) \rfloor + 1$. Hence, for the given problem, the number of digits is:

$$\lfloor 1000 \log_{10}(2) \rfloor + \lfloor 1000 \log_{10}(5) \rfloor + 2$$

Now,

$$\lfloor 1000 \log_{10}(2) \rfloor + \lfloor 1000 \log_{10}(5) \rfloor \leq 1000 \log_{10}(2) + 1000 \log_{10}(5) = 1000 \log_{10}(10) = 1000$$

Also,

$$\lfloor 1000 \log_{10}(2) \rfloor + \lfloor 1000 \log_{10}(5) \rfloor > (1000 \log_{10}(2) - 1) + (1000 \log_{10}(5) - 1) = 1000 \log_{10}(10) - 2 = 998.$$

Hence,

$$\lfloor 1000 \log_{10}(2) \rfloor + \lfloor 1000 \log_{10}(5) \rfloor$$

can either be 999 or 1000. The equality

$$\lfloor a \rfloor + \lfloor b \rfloor = a + b$$

holds only when both a and b are integers. However, $1000 \log_{10}(2)$ and $1000 \log_{10}(5)$ are not integers because $\log_{10}(2)$ and $\log_{10}(5)$ are irrational. Thus,

$$\lfloor 1000 \log_{10}(2) \rfloor + \lfloor 1000 \log_{10}(5) \rfloor = 999.$$

Therefore, the required number of digits is:

$$999 + 2 = 1001.$$

Answer: 1001

506. Truck Parking

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

It is easier to calculate the probability that Bob cannot park. Bob cannot park if none of the remaining 4 spaces are adjacent. There is a one-to-one correspondence between all such cases and the number of ways to pick four spots out of 13. So, the probability Bob can park is:

$$1 - \frac{\binom{13}{4}}{\binom{16}{4}} = 1 - \frac{13 \cdot 12 \cdot 11 \cdot 10}{16 \cdot 15 \cdot 14 \cdot 13} = 1 - \frac{11}{28} = \frac{17}{28}$$

Answer: 0.60714

507. Unknown Balls

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $n = 100$. By symmetry, the probability that the first ball to roll out is black is $\frac{1}{2}$. The probability that there are exactly k black balls and the first ball to roll out is black is $\frac{k}{n(n+1)}$. Thus, the probability that there are k black balls given that the first ball is black is $\frac{2k}{n(n+1)}$. Therefore, the probability there are k black balls and the second ball is black given that the first one is black is

$$\frac{2k(k-1)}{n(n+1)(n-1)} = \frac{4\binom{k}{2}}{n(n+1)(n-1)}$$

The final probability is

$$\sum_{k=2}^n \frac{4\binom{k}{2}}{n(n+1)(n-1)} = \frac{4\binom{n+1}{3}}{n(n+1)(n-1)} = \frac{2}{3}$$

Answer:

508. Coin Transactions

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Define a cycle of length n to be a sequence of n transactions, starting and ending with the same person. For example, $A \rightarrow B, B \rightarrow C, C \rightarrow A$ would be a cycle of 3.

Case 1: Two different cycles of 2: There are $\binom{4}{2}$ ways to choose the people for each cycle, and $\binom{4}{2} = 15$ possible ways to select the two cycles. Additionally, there are $4! = 24$ ways to order the transactions. This results in $15 \cdot 24 = 360$ ways for this case.

Case 2: Two identical cycles of 2: There are $\binom{4}{2}$ ways to choose the two people involved in the cycle, and multiplying by $\binom{4}{2}$ gives the number of orderings of the four transactions. Thus, there are $6 \cdot 6 = 36$ ways for this case.

Case 3: One cycle of 4: We choose the person A gives a coin to in 3 ways, and the person who gives a coin to A in 2 ways. The remaining two transactions get determined automatically. There are $4! = 24$ ways to order the four transactions, giving $3 \cdot 2 \cdot 24 = 144$ ways for this case.

Adding these cases together, the total number of ways is:

$$360 + 36 + 144 = 540.$$

The total number of possible outcomes is $12^4 = 20736$, so the probability is:

$$\frac{540}{20736} = \frac{5}{192}.$$

Answer:

509. Red Blue Game

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We let $E(r, b)$ be the maximum expected dollars you can earn when there are r red marbles and b blue marbles remaining, and you must guess a color before each draw. At state (r, b) you compare two choices: Guess red, which yields the expectation:

$$R = \frac{r}{r+b}(1 + E(r-1, b)) + \frac{b}{r+b}(0 + E(r, b-1))$$

Guess blue, which yields:

$$B = \frac{b}{r+b}(1 + E(r, b-1)) + \frac{r}{r+b}(0 + E(r-1, b))$$

You choose whichever is larger, so the recurrence for $r, b \geq 1$ is

$$E(r, b) = \max\{R, B\}$$

The base cases are:

$$E(r, 0) = r, \quad E(0, b) = b$$

since with only one color left you always guess that color and succeed each draw. Filling in $0 \leq r, b \leq 3$ yields this table of optimal values:

$r \backslash b$	0	1	2	3
0	0	0	0	0
1	1	$\frac{3}{2}$	$\frac{7}{3}$	$\frac{13}{4}$
2	2	$\frac{7}{3}$	$\frac{17}{6}$	$\frac{18}{5}$
3	3	$\frac{13}{4}$	$\frac{18}{5}$	$\frac{41}{10}$

In particular, starting from three reds and three blues, the optimal expected payout is

$$E(3, 3) = \frac{41}{10}$$

Answer:

510. Hard Card Expectation

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

To solve this problem, we first consider the two possible outcomes: drawing a king or an ace first. The expected number of draws to obtain either a king or an ace is $\frac{44}{9} + 1$, as the remaining 44 cards are divided into 9 segments by the 4 kings and 4 aces. Thus, the first king or ace is drawn on average at the $\frac{44}{9} + 1$ -th card. If the first card is a king (with probability $\frac{1}{2}$), we then need to draw either another king or an ace. The expected number of additional draws in this case is $\frac{52-44-1-7}{8} + 1$, as there are 7 remaining kings and aces dividing the remaining 52 cards into 8 segments. If the first card is an ace (with probability $\frac{1}{2}$), we only need to draw a king. The expected number of additional draws here is $\frac{52-44-1-4}{5} + 1$, as the 4 remaining kings divide the remaining deck into 5 segments. Hence, the required expected number of draws is:

$$\frac{44}{9} + 1 + \frac{1}{2} \cdot \frac{424}{45} + \frac{1}{2} \cdot \frac{424}{72} = 13.54444$$

Answer:

511. Poisoned Wine I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We can label the bottles from 0 to $n - 1$ using their binary representation. Since there are 10 servants, each bottle can be labeled with up to 10 binary digits. Let each servant correspond to a binary digit (from 0 to 9). If bottle i is labeled as $a_0 + a_1 \cdot 2^1 + \dots + a_9 \cdot 2^9$, where each a_k is either 0 or 1, we give wine from bottle i to the servants whose index k satisfies $a_k = 1$. For example, if bottle $i = 17$, its binary representation is 0000010001_2 , meaning we give wine from bottle 17 to servants 0 and 4. After one month, the combination of which servants die will uniquely identify the poisoned bottle. If servants 0 and 4 die, this indicates bottle 17 is poisoned. This method creates a one-to-one mapping between the bottles and binary sequences of length 10. Therefore, the maximum number of bottles n is $2^{10} = 1024$, since we can only create 1024 unique binary sequences with 10 digits. If n were greater than 1024, it would be impossible to uniquely identify the poisoned bottle. Thus, the maximum n is 1024.

Answer: 1024**512. Non-negative Expectation**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability of getting two sixes is $\frac{1}{36}$. The probability of getting a six and a non-six is $\frac{10}{36}$, since there are 10 ways to get a six and a non-six (6 on the first die, 1-5 on the second die or 1-5 on the first die and 6 on the second die). Thus, the probability of getting two sixes before getting a six and a non-six is

$$\frac{\frac{1}{36}}{\frac{1}{36} + \frac{10}{36}} = \frac{1}{11}$$

Now, the expected payoff is

$$\frac{1}{11} \times 100 + \frac{10}{11} \times (-x)$$

For a non-negative expectation,

$$\frac{1}{11} \times 100 + \frac{10}{11} \times (-x) \geq 0$$

This simplifies to $x \leq 10$. Hence, the maximum value of x for a non-negative expectation is 10.

Answer: 10**513. Regular 21-gon**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider the case where the diagonals we choose do not share vertices. In this case, the probability that they intersect is simply $\frac{1}{3}$, since if we select any 4 points, there are 3 ways to choose which pairs get connected, and only one of these has the diagonals intersecting. There are $n \cdot \binom{n-1}{2}$ ways to select pairs of segments that share a vertex, $\frac{\binom{n}{2} \cdot \binom{n-2}{2}}{2}$ ways to select pairs of segments that don't share a vertex, and $\binom{\binom{n}{2}}{2}$ ways to select distinct pairs of segments. The probability we select non-intersecting line segments is thus:

$$\frac{\frac{\binom{n}{2} \cdot \binom{n-2}{2}}{6} + n \cdot \binom{n-1}{2}}{\binom{\binom{n}{2}}{2}}$$

This simplifies to

$$\frac{(n-2)(n+9)}{6 \cdot \binom{n}{2} - 1} = \frac{19 \cdot 5}{209} = \frac{5}{11}$$

Answer:

514. Maximum Till Now I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For non-negative integer-valued random variables, we have $E[k] = \sum_{n=0}^{\infty} P(k > n)$ according to the tail sum formula. Clearly, $P(k > 0) = 1$. Also, $P(k > 1) = 1$. Now, for $k > n$, we must have $k_1 < k_2 < \dots < k_n$. Hence, the probability $P(k > n) = \frac{1}{n!}$, since there are $n!$ ways of arranging $k_1, k_2, \dots, k_n$ and only one of those ways makes $k_1 < k_2 < \dots < k_n$. Hence, the required expected value is:

$$E[k] = \sum_{n=0}^{\infty} P(k > n) = \sum_{n=0}^{\infty} \frac{1}{n!} = e$$

Answer:

515. Home Run

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p denote the probability that Alice hits more home runs than Bob. Alice hits more home runs if she hits a home run on her first try when Bob does not, or if they both hit home runs on their first try and Alice hits more home runs afterward. The probability of the first case is $\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$. The probability of the second case, where they both hit home runs and continue competing, is $\frac{1}{3} \times \frac{1}{2} \times p = \frac{p}{6}$. Thus, the total probability is given by the equation:

$$p = \frac{1}{6} + \frac{p}{6}$$

Solving for p :

$$p - \frac{p}{6} = \frac{1}{6}$$

$$\frac{5p}{6} = \frac{1}{6}$$

$$p = \frac{1}{5}$$

Answer:

516. Game Of Placing Kings

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Alice should start by placing a king at the center of the grid. Then, for the rest of the game, she can mirror Bob's moves across the center square. For example, if Bob places a king one square to the left and two squares up from the center, Alice places her king one square to the right and two squares down from the center. This way, after every pair of moves, the board stays symmetric about the center square. So, if Bob makes a legal move, then the mirrored move is also legal for Alice. Hence, Alice has a legal move whenever Bob does and thus, Alice has a winning strategy.

Answer:

517. Same Temperature

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The temperature function is continuous and cannot continuously increase or decrease along a circle. Consider any great circle, such as the Equator. Let $t(\theta)$ denote the temperature at point θ . Define $f(\theta) = t(\theta) - t(\theta + \pi)$. Observe that: $f(0) = t(0) - t(\pi)$ $f(\pi) = t(\pi) - t(2\pi) = t(\pi) - t(0) = -f(0)$ This implies:

- If $f(0) = 0$, then $f(\pi) = 0$, so $t(0) = t(\pi)$.
- If $f(0) \neq 0$, then $f(0)$ and $f(\pi)$ have opposite signs. By the Mean Value Theorem, since $f(\theta)$ changes sign between $\theta = 0$ and $\theta = \pi$, there must be some point where $f(\theta) = 0$.

Thus, there exist pairs of diametrically opposite points on the Earth's surface with the same temperature.

518. 4x4 Array

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Call a filled 4×4 array satisfying the given conditions cool. There are $4!$ possibilities for the first row; without loss of generality, let it be 1 2 3 4. Since each quadrant must contain all four numbers, we have exactly four possibilities for the second row: (i) 3 4 1 2 (ii) 3 4 2 1 (iii) 4 3 1 2 (iv) 4 3 2 1 I claim that the number of cool arrays with (i) is equal to those with (iv), and that the number of cool arrays with (ii) is equal to those with (iii). Let's first consider (i) and (iv). Now, (i) is:

$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{array}$$

while (iv) is:

$$\begin{array}{cccc} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{array}$$

In (iv), switch 3 and 4 (relabeling doesn't affect the coolness of the array); then, it becomes:

$$\begin{array}{cccc} 1 & 2 & 4 & 3 \\ 3 & 4 & 2 & 1 \end{array}$$

Now, interchange the last two columns, which also does not affect the coolness. This gives us (i). Hence, the cool arrays with (i) and the cool arrays with (iv) have a 1 : 1 correspondence. Using the exact same argument, we can show that the number of cool arrays with (ii) equals those with (iii). So we only need to consider cases (i) and (ii). It is easy to verify that there are four cool arrays with (i), determined precisely by, say, the first two entries of the third row; and two with (ii), determined precisely by, say, the first entry of the third row. Hence, the answer is $4! \times (4 + 2) \times 2 = 288$.

Answer: 288

519. Chess Tournament VII

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For the top 2 rated players to play in the final, they must be placed in opposite halves of the bracket in the first round. This happens with probability:

$$\frac{64}{127}$$

Then, both players must win all of their matches to reach the final. The probability that one player wins 6 consecutive matches (assuming each match is independently won with probability $\frac{2}{3}$) is:

$$\left(\frac{2}{3}\right)^6$$

So, the probability that both top-rated players win all their matches is:

$$\left(\frac{2}{3}\right)^6 \cdot \left(\frac{2}{3}\right)^6 = \left(\frac{2}{3}\right)^{12}$$

Finally, the second-highest rated player must defeat the highest rated player in the final. The probability of this is:

$$\frac{1}{3}$$

Therefore, the required probability is:

$$\frac{64}{127} \cdot \left(\frac{2}{3}\right)^{12} \cdot \frac{1}{3} \approx 0.00129$$

Answer:

520. Final Number Possibilities

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the numbers written on the board be $a_1, a_2, a_3, \dots, a_n$. The product $(a_1 + 1)(a_2 + 1)(a_3 + 1) \dots (a_n + 1)$ is invariant. Initially, this product is $2 \times 3 \times \dots \times 101 = 101!$. Let the final number be x . Thus, $x + 1 = 101! \Rightarrow x = 101! - 1$. So, there is only one possible final number.

Answer:

521. Car Clusters

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Define X_i as an indicator variable such that $X_i = 1$ if the i -th car is not blocked by any slower car ahead, meaning it forms its own cluster, and $X_i = 0$ otherwise. The i -th car forms a new cluster if it's the fastest among the cars from its position onward. The probability of this happening is $\frac{1}{i}$. Thus,

$$E[X_i] = \frac{1}{i}$$

The total expected number of clusters, C , is the sum of the expected values of X_i for all cars:

$$E[C] = \sum_{i=1}^N E[X_i] = \sum_{i=1}^{10} \frac{1}{i}$$

This sum equal to 2.92897.

Answer:

522. Strange Party

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If p denotes the total number of people present before the bus arrives, then p must satisfy: $p \equiv 0 \pmod{12}$
 $p + 50 \equiv 0 \pmod{25} \Rightarrow p \equiv 0 \pmod{25}$ So, the minimum possible value of p is $\text{LCM}(12, 25) = 300$. Hence, the required number of adults before the bus arrived is $\frac{5}{12} \cdot 300 = 125$.

Answer:

523. Blackboard[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Read Hint 1 and Hint 2 before proceeding. By the result in Hint 2, starting from any $m \geq x$, the chance to ever hit $x - 1$ is $\frac{1}{x}$. We prove this claim by induction: Base case $m = x$: the chance to hit $x - 1$ is $\frac{1}{x}$. Inductive step: assume the claim holds for all k with $x \leq k < m$. Starting at m , the next number is uniform in $\{0, 1, \dots, m - 1\}$. - If the next number is $x - 1$, you succeed immediately. - If it is less than $x - 1$, you fail. - If it is $k \geq x$, by inductive hypothesis the chance to hit $x - 1$ from there is $\frac{1}{x}$. Total probability:

$$\frac{1}{m} + \frac{m - x}{m} \cdot \frac{1}{x} = \frac{1}{x}$$

Let A_k be the event " k appears at some time." Since the sequence is strictly decreasing, to have $A_{15} \cap A_{30} \cap A_{45}$ we must hit 45, then 30, then 15 in order. Therefore,

$$\Pr(A_{15} \cap A_{30} \cap A_{45}) = \Pr(A_{45}) \cdot \Pr(A_{30} \mid A_{45}) \cdot \Pr(A_{15} \mid A_{30}).$$

Using the induction result: - $\Pr(A_{45}) = \frac{1}{46}$ - $\Pr(A_{30} \mid A_{45}) = \frac{1}{31}$ - $\Pr(A_{15} \mid A_{30}) = \frac{1}{16}$ Multiplying, the final probability is:

$$\Pr(A_{15} \cap A_{30} \cap A_{45}) = \frac{1}{46} \times \frac{1}{31} \times \frac{1}{16}.$$

Answer: **524. Phone Connection**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Model the phones as vertices of a graph and each wire as an edge between the two phones it joins. Let d_i be the degree of vertex i , the number of wires incident to phone i . The total number of wire-ends counted over all phones is $\sum_{i=1}^{25} d_i$, and every wire contributes exactly two ends, so the number of wires equals $\frac{1}{2} \sum_{i=1}^{25} d_i$. If every phone has degree 15 then $\sum_{i=1}^{25} d_i = 25 \cdot 15 = 375$. That would force the number of wires to be $\frac{375}{2} = 187.5$, which is not an integer, so such a wiring cannot exist.

Answer: **525. Triangle Formation I**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

A triangle will not be valid if any one of $X > Y + Z$, $Y > X + Z$, or $Z > X + Y$ holds. These events are mutually exclusive, so by symmetry, we only need to find the probability of one and then multiply by 3. To calculate $P[Z > X + Y]$, consider the region where $z = x + y$ within the unit cube. This forms a tetrahedron with vertices at $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$. The volume of this tetrahedron is given by $\frac{1}{6}$, as the base is a right triangle with side lengths of 1 and the height is also 1. Since there are three such regions, the total probability that a triangle can't be formed is $\frac{1}{2}$, which means the probability that a triangle can be formed is also $\frac{1}{2}$.

Answer: **526. 3 Consecutive Heads II**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The sequences can be of the forms HHH _ _ _ HHH _ _ _ HHH, and _ _ _ HHH. The 1st and 4th cases are symmetric, while the 2nd and 3rd cases are also symmetric. Now, for the 1st case, the fourth spot must be a tail, and the other 2 blank spots can be heads or tails. So the 1st case contributes 4 sequences. Hence, the 4th case will also contribute 4 sequences. For the 2nd case, the 1st and 5th spots must be heads. The 6th spot can be heads or tails, so the 2nd case contributes 2 sequences. Hence, the 3rd case will contribute 2 sequences as well. So, the number of favorable sequences is $4 \times 2 + 2 + 2 = 12$. The required probability is

$$\frac{12}{2^6} = \frac{3}{16}$$

Answer: 0.18750

527. Too Many Unknowns

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The sum of the two rolls each person sees is what they say minus 14 (the expected sum of the rolls they don't see). Since the stated numbers are all different, the sum of the two rolls each person sees is a different number, which means that no two people look at each other's dice, so everyone looks in the same direction. Assume that Alice looks at Bob, Bob looks at Charlie, and so on. (The other case is identical.) The sum of the two rolls that Alice, Bob, Charlie, and Diana see, respectively, is 5, 8, 7, and 9. If we let a through f be the rolls of Alice through Fred, this means that $a + b = 5$, $b + c = 8$, $c + d = 7$, and $d + e = 9$. Adding the second and fourth equations and subtracting the first and third gives $e - a = 5$, meaning that $e = 6$ and $a = 1$. Thus, $b = 4$, $c = 4$, and $d = 3$. It remains to determine f , and note that $e + f$ and $f + a$ do not belong to the set $\{5, 7, 8, 9\}$, for this would violate the condition that everyone says a different number. This rules out everything but $f = 5$. Thus, we have

$$abcdef = 1 \cdot 4 \cdot 4 \cdot 3 \cdot 6 \cdot 5 = 1440.$$

Answer: 1440

528. Dice Triangle

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p denote the probability, and let the three numbers rolled be x , y , and z . We will instead calculate $1 - p$, which represents the probability that $x \geq y + z$, $y \geq z + x$, or $z \geq x + y$. Since these events are mutually exclusive, $1 - p$ equals 3 times the probability that $x \geq y + z$. This event occurs with probability

$$\frac{0 + 1 + 3 + 6 + 10 + 15}{216} = \frac{35}{216}$$

Therefore, the final result is:

$$1 - 3 \cdot \left(\frac{35}{216} \right) = 1 - \frac{35}{72} = \frac{37}{72}$$

Answer: 0.51389

529. Triangle Sum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The total sum of the numbers from 1 to 9 is 45. Since the sum of the integers on each side of the triangle is 17, the total sum across all three sides is 51. However, the integers at the vertices are counted twice (once for each adjoining side). Therefore, the sum of the numbers at the vertices must be $51 - 45 = 6$. Hence, the numbers on the vertices must be 1, 2, and 3. Since the rotations are considered similar, there are $(3 - 1)!$

ways to arrange these numbers on the vertices. Now, let side a have the numbers 1 and 2, side b have the numbers 1 and 3, and side c have the numbers 2 and 3 on their vertices. Hence, the sum of the numbers on side a (excluding the vertices) must be 14. Similarly, it should be 13 for side b and 12 for side c . To get the sum 14 on side a , it must either have the numbers 8 and 6 or 9 and 5. Once you put the numbers on side a , the entire triangle would be determined (why?). Hence, there are 2 ways of arranging the 6 numbers on the sides once the numbers on the vertices are arranged (remember, the problem states that the order of numbers on each side is irrelevant). Hence, the total number of arrangements are $(3 - 1)! \cdot 2 = 4$.

Answer: 4

530. Is It Fibonacci?

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let's express each of the first 10 terms using only f_1 and f_2 :

$$f_1 = f_1$$

$$f_2 = f_2$$

$$f_3 = f_2 + f_1$$

$$f_4 = 2f_2 + f_1$$

$$f_5 = 3f_2 + 2f_1$$

$$f_6 = 5f_2 + 3f_1$$

$$f_7 = 8f_2 + 5f_1$$

$$f_8 = 13f_2 + 8f_1$$

$$f_9 = 21f_2 + 13f_1$$

$$f_{10} = 34f_2 + 21f_1$$

Adding both sides, we see that the sum of the first 10 terms is equal to $88f_2 + 55f_1$. Therefore, the sum of the first 10 terms is:

$$88f_2 + 55f_1 = 11(8f_2 + 5f_1) = 11f_7 = 11 \cdot 83 = 913$$

Answer: 913

531. Dice Triangle II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If $t = 7$, then the triangle is a right triangle. This occurs with probability $\frac{1}{6}$. If $t > 7$, it is not possible to be a right triangle because a right triangle cannot have an obtuse angle. If $t < 7$, either the side of length a or the side of length b would have to be the hypotenuse, so that either $a \cos((t - 1)15^\circ) = b$ or $b \cos((t - 1)15^\circ) = a$, which means that $\cos((t - 1)15^\circ)$ must be rational. Of the possible angle measures 15° , 30° , 45° , 60° , and 75° , only $\cos(60^\circ) = \frac{1}{2}$ is rational, which corresponds to $t = 5$. When $t = 5$, either $a = 2b$ or $b = 2a$. The probability that $t = 5$ is $\frac{4}{36}$. The probability that $a = 2b$ is

$$\frac{3(1) + 5(2) + 5(3) + 3(4) + (1)(5)}{36^2} = \frac{45}{36^2}$$

This is also the probability that $b = 2a$. Since $a = 2b$ and $b = 2a$ are mutually exclusive, the probability that $t = 5$ and the triangle is right is

$$2 \cdot \frac{4}{36} \cdot \frac{45}{36^2}$$

We conclude that the desired probability is

$$\frac{1}{6} + 2 \cdot \frac{4}{36} \cdot \frac{45}{36^2} = \frac{113}{648}$$

Answer: 0.17438

532. Marble Swap III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We only need to track whether the blue marble is in the center, because if it is, the order is back to red-blue-red, and if it isn't, it must be on one of the edges (and the two reds are on the other two spots). Let p_n be the probability that after n moves the blue marble sits in the center. On move $n + 1$, you choose uniformly between the two edges to swap with the center. If the blue marble was on an edge (probability $1 - p_n$), then with probability $\frac{1}{2}$ you swap it into the center. If it was already in the center (probability p_n), then you swap it out to an edge for sure. Hence by the law of total probability,

$$p_{n+1} = \frac{1 - p_n}{2}$$

We also note $p_1 = 0$, since after the very first swap the blue must leave the center. Thus the linear recurrence is

$$p_{n+1} = \frac{1 - p_n}{2}, \quad p_1 = 0$$

This is an inhomogeneous first-order recurrence. Its steady-state p^* satisfies

$$p^* = \frac{1 - p^*}{2} \implies 3p^* = 1 \implies p^* = \frac{1}{3}$$

The general solution is

$$p_n = p^* + A \left(-\frac{1}{2} \right)^n$$

Enforcing $p_1 = 0$ gives

$$0 = \frac{1}{3} + A \left(-\frac{1}{2} \right) \implies A = \frac{2}{3}$$

so

$$p_n = \frac{1}{3} + \frac{2}{3} \left(-\frac{1}{2} \right)^n$$

In particular, after 15 moves,

$$p_{15} = \frac{1}{3} + \frac{2}{3} \left(-\frac{1}{2} \right)^{15} = \frac{1}{3} - \frac{1}{3 \times 2^{14}}$$

Hence the probability the marbles are back in red-blue-red after 15 moves is

$$\frac{1}{3} - \frac{1}{3 \times 2^{14}}$$

Answer: 0.33331

533. Jet Trail Intersection

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If the trails are extended infinitely in both directions, they will intersect with probability 1 (since the lines lie in the same plane, and the probability of them being parallel is negligible). Let the intersection point of the lines be X . The probability that the first ray passes through X is $\frac{1}{2}$ (since the ray can either go towards X or away from X , and both of these are equally likely). Similarly, the probability that the second ray also

passes through X is $\frac{1}{2}$. Hence, the required probability is

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Answer:

534. Increasing Decreasing I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The middle digit must be greater than the digits to its left and its right, so it must be 9. Now, there are $\binom{8}{4}$ ways to choose which four digits go before 9 and which four digits go after 9. For each of these possibilities, there is exactly one way to arrange all the digits so that the first four are increasing and the last four are decreasing. Therefore, the answer is 70.

Answer:

535. Game On Circle

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For an even number of points, Bob can always ensure a tie by using a mirroring strategy. For instance, for $n = 8$, divide the circle into two segments like $1 - 2 - 3 - 4$ and $5 - 6 - 7 - 8$. Mirror Alice's moves from one segment to the other: if Alice colors point 1, Bob colors point 5, and so on. This mirroring ensures that, by the end of the game, the longest consecutive segments of red and blue will be equal. Thus, Alice cannot guarantee a win.

Answer:

536. Tetrahedral Die I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the desired probability be p , so that p satisfies

$$p = \frac{1}{4}(1 + p + p^2 + p^3).$$

This equation simplifies to

$$(p^2 + 2p - 1)(p - 1) = 0,$$

so either $p = -1 \pm \sqrt{2}$ or $p = 1$. Since $-1 - \sqrt{2} < 0$, it can be rejected. Imagine that the dice-rolling process takes place in discrete units of time. Each minute, for example, every die is rolled, and the new copies are placed to the side to be rolled during the next minute. Let p_k be the probability any given die and all its copies vanish after k minutes. We will show by induction on k that $p_k < \sqrt{2} - 1$, so that $p = 1$ can also be rejected, and we can conclude $p = -1 + \sqrt{2}$. For the base case, $p_0 = 0 < \sqrt{2} - 1$. For the inductive step, suppose $p_k < \sqrt{2} - 1$. Then,

$$p_{k+1} = \frac{1}{4}(1 + p_k + p_k^2 + p_k^3) < \frac{1}{4}(1 + (-1 + \sqrt{2}) + (-1 + \sqrt{2})^2 + (-1 + \sqrt{2})^3) = \sqrt{2} - 1.$$

Thus, $p = \sqrt{2} - 1$.

Answer:

537. Highest Roll EV[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Because you will eventually see a 4, the possible highest numbers are 4, 5, and 6. Suppose k is the highest:

$$P(k = 4) = \frac{1}{3} \text{ since you see 4, 5, 6 first with equal probability.}$$

$$P(k = 6) = \frac{1}{2} \text{ since you will see 6 before 4 with probability } \frac{1}{2}.$$

$$P(k = 5) = \frac{1}{6} \text{ as only 1 of the } 3! \text{ arrangements works in 5's favor (see 5 before 4 and 4 before 6).}$$

You can illustrate this further by writing out all 6 permutations and evaluating which number would be the greatest in that given scenario.

$$E[k] = \frac{1}{3}(4) + \frac{1}{6}(5) + \frac{1}{2}(6) = \frac{31}{6}$$

Answer: **538. Chords Intersection II**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let X be the total number of interior intersections among the 9 chords. For each pair of chords, define an indicator variable X_{ij} , where $X_{ij} = 1$ if the i -th and j -th chords intersect, and $X_{ij} = 0$ otherwise. Then

$$X = \sum_{1 \leq i < j \leq 9} X_{ij}.$$

By linearity of expectation,

$$\mathbb{E}[X] = \sum_{1 \leq i < j \leq 9} \mathbb{E}[X_{ij}].$$

There are $\binom{9}{2} = 36$ pairs of chords in total. Among these, the 8 consecutive pairs $(1, 2), (2, 3), \dots, (8, 9)$ never contribute an interior intersection, so their expectation is 0. That leaves $36 - 8 = 28$ relevant pairs.

For any one of these 28 pairs, the four endpoints of the two chords are in random order around the circle. Among the three possible ways to pair four points into two chords, exactly one pairing gives a crossing. So for each such pair,

$$\Pr(X_{ij} = 1) = \frac{1}{3}, \quad \text{hence} \quad \mathbb{E}[X_{ij}] = \frac{1}{3}.$$

Therefore,

$$\mathbb{E}[X] = 28 \cdot \frac{1}{3} = \frac{28}{3}.$$

So the expected number of chord intersections is $\frac{28}{3}$.

Answer: **539. Increasing Birthdays**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To find the probability of the seven individuals sitting in age order, we first determine the total number of arrangements. Since the seating is circular, we can fix one person to eliminate equivalent arrangements due to rotation. Thus, we arrange the remaining six people, resulting in:

$$6! = 720 \quad (\text{total arrangements})$$

Next, we consider the successful arrangements where the individuals are seated in age order. There are two valid arrangements for age order:

* Increasing order in a clockwise direction. * Increasing order in a counterclockwise direction.

Thus, there are 2 favorable arrangements. The probability P that the seven people sit in age order is given by the ratio of favorable outcomes to total outcomes:

$$P = \frac{\text{Number of favorable arrangements}}{\text{Total arrangements}} = \frac{2}{720} = \frac{1}{360}$$

Answer:

540. Binary Trailing Zeros

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The number of trailing zeros in the binary representation of $142!$ corresponds to the highest power of 2 dividing $142!$. This is determined by the power of 2 in its prime factorization. To calculate this, we sum the contributions of multiples of 2, 4, 8, and so on, up to 142. The power of 2 is given by:

$$\left\lfloor \frac{142}{2} \right\rfloor + \left\lfloor \frac{142}{4} \right\rfloor + \left\lfloor \frac{142}{8} \right\rfloor + \dots$$

continuing until the division yields zero. Computing each term:

$$\left\lfloor \frac{142}{2} \right\rfloor = 71, \quad \left\lfloor \frac{142}{4} \right\rfloor = 35, \quad \left\lfloor \frac{142}{8} \right\rfloor = 17$$

$$\left\lfloor \frac{142}{16} \right\rfloor = 8, \quad \left\lfloor \frac{142}{32} \right\rfloor = 4, \quad \left\lfloor \frac{142}{64} \right\rfloor = 2, \quad \left\lfloor \frac{142}{128} \right\rfloor = 1.$$

Adding these values gives:

$$71 + 35 + 17 + 8 + 4 + 2 + 1 = 138.$$

Therefore, $142!$ has 138 trailing zeros in its binary representation.

Answer:

541. Vertical Hit

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If the frog is on one of the 2 diagonals, the probability of landing on vertical or horizontal each becomes $\frac{1}{2}$. Since it starts on $(1, 2)$, there is a $\frac{3}{4}$ chance (up, down, or right) it will reach a diagonal on the first jump and a $\frac{1}{4}$ chance (left) it will reach the vertical side. The probability of landing on a vertical is

$$\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8}$$

Answer:

542. Bacteria Population I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For a colony of three cells to become extinct, each of the three independent lineages must individually go

extinct. Since each cell has a $\frac{1}{3}$ chance of extinction, the probability that all three lineages go extinct is:

$$\left(\frac{1}{3}\right)^3 = \frac{1}{27}$$

Answer:

543. Count Regions IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Suppose that k spheres have already been drawn. Let us determine by how much the $(k+1)$ st sphere increases the number of regions. The $(k+1)$ st sphere intersects each of the first k spheres in a circle. These circles of intersection will all be different, no two of them will be tangent, and—viewed as curves on the $(k+1)$ st sphere—none will lie entirely inside or outside another. In "Count Regions II", it was shown that k circles in a plane divide the plane into $k^2 - k + 2$ regions. The same reasoning applies to circles on a sphere. Therefore, the surface of the $(k+1)$ st sphere is divided into $k^2 - k + 2$ regions by its intersections with the previous k spheres. Each of these regions bisects a 3D region formed by the previous k spheres. Thus, the $(k+1)$ st sphere increases the number of regions by $k^2 - k + 2$. Since a single sphere divides space into 2 regions, the total number of regions after drawing n spheres is:

$$2 + (1^2 - 1 + 2) + (2^2 - 2 + 2) + \cdots + ((n-1)^2 - (n-1) + 2) = \sum_{k=1}^{n-1} (k^2 - k + 2) + 2$$

This simplifies to:

$$\frac{n(n^2 - 3n + 8)}{3}$$

So, the maximum number of regions into which n spheres can divide 3D space is:

$$\frac{n(n^2 - 3n + 8)}{3}$$

Answer:

544. Expected Ping-pong Games

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The expected value of the ratio of A's to B's score is 3:7. So A is expected to win 9 games for each of B's 21. Thus the expected number of games in a match is 30.

Answer:

545. Small Chords

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the radius of the circle be R . A chord that subtends a central angle θ has length $2R \sin(\theta/2)$. For this to be less than R , we need

$$2R \sin(\theta/2) < R \implies \sin(\theta/2) < \frac{1}{2},$$

which means $\theta < 60^\circ$.

So all six chords are shorter than the radius exactly when every pair of chosen points is separated by less than 60° . That is equivalent to saying that all four points lie inside some arc of length 60° .

Now define E_i to be the event that, starting from point i and moving clockwise, all the other three points lie within the next 60° . For a fixed point i , each of the other three points has probability $\frac{1}{6}$ of landing in that 60° arc, so

$$P(E_i) = \left(\frac{1}{6}\right)^3$$

These events are disjoint. If all four points lie in some arc of length 60° , then there is only one point that can serve as the starting point of that arc. So at most one of the events E_1, E_2, E_3, E_4 can occur.

Therefore,

$$P(\text{all six chords are shorter than the radius}) = \sum_{i=1}^4 P(E_i) = 4 \left(\frac{1}{6}\right)^3 = \frac{1}{54}$$

So the probability is $\boxed{\frac{1}{54}}$.

Answer: 0.01852

546. Large Cycles

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Fix some k with $n < k \leq 2n$. Let Y_k be the number of k -cycles in the permutation. For each element $i \in \{1, \dots, 2n\}$, define the indicator

$$X_i = \begin{cases} 1, & \text{if } i \text{ lies in a cycle of length } k, \\ 0, & \text{otherwise.} \end{cases}$$

Since each k -cycle contains exactly k elements, we have

$$Y_k = \frac{1}{k} \sum_{i=1}^{2n} X_i.$$

By linearity of expectation,

$$E[Y_k] = \frac{1}{k} \sum_{i=1}^{2n} E[X_i].$$

All elements i are symmetric, so it suffices to compute $P(X_1 = 1)$. We count how many permutations of $\{1, \dots, 2n\}$ place element 1 in a cycle of length k : * Choose the other $k-1$ members of that cycle from the remaining $2n-1$ elements: $\binom{2n-1}{k-1}$ ways. * Arrange those k elements into a cycle containing 1: $(k-1)!$ ways (there are $(k-1)!$ cyclic orderings of the chosen k elements that fix 1's position). * Permute the remaining $2n-k$ elements arbitrarily: $(2n-k)!$ ways. Hence, the total number of permutations with 1 in a k -cycle is

$$\binom{2n-1}{k-1} (k-1)! (2n-k)!$$

and since there are $(2n)!$ permutations overall,

$$P(X_1 = 1) = \frac{\binom{2n-1}{k-1} (k-1)! (2n-k)!}{(2n)!}$$

Simplifying,

$$P(X_1 = 1) = \frac{1}{2n}.$$

Thus, $E[X_i] = \frac{1}{2n}$ for every i . So, $E[Y_k] = \frac{1}{k} \sum_{i=1}^{2n} E[X_i] = \frac{1}{k}$. Since C_n counts cycles of all lengths $k = n+1, n+2, \dots, 2n$,

$$E[C_n] = \sum_{k=n+1}^{2n} E[Y_k] = \sum_{k=n+1}^{2n} \frac{1}{k}.$$

We know that $\sum_{i=1}^n \frac{1}{i} \approx \ln(n)$ for large n . So,

$$\sum_{i=n+1}^{2n} \frac{1}{i} = \sum_{i=1}^{2n} \frac{1}{i} - \sum_{i=1}^n \frac{1}{i} \approx \ln(2n) - \ln(n) = \ln 2$$

for large n .

Answer: 0.69315

547. Cheating Husbands

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

It must be 10 husbands kicked out. If there was only 1 cheating husband in the town, there will be 99 women who know exactly who the cheater is. The 1 remaining woman, who is being cheated on, would have assumed there are no cheaters. But now that the mayor has confirmed that there is at least one cheater, she realizes that her own husband must be cheating on her. So her husband gets kicked on the day of the announcement. Now let's assume there are 2 cheaters in the town. There will be 98 women in the town who know who the 2 cheaters are. The 2 wives, who are being cheated on, would think that there is only 1 cheater in the town. Since neither of these 2 women know that their husbands are cheaters, they both do not report their husbands on the day of the announcement. The next day, when the 2 women see that no husband was kicked, they realize that there could only be one explanation – both their husbands are cheaters. Thus, on the second day, 2 husbands are kicked. Through induction, it can be proved that when this logic is applied to n cheating husbands, they are all kicked on the n^{th} day after the mayor's announcement. Hence it must be 10 husbands kicked in our case.

Answer: 10

548. Largest Prime Probability

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To get prime sums other than 2, we need to have both even and odd faces. If three faces are even and three are odd, then there will be 18 odd sums out of the 36 possible combinations. If two faces are odd and four are even, there will be 16 odd sums, and if one face is odd and five are even, then there will be 10 odd sums. So, the maximum number of prime sums out of 36 is 19 (if all odd sums are prime, and the sum 2 is achievable). This is achieved with the die $\{1, 2, 3, 4, 9, 10\}$, which yields the sum set $\{2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 18, 19, 20\}$, in which all odd numbers are prime. Thus, this die has a $\frac{19}{36} = 0.527$ probability of throwing a prime sum when rolled twice, compared to $\frac{5}{12} = 0.416$ for a standard die.

Answer: 0.52778

549. Jumping Frog VI

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Any valid path looks like:

$$0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_n \rightarrow 10$$

The first and last points are fixed (the first point is 0, and the last point is 10). The points $x_1, x_2, \dots, x_n$

can be any values in the closed interval $[1, 9]$. They must be distinct ($x_1 \neq x_2 \neq \dots \neq x_n$) and in increasing order ($x_1 < x_2 < \dots < x_n$).

Hence, the problem is equivalent to finding a (possibly empty) subset of the set:

$$\{1, 2, 3, \dots, 9\}$$

There are 2^9 such subsets because, for each element, you can decide either to include it in the subset or exclude it (2 choices for each element).

Thus, the number of ways in which the frog can reach point 10 is:

$$2^9 = 512$$

Answer: 512

550. Dice Game VIII

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Continue rolling until a number x or greater appears, at which point you take the money or re-roll. The probability of rolling x or higher is $\frac{21-x}{20}$. The expected number of rolls required to achieve a number x or greater is $\frac{20}{21-x}$. Therefore, the total number of turns used (including one to take the money) is

$$\frac{20}{21-x} + 1.$$

The average payoff for each successful roll is the average of the range from x to 20:

$$\text{Payoff} = \frac{20 \cdot 21/2 - (x-1) \cdot x/2}{21-x}$$

The total number of payoffs is given by

$$\frac{100}{1 + \frac{20}{21-x}}$$

Thus, the overall payoff can be expressed as

$$\frac{100}{1 + \frac{20}{21-x}} \cdot \frac{20 \cdot 21/2 - (x-1) \cdot x/2}{21-x}$$

By taking the derivative of this expression and calculating its maximum, we find that the optimal value occurs at $x = 6$, yielding a maximum payoff of approximately \$557.14286.

Answer: 557.14286

551. Simulate 1/3

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let e denote the expected number of coin flips required. In each iteration of 2 flips, either we get HH, TH, or HT (and the simulation ends), or we get TT (and the process continues). Hence,

$$e = \frac{3}{4} \times 2 + \frac{1}{4} \times (2 + e)$$

Therefore,

$$e = \frac{8}{3}$$

Answer: 2.66667

552. Count Sequences I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

You start at 0 and end at 4. At each stage, you add or subtract 1 from the previous number to get the current number. We perform these operations (+1 or −1) a total of 10 times. If we choose to +1 by x times, then −1 should be $(10 - x)$ times, and the result is:

$$x \cdot (+1) + (10 - x) \cdot (-1) = 2x - 10$$

To ensure this equals 4, we must have $x = 7$. So, the problem is equivalent to finding the number of ways to arrange 7 +1s and 3 −1s, which is:

$$\frac{10!}{3!7!} = 120$$

Answer: 120**553. Diagonal Cut II**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

From diagonal cut I, we know that for an $m \times n$ grid with $\gcd(m, n) = 1$, the answer is $m + n - 1$. For the current problem, the gcd is 18. We can break up the problem into 18 smaller rectangles, each with dimensions $\frac{234}{18} = 13$ and $\frac{306}{18} = 17$. For each of these smaller rectangles, the gcd of width and height is 1. Hence, for each of the rectangles, the number of squares cut is $13 + 17 - 1 = 29$. There are 18 such rectangles. Hence, the required answer is $18 \times 29 = 522$.

Answer: 522**554. Last Ball**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In a scenario with 20 blue balls and 14 red balls, we draw two balls at random without replacement. If both balls are blue, we add a blue ball to the bag. If the balls are of different colors, we add a red ball. This process continues until only one ball remains. The goal is to determine the color of the final ball.

Denote the number of blue and red balls as (B, R) . After drawing two balls, the following changes occur:

- If both balls are blue: $(B, R) \rightarrow (B - 1, R)$
- If both balls are red: $(B, R) \rightarrow (B + 1, R - 2)$
- If one ball is blue and one is red: $(B, R) \rightarrow (B - 1, R)$

We observe that the number of red balls R either remains unchanged or decreases by 2. Therefore, the total count of red balls will always be even. Since the total number of balls decreases by one in each operation, starting with an even number of red balls ensures that the final remaining ball must be blue. Conversely, if the initial count of red balls is odd, the last ball will be red.

Answer: 2**555. Treasure Island**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

It does not matter much where the grave is! There are only two places where the treasure can be. Let the unmarked grave be the origin. Let A and B denote vectors to the two oak trees. The first flag is at $A + iA$. The second flag is at $B - iB$. The mid-point is $\frac{1}{2}(A + B) + \frac{1}{2}(A - B)i$, which may be rewritten as $A + \frac{1}{2}(B - A) - \frac{1}{2}(B - A)i$. Note that $-iX$ denotes a counterclockwise rotation of vector X by 90 degrees.

Now the last expression has three summands, which can be visualized as follows: "A" denotes moving from the unmarked grave to the left oak tree, $\frac{1}{2}(B - A)$ denotes moving half the distance between the two oak trees from A to B, and $-\frac{1}{2}(B - A)i$ denotes turning counterclockwise by 90 degrees and moving the same amount as the previous step. Finally, the unmarked grave could be on either side of the line joining A and B. So there are two points where the treasure might be.

Answer: 2

556. Vector Toggle

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Model the set of $2^{10} = 1024$ binary vectors as the states of a Markov chain. From any state x the chain moves in one step to one of the ten neighbors that differ from x in exactly one coordinate, each chosen with probability $\frac{1}{10}$. Now note the two properties the chain satisfies: (i) whenever $p_{ij} > 0$ we also have $p_{ji} > 0$, and (ii) whenever $p_{ij} > 0$ then $p_{ij} = 1/d_i$, where d_i is the number of neighbors of state i . Kac's lemma states that for a finite irreducible Markov chain the expected return time to state j equals $1/\pi_j$, where π is the stationary distribution. Using the theorem described in the chapter "Expected First Passage and Return Times" together with properties (i) and (ii) we have for every state j

$$\pi_j = \frac{d_j}{\sum_k d_k}.$$

Here every state has degree $d_j = 10$, so

$$\sum_k d_k = 1024 \cdot 10 = 10240,$$

and hence $\pi_j = \frac{1}{1024}$ for every state j . By Kac's lemma the expected return time to any fixed state j is therefore

$$\frac{1}{\pi_j} = 1024.$$

Use this return time to find the desired hitting time. Let $t = (1, 0, 0, \dots, 0)$. Starting from t the chain makes one move and lands at one of the ten neighbors of t with probability $\frac{1}{10}$ each. If H denotes the expected time to hit t when started from any neighbor of t then the expected return time to t satisfies

$$\text{expected return to } t = 1 + \frac{1}{10} \sum_{\text{neighbors } k} \mathbb{E}_k[T_t] = 1 + H,$$

because the neighbors are all equivalent by symmetry and each appears with probability $\frac{1}{10}$ as the first step. Since the expected return time equals 1024, we get $1 + H = 1024$ and therefore $H = 1023$. The starting vector $(0, 0, 0, \dots, 0)$ is a neighbor of t , so the expected number of moves to reach $(1, 0, 0, \dots, 0)$ from $(0, 0, 0, \dots, 0)$ is 1023.

Answer: 1023

557. Prisoner Hats-multicolor

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Assign each of the 10 colors a unique value from 0 to 9. The last prisoner, who can see the hats of all others, will calculate the weighted sum of the number of visible hats of each color, using the assigned values as weights, and take this sum modulo 10. Let this result be v . The last prisoner will then announce the color corresponding to v . The second-last prisoner, knowing the declared value and the colors in front of them, can deduce their own hat color using this information. Each subsequent prisoner can use the previous declarations to calculate their hat color. This process ensures that 99 prisoners can be saved, while the first

prisoner survives with a probability of $\frac{1}{10}$.

Answer:

558. Tennis Odds

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will use the method of reduced sample space. The winning player must win 2 times consecutively. The probability that Alice wins 2 times consecutively is 0.6^2 . The probability that the other player wins 2 times consecutively is 0.4^2 . Hence, using the method of reduced sample space, the probability that Alice wins is,

$$\frac{0.6^2}{0.6^2 + 0.4^2} = \frac{9}{13}$$

Answer:

559. Pizza Standard Deviation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Given that the workers are independent, the variances add up. The variance of pizzas produced by one worker is $60^2 = 3600$. For 25 workers, the total variance is:

$$25 \times 3600 = 90,000$$

The standard deviation is the square root of this total variance:

$$\sqrt{90,000} = 300$$

Hence, the standard deviation of the total pizzas produced by all workers per day is 300 pizzas.

Answer:

560. Circuit

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider subsets of 4 bulbs. There are $\binom{12}{4} = 495$ such subsets. Let X_i be an indicator variable which takes the value 1 if the i -th subset forms a 4-circuit and 0 otherwise. So, the expected value of 4-circuits is:

$$E[X_1] + E[X_2] + \cdots + E[X_{495}]$$

By symmetry, each of these values is equal. Hence, the expected value of 4-circuits is:

$$495 \cdot E[X_1]$$

Now, $E[X_1] =$ the probability of a random subset of 4 bulbs forming a 4-circuit. There are $\binom{4}{2} = 6$ pairs of bulbs in total for a subset of 4 bulbs. Each of these pairs must be connected by a wire for a 4-circuit. Hence, the required probability of a random subset of 4 bulbs forming a 4-circuit is $\left(\frac{1}{2}\right)^6$. Thus, $E[X_1] = \left(\frac{1}{2}\right)^6$. Therefore, the expected number of 4-circuits is:

$$495 \cdot \left(\frac{1}{2}\right)^6 = \frac{495}{64}$$

Answer:

561. Expected Number Of Rectangles[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The expected number of rectangles added to the initial rectangle is the sum of the expected number of new rectangles obtained with each new line drawn for each of the 20 new lines. When the k -th line is drawn, there are already $k - 1$ lines across the post-it note. Suppose that v of these $k - 1$ lines are vertical and h are horizontal, so that $v + h = k - 1$. There is a 50% chance that $v + 1$ new rectangles are formed and a 50% chance that $h + 1$ new rectangles are formed, depending on whether the new line is horizontal or vertical, respectively. The expected number of new rectangles obtained when the k -th line is drawn is therefore:

$$\frac{1}{2}((v + 1) + (h + 1)) = \frac{k + 1}{2}.$$

Thus, the expected number of rectangles after 20 lines are drawn is:

$$1 + \sum_{k=1}^{20} \frac{k + 1}{2} = 116.$$

Answer: 116**562. Triangles Inside Triangle**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Altogether there are $3 + 2 + 3 + 7 = 15$ dots, and thus $\frac{15}{3} = 455$ combinations of 3 dots. Of these combinations,

$$\frac{2 + 2}{3} + \frac{2 + 3}{3} + \frac{2 + 7}{3} = \frac{4}{3} + \frac{5}{3} + \frac{9}{3} = 4 + 10 + 84 = 98$$

do not form triangles because they are collinear (the rest do form triangles). Thus, $455 - 98 = 357$ different triangles can be formed.

Answer: 357**563. Divisible By 37**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

$37xyz$ can be written as:

$$37000 + 100x + 10y + z$$

So, $37xyz$ is divisible by 37 if and only if xyz is divisible by 37. Now, consider $A = xyz$ and $B = yzx$, where x, y, z are digits. We will prove that if A is divisible by 37, then B is also divisible by 37.

$$A - B = 100x - x + 10y - 100y + z - 10z = 9(11x - 10y - z)$$

Now,

$$9(11x - 10y - z) \equiv 9(-26x - 10y - z) \pmod{37}$$

$$\Rightarrow 9(11x - 10y - z) \equiv -9(26x + 10y + z) \pmod{37}$$

$$\Rightarrow 9(11x - 10y - z) \equiv -9(100x + 10y + z) \pmod{37}$$

$$\Rightarrow 9(11x - 10y - z) \equiv -9(0) \pmod{37} \quad (\text{since } 100x + 10y + z \text{ is divisible by } 37)$$

So, if A is divisible by 37, then $A - B$ is also divisible by 37, which means B is also divisible by 37. Similarly, if yzx is divisible by 37, then zxy is also divisible by 37. In other words, if xyz is divisible by 37, both yzx and zxy will be divisible by 37. Thus, the condition that $37xyz, 37yzx$, and $37zxy$ are all divisible by 37 simplifies to $37xyz$ being divisible by 37, which further reduces to xyz being divisible by 37. How many

numbers of the form xyz are there that are divisible by 37? It is $\text{floor}(999/37) + 1 = 27 + 1 = 28$. The first number is $37 \times 0 = 000$ and the last number is $37 \times 27 = 999$. So, the required answer is: 28.

Answer: 28

564. 3 Heads 3 Tails III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The chance Alice wins is $\frac{1}{2}$, based on symmetry. The expected tosses for Alice and Bob to win are identical. Applying the law of total expectation, we need to double the expected tosses for Alice. Alice wins after 3 tosses with a probability of $\frac{1}{8}$, after 4 tosses with a probability of $\frac{3}{16}$ (as there are 3 ways to arrange 3 heads and 1 tail, ending in heads), and after 5 tosses with a probability of $\frac{6}{32}$ (6 ways to arrange 3 heads and 2 tails, ending in tails). The expected number of tosses for Alice is calculated as:

$$\frac{1}{8} \cdot 3 + \frac{3}{16} \cdot 4 + \frac{6}{32} \cdot 5 = \frac{66}{32}$$

Multiplying by 2 gives $\frac{33}{8}$.

Answer: 4.12500

565. Unlucky Seven

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

There is no point in re-rolling if the first turn yields 3, 4, 5, or 6 because we can improve our payoff only by rolling a larger number on the second turn, but this would make the sum of the rolls greater than 7 and the payoff will become negative. If we get 1 on the first roll, we would lose 1 dollar with a probability $\frac{1}{6}$ (by rolling a 6 on the second roll) or get the second roll if we roll 1, 2, 3, 4, or 5 on the second roll. Hence, the expected payoff if we roll again is,

$$\frac{1}{6}(-1) + \frac{1}{6}(1 + 2 + 3 + 4 + 5) = \frac{7}{3}$$

Since $\frac{7}{3} > 1$, we should re-roll if we get 1 on the first roll. Similarly, if we get 2 on the first roll, we would lose 2 dollars with probability $\frac{2}{6}$ (by rolling a 5 or a 6 on the second roll) or get the second roll if we roll 1, 2, 3, or 4 on the second roll. Hence, the expected payoff if we roll again is,

$$\frac{2}{6}(-2) + \frac{1}{6}(1 + 2 + 3 + 4) = 1.$$

Since $1 < 2$, we should not re-roll if we get 2 on the first roll. Hence, the expected payoff of the entire game is,

$$\frac{\frac{7}{3} + 2 + 3 + 4 + 5 + 6}{6} = \frac{67}{18}$$

Answer: 3.72222

566. Toss Exceedance

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Model the difference between the number of heads and tails after each toss as a simple symmetric random walk on the integers, taking a step of +1 when a head appears and a step of -1 when a tail appears, so the process starts at 0. Absorption occurs when this walk first reaches +4, which corresponds to heads exceeding tails by 4, or when it first reaches -6, which corresponds to tails exceeding heads by 6. Recall that the

expected number of steps until absorption for a simple symmetric random walk starting at the origin, with absorbing boundaries at $-a$ and $+b$, is given by ab . So, for the given problem, the expected number of steps (tosses) is

$$4 \times 6 = 24.$$

Answer:

567. Tennis Tournament

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let x and y denote the probabilities of winning against Alice and Bob, respectively, where it's given that $x < y$. For the game sequence ABA , the probability of winning the tournament is calculated as follows:

$$xy + xy - x^2y = 2xy - x^2y$$

For the sequence BAB , the probability becomes:

$$xy + xy - y^2x = 2xy - y^2x$$

Since $x < y$, we know that $y^2x > x^2y$, meaning the probability of winning in the sequence BAB is lower because a larger value is subtracted. Thus, the order ABA yields a higher probability of winning.

Answer:

568. Chess Tournament VI

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The given condition can occur only if either player 50 does not play till the last match OR player 49 does not play till the last match. This is true because if both of them play some match other than the last match, then player 49 will be eliminated as he will have to face player 50 in some match before the final match.

We can consider the tournament as a permutation of the 50 players, where the first 2 players play the first match. The winner of the first match plays the 3rd player in the permutation, and so on. Player 50 does not play till the last match if it is the last player in the permutation. The probability of this is $\frac{1}{50}$, since all players are equally likely to be the last player. Similarly, player 49 does not play till the last match if it is the last player in the permutation, and the probability of this is also $\frac{1}{50}$.

So, the probability that either player 50 does not play till the last match OR player 49 does not play till the last match is:

$$\frac{1}{50} + \frac{1}{50} = \frac{1}{25}$$

(since both these events are disjoint).

Answer:

569. Expected Wait Time II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will use the law of total expectation. On average, there are 9 arrivals after a gap of 10 minutes and 1 arrival after a gap of 70 minutes. Therefore, in 160 minutes, the bus is on a refill trip for 70 minutes on average. Thus, the probability of arriving during a refill trip is

$$\frac{70}{160}$$

while the probability of arriving during a normal trip is

$$\frac{90}{160}$$

The expected wait time during a normal trip is 5 minutes, while the expected wait time during a refill trip is 35 minutes. Hence, the expected wait time is

$$\frac{70}{160} \times 35 + \frac{90}{160} \times 5 = \frac{145}{8} \text{ minutes.}$$

Answer: 18.12500

570. Weighted Die

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Define p_i as the probability of rolling a value i on the die. To achieve a sum of 2, both dice must show 1, thus $p_2 = p_1^2$. For a sum of 3, the dice must show a combination of 1 and 2, hence $p_3 = 2p_1p_2$. For a sum of 4, the combinations are (1, 3) and (2, 2), leading to $p_4 = 2p_1p_3 + p_2^2$. For a sum of 5, the possible combinations are (1, 4) and (2, 3), so $p_5 = 2p_1p_4 + 2p_2p_3$. To get a sum of 6, the combinations are (2, 4), (3, 3), and (1, 5), resulting in $p_6 = p_3^2 + 2p_2p_4 + 2p_1p_5$. By substituting $p_2 = p_1^2$, $p_3 = 2p_1p_2$, and following this pattern, we get:

$$p_4 = 5p_1^4$$

$$p_5 = 14p_1^5$$

$$p_6 = 42p_1^6$$

Since $p_1 + p_2 + p_3 + p_4 + p_5 + p_6 = 1$, substituting the values, we have:

$$p_1 + p_1^2 + 2p_1^3 + 5p_1^4 + 14p_1^5 + 42p_1^6 = 1$$

This simplifies to:

$$42p_1^6 + 14p_1^5 + 5p_1^4 + 2p_1^3 + p_1^2 + p_1 = 1$$

Thus, the sum of the constants c_k is 65.

Answer: 65

571. Ordered Pairs In Permutation

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

First, observe that the total number of ordered pairs a_i, a_j satisfying $i - j > 155$ is equal to

$$(2015 - 156) + (2015 - 157) + \cdots + 1 = \binom{2015 - 155}{2} = 1728870,$$

where we count by casework on $j = 1, 2, \dots, 1859$. Since the permutation is random, the probability that any arbitrary ordered pair of elements a_i, a_j satisfy $a_i - a_j > n$ for some n is the same for any i, j . Furthermore, this probability is equal to the number of ordered pairs x, y satisfying $x - y > n$ and the total number of ordered pairs (x, y) , as (a_i, a_j) is equally likely to be any ordered pair x, y (the former counted similarly as above). Thus, when $n = 266$, the probability is:

$$p = \frac{\binom{2015-266}{2}}{\binom{2015}{2}} = \frac{759}{2015}.$$

By linearity of expectation, we have that the expected total number of ordered pairs a_i, a_j satisfying $i - j >$

155 that also satisfy $a_i - a_j > 266$ is:

$$E = 1728870 \cdot \frac{759}{2015} = 651222.$$

Answer: 651222

572. Variance Maximum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To determine the maximum possible value of $\text{Var}(X)$ for a random variable X defined on the interval $[-1, 1]$, we start by noting that $X^2 \leq 1$ for all X in this range, which implies $E[X^2] \leq 1$. The variance can be expressed as

$$\text{Var}(X) = E[X^2] - (E[X])^2,$$

leading to the inequality

$$\text{Var}(X) \leq 1 - z^2,$$

where $z = E[X]$. To maximize $1 - z^2$, we minimize z^2 , achieving its minimum value of 0 when $z = 0$. Thus, $\text{Var}(X)$ can reach a maximum of 1. This maximum can be achieved. Equality occurs when $E[X^2] = 1$ and $E[X] = 0$, which is satisfied when X takes values -1 and 1 with equal probability, specifically $P(X = -1) = P(X = 1) = \frac{1}{2}$. Therefore, the maximum possible value of $\text{Var}(X)$ is 1, achieved under these conditions.

Answer: 1

573. Additional Info Die Rolls

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let A be the event that the first 6 appears before the first 5. By definition,

$$E[N|A] = \sum_{n=1}^{\infty} n \cdot P(N = n|A)$$

From above,

$$E[N|A] = \sum_{n=1}^{\infty} n \cdot \left(\frac{2}{3}\right)^{n-1} \cdot \left(\frac{1}{3}\right) = \frac{1}{3} \sum_{n=1}^{\infty} nr^{n-1}, \text{ where } r = \frac{2}{3}$$

Let

$$S = \sum_{n=1}^{\infty} nr^{n-1}$$

Then

$$rS = \sum_{n=1}^{\infty} nr^n = \sum_{m=2}^{\infty} (m-1)r^{m-1}$$

So

$$S - rS = 1 + \sum_{n=2}^{\infty} r^{n-1} = 1 + \frac{r}{1-r} = \frac{1}{1-r}$$

Thus

$$S = \frac{1}{(1-r)^2} = \frac{1}{(1/3)^2} = 9$$

and finally

$$E[N|A] = \frac{1}{3} \cdot 9 = 3$$

Answer: 3

574. Word Problem I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Clearly, the total number of possible words is 5^5 . Let's find the number of favorable words. First, calculate the total number of strings where no two consecutive characters are the same. There are 5 choices for the first character, and for each of the remaining 4 positions, there are 4 choices (excluding the previous character). Thus, the total number of such strings is

$$5 \times 4^4 = 1280$$

Next, exclude the sequences where any character appears 3, 4, or 5 times. It is impossible to have a sequence with 4 or 5 occurrences of the same character while maintaining the no-consecutive-characters-same condition. For sequences with exactly 3 occurrences of the same character, the pattern would be of the form $ABACA$, where A is not equal to B and C . There are 5 choices for A and 4 choices each for B and C , giving

$$5 \times 4 \times 4 = 80$$

such sequences to remove. Therefore, the number of valid sequences is

$$1280 - 80 = 1200$$

Hence, the desired probability is $\frac{1200}{3125} = \frac{48}{125}$.

Answer: 0.38400**575. Truth**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume that x statements are true. This implies that the $(x+1)$ -th statement through the 100th statement are true. Hence,

$$x = 100 - (x + 1) + 1$$

$$\Rightarrow x = 100 - x$$

$$\Rightarrow x = 50$$

Answer: 50**576. Normal Distributions**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let d_1 and d_2 represent the probability density functions (PDFs) of the random variables X_1 and X_2 , respectively. Since X_1 and X_2 are independent, their sum $X_1 + X_2$ follows a normal distribution given by $N(0, 25)$ by adding the variances. Now, let's determine the conditional probability distribution of X_1 given $X_1 + X_2 = 5$. We have:

$$f_{X_1|X_1+X_2=5}(x) = \frac{f_{X_1, X_1+X_2}(x, 5)}{f_{X_1+X_2}(5)}$$

Note that if $X_1 = x$, then $X_2 = 5 - x$ to satisfy the condition $X_1 + X_2 = 5$. Hence, the joint PDF is:

$$f_{X_1, X_1+X_2}(x, 5) = d_1(x) \cdot d_2(5 - x)$$

This implies:

$$f_{X_1|X_1+X_2=5}(x) = \frac{d_1(x) \cdot d_2(5 - x)}{f_{X_1+X_2}(5)}$$

Now, plugging in the respective PDFs:

$$d_1(x) = \frac{1}{\sqrt{2\pi \cdot 18}} e^{-\frac{x^2}{18}}, d_2(5-x) = \frac{1}{\sqrt{2\pi \cdot 32}} e^{-\frac{(5-x)^2}{32}}, f_{X_1+X_2}(5) = \frac{1}{\sqrt{2\pi \cdot 50}} e^{-\frac{25}{50}}$$

This results in:

$$f_{X_1|X_1+X_2=5}(x) = \frac{\frac{1}{\sqrt{2\pi \cdot 18}} e^{-\frac{x^2}{18}} \cdot \frac{1}{\sqrt{2\pi \cdot 32}} e^{-\frac{(5-x)^2}{32}}}{\frac{1}{\sqrt{2\pi \cdot 50}} e^{-\frac{25}{50}}}$$

After some simplification, we find that:

$$X_1|(X_1 + X_2 = 5) \sim N\left(\frac{9}{5}, \frac{144}{25}\right)$$

Thus, the required mean is

$$\frac{9}{5}.$$

Answer:

577. Sports Betting

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If Bob refuses Alice's raised bet, his expected outcome is -10, as he loses his initial bet. However, if Bob decides to accept the bet, then with probability p , he wins 20 dollars, and with probability $1 - p$, he loses 20 dollars. We need to determine the smallest p such that:

$$20p - 20(1 - p) \geq -10$$

Solving this, we find $p \geq \frac{1}{4}$. Thus, the smallest value of p where Bob should accept the bet is $\frac{1}{4}$.

Answer:

578. Even Head Stop I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To determine the probability that the total number of tosses is even if we stop after the first head, we need to consider sequences like "TH", "TTTH", "TTTTTH", etc. This is a geometric series:

$$\text{Probability} = \sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^{2i} = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}$$

Now, to find the probability that the total number of tosses is even if we stop after getting two heads, consider two scenarios: 1. We have an even number of tosses for the first head followed by an even number of tosses for the second head. The probability for this is:

$$\frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

2. We have an odd number of tosses for the first head followed by an odd number of tosses for the second head. The probability for this is:

$$\frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$$

Thus, the total probability that the number of tosses is even when stopping after getting two heads is:

$$\frac{1}{9} + \frac{4}{9} = \frac{5}{9}$$

Answer:

579. Interesting Constraints

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To simplify the given equation $x_i + 2x_{i+1} = 1$, let $y_i = x_i - \frac{1}{3}$. Then the equation becomes

$$y_i + 2y_{i+1} = 0, \text{ or } y_i = -2y_{i+1}.$$

Unwinding, we get

$$y_1 = -2y_2 = (-2)^2 y_3 = (-2)^3 y_4 = \cdots = (-2)^9 y_{10}.$$

So

$$y_1 + 512y_{10} = 0.$$

Going back to x_i , we get

$$x_1 + 512x_{10} = \left(y_1 + \frac{1}{3}\right) + 512\left(y_{10} + \frac{1}{3}\right) = (y_1 + 512y_{10}) + \frac{1}{3} + \frac{512}{3} = 171.$$

Answer:

580. Mathematical Expression I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

After the plus signs and multiplication signs are inserted, we obtain a sum of powers of $\sqrt{3}$, which has the form $a + b\sqrt{3}$, where a and b are nonnegative integers. If any power is an even power of $\sqrt{3}$, we would have $a > 0$, which cannot occur.

Now observe that $(\sqrt{3})^1 = \sqrt{3}$ and $(\sqrt{3})^3 = 3\sqrt{3}$, but $(\sqrt{3})^{2m+1} > (2m+1)\sqrt{3}$ for $m > 1$ since this inequality is equivalent to $3^m > 2m + 1$, which follows from the binomial theorem applied to $(1 + 2)^m$.

Therefore, the only expressions that evaluate to $12\sqrt{3}$ are those which result in a sum of first and third powers of $\sqrt{3}$ only.

In other words, we must count the number of ways to assign the arithmetic symbols so that the multiplication symbols occur only as isolated consecutive pairs. One approach is to observe that for a fixed number of multiplication symbol pairs, the number of ways to place the pairs can be treated as a “balls in urns” combinatorial problem, where the balls are plus signs and the urns are the spaces before, between, and after pairs of consecutive multiplication symbols.

- For 0 pairs, there are 11 balls and 1 urn, yielding 1 expression.
- For 1 pair, there are 9 balls and 2 urns, yielding 10 expressions.
- For 2 pairs, there are 6 balls (because we must insist that one plus sign goes in between the two pairs of consecutive multiplication signs) and 3 urns, yielding 28 expressions.
- For 3 pairs, there are 3 balls and 4 urns, yielding 20 expressions.
- For 4 pairs, there are 0 balls and 5 urns, yielding 1 expression.

No further pairs are possible, giving us a grand total of $1 + 10 + 28 + 20 + 1 = 60$ expressions and a probability of $\frac{60}{2^{11}} = \frac{15}{512}$.

Answer:

581. Inside The Range[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To solve the inequality $6 \leq x_1 + x_2 + x_3 + x_4 + x_5 \leq 10$, we first note that the number of non-negative integer solutions to $x_1 + x_2 + \cdots + x_n \leq k$ is equivalent to the number of solutions to $x_1 + x_2 + \cdots + x_n + x_{n+1} = k$, where x_{n+1} is an additional variable representing the "slack" in the inequality.

Now, we calculate the number of solutions for $x_1 + x_2 + x_3 + x_4 + x_5 \leq 10$. Introducing x_6 , the equation becomes $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 10$. The number of non-negative integer solutions is given by the stars and bars formula,

$$\binom{10+5}{5} = \binom{15}{5} = 3003.$$

Similarly, for $x_1 + x_2 + x_3 + x_4 + x_5 \leq 5$, introducing x_6 , we solve $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 5$, yielding

$$\binom{5+5}{5} = \binom{10}{5} = 252.$$

The number of solutions satisfying $6 \leq x_1 + x_2 + x_3 + x_4 + x_5 \leq 10$ is the difference between these two results:

$$3003 - 252 = 2751.$$

Thus, there are 2751 non-negative integer solutions to the given inequality.

Answer: 2751**582. Increasing Rolls Expectation**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E_k be the expected sum of the rolls from the current roll onward, including the current roll itself, given that the current roll is k .

Then $E_{10} = 10$, since once you roll a 10, no later roll can be larger.

For $k < 10$, the next roll must be one of $k+1, k+2, \dots, 10$ in order for the process to continue. So

$$E_k = k + \frac{1}{10}(E_{k+1} + E_{k+2} + \cdots + E_{10}).$$

Now compute backward:

$$E_9 = 9 + \frac{1}{10}E_{10} = 9 + 1 = 10.$$

$$E_8 = 8 + \frac{1}{10}(E_9 + E_{10}) = 8 + \frac{1}{10}(10 + 10) = 10.$$

The same pattern continues. In fact, if $E_{k+1}, \dots, E_{10}$ are all 10, then

$$E_k = k + \frac{1}{10}(10 - k) \cdot 10 = k + (10 - k) = 10.$$

So by backward induction, $E_k = 10$ for every k .

That means no matter what the first roll is, the expected sum of all rolls except the last roll is 10.

So the answer is 10.

Answer: 10**583. Card Picking II**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Using conditional probability, the solution is simply the number of ways to arrange the cards such that the first ace is at the 27th position and the 2 of hearts is at the 28th position divided by the number of ways to arrange the cards such that the first ace is at the 27th position.

The denominator represents the total number of ways to position the first ace as the 27th card dealt. First, we need to place 26 of the 48 non-aces in the first 26 spots. This can be done in ${}^{48}P_{26}$ ways. Next, we place one of the 4 aces in the 27th position, leaving 25 remaining cards to arrange in any order, which gives us ${}^{48}P_{26} \times 4 \times 25!$ possible configurations where the first ace is the 27th card.

For the numerator, we must satisfy the additional condition that the 28th card is the 2 of hearts. Therefore, the first 26 cards must exclude both aces and the 2 of hearts. We place 26 of the remaining 47 cards in the first 26 spots, which can be done in ${}^{47}P_{26}$ ways. Then, one of the 4 aces is placed in the 27th position, and the 2 of hearts must be the 28th card. Finally, the remaining 24 cards can be arranged in any order, giving us ${}^{47}P_{26} \times 4 \times 24!$.

The probability is the ratio of these values:

$$P = \frac{4 \times {}^{47}P_{26} \times 24!}{4 \times {}^{48}P_{26} \times 25!} = \frac{11}{600}$$

Answer:

584. Prime Dice

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The numbers on the dice are $\{2, 3, 5, 7, 11, 13\}$. For the sum to be prime, it must be odd, since 2 is the only even prime number, and achieving a sum of 2 is impossible when both dice display prime numbers. To obtain an odd sum, one of the numbers must be 2, while the other can be 3, 5, or 11. The favorable outcomes are (2, 3), (2, 5), (2, 11), (3, 2), (5, 2), and (11, 2)—yielding a total of 6 favorable outcomes. The total number of possible outcomes is $6^2 = 36$. Therefore, the required probability is

$$\frac{6}{36} = \frac{1}{6}$$

Answer:

585. Fish Estimate

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since 10 out of the 40 captured fish were tagged, we have:

$$\frac{10}{40} = \frac{1}{4}$$

This indicates that approximately $\frac{1}{4}$ of the fish in the pond were tagged. Given that 50 fish were tagged, we can estimate the total number of fish in the pond as follows:

$$50 \times 4 = 200$$

Thus, the estimated total number of fish in the pond is 200.

Answer:

586. Proportional Dice

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let die A be the fair die and die B be the weighted die. Denote the probability of die B rolling an even number as x (so the probability of it rolling odd is $1 - x$). By the law of total probability, the probability that the sum of the rolls is odd can be calculated as:

$$x \cdot P(A \text{ shows odd}) + (1 - x) \cdot P(A \text{ shows even}) = x \cdot \frac{1}{2} + (1 - x) \cdot \frac{1}{2} = \frac{1}{2}$$

Answer:

587. Moving Ants II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

With 4 steps, the ant has $2^4 = 16$ possible paths, as it can choose between either of the two adjacent vertices at each step. Only two paths, $A_1 \rightarrow A_2 \rightarrow A_1 \rightarrow A_4 \rightarrow A_3$ and $A_1 \rightarrow A_4 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3$, visit all 4 vertices. Therefore, the probability is $\frac{2}{16} = \frac{1}{8}$.

Answer:

588. Red Card Prediction

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We label the deck's cards $1, 2, \dots, 52$ in the order they will be revealed, and let $\mathcal{F}_n$ be the information available after the first n cards have been turned face up. At that time there remain $52 - n$ cards, of which r_n are red. Define the stochastic process:

$$X_n = \frac{r_n}{52 - n}$$

Intuitively, X_n is exactly the probability that the next card (card $n + 1$) is red, given what we know so far. Claim: $\{X_n\}$ is a martingale. Proof: Conditioning on $\mathcal{F}_n$, the $(n + 1)$ th card is equally likely to be any of the $52 - n$ remaining cards.

- If it is red (probability $\frac{r_n}{52 - n}$), then $r_{n+1} = r_n - 1$ and the new denominator is $52 - (n + 1)$.
- If it is black (probability $\frac{52 - n - r_n}{52 - n}$), then $r_{n+1} = r_n$.

Hence

$$E[X_{n+1} | \mathcal{F}_n] = \frac{r_n}{52 - n} \cdot \frac{r_n - 1}{52 - n - 1} + \frac{52 - n - r_n}{52 - n} \cdot \frac{r_n}{52 - n - 1} = \frac{r_n(r_n - 1) + (52 - n - r_n)r_n}{(52 - n)(52 - n - 1)} = \frac{r_n(r_n - 1 + 52 - n - r_n)}{(52 - n)(52 - n - 1)} =$$

This verifies the martingale property $E[X_{n+1} | \mathcal{F}_n] = X_n$. Any strategy for deciding when to stop—that is, any stopping time T taking values in $\{0, 1, \dots, 51\}$ —has the property that you only stop while there is still at least one card left. At that stopping time T , your chance of winning is

$$P(\text{extwin} | \mathcal{F}_T) = X_T.$$

By the Optional Stopping Theorem (which applies here since T is bounded and $\{X_n\}$ is bounded),

$$E[X_T] = E[X_0].$$

Initially no cards have been revealed, so $r_0 = 26$ and

$$X_0 = \frac{26}{52} = \frac{1}{2}.$$

Therefore

$$E[\text{win probability}] = E[X_T] = \frac{1}{2}.$$

No matter how cleverly you choose T , the highest probability of winning is $\frac{1}{2}$.

Answer:

589. Infection Spread

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The perimeter of the infected region either remains the same or decreases over time. Initially, if k blocks are infected, the maximum perimeter is $4k$. To infect the entire board, the perimeter would need to expand to $4n$, where $k < n$. This is not feasible.

Answer:

590. Win-win

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Each of the five participants has

$$\binom{4}{2} = 6$$

choices of where to point. So the total number of choices is 6^5 . Consider a directed graph whose vertices are the five participants. Draw an edge from one participant to another if the first points to the second. The number of edges is $5 \cdot 2 = 10$. The number of unordered pairs of vertices is

$$\binom{5}{2} = 10.$$

In an all-winning game, each unordered pair has at most one edge. Because the number of edges equals the number of unordered pairs, each unordered pair must have exactly one edge. In human terms, that means for each pair of participants, exactly one of them points to the other. Next, let's count the number of all-winning outcomes. Alice has

$$\binom{4}{2} = 6$$

choices of where to point. For concreteness, suppose that Alice points to Beth and Carla. That means Dana and Eden will point to Alice. Between Beth and Carla, one will point to the other (2 choices); for concreteness, suppose that Beth points to Carla. Between Dana and Eden, one will point to the other (2 choices); for concreteness, suppose that Dana points to Eden. Now there is exactly one way to complete the picture. So the total number of all-winning outcomes is

$$6 \cdot 2 \cdot 2 = 24.$$

Therefore, the probability of an all-winning outcome is

$$\frac{24}{6^5} = \frac{1}{324}.$$

Answer:

591. Parking Lot

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For $1 \leq i < j \leq 4$, let (i, j) be the situation where the cars are in spaces i and j . During the process described in the problem, we will definitely be in the states $(1, 2)$, $(1, 3)$, $(2, 4)$, and $(3, 4)$. We may or may not be in one (and only one) of the states $(1, 4)$ and $(2, 3)$. On average, it takes two minutes to get from the initial state $(1, 2)$ to $(1, 3)$. From $(1, 3)$, in any given minute, there is a $\frac{3}{4}$ probability of leaving that state. So, on average, it takes $\frac{4}{3}$ minutes to leave state $(1, 3)$. The next (new) state after $(1, 3)$ is equally likely to be any of $(1, 4)$, $(2, 3)$, or $(2, 4)$. From either of states $(1, 4)$ or $(2, 3)$, the next (new) state will be $(2, 4)$, which will take an average of 2 minutes to get to. Of course, if our next new state after $(1, 3)$ is $(2, 4)$, then there are 0 additional minutes after leaving $(1, 3)$ to get to $(2, 4)$. So, getting from state $(1, 3)$ to state $(2, 4)$, on average, will take $\frac{4}{3} + 2 \times \frac{2}{3} + 1 \times 0 = \frac{8}{3}$ minutes. And getting from state $(2, 4)$ to state $(3, 4)$ will take an average of 2 minutes. Putting this all together gives an expected total time of $2 + \frac{8}{3} + 2 = \frac{20}{3}$ minutes to get from state $(1, 2)$ to state $(3, 4)$.

Answer: 6.66667

592. Head-tail Product

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Since the total number of tosses is n , $Y = n - X$. Hence,

$$\begin{aligned} XY &= X(n - X) = nX - X^2 \\ \Rightarrow E[XY] &= E[nX - X^2] = nE[X] - E[X^2] \end{aligned}$$

(using linearity of expectation) The number of heads X follows a binomial distribution with parameter $p = \frac{1}{2}$. Hence, the mean $E[X] = np = \frac{n}{2}$ and the variance $E[X^2] - E[X]^2 = np(1 - p) = \frac{n}{4}$, which implies

$$E[X^2] = \frac{n}{4} + E[X]^2 = \frac{n}{4} + \left(\frac{n}{2}\right)^2$$

So,

$$\begin{aligned} E[XY] &= n \cdot \frac{n}{2} - \left(\frac{n}{4} + \left(\frac{n}{2}\right)^2\right) \\ &= \frac{n(n-1)}{4} \end{aligned}$$

Answer: 22.50000

593. Guess The End

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

The number of ants falling off the east end of the rod is the same as the number of ants facing east at the start, since the number of ants facing east never changes. Additionally, if k ants fall off the east end, it is exactly the k easternmost ants who do so, as the ants maintain their order. Assuming, by symmetry, that Alice is facing east at the start, we know that she goes off the east end when the number of east-facing ants is at least 13. This implies that 12 or more of the other 24 ants are also facing east. The probability that 13 or more of 24 ants face east is equivalent to the probability that 11 or fewer face east. Therefore, the probability of the event of interest is one-half plus one-half times the probability of exactly 12 of the 24 ants facing east. The latter probability is given by $\binom{24}{12}/2^{24}$, which calculates to approximately 0.161180258. Thus, the final answer is approximately 0.580590129.

Answer: 0.58059

594. Optimal Die Choice

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If our strategy is to keep re-rolling until we get at least n , the expected value of the rolled value when we stop re-rolling is $\frac{n+20}{2}$. Also, the expected number of turns to get at least n is $\frac{20}{20-n+1}$. This is because the probability of seeing a value at least n is $\frac{20-n+1}{20}$ on any given turn, and hence, the number of turns to get at least n follows a geometric distribution with parameter $\frac{20-n+1}{20}$. So, on average, we would "take" for $100 - \frac{20}{20-n+1}$ moves. Thus, the expected payoff if you start taking when the rolled value is at least n is

$$E[n] = \frac{n+20}{2} \times \left(100 - \frac{20}{20-n+1}\right)$$

Differentiating the above expression,

$$E'[n] = \frac{10}{(n-21)^2} \times (5n^2 - 210n + 2164)$$

Setting $E'[n] = 0$ gives

$$5n^2 - 210n + 2164 = 0$$

Solving this quadratic equation results in

$$n \approx 18.14$$

Hence, the optimal integral n is either 18 or 19. Plugging the above values, we see that $n = 18$ gives a higher value of $\frac{5320}{3}$.

Answer: 1773.33333

595. Two Random Points

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Because R_1 is uniform on $[0, 1]$, the chance it lies in the tiny interval $[r, r + dr]$ is proportional to that interval's length. Since the total length of $[0, 1]$ is 1,

$$P\{R_1 \in [r, r + dr]\} = 1 \cdot dr = dr,$$

so the probability density of R_1 is

$$f_{R_1}(r) = 1, \quad 0 \leq r \leq 1.$$

Once the first point is at distance r_1 , the second must lie in the annulus between radii r_1 and 1, which has area $\pi(1 - r_1^2)$. A thin ring at radius r_2 of width dr_2 in that annulus has area $2\pi r_2 dr_2$. Uniform selection over the annulus gives

$$P\{R_2 \in [r_2, r_2 + dr_2] \mid R_1 = r_1\} = \frac{2\pi r_2 dr_2}{\pi(1 - r_1^2)} = \frac{2r_2}{1 - r_1^2} dr_2, \quad r_1 \leq r_2 \leq 1,$$

so the conditional density is

$$f_{R_2|R_1}(r_2 \mid r_1) = \frac{2r_2}{1 - r_1^2}.$$

By the law of total probability,

$$P\left(R_2 > \frac{1}{2}\right) = \int_0^1 P\left(R_2 > \frac{1}{2} \mid R_1 = r_1\right) f_{R_1}(r_1) dr_1 = \int_0^1 P\left(R_2 > \frac{1}{2} \mid R_1 = r_1\right) dr_1.$$

If $r_1 < \frac{1}{2}$, then

$$P\left(R_2 > \frac{1}{2} \mid R_1 = r_1\right) = \int_{1/2}^1 \frac{2r_2}{1 - r_1^2} dr_2 = \frac{1}{1 - r_1^2} [r_2^2]_{1/2}^1 = \frac{1 - (1/2)^2}{1 - r_1^2} = \frac{3/4}{1 - r_1^2}.$$

If $r_1 \geq \frac{1}{2}$, then automatically $R_2 \geq r_1 \geq \frac{1}{2}$, so the conditional probability is 1. Splitting the integral,

$$P\left(R_2 > \frac{1}{2}\right) = \int_0^{1/2} \frac{3/4}{1-r_1^2} dr_1 + \int_{1/2}^1 1 dr_1.$$

The second term is

$$\int_{1/2}^1 1 dr_1 = \frac{1}{2}.$$

For the first term, note

$$\int \frac{1}{1-r^2} dr = \frac{1}{2} \ln\left(\frac{1+r}{1-r}\right),$$

so

$$\int_0^{1/2} \frac{1}{1-r_1^2} dr_1 = \frac{1}{2} \ln(3).$$

Multiplying by $\frac{3}{4}$ gives $\frac{3}{8} \ln(3)$. Therefore,

$$P\left(R_2 > \frac{1}{2}\right) = \frac{1}{2} + \frac{3}{8} \ln(3) = \frac{4 + 3 \ln(3)}{8} \approx 0.91198.$$

Answer: 0.91198

596. Passing The Coin I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p represent the probability that Player 1 wins. If Player 1 tosses heads on the first turn (probability $\frac{1}{3}$), he wins immediately. If he tosses tails (probability $\frac{2}{3}$), the turn passes to players 2, 3, 4, and 5 in succession. For Player 1 to get the coin back, each of those four must also toss tails, which occurs with probability $(\frac{2}{3})^4$. Once the coin returns to Player 1, his chance to eventually win is again p . By the law of total probability:

$$p = \frac{1}{3} + \frac{2}{3} \times \left(\frac{2}{3}\right)^4 \times p$$

$$p = \frac{81}{211}$$

Answer: 0.38389

597. Jumping Frog IX

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let p_n be the probability of landing on lily pad n . Observe that if there are no restrictions, we would have

$$p_n = \frac{1}{2} \cdot p_{n-1} + \frac{1}{2} \cdot p_{n-2}.$$

This is because, given that the frog is at lily pad $n-2$, there is a $\frac{1}{2}$ probability that it will make a 2-jump to reach lily pad n , and the same applies for a 1-jump to reach lily pad $n-1$. We will now compute the values of p_n recursively, but we will skip over 3 and 6. That is, we will not consider any jumps from lily pads 3 or 6 when considering the probabilities. We compute the values of p_0 to p_{10} as follows:

n	0	1	2	3	4	5	6	7	8	9	10
p_n	1	$\frac{1}{2}$	$\frac{3}{4}$	X (skipped)	$\frac{3}{8}$	X (skipped)	$\frac{3}{16}$	$\frac{3}{32}$	$\frac{3}{64}$	$\frac{9}{128}$	$\frac{15}{256}$

Hence, the answer is $\frac{15}{256}$.

Answer: 0.05859

598. Expected Determinant

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If we swap the first 2 rows, the determinant simply changes its sign. Let f be a function that swaps the first two rows. Notice that $f(A)$ and A have the same distribution, and thus:

$$E[\det A] = E[\det f(A)] = E[-\det A] = -E[\det A]$$

This implies $E[\det A]$ is 0.

Answer: 0

599. Least N

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $A = \{18a + 1, 18a + 2, \dots, 18a + 9 : a = 0, 1, 2, 3, 4, 5\}$. Then, notice that A has 54 elements, and no two elements have a difference of 9. So, we know $n \geq 55$. Now, we just need to convince ourselves that any subset with 55 elements must have two elements with a difference of 9. To do this, take the sets $\{x, x + 9\}$ for $x = 1, 2, \dots, 91$. Notice that there are 91 such sets. The numbers $1, 2, \dots, 9$ and $92, \dots, 100$ each fall into exactly one of the sets, while all other numbers fall into two of them. So, there are 18 numbers that fall into exactly one set, and then there are 73 other sets. That means that, in a set of 55 elements, at least 37 fall into two sets. As $37 \times 2 + 18 = 92 > 91$ (the number of sets), by the pigeonhole principle, two of them fall in the same set, and thus have a difference of 9.

Answer: 55

600. Forming Quadratic Polynomial II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The roots of a quadratic with integer coefficients are real when its discriminant is nonnegative. The discriminant of the quadratic $x^2 + bx + c$ is $b^2 - 4c$. We therefore need to determine the number of lattice points (b, c) with $1 \leq b, c \leq 100$ such that $b^2 - 4c \geq 0$, or $c \leq \frac{b^2}{4}$. If $b \geq 20$, then $\frac{b^2}{4} \geq 100$, so all the lattice points (b, c) for which $b \geq 20$ and $1 \leq c \leq 100$ contribute, and these total 8100. If $b = 1$, then $\frac{b^2}{4} = \frac{1}{4}$, so there is no lattice point of the form $(1, c)$ that contributes to our count. For $1 < b < 20$, note that the fractional part of $\frac{b^2}{4}$ is either 0 or $\frac{1}{4}$ depending on whether b is even or odd, respectively. Thus, the contribution from lattice points (b, c) with $1 < b < 20$ can be determined by tallying up the values of $\frac{b^2}{4}$, but then adjusting for the fractional excesses by subtracting $\frac{1}{4}$ for each of the 9 odd numbers b in the range $1 < b < 20$. Using the algebraic identity for the sum of squares, namely $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$, we compute the contribution from lattice points (b, c) with $1 < b < 20$ to be:

$$\left(\sum_{b=2}^{19} \frac{b^2}{4} \right) - 9 \left(\frac{1}{4} \right) = \frac{1}{4} \left(\frac{19 \cdot 20 \cdot 39}{6} - 9 \right) - \frac{9}{4} = 615.$$

Thus, the total is $8100 + 615 = 8715$, and since each lattice point is equally likely, the probability is $\frac{8715}{10000} = \frac{1743}{2000}$.

Answer: 0.87150

601. Crazy Weights Arrangement[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $f(n-1)$ represent the total number of valid ways to place $n-1$ weights. For the set of n weights, dividing the weights by 2 yields the same set as $n-1$ weights, with an additional weight of 1. Now consider the placement of weight 1. If placed first, it must go on the left pan. After the first move, the difference between the left and right pans will always be at least 2, allowing weight 1 to be placed on either side in subsequent moves. Thus, there are $2n-1$ possible positions for weight 1 in each of the $f(n-1)$ valid sequences, resulting in

$$f(n) = (2n-1) \times f(n-1).$$

Solving it gives

$$f(n) = 1 \times 3 \times 5 \times \cdots \times (2n-1).$$

Answer:

602. Vending Machine I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let us define an indicator variable I_i for each of the 5 unique stickers, where $I_i = 1$ if the i -th sticker is among the 7 stickers purchased and $I_i = 0$ otherwise. The total number of distinct stickers collected can then be expressed as the sum of these indicator variables:

$$X = I_1 + I_2 + I_3 + I_4 + I_5.$$

Using the linearity of expectation, we find that:

$$E[X] = E[I_1] + E[I_2] + E[I_3] + E[I_4] + E[I_5].$$

For any specific sticker i , the probability that it appears at least once in the 7 stickers is:

$$1 - \left(\frac{4}{5}\right)^7,$$

since each sticker has a $\frac{1}{5}$ chance of being selected independently. Thus, $E[I_i] = 1 - \left(\frac{4}{5}\right)^7$. Summing this over all 5 stickers, the expected number of distinct stickers is:

$$5 \times \left(1 - \left(\frac{4}{5}\right)^7\right).$$

Answer:

603. Early Head Probability[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability that Alice gets her first heads in fewer flips than Bob is $\frac{1-p}{2}$, where p is the probability that they obtain their first heads on the same toss. To find p , we sum the probability that they both get their first heads on the n -th toss for $n = 1, 2, \dots$:

$$p = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{2(n-1)} \cdot \frac{1}{3} \cdot \frac{1}{3} = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{2(n-1)} \cdot \frac{1}{9}$$

This simplifies to an infinite geometric series:

$$p = \frac{1}{9} \sum_{n=0}^{\infty} \left(\frac{4}{9}\right)^n = \frac{1}{9} \cdot \frac{1}{1 - \frac{4}{9}} = \frac{1}{9} \cdot \frac{1}{\frac{5}{9}} = \frac{1}{9} \cdot \frac{9}{5} = \frac{1}{5}$$

Therefore, the probability that Alice obtains her first heads in fewer flips than Bob is

$$\frac{1-p}{2} = \frac{1 - \frac{1}{5}}{2} = \frac{\frac{4}{5}}{2} = \frac{4}{10} = \frac{2}{5}$$

Thus, the answer is $\frac{2}{5}$.

Answer:

604. Shadow

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

By rotating the cube, you can create various distinct n -sided polygons as shadows. Starting with the light directed at one face of the cube, the shadow will be a square, as only one face is visible. When rotating around one of the three axes, two faces become visible, resulting in a rectangular shadow. When rotating along two axes simultaneously, three faces of the cube are visible. In this configuration, you can observe nine distinct edges: three shared by exactly two faces and six that touch only a single face. These edges outline the shadow, which forms a 6-sided polygon, specifically a hexagon. Thus, the largest-sided polygon that can be created as the shadow of a rotating cube is a hexagon, making the maximum number of sides equal to 6.

Answer:

605. Rabbit On A Square Grid

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Suppose the rabbit starts at $(0,0)$ and goes to sleep at (x,y) . It is easy to see that the coordinates of the frog's final position must have the same parity. Then $-100 \leq y \leq 100$. We have $|x| \leq 100 - |y|$, so x can take on the values $-100 + |y|, -98 + |y|, \dots, 100 - |y|$. There are $101 - |y|$ such values, so the total number of such locations is

$$\sum_{y=-100}^{100} (101 - |y|) = 201 \cdot 101 - 2 \cdot \frac{100(100+1)}{2} = 1012 = 10201.$$

Answer:

606. Game Of Moving Rooks

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

On a 2×2 chessboard, for Bob to win, the rook must be moved to either the top-right or bottom-left square. Alice will be forced to move the rook to one of these squares on her first move, ensuring Bob's win.

This strategy can be extended to larger chessboards. Bob always moves the rook to a square on the diagonal line extending from the bottom-right to the top-left corner. By maintaining this diagonal, Bob can guarantee a win regardless of Alice's moves. Hence, Bob has a winning strategy.

Answer:

607. Origin In Between[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability that the first point is to the left of the origin is $\frac{2}{6}$. The probability that the second point is to the right of the origin is $\frac{4}{6}$. Similarly, the probability that the second point is to the left of the origin is $\frac{2}{6}$ and the probability that the first point is to the right of the origin is $\frac{4}{6}$. Now, using the law of total probability, the probability that the origin is between the two points is:

$$\frac{2}{6} \times \frac{4}{6} + \frac{2}{6} \times \frac{4}{6} = \frac{4}{9}$$

Answer: **608. Increasing Dice Rolls I**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The probability of getting 3 strictly increasing numbers can be expressed as:

$$P(3 \text{ strictly increasing numbers}) = P(3 \text{ different numbers}) \times P(\text{increasing order} \mid 3 \text{ different numbers})$$

First, we need to calculate the probability of getting 3 different numbers:

$$P(3 \text{ different numbers}) = \frac{6}{6} \times \frac{5}{6} \times \frac{4}{6}$$

Next, we calculate the probability that these 3 different numbers are in strictly increasing order. Since there is only one way to arrange three distinct numbers in increasing order, this probability is:

$$P(\text{increasing order} \mid 3 \text{ different numbers}) = \frac{1}{3!} = \frac{1}{6}$$

Combining these probabilities, we find:

$$P(3 \text{ strictly increasing numbers}) = \frac{6}{6} \times \frac{5}{6} \times \frac{4}{6} \times \frac{1}{6} = \frac{5}{54}$$

Therefore, the probability of getting 3 strictly increasing numbers is $\frac{5}{54}$.

Answer: **609. Tenth Toss**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The last two tosses must be heads, and the 8th toss must be a tail—the probability of this is $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$. The first seven tosses must be such that no two consecutive heads appear. Let the probability of this be p . So, our answer is $\frac{1}{8} \times p$. We will calculate p by counting the number of sequences of length 7 such that no heads occur simultaneously and then dividing it by the total number of sequences— 2^7 . The desired number of sequences can be counted using recursion. Let f_i denote the number of sequences of length i such that no heads occur simultaneously. If the i -th flip is tails, the $(i-1)$ -th flip can be anything, and the problem simply reduces to $i-1$ flips. If the i -th flip is heads, the $(i-1)$ -th flip must be tails, and hence the problem reduces to $i-2$ flips. So,

$$f_i = f_{i-1} + f_{i-2}$$

Also, $f_1 = 2$ and $f_2 = 3$. Using the base conditions and the recurrence relation, we get $f_7 = 34$. Hence,

$$p = \frac{34}{2^7}$$

and the required answer is:

$$\frac{1}{8} \times p = \frac{34}{2^{10}} = \frac{17}{512}$$

Answer:

610. Three In A Line

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Alice can guarantee a win using a symmetry-based strategy. She begins by placing a 1 in the center square of the grid. After that, she mirrors each of Bob's moves across the center, except when Bob's next move could immediately create three identical numbers in a line. However, note that we must ensure that if Bob's move does not create a position where the next move can complete a line of three identical numbers, then Alice's mirrored move also does not create such a position. This is true by symmetry whenever Bob's move does not involve the center square. However, if Bob's move does touch the center, then he must place a 0, since placing a 1 would immediately form two identical numbers in a line, allowing Alice to complete the line on her next turn. By placing a 0, Bob avoids this, and Alice's mirrored move will likewise not create a position where a line of three can be completed next (provided that, before her move, there is no position where a line of three identical numbers can be completed on the next turn; if such a position does exist, she can simply complete the line herself and win). It is also clear that someone will eventually win, since it is impossible for the entire grid to be filled without having three identical numbers in a straight line (diagonals included). We have shown that by using the mirror strategy, Alice cannot lose. Therefore she will win.

Answer: 1

611. Matching Cards Removal

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let D_i represent the number of turns needed to remove the i th pair of cards given that $i - 1$ pairs have already been removed. The overall number of turns is $D = \sum_{i=1}^{15} D_i$. Using the linearity of expectation, the expected number of turns is:

$$E(D) = E\left(\sum_{i=1}^{15} D_i\right) = \sum_{i=1}^{15} E(D_i).$$

When $i - 1$ pairs are already removed, there are $2[15 - (i - 1)]$ cards remaining. For each turn, the chance that the second card matches the first is $\frac{1}{2[15 - (i - 1)] - 1}$. Thus, the number of turns required to find a matching pair follows a geometric distribution with this probability. Therefore, the expected number of turns for the i th pair is $\frac{2}{2[15 - (i - 1)] - 1}$. Summing these expectations, the total expected number of turns to remove all 15 pairs is:

$$\sum_{i=1}^{15} \left(\frac{2}{2[15 - (i - 1)] - 1} \right) = 15^2 = 225.$$

Answer: 225

612. Blue Marble Remaining

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let k be an integer between 1 and 5 (inclusive). We will first compute the probability that at the end there are exactly k marbles in bag Y. The final marble removed must have been from bag X. Before that, 4 marbles were removed from bag X and $5 - k$ marbles were removed from bag Y. There are $\binom{9-k}{4}$ ways of choosing the bag sequence. The probability of making a particular choice is $\left(\frac{1}{2}\right)^{10-k}$. So the probability that at the end there are exactly k marbles in bag Y is

$$\binom{9-k}{4} \cdot \frac{2^k}{2^{10}}$$

We can use this result to find the expected value of the number of marbles remaining in bag Y. The expected

value is

$$\frac{1}{2^{10}} \sum_{k=1}^5 \left(\binom{9-k}{4} \cdot 2^k \cdot k \right)$$

Let's compute the sum:

$$\begin{aligned} \sum_{k=1}^5 \left(\binom{9-k}{4} \cdot 2^k \cdot k \right) &= 2 \cdot \binom{8}{4} + 8 \cdot \binom{7}{4} + 24 \cdot \binom{6}{4} + 64 \cdot \binom{5}{4} + 160 \cdot \binom{4}{4} \\ &= 2 \cdot 70 + 8 \cdot 35 + 24 \cdot 15 + 64 \cdot 5 + 160 \cdot 1 \\ &= 140 + 280 + 360 + 320 + 160 \\ &= 1260 \end{aligned}$$

Hence, the expected number of marbles remaining in bag Y is

$$\frac{1260}{2^{10}}$$

The blue marble is 1 of the 5 marbles in bag Y. So the probability that the blue marble remains in bag Y is $\frac{1}{5}$ of the expected number of marbles remaining in bag Y:

$$\frac{1260}{5 \cdot 2^{10}} = \frac{252}{2^{10}} = \frac{63}{2^8} = \frac{63}{256}$$

Answer: 0.24609

613. Expected Collisions II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume each ant is carrying a flag. Imagine that at each collision, flags are exchanged between the colliding ants. Thus, at all times, each ant is carrying some flag (not necessarily the one it started with), and the flags march straight past one another. This simplifies the problem, as the flags simply go past each other. It's easy to compute the number of collisions Alice's flag has. Assuming that Alice initially faces east, there will be an average of 6 (out of 12) flags ahead of Alice facing west; hence, her flag expects to pass six other flags. However, Alice is not always carrying her original flag, and we expect Alice to have more than six collisions on average. Why? Because the average ant has six collisions ($75 \times \frac{2}{25}$), and Alice, being the middle ant, should have more than the average number of collisions. Now, any given ant collides only with its two neighbors, alternating between them (since its direction alternates between collisions). An ant's last collision will be with its western neighbor if it falls off the east end, and with its eastern neighbor if it drops off the west end. Suppose that k ants face west initially. Since their flags march off the west end of the rod, the k westernmost ants drop off the west end. Each ant facing west initially will have an equal number of collisions on each side, while those facing east will have one extra collision on the east side. The number of collisions between ant j (counting from the west) and ant $j+1$ is equal to the number of east-facing ants among ants 1 through j , as long as $j < k$. By symmetry, we may assume that k is between 13 and 25 (i.e., Alice herself drops off the west end). The number of collisions between Alice and her western neighbor is exactly the number of east-facing ants west of Alice; call this number x . The total number of collisions experienced by Alice would then be $2x$ or $2x+1$, depending on whether she herself faced west or east at the start. A priori, the expected value $E[x]$ is 6, since there are 12 ants west of Alice, and each could face either way. However, we assumed that more than half the ants faced west. The expected number of east-facing ants east of Alice is the same as $E[x]$, and the number $2E[x+1]$ is the total expected number of east-facing ants, given that they are in the minority. Suppose the ants were assigned directions in alphabetical order, with Ant Zelda last. There are $\frac{2^{25}}{2} = 2^{24}$ ways to assign directions such that there is a west-facing majority; of those, $\binom{24}{12}$ result in 12 east-facers among the first 24 choices. In those cases, Zelda is forced to face west; in the others, she is equally likely to face west or east. The probability that she faces east is $\frac{1}{2} - \frac{1}{2} \times \frac{\binom{24}{12}}{2^{24}} \approx 0.419409871$. Since Zelda's probability of facing east is the same as any other ant's, multiplying this by 25 gives the expected number of east-facing ants, approximately 10.4852468. This is the average number of collisions experienced

by Alice.

Answer: 10.48525

614. Minimum Correlation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the random variables be $Y_1, Y_2, \dots, Y_{10}$, and let the minimum pairwise correlation be m . For 10 random variables, each with mean 0 and variance 1, let $Y = \frac{1}{10}(Y_1 + Y_2 + \dots + Y_{10})$. Hence, we have:

$$\text{Var}(Y) = \frac{1}{100} \left(\sum_{i=1}^{10} \text{Var}(Y_i) + \sum_{i \neq j} \text{Cov}(Y_i, Y_j) \right)$$

The first term is 10, and the second, with $\text{Cov}(Y_i, Y_j) = m$, contributes $90m$. Hence, we have:

$$\text{Var}(Y) = 10 + 90m$$

Since variance is non-negative,

$$10 + 90m \geq 0$$

giving

$$m \geq -\frac{1}{9}$$

The minimum pairwise correlation is $-\frac{1}{9}$.

Answer: -0.11111

615. Complex Circle

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The unit circle maps onto itself under the function $f(z) = z^2$, since every point on the circle can be represented as $e^{i\theta}$ for some θ . Thus, $Z^2 = e^{2i\theta}$, which also lies on the unit circle. Given that the circle is symmetric, the expected value of Z^2 is the same as the expected value of Z , which is 0. Therefore, $E[Z^2] = 0$.

Answer: 0

616. Higher Sum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To solve this, we first define the two dice rolls as x and y , where both x and y are integers between 1 and 6. We want to find the probability that the sum $x + y$ is greater than or equal to the product $x \times y$. This can be written as the inequality $x + y \geq x \times y$. Rewriting this inequality, we get:

$$x + y - x \times y \geq 0$$

or equivalently:

$$(x - 1)(y - 1) \leq 1$$

We now analyze this inequality for all possible values of x and y . By checking all pairs of (x, y) where $x, y \in \{1, 2, 3, 4, 5, 6\}$, we count the number of successful outcomes where $(x - 1)(y - 1) \leq 1$. * If $x = 1$, this holds for any y . * If $y = 1$, this holds for any x . * There are 11 such pairs. * This also holds if $x = 2$ and $y = 2$. Hence, the total number of favorable pairs is 12. The total number of outcomes is $6 \times 6 = 36$. Hence,

the required probability is:

$$\frac{12}{36} = \frac{1}{3}$$

Answer:

617. 5 Points On A Sphere

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We claim that the convex hull of the 5 points is a triangular bipyramid. To see this, note that having 4 coplanar points is a probability-zero event, so all the faces of the convex hull are triangles. If E is the number of edges of the convex hull and F is the number of faces, then $E = \frac{3F}{2}$. Euler's formula tells us that $5 - E + F = 2$, from which we find $F = 6$ and $E = 9$. Thus, the degree of the average vertex is $\frac{18}{5}$, and hence, at least one vertex must have a degree greater than 3. Since there are only 5 vertices, there is a vertex with degree 4. Thus, there are 4 triangular faces that share a common vertex A . The 4 vertices other than A form a nonplanar quadrilateral, contributing 2 more faces and 1 more edge, which must connect two opposite vertices of the quadrilateral. The 2 unconnected vertices of the quadrilateral thus have degree 3 and are the apices of triangular pyramids that share the common triangular base formed by the other 3 vertices. In a convex triangular bipyramid, there are 6 edges emanating from the two degree-3 vertices. If we pick 2 vertices that are endpoints of such an edge, they cannot be split by the plane through the complementary vertices since the complementary vertices are the vertices of a triangular face. However, any 2 vertices of degree 4 must be split by the plane containing the complementary vertices, since if not, then the triangle whose vertices are the complementary vertices would have to be a face, but that's not possible since among those vertices are the two unconnected vertices of degree 3. The only pair of vertices left to consider are the two degree-3 vertices not connected by an edge. These two vertices must be split by the plane containing the complementary vertices since the triangle whose vertices are the complementary vertices is also not a face. Hence, 4 of the 10 pairs of points are split by the plane containing the other points, and the answer is

$$\frac{4}{10} = \frac{2}{5}$$

Answer:

618. Count Sequences II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Suppose the number of 0s is n . We can construct the sequence in two steps: Step 1: Put $n - 1$ of 1s between the 0s; Step 2: Put the rest $19 - n - (n - 1) = 20 - 2n$ of 1s in the $n - 1$ spots where there is a 1. There are $\binom{n-1}{20-2n}$ ways of doing this. Therefore, the answer is

$$\sum_n \binom{n-1}{20-2n} = \binom{6}{6} + \binom{7}{4} + \binom{8}{2} + \binom{9}{0} = 65.$$

Answer:

619. Remove Or Flip

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

4013 is odd. So, in the beginning the number of red and black tokens is different. Then Alice can remove some number of tokens of the color with more tokens so that the numbers of tokens of different colors become equal. If Bob plays, he cannot leave the same number of tokens of both colors, so he cannot win. Also, he

permits Alice to repeat the process while reducing the number of tokens. Thus Alice wins after at most 4013 moves.

Answer:

620. Unit Digit Probability I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To ensure the product ends in those digits, no selected number can be even, as that would result in an even product. Additionally, the digit 5 cannot be selected, as it would cause the product to end in 5. Thus, the problem is equivalent to finding the probability that each selected number is from the set $\{1, 3, 7, 9\}$. Since there are 4 favorable choices out of 9 possible digits, the probability for each number is $\frac{4}{9}$. Therefore, the total probability is $\left(\frac{4}{9}\right)^4 = \frac{256}{6561}$.

Answer:

621. Interview III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

When we need to hire two people—specifically the overall best and the overall worst out of n candidates—and let n grow without bound, we can adapt the single-hire “skip then pick” idea in a straightforward way. Number the interview positions 1 through n . We decide on a fraction $\alpha \in (0, 1)$, skip the first αn candidates without hiring anyone, and then use the remainder of the sequence to fill our two slots: one for the best and one for the worst. In the “selection phase” (positions $\alpha n + 1$ through n), we watch for record-highs to fill our “best” slot and record-lows to fill our “worst” slot. Because the best and worst orders evolve independently (records for maxima and minima don’t interfere in the large- n limit), the probability of correctly selecting the absolute maximum in that window tends to

$$-\alpha \ln \alpha$$

and likewise the probability of correctly selecting the absolute minimum also tends to

$$-\alpha \ln \alpha.$$

These two events become asymptotically independent as $n \rightarrow \infty$, so the joint probability of getting both the best and the worst is

$$(-\alpha \ln \alpha) \times (-\alpha \ln \alpha) = (-\alpha \ln \alpha)^2.$$

We now choose α to maximize $(-\alpha \ln \alpha)^2$. Since $-\alpha \ln \alpha$ itself is maximized at $\alpha = 1/e$, its square is also maximized there. Substituting $\alpha = 1/e$ gives

$$\left(-\frac{1}{e} \ln \frac{1}{e}\right)^2 = \left(\frac{1}{e}\right)^2 = \frac{1}{e^2}.$$

In conclusion, with two positions (best and worst) and $n \rightarrow \infty$, the optimal strategy is: 1. Skip roughly the first $\frac{n}{e}$ candidates outright. 2. Then in the remaining stream, hire the first record-high for your “best” slot and the first record-low for your “worst” slot. This achieves an asymptotic probability of

$$\frac{1}{e^2} \approx 0.13534.$$

Answer:

622. Expected Value[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

By linearity of expectation, we have that $E[r] = E[\frac{p_0}{10}] + E[\frac{p_1}{10^2}] + \cdots = \sum_{i=0}^{\infty} \frac{E[p_i]}{10^{i+1}}$, so it suffices to compute the expected value of each digit. We calculate $E[p_i]$ inductively. First, note that $E[p_0] = 0 \cdot \frac{1}{10} + 1 \cdot \frac{1}{10} + \cdots + 9 \cdot \frac{1}{10} = \frac{9}{2}$. Now, we try to compute $E[p_{i+1}]$ in terms of $E[p_i]$. Suppose we know that $p_i = d$, where d is a digit from 0 to 9. Then p_{i+1} is randomly generated from d to 9, inclusive, so

$$E[p_{i+1}|p_i = d] = d \cdot \frac{1}{10-d} + (d+1) \cdot \frac{1}{10-d} + \cdots + 9 \cdot \frac{1}{10-d} = \frac{9+d}{2}$$

Now, to compute the expected value of $E[p_{i+1}]$, we consider all cases for p_i . We have that

$$\begin{aligned} E[p_{i+1}] &= \sum_{d=0}^9 E[p_{i+1}|p_i = d]P(p_i = d) = \sum_{d=0}^9 \frac{9+d}{2} P(p_i = d) \\ &= \sum_{d=0}^9 \frac{9}{2} P(p_i = d) + \sum_{d=0}^9 \frac{d}{2} P(p_i = d) = \frac{9}{2} + \frac{1}{2} E[p_i] \end{aligned}$$

by the law of total probability and definition of expectation. Thus, we get the recurrence relation

$$E[p_0] = \frac{9}{2}$$

$$E[p_{i+1}] = \frac{9}{2} + \frac{1}{2} E[p_i]$$

Solving this recurrence gives

$$E[p_i] = 9 - \frac{9}{2} \left(\frac{1}{2} \right)^i$$

so we have that

$$\mathbb{E}[r] = \sum_{i=0}^{\infty} \frac{9 - \frac{9}{2} \left(\frac{1}{2} \right)^i}{10^{i+1}} = \sum_{i=0}^{\infty} \frac{9}{10} \left(\frac{1}{10} \right)^i - \sum_{i=0}^{\infty} \frac{9}{20} \left(\frac{1}{20} \right)^i.$$

Evaluating each sum, we get

$$E[r] = \frac{\frac{9}{10}}{1 - \frac{1}{10}} - \frac{\frac{9}{20}}{1 - \frac{1}{20}} = \frac{9}{10} \cdot \frac{10}{9} - \frac{9}{20} \cdot \frac{20}{19} = 1 - \frac{9}{19} = \frac{10}{19}$$

Answer: 0.52632

623. Forming Quadratic Polynomial I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We compute the complement: namely, we compute the probability that regardless of the ordering of a , b , and c , no quadratic equation Alice creates will have 2 distinct real roots. For this to be the case, it is sufficient that the largest possible discriminant is nonpositive. Without loss of generality, assume $b = \max(a, b, c)$, so that the largest possible discriminant is $b^2 - 4ac$. Now, let $x = \log_2 a$, $y = \log_2 b$, and $z = \log_2 c$, so that x , y , and z are integers satisfying $y = \max(x, y, z)$. Then

$$0 \geq b^2 - 4ac = 2^{2y} - 4 \cdot 2^x \cdot 2^z$$

so

$$x + z + 2 \geq 2y \geq x + z.$$

Thus, $x + z \in \{2y, 2y - 1, 2y - 2\}$, so the unordered pair $\{x, z\}$ is one of $\{y, y\}$, $\{y, y - 1\}$, $\{y, y - 2\}$, or $\{y - 1, y - 1\}$.

Now, we lift our assumption that $y = \max(x, y, z)$ to compute the answer. We have four cases: * Suppose (x, y, z) is some ordering of (t, t, t) . There are 8 such ordered triples. * Suppose (x, y, z) is some ordering of $(t, t, t - 1)$. There are $3 \cdot 7 = 21$ such ordered triples. * Suppose (x, y, z) is some ordering of $(t, t - 1, t - 1)$. There are $3 \cdot 7 = 21$ such ordered triples. * Suppose (x, y, z) is some ordering of $(t, t, t - 2)$. There are $3 \cdot 6 = 18$ such ordered triples.

Subtracting, the probability is

$$1 - \frac{8 + 21 + 21 + 18}{8^3} = \frac{111}{128}.$$

Answer:

624. Rope Variance

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Denote the length of the shorter piece as X , where X is uniformly distributed between 0 and 0.5 meters. This is because the cut is random, and the shorter piece can range from 0 to 0.5 meters, with equal probability across this interval. The length of the longer piece is then $1 - X$, which lies between 0.5 and 1 meter. Thus, the length of the longer piece follows a uniform distribution on the interval $(0.5, 1)$. The variance of a uniform distribution on the interval (a, b) is given by:

$$\text{Var}(L) = \frac{(b - a)^2}{12}$$

For L , where $a = 0.5$ and $b = 1$, the variance is:

$$\text{Var}(L) = \frac{(1 - 0.5)^2}{12} = \frac{0.25}{12} = \frac{1}{48}$$

Therefore, the variance of the length of the longer piece is $\frac{1}{48}$.

Answer:

625. Triangular Grid

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $P(0, n)$ denote the probability that you are at the starting point after n moves without ever going beyond 1 unit from your starting point. Let $P(1, n)$ denote the probability that you are at a distance of 1 unit from the starting point after n moves without ever going beyond 1 unit from your starting point. Clearly, $P(0, 0) = 1, P(1, 0) = 0, P(0, 1) = 0, P(1, 1) = 1$. Now,

$$P(0, n) = \frac{1}{6} \times P(1, n - 1)$$

$$P(1, n) = \frac{1}{3} \times P(1, n - 1) + P(0, n - 1)$$

Using the above recurrence and dynamic programming,

$$P(0, 5) = \frac{2}{81}$$

$$P(1, 5) = \frac{31}{324}$$

Hence, the required probability is $1 - \frac{2}{81} - \frac{31}{324} = \frac{95}{108}$.

Answer:

626. Strange Comparison I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since $31 < 32$, $31^{11} < 32^{11} = (2^5)^{11} = 2^{55}$. Since $17 > 16$, $17^{14} > 16^{14} = (2^4)^{14} = 2^{56}$. Therefore, $31^{11} < 2^{55} < 2^{56} < 17^{14}$, so $17^{14} > 31^{11}$.

Answer: 2**627. Mathematical Expression II**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Martha can produce four different expression types, each equally likely: $a + b + c$, $a + bc$, $ab + c$, and abc . Therefore, to find the expected value, we can take the average of the expected values for each of these expression types. The expected value of a sum is the sum of the expected values, and the expected value of a product of independent random variables is the product of their expected values.

Since the expected value of a randomly selected digit is 5, the expected values of each expression type are:

- $a + b + c = 5 + 5 + 5 = 15$,
- $a + bc = 5 + 5 \times 5 = 30$,
- $ab + c = 5 \times 5 + 5 = 30$,
- $abc = 5 \times 5 \times 5 = 125$.

Thus, the average expected value is:

$$\frac{15 + 30 + 30 + 125}{4} = 50.$$

Hence, the expected value is 50.

Answer: 50**628. Boy-girl Adjacent Pairs**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To determine the expected number of times a boy and a girl are adjacent in a random lineup of 15 boys and 10 girls, we first define a random variable X that represents the total number of such adjacent pairs. To solve this, we use indicator variables. Let I_i be an indicator variable for the i -th pair of adjacent positions, where $I_i = 1$ if the i -th and $(i + 1)$ -th positions contain a boy and a girl, and $I_i = 0$ otherwise. Since there are 25 people in total, there are 24 possible adjacent pairs. Next, we calculate the probability that a specific adjacent pair consists of a boy and a girl. The probability of having a boy in the first position and a girl in the second is given by:

$$P(BG) = \frac{15}{25} \cdot \frac{10}{24}$$

Similarly, the probability of having a girl in the first position and a boy in the second is:

$$P(GB) = \frac{10}{25} \cdot \frac{15}{24}$$

Adding these probabilities, we get:

$$P(BG \text{ or } GB) = \frac{15}{25} \cdot \frac{10}{24} + \frac{10}{25} \cdot \frac{15}{24} = \frac{1}{2}$$

Therefore, the expected value of X , which is the sum of the 24 indicator variables I_i , is:

$$E[X] = 24 \cdot \frac{1}{2} = 12$$

Thus, the expected number of times a boy and a girl are standing next to each other in the lineup is 12.

Answer:

629. Optimal Bid

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If you bet B and get the item, the owner's value for it, V , is uniformly distributed between 0 and B . Consequently, its value to you is $1.8 \times V$. The expected value of V is $\frac{B}{2}$. Thus, the expected value to you is:

$$1.8 \times \frac{B}{2} = 0.9 \times B$$

Since you are paying B for the item, and the expected value to you is $0.9 \times B$, which is less than the amount you pay, it is better not to place a bet at all. Therefore, the optimal bid is 0.

Answer:

630. 3 Consecutive Heads I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The sequence should appear as HHH–, –HHH–, or –HHH. For the HHH– pattern, the first dash must be T to prevent extension of the sequence, leaving 2 choices for the second dash. For –HHH–, both dashes must be T to avoid overlap with HHH, giving 1 option. For –HHH, there are 2 choices for the first dash and 1 choice for the second dash. Hence, the total number of sequences is 5. The total number of outcomes for 5 tosses is $2^5 = 32$. So, the required probability is

$$\frac{5}{32}$$

Answer:

631. Two Rolls Reward

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E be the expected total dollars you will earn starting from the beginning of a fresh round. In any given round, you roll twice: With probability $\frac{1}{8}$ the two rolls match and the game ends immediately with no further earnings. Otherwise you continue, and you earn \$1 if the second roll exceeds the first by at least 2, or \$0 if it does not. Let's compute the overall probabilities of those "continue" outcomes: Probability of winning \$1 in a round: Label the first roll $i \in \{1, \dots, 8\}$. To earn \$1, we need $j \geq i + 2$ on the second roll. Note that for $i = 7, 8$ there are no such j , so the probability is 0. For $i = 1$ through 6, there are $7 - i$ winning values of j out of 8. Hence

$$P_{\text{win}} = \frac{1}{8} \sum_{i=1}^6 \frac{7-i}{8} = \frac{1}{64} \sum_{i=1}^6 (7-i) = \frac{21}{64}$$

Probability the game continues (without winning \$1): The game ends only when the two rolls are equal, which happens with probability $\frac{1}{8}$. Hence the probability we continue (win \$1 or win \$0) is

$$1 - \frac{1}{8} = \frac{7}{8}$$

Of that, the fraction that pays \$1 is $\frac{21}{64}$, so the fraction that pays \$0 but continues is $\frac{35}{64}$. Now, let's setup a

recursion. On the first round: * With probability $\frac{1}{8}$, you earn nothing and the game stops, contributing 0 to the expectation. * With probability $\frac{21}{64}$, you earn \$1 now and then, because the game continues in exactly the same situation you started in, expect to earn another E on average. * With probability $\frac{35}{64}$, you earn \$0 now but still continue, so you again expect to earn another E . Putting these together:

$$E = \frac{1}{8} \cdot 0 + \frac{21}{64}(1 + E) + \frac{35}{64}(0 + E).$$

$$E = \frac{21}{8}$$

Answer: 2.62500

632. Broken Elevator

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Suppose there are u ups, d downs, and s seconds at which the elevator stays still. Since the elevator returns to its original height, $u = d$. Since 8 seconds elapse, $u + d + s = 2u + s = 8$. It is clear that s is even, so $s \in \{0, 2, 4, 6, 8\}$, and $u = d = \frac{8-s}{2}$. We do casework based on the value of s . Given a path with s stops, delete all seconds at which the elevator stays still to obtain a "reduced" path. The resulting path corresponds to a walk in the $u - d$ plane by sending each up step to $(1, 0)$ and each down step to $(0, 1)$, such that the walk goes from $(0, 0)$ to $(\frac{8-s}{2}, \frac{8-s}{2})$ without crossing the line $u = d$. The number of such paths is the Catalan number $C_{\frac{8-s}{2}}$. Now, we count how many ways there are to insert the stops into the path. Suppose we insert x_i stops before the i -th move in the reduced path, as well as x_{9-s} after the last move. We must count the number of solutions to $x_1 + \cdots + x_{9-s} = s$ over the nonnegative integers. By stars and bars, this is

$$\binom{8}{s}$$

We can explicitly compute that $C_0 = 1, C_1 = 1, C_2 = 2, C_3 = 5, C_4 = 14$. Then, the total number of possible paths is

$$\sum_{s \in \{0, 2, 4, 6, 8\}} \binom{8}{s} C_{\frac{8-s}{2}} = 323.$$

The number of paths with no stops is simply the term corresponding to $s = 0$, namely

$$\binom{8}{0} C_4 = 14.$$

It follows that the desired probability is $\frac{14}{323}$.

Answer: 0.04334

633. 3 Lights

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

By symmetry, the probabilities of the rightmost and leftmost lights being on are the same. Let's calculate the probability for the leftmost light. It changes its state if either it or the middle light is toggled, which occurs with a probability of $\frac{2}{3}$.

Thus, the probability of the leftmost light being on after three seconds is:

$$\left(\frac{2}{3}\right)^3 + 3 \cdot \frac{2}{3} \cdot \left(\frac{1}{3}\right)^2 = \frac{14}{27}$$

Additionally, the middle light always changes its state, so its probability of being on is 1.

Then use the linearity of expectation, the expected number of lights on after three seconds is:

$$\frac{14}{27} + 1 + \frac{14}{27} = \frac{55}{27}$$

Answer:

634. Dice Game II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For a single roll, the expected payoff is the average of all possible outcomes, which is 3.5. In a two-roll scenario, if you roll a 6 first, you wouldn't roll again because the only outcome is keeping the 6, which is the highest possible value. Similarly, if you roll a 4 or 5, you also wouldn't continue since the expected value of a reroll is only 3.5. But if you roll a 1, 2, or 3, it's worth rolling again to aim for a higher number. Hence, the expected payoff for 2 rolls is:

$$\frac{1}{6} \times 4 + \frac{1}{6} \times 5 + \frac{1}{6} \times 6 + \frac{3}{6} \times 3.5 = 4.25$$

Now, consider three rolls. If you roll a 5 or 6 first, you keep it. But even a 4 becomes worth rerolling since the expected value of the remaining two rolls is 4.25. Hence, the expected payoff is:

$$\frac{1}{6} \times 5 + \frac{1}{6} \times 6 + \frac{4}{6} \times 4.25 = 4.67$$

Answer:

635. Checks

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let K_1, K_2, K_3 denote the three knights and k_1, k_2, k_3 the three kings. For each pair (i, j) with $i, j \in \{1, 2, 3\}$, define the indicator variable

$$I_{ij} = \begin{cases} 1, & \text{if knight } K_i \text{ attacks king } k_j, \\ 0, & \text{otherwise.} \end{cases}$$

The total number of checks is then

$$T = \sum_{i=1}^3 \sum_{j=1}^3 I_{ij}.$$

By linearity of expectation,

$$\mathbb{E}[T] = \sum_{i=1}^3 \sum_{j=1}^3 \mathbb{E}[I_{ij}] = \sum_{i=1}^3 \sum_{j=1}^3 \mathbb{P}(I_{ij} = 1).$$

To find $\mathbb{P}(I_{ij} = 1)$, observe that choosing distinct squares for K_i and k_j out of 16 on the board gives $16 \times 15 = 240$ equally likely ordered placements. Now, summing over all 16 possible knight positions, the total number of squares they can attack is 48, so there are 48 such knight-king placements. Hence

$$\mathbb{P}(I_{ij} = 1) = \frac{48}{240} = \frac{1}{5}.$$

Since there are $3 \times 3 = 9$ independent knight-king pairs,

$$\mathbb{E}[T] = 9 \times \frac{1}{5} = \frac{9}{5} = 1.8.$$

Thus, the expected number of checks is 1.8.

Answer:

636. Arrange Teams

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For $n = 1$, the problem is solved trivially. Address the problem recursively for a selected group of $n - 1$ teams. After obtaining the list of these $n - 1$ teams, insert the excluded team before the first team in the list that it defeated in the tournament. If the excluded team lost to all the teams on the list, place it at the end of the list.

Answer: 1

637. Interview IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

It is clearly never optimal to select someone who is not a “record extreme” (i.e., neither the best nor the worst among those seen so far): if they aren’t a record high or low up to that point, they cannot turn out to be the overall extreme. Just like in “Interview I”, if you select the i^{th} candidate who is either the best or the worst among the first i interviewed, then the probability that this candidate is actually the overall best or worst is $\frac{2}{i}$. Here’s the proof: Define E_i : “Candidate i is either the best or the worst of the first i interviewed,” E : “Candidate i is either the overall best or the overall worst of all six.” We want

$$P(E \mid E_i) = \frac{P(E \cap E_i)}{P(E_i)}$$

Because being the overall extreme (E) necessarily makes you a record extreme among the first i (E_i), we have $E \subseteq E_i$, and so

$$P(E \cap E_i) = P(E) = \frac{2}{6} = \frac{1}{3}$$

since each of the six could equally likely be the best or the worst (two favorable out of six). Meanwhile, among the first i , each position is equally likely to furnish a new high or new low, so

$$P(E_i) = \frac{2}{i}$$

Hence

$$P(E \mid E_i) = \frac{1}{3} \cdot \frac{i}{2} = \frac{i}{6}$$

Because $i/6$ grows with i , record extremes become strictly more likely to be true extremes the later they occur. So, the rule—“skip first k candidates, then select the first best or worst candidate”—works here as well. If the true “best or worst” candidate appears in position i , that event has probability $\frac{2}{6}$. Given they appear at $i \geq k + 1$, we will select them exactly when neither a record-high nor a record-low has occurred in positions $k + 1, \dots, i - 1$. Equivalently, both the overall maximum and minimum among the first $i - 1$ must lie in the initial skipped block of size k , which has probability

$$\frac{k}{i-1} \cdot \frac{k-1}{i-2}$$

Hence for each $k = 0, 1, \dots, 5$, the success probability is

$$P(k) = \sum_{i=k+1}^6 \frac{2}{6} \frac{k(k-1)}{(i-1)(i-2)}$$

The term under summation can be viewed as: (the best or worst candidate overall appears at position i) $\times$ (both the overall maximum and minimum among the first $i - 1$ lie in the initial skipped block of size k —so

that the candidate at position i is indeed selected). $k = 0$ (hire candidate 1 immediately):

$$P(0) = \frac{2}{6} = \frac{1}{3}$$

$k = 1$ (skip one, then first extreme):

$$P(1) = \frac{2}{6} = \frac{1}{3}$$

$k = 2$ (skip two):

$$P(2) = \sum_{i=3}^6 \frac{2}{6} \frac{2 \cdot 1}{(i-1)(i-2)} = \frac{8}{15}$$

$k = 3$ (skip three):

$$P(3) = \sum_{i=4}^6 \frac{2}{6} \frac{3 \cdot 2}{(i-1)(i-2)} = \frac{3}{5}$$

$k = 4$ (skip four):

$$P(4) = \sum_{i=5}^6 \frac{2}{6} \frac{4 \cdot 3}{(i-1)(i-2)} = \frac{8}{15}$$

$k = 5$ (skip five, must take sixth):

$$P(5) = \frac{1}{3}$$

Comparing these fractions, the maximum occurs at $k = 3, P(3) = \frac{3}{5}$.

Answer:

638. Coin And Die Together

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If all the coin tosses come out tails, then the die will remain at 1. The probability that all 5 tosses will be tails is

$$\frac{1}{2^5} = \frac{1}{32}$$

Otherwise, if at least one of the tosses comes out heads, then at the end the die will be in a random position, because each roll has an equal probability of showing any particular number. The probability that at least one toss is heads is $1 - \frac{1}{32}$, or $\frac{31}{32}$. Given that condition, the probability that the die will be 1 is $\frac{1}{6}$. So the probability that at least one toss is heads and the die ends at 1 is:

$$\frac{31}{32} \times \frac{1}{6} = \frac{31}{192}$$

Hence, adding the two cases together, we get that the total probability is:

$$\frac{1}{32} + \frac{31}{192} = \frac{37}{192}$$

Answer:

639. Finding Seven

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If the sum of the dice is 7, we reroll no dice. If the sum of any two dice is ≤ 6 , we reroll 1 die because rolling 1 die will give Alice a $\frac{1}{6}$ chance of winning. Thus, we consider the scenario where the sum of any two dice is 7 or greater.

If the smallest die is 1, the other two dice must be 6, which has 3 possible permutations. If the smallest die is 2, the other two dice can be 5 or 6. There are 3 ways to choose the position of the smallest die, 2 ways each to choose the value of the other dice, so there are $2 \times 2 \times 3 = 12$ possible permutations. If the smallest die is 3, the other two dice can be 4, 5, or 6. Similarly, there are 3 ways to choose the position of the smallest die, and 3 ways each to choose the value of the other dice, so there are $3 \times 3 \times 3 = 27$ possible permutations. If all dice are 4 or above, then we reroll 3 dice. When we reroll all 3 dice, the winning rolls are 1,1,5; 1,2,4; 1,3,3; 2,2,3 in some arrangement, for a total of 15 permutations. Thus, the probability of winning after rerolling 3 dice is $\frac{15}{216} = \frac{5}{72}$. If we reroll only 2 dice with the smallest die being 4, we would need to roll 1,2 or 2,1 to win, which gives a $\frac{2}{36} = \frac{1}{18}$ chance of winning, so this is worse. If the smallest die is 5 or 6, rerolling 2 dice gives us even worse odds. Therefore, the cases where 1, 2, or 3 are the smallest die covers all cases where 2 rerolls is optimal. There are $3 + 12 + 27 = 42$ permutations out of 216 total permutations, giving us a probability of $\frac{42}{216} = \frac{7}{36}$.

Answer: 0.19444

640. Choose Values III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Because Alice and Bob end up using exactly the same die (same face values and probabilities), symmetry forces

$$P(\text{Alice wins}) = P(\text{Bob wins}).$$

Since one of "Alice wins," "Bob wins," or "tie" must happen, we have

$$2P(\text{Alice wins}) + P(\text{tie}) = 1,$$

so Alice's win probability is

$$P(\text{Alice wins}) = \frac{1}{2}(1 - P(\text{tie})).$$

Thus maximizing her chance of victory is equivalent to minimizing the probability of a tie. Let p_i be the probability that the die shows face value i , for $i = 1, 2, \dots, 6$. A tie occurs exactly when both players roll the same i , so

$$P(\text{tie}) = \sum_{i=1}^6 p_i^2.$$

We must choose the p_i so that $\sum_{i=1}^6 p_i = 1$ (they form a probability distribution) and each face value lies between 1 and 6. By the Cauchy-Schwarz (or QM-AM) inequality,

$$\sum_{i=1}^6 p_i^2 \geq \frac{1}{6} \left(\sum_{i=1}^6 p_i \right)^2 = \frac{1}{6},$$

with equality if and only if all the p_i are equal. Therefore, the tie probability is minimized at $\frac{1}{6}$, achieved by setting

$$p_1 = p_2 = \dots = p_6 = \frac{1}{6}.$$

Alice should choose a six-sided die with distinct values and equal probabilities for each face. Specifically, she should select a die with values 1, 2, 3, 4, 5, 6, each having a probability of $\frac{1}{6}$. Plugging $P(\text{tie}) = \frac{1}{6}$ into our formula gives

$$P(\text{Alice wins}) = \frac{1}{2} \left(1 - \frac{1}{6} \right) = \frac{5}{12}.$$

Answer: 0.41667

641. Cherries

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To ensure k cherries are left in the bowl, they must be purple, and the $k + 1$ -th cherry from the end must be red. Thus, the problem reduces to arranging $m - 1$ red cherries and $n - k$ purple cherries in any order. We account for indistinguishable cherries by using the formula:

$$\frac{(m + n - k - 1)!}{(m - 1)!(n - k)!}$$

The total number of arrangements of m red and n purple cherries is:

$$\frac{(m + n)!}{m!n!}$$

Thus, the probability is:

$$\frac{\frac{(m+n-k-1)!}{(m-1)!(n-k)!}}{\frac{(m+n)!}{m!n!}} = \frac{70}{429}$$

Answer: 0.16317

642. Even Tails Run

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let a_n be the number of length- n sequences with every run of tails even. Think about how such a sequence starts: - If the first flip is H, then the remaining $n - 1$ flips can be any valid sequence of length $n - 1$, giving a_{n-1} possibilities. - If the first flip is T, you must immediately flip T again, and the rest $n - 2$ flips form any valid sequence of length $n - 2$, contributing a_{n-2} possibilities. So:

$$a_n = a_{n-1} + a_{n-2}$$

To use this recurrence, we need initial values: - For $n = 1$, the only valid sequence is H, since a single T would be an odd run. So $a_1 = 1$. - For $n = 2$, the valid sequences are HH, HT, and TT. TH is invalid because the single T is an odd-length run. So $a_2 = 2$. Thus:

$$a_1 = 1, \quad a_2 = 2, \quad a_n = a_{n-1} + a_{n-2} \quad (n \geq 3)$$

Starting from $a_1 = 1$ and $a_2 = 2$, the recurrence produces:

$$a_3 = 3, \quad a_4 = 5, \quad \dots, \quad a_{10} = 89$$

Since there are $2^{10} = 1024$ equally likely sequences of 10 fair coin flips, the probability of having no run of tails of odd length is:

$$\frac{a_{10}}{2^{10}} = \frac{89}{1024}$$

Therefore, the answer is

$$\frac{89}{1024}$$

Answer: 0.08691

643. Same Birthday Probability

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will calculate the probability that all students have unique birthdays and then subtract it from 1 to get the answer. The probability that all of them have unique birthdays is:

$$\frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \dots \times \frac{343}{365}$$

This is because the first student can have any of the 365 days as a birthday ($\frac{365}{365}$), the second student must have a birthday different from the first, leaving 364 choices out of 365 ($\frac{364}{365}$), and so on, until the 23rd student, who has 343 available days. Hence, the required probability is:

$$1 - \left(\frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \cdots \times \frac{343}{365} \right)$$

This is approximately equal to 0.507 (of course, you are not expected to calculate it on your own).

Answer:

644. Matrix Attack I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In a 3×3 grid, where the total sum of all numbers (1 through 9) is odd, achieving all rows and columns with even sums is not feasible. This is because an odd total sum cannot be split evenly into rows and columns where each row and column's sum must be even. Thus, it's impossible to satisfy the condition, resulting in a probability of 0.

Answer:

645. Count Of Rational Numbers

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

From $\gcd(a, b) = 1$, we note that for each prime p dividing $25!$, either $p \mid a$ or $p \mid b$. As a result, for each prime p dividing $25!$, we have two choices: whether p divides a or p divides b . Since there are nine primes less than 25, the total number of ways to choose is $2^9 = 512$.

Answer:

646. Shooting Arrows I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The circular target has three concentric rings with radii 1 meter, 2 meters, and 3 meters. The total area of the target is the area of the outermost circle, which is:

$$\pi(3)^2 = 9\pi m^2$$

The area of the innermost circle (radius 1 meter) is:

$$\pi(1)^2 = \pi m^2$$

Thus, the probability that a single arrow lands within this innermost circle is:

$$\frac{\pi}{9\pi} = \frac{1}{9}$$

Now, the probability that an arrow does not land in the innermost circle is:

$$1 - \frac{1}{9} = \frac{8}{9}$$

Since the arrows land independently, the probability that none of the 3 arrows land within the innermost circle is:

$$\left(\frac{8}{9}\right)^3 = \frac{512}{729}$$

Therefore, the probability that at least one of the arrows lands within the innermost circle is:

$$1 - \frac{512}{729} = \frac{217}{729}$$

Hence, the required probability is $\frac{217}{729}$.

Answer:

647. Card Picking I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider only the king (all 4 suits), queen (all 4 suits), jack (all 4 suits), and ace of diamonds. The favorable arrangements are those where exactly 1 king, 1 queen, and 1 jack appear before the ace of diamonds, followed by the remaining 9 cards. The total number of arrangements of these 13 cards is $13!$. To find the number of favorable arrangements, imagine the following: 1. Choose the suit of the king that appears before the ace of diamonds (4 choices). 2. Choose the suit of the queen that appears before the ace of diamonds (4 choices). 3. Choose the suit of the jack that appears before the ace of diamonds (4 choices). 4. Arrange these three chosen cards ($3!$ ways). 5. Place the ace of diamonds (1 way). 6. Arrange the remaining 9 cards ($9!$ ways). Thus, the number of favorable arrangements is: $4 \cdot 4 \cdot 4 \cdot 3! \cdot 1 \cdot 9!$. Hence, the required probability is:

$$\frac{4 \cdot 4 \cdot 4 \cdot 3! \cdot 9!}{13!} = \frac{16}{715}$$

Answer:

648. Game On 753 Coins

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

After each operation, the number of piles increases by one and the number of coins in play decreases by one. Suppose it is possible for Bob to win, i.e., there exists some number of operations n such that, after n operations, Bob wins. Importantly, there must be $n + 1$ piles of coins after these n operations, each of which has 3 coins. If this is right, then the following equation should hold:

$$753 = 3(n + 1) + n$$

Simplifying this equation, we get:

$$750 = 4n$$

$$n = 187.5$$

which is a contradiction. Hence, Bob cannot win.

Answer:

649. Triangle Inside Circle III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The triangle is created by linking pairs of points with chords. Since the points are drawn independently, each side length has the same expected value. Thus, to find the expected perimeter, we need to multiply the expected length of one side by 3. Let P be the perimeter, and S_{AB} the length of the chord between points A and B . The expected perimeter $E[P]$ equals $3E[S_{AB}]$. This leads us to find the expected length of a chord drawn between two random points on a circle. Let angles θ and ϕ be independently drawn from $Unif(0, 2\pi)$, representing two random points on the circle. The angle between these points is $\theta - \phi$. To simplify the problem, we can rotate the circle so that one of the points is fixed at $(1, 0)$, i.e., $\theta_1 = 0$. The distance between $\theta_1 = 0$ and the other point $\theta_2 = \theta$ is given by $\sqrt{(1 - \cos(\theta))^2 + \sin^2(\theta)} = \sqrt{2(1 - \cos(\theta))}$. So, the expected length of the chord becomes $2E[1 - \cos(\theta)]$, where $\theta \sim Unif(0, 2\pi)$. Now, $2E[1 - \cos(\theta)] = \frac{2}{2\pi} \int_0^{2\pi} (1 - \cos(\theta)) d\theta$. This integral simplifies to $\frac{4}{\pi}$. Thus, the expected length of the chord is $\frac{4}{\pi}$. Finally, the expected perimeter of the triangle is $3 \times \frac{4}{\pi} = \frac{12}{\pi}$.

Answer:

650. Coin Toss Game VI

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For $1 \leq n \leq 6$, the expected reward is $\frac{1}{2^n} \times 2^n = 1$ for each n . For instance, when $n = 1$ (which occurs with probability $\frac{1}{2}$), Alice receives 2 dollars, so the expected reward is 1 dollar. Similarly, for $n = 2$, which happens with probability $\frac{1}{4}$, Alice receives 4 dollars, making the expected reward 1 dollar. This pattern holds up to $n = 6$, after which the reward is capped at 64 dollars.

For $n \geq 7$, the expected reward becomes:

$$\frac{1}{2^7} \times 64 + \frac{1}{2^8} \times 64 + \cdots = \frac{1}{2} + \frac{1}{4} + \cdots = \sum_{t \geq 0} \frac{1}{2^t} = 1$$

Thus, the total expected reward is $6 \times 1 + 1 = 7$.

Answer:

651. Count Paths I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label each upward step as U and each rightward step as R . There is a one-to-one correspondence between each valid path and a string consisting of 8 R 's and 10 U 's. For instance, the path that moves from $(0, 0)$ to $(8, 0)$, and then to $(8, 10)$, can be represented as $RRRRRRRRUUUUUUUUUU$. Hence, the required number of paths is:

$$\frac{(8 + 10)!}{8! \cdot 10!} = 43758$$

Answer:

652. Game On Divisors II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Suppose Alice doesn't have a winning strategy. Let's assume she removes 1. Then Bob will have to remove some other number (say p), along with its divisors, and this must lead to a losing state for Alice when it's her turn. However, Alice could have given Bob this state in her first turn by selecting p . This leads to a contradiction.

Hence, Alice has a winning strategy.

Answer:

653. Polygon Triangle

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the 17-sided polygon be P , and choose an arbitrary vertex A . Assume without loss of generality that A is one of the selected vertices. Let the other two vertices be B and C , chosen in a clockwise direction. Suppose there are k edges of P contained in the minor arc BAC . For the triangle $\triangle ABC$ to contain the center of P , the angle $\angle BAC$ must be less than 90° , which implies that the number of edges k in the minor arc BAC must satisfy: $1 \leq k \leq 8$. For each value of k between 1 and 8, there are k possible choices for vertices B and C , giving a total of:

$$\sum_{k=1}^8 k = \frac{8 \times (8+1)}{2} = 36$$

valid triangles. Since there are $\binom{16}{2} = \frac{16 \times 15}{2} = 120$ ways to choose the vertices B and C from all the remaining vertices of P , the probability that a randomly chosen triangle contains the center of P is: $\frac{36}{120} = \frac{3}{10}$

Answer:

654. Ants On A Wire

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Instead of assuming the ants change direction upon collision, we can think of them as simply passing through one another. This approach works because, after each collision, the ants will continue moving in the direction the other was originally heading. In this sense, it is as if no collision occurred. With this perspective, it's clear that each of Alice's 40 ants will collide with each of Bob's 80 ants. Therefore, the total number of collisions is $40 \times 80 = 3200$.

Answer:

655. Clock Angle

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The minute hand, at 15 minutes past the hour, points at the 3 on the clock. The hour hand, at 3 : 15, is slightly ahead of the 3 because it moves continuously throughout the hour. Specifically, the hour hand advances 0.5 degrees per minute, so in 15 minutes, it moves $15 \times 0.5 = 7.5$ degrees past the 3 o'clock position. Hence, the required angle is 7.5 degrees.

Answer:

656. Weighted Coins

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We note that these weights form binary numbers, except the "1" is omitted. Thus the numbers that are generated are exactly the even numbers between 2 and 2046, inclusive. Thus the number of possibilities is the number of even numbers between 138 and 1060, inclusive, which is exactly 462. There are $2^{10} = 1024$ possible weights, which gives us a probability of $\frac{462}{1024} = \frac{231}{512}$.

Answer:

657. 5000 Sets[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

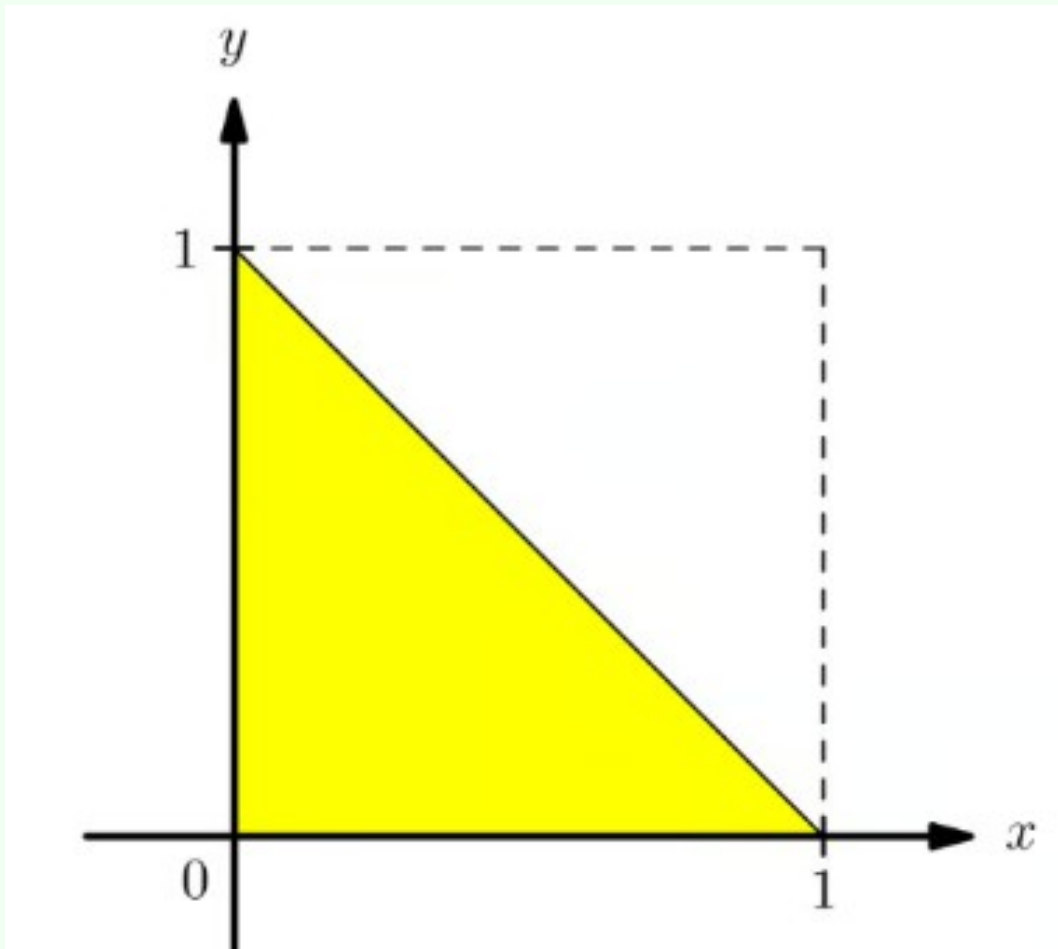
Let U be the union of the finite sets. A 12-element set has $2^{12} = 4096$ subsets, therefore, there must be at least 13 elements in U . Suppose U has 13 elements. We can pair each subset of U with its complement. There are 4096 such pairs. Since there are 5000 sets in the collection, there must be a pair consisting of a set and its complement among the sets in the collection. However, these two sets have an empty intersection, contradicting the requirement of the problem. Now, suppose U has 14 elements. There are $\frac{2^{14} - \binom{14}{7}}{2} = 6476$ subsets of U which contain 8 or more elements. We create a collection of 5000 subsets of U by taking the 14 subsets that have 13 elements and any $5000 - 14$ other subsets that have at least 8 elements, but not all 14 elements. Then, any two sets in this collection will have a nonempty intersection because otherwise U would need to have at least 16 elements. Additionally, the intersection of all 5000 sets is empty since no element is present in all 14 of the subsets with 13 elements.

Answer: 14**658. Three Steps**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let x and y be random variables that are independently and uniformly distributed in the interval $(0, 1)$. Note that:

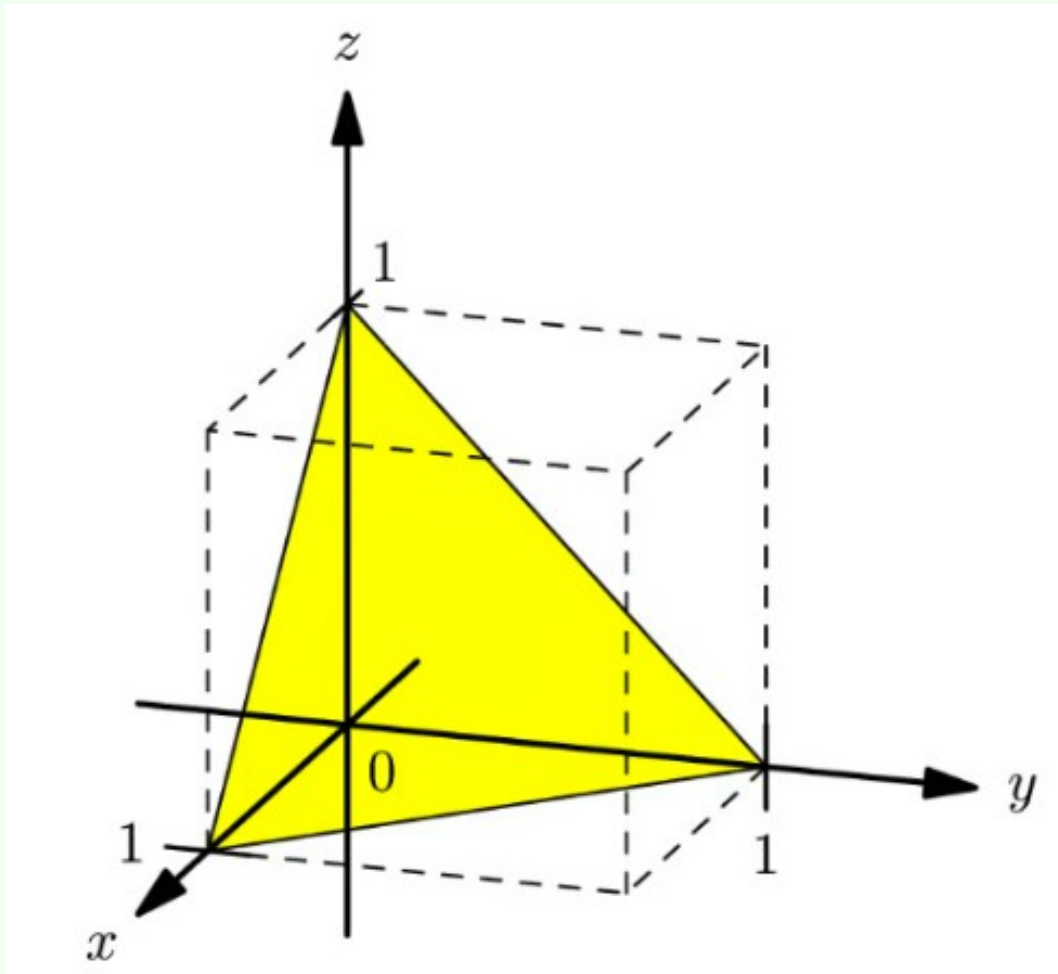
$$P(x + y \leq 1) = \frac{\frac{1}{2} \cdot 1^2}{1^2} = \frac{1}{2},$$

as shown below:



$$P(x + y + z \leq 1) = \frac{\frac{1}{3} \cdot \left(\frac{1}{2} \cdot 1^2\right) \cdot 1}{1^3} = \frac{1}{6},$$

as shown below:



Case 1: The ant takes exactly 2 steps We need $x_1 + x_2 > 1$ and $t_1 + t_2 > 1$. The probability is:

$$P(x_1 + x_2 > 1) \cdot P(t_1 + t_2 > 1) = \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{2}\right) = \frac{1}{4}$$

Case 2: The ant takes exactly 3 steps We need $x_1 + x_2 + x_3 > 1$ and $t_1 + t_2 \leq 1$. The probability is:

$$P(x_1 + x_2 + x_3 > 1) \cdot P(t_1 + t_2 \leq 1) = \left(1 - \frac{1}{6}\right) \cdot \frac{1}{2} = \frac{5}{12}$$

Final Calculation Adding the probabilities from both cases, the total is:

$$\frac{1}{4} + \frac{5}{12} = \frac{2}{3}$$

Answer:

659. Just Even Stop

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Use recursion. On the first toss, you can get either heads or tails. On the second toss, there is a $\frac{1}{2}$ probability of getting the same outcome as the first toss, resulting in an even number of tosses. If the second toss differs

from the first, the problem is reversed, as the first toss becomes irrelevant, and now you need to end with an odd number of tosses. Let the required probability be p . We have the following equation:

$$p = \frac{1}{2} + \frac{1}{2}(1 - p)$$

$$\Rightarrow p = \frac{2}{3}$$

Answer:

660. Shooting Stacks

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Consider each target in a stack as labeled by the stack it belongs to: 1 for the first stack, 2 for the second stack, 3 for the third stack, and 4 for the fourth stack. The task then becomes finding the number of distinct ways to arrange the sequence of targets while respecting the rule that within each stack, lower targets must be hit before higher ones. This is equivalent to calculating the number of permutations of a multiset consisting of 2 occurrences of 1, 3 occurrences of 2, 3 occurrences of 3, and 4 occurrences of 4. Using the formula for permutations of a multiset, the total number of arrangements is given by:

$$\frac{12!}{2! \cdot 3! \cdot 3! \cdot 4!} = 277200$$

Thus, there are 277200 distinct sequences in which the shooter can hit all the targets.

Answer:

661. Segments Inside Square I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let L denote the straight line through the random points P and Q . For almost every choice of P and Q the line L meets the boundary of the square in exactly two points and those two intersection points lie on two of the four sides of the square. Let I_{bottom} be the indicator that the line meets the bottom side. Similarly define indicators for the four sides. For each pair of points the sum of these four indicators equals 2 (the line meets exactly two sides). By rotational symmetry all four sides are equally likely to be hit, so the expected value of each indicator is the same. Taking expectations,

$$4 \cdot E[I_{\text{bottom}}] = E[\text{sum of indicators}] = 2$$

hence

$$E[I_{\text{bottom}}] = \frac{1}{2}$$

which is the probability that the line L intersects a given side. So the probability that L intersects AB is $\frac{1}{2}$. When the line L does intersect AB , the points A and B lie on opposite sides of L . In that situation the segments AP and BQ lie on opposite sides of L and therefore cannot cross inside the square. Consequently an intersection of the segments is only possible when the line L does not meet AB . The probability of that complementary event is also $\frac{1}{2}$. Condition now on the event that L does not intersect AB . In this case the four points A, B, P, Q are the vertices of a convex quadrilateral. For any convex quadrilateral the two segments joining opposite vertices cross exactly when the four vertices appear in alternating order around the boundary. Here A and B are adjacent vertices of the square, so the diagonals we consider are AP and BQ . Given L does not hit AB , the two possible circular orders of the four vertices that keep A and B adjacent are equally likely by symmetry of choosing P and Q independently. One of those orders makes the segments AP and BQ intersect and the other does not, so the conditional probability of intersection given that L misses AB is $\frac{1}{2}$. Combine the two facts by the multiplication rule. The probability that AP and BQ intersect equals the probability that L does not hit AB multiplied by the conditional probability

of intersection in that case. This equals

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Therefore the two random segments intersect with probability $\frac{1}{4}$.

Answer:

662. Dice Game VI

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Calculate the expected payout for rolling again for each possible result of the first roll. If the first roll is 1, rolling again would yield:

$$\frac{1}{6} \times (-1) + \frac{1}{6} \times (2 + 3 + 4 + 5 + 6) = \frac{19}{6}$$

Since the expected payout if we re-roll $\frac{19}{6}$ is higher than the expected payout if we keep the first roll (1), we should roll again if the first roll is 1. If the first roll is 2, rolling again would yield:

$$\frac{1}{3} \times (-2) + \frac{1}{6} \times (3 + 4 + 5 + 6) = \frac{7}{3}$$

Hence, roll again if the first roll is 2. If the first roll is 3, rolling again would yield:

$$\frac{1}{2} \times (-3) + \frac{1}{6} \times (3 + 4 + 5 + 6) = 1$$

Therefore, keep the roll if the first roll is 3 or more. Overall, the expected payout with this strategy is:

$$\frac{1}{6} \times \frac{19}{6} + \frac{1}{6} \times \frac{7}{3} + \frac{1}{6} \times (3 + 4 + 5 + 6) = \frac{47}{12}$$

Answer:

663. Fibonnaci Error

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The calculation for G_{101} is $G_{100} + F_{99} = 1 + (F_{100} + F_{99}) = 1 + F_{101}$. Subsequently, G_{102} is calculated as $G_{101} + G_{100} = 2 + (F_{101} + F_{100}) = 2 + F_{102}$. Generally, the error propagates similarly to the Fibonacci sequence itself, since each error is derived from the sum of errors in the two preceding terms. Initially, the errors are 1 and 1 instead of 0 and 1. Therefore, the error term follows F_{n+1} . For $n = 10$, where calculations are made for 10 terms beyond F_{100} , the error is $F_{11} = 89$.

Answer:

664. Digit Sum V

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To minimize $S(n+1)$, when we add 1 to n , we should change as many digits as possible from 9 to 0 due to the carry. For example, if we have $n = 199$, adding 1 gives us $n+1 = 200$, changing two nines to zeros. Since $2018 = 224 \cdot 9 + 2$, we can let n be a number starting with the digit 2 followed by 224 nines. Then adding 1 to n gives us the digit 3 followed by 224 zeros. Hence, $S(n+1) = 3$.

Answer:

665. Expected Swaps[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

An inversion is a pair of indices (i, j) where $i < j$ and $a[i] > a[j]$. Each adjacent swap in bubble sort resolves one inversion, meaning the total number of swaps performed by the algorithm corresponds directly to the number of inversions that need to be resolved. For a permutation of length n , let I_{ij} be an indicator variable for the inversion (i, j) . Specifically, $I_{ij} = 1$ if (i, j) is an inversion and $I_{ij} = 0$ otherwise.

To compute the expected value of I_{ij} , we find that

$$E[I_{ij}] = P((i, j) \text{ is an inversion}) = \frac{1}{2}.$$

This probability arises from symmetry: for any pair (i, j) that is an inversion, the pair (j, i) is not an inversion.

The total number of pairs (i, j) with $i < j$ in an array of length n is given by

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

By applying the linearity of expectation, the expected number of inversions in the entire array can be calculated as follows:

$$E[\text{Total Inversions}] = \sum_{1 \leq i < j \leq n} E[I_{ij}] = \sum_{1 \leq i < j \leq n} \frac{1}{2} = \frac{1}{2} \cdot \binom{n}{2} = \frac{1}{2} \cdot \frac{n(n-1)}{2}.$$

This leads to the conclusion that the expected number of inversions is

$$E[\text{Total Inversions}] = \frac{n(n-1)}{4}.$$

Since each swap resolves one inversion, the expected number of swaps performed by the bubble sort algorithm is also

$$\frac{n(n-1)}{4}.$$

Answer: 22.50000**666. Interesting Average I**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We calculate the expected value of each digit, then use linearity of expectation to find the total expected value. Let $a_1, a_2, a_3, a_4, a_5, a_6$ denote the expected value of each digit and let the expected value be $a = a_1 a_2 a_3 a_4 a_5 a_6$. By symmetry, we have $a_2 = a_5 = 5$, and $a_1 + a_3 = a_4 + a_6 = 10$. If $a_1 = k$, then $a_2, a_3 \in \{k+1, \dots, 9\}$ and (a_2, a_3) correspond to the number of ways: $(9-k-1) + (9-k-2) + \dots + 1 = \binom{9-k}{2}$. To choose two numbers among $\{k+1, \dots, 9\}$, so calculating and using the hockey stick identity

$$a_1 = \frac{1 \cdot \binom{8}{2} + 2 \cdot \binom{7}{2} + \dots + 7 \cdot \binom{2}{2}}{\binom{8}{2} + \binom{7}{2} + \dots + \binom{2}{2}} = \frac{\binom{9}{3} + \binom{8}{3} + \dots + \binom{3}{3}}{\binom{9}{3}} = \frac{10}{4} = \frac{5}{2}$$

If $a_6 = k$, then $a_5, a_4 \in \{k, \dots, 9\}$ and (a_5, a_4) correspond to the number of ways: $(9-k+1) + (9-k) + \dots + 1 = \binom{9-k+2}{2}$. To choose two numbers among $\{k, \dots, 9\}$, allowing replacement, so calculating and using the hockey stick identity:

$$a_6 = \frac{1 \cdot \binom{10}{2} + 2 \cdot \binom{9}{2} + \dots + 9 \cdot \binom{2}{2}}{\binom{10}{2} + \binom{9}{2} + \dots + \binom{2}{2}} = \frac{\binom{11}{3} + \binom{10}{3} + \dots + \binom{3}{3}}{\binom{11}{3}} = \frac{12}{4} = 3$$

Then it follows that $a = a_1 a_2 a_3 a_4 a_5 a_6 = 250000 + 50000 + 7500 + 700 + 70 + 3 = 308253$.

Answer: 308253

667. Changing Number

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We seek a positive integer N ending in the digit 4 such that when we move that final 4 to the front, the new number is exactly four times N . Since N ends with 4, we can write $N = 10A + 4$, where A is the integer formed by all but the last digit of N . If N has $n + 1$ digits, then A has n . Moving the final 4 to the front produces the number $M = 4 \cdot 10^n + A$. Our requirement $M = 4N$ becomes $4 \cdot 10^n + A = 4(10A + 4) = 40A + 16$, so $39A = 4 \cdot 10^n - 16$, and hence $A = \frac{4 \cdot 10^n - 16}{39}$. We must find the smallest n for which the right-hand side is an integer with exactly n digits (so that N really has $n + 1$ digits and ends in 4). If $n = 1$, we get $4 \cdot 10 - 16 = 24$, but $\frac{24}{39}$ is not an integer. If $n = 2$, $4 \cdot 100 - 16 = 384$, and $\frac{384}{39}$ is not integral. If $n = 3$, $4 \cdot 1000 - 16 = 3984$, and $\frac{3984}{39}$ is not integral. If $n = 4$, $4 \cdot 10000 - 16 = 39984$, and $\frac{39984}{39}$ is not integral. When $n = 5$, however, $4 \cdot 100000 - 16 = 399984$, $\frac{399984}{39} = 10256$, which is indeed a 5-digit integer. Thus $A = 10256$, and $N = 10A + 4 = 102564$.

Answer: 102564

668. Acute Triangle

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Thales' theorem says that if a side AB of an inscribed triangle is a diameter, then the angle at the opposite vertex C is a right angle. If the circle's center lies outside the triangle, there is some side (say AB) whose supporting line separates the center from the opposite vertex C , in that case the angle $\angle ACB$ is obtuse. Consequently, a triangle inscribed in a circle is acute-angled if and only if this never happens for any of its sides. This condition is equivalent to saying that the center of the circle lies strictly inside the triangle. Thus, the problem reduces to finding the probability that the center of the circle lies inside the triangle formed by three random points on the circumference.

Fix the circle's circumference length and call it C . Place the first chosen point (call it A) at a reference location on the circle. Let X denote the length (along the circle) of the minor arc between the first and the second point (call it B), by definition $0 \leq X \leq C/2$. To find the distribution of X , note that when the second point is uniformly distributed on the circle, the event $X \leq t$ (for $0 \leq t \leq C/2$) is equivalent to the second point falling within the union of two arcs of total length $2t$. Hence the cumulative distribution function is given by

$$F_X(t) = P(X \leq t) = \frac{2t}{C} \text{ for } 0 \leq t \leq C/2$$

and differentiating gives the probability density

$$f_X(x) = \frac{2}{C} \text{ for } 0 < x < C/2$$

Next, condition on $X = x$. With A and B fixed at minor-arc separation x , the set of positions for the third point that make the center of the circle lie inside the triangle is exactly the arc opposite the minor arc AB , and that opposite arc has length x . Therefore, given $X = x$, the conditional probability that the third point produces an acute triangle is x/C .

Apply the law of total probability (continuous form): the required probability equals

$$\int_0^{C/2} P(\text{acute} \mid X = x) f_X(x) dx = \int_0^{C/2} \frac{x}{C} \cdot \frac{2}{C} dx = \frac{1}{4}$$

Hence, the probability that a triangle formed by three random points on a circle is acute-angled is $\frac{1}{4}$.

Answer: 0.25

669. Segment Intersection[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Model each line segment as a vertex of an undirected graph, and draw an edge between two vertices if the corresponding segments intersect. Since each segment intersects exactly three others, every vertex has degree 3. Hence, the sum of the degrees is $693 = 207 \cdot 3$.

In any undirected graph, each edge contributes 2 to the total degree count, so the number of edges must be half the sum of the degrees (over all vertices). This would give $\frac{207}{2} = 103.5$ edges, which is impossible since the number of edges must be an integer. So, such a configuration cannot exist.

Answer: 2**670. Powerful Three**[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Note that $3^{4n} = (3^2)^{2n} = 9^{2n}$, and since $9 = -1 + 10$, we may write $3^{4n} = (-1 + 10)^{2n}$. Expanding this using the binomial theorem gives

$$(-1 + 10)^{2n} = \binom{2n}{0}(-1)^{2n} + \binom{2n}{1}(-1)^{2n-1}10 + \binom{2n}{2}(-1)^{2n-2}10^2 + \cdots + \binom{2n}{2n}10^{2n}.$$

The first term is exactly 1. Every term from the 102nd term onward contains a factor of 10^{101} or higher, so such terms automatically end with at least 101 zeros. We now focus on the remaining terms, namely the terms involving $10, 10^2, \dots, 10^{100}$. Each of these terms has the form $\binom{2n}{k}(-1)^{2n-k}10^k$ with $1 \leq k \leq 100$. If we choose n so that $2n$ is a very large power of 10, then each binomial coefficient $\binom{2n}{k}$ for $1 \leq k \leq 100$ is divisible by a very large power of 10. In particular, we can choose n so that $\binom{2n}{k}(-1)^{2n-k}10^k$ has at least 101 trailing zeros for $k = 1$ to 100. Consequently, each of these terms (except the first) is divisible by 10^{101} and hence ends with at least 101 zeros. Thus, apart from the first term 1, every other term in the expansion ends with at least 101 zeros. Since the entire sum is positive, the decimal representation of 3^{4n} for such a choice of n ends with a 100 zeros followed by a 1, and therefore there does exist such a positive integral power of 3.

Answer: 1**671. Socks Remaining II**[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let T_r, T_b, T_g be the draw positions (from 1 to 60) at which the last red, last blue, and last green sock are removed respectively. We want the probability that when the last red sock is removed there still remains at least one blue and one green sock in the drawer; in terms of the last-occurrence times this is the event $T_r < T_b$ and $T_r < T_g$ (i.e. the last red appears earlier than the last blue and the last green). The three times T_r, T_b, T_g give a random ordering, one convenient way to evaluate the desired probability is to condition on which color is the very last sock drawn (the color occupying position 60).

If the last sock is green (this happens with probability $30/60$), then $T_g = 60$ and the remaining 59 positions are a random arrangement of the other 29 greens together with the 10 reds and 20 blues. Among the 30 red-or-blue socks that lie somewhere in those first 59 positions, every such sock is equally likely to be the last one among red and blue, therefore the probability that the last among red-and-blue is blue equals $20/30$. Conditioned on $T_g = 60$, if the last among red-and-blue is blue then $T_r < T_b$, so $\Pr(T_r < T_b \mid T_g = 60) = 20/30$.

If instead the last sock is blue (probability $20/60$), then by the same reasoning among the 10 reds and 30 greens in the first 59 positions the last of those is equally likely to be any of the 40 red-or-green socks, so the probability the last among red-and-green is green is $30/40$; when that happens the last green comes after

the last red, so $T_r < T_g$.

If the last sock is red (probability $10/60$) then $T_r = 60$ and it cannot be true that $T_r < T_b$ and $T_r < T_g$. Putting these conditioned pieces together and using total probability gives

$$\Pr(T_r < T_b, T_r < T_g) = \frac{30}{60} \cdot \frac{20}{30} + \frac{20}{60} \cdot \frac{30}{40} = \frac{7}{12}.$$

Hence the probability that after all red socks have been taken out there still remains at least one blue and at least one green sock is $7/12$.

Answer:

672. Tetrahedron Coloring I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Assume we have 4 colors - 1, 2, 3, and 4. There are $4! = 24$ ways to assign four distinct colors to the four faces if we ignore symmetry, but many of these assignments are equivalent under rotations of the tetrahedron. One quick way to account for rotations is to fix the bottom as color 1. On the remaining sides, the colors can be arranged as 2,3,4 (in that order) or as 2,4,3 - these two arrangements are not rotationally identical. Hence, there are exactly 2 valid colorings.

Answer:

673. Domino Tiling I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

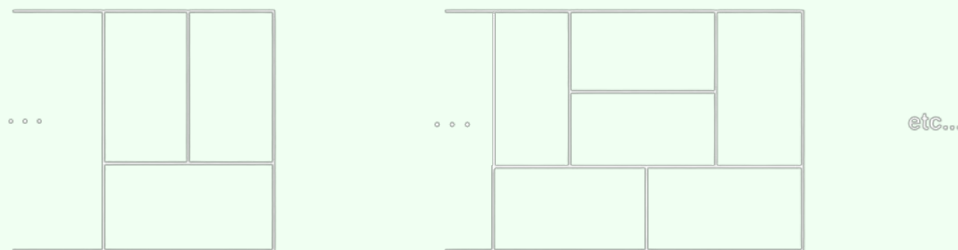
Let a_n be the number of tilings of a $2 \times n$ rectangle. For $n = 1$ there is only one tiling (one vertical domino), so $a_1 = 1$; for $n = 2$ there are two tilings (two vertical dominos or two horizontal dominos), so $a_2 = 2$. For $n \geq 3$ consider the leftmost column: either it is covered by a single vertical domino (leaving a $2 \times (n - 1)$ region, which can be tiled in a_{n-1} ways), or it is covered by two horizontal dominos that occupy the first two columns (leaving a $2 \times (n - 2)$ region, which can be tiled in a_{n-2} ways). These two cases are disjoint and exhaust all possibilities, so $a_n = a_{n-1} + a_{n-2}$. This recurrence with the initial values $a_1 = 1, a_2 = 2$ produces the sequence 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ..., hence $a_{10} = 89$. Therefore there are 89 distinct tilings of a 2×10 grid by 2×1 dominos.

Answer:

674. Domino Tiling II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Trivially, there is exactly 1 way to tile a 3×0 rectangle, since it contains no squares. It is also easy to verify by direct inspection that a 3×2 rectangle can be tiled in exactly 3 different ways using 2×1 dominos. Let T_n denote the number of tilings of a $3 \times n$ rectangle, where n must be even, since an odd number of columns cannot be fully covered by dominos.



By examining the possible ways dominos can occupy the leftmost part of the rectangle (as illustrated in the diagrams), we see two types of configurations. Either the first two columns are completely filled in one of the 3 basic ways, leaving a $3 \times (n-2)$ rectangle, or a more complicated interlocking pattern appears that forces an extended structure, contributing terms involving $T_{n-4}, T_{n-6}, \dots, T_0$. Accounting for all such possibilities leads to the recurrence relation

$$T_n = 3T_{n-2} + 2(T_{n-4} + T_{n-6} + \dots + T_2 + T_0)$$

Using the initial values $T_0 = 1$ and $T_2 = 3$, we can compute successive values: $T_4 = 11$ and $T_6 = 41$. Substituting these into the recurrence once more gives $T_8 = 153$. Therefore, there are exactly 153 distinct ways to tile a 3×8 rectangular grid with 2×1 dominos.

Answer: 153

675. Largest Mean Median Difference

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let the five scores, arranged in nondecreasing order, be $a \leq b \leq c \leq d \leq e$, where each score is an integer between 0 and 100. Since there are five scores, the median is the middle one, namely c , and the mean is $\frac{a+b+c+d+e}{5}$. We want to make the absolute difference between the mean and the median as large as possible.

First, suppose we want the mean to be much larger than the median. To keep the median small, we can take the first three scores as low as possible by setting $a = b = c = 0$. To make the mean large, we take the remaining two scores as large as possible, setting $d = e = 100$. In this case, the mean is $\frac{0+0+0+100+100}{5} = \frac{200}{5} = 40$, while the median is 0, giving a difference of 40.

Next, suppose we try to make the median large while keeping the mean relatively small. To force the median to be large, we must have at least three scores equal to 100, so take $c = d = e = 100$. To reduce the mean, we choose the remaining two scores as small as possible, setting $a = b = 0$. The mean is then $\frac{0+0+100+100+100}{5} = 60$, and the median is 100, giving a difference of 40 again.

No other arrangement can produce a larger separation, since the scores are bounded between 0 and 100. Therefore, the largest possible difference between the mean and the median is 40.

Answer: 40

676. New Language

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

There are 5 basic letters, and each letter can appear in two forms: emphasized or not. Thus each position in a word has $5 \times 2 = 10$ possible choices. For a word of length k , each of the k positions can be filled independently in 10 ways. Hence there are 10^k distinct words of length k . Words may have lengths from 1 through 5, so the total number of distinct words is $10^1 + 10^2 + 10^3 + 10^4 + 10^5 = 111110$

Answer:

677. Constrained Subsets III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The sum of all numbers 1 through 64 is $\frac{64 \cdot 65}{2} = 2080$. A subset of $\{1, \dots, 64\}$ has sum 2072 exactly when its complement has sum $2080 - 2072 = 8$. So there is a one-to-one correspondence between subsets summing to 2072 and subsets summing to 8. Any subset summing to 8 can only use numbers 1 through 8, the distinct subsets of $\{1, \dots, 8\}$ with sum 8 are $\{8\}, \{1, 7\}, \{2, 6\}, \{3, 5\}, \{1, 2, 5\}, \{1, 3, 4\}$, which gives 6 possibilities. Therefore the number of subsets of $\{1, \dots, 64\}$ whose elements sum to 2072 is 6.

Answer:

678. Count Rectangles II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If the grid were complete, any rectangle would be determined by choosing two of the 7 vertical lines and two of the 7 horizontal lines, so there would be $\binom{7}{2}^2 = 21^2 = 441$ axis-aligned rectangles. A rectangle is invalid exactly when its boundary passes through the removed central intersection $(3, 3)$ or when one of its corners lies at $(3, 3)$. If $(3, 3)$ lies in the interior of a side (say the south side), there are $3 \cdot 3 \cdot 3 = 27$ choices (three choices for the left vertical line, three for the right vertical line, and three for the top horizontal line). Accounting for all four sides gives $4 \cdot 27 = 108$ such rectangles. If $(3, 3)$ is a corner of the rectangle (for instance, the south-west corner), then there are $3 \cdot 3 = 9$ possible choices for the remaining corner positions. Accounting for all four possible orientations of the rectangle, this gives a total of $4 \cdot 9 = 36$ such rectangles. Thus the number of invalid rectangles is $108 + 36 = 144$, and the number of allowed rectangles is $441 - 144 = 297$.

There are 297 such rectangles.

Answer:

679. Tricky Palindrome I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Break the palindromes under 1000 into cases by their number of digits.

One-digit palindromes. Every single-digit positive integer $1, 2, \dots, 9$ is a palindrome (we exclude 0 because the definition forbids divisibility by 10). That gives 9 palindromes.

Two-digit palindromes. A two-digit palindrome must have the form $\overline{aa}$, where the tens and units digits are equal. The leading digit a cannot be 0, so $a \in 1, \dots, 9$. Thus there are 9 two-digit palindromes.

Three-digit palindromes. A three-digit palindrome has the form $\overline{aba}$: the first and last digits are the same nonzero digit $a \in 1, \dots, 9$, and the middle digit b can be any digit $0, 1, \dots, 9$. This yields $9 \times 10 = 90$ three-digit palindromes.

Adding the three cases gives the total number of palindromes less than 1000: $9 + 9 + 90 = 108$.

So there are 108 palindromes less than 1000.

Answer:

680. Tricky Palindrome II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write a generic four-digit palindrome as $\overline{abba}$, where a is the thousands (and units) digit and b is the hundreds (and tens) digit. Here $a \in 1, 2, \dots, 9$ (cannot be 0) and $b \in 0, 1, \dots, 9$.

Multiply $\overline{abba}$ by 2. We must keep track of carries:

•The units digit of $2n$ equals $2a \pmod{10}$; its carry to the tens place is $c_1 = \lfloor 2a/10 \rfloor$ (so $c_1 \in 0, 1$). •The tens digit of $2n$ equals $(2b + c_1) \pmod{10}$ and produces a carry $c_2 = \lfloor (2b + c_1)/10 \rfloor$. •The hundreds digit of $2n$ equals $(2b + c_2) \pmod{10}$ and produces a carry $c_3 = \lfloor (2b + c_2)/10 \rfloor$. •The thousands digit of $2n$ equals $(2a + c_3) \pmod{10}$, and any further carry would make $2n$ five-digit.

Two observations force restrictions on a and b .

1. If $a \geq 5$ then $2a \geq 10$, so doubling $\overline{abba}$ produces a five-digit number (a carry into a new leading digit). In that case the leading digit of $2n$ would be 1 (from the carry) while the last digit is $2a \pmod{10}$, which is even; these cannot be equal, so $2n$ cannot be a palindrome. Hence $a \leq 4$. 2. With $a \leq 4$ we have $c_1 = 0$. For $2n$ to be a four-digit palindrome its tens and hundreds digits must match:

$$(2b + c_1) \pmod{10} = (2b + c_2) \pmod{10}$$

Since $c_1 = 0$ and $c_2 \in 0, 1$, this equality forces $c_2 = 0$. But $c_2 = 0$ means $2b < 10$, so $b \leq 4$.

Thus both digits satisfy $a \in 1, 2, 3, 4$ and $b \in 0, 1, 2, 3, 4$. Any choice of such a, b produces $\overline{abba}$ with no carries from the ones or tens places, so doubling preserves the palindromic pattern.

Counting choices: 4 choices for a and 5 choices for b , giving $4 \times 5 = 20$ valid four-digit integers.

Therefore there are 20 four-digit integers n for which both n and $2n$ are palindromes.

Answer: 20

681. Tricky Palindrome III

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Write the decimal representation of n . Because n itself is a palindrome, its first (most significant) digit equals its last (units) digit - call that common digit a . We show first that $a = 1$. 1. a cannot be even: if a were even then $5n$ would end in 0, so $5n$ would be divisible by 10, contradicting the hypothesis that $5n$ is a palindrome (palindromes here are not divisible by 10). 2. a cannot be 5: if $a = 5$ then $2n$ ends with 0 (since $2 \cdot 5 = 10$), again impossible. 3. a cannot be 3: if $a = 3$ then $4 \cdot a = 12$, so $4n$ acquires an extra leading digit 1 while its units digit is 2; these disagree, so $4n$ cannot be a palindrome. 4. a cannot be 3 or 7: if $a \in \{7, 9\}$ then $2 \cdot a \geq 14$ or 18, so $2n$ gains a leading 1 while its units digit is even, a contradiction. Having excluded all possibilities except $a = 1$, we conclude the first (and last) digit of n is 1.

Next, assume for contradiction that n contains some digit that is neither 0 nor 1. Let b be the leftmost (most-significant) digit of n that is not 0 or 1. Thus the left end of n looks like

$$\alpha_1 \alpha_2 \cdots \alpha_r b \cdots$$

with each $\alpha_i \in 0, 1$, $r \geq 1$, and $b \geq 2$. Because n is a palindrome, the right end of n has the corresponding block

$$\cdots \alpha_r \alpha_{r-1} \cdots \alpha_1.$$

Now multiply n by 9. Look at the rightmost r columns (the columns corresponding to $\alpha_1, \dots, \alpha_r$). For each $\alpha_i \in 0, 1$ we have $9\alpha_i \in 0, 9$, so multiplying these columns by 9 produces no carry to the next column to the left. Therefore the r -th digit from the right in the product $9n$ is simply $9\alpha_r$ (either 0 or 9).

On the other hand, consider the corresponding position counted from the left: the block on the left that contains b is being multiplied by 9, and since $b \geq 2$ we have $9b \geq 18$, so the column containing b creates a carry into the next more-significant place. That carry changes the digit that appears in the r -th position from the left of $9n$; in particular that digit cannot equal $9\alpha_r$ (which is 0 or 9) because of the carry produced by $9b$. Hence the r -th digit from the left and the r -th digit from the right in $9n$ are different, so $9n$ is not a palindrome - contradicting the hypothesis.

This contradiction shows there is no digit $b \geq 2$ in n . Therefore every decimal digit of n must be either 0 or

1, as required.

682. Skipping 2s

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Numbers that do not contain the digit 2 use exactly nine possible digits in each decimal place: 0, 1, 3, 4, 5, 6, 7, 8, 9. This gives a natural bijection with base-9 representations: think of the allowed digits as a “shifted” set of base-9 digits. Concretely, write the index k (here $k = 2026$) in base 9 and then replace each base-9 digit d by

$$\begin{cases} d, & \text{if } d \in 0, 1, \\ d + 1, & \text{if } d \in 2, 3, 4, 5, 6, 7, 8, \end{cases}$$

to obtain the corresponding decimal number that avoids the digit 2. This mapping is a one-to-one correspondence between positive integers (counting in base 9) and the sequence of integers with no digit 2.

Step 1 - write 2026 in base 9.

$$2026_{10} = 2701_9.$$

Step 2 - apply the digit replacement rule to the base-9 string 2701_9 :

$$2 \mapsto 3, \quad 7 \mapsto 8, \quad 0 \mapsto 0, \quad 1 \mapsto 1.$$

Thus 2701_9 corresponds to the decimal number 3801.

Therefore the 2026th number Ava types is 3801.

Answer: 3801

683. Arrow On Dice

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

There are 4 choices for where to go from face 1. Consider the 4 faces adjacent to 1. We can visit either 1, 2, or 3 of them before visiting the face opposite 1. If we only visit one of these adjacent faces, we have 4 choices for which one, then we visit face 6, opposite face 1, then we visit the remaining 3 faces in one of 2 orders, for a total of 8 ways. If we visit 2 adjacent faces first, there are 8 choices for these two faces, then 2 choices for the path back from face 6 to face 1. Lastly, there are 8 ways to visit three of the adjacent faces before visiting the opposite face. These choices give 32 total.

Answer: 32

684. Grid Arrangement III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We factor 171 as $3^2 \cdot 19$ and place the 19's and the 3's in the squares independently. The two primes act independently, so the total number of fillings is

$$(\text{number of valid patterns for } 19) \times (\text{number of valid patterns for } 3).$$

Let a_i be the number of 19's in the i th column. The number of ways to fill the grid 19's so that the divisibility property is satisfied is equal to the number of nondecreasing sequences a_1, a_2, a_3 where each $a_i \in 0, 1, 2, 3$ and the sequence is not $(0, 0, 0)$ and not $(1, 1, 1)$. Thus there are

$$\binom{3+4-1}{3} - 2 = 18$$

ways to choose which squares are divisible by 19.

To count the arrangements of divisibility by 3, we consider three cases-

If 9 divides the middle square, then each of the squares to the right and below it are divisible by 9. The two squares in the top row (besides the upper left) can be (1, 1), (1, 3), (1, 9), (3, 3), (3, 9), or (9, 9) (in terms of the highest power of 3 dividing the square). The same is true, independently, for the two blank squares on the left column, for a total of $6^2 = 36$ possibilities in this case.

If 3 does not divide middle square, there are also 36 possibilities by a similar argument.

If 3 divides the middle square (and 9 doesn't). Name the (horizontal) rows as a, b, c and (vertical) columns as 1, 2, 3. Consider the cells a_2, a_3 and b_1 . There are 8 ways to fill these cells. Similarly, there are 8 ways to fill the cells b_1, c_1 and c_2 . Thus there are 64 possibilities in this case.

Thus there are a total of 136 options for the divisibility of each number by 3 and 9, and 18 options for the divisibility of the numbers by 19. Since each number divides 171, this uniquely determines the numbers, and so there are a total of $18 \cdot 136 = 2448$ possibilities.

Answer: 2448

685. Tetrahedron Coloring II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

No face may be monochromatic, so no color can appear three or four times among the four vertices. Thus the only possible color-count patterns are $2 + 2$ (two colors used twice) or $2 + 1 + 1$ (one color used twice, the other two used once).

If exactly two colors are used, the counts must be $2 + 2$. Choose which two colors appear: $\binom{3}{2} = 3$ choices. For any fixed pair of colors, every way of putting the two copies of one color and the two copies of the other on the tetrahedron is equivalent to every other such way by a rotation (all perfect matchings of the four vertices are equivalent under the tetrahedron's rotation group). So each choice of two colors gives one distinct coloring up to rotation, contributing 3 colorings total.

If all three colors appear, the only allowed multiset of counts is $2 + 1 + 1$. Choose the color that appears twice: 3 choices. For a fixed choice (say red twice, green and blue once), put the two red vertices anywhere - any pair of vertices is an edge and all such pairs are equivalent by rotation, so the position of the red pair is unique up to rotation. The remaining two vertices receive green and blue; swapping those two singletons can be realized by a 180° rotation that fixes the red pair setwise and interchanges the other two vertices, so the two assignments of green/blue are the same up to rotation. Hence for each choice of the doubled color there is exactly one distinct coloring up to rotation, giving another 3 colorings.

Adding the two cases gives $3 + 3 = 6$ distinct colorings.

Answer: 6

686. Logarithm Comparison

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To compare these two quantities, consider the function

$$f(x) = \log_x(x+1) \text{ for } x > 1$$

The two expressions in the problem are simply $f(1111)$ and $f(1112)$. So it is enough to understand how $f(x)$ behaves as x increases.

Rewrite $f(x)$ using natural logarithms:

$$f(x) = \frac{\ln(x+1)}{\ln x}.$$

Differentiate:

$$f'(x) = \frac{\ln x \cdot \frac{1}{x+1} - \ln(x+1) \cdot \frac{1}{x}}{(\ln x)^2} = \frac{x \ln x - (x+1) \ln(x+1)}{x(x+1)(\ln x)^2}.$$

For $x > 1$, the denominator is positive, so the sign of $f'(x)$ is determined by the numerator

$$x \ln x - (x+1) \ln(x+1).$$

This expression is negative for all $x > 1$. Hence $f'(x) < 0$ for all $x > 1$, and therefore $f(x)$ is strictly decreasing on $(1, \infty)$.

Because $1111 < 1112$ and f is decreasing, we conclude that $f(1111) > f(1112)$, that is,

$$\log_{1111}(1112) > \log_{1112}(1113).$$

So the greater quantity is $\log_{1111}(1112)$.

Answer: 1

687. Circular Frog

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let E_k be the expected number of steps needed to increase the size of the visited contiguous arc from $k-1$ points to k points, given the frog starts at one of the arc's end points. By linearity of expectation the total expected time to visit all 10 points is

$$\sum_{k=2}^{10} E_k,$$

since we begin with an arc of size 1 and need nine expansions to reach size 10. To compute E_k use the standard formula for the expected hitting time of either boundary in a simple symmetric random walk on a finite interval: if the walk runs on $1, 2, \dots, n$ and starts at i , the expected time to hit 1 or n is

$$(i-1)(n-i).$$

Now map our situation for a visited arc of size $k-1$. The two new points that expand the arc are the two neighbors outside the arc, so consider the interval from the left neighbor to the right neighbor; this interval has length $n = k+1$. Without loss of generality we can assume that the frog starts at index $i = k$ (recall that, in our setup, E_k denotes the time required to expand the set of visited points to k , starting from $k-1$, given that we begin at one of the endpoints of the arc.). If the frog starts at the endpoint corresponding to index $i = k$, the formula gives

$$E_k = (i-1)(n-i) = (k-1)((k+1)-k) = k-1.$$

Therefore the total expected time is

$$\sum_{k=2}^{10} E_k = \sum_{k=2}^{10} (k-1) = 1 + 2 + \dots + 9 = 45.$$

Hence the expected time to visit all ten points is 45 seconds.

Answer: 45

688. Square Factor

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Total number of positive divisors equals

$$(3+1)(5+1)(7+1)(9+1) = 4 \cdot 6 \cdot 8 \cdot 10 = 1920.$$

A divisor is a perfect square exactly when each prime's exponent in the divisor is an even integer between 0 and the prime's exponent in the original number (inclusive). Thus the number of perfect-square divisors is

$$2 \cdot 3 \cdot 4 \cdot 5 = 120,$$

because for 2^3 the allowed exponents are 0, 2 (2 choices), for 3^5 they are 0, 2, 4 (3 choices), for 5^7 they are 4 choices, and for 7^9 they are 5 choices.

Therefore the desired probability is

$$\frac{120}{1920} = \frac{1}{16}.$$

So the probability that a randomly chosen divisor is a perfect square is $\frac{1}{16}$.

Answer: 0.0625

689. Sum Sum Sum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The sum of digits is invariant modulo 9: write any integer

$$N = \sum_{i=0}^k a_i 10^i$$

with decimal digits a_i . Since $10 \equiv 1 \pmod{9}$ we have $10^i \equiv 1 \pmod{9}$ for every i , so

$$N = \sum_{i=0}^k a_i 10^i \equiv \sum_{i=0}^k a_i \pmod{9}.$$

Thus replacing a number by the sum of its digits does not change its residue modulo 9 - that residue is an invariant of the digit-sum operation. Repeating the digit-sum preserves the same invariant, so the final single digit (the digital root) equals the remainder of the original number modulo 9, except that a remainder of 0 corresponds to the digit 9.

Now compute $19^{100} \pmod{9}$. We have $19 \equiv 1 \pmod{9}$, so

$$19^{100} \equiv 1^{100} \equiv 1 \pmod{9}.$$

The remainder is 1, so the repeated digit-sum ends at the single digit 1.

Answer: 1

690. Constrained Combinatorics

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Choose any set of four distinct numbers and write them in increasing order $t_1 < t_2 < t_3 < t_4$. For a fixed set t_1, t_2, t_3, t_4 we count how many permutations (w, x, y, z) of these values meet the three inequalities. If $y = t_4$ then x and w must be two of t_1, t_2, t_3 with $x < w$ (three choices for the ordered pair $x < w$) and z is the remaining one, giving 3 arrangements. If $y = t_3$ then z must be one of t_1, t_2 (2 choices) and the remaining two values must be placed as $x < w$ (only one order for each choice of z), giving 2 arrangements. No valid arrangement exists when $y = t_1$ or $y = t_2$. Thus exactly $3 + 2 = 5$ permutations work for each 4-element set.

Since there are $\binom{9}{4} = 126$ choices of the set, the total number of ordered quadruples is $126 \cdot 5 = 630$.

Answer: 630

691. Evenbin I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Observe that for every $k \geq 1$ the numbers $2k$ and $2k + 1$ have binary representations that differ only in the last digit, so their counts of 1's have opposite parity. Therefore exactly one number in each pair $(2k, 2k + 1)$ is evenbin. Also, 1 is not evenbin. So among the integers $1, 2, \dots, 2m + 1$ there are exactly m evenbin numbers (coming from the m pairs $(2, 3), (4, 5), \dots, (2m, 2m + 1)$). To get 2050 evenbin numbers we take $m = 2050$, so the relevant pair to check is $(4100, 4101)$: there are 2049 evenbin numbers ≤ 4099 , and exactly one of 4100 or 4101 will be the 2050th evenbin number. Now check which of them is evenbin:

$$4100_{10} = 1000000000100_2,$$

which has exactly 2 ones, an even number. Hence 4100 is evenbin, so the 2050th smallest evenbin number is 4100.

Answer: 4100

692. Prime Set

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the set be denoted by A , containing n primes with average 27, so the total sum is $27n$, and let the largest prime be p . Since 27 is odd, the sum $27n$ is odd, which forces n to be odd. If 2 were included in the set, then the remaining $n - 1$ primes would all be odd. But $n - 1$ is even, so the sum of those odd primes would be even, and adding 2 would still give an even total. This contradicts the fact that $27n$ is odd. Hence $2 \notin A$, and all primes in A are odd.

The primes lesser than 27 (and ≥ 3) are 3, 5, 7, 11, 13, 17, 19 and 23. It is best to include all of these numbers in the set (why?). Thus the largest prime p is at most $27 \cdot 9 - 3 - 5 - 7 - 11 - 13 - 17 - 19 - 23 = 145$, so $p \leq 145$. When the primes are 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, and 139, their average is 27. Therefore $p = 139$.

Answer: 139

693. Catch Me If You Can I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Such a sequence exists. Fire at the perfect squares: choose the firing sequence $a_n = n^2$ for $n = 1, 2, 3, \dots$. If the thief's (integer) speed is v , then after t seconds his position is vt . At time $t = v$ his position is $v \cdot v = v^2$, and the officer is aiming at $a_v = v^2$ at that instant. Therefore the thief is caught at time v . Hence the sequence $1, 4, 9, 16, \dots$ guarantees a capture.

Answer: 1

694. Catch Me If You Can II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Such a sequence exists. Represent the thief's motion by an ordered pair (a, b) , where a is the starting point of the thief and b is its velocity. We want to find a way to map each positive integer to a possible ordered pair so that every ordered pair is covered. This way, if we fire at $b_n n + a_n$ at time n , where (a_n, b_n) is the point that n maps to, then we will eventually catch the thief. (Keep in mind that $b_n n + a_n$ would be the location of the thief at time n .) There are many such ways to map the positive integers to possible points; here is one way:

$$1 \rightarrow (1, 1), 2 \rightarrow (2, 1), 3 \rightarrow (1, 2), 4 \rightarrow (3, 1), 5 \rightarrow (2, 2), 6 \rightarrow (1, 3), 7 \rightarrow (4, 1), 8 \rightarrow (3, 2), \\ 9 \rightarrow (2, 3), 10 \rightarrow (1, 4), 11 \rightarrow (5, 1), 12 \rightarrow (4, 2), 13 \rightarrow (3, 3), 14 \rightarrow (2, 4), 15 \rightarrow (1, 5), \dots$$

(The path of points trace out diagonal lines that sweep every lattice point in the coordinate plane.) Since we cover every point, we will eventually catch the thief.

Answer: 1

695. Game On Divisors IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We prove that Alice wins exactly for every even N except when $N = 2^k$ with k odd.

First, note that every odd N is a losing start, because if N is odd and composite, every proper divisor is odd, so any move gives the opponent an even number of the form $d(b-1)$ (when $N = db$ with $b > 1$). That even number cannot be a power of two, so the opponent has an odd proper divisor to subtract and returns a strictly smaller odd number to you; repeating this forces the player who started with the odd N to eventually be handed an odd prime and lose. If N is even but not a power of two, it has an odd proper divisor $o > 1$; subtracting o hands the opponent an odd number, which from the previous argument is a losing position for whoever receives it, so the starter wins.

Now, we prove that even numbers that are not powers of 2 are winning. If N is even but not a pure power of two, then it has an odd divisor $o > 1$ (there is an odd prime factor). Take that odd proper divisor o and subtract it: you hand your opponent $N - o$, which is odd. We just showed odd positions are losing for the player who receives them. So by passing an odd number to the opponent on the very first move, you hand them a losing position. Hence, Alice wins in this case. If $N = 2^k$, every proper divisor is also a power of two. If you ever subtract a power-of-two divisor smaller than 2^{k-1} you hand the opponent an even number that is not a power of two — that would let the opponent use the argument from above and hand you an odd number, which would be bad for you. So optimal play on powers of two is to always subtract 2^{k-1} (the largest proper divisor), which halves the number. Thus with optimal play the sequence is forced:

$$2^k \rightarrow 2^{k-1} \rightarrow 2^{k-2} \rightarrow \dots \rightarrow 2.$$

The player who is given 2 loses (because 2 is prime). So, if k is even, Alice wins as Bob gets 2. If k is odd, Alice loses as she eventually gets 2.

Now, there are 995 even numbers that are not odd powers of 2 in $\{1, 2, \dots, 2000\}$. So, the answer is 995.

Answer: 995

696. Evenbin II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

To find the sum, look at each bit position separately. Fix a bit position i (where $i = 0, 1, \dots, 8$). Count how many of the even-parity numbers have a 1 in that position. Set bit i to 1 and consider all assignments of the other 8 bits: there are $2^8 = 256$ assignments. For the whole 9-bit string to have an even number of 1s, the remaining 8 bits must have an odd number of 1s, and exactly half of the 256 assignments satisfy that. So there are 128 even-parity numbers with bit i equal to 1. Each such number contributes 2^i to the total from that bit, so the contribution from bit i is $128 \cdot 2^i$.

Add the contributions of all nine bit positions:

$$\sum_{i=0}^8 128 \cdot 2^i = 128 \cdot (2^0 + 2^1 + \cdots + 2^8) = 128 \cdot (2^9 - 1) = 65408.$$

Answer: 65408

697. Hexagon Inside Hexagon

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the original regular hexagon H . The area of the inner hexagon equals the area of H minus the total area of the six outer corner regions that get cut away. Each of those six cut-away regions is a triangle formed at a vertex of H by the two chosen points on the two sides meeting at that vertex. Because the hexagon is regular, all six of these triangles are identically distributed, so it suffices to compute the expected area of one such triangle and multiply by 6.

Fix one vertex and let the two adjacent sides meeting there have length 1. Let a and b be the distances from the vertex to the chosen points along those two sides. Because each point is chosen uniformly on a side of length 1, the random variables a and b are independent and each has mean $1/2$. The angle between the two sides at the vertex is 60° . For a triangle with adjacent side lengths a and b and included angle θ , the area is $\frac{1}{2}ab \sin \theta$. Here $\theta = 60^\circ$, so the area of the triangle is

$$\frac{1}{2}ab \sin 60^\circ$$

Taking expectation, and using independence so that $E[ab] = E[a]E[b]$, $E[\text{area of one triangle}] = \frac{1}{2} \sin 60^\circ E[a]E[b] = \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{16}$.

Thus the expected total area removed by the six corner triangles is

$$6 \cdot \frac{3}{16} = \frac{3\sqrt{3}}{8}$$

The area of the original regular hexagon of side 1 is

$$\text{Area}(H) = \frac{3\sqrt{3}}{2}$$

Subtracting the expected removed area gives the expected area of the inner hexagon:

$$E[\text{area of inner hexagon}] = \frac{3\sqrt{3}}{2} - \frac{3\sqrt{3}}{8} = \frac{9\sqrt{3}}{8}$$

Answer: 1.94856

698. Subset Chain

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The idea is to look at each element individually, rather than each subset. For each $k \in 1, 2, \dots, 999$, there are 1000 choices for the first subset in the chain that contains k . This count includes the possibility that k doesn't appear in any of the subsets. If S_i is the first subset containing k , for some $i \in \{1, 2, \dots, 999\}$, then k is also in S_j , for all $i < j \leq 999$. As a result, picking the first subset that contains k uniquely determines the appearance of k in all the subsets. It follows that there are $1000^{999} = 10^{2997}$ such subset chains.

Answer: 2997

699. Triangle Tiling

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Comparing areas, we find that there are 25 tiles in a tiling. $\sqrt{2}$ is irrational. If a, b, c, d are rational, then $a + b\sqrt{2} = c + d\sqrt{2}$ implies $a = c$ and $b = d$. Therefore, the hypotenuse of the tiling area must be lined with 5 hypotenuses of tiles. Removal of these 5 tiles on the hypotenuse leaves a shape consisting of 10 unit squares in a staircase shape. We claim that these squares must each be filled with a pair of tiles joined along their hypotenuses. To see this, note that a unit-length side along the stairs must be attached to the leg of a tile, and the hypotenuse of that tile must be attached to the hypotenuse of another tile. By filling in the 4 squares abutting the stairs, there results a smaller staircase consisting of 6 unit squares, which we continue filling with tiles in a similar manner. Since there are 2 ways to place each pair of tiles into a unit square, there are a total of $2^{10} = 1024$ tilings.

Answer: 1024

700. Constrained Subsets IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Divide the numbers $1, 2, \dots, 2000$ into consecutive blocks of three: $\{1, 2, 3\}$, $\{4, 5, 6\}$, $\{7, 8, 9\}$, $\dots$

Claim: each block can contribute at most one element to S . Indeed, take any block $\{m, m+1, m+2\}$. -If we pick m and $m+1$, their difference is 1, which divides every integer, so $1 \mid (m + (m+1))$ - forbidden. -If we pick $m+1$ and $m+2$, the same reasoning applies, since their difference is again 1. -If we pick m and $m+2$, their difference is 2 and their sum is $2m+2 = 2(m+1)$, so $2 \mid (m + (m+2))$ - also forbidden.

Thus no two elements from the same block can both lie in S . Therefore

$$|S| \leq \left\lceil \frac{2000}{3} \right\rceil = 667.$$

We can achieve this upper bound by taking all numbers congruent to 1 (mod 3):

$$S = \{1, 4, 7, \dots, 1999\}$$

Answer: 667

701. Rook Moves

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the expected number of minutes it will take the rook to reach the upper right corner from the top or right edges be E_e , and let the expected number of minutes it will take the rook to reach the upper right corner from any other square be E_c . Note that this is justified because the expected time from any square on the top or right edges is the same, as is the expected time from any other square (this is because swapping any two rows or columns does not affect the movement of the rook). This gives us two linear equations:

$$\begin{aligned} E_c &= \frac{2}{14}(E_e + 1) + \frac{12}{14}(E_c + 1) \\ E_e &= \frac{1}{14}(1) + \frac{6}{14}(E_e + 1) + \frac{7}{14}(E_c + 1) \end{aligned}$$

Solving these gives $E_e = 63$ and $E_c = 70$.

Answer: 70

702. Real Addition I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Each turn of the game is equivalent to adding a number between 0 and 1 to the number on the board. Bob (the second player) has a winning strategy. He uses the following pairing strategy: whenever Alice increases the number by some amount z with $0 < z < 1$, Bob responds by increasing the number by $1 - z$ (which also lies in $(0, 1)$). Thus the total increase in each full round (Alice + Bob) is exactly 1.

After Bob's i th turn, the number on the board will be i . Therefore, after Bob's 999th turn, Alice will be forced to write a number between 999 and 1000, after which Bob can write 1000 and win the game.

Answer: 2

703. Real Addition II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Alice wins. She first writes any $a \in (0, 1)$. Thereafter, whenever Bob adds $z \in (0, 1)$, Alice replies with $1 - z$. Hence after Alice's k -th move the board equals $(k - 1) + a$. After her 1000th move it is $999 + a < 1000$; Bob cannot safely reach ≥ 1000 on his 1000th turn, so his move leaves the board < 1000 . On her 1001st turn Alice then adds the legal amount to reach 1000 and wins.

Answer: 1

704. Strange Comparison II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Use the binomial expansion for $(7 + 1)^{91}$. Since every binomial term is positive,

$$8^{91} = (7 + 1)^{91} = \sum_{k=0}^{91} \binom{91}{k} 7^k > \binom{91}{91} 7^{91} + \binom{91}{90} 7^{90} = 7^{91} + 91 \cdot 7^{90}.$$

Now, $7^{91} + 91 \cdot 7^{90} = 2 \cdot 7^{92} > 7^{92}$.

So, $8^{91} > 7^{92}$.

Answer: 2

705. GCD Expectation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write 2025 as a product of prime powers: $2025 = 3^4 \cdot 5^2$. Hence $2025^{2025} = 3^{8100} \cdot 5^{4050}$, and any divisor d can be written uniquely as $d = 3^a 5^b$ with integers $0 \leq a \leq 8100, 0 \leq b \leq 4050$. Choosing d uniformly is equivalent to choosing a uniformly from $\{0, \dots, 8100\}$ and b uniformly from $\{0, \dots, 4050\}$, independently.

The greatest common divisor with 2025 is $\gcd(d, 2025) = 3^{\min(a, 4)} \cdot 5^{\min(b, 2)}$. Because a and b are independent, the expectation splits:

$$E[\gcd(d, 2025)] = E[3^{\min(a, 4)}] \cdot E[5^{\min(b, 2)}]$$

For a there are 8101 equally likely values, so

$$\sum_{a=0}^{8100} 3^{\min(a, 4)} = \sum_{a=0}^4 3^a + (8101 - 5) \cdot 3^4 = 121 + 8096 \cdot 81 = 655897$$

giving $E[3^{\min(a, 4)}] = \frac{655897}{8101}$

For b there are 4051 equally likely values, so

$$\sum_{b=0}^{4050} 5^{\min(b,2)} = \sum_{b=0}^2 5^b + (4051 - 3) \cdot 5^2 = 31 + 4048 \cdot 25 = 101231$$

giving $E[5^{\min(b,2)}] = \frac{101231}{4051}$

Multiplying these yields the exact value

$$E[\gcd(d, 2025)] = \frac{655897}{8101} \cdot \frac{101231}{4051} = \frac{66397109207}{32817151} \approx 2023.24416$$

Answer: 2023.24416

706. Expected Good Points

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the ordered sample by $X_{(1)} < X_{(2)} < \dots < X_{(50)}$ and write the consecutive gaps $G_i = X_{(i+1)} - X_{(i)}$ for $i = 1, \dots, 49$. The nearest neighbor of any interior point $X_{(i)}$ must be one of its two adjacent points, so two adjacent points $X_{(i)}$ and $X_{(i+1)}$ form a mutual nearest-neighbor pair exactly when their gap G_i is smaller than both neighboring gaps G_{i-1} and G_{i+1} . For the first and last gaps the condition is similar: the pair $(X_{(1)}, X_{(2)})$ is mutual iff $G_1 < G_2$, and $(X_{(49)}, X_{(50)})$ is mutual iff $G_{49} < G_{48}$.

Because the spacings G_i are exchangeable and continuous, the probability that an interior gap is the smallest among its triple (G_{i-1}, G_i, G_{i+1}) equals $1/3$, while the probability an endpoint gap is smaller than its single neighbor equals $1/2$. There are 49 gaps in total, with 47 interior gaps and 2 endpoint gaps. Linearity of expectation gives the expected number of mutual nearest-neighbor pairs $E[\text{mutual pairs}] = 2 \cdot \frac{1}{2} + 47 \cdot \frac{1}{3} = \frac{50}{3}$.

Each mutual pair contributes two good points, so the expected number of good points is twice this: $E[\text{good points}] = 2 \cdot \frac{50}{3} = \frac{100}{3} \approx 33.3333$.

Answer: 33.33333

707. Count Triangular Regions I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $m = 10$ be the number of segments from A and $n = 10$ the number from B . Draw all m segments from A first: these radiate from A to the opposite side and split the triangle into $m + 1$ regions. Each segment from B meets every segment from A exactly once (by the general-position assumption), so a single segment from B is cut into $m + 1$ pieces and thus passes through $m + 1$ of the current regions, increasing the region count by $m + 1$. Repeating this for all n segments from B adds $n(m + 1)$ regions.

So, the total number of regions is $1 + m + n(m + 1) = 1 + m + n + mn$. Substituting $m = n = 10$ gives $1 + 10 + 10 + 10 \cdot 10 = 121$.

Answer: 121

708. Count Triangular Regions II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let m, n, p be the numbers of segments from the three vertices (here $m = n = p = 10$). Drawing the m segments from the first vertex splits the triangle into $m + 1$ regions. Each segment from the second vertex meets every one of those m segments exactly once, so a single second-vertex segment is cut into $m + 1$ pieces and therefore passes through $m + 1$ existing regions, increasing the count by $m + 1$. Doing this for all n segments adds $n(m + 1)$ regions, bringing the total to $1 + m + n + mn$.

Each segment from the third vertex meets every segment drawn so far (all $m + n$ of them), so each third-vertex segment is split into $(m + n) + 1$ pieces and adds $m + n + 1$ regions; all p such segments add $p(m + n + 1)$ regions. Summing everything gives regions = $1 + m + n + mn + p(m + n + 1) = 1 + m + n + p + mn + np + pm$. With $m = n = p = 10$ this becomes $1 + 10 + 10 + 10 + 100 + 100 + 100 = 331$.

Answer: 331

709. Repeated Erase

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $P_n = \Pr(A \text{ is erased before } B)$ for the size- n problem. Focus on the first erasure: call K the value of the number erased in that first step. The first-erased number is the smaller member of the first randomly chosen unordered pair, so for $k = 1, \dots, n - 1$,

$$\Pr(K = k) = \frac{\text{pairs whose smaller element is } k}{\binom{n}{2}} = \frac{n - k}{\binom{n}{2}}$$

Condition on the event $K = k$. Three mutually exclusive outcomes for the pre-chosen circled pair A, B cover the possibilities:

* the circled pair is k, j with $j > k$. Then $A = k$ and it is erased immediately, so success occurs. The conditional probability of this is $\frac{n-k}{\binom{n}{2}}$. * the circled pair is i, k with $i < k$. Then $B = k$ and it is erased immediately, so failure occurs. The conditional probability is $\frac{k-1}{\binom{n}{2}}$. * the circled pair does not involve k . In that case neither circled element is erased at the first step and the remaining process (after relabeling the survivors in increasing order) is distributionally the same as the size- $(n - 1)$ problem, so conditional on this event the probability of success equals P_{n-1} . The probability of this third case equals

$$1 - \frac{n - 1}{\binom{n}{2}}$$

since the first two cases together have total probability $\frac{n-k}{\binom{n}{2}} + \frac{k-1}{\binom{n}{2}} = \frac{n-1}{\binom{n}{2}}$.

Putting these together, for fixed k the conditional probability of success is

$$\Pr(\text{success} \mid K = k) = \frac{n - k}{\binom{n}{2}} + \left(1 - \frac{n - 1}{\binom{n}{2}}\right) P_{n-1}$$

Now average over K (law of total probability), using $\Pr(K = k) = \frac{n-k}{\binom{n}{2}}$. Because the probabilities $\Pr(K = k)$ sum to 1, the term multiplying P_{n-1} stays the same after averaging, and we obtain

$$P_n = \sum_{k=1}^{n-1} \Pr(K = k) \Pr(\text{success} \mid K = k) = \frac{1}{\binom{n}{2}^2} \sum_{k=1}^{n-1} (n - k)^2 + \left(1 - \frac{n - 1}{\binom{n}{2}}\right) P_{n-1}$$

Simplifying gives the one-step recurrence

$$P_n = \frac{2(2n - 1)}{3n(n - 1)} + \frac{n - 2}{n} P_{n-1}, n \geq 3$$

with base case $P_2 = 1$.

To identify the pattern, evaluate a few initial cases: * $n = 2$: $P_2 = 1$. * $n = 3$: $P_3 = \frac{1}{3}P_2 + \frac{2 \cdot 5}{3 \cdot 3 \cdot 2} = \frac{1}{3} \cdot 1 + \frac{10}{18} = \frac{8}{9}$. * $n = 4$: $P_4 = \frac{2}{4}P_3 + \frac{2 \cdot 7}{3 \cdot 4 \cdot 3} = \frac{1}{2} \cdot \frac{8}{9} + \frac{14}{36} = \frac{4}{9} + \frac{7}{18} = \frac{5}{6}$. * $n = 5$: $P_5 = \frac{3}{5}P_4 + \frac{2 \cdot 9}{3 \cdot 5 \cdot 4} = \frac{3}{5} \cdot \frac{5}{6} + \frac{18}{60} = \frac{1}{2} + \frac{3}{10} = \frac{4}{5}$.

So, we can guess the following pattern:

$$P_n = \frac{2(n + 1)}{3n}$$

To verify, substitute $P_{n-1} = \frac{2n}{3(n-1)}$ into the right-hand side:

$$\frac{2(2n-1)}{3n(n-1)} + \frac{n-2}{n} \cdot \frac{2n}{3(n-1)} = \frac{2(n+1)}{3n}$$

so the formula holds for all $n \geq 2$.

For $n = 1000$ this yields $P_{1000} = \frac{2(1001)}{3 \cdot 1000} \approx 0.66733$.

Answer: 0.66733

710. Everything Odd

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We will use recursion. Use the number of faces that have appeared an odd number of times as the state. For example, if the roll sequence is 112133, then the state equals 2 because two faces, 1 and 2, have appeared an odd number of times. Call this k .

k moves on $0, 1, 2, 3$ with the rule that a single roll increases k by 1 with probability $(3-k)/3$ (you hit one of the currently-even faces) and decreases k by 1 with probability $k/3$ (you hit one of the currently-odd faces). Let E_k be the expected remaining rolls to hit $k = 3$ starting from k . Then $E_3 = 0$ and for $k = 0, 1, 2$

$$E_k = 1 + \frac{3-k}{3}E_{k+1} + \frac{k}{3}E_{k-1}.$$

From $k = 0$ we get $E_0 = 1 + E_1$. From $k = 2$ we get $E_2 = 1 + \frac{2}{3}E_1$. Substitute these into the $k = 1$ equation:

$$E_1 = 1 + \frac{2}{3}E_2 + \frac{1}{3}E_0 = 1 + \frac{2}{3}\left(1 + \frac{2}{3}E_1\right) + \frac{1}{3}(1 + E_1)$$

Solving this linear equation gives $E_1 = 9$. Hence $E_0 = 1 + E_1 = 10$ and $E_2 = 1 + \frac{2}{3} \cdot 9 = 7$.

Thus the expected number of rolls starting from no occurrences (the actual start) is 10.

Answer: 10

711. Expectation Lower Bound

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Use the tail-integral identity $E[X] = \int_0^\infty \Pr(X > t)dt$. For any $t \geq 0$ we have

$$\int_0^t f(x)dx \leq \int_0^t 1dx = t$$

because $f \leq 1$. Therefore

$$\Pr(X > t) = 1 - \int_0^t f(x)dx \geq 1 - t$$

Putting this into the tail integral gives

$$E[X] = \int_0^\infty \Pr(X > t)dt \geq \int_0^1 (1-t)dt = \frac{1}{2}$$

Thus $E[X] \geq \frac{1}{2}$. Equality holds when $\Pr(X > t) = (1-t)$ for all t , i.e. when $f(x) = 1$ on $[0, 1]$ and $f(x) = 0$ for $x > 1$. Hence the minimum possible expected value is $\frac{1}{2}$, attained exactly by the uniform density on $[0, 1]$.

Answer: 0.50000

712. Expected Stock Price[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let M_i be the daily multiplicative factor on day i (so $M_i \in 1.05, 0.97$). The price after 10 days is

$$S_{10} = 100 \prod_{i=1}^{10} M_i$$

Taking expectations,

$$E[S_{10}] = 100E\left[\prod_{i=1}^{10} M_i\right]$$

Recall that $E[XY] = E[X]E[Y]$ if X and Y are independent. So,

$$E[S_{10}] = 100E\left[\prod_{i=1}^{10} M_i\right] = 100 \prod_{i=1}^{10} E[M_i]$$

Now, the expected value of each M_i is,

$$E[M_i] = \frac{3}{5} \cdot 1.05 + \frac{2}{5} \cdot 0.97 = 1.018$$

Thus $E[S_{10}] = 100 \cdot 1.018^{10} = 100 \cdot (1.018)^{10} \approx 119.53024$.

Answer:

713. Even Lattice Points[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For two lattice points (x_1, y_1) and (x_2, y_2) the number of lattice points on the line segment joining them equals $\gcd(|x_2 - x_1|, |y_2 - y_1|) + 1$. Applying this to the segment from $(0, 2004)$ to a chosen interior point (a, b) gives the number of lattice points $N = \gcd(a, 2004 - b) + 1$.

Because 2004 is even, $2004 - b$ is even exactly when b is even. A greatest common divisor is even precisely when both its arguments are even, so $\gcd(a, 2004 - b)$ is even exactly when a and b are both even. When the gcd is even, $N = \gcd(\cdot) + 1$ is odd; conversely, N is even exactly when it is not the case that both a and b are even.

There are 98 choices for each coordinate a and b , so $98^2 = 9604$ interior lattice points in all. The even integers in $\{1, \dots, 98\}$ are $2, 4, \dots, 98$, a total of 49 values. So the number of interior points with both coordinates even is $49^2 = 2401$. Thus the probability both coordinates are even is $2401/9604 = 1/4$, and the probability N is even equals $1 - 1/4 = 3/4$.

Answer:

714. Polygon Constraints[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Fix a particular label, say $x \in \{1, \dots, 10\}$. Every occurrence of x at a vertex can contribute at most two distinct neighbor labels (one on each side). To ensure that x appears adjacent to each of the other nine labels at least once, the label x must occur on at least $\lceil 9/2 \rceil = 5$ different vertices. This argument applies to every label $1, \dots, 10$, so there must be at least $10 \cdot 5 = 50$ vertices in total. Hence $n \geq 50$.

To see that $n = 50$ is feasible, let the numbers $1, \dots, 10$ be the vertices of a complete graph. Then each vertex has degree 9, and there are $\binom{10}{2} = 45$ edges. If we attach extra copies of the edges 1-2, 3-4, 5-6, 7-8, and 9-10, then every vertex will have degree 10.

In particular, the graph has an Eulerian tour, so we can follow this tour, successively numbering vertices of the 50-gon according to the vertices of the graph we visit. Then, for each edge of the graph, there will be a corresponding edge of the polygon with the same two vertex labels on its endpoints. It follows that every pair of distinct numbers occurs at the endpoints of some edge of the polygon, and so $n = 50$ is the answer.

Answer: 50

715. Divisible Permutation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The number of pairs (n, m) with $n \mid m$ must equal the number of pairs with $a_n \mid a_m$, but since each pair of the former type is also of the latter type, the converse must be true as well. Thus, $n \mid m$ if and only if $a_n \mid a_m$. Now for each $n = 1, 2, \dots, 6$, the number of values divisible by n uniquely determines n , so $n = a_n$. Similarly, 7, 8 must either be kept fixed by the rearrangement or interchanged, because they are the only values that divide exactly 2 other numbers in the sequence; since 7 is prime and 8 is not, we conclude they are kept fixed. Then we can easily check by induction that $n = a_n$ for all larger composite numbers $n \leq 25$ (by using $m = a_m$ for all proper factors m of n) and $n = 11$ (because it is the only prime that divides exactly 1 other number). So we have only the primes $n = 13, 17, 19, 23$ left to rearrange, and it is easily seen that these can be permuted arbitrarily, leaving $4! = 24$ possible orderings altogether.

Answer: 24

716. Lottery

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Notice that your chances of winning remain the same if we imagine the lottery instead drawing all tickets in random order, and awarding prizes to the holder of the earliest odd-numbered ticket drawn and to the holders of all even-numbered tickets drawn before it. Thus, the probability of your winning is the probability that, in a random ordering of the tickets, your ticket precedes all the odd tickets. This, in turn, equals the probability that your ticket is the first in the set consisting of your own and all odd-numbered tickets, irrespective of the other even-numbered tickets. Since all $501!$ orderings of these tickets are equally likely, the desired probability is $1/501$.

Answer: 0.002

717. Central Cities

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

If there are h central cities, every pair of central cities must be joined by a road, and every central city must connect to each of the $10 - h$ non-central cities. Thus even if we put no roads among the non-central cities, the roads forced by the central cities already number

$$\binom{h}{2} + h(10 - h) = \frac{h(19 - h)}{2}$$

This is the minimum possible number of roads a configuration with h central cities can have. Since the total available roads is 40 we need

$$\frac{h(19 - h)}{2} \leq 40 \implies h(19 - h) \leq 80$$

$h = 7$ gives $7 \cdot 12 = 84 > 80$, so $h \geq 7$ is impossible, while $h = 6$ gives $6 \cdot 13 = 78 \leq 80$. Hence $h \leq 6$.

It remains to show $h = 6$ is attainable. Take six central cities and connect them pairwise (this gives $\binom{6}{2} = 15$ edges) and connect each of those six to each of the remaining four cities (additional $6 \cdot 4 = 24$ edges), totalling $15 + 24 = 39$ roads. Add one more road between any two of the four non-central cities to reach 40 roads. In this graph the six designated cities have degree 9 (they are central), while none of the other four reaches degree 9, so we indeed have exactly six central cities.

Therefore the maximum possible number of central cities is 6.

Answer: 6

718. Expected Wait Time III

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Measure time from 10:00. Let $T_A \sim \text{Unif}(0, 60)$ and $T_B \sim \text{Unif}(0, 60)$ be the two independent start offsets, and let your arrival U be uniform on $[60, 240]$ (11:00–14:00). Because the arrival-window length $240 - 60 = 180$ is a multiple of both periods 3 and 5, the remainder of U modulo 3 is uniform on $[0, 3)$ and the remainder modulo 5 is uniform on $[0, 5)$. Moreover these remainders are independent of T_A and T_B . Therefore the residual time until the next Bus A at your arrival is distributed as $X \sim \text{Unif}(0, 3)$ and the residual until the next Bus B is $Y \sim \text{Unif}(0, 5)$, with X and Y independent. Your wait is $W = \min(X, Y)$.

Use the tail-integral formula $E[W] = \int_0^\infty \Pr(W > t) dt$. For $0 \leq t \leq 3$ we have $\Pr(X > t) = 1 - \frac{t}{3}$ and $\Pr(Y > t) = 1 - \frac{t}{5}$, so independence gives

$$\Pr(W > t) = \left(1 - \frac{t}{3}\right) \left(1 - \frac{t}{5}\right)$$

For $t \geq 3$ the probability $\Pr(W > t) = 0$ because $X \leq 3$. Hence

$$E[W] = \int_0^3 \left(1 - \frac{t}{3}\right) \left(1 - \frac{t}{5}\right) dt = 1.2$$

Answer: 1.2

719. Expected Wait Time IV

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Because both schedules start together at 10:00 and are periodic, their combined pattern repeats every $\text{lcm}(3, 5) = 15$ minutes. The arrival window from 11:00 to 14:00 is 3 hours = 180 minutes, which is an integer multiple of 15. Thus it suffices to compute the expected wait when your arrival is uniform in a single 15-minute cycle.

Inside one 15-minute cycle (measuring minutes after the cycle start) Bus A arrives at 0, 3, 6, 9, 12 and Bus B at 0, 5, 10. The ordered event times are 0, 3, 5, 6, 9, 10, 12, 15, so the open gaps between consecutive arrivals have lengths 3, 2, 1, 3, 1, 2, 3, which indeed sum to 15. If you land uniformly in a gap of length g , your waiting time is uniform on $[0, g]$ so its conditional mean is $g/2$. Using the law of total expectation,

$$E[\text{wait}] = \sum_i \frac{g_i}{15} \cdot \frac{g_i}{2} = \frac{1}{30} \sum_i g_i^2$$

Compute the sum of squares: $3^2 + 2^2 + 1^2 + 3^2 + 1^2 + 2^2 + 3^2 = 37$. Therefore, $E[\text{wait}] = \frac{37}{30}$ minutes ≈ 1.2333 min.

Answer: 1.23333

720. Expected Occupied Seats[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $E(n)$ be the expected final number of occupied seats when we begin with a contiguous block of n available seats. If the first arrival sits in position i (with $1 \leq i \leq n$), that person contributes 1 occupied seat and splits the remaining available seats into two independent blocks: a left block of length $i - 2$ and a right block of length $n - i - 1$ (either length may be ≤ 0 , which we treat as producing no further occupied seats). Thus the conditional expected total after choosing i is

$$1 + E(i - 2) + E(n - i - 1),$$

and since the first seat is chosen uniformly among the n positions,

$$E(n) = \frac{1}{n} \sum_{i=1}^n (1 + E(i - 2) + E(n - i - 1))$$

Set $E(k) = 0$ for $k \leq 0$. Now compute values up to $n = 6$.

$E(0) = 0$. For $n = 1$ the first (and only) arrival occupies the seat, so $E(1) = 1$.

For $n = 2$ whichever seat is chosen blocks the other, so $E(2) = 1$.

For $n = 3$ the three choices give outcomes $1 + E(1) = 2$ (if $i = 1$ or 3) or $1 + E(0) + E(0) = 1$ (if $i = 2$). Hence

$$E(3) = \frac{2 + 1 + 2}{3} = \frac{5}{3}$$

For $n = 4$ the four choices all yield $1 + E(\text{remaining}) = 2$, so $E(4) = 2$.

For $n = 5$ the outcomes for $i = 1, \dots, 5$ are $1 + E(3) = 1 + \frac{5}{3} = \frac{8}{3}$, $1 + E(0) + E(2) = 2$, $1 + E(1) + E(1) = 3$, 2 , $\frac{8}{3}$. Summing and dividing by 5 gives

$$E(5) = \frac{8/3 + 2 + 3 + 2 + 8/3}{5} = \frac{37}{15}$$

Finally for $n = 6$ the six outcomes are $1 + E(4) = 3$, $1 + E(0) + E(3) = 1 + \frac{5}{3} = \frac{8}{3}$, $1 + E(1) + E(2) = 3$, 3 , $\frac{8}{3}$, 3 . Summing,

$$\sum_{i=1}^6 (1 + E(i - 2) + E(6 - i - 1)) = 3 + \frac{8}{3} + 3 + 3 + \frac{8}{3} + 3 = \frac{52}{3}$$

so

$$E(6) = \frac{1}{6} \cdot \frac{52}{3} = \frac{52}{18} \approx 2.88889$$

Answer: 2.88889

721. N-Palindrome[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

In a palindrome, every digit must occur an even number of times except possibly one digit, which may occur an odd number of times if the total length is odd.

If $n \leq 9$, then (say) the digits 1, 2 each appear once in any n -string, so we cannot have a palindrome. If $10 \leq n \leq 18$, then 0, 9 each appear once, and we again cannot have a palindrome.

The following string works for $n = 19$:

$$9|18|7|16|5|14|3|12|1|10|11|2|13|4|15|6|17|8|19$$

So, 19 is the smallest possible n .

Answer: 19

722. Maximum Fraction Sum[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For fixed denominators $a < b$ (with sum 10), we maximize the sum of the fractions by giving the smaller denominator as large a numerator as possible: $8/a + 1/b$. Then, if $a \geq 2$, this quantity is at most $8/2 + 1/1 = 5$, which is clearly smaller than the sum we get by setting $a = 1$, namely $8/1 + 1/9 = 73/9$. So this is the answer.

Answer: **723. Circular Seating**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

This process is equivalent to a process where everyone pre-selects a seat and then each guest sits in their pre-selected seat in a random order. So let everyone sit at the table using this equivalent process. Consider this set of three people: Alice, the guest to her left, and the guest to her right. By symmetry, there are the same number of seating orders where Alice sits down last as seating orders where the guest to her left sits last, and there are also the same number of seating orders where the guest to her right sits last. Therefore there is $\frac{1}{3}$ probability that Alice sits down between two occupied seats.

Answer: **724. 2, 4 and 6**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The polynomial simplifies in a convenient way:

$$F(x, y, z) = (x + 1)(y + 1)(z + 1) - 1.$$

Thus when a number n appears on the board we automatically have $n + 1$ appearing in the multiplicative set generated by the numbers $2 + 1, 4 + 1, 6 + 1 = 3, 5, 7$. Conversely, whenever a product of three already-available “ $n + 1$ ” values appears, the corresponding product minus one is produced by the rule. It is therefore natural to work with the shifted set

$$S = \{m = n + 1 : n \text{ appears on the board}\}.$$

Initially S contains 3, 5, 7. The rule says that if $a, b, c \in S$ then $abc \in S$ (because the corresponding $(abc) - 1$ will be written on the board). So S is closed under multiplying any three of its elements.

Write every element of S as $3^\alpha 5^\beta 7^\gamma$ (unique prime factorization). Initially the exponent triples (α, β, γ) present are $(1, 0, 0), (0, 1, 0), (0, 0, 1)$, each having coordinate sum 1. Multiplying three elements corresponds to adding their exponent triples. Therefore the set of exponent triples that appear is closed under “add any three existing triples”. From the initial triple $(1, 0, 0)$ etc. one easily checks by induction on the total exponent sum that every triple (α, β, γ) with $\alpha, \beta, \gamma \geq 0$ whose sum $\alpha + \beta + \gamma$ is an odd positive integer does occur in S . (The induction step uses that for any odd sum $s \geq 3$ we can write $s = 1 + 1 + (s - 2)$ and produce the triple for s by adding two unit-basis triples to the triple of sum $s - 2$.)

Hence elements of S are exactly the numbers of the form

$$3^\alpha 5^\beta 7^\gamma$$

with $\alpha, \beta, \gamma \geq 0$ and $\alpha + \beta + \gamma$ odd. A number n appears on the board iff $n + 1$ is such a product and $n \leq 2000$ (i.e. $3^\alpha 5^\beta 7^\gamma \leq 2001$).

So the task reduces to counting nonnegative integer triples (α, β, γ) with $\alpha + \beta + \gamma$ odd and

$$3^\alpha 5^\beta 7^\gamma \leq 2001$$

It can be easily checked that there are 22 such numbers.

Answer: 22

725. Half Permutation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

1 must be the first number in the permutation, this happens with probability $\frac{1}{14}$. 2 must come before 4, 5, 8, 9, 10, and 11, this happens with probability $\frac{1}{7}$. 3 must come before 6, 7, 12, 13, and 14, this happens with probability $\frac{1}{6}$. 4 must come before 8 and 9, this happens with probability $\frac{1}{3}$. 5 must come before 10 and 11, this happens with probability $\frac{1}{3}$. 6 must come before 12 and 13, this happens with probability $\frac{1}{3}$. 7 must come before 14, this happens with probability $\frac{1}{2}$.

All these events are independent, so the answer is the product of the above probabilities, or $\frac{1}{31752}$.

Answer: 0.00003

726. Proportional Coordinates

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Treat the set of allowed values as $1, 2, \dots, 100$ and let the 99 rows (and 99 columns) be indexed by $1, 2, \dots, 99$. By the row condition, the numbers appearing in row r are exactly the 100 integers with r removed, so the sum of values in row r is

$$S_r = \sum_{t=1}^{100} t - r = 5050 - r.$$

Hence the total weight (sum of all assigned integers over the whole 99×99 grid) is

$$W = \sum_{r=1}^{99} S_r = \sum_{r=1}^{99} (5050 - r) = 99 \cdot 5050 - \sum_{r=1}^{99} r = 495000.$$

By linearity and symmetry it suffices to compute $E[y]$ and then double it, since $E[x] = E[y]$ and $E[x + y] = 2E[y]$. The weighted contribution of row r to the numerator of $E[y]$ is $r \cdot S_r$, so

$$E[y] = \frac{1}{W} \sum_{r=1}^{99} r(5050 - r) = \frac{5050 \sum_{r=1}^{99} r - \sum_{r=1}^{99} r^2}{W}.$$

Use $\sum_{r=1}^{99} r = 4950$ and $\sum_{r=1}^{99} r^2 = \frac{99 \cdot 100 \cdot 199}{6} = 328350$ to get

$$E[y] = \frac{5050 \cdot 4950 - 328350}{495000} = \frac{14951}{300}.$$

Therefore

$$E[x + y] = 2E[y] = \frac{14951}{150} \approx 99.67333.$$

Answer: 99.67333

727. Grid Frog II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Similar to "Grid Frog I", the answer is simply the square of the number of domino tilings of a 3×8 grid. From "Domino Tiling II", we know that the number of such tilings is 153. Hence, the required answer is $153^2 = 23409$.

Answer: 23409

728. Max Min Ratio

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $n = 10$. Use the law of total expectation conditioning on the maximum:

$$E\left[\frac{m}{M}\right] = E\left[E\left[\frac{m}{M} \mid M\right]\right].$$

For continuous iid samples the maximum is (with probability 1) attained at a single sample. Conditioning on $M = b$ fixes one sample at b and leaves the other $n - 1$ samples independent and uniformly distributed on $[0, b]$. Thus, given $M = b$, the minimum m is the minimum of $n - 1$ iid $\text{Unif}(0, b)$ variables, whose expectation is

$$E[m \mid M = b] = \frac{b}{(n - 1) + 1} = \frac{b}{n}.$$

Therefore

$$E\left[\frac{m}{M} \mid M = b\right] = \frac{E[m \mid M = b]}{b} = \frac{b/n}{b} = \frac{1}{n},$$

which does not depend on b . Taking the outer expectation yields

$$E\left[\frac{m}{M}\right] = \frac{1}{n} = \frac{1}{10}.$$

In this instance, the result happens to equal $E[m]/E[M]$, but this is purely coincidental. In general, the expectation of a ratio is not the ratio of expectations, so $E[X/Y] \neq E[X]/E[Y]$ in general.

Answer: 0.1

729. Increasing Random Variables

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The overall sample space is a cuboid with dimensions 1, 2, and 3, giving a total volume of

$$1 \times 2 \times 3 = 6.$$

To find the probability that a_1, a_2, a_3 are in ascending order, we set up a triple integral representing the volume of the region where $a_1 < a_2 < a_3$:

$$\int_0^1 \int_{a_1}^2 \int_{a_2}^3 1 da_3 da_2 da_1.$$

Evaluating the innermost integral with respect to a_3 :

$$\int_{a_2}^3 1 da_3 = 3 - a_2.$$

Substitute into the next integral:

$$\int_{a_1}^2 (3 - a_2) da_2.$$

Evaluating:

$$\left[3a_2 - \frac{a_2^2}{2}\right]_{a_1}^2 = (6 - 2) - \left(3a_1 - \frac{a_1^2}{2}\right) = 4 - 3a_1 + \frac{a_1^2}{2}.$$

Now integrate with respect to a_1 :

$$\int_0^1 \left(4 - 3a_1 + \frac{a_1^2}{2}\right) da_1.$$

Evaluate:

$$\left[4a_1 - \frac{3a_1^2}{2} + \frac{a_1^3}{6}\right]_0^1 = 4 - \frac{3}{2} + \frac{1}{6} = \frac{8}{3}.$$

So the volume where $a_1 < a_2 < a_3$ is $\frac{8}{3}$.

The total volume is 6, so the probability is

$$\frac{8/3}{6} = \frac{8}{18} = \frac{4}{9}.$$

Therefore, the probability that $a_1 < a_2 < a_3$ is $\frac{4}{9}$.

Answer: 0.44444

730. Largest In Smallest

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Condition on the value of $A_1 = x$. Since the variables are independent,

$$\Pr(A_1 > \max(A_2, A_3, A_4)) = \int_0^2 f_{A_1}(x) \Pr(A_2 < x) \Pr(A_3 < x) \Pr(A_4 < x) dx.$$

Here $f_{A_1}(x)$ is the density of a $\text{Unif}(0, 2)$ random variable, so

$$f_{A_1}(x) = \begin{cases} \frac{1}{2}, & 0 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

For $0 \leq x \leq 2$,

$$\Pr(A_2 < x) = \frac{x}{4}, \quad \Pr(A_3 < x) = \frac{x}{6}, \quad \Pr(A_4 < x) = \frac{x}{8}.$$

Thus

$$\Pr(A_1 \text{ is largest}) = \int_0^2 \frac{1}{2} \cdot \frac{x}{4} \cdot \frac{x}{6} \cdot \frac{x}{8} dx = \frac{1}{384} \int_0^2 x^3 dx = \frac{1}{384} \cdot \frac{2^4}{4} = \frac{4}{384} = \frac{1}{96}.$$

So the probability that A_1 is the largest equals $\frac{1}{96}$ (about 0.01042).

Answer: 0.01042

731. Alice Rolled The 6

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label by $k \geq 1$ the event that Alice's first 6 occurs on HER k -th turn. The game then ends on roll number $2k - 1$. For there to be no 6 in the first $2k - 2$ rolls and then a 6 on $(2k - 1)$ -th roll we need

$$\Pr(\text{Alice first 6 at } 2k - 1) = \left(\frac{5}{6}\right)^{2(k-1)} \cdot \frac{1}{6}.$$

Summing over k gives the unconditional probability Alice wins:

$$\Pr(\text{Alice wins}) = \sum_{k \geq 1} \left(\frac{5}{6}\right)^{2(k-1)} \frac{1}{6} = \frac{1/6}{1 - (5/6)^2} = \frac{6}{11}.$$

Thus the conditional distribution of the finishing roll (given Alice wins) is

$$\Pr(\text{total} = 2k - 1 \mid \text{Alice wins}) = \frac{\left(\frac{5}{6}\right)^{2(k-1)} \frac{1}{6}}{6/11} = \frac{11}{36} \left(\frac{25}{36}\right)^{k-1}, \quad k \geq 1.$$

Let $r = \frac{25}{36}$. The expected total number of rolls given Alice wins is

$$E[\text{total} \mid \text{Alice wins}] = \sum_{k \geq 1} (2k - 1) \frac{11}{36} r^{k-1}.$$

Use the standard sums $\sum_{k \geq 1} r^{k-1} = \frac{1}{1-r}$ and $\sum_{k \geq 1} k r^{k-1} = \frac{1}{(1-r)^2}$. With $1 - r = \frac{11}{36}$ we get

$$E[\text{total} \mid \text{Alice wins}] = \frac{11}{36} \left(2 \cdot \frac{1}{(1-r)^2} - \frac{1}{1-r} \right) = 2 \cdot \frac{1}{1-r} - 1 = 2 \cdot \frac{36}{11} - 1 = \frac{61}{11}.$$

Therefore the expected total number of rolls, conditioned on Alice being the first to roll a 6, is $\frac{61}{11}$.

Answer:

732. Given LCM

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that $216 = 2^3 3^3$. First, consider the power of 2. x must be divisible by one of $2^0 = 1, 2^1 = 2, 2^2 = 4$, and $2^3 = 8$; similarly, y must be divisible by one of these four powers of 2. However, we cannot have both x and y be divisible by only $2^0, 2^1$, or 2^2 and not 2^3 . Hence, there are $4 \cdot 4 - 3 \cdot 3 = 7$ ways of distributing the powers of 2. Similarly, there are 7 ways of distributing the powers of 3. Hence, the number of ways to choose positive integers x and y is $7 \cdot 7 = 49$.

Answer:

733. Conditional Normal I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that $-X$ has the same distribution as X (since $X \sim N(0, 1)$). So,

$$\begin{aligned} E[X \mid X^4] &= E[-X \mid (-X)^4] = -E[X \mid X^4] \\ &\Rightarrow E[X \mid X^4] = 0. \end{aligned}$$

Answer:

734. Conditional Normal II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that $-X$ has the same distribution as X (since $X \sim N(0, 1)$). Also, note that since X and Y are independent, $-X$ and Y are also independent. So, we can replace X with $-X$ in the required expression.

So,

$$\begin{aligned} E[X \mid X^4 + Y^4] &= E[-X \mid (-X)^4 + Y^4] = -E[X \mid X^4 + Y^4] \\ &\Rightarrow E[X \mid X^4 + Y^4] = 0. \end{aligned}$$

Answer:

735. Conditional Normal III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Note that $-X$ has the same distribution as X (since $X \sim N(0, 1)$). Also, note that since X and Y are independent, $-X$ and Y are also independent. So, we can replace X with $-X$ in the required expression. So,

$$\begin{aligned} E[X \mid X^4 + Y] &= E[-X \mid (-X)^4 + Y] = -E[X \mid X^4 + Y] \\ &\Rightarrow E[X \mid X^4 + Y] = 0. \end{aligned}$$

Answer:

736. Triangle Formation III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For positive numbers $X, Y, 1$ to form a triangle, the triangle inequalities require

$$X + Y > 1, \quad X + 1 > Y, \quad Y + 1 > X.$$

Since $0 \leq X, Y \leq 1$, the last two inequalities automatically hold: indeed $X + 1 > Y$ and $Y + 1 > X$ are always true because both X and Y are at most 1. Hence the only condition needed for a triangle to exist is

$$X + Y > 1.$$

Because $X, Y \leq 1$, the side of length 1 is always the largest side. By the cosine law, a triangle is obtuse precisely when the square of the largest side exceeds the sum of squares of the other two. Hence the triangle is obtuse exactly when

$$1^2 > X^2 + Y^2.$$

Therefore the required event is

$$X + Y > 1 \text{ and } X^2 + Y^2 < 1.$$

Under the uniform distribution on the unit square $[0, 1]^2$, the probability equals the area of the region

$$(x, y) : x + y > 1, ; x^2 + y^2 < 1.$$

Hence

$$\Pr(\text{obtuse}) = \int_0^1 (\sqrt{1-x^2} - (1-x)) dx = \int_0^1 \sqrt{1-x^2} dx - \int_0^1 (1-x) dx.$$

The first integral equals the area of a quarter unit circle, $\pi/4$, and the second equals $1/2$. Thus

$$\Pr(\text{obtuse}) = \frac{\pi}{4} - \frac{1}{2}.$$

Answer:

737. Same Part

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

All orderings of the six distinct points along the stick are equally likely. The two marked points end up in the same piece exactly when no cut point lies between them, i.e. when the two marks are adjacent in the sorted order of the six points.

We use the block method to count the favorable arrangements. Treat the two initial marked points as a single block. Then there are 5 entities in total: this block and the four breaking points, which can be arranged in $5!$ ways. The two marked points within the block can also be arranged in $2!$ ways. Hence, the total number of favorable permutations is $5! \cdot 2!$.

Thus, the required probability is,

$$P = \frac{2 \cdot 5!}{6!} = \frac{1}{3}.$$

Answer:

738. Elevator I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Arriving at a random time is the same as sampling the elevator's current position uniformly along its path between floors 1 and 50 (the elevator moves at constant speed and only reverses at the endpoints). If the elevator is currently above floor 40 (i.e. somewhere in the interval $(40, 50]$) then the next time it reaches floor 40 it must be coming down; if it is currently below floor 40 it will next arrive up. The length of the interval above you is $50 - 40 = 10$ floor-units, while the total travel length between the end floors is $50 - 1 = 49$ floor-units, so the probability the elevator is above you now - and hence that your next pickup is from above (going down) - is $\frac{10}{49} \approx 0.20408$.

Answer:

739. Unfair Advantage

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Partition outcomes into the four mutually exclusive, exhaustive events:

Event A: Player 1 has at least one ace among his 13 cards and gets his first ace before Player 2 gets any ace. Call its probability p . Event B: Player 2 has at least one ace among his first 13 cards and gets his first ace before Player 1 gets any ace. By symmetry $P(B) = p$. Event T (tie): Both players have at least one ace among their first 13 cards and their first aces occur in the same simultaneous round. Call its probability t . Event N: Neither player has any ace among the first 13 cards (all 4 aces lie in the final 26 cards). Its probability is $\frac{\binom{26}{4}}{\binom{52}{4}}$.

These four probabilities sum to 1:

$$p + p + t + \frac{\binom{26}{4}}{\binom{52}{4}} = 1.$$

Now compute the tie probability t . If the tie happens on simultaneous round k (both players' first ace appears exactly in the two cards shown on round k), place the remaining two aces anywhere among the remaining $52 - 2k$ positions, giving $\binom{52-2k}{2}$ placements. Summing over $k = 1$ to 13 and dividing by $\binom{52}{4}$ gives

$$t = \frac{1}{\binom{52}{4}} \sum_{k=1}^{13} \binom{52-2k}{2} = \frac{9503}{270725} = \frac{43}{1225}.$$

Now solve for p :

$$2p = 1 - t - \frac{\binom{26}{4}}{\binom{52}{4}} = 1 - \frac{43}{1225} - \frac{46}{833}.$$

$$\Rightarrow p = \frac{9472}{20825}.$$

Answer:

740. Monotonic Subsequences

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Choose which 6 of the 12 positions will hold the even numbers: $\binom{12}{6} = 924$ choices.

For any fixed choice of positions, the six even values can appear only in two orders (increasing or decreasing), and the six odd values likewise only in two orders, so each position-choice yields $2 \times 2 = 4$ valid fillings.

Therefore the total number is $\binom{12}{6} \cdot 4 = 924 \cdot 4 = 3696$.

Answer:

741. Possible Cover

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Color the board in the usual alternating pattern so adjacent squares have opposite colors - with row and column indices from 1 to 8, the squares (1, 1) and (8, 8) have the same parity and therefore the same color. Originally the board has 32 squares of each color; removing two same-colored corners leaves 32 squares of one color and 30 of the other. Any 1×2 domino placed on the board always covers two adjacent squares, which are necessarily of opposite colors, so every domino covers exactly one square of each color. Thirty-one dominos would therefore cover 31 squares of each color, but the board after removal has 32 of one color and 30 of the other, so the counts cannot match. Because the color counts are inconsistent with what dominos must cover, an exact tiling with thirty-one dominos is impossible.

Answer:

742. Alternating Sum Sum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the subsets of I be divided into two classes A and B such that each subset in A contains the number 30 and each subset in B does not contain the number 30, and define the alternating sum of the empty set to be zero. Let the set $a_1 < a_2 < \dots < a_k < 30$ in A and the set $a_1 < a_2 < \dots < a_k$ in B be paired. Then the number of such pairs is 2^{29} , and the total of the alternating sums of any pair $a_1 < a_2 < \dots < a_k < 30$ and $a_1 < a_2 < \dots < a_k$ is 30. Therefore, the total of all the alternating sums is $S = 2^{29} \cdot 30$.

Answer:

743. 3 Eggs

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First decide where to drop the first egg. We may use at most 10 drops, so consider how many floors can be covered after any single drop in the worst case. If the first egg breaks at that first drop, we are left with 2 eggs and 9 drops; with 2 eggs and 9 drops we can test $\frac{10 \cdot 9}{2}$ floors (using the result from the problem "2 Eggs", note that we don't add 1 here because we haven't necessarily reached the top floor yet). Hence the first drop should be at floor $\frac{10 \cdot 9}{2} + 1$: if the egg breaks there, the remaining 9 drops and 2 eggs allow us to search the preceding $\frac{10 \cdot 9}{2}$ floors.

If the egg does not break at that floor, we use a second drop. After that second drop there are 8 drops

remaining, so the next jump should be $\frac{9 \cdot 8}{2} + 1$ floors (so that, if it breaks, the intervening $\frac{9 \cdot 8}{2}$ floors can be explored with 2 eggs and the remaining drops). More generally, when k drops remain we should advance by $\frac{k(k-1)}{2} + 1$ floors from the previous drop point, because with $k-1$ drops and 2 eggs we can then search the $\frac{k(k-1)}{2}$ floors below—using the result from the problem "2 Eggs".

Therefore the highest floor we can reach with this scheme is

$$\left(\frac{10 \cdot 9}{2} + 1\right) + \left(\frac{9 \cdot 8}{2} + 1\right) + \cdots + (3 + 1) + (1 + 1) + (0 + 1) = \frac{10(10^2 + 5)}{6}.$$

If the egg does not break at floor $\frac{10(10^2+5)}{6}$, the next floor must be the threshold, so the maximum height for which the threshold can be uniquely determined is $\frac{10(10^2+5)}{6} + 1$. So, the answer 176.

Answer: 176

744. Expected Minimum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let M be the minimum of the five independent uniform draws from 1, 2, 3, 4, 5. The tail-sum formula for a nonnegative integer-valued random variable says

$$E[M] = \sum_{k=1}^{\infty} P(M \geq k).$$

Here M takes values in $1, \dots, 5$, so the sum truncates at $k = 5$. For a fixed $k \in 1, \dots, 5$ the event $M \geq k$ means every chosen number is at least k ; each pick has probability $\frac{6-k}{5}$ of being $\geq k$, so

$$P(M \geq k) = \left(\frac{6-k}{5}\right)^5.$$

Hence

$$E[M] = \sum_{k=1}^5 \left(\frac{6-k}{5}\right)^5.$$

Changing variables by $j = 6 - k$ (so $j = 1, \dots, 5$) gives

$$E[M] = \sum_{j=1}^5 \left(\frac{j}{5}\right)^5 = \frac{1^5 + 2^5 + 3^5 + 4^5 + 5^5}{5^5} = \frac{4425}{3125} = \frac{177}{125} = 1.416.$$

Answer: 1.416

745. Grid Arrangement IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since only the parity of the sums matters, reduce all numbers modulo 2. Among 1, ..., 9, there are five odd numbers and four even numbers. Thus we are placing five 1s and four 0s into the nine cells of the grid. The total number of such arrangements is $\binom{9}{5}$, since we choose 5 of the 9 positions to contain 1s (the remaining 4 positions will contain 0s).

For a row sum to be odd, the row must contain either three 1s or exactly one 1. Since we have 5 1s, the only possibility is that one row contains three 1s and the other two rows each contain one 1. The same argument applies to the columns: exactly one column contains three 1s and the other two columns each contain one 1.

Once we choose which row contains three 1s (3 choices) and which column contains three 1s (3 choices), the positions of the five 1s are uniquely determined: they occupy all cells in the chosen row and all cells in the

chosen column. Hence, the number of favorable arrangements is $3 \times 3 = 9$.

Therefore, the required probability is $\frac{9}{\binom{9}{5}} = \frac{1}{14}$.

Answer: 0.07143

746. Subset Length Sum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The sum of all numbers from 1 to 13 is $1 + 2 + \dots + 13 = 91$. Take any subset of $\{1, \dots, 13\}$ whose elements add up to some value k . The elements not chosen form its complement, and their sum is $91 - k$. Thus every subset with total k naturally pairs with another subset whose total is $91 - k$. In other words, the number of subsets giving sum k is exactly the same as the number giving sum $91 - k$.

There are $2^{13} = 8192$ subsets in total. Since 91 is odd, no subset can have sum exactly half of 91. Hence, within each complementary pair, one subset has sum less than 46 and the other has sum at least 46. This means exactly half of all subsets have sum at least 46.

So the number of subsets whose sums lie from 46 up to 91 (inclusive) equals $\frac{8192}{2} = 4096$.

We only need sums from 46 through 86, so we subtract those with sums 87, 88, 89, 90, and 91. By the complement pairing described above, counting subsets with these large sums is the same as counting subsets whose sums are 4, 3, 2, 1, and 0 respectively.

These are easy to enumerate: *Sum 0: only the empty set, so 1 subset. *Sum 1: 1, so 1 subset. *Sum 2: 2, so 1 subset. *Sum 3: 3 and 1, 2, so 2 subsets. *Sum 4: 4 and 1, 3, so 2 subsets.

Altogether this contributes $1 + 1 + 1 + 2 + 2 = 7$ subsets.

Therefore, $\sum_{n=46}^{86} s(n) = 4096 - 7 = 4089$.

Answer: 4089

747. Extended Rock-Paper-Scissors

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Call E_n the expected number of additional rounds until absorption at one player when there are currently n players. Clearly $E_1 = 0$. Fix some $n \geq 2$. In one round the number of survivors can be any $m \in 1, 2, \dots, n$. Let $P_{n \rightarrow m}$ be the probability that after the round exactly m players survive. By the multinomial law the probability that the three gesture counts are (i, j, k) with $i + j + k = n$ is $\frac{n!}{i!j!k!} 3^{-n}$. A nontrivial elimination occurs precisely when exactly two gestures appear (the third count is zero) and in that case the winner is predetermined by which two gestures are present, so for any $m \in 1, \dots, n - 1$ the unconditional probability that the number of survivors equals m is

$$P_{n \rightarrow m} = \frac{3 \binom{n}{m}}{3^n},$$

because there are three ordered winning–losing pairs (Rock beats Scissors, Scissors beats Paper, Paper beats Rock) and for each such ordered pair the number of assignments with the winner appearing m times is $\binom{n}{m}$. The probability that no one is eliminated (so the number of survivors stays n) equals the complement

$$P_{n \rightarrow n} = 1 - \sum_{m=1}^{n-1} P_{n \rightarrow m} = 1 - \frac{3(2^n - 2)}{3^n}.$$

Conditioning on the outcome of a single round gives

$$E_n = 1 + \sum_{m=1}^n P_{n \rightarrow m} E_m,$$

because we always use one round and then, depending on the survivors, need the corresponding remaining expected rounds. Rearranging to isolate E_n and using $\sum_{m=1}^{n-1} P_{n \rightarrow m} = 1 - P_{n \rightarrow n}$ yields

$$(1 - P_{n \rightarrow n})E_n = 1 + \sum_{m=1}^{n-1} P_{n \rightarrow m} E_m$$

Substituting $P_{n \rightarrow m} = \frac{3\binom{n}{m}}{3^n}$ and simplifying gives the convenient recurrence

$$E_n = \frac{3^{n-1} + \sum_{m=1}^{n-1} \binom{n}{m} E_m}{2^n - 2}$$

This formula is computationally efficient for small n . Starting from $E_1 = 0$ we compute step by step. For $n = 2$ the recurrence yields

$$E_2 = \frac{3^1 + \binom{2}{1} E_1}{2^2 - 2} = \frac{3}{2} = 1.5$$

For $n = 3$ we have

$$E_3 = \frac{3^2 + \binom{3}{1} E_1 + \binom{3}{2} E_2}{2^3 - 2} = \frac{9 + 3 \cdot 0 + 3 \cdot \frac{3}{2}}{6} = \frac{9}{4}$$

For $n = 4$ we get

$$E_4 = \frac{3^3 + \binom{4}{1} E_1 + \binom{4}{2} E_2 + \binom{4}{3} E_3}{2^4 - 2} = \frac{27 + 6 \cdot \frac{3}{2} + 4 \cdot \frac{9}{4}}{14} = \frac{45}{14}.$$

For $n = 5$,

$$E_5 = \frac{3^4 + \binom{5}{1} E_1 + \binom{5}{2} E_2 + \binom{5}{3} E_3 + \binom{5}{4} E_4}{2^5 - 2} = \frac{157}{35}.$$

Answer: 4.48571

748. King Before Ace

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Think only about the eight “special” cards (the four kings and the four aces) and how they sit among the 52 positions. In a random shuffle the 8 special cards occupy 8 random distinct positions and their relative order is uniformly random. The 44 non-special cards (the other ranks) are distributed into the 9 gaps determined by the special cards (before the first special, between consecutive specials, and after the last special); by symmetry the expected number of non-special cards in any one of these 9 gaps is $44/9$.

Condition on the event that the very first special card (the earliest among the eight) is a king; this is exactly the given conditioning that the first king comes before the first ace. Let the special cards in increasing deck order be $S_1, S_2, \dots, S_8$ and suppose S_1 is a king. Let j be the index of the first ace among these specials (so $j \geq 2$). There are then $j - 2$ special cards strictly between S_1 and S_j , and there are $j - 1$ gaps of non-special cards between S_1 and S_j (the gaps between $S_1 - S_2, S_2 - S_3, \dots, S_{j-1} - S_j$). Hence the number of cards strictly between the first king and the first ace equals

$$(\text{specials between}) + (\text{non-specials between}) = (j - 2) + \sum_{k=2}^j G_k,$$

where the G_k are the non-special counts in those gaps. Taking expectations and using linearity, and the fact

$E[G_k] = 44/9$ for each gap, gives

$$E[\text{cards between} \mid S_1 \text{ is a king}] = E[j - 2] + \frac{44}{9}E[j - 1]$$

So we only need the distribution of j given S_1 is a king. After S_1 is a king, the remaining 7 special cards consist of 3 kings and 4 aces in random order. The first ace among those 7 can occur immediately (probability $4/7$), or after one king (probability $(3/7)(4/6) = 2/7$), or after two kings (probability $(3/7)(2/6)(4/5) = 4/35$), or after three kings (probability $(3/7)(2/6)(1/5)(4/4) = 1/35$). These four cases correspond to $j = 2, 3, 4, 5$ respectively. From these probabilities, we can compute

$$E[j] = 2 \cdot \frac{4}{7} + 3 \cdot \frac{2}{7} + 4 \cdot \frac{4}{35} + 5 \cdot \frac{1}{35} = \frac{13}{5}$$

Thus $E[j - 1] = \frac{8}{5}$ and $E[j - 2] = \frac{3}{5}$. Plugging into the formula above yields

$$E[\text{cards between} \mid \text{first king before first ace}] = \frac{3}{5} + \frac{44}{9} \cdot \frac{8}{5} = \frac{379}{45} \approx 8.4222 \dots$$

Answer: 8.42222

749. To Infinity And Beyond II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write the increments $U_n := X_n - X_{n-1}$ for $n \geq 2$. By construction, conditional on $X_{n-1} = x$ the increment U_n is $\text{Uniform}(0, 2^{-n})$; in particular its law does not depend on x . Hence each U_n (for $n \geq 2$) is independent of the past and of X_1 , and we have

$$X = X_1 + \sum_{n=2}^{\infty} U_n$$

Because the U_n are independent, variances add. Compute the variances: for a $\text{Uniform}(0, a)$ variable, $\text{Var} = \frac{a^2}{12}$. Thus

$$\text{Var}(X_1) = \frac{(1/2)^2}{12} = \frac{1}{48}, \quad \text{Var}(U_n) = \frac{(2^{-n})^2}{12} = \frac{2^{-2n}}{12}$$

Therefore

$$\begin{aligned} \sum_{n=2}^{\infty} \text{Var}(U_n) &= \frac{1}{12} \sum_{n=2}^{\infty} 2^{-2n} = \frac{1}{12} \sum_{n=2}^{\infty} 4^{-n} = \frac{1}{12} \cdot \frac{(1/4)^2}{1 - 1/4} = \frac{1}{144} \\ \Rightarrow \text{Var}(X) &= \text{Var}(X_1) + \sum_{n=2}^{\infty} \text{Var}(U_n) = \frac{1}{48} + \frac{1}{144} = \frac{1}{36} \end{aligned}$$

Answer: 0.02778

750. Everything Contained

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the intervals $[x_i, y_i]$ for $i = 1, \dots, 8$. After sampling we have 16 labeled endpoints, the total number of possible orderings of these 16 distinct labeled points along $(0, 1)$ is $16!$. For a given interval $[x_i, y_i]$ we regard its left endpoint as the one that appears earlier in the order and its right endpoint as the one that appears later. For each interval there are two possible orderings of its two labeled endpoints in a permutation, but only the ordering with its left endpoint before its right endpoint matches the geometric interval orientation we use, enforcing this for all eight intervals means we keep exactly one of the two orderings per interval. Thus the number of orderings consistent with the usual left/right assignment is $\frac{16!}{2^8}$.

Now count the orderings where some interval contains all others. If an interval contains every other interval

its two endpoints must occupy the extreme positions (leftmost and rightmost) among the 16 points. There are 8 choices for which interval is the containing one. Once we fix that interval to be the extremes, the remaining 14 endpoints (from the other 7 intervals) can be arranged arbitrarily among the middle 14 positions, but again for each of those 7 intervals only one of the two internal orderings is valid. Hence the number of favorable orderings equals $\frac{8 \cdot 14!}{2^7}$.

Therefore the desired probability is

$$\frac{8 \cdot \frac{14!}{2^7}}{\frac{16!}{2^8}} = \frac{8 \cdot 14! \cdot 2^8}{2^7 \cdot 16!} = \frac{16 \cdot 14!}{16 \cdot 15 \cdot 14!} = \frac{1}{15}.$$

So the probability that one of the eight random intervals contains all the others is $\frac{1}{15}$.

Answer:

751. Disjoint Arcs

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Fix A as a reference point on the circle (this is allowed by rotational symmetry). The remaining three points B, C, D appear in a random counterclockwise order around the circle; all $3! = 6$ orders are equally likely. List the six possible orders (in COUNTER-CLOCKWISE direction) starting from A :

$A - B - C - D$ $A - B - D - C$ $A - C - B - D$ $A - C - D - B$ $A - D - B - C$ $A - D - C - B$

Now determine when the two counterclockwise arcs are disjoint. The red arc is the CCW stretch from A to B ; its complement on the circle is the arc from B back to A . For the green arc (the CCW arc from C to D) to be entirely disjoint from the red arc it must lie completely inside the complementary arc $B \rightarrow A$. That requires both C and D to appear after B and before returning to A , and moreover C must come before D in that segment so the CCW arc from C to D stays inside the complementary segment.

Among the six orders above only the first order, $A - B - C - D$, satisfies those conditions.

Since exactly 1 of the 6 equally likely circular orders gives disjoint arcs, the probability is $\frac{1}{6}$.

Answer:

752. Maximize RPS Wins

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We can achieve 7 wins using the following construction:

- Alice plays 2 Rocks against Bob's 2 Scissors $\rightarrow$ 2 wins.
- Alice plays 1 Rock against Bob's 1 Paper $\rightarrow$ 1 loss.
- Alice plays 3 Scissors against Bob's 3 Papers $\rightarrow$ 3 wins.
- Alice plays 2 Scissors against Bob's 2 Rocks $\rightarrow$ 2 losses.
- Alice plays 2 Papers against Bob's 2 Rocks $\rightarrow$ 2 wins.

Now we show that achieving 8 (or more) wins is impossible. Alice can only win in three types of pairings: $R_i S$, $S_i P$, $P_i R$. Upper bounds per type are

$$R \rightarrow S \leq \min(3, 2) = 2, S \rightarrow P \leq \min(5, 4) = 4, P \rightarrow R \leq \min(2, 4) = 2,$$

so the absolute upper bound on wins is $2 + 4 + 2 = 8$. Thus to get 8 wins all three bounds must be attained simultaneously. That means the winning match counts would be

$$R \rightarrow S = 2, S \rightarrow P = 4, P \rightarrow R = 2.$$

Those consume Alice's moves $R : 2, S : 4, P : 2$ (so Alice has $R : 1, S : 1, P : 0$ left) and Bob's moves $S : 2, P : 4, R : 2$ (so Bob has $R : 2, S : 0, P : 0$ left). The two remaining Bob moves are both Rock, while

Alice's remaining moves are one Rock and one Scissors. Any pairing now forces Alice's remaining Rock to face one of Bob's Rocks (a tie), which is not allowed. Therefore the simultaneous attainment of the three maxima is impossible, so 8 wins cannot occur.

Hence the maximum achievable is 7.

Answer: 7

753. Maximum Within An Interval

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $X_1, \dots, X_5$ be iid $\text{Unif}(0, 5)$. For any $t \in [0, 5]$ the CDF is $F(t) = t/5$. The event $\max_i X_i \leq t$ is the same as all five points being $\leq t$, so

$$\Pr(\max X_i \leq t) = (F(t))^5 = \left(\frac{t}{5}\right)^5.$$

Hence

$$\Pr(3 \leq \max X_i \leq 4) = \Pr(\max X_i \leq 4) - \Pr(\max X_i \leq 3) = \left(\frac{4}{5}\right)^5 - \left(\frac{3}{5}\right)^5.$$

So, the required probability is 0.24992.

Answer: 0.24992

754. Bulb Toggle I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

A bulb numbered n is toggled once for every positive divisor of n . (Person d toggles bulb n exactly when $d \mid n$.) Since bulbs start off, the final state of bulb n is:

- on if n has an odd number of divisors,
- off if n has an even number of divisors.

Every divisor of n is obtained by choosing, for each i , an exponent e_i with $0 \leq e_i \leq a_i$. Thus the total number of positive divisors is

$$d(n) = \prod_{i=1}^k (a_i + 1).$$

Now look at the parity of $d(n)$. A product of integers is odd precisely when every factor is odd, so $d(n)$ is odd iff each $a_i + 1$ is odd. That means every a_i is even and this implies n is a perfect square.

So we need the count of perfect squares ≤ 1000 . The largest integer whose square is ≤ 1000 is $\lfloor \sqrt{1000} \rfloor = 31$ (since $31^2 = 961$ and $32^2 = 1024 > 1000$). Therefore the bulbs that are finally on are

$$1^2, 2^2, 3^2, \dots, 31^2$$

So the number of bulbs that are on is 31.

Answer: 31

755. Ten Cars Meet

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

All ten cars are present at some common time iff the earliest arrival and the latest arrival are within 15 minutes, i.e.

$$\max_i X_i - \min_i X_i \leq 15,$$

where $X_1, \dots, X_{10}$ are iid Uniform $[0, 60]$.

Let $L = 60$, $t = 15$, and $n = 10$. Using the approach given in hint 3, the joint density of the minimum m and maximum M of n iid Uniform $[0, L]$ is

$$f_{m,M}(u, v) = \frac{n(n-1)}{L^n} (v-u)^{n-2}, \quad 0 \leq u \leq v \leq L.$$

Thus

$$\Pr(M - m \leq t) = \int_{u=0}^L \int_{v=u}^{\min(u+t, L)} \frac{n(n-1)}{L^n} (v-u)^{n-2} dv du.$$

Evaluate the inner integral by the substitution $w = v - u$ to get

$$\Pr(M - m \leq t) = \frac{n}{L^n} \int_0^L (\min(t, L-u))^{n-1} du.$$

Splitting the integral at $u = L - t$ and simplifying yields the closed form

$$\Pr(M - m \leq t) = \frac{t^{n-1}(nL - (n-1)t)}{L^n}.$$

Plug in $n = 10$, $L = 60$, $t = 15$:

$$\Pr = \frac{15^9(10 \cdot 60 - 9 \cdot 15)}{60^{10}} = \frac{15^9 \cdot 465}{60^{10}} = \frac{31}{4^{10}}.$$

Answer: 0.00003

756. Perfect Rank Guess

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $n = 4$ for brevity. We will use recursion/dynamic programming. Note that the optimal probability of winning does not depend on the length of the interval, it depends only on the number of points. Thus, even if the numbers were chosen from the interval $[a, b]$, the optimal probability of winning would remain the same. Let P_n denote the optimal probability of winning when there are n points in the interval under the rules of the problem.

Suppose the first observed value is t . Also, suppose that you choose some rank r right now. Let $a = r - 1$ and $b = n - r$. The probability your choice is correct and you then succeed on the remaining groups can be at most

$$\Pr(\text{true rank} = r \mid t) \times (\text{best possible continuation success}).$$

But the best possible continuation success - no matter how clever you are later - is exactly $P_a P_b$ (the optimal success probabilities on the left-interval group and right-interval group). Therefore for any strategy that chooses r at this point,

$$\text{conditional success} \leq \Pr(\text{rank} = r \mid t) \cdot P_a P_b$$

Denote $\Pr(\text{rank} = r \mid t) \cdot P_a P_b$ by $S_r(t)$. Therefore,

$$\text{conditional success} \leq S_r(t)$$

That is true pointwise for every r . So whatever your strategy does at this moment, its conditional success $v(t)$ cannot exceed $\max_r S_r(t)$. Now define the strategy that, for this t , chooses

$$r^*(t) = \arg \max_r S_r(t)$$

and thereafter plays optimally on the two subproblems (using the same strategy). This is the optimal strategy.

Now, let's calculate the optimal winning probability for $n = 4$. We will calculate P_1 through P_4 sequentially as we need the values from smaller number to evaluate for larger numbers. We compute P_1 through P_4 sequentially, since evaluating P_n requires the values for smaller n .

P_1 : Trivial: with one item you always win. So, $P_1 = 1$.

P_2 : Observe $x_1 = t$. The chance its true rank is 1 is $1 - t$, and rank 2 is t . If you call the more likely rank (threshold at $t = \frac{1}{2}$) the conditional win probability is $\max\{1 - t, t\}$. Hence

$$P_2 = \int_0^1 \max\{1 - t, t\} dt = \int_0^{1/2} (1 - t) dt + \int_{1/2}^1 t dt = \frac{3}{4}.$$

$$\implies P_2 = \frac{3}{4} = 0.75.$$

P_3 : Fix $x_1 = t$. For guesses $r = 1, 2, 3$ the conditional success probabilities are

- $S_1(t) = \frac{3}{4}(1 - t)^2$ (both remaining $> t \Rightarrow P_2 = \frac{3}{4}$),
- $S_2(t) = 2t(1 - t)$ (one below, one above $\Rightarrow$ forced),
- $S_3(t) = \frac{3}{4}t^2$ (both remaining $< t$).

Compare S_1 and S_2 : $\frac{3}{4}(1 - t)^2 = 2t(1 - t) \Rightarrow t = \frac{3}{11}$. By symmetry $S_2 = S_3$ at $t = \frac{8}{11}$. So the optimal first call is

- rank 1 for $t \in (0, \frac{3}{11})$,
- rank 2 for $t \in [\frac{3}{11}, \frac{8}{11}]$,
- rank 3 for $t \in (\frac{8}{11}, 1)$.

Integrate:

$$P_3 = \int_0^{3/11} \frac{3}{4}(1 - t)^2 dt + \int_{3/11}^{8/11} 2t(1 - t) dt + \int_{8/11}^1 \frac{3}{4}t^2 dt = \frac{377}{726}$$

P_4 : Use the same pointwise- $S_r(t)$ recipe with $P_0 = P_1 = 1$, $P_2 = \frac{3}{4}$, $P_3 = \frac{377}{726}$. For $n = 4$ the relevant first-step S -values are

$$S_1(t) = P_3(1 - t)^3, \quad S_2(t) = 3P_2t(1 - t)^2, \quad S_3(t) = 3P_2t^2(1 - t), \quad S_4(t) = P_3t^3.$$

By symmetry the inner two regions meet at $t = \frac{1}{2}$: the left cutoff t^* between S_1 and S_2 satisfies

$$(1 - t^*)^3 P_3 = 3t^*(1 - t^*)^2 P_2 \Rightarrow t^* = \frac{P_3}{P_3 + 3P_2} = \frac{754}{4021}$$

So optimal first calls are: rank 1 on $[0, t^*]$, rank 2 on $[t^*, 1/2]$, rank 3 on $[1/2, 1 - t^*]$, rank 4 on $(1 - t^*, 1]$. Therefore

$$P_4 = 2 \left(\int_0^{t^*} P_3(1 - t)^3 dt + \int_{t^*}^{1/2} 3P_2t(1 - t)^2 dt \right)$$

Evaluating those integrals (exact arithmetic) yields

$$P_4 = \frac{1037891564126671}{3020778029791104} \approx 0.34358$$

Answer: 0.34358

757. Interesting Average III

[[Question](#) $\uparrow$ | [Hint](#) $\uparrow$ | $\uparrow$]

We are interested in finding the expected value of $\sum_{i=0}^9 d_i \cdot 10^i$ where each digit d_i is chosen independently with a uniform distribution over its possible values. The expected value for each d_i is $\mathbb{E}[d_i] = i$ for $1 \leq i \leq 8$,

due to the uniform distribution. For $i = 0$ and $i = 9$, the expected values are 0.5 and 8.5, respectively.

By the linearity of expectation: $\mathbb{E} \left[\sum_{i=0}^9 d_i \cdot 10^i \right] = \sum_{i=0}^9 \mathbb{E}[d_i] \cdot 10^i$.

This gives: $\mathbb{E} \left[\sum_{i=0}^9 d_i \cdot 10^i \right] = 8.5 \cdot 10^9 + \sum_{i=1}^8 i \cdot 10^i + 0.5 \cdot 10^0 = 9376543210.5$.

Answer: 9376543210.5

758. Seven Colors II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let C be the information that the first 350 positions contain 15 more red balls than yellow balls. That implies the second half (positions 351–700) contains 15 more yellow balls than red balls, so the imbalance of red versus yellow is reversed between the two halves. Apart from the labels “red” and “yellow,” the collection of balls of the other five colours is identically distributed between the halves; the condition C only distinguishes the halves by swapping which of the two colours (red or yellow) is in excess. Because the original random ordering treats all colours and positions symmetrically, and because the constraint C simply exchanges the roles of red and yellow between the halves while leaving the other colours untouched, each particular violet ball has the same conditional chance of being in the first half as in the second half. In other words, conditioned on C a given violet is equally likely to lie among positions 1 – 350 or among positions 351 – 700. Note that this occurs only because the added condition applies to the entire first half (the first 350 balls). If the condition were imposed on some other initial segment instead (for example, only the first 8 balls as in the problem Seven Colors I), this symmetry would not hold.

Since there are 100 violet balls, the expected number of violets in the second half equals $\frac{100}{2} = 50$. Conditioned on C , every ball that lies in the second half is equally likely to occupy any one of the 350 second-half positions, so the probability that position 351 is violet is the expected fraction of violets in that half, namely $\frac{50}{350}$, or $\frac{1}{7}$.

An important note: For some reason, many candidates argue that the additional information concerns only the red and yellow balls, so the probability involving a violet ball should remain unaffected. Since the initial probability that the 351st position is occupied by a violet ball is $100/700 = 1/7$, they conclude that it must still be $1/7$ after conditioning. However, this reasoning is incorrect, even though it leads to the right answer here. The claim that “the additional information is about red and yellow balls, therefore violet balls remain unaffected” is false in general. For example, in the problem Seven Colors I, a similar additional information is given (concerning red and yellow balls only), yet the resulting probability is not equal to $1/7$.

Answer: 0.14286

759. Seven Colors III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let C be the information that the first 350 positions contain exactly 15 more red balls than yellow balls. Using the approach from problem Seven Colors II, we know that each of the five non-(red,yellow) colours is, conditional on C , equally likely to fall in the first half or the second half. Hence the expected number of balls of each of those five colours in the first half is $\frac{100}{2} = 50$, so the expected total number of non-(red,yellow) balls in the first half is $5 \cdot 50 = 250$. Since the first half contains 350 balls total, the expected number of red-plus-yellow balls in the first half is $350 - 250 = 100$.

Because the difference between red and yellow in the first half is fixed at 15, the expected number of red balls in the first half equals half the expected sum plus half the difference, i.e. $\frac{100+15}{2} = \frac{115}{2}$. Therefore the expected number of red balls in the second half is $100 - \frac{115}{2} = \frac{85}{2}$. Position 351 is one of the 350 spots in the second half and, conditional on C , every ball that lies in the second half is equally likely to occupy any particular second-half position; thus the probability that position 351 is red equals the expected fraction of

red balls in that half:

$$\frac{\frac{85}{2}}{350} = \frac{85}{700} = \frac{17}{140}$$

Answer: 0.12143

760. Maximum Till Now II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Fix an integer $n \geq 1$ and consider the event that $\tau > n$. The event $\tau > n$ means that none of $X_2, \dots, X_n$ is smaller than X_1 , equivalently X_1 is the minimum among the sample $X_1, \dots, X_n$. Because the variables are i.i.d. and the distribution is continuous so ties have probability zero, each of the n coordinates is equally likely to be the minimum. Therefore

$$P(\tau > n) = P(X_1 = \min X_1, \dots, X_n) = \frac{1}{n}$$

Use the tail sum formula for a positive integer valued random variable, which states

$$\mathbb{E}[\tau] = \sum_{k=0}^{\infty} P(\tau > k)$$

Here $P(\tau > 0) = 1$ and for every integer $n \geq 1$ we have $P(\tau > n) = \frac{1}{n}$, so

$$\mathbb{E}[\tau] = 1 + \sum_{n=1}^{\infty} \frac{1}{n}$$

The harmonic series diverges, hence

$$\mathbb{E}[\tau] = \infty$$

Thus the stopping time has infinite expectation.

Answer: -1

761. Partial Exponential I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

The key is symmetry and linearity of expectation. For general $k \leq 10$ write

$$\mathbb{E}[S_k \mid S_{10}] = \sum_{i=1}^k \mathbb{E}[X_i \mid S_{10}]$$

Because the X_i are iid the information S_{10} treats every coordinate the same, so the conditional expectations $\mathbb{E}[X_i \mid S_{10}]$ are all equal. Hence

$$\mathbb{E}[S_{10} \mid S_{10}] = \sum_{i=1}^{10} \mathbb{E}[X_i \mid S_{10}] = 10\mathbb{E}[X_1 \mid S_{10}]$$

But $\mathbb{E}[S_{10} \mid S_{10}] = S_{10}$ since S_{10} is known when conditioning, so

$$\mathbb{E}[X_1 \mid S_{10}] = \frac{S_{10}}{10}$$

Substituting this into the expression for $\mathbb{E}[S_k | S_{10}]$ gives

$$\mathbb{E}[S_k | S_{10}] = k \frac{S_{10}}{10} = \frac{k}{10} S_{10}$$

Now set $k = 4$ and $S_{10} = 15$ to obtain

$$\mathbb{E}[S_4 | S_{10} = 15] = \frac{4}{10} \cdot 15 = 6$$

Answer: 6

762. Partial Exponential II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write $S_{10} = S_4 + T$ with $T = \sum_{i=5}^{10} X_i$. Independence of the X_i implies the block $X_5, \dots, X_{10}$ (hence T) is independent of S_4 , so $\mathbb{E}[T | S_4] = \mathbb{E}[T]$. Because each remaining X_i has mean 1 (since each X_i is independent identically distributed exponential random variable with rate 1),

$$\mathbb{E}[T] = 6 \cdot 1 = 6$$

Therefore

$$\mathbb{E}[S_{10} | S_4] = S_4 + \mathbb{E}[T | S_4] = S_4 + 6$$

Plugging $S_4 = 5$ gives $\mathbb{E}[S_{10} | S_4 = 5] = 5 + 6 = 11$.

Answer: 11

763. Signal To Noise Ratio

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

From the given ratios write $\mathbb{E}[X_1] = 4\sigma$ and $\mathbb{E}[X_2] = \sigma$. Linearity and independence give

$$\mathbb{E}[Z_p] = p\mathbb{E}[X_1] + (1-p)\mathbb{E}[X_2] = \sigma(1+3p), \quad \text{Var}(Z_p) = \sigma^2(p^2 + (1-p)^2).$$

The objective to maximize is

$$g(p) = \frac{\mathbb{E}[Z_p]}{\sqrt{\text{Var}(Z_p)}} = \frac{1+3p}{\sqrt{p^2 + (1-p)^2}}.$$

Maximizing g is equivalent to maximizing $G(p) = g(p)^2 = \frac{(1+3p)^2}{2p^2-2p+1}$. Differentiate and set numerator of the quotient rule to zero:

$$\frac{d}{dp}G(p) = 0 \iff (6+18p)(2p^2-2p+1) - (1+6p+9p^2)(4p-2) = 0.$$

This factors as $-2(3p+1)(5p-4) = 0$, so the critical points are $p = -\frac{1}{3}$ (outside $[0, 1]$) and $p = \frac{4}{5}$. Evaluate at endpoints and the admissible critical point: $g(0) = 1$, $g(1) = 4$,

$$g\left(\frac{4}{5}\right) = \frac{1+3(4/5)}{\sqrt{(4/5)^2 + (1/5)^2}} = \frac{3.4}{\sqrt{0.68}} = \sqrt{\frac{17}{4}} \approx 4.123 > 4$$

Hence the maximum on $[0, 1]$ occurs at $p = \frac{4}{5}$.

Answer: 0.8

764. Your Choice[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let k_n be the number of cards showing n . If the friend picks n his expected payout equals $\frac{n \cdot k_n}{100}$. To minimize the friend's best choice we must minimize

$$M = \max_{1 \leq n \leq 100} n \cdot k_n$$

subject to the integer constraints $k_n \geq 0$ and $\sum_{n=1}^{100} k_n = 100$. For a candidate value M we have the constraint $k_n \leq \lfloor M/n \rfloor$ for every n , so a necessary and sufficient feasibility condition is

$$S(M) = \sum_{n=1}^{100} \left\lfloor \frac{M}{n} \right\rfloor \geq 100.$$

We seek the smallest integer M with $S(M) \geq 100$. Computing $S(M)$ shows $S(27) = 95 < 100$ while $S(28) = 101 \geq 100$, so the minimal feasible M is 28. Therefore the minimal achievable value of $\max_n n \cdot k_n$ is 28, and the friend's expected payout (his optimal value divided by 100) is $\frac{28}{100} = 0.28$.

An explicit construction achieving $M = 28$ is obtained by taking $k_n = \lfloor 28/n \rfloor$ for $n \leq 28$ and $k_n = 0$ for $n \geq 29$, then removing one card from the 1's to make the total 100, this yields $\max_n n \cdot k_n = 28$. Since no $M < 28$ can be feasible, the value 0.28 is the expected payout under optimal play.

Answer: 0.28**765. Monochromatic Removal**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Think about the moment two turns before the process ends, when there are still marbles of both colors but the game will finish in exactly two more rounds. Call the remaining counts r reds and g greens. For the game to end after two rounds, the next two actions must be: remove the entire leftmost monochromatic block, then remove all of the remaining marbles (which must therefore all be the other color). That is only possible if, at that two-turns-from-end moment, the remaining deck looks either like one contiguous block of reds followed by one contiguous block of greens, or the same with the colors swapped. If the deck had more than two monochromatic blocks then removing the leftmost block and reshuffling could not yield a single-color pile in the next round, so finishing in two rounds would be impossible. Because each reshuffle produces a uniform random permutation of the $r + g$ remaining marbles, all orderings of those marbles are equally likely and among all permutations exactly two have the “all reds together and all greens together” property (the red-block-then-green-block ordering and the green-block-then-red-block ordering), and those two are symmetric. Therefore, conditioned on being two turns from the end, the last block's color is equally likely to be red or green. This symmetry does not depend on the specific numbers r and g , so it holds throughout the process and in particular for the original counts $r = 30$, $g = 60$. So, the probability the final marble is green is $\frac{1}{2}$.

Answer: 0.5**766. Exponential Inequality I**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Using the law of total probability for continuous case, we have

$$P(X_1 < X_2) = \int_0^\infty P(X_2 > x) f_{X_1}(x) dx$$

Since $f_{X_1}(x) = 4e^{-4x}$ and $P(X_2 > x) = e^{-6x}$ we get

$$P(X_1 < X_2) = \int_0^\infty 4e^{-4x}e^{-6x}dx = 4 \int_0^\infty e^{-10x}dx = 4 \cdot \frac{1}{10} = \frac{2}{5}$$

Note: In general, if $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$ are independent, then $\Pr(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$.

Answer: 0.4

767. Tired Rabbit

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let T be the (random) hitting time of $+2, -2$. We want $\mathbb{E}[\min(T, 8)]$. Using the tail sum formula, we have, $\mathbb{E}[\min(T, 8)] = \sum_{k=1}^8 \Pr(T \geq k)$, so we need the survival probabilities $\Pr(T \geq k)$.

Starting at 0 the walk alternates parity: after an odd number of steps the frog is at ± 1 , after an even number of steps it is at an even position. In particular the absorbing states ± 2 have even parity, so absorption can occur only on even steps. Hence $\Pr(T \geq 1) = 1$ and $\Pr(T \geq 2) = 1 - \Pr(\text{hit } \pm 2 \text{ in two steps})$. The two-step absorption probabilities are $\Pr(0 \rightarrow +1 \rightarrow +2) = (1/3)^2$ and $\Pr(0 \rightarrow -1 \rightarrow -2) = (2/3)^2$, so $\Pr(T \geq 2) = 1 - ((1/3)^2 + (2/3)^2) = \frac{4}{9}$.

If the process survives the first two steps it must be back at 0 (check the two-step transition probabilities: conditioned on not having hit ± 2 after two steps, the only possible position is 0). Thus the process “renews” every two steps: surviving $2m$ steps has probability $(4/9)^m$, and surviving $2m + 1$ steps has the same probability because no absorption can happen at odd times. Therefore

$$\Pr(T \geq 1) = 1, \quad \Pr(T \geq 2) = \frac{4}{9}, \quad \Pr(T \geq 3) = \frac{4}{9},$$

$$\Pr(T \geq 4) = \frac{16}{81}, \quad \Pr(T \geq 5) = \frac{16}{81}, \dots$$

and in general $\Pr(T \geq 2m) = \Pr(T \geq 2m + 1) = (4/9)^m$.

Now sum these up for $k = 1, \dots, 8$:

$$\mathbb{E}[\min(T, 8)] = \sum_{k=1}^8 \Pr(T \geq k) = 2 \sum_{m=0}^3 \left(\frac{4}{9}\right)^m = \frac{2522}{729} \approx 3.45953$$

Answer: 3.45953

768. Exponential Inequality II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Decompose the event by conditioning on whether X exceeds the first summand. If $X \leq X_1$ then certainly $X \leq X_1 + \dots + X_n$, so only the case $X > X_1$ can contribute. Hence

$$\mathbb{P}(X > X_1 + \dots + X_n) = \mathbb{P}(X > X_1 + \dots + X_n \mid X > X_1) \mathbb{P}(X > X_1).$$

When we condition on $X > X_1$ and view a fixed sample point, the memoryless property of the exponential says the residual $X - X_1$ has the same distribution as an independent copy of X . Therefore, given $X > X_1$ the inequality $X > X_1 + \dots + X_n$ is equivalent in distribution to the event that this fresh exponential exceeds $X_2 + \dots + X_n$. In other words if we set $P(n) = \mathbb{P}(X > X_1 + \dots + X_n)$ then the conditioning step yields the recursion

$$P(n) = P(n-1) \mathbb{P}(X > X_1).$$

From “exponential inequality I”, we know that for two independent exponentials $\text{Exp}(\lambda_X)$ and $\text{Exp}(\lambda_Y)$ the probability that the first exceeds the second equals $\lambda_Y/(\lambda_X + \lambda_Y)$. Applying this with $\lambda_X = \lambda_2$ (the rate of

X) and $\lambda_Y = \lambda_1$ (the rate of each X_i), we get $\mathbb{P}(X > X_1) = \lambda_1/(\lambda_1 + \lambda_2)$. With base $P(0) = 1$ the recursion gives

$$P(n) = \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^n.$$

Plugging $\lambda_1 = 1$, $\lambda_2 = 5$ and $n = 5$ yields

$$\mathbb{P}(X > X_1 + \cdots + X_5) = \left(\frac{1}{1+5} \right)^5 = \left(\frac{1}{6} \right)^5 = \frac{1}{7776}.$$

Answer: 0.00013

769. Give And Take

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Because you never see amounts, any strategy is a predetermined sequence of p Gives and $m = 100 - p$ Takes. First, any adjacent pair “Take then Give” can be swapped to “Give then Take” with a strict gain. Concretely, let T be the total in both boxes immediately before the pair and let M be the number of Takes remaining after the pair. Let $a = 2^{-M}$. Doing Take then Give yields expected future collection

$$\frac{T}{2} + \left(\frac{T}{2} + 1 \right) (1 - a) = T + 1 - \left(\frac{T}{2} + 1 \right) a,$$

whereas Give then Take gives

$$\frac{T+1}{2} + \left(\frac{T+1}{2} \right) (1 - a) = T + 1 - \frac{T+1}{2} a.$$

Their difference is $a/2 = 2^{-(M+1)} > 0$. Thus swapping increases expected payoff and repeating swaps moves all Gives to the front without decreasing expectation. Therefore it suffices to consider schedules that do p Gives first, then $m = 100 - p$ Takes.

If you Give p times first the boxes contain total p . Each Take collects, in expectation, half the current total and halves the remaining total. Therefore, m successive Takes yield

$$f(p) = p \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^m} \right) = p(1 - 2^{-m}), \quad m = 100 - p.$$

To find the integer p that maximizes $f(p)$ compare $f(p)$ and $f(p+1)$. Let $S_t = \sum_{i=1}^t 2^{-i} = 1 - 2^{-t}$. Then

$$f(p) = pS_m, \quad f(p+1) = (p+1)S_{m-1},$$

and

$$\begin{aligned} f(p) > f(p+1) &\iff pS_m > (p+1)S_{m-1} \iff p(S_{m-1} + 2^{-m}) > (p+1)S_{m-1} \\ &\iff p2^{-m} > S_{m-1} \iff p > 2^m - 2. \end{aligned}$$

Since $m = 100 - p$ this condition is equivalently

$$f(p) > f(p+1) \iff p > 2^{100-p} - 2.$$

Thus f increases while $p \leq 2^{100-p} - 2$ and decreases once $p > 2^{100-p} - 2$: the maximiser is the largest integer p for which $f(p) \leq f(p+1)$, i.e. the smallest integer satisfying $p > 2^{100-p} - 2$.

For $p = 93$ the right side is $2^7 - 2 = 126$ so $93 > 126$ is false. For $p = 94$ the right side is $2^6 - 2 = 62$ so $94 > 62$ is true. Hence the peak occurs at $p = 94$. Compute

$$f(94) = 94(1 - 2^{-6}) = 94 \cdot \frac{63}{64} = \frac{2961}{32} = 92.53125,$$

while $f(93) \approx 92.2734$ and $f(95) \approx 92.03125$ are smaller.

Therefore the optimal policy is to Give 94 times and then Take 6 times, yielding maximal expected payoff 92.53125.

Answer: 92.53125

770. Maxima Sum I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let N be the number of selected numbers among 3, 5, 7 that are greater than the number immediately to their left. The sum is odd exactly when N is odd, because only odd selected numbers affect parity. We do not include 1 here, since the first position is not part of the set of numbers being considered and hence, 1 cannot be greater than the number immediately preceding it in any case.

For any set $x_1 < x_2 < \dots < x_k$ chosen from 3, 5, 7, we claim that the probability that all of them are greater than the number immediately to their left is

$$\frac{x_1 - 1}{7} \cdot \frac{x_2 - 2}{6} \cdot \dots \cdot \frac{x_k - k}{8 - k}.$$

Here is a short proof for the above claim. Start with a random permutation of $1, 2, \dots, 7$, and remove x_1 from it. The other 6 numbers are still in some order, and they create 7 possible places where x_1 could be put back: before the first number, between the first and second, and so on, all the way to after the last number. These possible places are the insertion gaps. Since the permutation is random, x_1 is equally likely to lie in any one of these 7 gaps. Now x_1 is good exactly when the number immediately to its left is smaller than x_1 , which happens precisely when x_1 is inserted right after one of the $x_1 - 1$ numbers smaller than it. So the probability is $(x_1 - 1)/7$. After that, remove x_1 and x_2 . There are now 5 remaining numbers besides x_2 , so x_2 has 6 insertion gaps, and it is good exactly when it is inserted right after one of the $x_2 - 2$ smaller remaining numbers. Continuing in this way gives the product above.

Now apply this to the odd numbers 3, 5, 7. We get

$$P(3) = \frac{2}{7}, \quad P(5) = \frac{4}{7}, \quad P(7) = \frac{6}{7},$$

$$P(3, 5) = \frac{2}{7} \cdot \frac{3}{6} = \frac{1}{7}, \quad P(3, 7) = \frac{2}{7} \cdot \frac{5}{6} = \frac{5}{21}, \quad P(5, 7) = \frac{4}{7} \cdot \frac{5}{6} = \frac{10}{21},$$

and

$$P(3, 5, 7) = \frac{2}{7} \cdot \frac{3}{6} \cdot \frac{4}{5} = \frac{4}{35}.$$

By inclusion-exclusion, the probability that N is odd is

$$P(N \text{ is odd}) = S_1 - 2S_2 + 4S_3,$$

where

$$S_1 = P(3) + P(5) + P(7) = \frac{2}{7} + \frac{4}{7} + \frac{6}{7} = \frac{12}{7},$$

$$S_2 = P(3, 5) + P(3, 7) + P(5, 7) = \frac{1}{7} + \frac{5}{21} + \frac{10}{21} = \frac{6}{7},$$

$$S_3 = P(3, 5, 7) = \frac{4}{35}.$$

Therefore,

$$P(N \text{ is odd}) = \frac{12}{7} - 2 \cdot \frac{6}{7} + 4 \cdot \frac{4}{35} = \frac{16}{35}.$$

So the required probability is $\frac{16}{35}$.

Answer: 0.45714

771. Maxima Sum II[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let N be the number of selected numbers among 3, 5, 7 that are greater than all the numbers before them. The sum is odd exactly when N is odd, because only odd selected numbers affect parity. We do not include 1 here, since the first position is not part of the set of numbers being considered and hence, 1 cannot be greater than all the previous numbers in any case.

For any set $x_1 < x_2 < \dots < x_k$ chosen from 3, 5, 7, the probability that all of them are selected is

$$\frac{x_1 - 1}{7} \cdot \frac{1}{8 - x_1} \cdot \frac{1}{8 - x_2} \cdots \frac{1}{8 - x_k}.$$

This is because the first number in the permutation must be one of $1, 2, \dots, x_1 - 1$, otherwise x_1 can never be a record from the second position onward. That gives the factor $(x_1 - 1)/7$. Once the first number is smaller than x_1 , the relative order of $x_1, x_1 + 1, \dots, 7$ is still random, so x_1 is selected exactly when it appears before all larger numbers, which happens with probability $1/(8 - x_1)$. The same reasoning then applies to x_2, x_3 , and so on.

Now apply this to 3, 5, 7. We get

$$\begin{aligned} P(3) &= \frac{2}{7} \cdot \frac{1}{5} = \frac{2}{35}, & P(5) &= \frac{4}{7} \cdot \frac{1}{3} = \frac{4}{21}, & P(7) &= \frac{6}{7} \cdot 1 = \frac{6}{7}, \\ P(3, 5) &= \frac{2}{7} \cdot \frac{1}{5} \cdot \frac{1}{3} = \frac{2}{105}, & P(3, 7) &= \frac{2}{7} \cdot \frac{1}{5} \cdot 1 = \frac{2}{35}, & P(5, 7) &= \frac{4}{7} \cdot \frac{1}{3} \cdot 1 = \frac{4}{21}, \end{aligned}$$

and

$$P(3, 5, 7) = \frac{2}{7} \cdot \frac{1}{5} \cdot \frac{1}{3} \cdot 1 = \frac{2}{105}.$$

By inclusion-exclusion, the probability that N is odd is

$$P(N \text{ is odd}) = S_1 - 2S_2 + 4S_3,$$

where

$$\begin{aligned} S_1 &= P(3) + P(5) + P(7) = \frac{2}{35} + \frac{4}{21} + \frac{6}{7} = \frac{116}{105}, \\ S_2 &= P(3, 5) + P(3, 7) + P(5, 7) = \frac{2}{105} + \frac{2}{35} + \frac{4}{21} = \frac{4}{15}, \\ S_3 &= P(3, 5, 7) = \frac{2}{105}. \end{aligned}$$

Therefore,

$$P(N \text{ is odd}) = \frac{116}{105} - 2 \cdot \frac{4}{15} + 4 \cdot \frac{2}{105} = \frac{68}{105}.$$

So the required probability is $\frac{68}{105}$.

Answer: 0.64762

772. Nearest Diagonal II[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the two diagonals be the lines joining $(0, 0)$ to $(2, 1)$ and $(0, 1)$ to $(2, 0)$. Their equations are

$$x - 2y = 0 \quad \text{and} \quad x + 2y - 2 = 0.$$

So if D is the distance from a random point (X, Y) to the nearer diagonal, then

$$D = \min \left(\frac{|X - 2Y|}{\sqrt{5}}, \frac{|X + 2Y - 2|}{\sqrt{5}} \right).$$

Now define

$$U = X - 2Y, \quad V = X + 2Y - 2.$$

In these coordinates, the rectangle becomes the diamond

$$|U| + |V| \leq 2,$$

and the distance is simply

$$D = \frac{\min(|U|, |V|)}{\sqrt{5}}.$$

So it is enough to find the expected value of $\min(|U|, |V|)$. Using the tail integral formula,

$$\mathbb{E}[\min(|U|, |V|)] = \int_0^\infty \mathbb{P}(\min(|U|, |V|) > t) dt.$$

The largest possible value of $\min(|U|, |V|)$ inside the diamond is 1, so

$$\mathbb{E}[\min(|U|, |V|)] = \int_0^1 \mathbb{P}(|U| > t, |V| > t) dt.$$

For a fixed $t \in [0, 1]$, the condition $|U| > t$ and $|V| > t$ removes a central cross from the diamond. What remains has area

$$8(1 - t)^2,$$

while the whole diamond has area 8. Therefore,

$$\mathbb{P}(|U| > t, |V| > t) = (1 - t)^2.$$

Hence

$$\mathbb{E}[\min(|U|, |V|)] = \int_0^1 (1 - t)^2 dt = \frac{1}{3}.$$

Finally,

$$\mathbb{E}[D] = \frac{1}{\sqrt{5}} \cdot \frac{1}{3} = \frac{1}{3\sqrt{5}}.$$

So the expected distance is $\frac{1}{3\sqrt{5}}$.

Answer: 0.14907

773. Fixed Maximum

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let S be the sum of the 6 numbers and let M be their maximum. We want $\mathbb{E}[S \mid M = 15]$.

It is easier to first condition on the event that all 6 draws are at most m . If $M \leq m$, then each draw is uniformly distributed over $1, 2, \dots, m$, so the expected value of one draw is $(m + 1)/2$. Therefore

$$\mathbb{E}[S \mid M \leq m] = 6 \cdot \frac{m + 1}{2} = 3(m + 1).$$

Now the event $M = 15$ means $M \leq 15$ but not $M \leq 14$. So we use

$$\mathbb{E}[S \mid M = 15] = \frac{\mathbb{P}(M \leq 15)\mathbb{E}[S \mid M \leq 15] - \mathbb{P}(M \leq 14)\mathbb{E}[S \mid M \leq 14]}{\mathbb{P}(M = 15)}.$$

Here

$$\mathbb{P}(M \leq 15) = \left(\frac{15}{20}\right)^6, \quad \mathbb{P}(M \leq 14) = \left(\frac{14}{20}\right)^6,$$

and

$$\mathbb{E}[S \mid M \leq 15] = 3(16) = 48, \quad \mathbb{E}[S \mid M \leq 14] = 3(15) = 45.$$

So

$$\mathbb{E}[S \mid M = 15] = \frac{48 \left(\frac{15}{20}\right)^6 - 45 \left(\frac{14}{20}\right)^6}{\left(\frac{15}{20}\right)^6 - \left(\frac{14}{20}\right)^6}.$$

This simplifies to

$$\mathbb{E}[S \mid M = 15] = \frac{48 \cdot 15^6 - 45 \cdot 14^6}{15^6 - 14^6} = \frac{207920880}{3861089} \approx 53.85032.$$

Answer: 53.85032

774. It Took Ten Turns

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let $S_n = X_1 + \cdots + X_n$, where the X_i are i.i.d. Uniform(0, 1), and let T be the first time the running sum becomes greater than 1. We want $\mathbb{E}[S_{10} \mid T = 10]$.

Given $S_9 = x$ and $T = 10$, the 10th draw is uniform on $(1 - x, 1)$, so

$$\mathbb{E}[X_{10} \mid S_9 = x, T = 10] = \frac{(1 - x) + 1}{2} = 1 - \frac{x}{2}.$$

Hence

$$\mathbb{E}[S_{10} \mid T = 10] = \mathbb{E}[S_9 + X_{10} \mid T = 10] = \mathbb{E}\left[S_9 + 1 - \frac{S_9}{2} \mid T = 10\right] = 1 + \frac{1}{2}\mathbb{E}[S_9 \mid T = 10].$$

So, we only need to find $\mathbb{E}[S_9 \mid T = 10]$.

Now $T = 10$ means that after 9 draws the sum is still at most 1, but after the 10th draw it becomes greater than 1. Let us find the conditional density of S_9 given $T = 10$. By definition,

$$\mathbb{P}(S_9 < x \mid T = 10) = \frac{\mathbb{P}(S_9 < x, S_9 < 1, S_{10} > 1)}{\mathbb{P}(S_9 < 1, S_{10} > 1)}.$$

If $x < 1$, then $S_9 < x$ already implies $S_9 < 1$, so the numerator becomes

$$\mathbb{P}(S_9 < x, S_{10} > 1).$$

The denominator is just a constant, so it does not affect how the answer depends on x .

For the numerator, think of two parts. First, the chance that the first 9 numbers have sum less than x is proportional to the volume of a 9-dimensional simplex, so it is proportional to x^9 . Second, if the first 9 numbers sum to x , then the 10th number must be larger than $1 - x$, which happens with probability x .

That means

$$\mathbb{P}(S_9 < x, S_{10} > 1) = \int_0^x \mathbb{P}(S_9 \in dt) \mathbb{P}(X_{10} > 1 - t).$$

The first part is proportional to $t^8 dt$ (since we are interested in the probability density (not the cumulative distribution), we differentiate t^{10} , which reduces the power by one and thus removes one factor of t in the proportionality), and the second part is t . So

$$\mathbb{P}(S_9 < x, S_{10} > 1) \propto \int_0^x t^8 \cdot t dt = \int_0^x t^9 dt \propto x^{10}.$$

After dividing by the denominator (which is a constant), the conditional CDF is still proportional to x^{10} . Therefore the conditional density is its derivative, so it is proportional to x^9 . Thus

$$f_{S_9|T=10}(x) = cx^9, \quad 0 \leq x \leq 1.$$

To find c , normalize:

$$1 = \int_0^1 cx^9 dx = c \cdot \frac{1}{10},$$

so $c = 10$. Therefore

$$f_{S_9|T=10}(x) = 10x^9, \quad 0 \leq x \leq 1.$$

Now compute the conditional expectation of S_9 :

$$\mathbb{E}[S_9 | T = 10] = \int_0^1 x \cdot 10x^9 dx = 10 \int_0^1 x^{10} dx = \frac{10}{11}.$$

Hence,

$$\mathbb{E}[S_{10} | T = 10] = 1 + \frac{1}{2} \cdot \mathbb{E}[S_9 | T = 10] = \frac{16}{11}.$$

So the expected sum is $\frac{16}{11}$.

Answer: 1.45455

775. Increasing Decreasing II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let f_n be the number of permutations of $1, 2, \dots, n$ that satisfy the alternating pattern

$$a_1 < a_2 > a_3 < a_4 > \dots.$$

Then the required probability is $\frac{f_8}{8!}$.

We now use recursion. Place the largest number n . Since the peaks occur at the even positions, n must lie in one of those positions. If n is in position $2k$, then the numbers to its left and right form two smaller problems of the same type: the left side uses $2k - 1$ numbers and must satisfy the same alternating pattern, the right side uses the remaining $n - 2k$ numbers and also must satisfy the same alternating pattern.

Thus,

$$f_n = \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n-1}{2k-1} f_{2k-1} f_{n-2k}.$$

Now we compute the smaller values:

$$f_0 = 1, \quad f_1 = 1,$$

$$f_2 = \binom{1}{1} f_1 f_0 = 1,$$

$$f_3 = \binom{2}{1} f_1 f_1 = 2,$$

$$f_4 = \binom{3}{1} f_1 f_2 + \binom{3}{3} f_3 f_0 = 3 \cdot 1 \cdot 1 + 1 \cdot 2 \cdot 1 = 5,$$

$$f_5 = \binom{4}{1} f_1 f_3 + \binom{4}{3} f_3 f_1 = 4 \cdot 1 \cdot 2 + 4 \cdot 2 \cdot 1 = 16,$$

$$f_6 = \binom{5}{1} f_1 f_4 + \binom{5}{3} f_3 f_2 + \binom{5}{5} f_5 f_0 = 5 \cdot 5 + 10 \cdot 2 \cdot 1 + 1 \cdot 16 = 61,$$

$$f_7 = \binom{6}{1} f_1 f_5 + \binom{6}{3} f_3 f_3 + \binom{6}{5} f_5 f_1 = 6 \cdot 16 + 20 \cdot 2 \cdot 2 + 6 \cdot 16 = 272.$$

$$\text{Finally, } f_8 = \binom{7}{1} f_1 f_6 + \binom{7}{3} f_3 f_4 + \binom{7}{5} f_5 f_2 + \binom{7}{7} f_7 f_0 = 7 \cdot 1 \cdot 61 + 35 \cdot 2 \cdot 5 + 21 \cdot 16 \cdot 1 + 1 \cdot 272 \cdot 1 = 1385.$$

So the required probability is

$$\frac{f_8}{8!} = \frac{1385}{40320} = \frac{277}{8064}.$$

Answer: 0.03435

776. Swap Swap

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let I_j be the indicator variable for the event that the number j is in its original position after 5 swaps. Then the required expected number is

$$\mathbb{E} \left[\sum_{j=1}^{16} I_j \right] = \sum_{j=1}^{16} \mathbb{E}[I_j].$$

By symmetry, all 16 numbers have the same probability of being fixed, so if p_5 denotes the probability that number 1 is in its original position after 5 swaps, then the answer is $16p_5$.

Now define p_n to be the probability that after n swaps, number 1 is in its correct position. Then after one more swap, there are two cases.

If number 1 is already correct, it stays correct provided it is not selected for swapping. The number of pairs of positions not involving position 1 is $\binom{15}{2}$, out of the total $\binom{16}{2}$, so this happens with probability $\frac{\binom{15}{2}}{\binom{16}{2}} = \frac{7}{8}$.

If number 1 is not currently correct, then it becomes correct only if the swap chooses exactly the two positions consisting of its current position and the correct position. There is exactly one such pair, so this happens with probability $\frac{1}{\binom{16}{2}} = \frac{1}{120}$.

Therefore,

$$p_{n+1} = p_n \cdot \frac{7}{8} + (1 - p_n) \cdot \frac{1}{120}$$

This can be rewritten as

$$p_{n+1} = \frac{13}{15}p_n + \frac{1}{120}$$

With $p_1 = \frac{13}{15}$, this linear recurrence gives

$$p_n = \frac{1}{16} + \frac{193}{240} \left(\frac{13}{15} \right)^{n-1}$$

So

$$p_5 = \frac{1}{16} + \frac{193}{240} \left(\frac{13}{15} \right)^4$$

Hence the expected number of numbers in their original positions is

$$16p_5 = 16 \left(\frac{1}{16} + \frac{193}{240} \left(\frac{13}{15} \right)^4 \right) \approx 8.25896$$

Answer: 8.25896

777. Exponential Pentagon

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Five positive numbers can form a pentagon exactly when no one of them is at least the sum of the other four. So the only way the five lengths fail to form a pentagon is if one side is too long, meaning that for some i we have $X_i > \sum_{j \neq i} X_j$.

These five bad events are disjoint, because two different sides cannot both be larger than the sum of the other four. Therefore, by symmetry,

$$P(\text{pentagon}) = 1 - 5P(X_1 > X_2 + X_3 + X_4 + X_5).$$

Now use the result from “Exponential Inequality II”: if $Y \sim \text{Exp}(\lambda_2)$ and $Y_1, \dots, Y_n$ are IID $\text{Exp}(\lambda_1)$, then

$$P(Y > Y_1 + \dots + Y_n) = \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^n.$$

Here all five variables are $\text{Exp}(1)$, so $\lambda_1 = \lambda_2 = 1$, and we take $n = 4$. Thus,

$$P(X_1 > X_2 + X_3 + X_4 + X_5) = \left(\frac{1}{1+1}\right)^4 = \frac{1}{16}.$$

Hence,

$$P(\text{pentagon}) = 1 - 5 \cdot \frac{1}{16} = \frac{11}{16}.$$

So the required probability is $\frac{11}{16}$.

Answer: 0.6875

778. Constrained Maximum

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $p = P(X = 0)$. Condition on whether X is zero or not. If $X \neq 0$, let Y denote the random variable describing the distribution of X given that X is nonzero. Note that Y may be continuous or discrete. Then X equals 0 with probability p , and has mean $E[Y]$ with probability $1 - p$.

Using the law of total expectation,

$$\begin{aligned} \mathbb{E}[X] &= p \cdot 0 + (1 - p)\mathbb{E}[Y] = 5, \\ \implies \mathbb{E}[Y] &= \frac{5}{1 - p}. \end{aligned}$$

Next, since $\text{Var}(X) = 25$ and $\mathbb{E}[X] = 5$,

$$\mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}[X])^2 = 25 + 25 = 50.$$

Also,

$$\begin{aligned} \mathbb{E}[X^2] &= p \cdot 0^2 + (1 - p)\mathbb{E}[Y^2], \\ \implies \mathbb{E}[Y^2] &= \frac{50}{1 - p}. \end{aligned}$$

Now we use the fact that variance is always nonnegative,

$$\text{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 \geq 0.$$

Substituting the values above,

$$\frac{50}{1 - p} - \left(\frac{5}{1 - p}\right)^2 \geq 0.$$

Multiplying by $(1 - p)^2$ gives $50(1 - p) - 25 \geq 0$, and hence, $p \leq \frac{1}{2}$.

Therefore, the maximum possible value of $P(X = 0)$ is $\frac{1}{2}$.

This bound is attainable, for example by taking $P(X = 0) = \frac{1}{2}$, $P(X = 10) = \frac{1}{2}$, which gives mean 5 and variance 25.

Answer: 0.5

779. Strange Product

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

It is natural to start by conditioning on X_1 and using the law of total expectation. If $X_1 = u$, then

$$\mathbb{E}[P] = \int_0^1 (1+u)\mathbb{E}[(1+uX_2)(1+uX_2X_3)\cdots]du.$$

At this stage, we are stuck. But if we look closely at

$$\mathbb{E}[(1+uX_2)(1+uX_2X_3)\cdots],$$

we see that it is the same problem with u inserted at the front. That suggests introducing the more general function

$$f(x) = \mathbb{E}[(1+xX_1)(1+xX_1X_2)(1+xX_1X_2X_3)\cdots].$$

We do this because (as shown below) it leads to the same function appearing again inside the integral (with a different parameter), which in turn yields a nice differential equation.

Condition on X_1 in $f(x)$. Since X_1 is uniform on $(0, 1)$, we get

$$f(x) = \int_0^1 (1+xu)\mathbb{E}[(1+xuX_2)(1+xuX_2X_3)\cdots]du.$$

But the expectation on the right is exactly $f(xu)$, because $(X_2, X_3, \dots)$ has the same distribution as $(X_1, X_2, \dots)$. So

$$f(x) = \int_0^1 (1+xu)f(xu)du.$$

Now let $t = xu$. Then

$$\begin{aligned} f(x) &= \frac{1}{x} \int_0^x (1+t)f(t)dt. \\ \implies xf(x) &= \int_0^x (1+t)f(t)dt. \end{aligned}$$

Differentiate both sides with respect to x :

$$f(x) + xf'(x) = (1+x)f(x).$$

Hence $f'(x) = f(x)$.

Also, $f(0) = 1$, since every factor becomes 1 when $x = 0$. Therefore $f(x) = e^x$.

Finally, $\mathbb{E}[P] = f(1) = e$.

Answer: 2.71828

780. Keep Multiplying

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let $f(x)$ be the expected additional time needed to go below $\frac{1}{2}$, given that the current value is x , where $\frac{1}{2} \leq x \leq 1$.

The idea is to write a recursion in terms of the current value. After one second, the new value is xU . If $xU < \frac{1}{2}$, then the process stops immediately. Otherwise, we restart from the new value xU .

So for $\frac{1}{2} \leq x \leq 1$,

$$f(x) = 1 + \int_{1/(2x)}^1 f(xu)du.$$

Now make the substitution $t = xu$. Then $du = \frac{dt}{x}$, and the lower limit becomes $\frac{1}{2}$, so

$$f(x) = 1 + \frac{1}{x} \int_{1/2}^x f(t)dt.$$

Multiplying by x ,

$$xf(x) = x + \int_{1/2}^x f(t)dt.$$

Differentiating both sides with respect to x , we get

$$f(x) + xf'(x) = 1 + f(x).$$

Hence, $xf'(x) = 1 \implies f'(x) = \frac{1}{x}$.

Integrating, $f(x) = \ln x + C$.

Now use the boundary condition at $x = \frac{1}{2}$. If the current value is exactly $\frac{1}{2}$, then after one more multiplication by $U \in (0, 1)$, the result is automatically smaller than $\frac{1}{2}$, so $f(\frac{1}{2}) = 1$.

Thus $1 = \ln(\frac{1}{2}) + C$, and hence, $C = 1 + \ln 2$. Therefore, $f(x) = 1 + \ln(2x)$.

Starting from $x = 1$, the expected time is $f(1) = 1 + \ln 2$.

So the answer is $1 + \ln 2$.

Answer: 1.69315

781. Maximum Exponential

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $M = \max(X_1, X_2, X_3, X_4)$. By the tail integral formula,

$$\mathbb{E}[M] = \int_0^\infty \Pr(M > t)dt.$$

Now

$$\Pr(M \leq t) = \prod_{i=1}^4 \Pr(X_i \leq t) = \prod_{i=1}^4 (1 - e^{-it}),$$

so

$$\mathbb{E}[M] = \int_0^\infty \left(1 - \prod_{i=1}^4 (1 - e^{-it}) \right) dt.$$

When the product is expanded, each nonempty subset $S \subseteq 1, 2, 3, 4$ contributes the term

$$(-1)^{|S|+1} e^{-(\sum_{i \in S} i)t},$$

hence

$$\mathbb{E}[M] = \sum_{\emptyset \neq S \subseteq 1, 2, 3, 4} \frac{(-1)^{|S|+1}}{\sum_{i \in S} i}.$$

So

$$\mathbb{E}[M] = \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) - \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} \right) + \left(\frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} \right) - \frac{1}{10}.$$

This is equal to $\frac{89}{72}$.

Therefore,

$$\mathbb{E}[\max(X_1, X_2, X_3, X_4)] = \frac{89}{72}.$$

Answer: 1.23611

782. Winner Takes All

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let N be the total number of rounds played, and let M_i be the number of rounds in which player i takes part. Since every round involves exactly two players, we have

$$2N = M_1 + M_2 + M_3 + M_4.$$

Taking expectations gives

$$2\mathbb{E}[N] = \mathbb{E}[M_1] + \mathbb{E}[M_2] + \mathbb{E}[M_3] + \mathbb{E}[M_4].$$

So it remains to find $\mathbb{E}[M_i]$ for each player.

Now focus on one player, say player i , who starts with x dollars. Each time this player plays, their fortune goes up by 1 or down by 1 with equal probability, because the game is fair. If we look only at the rounds involving this player, their money follows a simple symmetric random walk starting at x . This walk stops when the player reaches 0, or when they reach 19, since the total amount of money in the game is $1 + 4 + 6 + 8 = 19$.

For a simple symmetric random walk on $\{0, 1, \dots, 19\}$, the expected time to hit one of the boundaries starting from x is $\mathbb{E}[M_i] = x(19 - x)$.

Applying this to the four players gives $\mathbb{E}[M_1] = 1 \cdot 18 = 18$, $\mathbb{E}[M_2] = 4 \cdot 15 = 60$, $\mathbb{E}[M_3] = 6 \cdot 13 = 78$, $\mathbb{E}[M_4] = 8 \cdot 11 = 88$.

Therefore, $2\mathbb{E}[N] = 18 + 60 + 78 + 88 = 244$, $\implies \mathbb{E}[N] = 122$.

The expected number of rounds until the game ends is 122.

Answer: 122

783. Spaced Uniforms I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Think of placing the points from left to right. After the first point is chosen, reserve a segment of length $1/20$ immediately to its right (no point can lie in this reserved segment if the condition of the problem is to be satisfied). After the second point is chosen, reserve another segment of length $1/20$ to its right (again, no point can be placed in this reserved segment), and continue this way.

For all 10 points to be at least $1/20$ apart, there must be 9 gaps of length $1/20$ between consecutive points. So the total length that must be reserved is $9 \times \frac{1}{20} = \frac{9}{20}$.

That leaves an unreserved length of $1 - \frac{9}{20} = \frac{11}{20}$.

So the 10 points must fit into an effective interval of length $\frac{11}{20}$ instead of length 1. Since each point contributes one factor of $\frac{11}{20}$, the probability is $\left(\frac{11}{20}\right)^{10}$.

Therefore, the required probability is $\left(\frac{11}{20}\right)^{10} \approx 0.00253$.

Answer: 0.00253

784. Strictly Behind

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let D_t = frog position – rabbit position after t minutes. Then $D_0 = 4$, and each minute D_t changes by $+2$, 0 , or -2 . The value stays positive exactly when the rabbit is strictly behind the frog.

We want two things to happen by time 10: both the rabbit and the frog return to where they started, and D_t never drops to 0 or below. Since both end where they started, $D_{10} = 4$. For this to happen, the number of $+2$ changes in D_t must equal the number of -2 changes. Let that number be k .

So for a fixed k , there are $2k$ minutes when the relative distance actually changes, and $10 - 2k$ minutes when both move in the same direction. The same-direction minutes do not affect D_t , so we handle them separately.

First choose which $2k$ of the 10 minutes are the changing minutes: $\binom{10}{2k}$ ways.

Among the remaining $10 - 2k$ minutes, the rabbit and frog must move the same way. To make both return to their starting positions, we need exactly $5 - k$ minutes of rabbit-right/frog-right and $5 - k$ minutes of rabbit-left/frog-left. That gives $\binom{10-2k}{5-k}$ ways.

Now look only at the $2k$ changing minutes. After dividing the distance by 2, this becomes a walk that starts at 2, ends at 2, uses k up-steps and k down-steps, and must stay above 0 the whole time. We can use reflection principle for this. Count all walks from 2 to 2 with k up-steps and k down-steps: $\binom{2k}{k}$. Now subtract the “bad” walks that ever hit 0. For such a bad walk, reflect the part of the path up to its first visit to 0. That turns it into an unrestricted walk from -2 to 2 with k up-steps and k down-steps, and the number of those is $\binom{2k}{k+2}$. So the number of good changing-step patterns is $\binom{2k}{k} - \binom{2k}{k+2}$.

Therefore the total number of favorable 10-minute paths is

$$\sum_{k=0}^5 \binom{10}{2k} \binom{10-2k}{5-k} \left[\binom{2k}{k} - \binom{2k}{k+2} \right].$$

Evaluating this sum gives 49104 favorable paths. Since each minute has 4 equally likely joint moves, the total number of outcomes is $4^{10} = 1048576$.

So the required probability is $\frac{49104}{1048576} = \frac{3069}{65536}$.

Answer: 0.04683

785. Spaced Uniforms II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

From “Spaced Uniforms I”, if M is the minimum distance between any two of the 10 points, then for $0 \leq t \leq \frac{1}{9}$,

$$\Pr(M > t) = (1 - 9t)^{10}.$$

Also, for $t > \frac{1}{9}$, this probability is 0.

Now use the tail sum formula for a nonnegative random variable:

$$\mathbb{E}[M] = \int_0^\infty \Pr(M > t) dt.$$

$$\implies \mathbb{E}[M] = \int_0^{1/9} (1 - 9t)^{10} dt.$$

Let $u = 1 - 9t$. Then $du = -9dt$, and the integral becomes

$$\mathbb{E}[M] = \frac{1}{9} \int_0^1 u^{10} du = \frac{1}{9} \cdot \frac{1}{11} = \frac{1}{99}.$$

So the expected minimum distance is $\frac{1}{99}$.

Answer: 0.01010

786. Second Last Roll

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the completed sequence of rolls be $x_1, x_2, \dots, x_n$, where x_n is the final roll that first makes the sum exceed 100. We want the expected value of x_{n-1} , the second last roll.

The key idea is to compare the first roll and the second last roll. We claim they have the same probability distribution. To see why, fix any valid stopped sequence and look at the value in the first position and the value in the second last position. Now swap those two entries. This gives a natural one-to-one correspondence

between the set of valid sequences whose first roll is k and the set of valid sequences whose second last roll is k , for each $k \in 1, 2, 3, 4, 5, 6$. The swap is reversible, so this is a bijection. In other words, for every value k , the number of stopped sequences beginning with k is exactly the same as the number of stopped sequences whose second last roll is k .

Therefore, the first roll and the second last roll have the same distribution. But the first roll of a fair die is uniform on $1, 2, 3, 4, 5, 6$, so its expected value is $\frac{1+2+3+4+5+6}{6} = 3.5$.

Since the second last roll has the same distribution as the first roll, it has the same expected value. So the expected value of the second last roll is 3.5.

Note: If you simply assumed the answer is 3.5 without a proper proof, that does not count. For example, the expected value of the final roll is not 3.5 - it is actually higher, since exceeding 100 is more likely with larger rolls. The point is that you must rigorously justify the claim that the expected value of the second-to-last roll is 3.5.

Answer: 3.5

787. All Suits

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let T be the number of cards needed. If we look at the cards one at a time, then $T = \sum_x I_x$, where $I_x = 1$ if the card x is revealed before all four suits have appeared, and $I_x = 0$ otherwise. By symmetry, every card has the same probability of having this property, so using linearity of expectation

$$\mathbb{E}[T] = 52p,$$

where p is the probability that a fixed card is revealed before all four suits have been seen.

Now fix one particular card, and suppose it is a heart. Let A_S be the event that no spade has appeared before this card, and define A_H, A_C, A_D, A_S similarly for the four suits. Then the chosen card is revealed before all four suits have been seen exactly when at least one of these events occurs. So, $p = \Pr(A_H \cup A_C \cup A_D \cup A_S)$.

We now compute the probabilities of intersections.

If the chosen card is a heart, then

$$\Pr(A_H) = \frac{1}{13}, \quad \Pr(A_C) = \Pr(A_D) = \Pr(A_S) = \frac{1}{14}.$$

Indeed, for A_H , the chosen card must be the first heart among the 13 hearts. For any other suit, the chosen card must come before all 13 cards of that suit, so it is first among $13 + 1 = 14$ cards.

For two-suit intersections, we get

$$\Pr(A_H \cap A_C) = \Pr(A_H \cap A_D) = \Pr(A_H \cap A_S) = \frac{1}{26},$$

since the chosen card must be first among the 13 hearts and 13 cards of the other suit. Also,

$$\Pr(A_C \cap A_D) = \Pr(A_C \cap A_S) = \Pr(A_D \cap A_S) = \frac{1}{27},$$

since the chosen card must be first among the 13 cards of each of those two suits, together with itself.

For three-suit intersections,

$$\Pr(A_H \cap A_C \cap A_D) = \Pr(A_H \cap A_C \cap A_S) = \Pr(A_H \cap A_D \cap A_S) = \frac{1}{39},$$

and

$$\Pr(A_C \cap A_D \cap A_S) = \frac{1}{40}.$$

Finally,

$$\Pr(A_H \cap A_C \cap A_D \cap A_S) = \frac{1}{52}.$$

So inclusion-exclusion gives

$$p = \frac{1}{13} + \frac{3}{14} - \frac{3}{26} - \frac{3}{27} + \frac{3}{39} + \frac{1}{40} - \frac{1}{52}.$$

This simplifies to $p = \frac{4829}{32760}$.

Therefore, $\mathbb{E}[T] = 52p = 52 \cdot \frac{4829}{32760} = \frac{4829}{630}$.

So the expected number of cards needed is $\frac{4829}{630} \approx 7.66508$.

Answer: 7.66508

788. Subset Divisibility

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Instead of working with the set $1, 2, 3, \dots, 29$, we consider the set $0, 1, 2, \dots, 28$. This does not change the problem, since we only care about divisibility by 29, and replacing each element with its residue modulo 29 preserves whether a sum is congruent to 0 (mod 29).

Now take any 6-element subset S of $\{0, 1, 2, \dots, 28\}$. Let its sum be T (mod 29). For each $k = 0, 1, 2, \dots, 28$, define $S[k]$ to be the subset obtained by adding k to every element of S modulo 29.

As k increases by 1, every element increases by 1 modulo 29, so the total sum increases by 6 modulo 29. Therefore the sum of $S[k]$ is $T + 6k$ (mod 29).

Now, since 6 and 29 are relatively prime, the residues $6k$ (mod 29) for $k = 0, 1, \dots, 28$ are all distinct. So the 29 subsets $S[0], S[1], \dots, S[28]$ have 29 different sums modulo 29. That means they represent all 29 possible residues exactly once.

In particular, exactly one of these 29 subsets has sum congruent to 0 (mod 29). This argument applies to every group of 29 subsets formed by cyclic shifting. Therefore, one out of every 29 subsets has sum divisible by 29. Hence the required probability is $\frac{1}{29}$.

Answer: 0.03448

789. Hit 995

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For a simple symmetric random walk on the integers, if the walk starts at x with $-a < x < b$, then the probability of hitting b before $-a$ is

$$\frac{x + a}{a + b}.$$

Now let N be the number of times the rabbit lands on 995 before it first lands on -5 . We will compute $\mathbb{E}[N]$.

First, starting from 0, the probability of hitting 995 before -5 is

$$\frac{0 + 5}{995 + 5} = \frac{5}{1000} = \frac{1}{200}.$$

$$\implies \mathbb{P}(N = 0) = 1 - \frac{1}{200} = \frac{199}{200}.$$

Next, once the rabbit has reached 995, we ask for the probability that it returns to 995 again before hitting -5 . From 995, the first step is either to 996 or to 994, each with probability $\frac{1}{2}$.

If it steps to 996, then it must hit 995 before it can ever reach -5 , so this contributes probability 1.

If it steps to 994, then by the theorem above, the probability of hitting 995 before -5 is

$$\frac{994 + 5}{1000} = \frac{999}{1000}.$$

Therefore the probability of returning to 995 once it has been hit is

$$\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{999}{1000} = \frac{1999}{2000}.$$

So after each visit to 995, the chance of getting another visit before hitting -5 is $\frac{1999}{2000}$. This means that, conditioned on $N > 0$, the number of visits to 995 has a geometric distribution with success parameter $1 - \frac{1999}{2000} = \frac{1}{2000}$.

Hence,

$$\mathbb{E}[N \mid N > 0] = 2000.$$

Finally,

$$\mathbb{E}[N] = \mathbb{E}[N \mid N = 0]\mathbb{P}(N = 0) + \mathbb{E}[N \mid N > 0]\mathbb{P}(N > 0).$$

Since $\mathbb{E}[N \mid N = 0] = 0$ and $\mathbb{P}(N > 0) = \frac{1}{200}$, we get

$$\mathbb{E}[N] = 2000 \cdot \frac{1}{200} = 10.$$

So the expected number of times the rabbit lands on 995 before landing on -5 is 10.

Answer: 10

790. Penguin Walk III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

From Penguin Walk II, the probability that the penguin ever reaches 1 starting from 0 is $\frac{1}{2}$ when $p = \frac{1}{3}$. By strong markov property, the probability of ever reaching $i + 1$ starting from i is the same for every integer i , so each step from 0 to 1, 1 to 2, 2 to 3, and so on up to 6 to 7 has probability $\frac{1}{2}$.

Therefore, the probability that the penguin reaches 7 starting from 0 is $\left(\frac{1}{2}\right)^7 = \frac{1}{128}$.

So the answer is $\frac{1}{128}$.

Answer: 0.00781

791. Penguin Walk IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We first solve the more general version of the problem. Suppose $p < \frac{1}{2}$, the penguin is currently at position i , and we are told that it will eventually reach position n , where $i < n$. We want the probability that the next move is to position $i + 1$.

By conditioning on the first step, we look at the event that the penguin moves right immediately and still eventually reaches n . The probability of eventually reaching n starting from i is the denominator of the conditional probability. Since $p < \frac{1}{2}$, the probability of ever reaching n from i is

$$\frac{p^{n-i}}{(1-p)^{n-i}}.$$

Now for the numerator. The penguin first moves from i to $i + 1$, which happens with probability p . After

that, it still needs to reach n from $i + 1$. By the same formula, this happens with probability

$$\frac{p^{n-(i+1)}}{(1-p)^{n-(i+1)}}.$$

So the numerator is

$$p \cdot \frac{p^{n-(i+1)}}{(1-p)^{n-(i+1)}} = \frac{p^{n-i}}{(1-p)^{n-(i+1)}}.$$

Dividing this by the denominator gives

$$\frac{\frac{p^{n-i}}{(1-p)^{n-(i+1)}}}{\frac{p^{n-i}}{(1-p)^{n-i}}} = 1 - p.$$

So the probability that the penguin moves to the right in the next step, given that it will eventually reach n , is simply $1 - p$.

Here $p = \frac{1}{3}$, so the required probability is $1 - \frac{1}{3} = \frac{2}{3}$.

Answer: 0.66667

792. Uniform Product

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

A clean way to handle a product of independent Uniform(0, 1) variables is to take logarithms.

Let $S = -\ln Y$. Since $Y = X_1 X_2 X_3 X_4 X_5 X_6 X_7$, we have

$$S = -\ln X_1 - \ln X_2 - \ln X_3 - \ln X_4 - \ln X_5 - \ln X_6 - \ln X_7.$$

Now for each $X_i \sim \text{Unif}(0, 1)$, the random variable $-\ln X_i$ has the Exp(1) distribution. Because the X_i are independent, the seven terms are independent Exp(1) random variables, so S has a Gamma distribution with shape 7 and rate 1. Its probability density function is $f_S(s) = \frac{s^6 e^{-s}}{6!}$ for $s > 0$.

Since $Y = e^{-S}$, we have $S = -\ln Y$ and

$$\frac{ds}{dy} = -\frac{1}{y}.$$

So for $0 < y < 1$,

$$\begin{aligned} f_Y(y) &= f_S(-\ln y) \cdot \frac{1}{y} \\ &= \frac{(-\ln y)^6 e^{\ln y}}{6!} \cdot \frac{1}{y} \\ &= \frac{(-\ln y)^6 y}{6!} \cdot \frac{1}{y} \\ &= \frac{(-\ln y)^6}{6!}. \end{aligned}$$

Therefore,

$$f_Y\left(\frac{1}{2}\right) = \frac{(\ln 2)^6}{6!} = \frac{(\ln 2)^6}{720} \approx 0.00015$$

Answer: 0.00015

793. Last Uniform

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

For $0 \leq x \leq 1$, let $f(x)$ be the expected value of the last draw if we use the stopping rule “keep drawing until the sum exceeds x ” instead of 1. We want $f(1)$.

First note that $f(0) = \frac{1}{2}$, because when the target is 0, the very first draw is already the last draw, and a uniform $(0, 1)$ variable has mean $\frac{1}{2}$.

Now condition on the first draw $Y_1 = t$. There are two cases.

If $t \leq x$, then the process continues, and the remaining draws must now exceed $x - t$. Since the draws are independent and identically distributed, the expected final draw in that case is $f(x - t)$.

If $t > x$, then the process stops immediately, so the last draw is just t .

Averaging over all possible values of t , we get

$$f(x) = \int_0^x f(x-t)dt + \int_x^1 tdt.$$

Now change variables in the first integral by setting $u = x - t$. This gives

$$\int_0^x f(x-t)dt = \int_0^x f(u)du.$$

Also,

$$\int_x^1 tdt = \frac{1-x^2}{2}.$$

So

$$f(x) = \int_0^x f(u)du + \frac{1-x^2}{2}.$$

Differentiate both sides with respect to x :

$$f'(x) = f(x) - x.$$

This is a first-order linear ODE. Solving it with the condition $f(0) = \frac{1}{2}$ gives

$$f(x) = x + 1 - \frac{e^x}{2}.$$

Therefore,

$$f(1) = 2 - \frac{e}{2}.$$

So the expected value of the last draw is $\boxed{2 - \frac{e}{2}}$.

Answer: $\boxed{0.64086}$

794. Consecutive Stocks

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

We can always find a consecutive block of days during which the trader bought exactly 129 shares.

Let $S_0 = 0$, and for each $k = 1, 2, \dots, 365$, let S_k be the total number of shares bought in the first k days. Then

$$0 = S_0 < S_1 < \dots < S_{365} = 600,$$

because the trader buys at least one share each day.

Now consider the two sets of numbers

$$S_0, S_1, \dots, S_{365}$$

and

$$S_0 + 129, S_1 + 129, \dots, S_{365} + 129.$$

Altogether, this gives $366 + 366 = 732$ integers. Every one of them lies between 0 and 729, so there are only 730 possible values.

By the pigeonhole principle, two of these 732 numbers must be equal. They cannot both come from the first set, since $S_0 < S_1 < \dots < S_{365}$. They cannot both come from the shifted set either, since adding 129 preserves strict increase. So the equality must be between one number from the first set and one from the shifted set. That means for some i and j ,

$$S_j = S_i + 129.$$

Since the prefix sums are increasing, we have $i < j$. Therefore, the total number of shares bought from day $i + 1$ through day j is

$$S_j - S_i = 129.$$

So there must be a consecutive block of days with exactly 129 shares.

Answer: 1

795. Penguin Walk V

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

From "Penguin Walk I", we know that when $p < \frac{1}{2}$, the penguin does not reach position 1 with probability 1. In fact, the probability that it ever reaches position 1 is strictly less than 1.

That means there is a positive probability that the penguin never reaches position 1 at all. On that event, the waiting time to reach 1 is infinite. Since an infinite waiting time happens with positive probability, the expected number of steps is infinite.

Therefore, the required answer is -1 .

Answer: -1

796. Penguin Walk VI

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let T be the number of steps needed for the penguin to first reach position 1 starting from 0. We find $\mathbb{E}[T]$ by conditioning on the first step.

If the penguin moves right on the first step, which happens with probability p , then it reaches position 1 immediately, so $T = 1$.

If it moves left on the first step, which happens with probability $1 - p$, then it goes from 0 to -1 . From there, by the Markov property, the expected time to reach 1 is the same as going from -1 to 0 and then from 0 to 1, giving an expected total of $1 + 2\mathbb{E}[T]$ in this case.

So,

$$\mathbb{E}[T] = p \cdot 1 + (1 - p) \cdot (1 + 2\mathbb{E}[T])$$

Thus,

$$\mathbb{E}[T] = \frac{1}{2p - 1}$$

For $p = \frac{2}{3}$, this gives

$$\mathbb{E}[T] = \frac{1}{2 \cdot \frac{2}{3} - 1} = 3$$

So the answer is 3.

Answer: 3

797. Penguin Walk VII[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let T_1 be the number of steps needed for the penguin to move from position 0 to position 1. We condition on the first step and use the law of total expectation.

If the penguin moves right on the first step, which happens with probability p , then it reaches position 1 immediately, so $T_1 = 1$.

Otherwise, with probability $1 - p$, it moves left to position -1 . From there, by the Markov property, the expected time to go from -1 to 0 is the same as the expected time to go from 0 to 1, namely $\mathbb{E}[T_1]$. After that, it still needs another $\mathbb{E}[T_1]$ steps on average to reach 1. So in this case, the expected total time is $1 + 2\mathbb{E}[T_1]$.

Therefore,

$$\mathbb{E}[T_1] = p \cdot 1 + (1 - p) \cdot (1 + 2\mathbb{E}[T_1])$$

Solving this gives

$$\mathbb{E}[T_1] = \frac{1}{2p - 1}$$

Now set $p = \frac{1}{2}$. The denominator becomes 0, so the expected number of steps is infinite. Hence the answer is -1 .

Answer: -1

798. Penguin Walk VIII[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let T be the total number of steps needed for the penguin to go from position 0 to position 7. Write this as a sum of the time needed to move across each unit interval:

$$T = T_1 + T_2 + \cdots + T_7,$$

where T_i is the number of steps needed to move from position $i - 1$ to position i .

By the Markov property, each of these crossing times has the same expected value, so

$$\mathbb{E}[T] = 7\mathbb{E}[T_1]$$

From Penguin Walk VI, we know that when $p = \frac{2}{3}$, the expected number of steps to move from one position to the next is $\mathbb{E}[T_1] = 3$.

Therefore,

$$\mathbb{E}[T] = 7 \cdot 3 = 21$$

So the expected number of steps is 21.

Answer: 21

799. Row Expectation I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let X be the minimum entry in the second row.

Since the grid is filled by a random permutation of $1, 2, \dots, 36$, the twelve numbers in the second row are just a uniformly random set of 12 distinct numbers chosen from $1, 2, \dots, 36$.

Now use the tail sum formula:

$$\mathbb{E}[X] = \sum_{k=1}^{36} \Pr(X \geq k).$$

To have $X \geq k$, every number in the second row must come from the set $k, k+1, \dots, 36$, which has $37-k$ numbers. So

$$\Pr(X \geq k) = \frac{\binom{37-k}{12}}{\binom{36}{12}}.$$

Thus,

$$\mathbb{E}[X] = \frac{1}{\binom{36}{12}} \sum_{k=1}^{36} \binom{37-k}{12}.$$

Let $j = 37 - k$. Then

$$\mathbb{E}[X] = \frac{1}{\binom{36}{12}} \sum_{j=1}^{36} \binom{j}{12}.$$

By the hockey stick identity,

$$\sum_{j=1}^{36} \binom{j}{12} = \binom{37}{13}.$$

So

$$\mathbb{E}[X] = \frac{\binom{37}{13}}{\binom{36}{12}} = \frac{37}{13}.$$

So the expected smallest number in the second row is $\frac{37}{13}$.

Answer: 2.84615

800. Gojo's Six Eyes

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

There are $\binom{6}{2} = 15$ total segments. Label them 1 to 15, and let X_i be the indicator that the i -th segment is intersected by the plane. Then the total number of intersected segments is

$$T = X_1 + X_2 + \dots + X_{15},$$

so

$$\mathbb{E}[T] = \sum_{i=1}^{15} \mathbb{E}[X_i] = \sum_{i=1}^{15} \Pr(\text{segment } i \text{ is intersected})$$

Now consider any two points with position vectors v_1 and v_2 . The segment between them is intersected by the plane if the plane separates the two points, which depends only on the angle between v_1 and v_2 .

Let θ be the angle between v_1 and v_2 . Since both are unit vectors,

$$\cos \theta = v_1 \cdot v_2,$$

so

$$\theta = \arccos(v_1 \cdot v_2)$$

Because the normal direction is chosen uniformly, the probability that the plane separates the two points is

$$\frac{\theta}{\pi} = \frac{\arccos(v_1 \cdot v_2)}{\pi}$$

Among the 15 pairs, there are 3 opposite pairs: $(1, 0, 0)$ with $(-1, 0, 0)$, $(0, 1, 0)$ with $(0, -1, 0)$, $(0, 0, 1)$ with $(0, 0, -1)$. For these, $v_1 \cdot v_2 = -1$, so $\arccos(-1) = \pi$, and each contributes probability 1.

The remaining 12 pairs are orthogonal, so $v_1 \cdot v_2 = 0$, giving $\arccos(0) = \frac{\pi}{2}$, and each contributes probability $\frac{1}{2}$.

Therefore,

$$\mathbb{E}[T] = 12 \cdot \frac{1}{2} + 3 \cdot 1 = 6 + 3 = 9$$

So the expected number of intersected segments is 9.

Answer: 9

801. XY To The Power Z

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First set $U = XY$. To find the density of U , note that for $0 < u < 1$,

$$F_U(u) = \Pr(XY \leq u)$$

Now we condition $X = x$. Since $Y \sim \text{Unif}(0, 1)$,

$$\Pr(XY \leq u \mid X = x) = \Pr\left(Y \leq \frac{u}{x}\right)$$

If $x \leq u$, then $\frac{u}{x} \geq 1$, so this conditional probability is 1. If $x > u$, then it equals $\frac{u}{x}$. Therefore,

$$F_U(u) = \int_0^u 1 dx + \int_u^1 \frac{u}{x} dx = u + u \int_u^1 \frac{1}{x} dx = u - u \ln u$$

Differentiating gives

$$f_U(u) = \frac{d}{du}(u - u \ln u) = -\ln u, \quad 0 < u < 1.$$

Now condition on $U = u$. Then $W = u^Z$.

For a fixed $u \in (0, 1)$, the values of W lie in $(u, 1)$. For $u \leq w \leq 1$,

$$\Pr(W \leq w \mid U = u) = \Pr(u^Z \leq w)$$

Since $0 < u < 1$, taking logarithms and dividing by $\ln u$ reverses the inequality:

$$Z \geq \frac{\ln w}{\ln u}$$

Because $Z \sim \text{Unif}(0, 1)$,

$$\Pr(W \leq w \mid U = u) = 1 - \frac{\ln w}{\ln u}$$

Partially differentiating this with respect to w gives the conditional density,

$$f_{W|U=u}(w) = \frac{1}{w(-\ln u)}, \quad u < w < 1.$$

For $0 < w < 1$,

$$f_W(w) = \int_0^w f_{W|U=u}(w) f_U(u) du = \int_0^w \frac{1}{w(-\ln u)} (-\ln u) du = \int_0^w \frac{1}{w} du = 1$$

Thus W is uniformly distributed on $(0, 1)$ and hence,

$$f_W\left(\frac{1}{2}\right) = 1$$

Answer: 1

802. Penguin Walk IX[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let

$$m = \mathbb{E}[T], \quad s = \mathbb{E}[T^2]$$

From "penguin walk I", for $p > \frac{1}{2}$,

$$m = \frac{1}{2p-1}$$

So here,

$$m = \frac{1}{2 \cdot \frac{2}{3} - 1} = 3$$

Now condition on the first step.

If the penguin steps right, then $T = 1$, so $T^2 = 1$.

If the penguin steps left, then one step is used immediately, and the penguin must first return from -1 to 0 , and then go from 0 to 1 again. Let T_0 be the time to go from -1 to 0 , and let T_1 be an independent copy of T . Then in this case,

$$T = 1 + T_0 + T_1$$

By the Markov property,

$$\mathbb{E}[T_0] = \mathbb{E}[T_1] = m, \quad \mathbb{E}[T_0^2] = \mathbb{E}[T_1^2] = s$$

So

$$s = p \cdot 1 + (1-p)\mathbb{E}[(1 + T_0 + T_1)^2]$$

Expanding the square and using the facts above gives

$$s = p + (1-p)(1 + 4m + 2m^2 + 2s)$$

Now substitute $m = \frac{1}{2p-1}$ and solve for s . This yields

$$s = \frac{-4p^2 + 6p - 1}{(2p-1)^3}$$

Therefore,

$$\text{Var}(T) = s - m^2 = \frac{-4p^2 + 6p - 1}{(2p-1)^3} - \frac{1}{(2p-1)^2} = \frac{4p(1-p)}{(2p-1)^3}$$

Finally, with $p = \frac{2}{3}$,

$$\text{Var}(T) = \frac{4 \cdot \frac{2}{3} \cdot \frac{1}{3}}{(\frac{1}{3})^3} = 24$$

So the answer is 24.

Answer: 24**803. Penguin Walk X**[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

From penguin walk IX, we know that the variance of the number of steps needed to go from one integer to the next one to the right is

$$\text{Var}(T_1) = 24$$

Let T be the total number of steps needed for the penguin to reach 7 starting from 0. Then

$$T = T_1 + T_2 + \cdots + T_7,$$

where T_i is the number of steps needed for the penguin to go from position $i-1$ to position i .

These random variables are independent, because once the penguin reaches a new position, what happens next depends only on the current position and not on the past. So, we can split the sum

$$\text{Var}(T) = \text{Var}(T_1) + \text{Var}(T_2) + \cdots + \text{Var}(T_7) = 7\text{Var}(T_1)$$

Substituting $\text{Var}(T_1) = 24$,

$$\text{Var}(T) = 7 \cdot 24 = 168$$

So the answer is 168.

Answer: 168

804. Ten-Dimensional Ball

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

By symmetry, the distribution of each coordinate is centered at 0, so $\mathbb{E}[X_1] = 0$. Hence

$$\text{Var}(X_1) = \mathbb{E}[X_1^2].$$

Let $R = |X|$ be the distance of the random point from the origin. Since the point is uniformly distributed in a 10-ball, the probability that $R \leq r$ is proportional to the volume of a 10-ball of radius r . Therefore,

$$\Pr(R \leq r) = \left(\frac{r}{12}\right)^{10}, \quad 0 \leq r \leq 12,$$

so the density of R is

$$f_R(r) = \frac{10r^9}{12^{10}}, \quad 0 \leq r \leq 12.$$

Now we compute $\mathbb{E}[R^2]$:

$$\mathbb{E}[R^2] = \int_0^{12} r^2 \cdot \frac{10r^9}{12^{10}} dr = \frac{10}{12^{10}} \int_0^{12} r^{11} dr = \frac{10}{12^{10}} \cdot \frac{12^{12}}{12} = 120.$$

Also,

$$R^2 = X_1^2 + X_2^2 + \cdots + X_{10}^2.$$

Taking expectations and using symmetry, all ten coordinates have the same second moment, so

$$\mathbb{E}[R^2] = 10\mathbb{E}[X_1^2].$$

Thus,

$$\mathbb{E}[X_1^2] = \frac{120}{10} = 12.$$

Therefore,

$$\text{Var}(X_1) = 12.$$

Answer: 12

805. Row Expectation II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Think of the 6 numbers in the first row as 6 chosen positions among the numbers $1, 2, \dots, 24$. Equivalently, place 6 bars among 24 boxes, and let the remaining 18 boxes be stars. The third largest number in the first row is the same as the location of the fourth bar from the left, since there are 3 chosen numbers larger than it.

Now look at the 18 stars split by the 6 bars. There are 7 blocks of stars: one before the first bar, four

between consecutive bars, and one after the last bar. Since all 18 stars are distributed symmetrically among these 7 blocks, the expected number of stars in each block is $\frac{18}{7}$.

To reach the fourth bar, we pass through the first four star blocks and also cross 4 bars. So the expected location of the fourth bar is

$$4 \left(\frac{18}{7} + 1 \right)$$

Here, the +1 accounts for the bar itself.

Thus the expected value of the third largest number in the first row is

$$4 \left(\frac{18}{7} + 1 \right) = \frac{100}{7}$$

So the answer is $\frac{100}{7}$.

Answer: 14.28571

806. Conditional Normal IV

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let

$$R^2 = N_1^2 + N_2^2.$$

Because (N_1, N_2) is a standard bivariate normal vector, its distribution is rotationally symmetric in the plane. Conditioning on $R^2 = 6$ means we are looking at points on the circle of radius $\sqrt{6}$. By rotational symmetry, the direction on this circle is uniform, so we may write

$$(N_1, N_2) \mid (R^2 = 6) = (\sqrt{6} \cos \theta, \sqrt{6} \sin \theta),$$

where $\theta \sim \text{Unif}(0, 2\pi)$.

Then

$$N_1^4 \mid (R^2 = 6) = (\sqrt{6} \cos \theta)^4 = 36 \cos^4 \theta.$$

So

$$\mathbb{E}[N_1^4 \mid R^2 = 6] = 36 \mathbb{E}[\cos^4 \theta].$$

Now use the identity

$$\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}.$$

Squaring both sides gives

$$\cos^4 \theta = \left(\frac{1 + \cos(2\theta)}{2} \right)^2 = \frac{1}{4} (1 + 2 \cos(2\theta) + \cos^2(2\theta)).$$

Apply the same identity to $\cos^2(2\theta)$:

$$\cos^2(2\theta) = \frac{1 + \cos(4\theta)}{2}.$$

Substituting,

$$\cos^4 \theta = \frac{1}{4} \left(1 + 2 \cos(2\theta) + \frac{1 + \cos(4\theta)}{2} \right) = \frac{3}{8} + \frac{1}{2} \cos(2\theta) + \frac{1}{8} \cos(4\theta).$$

Since θ is uniform on $(0, 2\pi)$,

$$\mathbb{E}[\cos(2\theta)] = 0, \quad \mathbb{E}[\cos(4\theta)] = 0.$$

Therefore,

$$\mathbb{E}[\cos^4 \theta] = \frac{3}{8}.$$

So

$$\mathbb{E}[N_1^4 \mid N_1^2 + N_2^2 = 6] = 36 \cdot \frac{3}{8} = \frac{27}{2}.$$

Hence, the answer is $\frac{27}{2}$.

Answer: 13.5

807. Infinite Sum Expectation

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write the sum using an indicator:

$$S = \sum_{i=1}^{\infty} \frac{X_i}{2^i} \mathbf{1}_{N \geq i}.$$

So, by linearity of expectation,

$$\mathbb{E}[S] = \sum_{i=1}^{\infty} \frac{1}{2^i} \mathbb{E}[X_i \mathbf{1}_{N \geq i}].$$

Now fix i . To have $N \geq i$, the first i terms must be strictly decreasing:

$$X_1 > X_2 > \dots > X_i.$$

Condition on $X_i = t$. Then the event $N \geq i$ means that all of $X_1, \dots, X_{i-1}$ must lie above t , and among those $i-1$ variables, the order must be decreasing. The probability that each of $X_1, \dots, X_{i-1}$ exceeds t is $(1-t)^{i-1}$, and the probability that they appear in decreasing order is $\frac{1}{(i-1)!}$. Hence

$$\mathbb{P}(N \geq i \mid X_i = t) = \frac{(1-t)^{i-1}}{(i-1)!}.$$

Therefore,

$$\mathbb{E}[X_i \mathbf{1}_{N \geq i}] = \mathbb{E} \left[\frac{X_i (1 - X_i)^{i-1}}{(i-1)!} \right].$$

Since $X_i \sim \text{Unif}(0, 1)$,

$$\mathbb{E}[X_i \mathbf{1}_{N \geq i}] = \frac{1}{(i-1)!} \int_0^1 t(1-t)^{i-1} dt.$$

Using the Beta integral,

$$\int_0^1 t(1-t)^{i-1} dt = \frac{1}{(i+1)!} (i-1)!,$$

so

$$\mathbb{E}[X_i \mathbf{1}_{N \geq i}] = \frac{1}{(i+1)!}.$$

Thus,

$$\mathbb{E}[S] = \sum_{i=1}^{\infty} \frac{1}{2^i (i+1)!}.$$

Now rewrite this series:

$$\mathbb{E}[S] = 2 \sum_{i=1}^{\infty} \frac{1}{2^{i+1} (i+1)!} = 2 \sum_{k=2}^{\infty} \frac{(1/2)^k}{k!}.$$

Using the exponential series,

$$\sum_{k=0}^{\infty} \frac{(1/2)^k}{k!} = e^{1/2} = \sqrt{e},$$

so

$$\sum_{k=2}^{\infty} \frac{(1/2)^k}{k!} = \sqrt{e} - 1 - \frac{1}{2}.$$

Therefore,

$$\mathbb{E}[S] = 2 \left(\sqrt{e} - 1 - \frac{1}{2} \right) = 2\sqrt{e} - 3.$$

Hence,

$$\mathbb{E}[S] = 2\sqrt{e} - 3.$$

Answer: 0.29744

808. Secret Marbles I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $A(a, b)$ be the expected payout to player A from one draw when A chooses a and B chooses b .

The probability that a drawn marble belongs to A is $\frac{a}{a+b}$, so the expected payout to A from one draw is $A(a, b) = \frac{a}{a+b}(100 - a)$. Since the two draws are done with replacement, the expected total payout is just twice this value. So we first maximize the one-draw payoff.

Now fix b and view $A(a, b)$ as a function of a . Treat a as continuous for the moment. Differentiating gives

$$\frac{\partial}{\partial a} A(a, b) = \frac{100b - a^2 - 2ab}{(a + b)^2}$$

Setting the numerator equal to 0 gives

$$a^2 + 2ab - 100b = 0$$

Solving this quadratic for a , the positive root is

$$a^* = \sqrt{b^2 + 100b} - b$$

By symmetry, the best response of B is

$$b^* = \sqrt{a^2 + 100a} - a$$

At a symmetric equilibrium we have $a = b = x$. Substituting into the best-response equation gives

$$\begin{aligned} x &= \sqrt{x^2 + 100x} - x \\ \implies 2x &= \sqrt{x^2 + 100x} \\ \implies x(3x - 100) &= 0 \end{aligned}$$

Since $x > 0$, we get $x = \frac{100}{3}$.

Because the number of marbles must be an integer, we check the nearby values 33 and 34. This gives the two relevant equilibria (33, 33) and (34, 34), and the better one for player A is (33, 33).

For (33, 33), the expected payout to A from one draw is

$$\frac{33}{33 + 33}(100 - 33) = \frac{33}{66} \cdot 67 = \frac{67}{2}$$

Since there are two draws, the expected total payout is $2 \cdot \frac{67}{2} = 67$.

So the expected total payout of player A is 67.

Answer: 67

809. Secret Marbles II[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $A(a, b)$ denote the expected payoff of player A when A chooses a marbles and B chooses b marbles.

The probability that the drawn marble comes from A is $\frac{a}{a+b}$, and the probability that it comes from B is $\frac{b}{a+b}$. So

$$A(a, b) = \frac{a}{a+b}(100 - a) - \frac{b}{a+b}(100 - b)$$

Now fix b and optimize over a . Treat a as continuous for the moment. Differentiating with respect to a gives

$$\frac{\partial}{\partial a} A(a, b) = \frac{200b - (a+b)^2}{(a+b)^2}$$

At the best response, this derivative is 0. Hence

$$(a+b)^2 = 200b$$

so the best response of A is

$$a^* = 10\sqrt{2b} - b$$

By symmetry, the best response of B is

$$b^* = 10\sqrt{2a} - a$$

At equilibrium, the game is symmetric, so $a = b = x$. Substituting gives

$$x = 10\sqrt{2x} - x,$$

which yields $x = 50$.

Therefore, player A should choose 50 marbles.

Answer: 50

810. Second Last I[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $f(x)$ be the expected final record value when the current largest value is x . Since the variables are uniform on $(0, 1)$, the initial largest value is effectively 0, so the quantity we want is $f(0)$.

From state x , the next sample is less than x with probability x . In that case, the process stops and the final largest value is still x . If the next sample is $y > x$, then y becomes the new largest value, and the expected final record value from that point is $f(y)$. Therefore,

$$f(x) = x \cdot x + \int_x^1 f(y) dy$$

Now differentiate both sides with respect to x . This gives

$$f'(x) = 2x - f(x),$$

or equivalently,

$$f'(x) + f(x) = 2x$$

We also need a boundary condition. If the current largest value is 1, then the process is already at the top of the interval, so the final record value is just $f(1) = 1$.

To solve the differential equation, use an integrating factor e^x . Multiplying through by e^x ,

$$e^x f'(x) + e^x f(x) = 2xe^x,$$

$$\implies (e^x f(x))' = 2xe^x$$

Integrating,

$$e^x f(x) = \int 2xe^x, dx = 2e^x(x-1) + C$$

Hence

$$f(x) = 2x - 2 + Ce^{-x}$$

Apply the condition $f(1) = 1$:

$$1 = 2 - 2 + Ce^{-1},$$

so $C = e$. Thus

$$f(x) = 2x - 2 + e^{1-x}$$

Finally,

$$\mathbb{E}[X_{N-1}] = f(0) = e - 2$$

So the answer is $\boxed{e - 2}$.

Answer: $\boxed{0.71828}$

811. Second Last II

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let $f(x)$ be the expected final record low when the current minimum is x . Since the variables are uniform on $(0, 1)$, we start with $x = 1$, so the quantity we want is $f(1)$.

Now suppose the current minimum is x . The next draw is larger than x with probability $1 - x$, and then the process stops immediately, leaving the final minimum equal to x . With the remaining probability, the next draw falls below x . In that case, conditional on being smaller than x , the new value is uniformly distributed on $(0, x)$, so the expected final minimum from that point is $f(y)$ averaged over $0 < y < x$. This gives

$$f(x) = x(1 - x) + \int_0^x f(y)dy$$

Differentiate both sides with respect to x :

$$f'(x) = 1 - 2x + f(x)$$

$$\implies f'(x) - f(x) = 1 - 2x$$

This is a first-order linear differential equation. Using integrating factor e^{-x} , we get

$$(e^{-x} f(x))' = (1 - 2x)e^{-x}$$

Integrating,

$$e^{-x} f(x) = (2x + 1)e^{-x} + C$$

$$\implies f(x) = 2x + 1 + Ce^x$$

Since the expected minimum is 0 when the current minimum is 0, we have $f(0) = 0$. Thus $0 = 1 + C$, which gives $C = -1$. Therefore

$$f(x) = 2x + 1 - e^x$$

Finally, $f(1) = 3 - e$.

So the expected final record low is $\boxed{3 - e}$.

Answer: $\boxed{0.28172}$

812. Powerful Scaling[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write $X_{n+1} = X_n^3 U$, where $U \sim \text{Unif}(0, 1)$. This is because, given X_n , the random variable X_{n+1} is uniform on $(0, X_n^3)$, so it can be obtained by taking a uniform $(0, 1)$ variable and scaling it by X_n^3 .

Now condition on X_n . Since X_n is known inside the conditional expectation, it acts like a constant. Therefore,

$$\mathbb{E}[X_{n+1}^p \mid X_n] = \mathbb{E}[(X_n^3 U)^p \mid X_n] = \mathbb{E}[X_n^{3p} U^p \mid X_n] = X_n^{3p} \mathbb{E}[U^p].$$

The last step is valid because X_n^{3p} is determined once we condition on X_n , so it can be pulled outside the conditional expectation.

Now compute $\mathbb{E}[U^p]$ for $U \sim \text{Unif}(0, 1)$:

$$\mathbb{E}[U^p] = \int_0^1 u^p du = \frac{1}{p+1}.$$

So,

$$\mathbb{E}[X_{n+1}^p \mid X_n] = \frac{X_n^{3p}}{p+1}.$$

We now work backward from X_5 .

$$\mathbb{E}[X_5] = \mathbb{E}[\mathbb{E}[X_5 \mid X_4]] = \frac{1}{2} \mathbb{E}[X_4^3].$$

Next,

$$\mathbb{E}[X_4^3] = \mathbb{E}[\mathbb{E}[X_4^3 \mid X_3]] = \frac{1}{4} \mathbb{E}[X_3^9].$$

Then,

$$\mathbb{E}[X_3^9] = \mathbb{E}[\mathbb{E}[X_3^9 \mid X_2]] = \frac{1}{10} \mathbb{E}[X_2^{27}],$$

and

$$\mathbb{E}[X_2^{27}] = \mathbb{E}[\mathbb{E}[X_2^{27} \mid X_1]] = \frac{1}{28} \mathbb{E}[X_1^{81}].$$

Since $X_1 \sim \text{Unif}(0, 1)$,

$$\mathbb{E}[X_1^{81}] = \frac{1}{82}.$$

So,

$$\mathbb{E}[X_5] = \frac{1}{2 \cdot 4 \cdot 10 \cdot 28 \cdot 82} = \frac{1}{183680}.$$

So the answer is

$$\mathbb{E}[X_5] = \frac{1}{183680}.$$

Answer: 0.00001

813. One Before Ten[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let X be the value of the first card, and let A be the event that 1 appears before 10.

We compute $\mathbb{E}[X \mid A]$ by finding the conditional probability that each card is first.

If the first card is 1, then the event A certainly happens, so

$$P(X = 1 \mid A) = \frac{P(X = 1)}{P(A)} = \frac{\frac{1}{10}}{\frac{1}{2}} = \frac{1}{5}$$

If the first card is 10, then A cannot happen, so

$$P(X = 10 \mid A) = 0$$

Now suppose the first card is one of $2, 3, \dots, 9$. Once that card is fixed in the first position, the remaining 9 cards are still randomly arranged. Among those remaining cards, 1 and 10 are equally likely to appear in either order, so

$$P(A \mid X = k) = \frac{1}{2} \quad \text{for } k = 2, 3, \dots, 9$$

Therefore,

$$P(X = k \mid A) = \frac{P(X = k \text{ and } A)}{P(A)} = \frac{\left(\frac{1}{10}\right)\left(\frac{1}{2}\right)}{\frac{1}{2}} = \frac{1}{10} \quad \text{for } k = 2, 3, \dots, 9$$

So the conditional distribution of X is:

$$P(X = 1 \mid A) = \frac{1}{5}, \quad P(X = k \mid A) = \frac{1}{10} \quad \text{for } k = 2, \dots, 9$$

So the expected value of the first card is:

$$\mathbb{E}[X \mid A] = 1 \cdot \frac{1}{5} + \sum_{k=2}^9 k \cdot \frac{1}{10} = \frac{23}{5}$$

Answer: 4.6

814. Frog Distance I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let m be the minimum position and M be the maximum position of the 10 frog locations. Then the quantity we want is

$$P\left(\frac{1}{4} \leq M - m \leq \frac{1}{2}\right)$$

We first derive the joint density of m and M . Suppose $m \in [u, u + du]$ and $M \in [v, v + dv]$, where $0 \leq u \leq v \leq 1$. For this to happen, one of the 10 frogs must fall in the tiny interval $[u, u + du]$, one of the remaining 9 frogs must fall in $[v, v + dv]$, and the other 8 frogs must all lie between u and v . There are 10 choices for the leftmost frog and 9 choices for the rightmost frog, so the number of ways is $10 \cdot 9$. The probability that the 8 middle frogs land in $[u, v]$ is $(v - u)^8$, and the tiny end intervals contribute a factor $du dv$. Hence

$$P(m \in [u, u + du], M \in [v, v + dv]) \approx 10 \cdot 9(v - u)^8 du dv$$

Therefore the joint density is

$$f_{m,M}(u, v) = 90(v - u)^8, \quad 0 \leq u \leq v \leq 1$$

Now we integrate this density over the region where the gap $v - u$ lies between $\frac{1}{4}$ and $\frac{1}{2}$:

$$\begin{aligned} P\left(\frac{1}{4} \leq M - m \leq \frac{1}{2}\right) &= \int_0^1 \int_{u+\frac{1}{4}}^{\min(u+\frac{1}{2}, 1)} 90(v - u)^8 dv du \\ \implies P\left(\frac{1}{4} \leq M - m \leq \frac{1}{2}\right) &= \int_0^{1/2} \int_{u+\frac{1}{4}}^{u+\frac{1}{2}} 90(v - u)^8 dv du + \int_{1/2}^1 \int_{u+\frac{1}{4}}^1 90(v - u)^8 dv du. \end{aligned}$$

Now we can evaluate each part directly in v :

$$\int 90(v - u)^8 dv = 10(v - u)^9.$$

So

$$\int_0^{1/2} \int_{u+\frac{1}{4}}^{u+\frac{1}{2}} 90(v-u)^8 dv du = \int_0^{1/2} 10 \left[\left(\frac{1}{2}\right)^9 - \left(\frac{1}{4}\right)^9 \right] du = \frac{2555}{262144},$$

and

$$\int_{1/2}^{3/4} \int_{u+\frac{1}{4}}^1 90(v-u)^8 dv du = \int_{1/2}^{3/4} 10 \left[(1-u)^9 - \left(\frac{1}{4}\right)^9 \right] du = \frac{1013}{1048576}.$$

Adding them,

$$P\left(\frac{1}{4} \leq M - m \leq \frac{1}{2}\right) = \frac{11233}{1048576}.$$

Answer: 0.01071

815. Frog Distance II

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let m be the minimum landing position and M be the maximum landing position. As derived in "Frog Distance I", the joint density is

$$f_{m,M}(u,v) = 90(v-u)^8, \quad 0 \leq u \leq v \leq 1$$

We want

$$P\left(\frac{1}{4} < M - m < \frac{1}{2} \mid M - m > \frac{1}{4}\right) = \frac{P\left(\frac{1}{4} < M - m < \frac{1}{2}\right)}{P\left(M - m > \frac{1}{4}\right)}$$

We evaluated the numerator in "Frog Distance I" to be $\frac{11233}{1048576}$.

Now, the denominator is given by

$$\begin{aligned} \int_0^{\frac{3}{4}} \int_{u+\frac{1}{4}}^1 90(v-u)^8 dv du &= \int_0^{\frac{3}{4}} [10(v-u)^9]_{u+\frac{1}{4}}^1 du = \int_0^{\frac{3}{4}} 10 \left((1-u)^9 - \left(\frac{1}{4}\right)^9 \right) du \\ &\implies P\left(M - m > \frac{1}{4}\right) = \frac{1048545}{1048576} \end{aligned}$$

Therefore,

$$P\left(\frac{1}{4} < M - m < \frac{1}{2} \mid M - m > \frac{1}{4}\right) = \frac{\frac{11233}{1048576}}{\frac{1048545}{1048576}} = \frac{11233}{1048545}$$

So the answer is $\frac{11233}{1048545}$.

Answer: 0.01071

816. Additional Info Coin Flips II

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let X_A, X_B , and X_C be the number of tosses needed by A, B , and C . We want

$$P(X_C > X_B \mid X_B > X_A) = \frac{P(X_A < X_B < X_C)}{P(X_A < X_B)}$$

So first we find $P(X_A < X_B)$. Let $p = P(X_A < X_B)$.

Look at the first tosses of A and B .

With probability $\frac{1}{4}$, both get tails. Then the situation starts over exactly as before, so this contributes $\frac{1}{4}p$.

With probability $\frac{1}{4}$, A gets tails and B gets heads. Then A definitely takes more tosses than B , so this contributes $\frac{1}{4}$.

The remaining two cases do not result in B taking more tosses than A . So

$$p = \frac{1}{4}p + \frac{1}{4}$$

Solving gives $p = \frac{1}{3}$.

Now let $q = P(X_A < X_B < X_C)$. Now look at the first tosses of all three people.

With probability $\frac{1}{8}$, all three get tails. Then the whole problem restarts, so this contributes $\frac{1}{8}q$.

With probability $\frac{1}{8}$, A gets heads while B and C both get tails. Then A is already finished in one toss, and the remaining condition becomes $X_B < X_C$ for the future tosses of B and C . That probability is exactly p , which we already found to be $\frac{1}{3}$.

All other first-toss outcomes cannot lead to $X_A < X_B < X_C$. Therefore

$$q = \frac{1}{8}q + \frac{1}{8}p$$

Substituting $p = \frac{1}{3}$,

$$\begin{aligned} q &= \frac{1}{8}q + \frac{1}{24} \\ \implies \frac{7}{8}q &= \frac{1}{24} \implies q = \frac{1}{21} \end{aligned}$$

Finally,

$$P(X_C > X_B \mid X_B > X_A) = \frac{q}{p} = \frac{\frac{1}{21}}{\frac{1}{3}} = \frac{1}{7}$$

So the required probability is $\frac{1}{7}$.

Answer: 0.14286

817. Power Distance II

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let

$$S_{30} = X_1 + X_2 + \cdots + X_{30},$$

where each X_i is $+1$ or -1 with probability $\frac{1}{2}$. We want $\mathbb{E}[S_{30}^4]$.

Now expand:

$$S_{30}^4 = (X_1 + X_2 + \cdots + X_{30})^4$$

When we take expectation, any term with at least one variable appearing to an odd power has expectation 0. This is because each X_i is symmetric about 0, so $\mathbb{E}[X_i] = 0$. Since the variables are independent, the expectation of a product splits into the product of expectations, so if one factor appears to an odd power, that factor contributes a mean of 0, and the whole term vanishes. Thus only terms in which every variable appears with an even power survive.

So we only need to keep track of two types of contributions.

First, there are the terms where all four factors come from the same variable. This gives $\sum_{i=1}^{30} X_i^4$, whose expectation is 30.

Second, there are the terms where two different variables each appear twice, such as $X_i^2 X_j^2$. For each pair $i < j$, the coefficient of $X_i^2 X_j^2$ in the full expansion is 6. So this contribution is

$$6 \sum_{1 \leq i < j \leq 30} X_i^2 X_j^2$$

Since $X_i^2 X_j^2 = 1$, its expectation is

$$6 \binom{30}{2} = 6 \cdot 435 = 2610$$

All other terms have at least one variable appearing with an odd power, so their expectation is 0.

Therefore,

$$\mathbb{E}[S_{30}^4] = 30 + 2610 = 2640$$

So the expected fourth power of the position after 30 seconds is 2640.

Answer: 2640

818. Normal Ratio I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Since X and Y are independent standard normal random variables, the point (X, Y) has a radially symmetric joint distribution in the plane. That means the direction of the point is uniform, so the probability of an event depends only on the angle it cuts out at the origin.

Write the point in polar form as

$$X = R \cos \Theta, \quad Y = R \sin \Theta,$$

where $R \geq 0$ and Θ is uniform on $[0, 2\pi]$.

Then

$$\frac{X}{Y} = \cot \Theta.$$

So we want

$$1 < \cot \Theta < \sqrt{3}.$$

On the interval $(0, \pi/2)$, the cotangent decreases from ∞ to 0, so this is equivalent to

$$\frac{\pi}{6} < \Theta < \frac{\pi}{4}.$$

The same ratio also occurs in the third quadrant, where angles differ by π , so we also get

$$\frac{7\pi}{6} < \Theta < \frac{5\pi}{4}.$$

Thus the total angular size of the event is

$$\left(\frac{\pi}{4} - \frac{\pi}{6}\right) + \left(\frac{5\pi}{4} - \frac{7\pi}{6}\right) = \frac{\pi}{6}.$$

Since Θ is uniform on $[0, 2\pi]$, the required probability is $\frac{\pi/6}{2\pi} = \frac{1}{12}$.

So the answer is $\frac{1}{12}$.

Answer: 0.08333

819. Normal Ratio II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Write $X = \sqrt{2}Z$, where $Z \sim N(0, 1)$. Since X and Y are independent, Z and Y are independent standard normal random variables.

Then

$$\sqrt{2} < \frac{X}{Y} < \sqrt{6}$$

$$\implies \sqrt{2} < \frac{\sqrt{2}Z}{Y} < \sqrt{6}.$$

$$\implies 1 < \frac{Z}{Y} < \sqrt{3}.$$

Now use the radial symmetry of the bivariate standard normal (Z, Y) . If we write

$$Z = R \cos \Theta, \quad Y = R \sin \Theta,$$

then Θ is uniform on $[0, 2\pi]$, and $\frac{Z}{Y} = \cot \Theta$.

So we want

$$1 < \cot \Theta < \sqrt{3}.$$

This corresponds to

$$\frac{\pi}{6} < \Theta < \frac{\pi}{4}$$

in the first quadrant, and the same angular interval in the third quadrant. The total favorable angle is therefore

$$2 \left(\frac{\pi}{4} - \frac{\pi}{6} \right) = \frac{\pi}{6}.$$

Since the full circle has angle 2π , the probability is

$$\frac{\pi/6}{2\pi} = \frac{1}{12}.$$

So the answer is $\frac{1}{12}$.

Answer: 0.08333

820. Exponential Ratio

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Since $X \sim \text{Exp}(1)$ and $Y \sim \text{Exp}(2)$ are independent, their joint density is

$$f_{X,Y}(x, y) = 2e^{-x-2y}, \quad x > 0, y > 0.$$

The event $1 < \frac{X}{Y} < 2$ is the same as $Y < X < 2Y$.

So we integrate the joint density over the region where $y > 0$ and x lies between y and $2y$:

$$\mathbb{P} \left(1 < \frac{X}{Y} < 2 \right) = \int_0^\infty \int_y^{2y} 2e^{-x-2y} dx dy.$$

First, we evaluate the inner integral:

$$\int_y^{2y} 2e^{-x-2y} dx = 2e^{-2y} \int_y^{2y} e^{-x} dx = 2e^{-2y} (e^{-y} - e^{-2y}).$$

So the required probability is

$$\int_0^\infty (2e^{-3y} - 2e^{-4y}) dy = 2 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{1}{6}.$$

Answer: 0.16667

821. Square Intersection II

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let the square be the unit square positioned in the first quadrant, with its bottom-left vertex at the origin and its sides parallel to the coordinate axes.

For the line to intersect the left and bottom sides, both points must lie in the lower-left triangle

$$T = (x, y) : x \geq 0, y \geq 0, x + y \leq 1,$$

whose area is $\frac{1}{2}$. So

$$\Pr(\text{both points lie in } T) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}.$$

Now condition on both points being in T . Then we have two random points inside a triangle. The line through them can cross one of the three pairs of sides of that triangle, and by affine symmetry these three possibilities are equally likely. So the probability that it crosses the two sides meeting at the origin, namely the left and bottom sides of the square, is $\frac{1}{3}$.

Therefore,

$$\Pr(\text{line intersects left and bottom sides}) = \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}.$$

Answer: 0.08333

822. Increasing Dice Rolls IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

We count all nondecreasing sequences of length 10, and then count how many of those contain at least one 3.

A nondecreasing sequence of die rolls is determined only by how many 1s, 2s, 3s, 4s, 5s, and 6s appear. So we are counting the number of solutions to

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 10$$

with $x_i \geq 0$. By stars and bars, the total number of such sequences is

$$\binom{10+6-1}{6-1} = \binom{15}{5} = \binom{15}{10} = 3003.$$

Now count the sequences with no 3 at all. Then the rolls can only be 1, 2, 4, 5, 6, so we count solutions to

$$x_1 + x_2 + x_4 + x_5 + x_6 = 10$$

with all variables nonnegative. Again by stars and bars, the number of such sequences is

$$\binom{10+5-1}{5-1} = \binom{14}{4} = \binom{14}{10} = 1001.$$

So the number of nondecreasing sequences that do contain at least one 3 is $3003 - 1001 = 2002$.

Therefore the required probability is $\frac{2002}{3003} = \frac{2}{3}$.

Answer: 0.66667

823. Hexagon Intersection

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First, look at the rectangle inside the hexagon for which sides A and D are opposite sides. A line can hit both A and D only if both chosen points lie inside this rectangle.

The area of this rectangle is $\frac{2}{3}$ of the area of the whole hexagon, so the probability that both random points lie in it is $(\frac{2}{3})^2 = \frac{4}{9}$.

Now condition on both points being inside that rectangle. Similar to "Square Intersection II", the probability that the line joining them crosses the two opposite sides A and D is $\frac{1}{3}$.

So the required probability is $\frac{4}{9} \cdot \frac{1}{3} = \frac{4}{27}$.

Answer: 0.14815

824. Sum Over One II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let X be the smallest number and Y be the second largest number.

First, we derive the joint density of (X, Y) . Consider the probability that $X = x$ and $Y = y$ with small intervals of width dx and dy around these values, where $0 < x < y < 1$. For this to happen, one of the ten numbers must fall in the interval $[x, x + dx]$, one must fall in $[y, y + dy]$, seven of the remaining numbers must lie between x and y , and the last number must lie above y .

So the probability is

$$\frac{dx}{1} \cdot \frac{dy}{1} \cdot \left(\frac{y-x}{1}\right)^7 \cdot \frac{1-y}{1} \cdot 10 \cdot 9 \cdot 8.$$

Here, 10 counts the choice of which number is the smallest, 9 counts the choice of which number is the second largest, and 8 counts the choice of which number is the largest. The factor $(y-x)^7$ comes from the seven numbers that must fall between x and y , and the factor $(1-y)$ comes from the one number that must lie above y .

Therefore, the joint density is

$$f_{X,Y}(x, y) = 10 \cdot 9 \cdot 8(y-x)^7(1-y) = 720(y-x)^7(1-y), \quad 0 < x < y < 1.$$

So, the required probability is given by the following double integral,

$$P(X + Y > 1) = \int_{1/2}^1 \int_{1-y}^y 720(y-x)^7(1-y) dx dy.$$

For the inner integral,

$$\int_{1-y}^y (y-x)^7 dx = \int_0^{2y-1} u^7 du = \frac{(2y-1)^8}{8}.$$

So

$$P(X + Y > 1) = \int_{1/2}^1 720(1-y) \cdot \frac{(2y-1)^8}{8} dy = \int_{1/2}^1 90(1-y)(2y-1)^8 dy.$$

Now let $t = 2y - 1$. Then $y = \frac{t+1}{2}$, $1-y = \frac{1-t}{2}$, and $dy = \frac{dt}{2}$. This gives

$$P(X + Y > 1) = \frac{45}{2} \int_0^1 (1-t)t^8 dt = \frac{45}{2} \left(\int_0^1 t^8 dt - \int_0^1 t^9 dt \right).$$

$$\implies P(X + Y > 1) = \frac{45}{2} \left(\frac{1}{9} - \frac{1}{10} \right) = \frac{45}{2} \cdot \frac{1}{90} = \frac{1}{4}.$$

Answer: 0.25

825. Last Card Survivor

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

When two cards are compared, each card has probability $\frac{1}{2}$ of being the one that survives. So the expected label of the surviving card from a pair is just the average of the two labels. In other words, each time a card passes through one such comparison, its contribution to the final expected value is multiplied by $\frac{1}{2}$.

Now use the way the deck is arranged. The first 36 steps compare the pairs $(1, 2), (3, 4), \dots, (71, 72)$. After these 36 steps, the top 72 cards have each taken part in one averaging step, and 64 cards remain in the deck. Since $64 = 2^6$, those remaining cards go through 6 more averaging steps before one card is left.

So cards 1 through 72 are weighted by $2^{-7} = \frac{1}{128}$, since they pass through 7 averaging steps in total. Cards 73 through 100 are weighted by $2^{-6} = \frac{1}{64}$, since they pass through only 6 averaging steps.

Therefore,

$$\mathbb{E}[X] = \frac{1}{128}(1 + 2 + \dots + 72) + \frac{1}{64}(73 + 74 + \dots + 100) = \frac{467}{8}$$

Answer: 58.375

826. Triangle Formation IV

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the three numbers be X_1, X_2, X_3 , and let $X_{(1)} \leq X_{(2)} \leq X_{(3)}$ be their ordered values. We want

$$\mathbb{E}[X_{(1)} \mid X_1, X_2, X_3 \text{ form a triangle}].$$

For three positive numbers in increasing order, the triangle condition is simply

$$X_{(3)} < X_{(1)} + X_{(2)}.$$

So we will integrate over the region

$$0 < x < y < z < 1, \quad z < x + y,$$

where $x = X_{(1)}, y = X_{(2)}$, and $z = X_{(3)}$. Since the joint density of the order statistics is 6 on the region $0 < x < y < z < 1$, we have

$$\mathbb{E}[X_{(1)} \mid T] = \frac{\iiint_R x \cdot 6dzdydx}{\iiint_R 6dzdydx},$$

where T is the triangle event and R is the region described above. First, the probability of the triangle event is

$$\iiint_R 6dzdydx = \frac{1}{2}.$$

Now let's evaluate the numerator. We split according to whether $x + y \leq 1$ or $x + y > 1$. When $0 < x < \frac{1}{2}$, we have two ranges for y :

$$x < y < 1 - x \quad \text{and} \quad 1 - x < y < 1.$$

So

$$\iiint_R x \cdot 6dzdydx = 6 \left[\int_0^{1/2} \int_x^{1-x} \int_y^{x+y} xdzdydx + \int_0^{1/2} \int_{1-x}^1 \int_y^1 xdzdydx + \int_{1/2}^1 \int_x^1 \int_y^1 xdzdydx \right].$$

Evaluating these integrals gives

$$\begin{aligned} \iiint_R x \cdot 6dzdydx &= \frac{3}{16}. \\ \implies \mathbb{E}[X_{(1)} \mid T] &= \frac{3/16}{1/2} = \frac{3}{8}. \end{aligned}$$

So the expected value of the minimum is $\frac{3}{8}$.

Answer: 0.375

827. Random Tree Construction

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Label the vertices by the time they are created: $1, 2, \dots, 10$, where vertex 1 is the root.

For $i < j$, let $A_{i,j}$ be the event that vertex i is an ancestor of vertex j . We claim that

$$\Pr(A_{i,j}) = \frac{1}{i}.$$

This is true because when vertex j is added, its parent is chosen uniformly from the $j - 1$ earlier vertices. So vertex i is an ancestor of j exactly when the parent of j is one of the vertices that already has i as an ancestor. Using induction on j (with $j = i + 1$ as the base case), this gives the constant probability $\frac{1}{i}$ for any $j > i$.

Now fix a vertex i . Its subtree size is

$$S(i) = 1 + \sum_{j=i+1}^{10} \mathbf{1}_{A_{i,j}}.$$

Taking expectation,

$$\mathbb{E}[S(i)] = 1 + \sum_{j=i+1}^{10} \Pr(A_{i,j}) = 1 + \sum_{j=i+1}^{10} \frac{1}{i} = 1 + \frac{10-i}{i} = \frac{10}{i}.$$

Since v_1 is chosen uniformly from the 10 vertices,

$$\mathbb{E}[S(v_1)] = \frac{1}{10} \sum_{i=1}^{10} \mathbb{E}[S(i)] = \frac{1}{10} \sum_{i=1}^{10} \frac{10}{i} = \sum_{i=1}^{10} \frac{1}{i}.$$

So the expected value is

$$H_{10} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{10} = \frac{7381}{2520} \approx 2.92897.$$

Answer: 2.92897

828. Subset Subset

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let $n = 8$. We will use linearity of expectation.

For each non-empty subset $S \subseteq 1, 2, \dots, 8$, define an indicator variable I_S , where

$$I_S = \begin{cases} 1 & \text{if } S \text{ is ever chosen,} \\ 0 & \text{otherwise.} \end{cases}$$

Then the total number of days is

$$T = \sum_{S \neq \emptyset} I_S,$$

so

$$\mathbb{E}[T] = \sum_{S \neq \emptyset} \mathbb{E}[I_S] = \sum_{S \neq \emptyset} \Pr(S \text{ is ever chosen}).$$

Now fix a particular subset S . The only way S can be removed before it is chosen is if some superset of S gets chosen first, because choosing a set removes all of its subsets. So S is ever chosen exactly when it is the first chosen set among all supersets of S .

How many supersets does S have? If $|S| = k$, then each of the $8 - k$ elements outside S may be either included or not included, so the number of supersets is 2^{8-k} .

Since the process always picks uniformly among the sets still remaining, every one of these supersets is equally likely to be the first one chosen among them. Therefore,

$$\Pr(S \text{ is ever chosen}) = \frac{1}{2^{8-k}}.$$

Now sum over all non-empty subsets of size k . There are $\binom{8}{k}$ such subsets, so

$$\mathbb{E}[T] = \sum_{k=1}^8 \binom{8}{k} \frac{1}{2^{8-k}} = \frac{1}{2^8} \sum_{k=1}^8 \binom{8}{k} 2^k.$$

Using the binomial theorem,

$$\sum_{k=0}^8 \binom{8}{k} 2^k = (1+2)^8 = 3^8.$$

So

$$\sum_{k=1}^8 \binom{8}{k} 2^k = 3^8 - 1,$$

where the 1 is the $k = 0$ term. Hence

$$\mathbb{E}[T] = \frac{3^8 - 1}{2^8} = \frac{6561 - 1}{256} = \frac{6560}{256} = \frac{205}{8}.$$

So the expected number of days is $\boxed{\frac{205}{8}}$.

Answer: $\boxed{25.625}$

829. Socks Remaining III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let R, B , and G be the positions of the last red sock, last blue sock, and last green sock in a random ordering of all 60 socks.

The process stops when the first color runs out, so the stopping time is $\min(R, B, G)$. Therefore, the number of socks remaining is $60 - \min(R, B, G)$. So we only need $\mathbb{E}[\min(R, B, G)]$.

Now use the identity for three numbers:

$$\min(R, B, G) = R + B + G - \max(R, B) - \max(R, G) - \max(B, G) + \max(R, B, G).$$

Taking expectations gives

$$\mathbb{E}[\min(R, B, G)] = \mathbb{E}[R] + \mathbb{E}[B] + \mathbb{E}[G] - \mathbb{E}[\max(R, B)] - \mathbb{E}[\max(R, G)] - \mathbb{E}[\max(B, G)] + \mathbb{E}[\max(R, B, G)].$$

Each of the terms above is easy to compute. If a color has k socks among 60, then the expected position of its last sock is the expected maximum of a random k -subset of $1, 2, \dots, 60$, which is $\frac{k(61)}{k+1}$.

So

$$\mathbb{E}[R] = \frac{10 \cdot 61}{11}, \quad \mathbb{E}[B] = \frac{20 \cdot 61}{21}, \quad \mathbb{E}[G] = \frac{30 \cdot 61}{31}.$$

Also, $\max(R, B)$ is the last position among the $10 + 20 = 30$ red-or-blue socks, so

$$\mathbb{E}[\max(R, B)] = \frac{30 \cdot 61}{31}.$$

Similarly,

$$\mathbb{E}[\max(R, G)] = \frac{40 \cdot 61}{41}, \quad \mathbb{E}[\max(B, G)] = \frac{50 \cdot 61}{51},$$

and clearly $\mathbb{E}[\max(R, B, G)] = 60$.

Substituting the above values, the expected number of socks remaining is $60 - \mathbb{E}[\min(R, B, G)] = 5.76633$.

Answer: 5.76633

830. Expected Head-Tail Ratio

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let τ be the stopping time. Since the first moment when tails are strictly more than heads must occur after an odd number of tosses, write $\tau = 2k + 1$ for some $k \geq 0$.

To have $\tau = 2k + 1$, the first $2k$ tosses must have exactly k heads and k tails, and in every earlier initial segment, the number of heads must be at least the number of tails. The reflection principle counts these sequences by subtracting from all balanced length- $2k$ sequences those that violate the condition. The total number of balanced sequences is $\binom{2k}{k}$. A sequence that ever has more tails than heads can be reflected up to its first such violation, and this gives a bijection with sequences of $2k$ tosses having $k + 1$ tails and $k - 1$ heads. There are $\binom{2k}{k-1}$ of those. Hence the number of valid sequences is

$$\binom{2k}{k} - \binom{2k}{k-1} = \frac{1}{k+1} \binom{2k}{k}.$$

The last toss must then be a tail, so

$$\Pr(\tau = 2k + 1) = \frac{1}{2^{2k+1}} \cdot \frac{1}{k+1} \binom{2k}{k}.$$

If $\tau = 2k + 1$, then the stopping configuration has $k + 1$ tails and k heads, so the proportion of tails is $\frac{k+1}{2k+1}$. Therefore the expected proportion of tails is

$$\sum_{k=0}^{\infty} \frac{k+1}{2k+1} \Pr(\tau = 2k + 1) = \sum_{k=0}^{\infty} \frac{k+1}{2k+1} \cdot \frac{1}{2^{2k+1}} \cdot \frac{1}{k+1} \binom{2k}{k}.$$

The $k + 1$ cancels, giving

$$\sum_{k=0}^{\infty} \frac{1}{2^{2k+1}} \frac{1}{2k+1} \binom{2k}{k} = \frac{1}{2} \sum_{k=0}^{\infty} \frac{1}{4^k} \frac{1}{2k+1} \binom{2k}{k}.$$

Now use the Taylor expansion

$$\arcsin x = \sum_{k=0}^{\infty} \frac{1}{2k+1} \binom{2k}{k} \frac{x^{2k+1}}{4^k}.$$

Setting $x = 1$ gives

$$\sum_{k=0}^{\infty} \frac{1}{2k+1} \binom{2k}{k} \frac{1}{4^k} = \arcsin(1) = \frac{\pi}{2}.$$

So the desired expectation is $\frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$.

Answer: 0.78540

831. Too Many Slips

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let T be the stopping time, and measure the running total in cents. Then the four possible increments are 1, 5, 10, and 25, each chosen with probability $\frac{1}{4}$. Since we stop exactly when the total becomes an integer, what really matters is the fractional part of the sum, or equivalently the sum modulo 100.

So we look at a Markov chain on the states $0, 1, \dots, 99$, where state r means the current sum is congruent to $r \pmod{100}$. From any state, the chain moves by adding 1, 5, 10, or 25 modulo 100, each with probability $\frac{1}{4}$. This chain is finite and irreducible, so it has a unique stationary distribution. In this case, every state has exactly four incoming transitions and four outgoing transitions with equal probabilities, so using the theorem discussed in the chapter on return times in markov chains section, the stationary distribution is uniform:

$$\pi_r = \frac{1}{100} \quad \text{for all } r.$$

Kac's lemma (discussed in the chapter on return times in markov chains section) states that for a finite irreducible Markov chain, the expected return time to state i is

$$t_i^* = \frac{1}{\pi_i}.$$

Applying this to state 0, the expected time to return to an integer sum is

$$\mathbb{E}[T] = \frac{1}{\pi_0} = 100.$$

Now let $X_1, X_2, \dots$ be the chosen slip values in dollars. Each has mean

$$\mathbb{E}[X_1] = \frac{0.01 + 0.05 + 0.10 + 0.25}{4} = 0.1025.$$

By Wald's equality,

$$\mathbb{E}\left[\sum_{k=1}^T X_k\right] = \mathbb{E}[T] \cdot \mathbb{E}[X_1] = 100 \cdot 0.1025 = 10.25.$$

Answer: 10.25

832. Gold Rush

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

By symmetry, assume $0 \leq x \leq \frac{1}{2}$. Let player 1 place the first mark at x , and let player 2 place the second mark at $y > x$.

Now player 3 has three relevant choices:

If player 3 places a mark just to the left of x , then player 3 gets x .

If player 3 places a mark between x and y , then player 3 gets $\frac{y-x}{2}$.

If player 3 places a mark just to the right of y , then player 3 gets $1 - y$.

So player 3 chooses the largest of

$$x, \quad \frac{y-x}{2}, \quad 1-y.$$

First consider the case where player 2 wants player 3 to choose just to the left of x . For that, we need

$$x \geq \frac{y-x}{2} \quad \text{and} \quad x \geq 1-y.$$

These are equivalent to

$$y \leq 3x \quad \text{and} \quad y \geq 1-x.$$

So this strategy is possible exactly when $1-x < 3x$, that is, $x > \frac{1}{4}$.

In that case, player 2 chooses y just to the right of $1-x$, because that gives player 2 the largest share while

still keeping player 3 on the left option. Then player 1 receives the middle interval, whose length is

$$\frac{y-x}{2} \approx \frac{(1-x)-x}{2} = \frac{1}{2} - x.$$

Now consider the case $x < \frac{1}{4}$. Then the interval $(1-x, 3x)$ is empty, so player 2 cannot force player 3 to choose the left option. Instead, player 2 tries to force player 3 to choose a point between x and y . For that, we need

$$\frac{y-x}{2} \geq x \quad \text{and} \quad \frac{y-x}{2} \geq 1-y.$$

These become

$$y \geq 3x \quad \text{and} \quad y \geq \frac{2+x}{3}.$$

Since $x < \frac{1}{4}$, the stronger condition is

$$y > \frac{2+x}{3}.$$

So player 2 chooses y just to the right of $\frac{2+x}{3}$. Then player 3 is indifferent among all points in (x, y) , so it chooses uniformly there.

If player 3 chooses $z \in (x, y)$, then player 1 gets $\frac{x+z}{2}$. Hence

$$\mathbb{E}[\text{P1 share}] = \frac{x + \mathbb{E}[z]}{2} = \frac{x + \frac{x+y}{2}}{2} = \frac{3x+y}{4}.$$

With $y = \frac{2+x}{3}$, this becomes $\frac{5x+1}{6}$.

If $x > \frac{1}{4}$, then player 2 will force the game into the first case, and player 1 gets $\frac{1}{2} - x$. If $x < \frac{1}{4}$, then player 2 will force the second case, and player 1 gets an expected share of $\frac{5x+1}{6}$. So, the best choice for player 1 is to place the mark as close as possible to $\frac{1}{4}$ from the left, which would give player 1 a guaranteed share of $\frac{1}{4}$ and an expected share of $\frac{3}{8}$.

Therefore the first player should place the mark his point infinitesimally to the left $\frac{1}{4}$ and, by symmetry, he can also mark his point infinitesimally to the right of $\frac{3}{4}$.

Answer: 0.25

833. Expected Number After Swaps

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let E_p be the expected digit in position p after the swaps, where position 1 is the leftmost digit and position 9 is the rightmost digit.

For a fixed position, one swap does the following: with probability $\frac{7}{9}$, that position is not chosen, so its digit stays the same; with probability $\frac{2}{9}$, that position is chosen, and then it is swapped with one of the other 8 positions. So if the current expected digit in that position is x , after one swap the new expected digit is

$$\frac{7}{9}x + \frac{1}{36} \cdot (\text{sum of the other 8 digits}).$$

Since the digits are $1, 2, \dots, 9$, their total sum is 45, so the sum of the other 8 digits is $45 - x$. Therefore the update rule is

$$x \mapsto \frac{7}{9}x + \frac{45-x}{36} = \frac{3}{4}x + \frac{5}{4}.$$

Initially, position p contains the digit p . After one swap,

$$E_p^{(1)} = \frac{3}{4}p + \frac{5}{4}.$$

After the second swap,

$$E_p^{(2)} = \frac{3}{4} \left(\frac{3}{4}p + \frac{5}{4} \right) + \frac{5}{4} = \frac{9p+35}{16}.$$

Now the expected final number is the sum of each position's expected digit times its place value:

$$\sum_{p=1}^9 E_p^{(2)} \cdot 10^{9-p} = \sum_{p=1}^9 \frac{9p+35}{16} \cdot 10^{9-p}.$$

Split this into two parts:

$$\frac{9}{16} \sum_{p=1}^9 p \cdot 10^{9-p} + \frac{35}{16} \sum_{p=1}^9 10^{9-p}.$$

The first sum is the original number 123456789, and the second sum is 111111111.

So the expected value is

$$\frac{9}{16} \cdot 123456789 + \frac{35}{16} \cdot 111111111 = \frac{2499999993}{8}.$$

So the expected final number is $\frac{2499999993}{8} = 312499999.125$.

Answer: 312499999.125

834. Direction Change III

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let X be the number of direction changes in Alice's path.

A shortest path from the bottom-left corner to the top-right corner has exactly 5 right moves and 5 up moves, so every path is a string of 10 moves containing five R 's and five U 's. A direction change happens exactly when two consecutive moves are different, so counting direction changes is the same as counting how many times the path switches between blocks of R 's and blocks of U 's.

If a path has exactly k direction changes, then it has $k+1$ blocks. We now count the paths for each $k \geq 6$.

For $k=6$, there are 7 blocks. One direction must appear in 4 blocks and the other in 3 blocks. If the path starts with R , then the R -moves are split into 4 positive parts and the U -moves into 3 positive parts. The number of ways to split 5 into 4 positive parts is $\binom{4}{3} = 4$, and the number of ways to split 5 into 3 positive parts is $\binom{4}{2} = 6$. So there are $4 \cdot 6 = 24$ such paths starting with R , and the same number starting with U . Thus $N_6 = 48$.

For $k=7$, there are 8 blocks, so both directions appear in 4 blocks. The number of ways to split 5 into 4 positive parts is $\binom{4}{3} = 4$, so

$$N_7 = 2 \cdot 4 \cdot 4 = 32.$$

For $k=8$, there are 9 blocks, so one direction appears in 5 blocks and the other in 4 blocks. Splitting 5 into 5 positive parts can be done in only one way, and splitting 5 into 4 positive parts can be done in 4 ways. Hence

$$N_8 = 2 \cdot 1 \cdot 4 = 8.$$

For $k=9$, the path must alternate at every step, so there are only two possibilities:

$$RURURURURUR, \quad URURURURUR.$$

Thus $N_9 = 2$.

Now condition on the event $X \geq 6$. Writing the expectation in probability form,

$$\mathbb{E}[X \mid X \geq 6] = \sum_{k=6}^9 k \Pr(X = k \mid X \geq 6) = \sum_{k=6}^9 k \frac{\Pr(X = k)}{\Pr(X \geq 6)}.$$

Using the counts above,

$$\Pr(X = k \mid X \geq 6) = \frac{N_k}{N_6 + N_7 + N_8 + N_9},$$

$$\implies \mathbb{E}[X \mid X \geq 6] = \sum_{k=6}^9 k \frac{N_k}{N_6 + N_7 + N_8 + N_9} = \frac{6(48) + 7(32) + 8(8) + 9(2)}{48 + 32 + 8 + 2} = \frac{33}{5}.$$

So the expected number of direction changes is $\boxed{\frac{33}{5}}$.

Answer: $\boxed{6.6}$

835. Submarine Distance

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let the first submarine choose angle $\alpha \in [0, \pi/2]$, and let the second choose angle $\pi - \beta$ where $\beta \in [0, \pi/2]$. Then their positions after t seconds are

$$A(t) = (t \cos \alpha, t \sin \alpha)$$

and

$$B(t) = (10 - t \cos \beta, t \sin \beta).$$

So the squared distance between them is

$$d^2(t) = (10 - t(\cos \alpha + \cos \beta))^2 + t^2(\sin \beta - \sin \alpha)^2.$$

Write $d^2(t)$ as a quadratic in t :

$$d^2(t) = At^2 + Bt + 100,$$

where

$$A = (\cos \alpha + \cos \beta)^2 + (\sin \beta - \sin \alpha)^2, \quad B = -20(\cos \alpha + \cos \beta).$$

Since an upward opening quadratic (i.e. a quadratic where $A > 0$) attains its minimum at its vertex ($t = -\frac{B}{2A}$), the minimum value is $100 - \frac{B^2}{4A}$. Substituting A and B gives

$$d_{\min} = \frac{10|\sin \beta - \sin \alpha|}{\sqrt{(\cos \alpha + \cos \beta)^2 + (\sin \beta - \sin \alpha)^2}}.$$

Now use the identities

$$\sin \beta - \sin \alpha = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\beta - \alpha}{2} \right)$$

and

$$(\cos \alpha + \cos \beta)^2 + (\sin \beta - \sin \alpha)^2 = 4 \cos^2 \left(\frac{\alpha + \beta}{2} \right).$$

Substituting these into the expression for $d_{\min}$ gives

$$d_{\min} = 10 \left| \sin \left(\frac{\beta - \alpha}{2} \right) \right|.$$

So we want

$$10 \left| \sin \left(\frac{\beta - \alpha}{2} \right) \right| < 4.$$

Since $\beta - \alpha \in [-\pi/2, \pi/2]$, this is equivalent to

$$|\beta - \alpha| < 2 \arcsin \frac{2}{5}.$$

Now (α, β) is uniformly distributed over the square $[0, \pi/2] \times [0, \pi/2]$. The condition $|\beta - \alpha| < c$ means the point lies in the diagonal band of width c inside the square. The only excluded parts are two corner triangles, each with side length $\frac{\pi}{2} - c$, so the probability is

$$1 - \frac{\left(\frac{\pi}{2} - c\right)^2}{\left(\frac{\pi}{2}\right)^2}.$$

Here $c = 2 \arcsin \frac{2}{5}$. Therefore,

$$\Pr(d_{\min} < 4) = 1 - \left(1 - \frac{4}{\pi} \arcsin \frac{2}{5}\right)^2.$$

So the answer is

$$1 - \left(1 - \frac{4}{\pi} \arcsin \frac{2}{5}\right)^2 \approx 0.77339.$$

Answer: 0.77339

836. Nearest Paint I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the people be arranged from left to right, and let the 11 gaps be $G_0, G_1, \dots, G_{10}$, where G_0 and G_{10} are the two end gaps, and $G_1, \dots, G_9$ are the gaps between consecutive people.

The two end gaps are never painted, because no one paints beyond the nearest neighbor. Also, G_1 and G_9 are always painted. So the only gaps that can sometimes remain unpainted are the seven middle gaps $G_2, \dots, G_8$.

Now fix one middle gap, say G_i with $2 \leq i \leq 8$. Its left endpoint paints across it exactly when $G_i < G_{i-1}$, and its right endpoint paints across it exactly when $G_i < G_{i+1}$. So G_i is unpainted exactly when

$$G_i > G_{i-1} \quad \text{and} \quad G_i > G_{i+1}.$$

Basically, a middle gap is unpainted precisely when it is larger than the two gaps next to it.

So we only need the expected contribution of one middle gap. Consider three consecutive gaps with total length T . After dividing by T , we get three nonnegative numbers x, y, z with $x + y + z = 1$.

Because the original points are random, the point (x, y, z) is uniformly spread over the triangle $x + y + z = 1$. The middle gap is unpainted exactly when $y > x$ and $y > z$. So the expected unpainted part of the middle gap is T times the average value of y over the part of the triangle where y is the largest coordinate.

To compute this average value of y , write $z = 1 - x - y$. The triangle $x + y + z = 1$ has area $\frac{1}{2}$, so the uniform density is 2. The region where y is the largest is given by

$$y > x \quad \text{and} \quad y > 1 - x - y.$$

This becomes

$$x < y \quad \text{and} \quad x > 1 - 2y.$$

Hence the average value of y is

$$2 \left(\int_{1/3}^{1/2} \int_{1-2y}^y y dx dy + \int_{1/2}^1 \int_0^{1-y} y dx dy \right) = \frac{11}{54}.$$

So for a triple of consecutive gaps with total length T , the expected unpainted part of the middle gap is $T \cdot \frac{11}{54}$.

Now the expected length of each of the 11 gaps is $\frac{100}{11}$. So for any middle gap, the expected total of the three consecutive gaps around it is $3 \cdot \frac{100}{11} = \frac{300}{11}$. Therefore the expected unpainted length contributed by each of the seven middle gaps is $\frac{11}{54} \cdot \frac{300}{11} = \frac{50}{9}$. The two end gaps contribute $2 \cdot \frac{100}{11}$.

Thus the expected unpainted length is

$$2 \cdot \frac{100}{11} + 7 \cdot \frac{50}{9} = \frac{5650}{99}.$$

So the expected unpainted length is $\boxed{\frac{5650}{99}}$ meters.

Answer: 57.07071

837. Nearest Paint II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let the 10 people divide the circle into 10 consecutive gaps, and call them $G_1, G_2, \dots, G_{10}$. Because the circle has no endpoints, every gap has two neighboring gaps.

A gap G_i is painted from its left endpoint exactly when it is shorter than the gap just before it, and it is painted from its right endpoint exactly when it is shorter than the gap just after it. So G_i is unpainted exactly when it is longer than both neighboring gaps:

$$G_i > G_{i-1} \quad \text{and} \quad G_i > G_{i+1},$$

with indices taken cyclically mod 10.

So we only need the expected contribution of one gap. Since all gaps are symmetric, each gap has the same expected unpainted length.

Now consider any three consecutive gaps with total length T . After dividing by T , we get three nonnegative numbers x, y, z with $x + y + z = 1$.

The middle gap is unpainted exactly when $y > x$ and $y > z$. So the expected unpainted part of the middle gap is T times the average value of y over the part of the triangle where y is the largest coordinate.

To compute this average value of y , write $z = 1 - x - y$. The triangle $x + y + z = 1$ has area $\frac{1}{2}$, so the uniform density is 2. The region where y is the largest is given by

$$y > x \quad \text{and} \quad y > 1 - x - y.$$

This becomes

$$x < y \quad \text{and} \quad x > 1 - 2y.$$

Hence the average value of y over the part of the triangle where y is the largest coordinate is

$$2 \left(\int_{1/3}^{1/2} \int_{1-2y}^y y dx dy + \int_{1/2}^1 \int_0^{1-y} y dx dy \right) = \frac{11}{54}.$$

So for a triple of consecutive gaps with total length T , the expected unpainted part of the middle gap is $T \cdot \frac{11}{54}$.

Now the expected length of each of the 10 gaps is $\frac{100}{10} = 10$. So for any gap, the expected total of the three consecutive gaps around it is $3 \cdot 10 = 30$. Therefore the expected unpainted length contributed by each gap is $30 \cdot \frac{11}{54} = \frac{55}{9}$. Since there are 10 gaps, the expected total unpainted length is $10 \cdot \frac{55}{9} = \frac{550}{9}$.

So the expected unpainted length is $\boxed{\frac{550}{9}}$ meters.

Answer: 61.11111

838. Normal Inequality V

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Define

$$U_1 = N_1^5, \quad U_2 = N_2^5, \quad U_3 = N_3^5.$$

Then U_1, U_2 , and U_3 are still independent and identically distributed, and each has a distribution symmetric about 0, because N_i is symmetric about 0 and x^5 preserves the sign.

Now consider

$$S = U_1 + U_2 - U_3.$$

The event we want is exactly $S > 0$. Since U_1, U_2 , and U_3 are symmetric about 0, the random variable

$$-S = (-U_1) + (-U_2) + U_3$$

has the same distribution as S . So S is symmetric about 0. Also, S has a continuous distribution, so

$$\Pr(S = 0) = 0.$$

Therefore,

$$\Pr(S > 0) = \Pr(S < 0) = \frac{1}{2}.$$

So the required probability is $\boxed{\frac{1}{2}}$.

Answer: $\boxed{0.5}$

839. Expected Hits I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let N be the total number of visits to +4 before absorption.

First, we compute the probability that the walk ever reaches +4 starting from 0. Since +4 lies between 0 and the lower absorbing boundary -10, the walk can fail to reach +4 only by hitting -10 first. Using the standard absorbing-probability formula for a symmetric random walk,

$$P_0(T_4 < T_{-10}) = \frac{0 - (-10)}{4 - (-10)} = \frac{10}{14} = \frac{5}{7}.$$

Now suppose the walk has reached +4. Each visit to +4 is followed by one step to either +5 or +3, each with probability $\frac{1}{2}$. Let r be the probability that, after leaving +4, the walk ever comes back to +4 before being absorbed.

From +5, the only way not to return to +4 is to hit +20 first, so

$$P_5(T_4 < T_{20}) = \frac{20 - 5}{20 - 4} = \frac{15}{16}.$$

From +3, the only way not to return to +4 is to hit -10 first, so

$$P_3(T_4 < T_{-10}) = \frac{3 - (-10)}{4 - (-10)} = \frac{13}{14}.$$

Therefore

$$r = \frac{1}{2} \left(\frac{15}{16} + \frac{13}{14} \right) = \frac{209}{224}.$$

So, once the walk first reaches +4, the expected number of visits to +4 (including the first visit) is

$$1 + r + r^2 + \cdots = \frac{1}{1-r} = \frac{224}{15}.$$

Multiplying by the probability of ever reaching +4,

$$\mathbb{E}[N] = \frac{5}{7} \cdot \frac{224}{15} = \frac{32}{3}.$$

So the expected number of visits to +4 is $\boxed{\frac{32}{3}}$.

Answer: $\boxed{10.66667}$

840. Expected Hits II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Let N be the total number of visits to +4 before absorption.

We first find the probability that the walk ever reaches +4. On the interval $[-10, 4]$, the standard absorbing probability formula for a biased simple random walk says that if

$$p = \frac{1}{3}, \quad q = \frac{2}{3}, \quad r = \frac{q}{p} = 2,$$

then for $a < i < b$,

$$P_i(T_b < T_a) = \frac{1 - r^{i-a}}{1 - r^{b-a}}.$$

Here $a = -10$, $b = 4$, and $i = 0$, so

$$P_0(T_4 < T_{-10}) = \frac{1 - 2^{10}}{1 - 2^{14}} = \frac{2^{10} - 1}{2^{14} - 1} = \frac{1023}{16383}.$$

Now suppose the walk has reached +4. Each visit to +4 is followed by one step to +5 with probability $\frac{1}{3}$ or to +3 with probability $\frac{2}{3}$. We compute the probability of returning to +4 before absorption after that departure.

From +5, the walk returns to +4 before reaching +20 with probability

$$P_5(T_4 < T_{20}) = 1 - \frac{1 - 2^1}{1 - 2^{16}} = \frac{65534}{65535}.$$

From +3, the walk returns to +4 before reaching -10 with probability

$$P_3(T_4 < T_{-10}) = \frac{1 - 2^{13}}{1 - 2^{14}} = \frac{8191}{16383}.$$

So the probability of another visit to +4 after one visit is

$$r = \frac{1}{3} \cdot \frac{65534}{65535} + \frac{2}{3} \cdot \frac{8191}{16383}.$$

Thus, once the walk hits +4, the expected number of visits to +4 is

$$1 + r + r^2 + \cdots = \frac{1}{1-r}.$$

Multiplying by the probability of ever reaching +4,

$$\mathbb{E}[N] = \frac{1023}{16383} \cdot \frac{1}{1-r},$$

with

$$r = \frac{1}{3} \cdot \frac{65534}{65535} + \frac{2}{3} \cdot \frac{8191}{16383}.$$

This simplifies to

$$\mathbb{E}[N] = \frac{65535}{349867} \approx 0.18731.$$

Answer: 0.18731

841. Jumping Frog X

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

The key observation is that before the walk first hits 15, the only way to have visited every other point is to have visited both neighbors of 15, namely 14 and 16. Once both 14 and 16 have been visited, all points in between them on the circle have also been visited, so this is exactly the event we want.

Now cut the circle at 15 and unwrap the remaining 99 points into a line:

$$16, 17, \dots, 100, 1, 2, \dots, 14.$$

Call these positions $0, 1, \dots, 98$, so that 16 is 0, 14 is 98, and 1 is 85. On this line, the walk moves one step to the right with probability $\frac{1}{3}$ and one step to the left with probability $\frac{2}{3}$. The walk is absorbed as soon as it tries to step past 0 or past 98, which corresponds exactly to hitting 15.

So we need the probability that starting from 85, the walk visits both endpoints 0 and 98 before absorption.

Split according to which endpoint is hit first. By the strong Markov property,

$$P(\text{success}) = P_{85}(T_0 < T_{98})P_0(T_{98} < T_{-1}) + P_{85}(T_{98} < T_0)P_{98}(T_0 < T_{99}).$$

Now use the absorbing probability formula for biased random walk: with right-step probability $p = \frac{1}{3}$ and left-step probability $q = \frac{2}{3}$. Writing $\frac{q}{p} = 2$, for $0 \leq i \leq N$,

$$P_i(T_N < T_0) = \frac{1 - 2^i}{1 - 2^N}.$$

First, with $N = 98$ and $i = 85$,

$$\begin{aligned} P_{85}(T_{98} < T_0) &= \frac{1 - 2^{85}}{1 - 2^{98}} = \frac{2^{85} - 1}{2^{98} - 1}, \\ \implies P_{85}(T_0 < T_{98}) &= \frac{2^{98} - 2^{85}}{2^{98} - 1}. \end{aligned}$$

Next, starting from 0, to reach 98 before falling off the left end is the same as starting one step above the left boundary in an interval of length 99, so

$$P_0(T_{98} < T_{-1}) = \frac{1}{2^{99} - 1}.$$

Finally, starting from 98, the probability of hitting 0 before 99 is

$$P_{98}(T_0 < T_{99}) = \frac{2^{98}}{2^{99} - 1}.$$

Substituting,

$$P(\text{success}) = \frac{2^{98} - 2^{85}}{2^{98} - 1} \cdot \frac{1}{2^{99} - 1} + \frac{2^{85} - 1}{2^{98} - 1} \cdot \frac{2^{98}}{2^{99} - 1}.$$

$$\implies P(\text{success}) = \frac{2^{85}(2^{98} - 1)}{(2^{98} - 1)(2^{99} - 1)} = \frac{2^{85}}{2^{99} - 1}.$$

Thus the required probability is $\boxed{\frac{2^{85}}{2^{99} - 1}}$.

Answer: $\boxed{0.00006}$

842. Elevator II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

A full round trip for one elevator takes 98 time units. Since Bob leaves at a uniformly random time, the elevator's position in its cycle is uniformly random as well.

For one elevator, the 20th floor is visited twice in each cycle: once on the way up and once on the way down. These two visits split the cycle into two arcs of lengths 38 and 60. So the next time this elevator reaches the 20th floor is downward with probability $\frac{60}{98} = \frac{30}{49}$.

Conditioned on that event, the waiting time until that downward arrival is uniform on $[0, 60]$. Likewise, if the next arrival is upward, the waiting time is uniform on $[0, 38]$.

So for one elevator, the density of a downward-arrival time is

$$f_d(t) = \frac{1}{98}, \quad 0 \leq t \leq 60.$$

Its survival function is

$$S(t) = P(T > t) = \begin{cases} 1 - \frac{t}{49}, & 0 \leq t \leq 38, \\ \frac{60-t}{98}, & 38 \leq t \leq 60. \end{cases}$$

With 4 independent elevators, the first elevator to arrive is downward if one elevator has downward-arrival time t , while the other three have next-arrival times larger than t . So

$$P(\text{first arrival is down}) = 4 \int_0^{60} f_d(t) S(t)^3 dt.$$

Splitting at $t = 38$,

$$P = \frac{4}{98} \left(\int_0^{38} \left(1 - \frac{t}{49}\right)^3 dt + \int_{38}^{60} \left(\frac{60-t}{98}\right)^3 dt \right).$$

This simplifies to $P = \frac{2889721}{5764801} \approx 0.50127$.

Answer: $\boxed{0.50127}$

843. Unknown Gunmen I

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

At any position x , A's hit probability is simply x while B's is x^2 . Because the hit probabilities are strictly increasing in x , both players prefer to delay their shot as long as possible, but each must also guard against the other shooting first.

Consider the balancing point where $x = 1 - x^2$, let γ be the root of this equation in $(0, 1)$, so $\gamma = \frac{\sqrt{5}-1}{2}$. We prove below that the optimal strategy is for A to shoot exactly at γ and for B to wait completely. This leads to A winning with probability γ . For B, by waiting he wins with probability $1 - \gamma$ (if A misses, which happens with probability $1 - \gamma$, then B wins for sure by waiting until the target and shooting with probability 1).

Neither player benefits from shooting strictly before γ . For A any such shot yields win probability strictly less than γ . For B the situation requires an explicit comparison with his equilibrium payoff. In the proposed equilibrium B achieves win probability exactly γ^2 by staying silent and winning whenever A misses. If instead B deviates and shoots at some earlier position $s < \gamma$ (while A continues to follow the equilibrium strategy of waiting until γ), then B's win probability is exactly s^2 : he wins immediately with probability s^2 if he hits, but loses with certainty if he misses (because A will then shoot at γ and win with probability γ). Since the function s^2 is strictly increasing on $[0, \gamma]$, it follows that $s^2 < \gamma^2$ for every $s < \gamma$. Therefore every strict early deviation by B yields a win probability strictly smaller than the γ^2 he obtains in equilibrium, so such deviations are unprofitable.

It remains to confirm that B has no incentive to shoot at γ itself. If B stays silent he wins with probability γ^2 . If he shoots simultaneously with A, a rough calculation shows his expected win probability falls below this value. At $\gamma \approx 0.62$, A's accuracy is about 0.62 and B's about 0.38. The probability that B wins immediately in a simultaneous shot is only the chance that B hits while A misses, roughly $0.38 \times 0.38 = 0.144$. With probability roughly 0.47 the shots produce a reset (both hit or both miss), in which case B's continuation win probability is again about 0.38. B's overall win probability is then roughly $0.144 + 0.47 \times 0.38 \approx 0.32$, which is less than the 0.38 he obtains by staying silent. Hence B prefers not to shoot simultaneously.

Finally, A has no incentive to wait past γ . For every position $x > \gamma$ the inequality $x^2 > 1 - x$ holds, so B would immediately prefer to shoot as it gives him a higher winning probability than letting A shoot first. A would then receive only win probability $1 - x^2 < \gamma$, which is strictly worse than the γ secured by shooting at γ . Consequently neither player possesses a profitable deviation from the stated strategies. Under optimal play A therefore wins with probability exactly $\gamma = \frac{\sqrt{5}-1}{2}$.

Answer: 0.61803

844. Random And Rational

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Suppose A chooses the number a . Since B can read A's mind, B knows a before deciding what number b to choose.

First look at B's options. If B chooses $b < a$, then B never has the largest number unless C is smaller than both, but in that case A is largest, not B. So B simply receives b no matter what C does. Thus B can guarantee almost a by choosing just below a .

Now suppose B chooses $b > a$. Then B is largest whenever $C < b$, and in that case B must pay A and C. If $C > b$, then C is largest and pays B the amount b . So B's expected payoff is

$$\frac{1}{100} \left(\int_0^b -(a+x)dx + \int_b^{100} bdx \right).$$

This simplifies to

$$\frac{100b - ab - \frac{3b^2}{2}}{100}.$$

For a fixed a , this is maximized at

$$b = \frac{100 - a}{3}.$$

So B compares two strategies: choosing just below a , which gives him almost a , or choosing

$$b = \frac{100 - a}{3},$$

which gives him expected payoff

$$\frac{(100 - a)^2}{600}.$$

Therefore B will choose the risky larger number exactly when

$$\frac{(100 - a)^2}{600} > a.$$

Solving this,

$$\begin{aligned} (100 - a)^2 &> 600a, \\ \implies a^2 - 800a + 10000 &> 0. \end{aligned}$$

The relevant root between 0 and 100 is

$$a = 400 - 50\sqrt{60} = 50(8 - \sqrt{60}).$$

Thus, if

$$a < 50(8 - \sqrt{60}),$$

then B chooses a number larger than A 's number. In that case A is never the largest number, so A always receives his own chosen amount a . Hence A 's expected payoff is simply a .

If instead $a \geq 50(8 - \sqrt{60})$, then B chooses just below a . Now A is largest whenever $C < a$, and C is largest whenever $C > a$. Taking B 's choice as essentially a , A 's expected payoff becomes

$$\frac{1}{100} \left(\int_0^a -(a + x)dx + \int_a^{100} a dx \right) = a - \frac{a^2}{40}.$$

The maximum of this expression occurs at $a = 20$, giving payoff 10. But A can do better by choosing a just below $50(8 - \sqrt{60})$, because then B strictly prefers the risky larger number, and A 's expected payoff is almost

$$50(8 - \sqrt{60}) \approx 12.70167.$$

So, the optimal strategy is for A to choose a number just less than $50(8 - \sqrt{60}) \approx 12.70167$. This makes B choose a larger number, ensuring that A is never the largest and therefore receives approximately 12.70167 in expectation.

Answer: 12.70167

845. Line Switch

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Suppose it is worth switching when your current position is N , but not when your position is any smaller than that. Then exactly $100 - N + 1$ people, namely those in positions $N, N + 1, \dots, 100$, will switch.

So the new line will contain $100 - N + 1$ people, and since your spot is chosen at random, your expected position in that line is the average of the numbers 1 through $100 - N + 1$, which is

$$\frac{1 + (100 - N + 1)}{2} = \frac{102 - N}{2}.$$

You should switch exactly when this expected position is better than staying in position N , so we require

$$\frac{102 - N}{2} < N.$$

Solving gives

$$102 - N < 2N \quad \Rightarrow \quad 102 < 3N \quad \Rightarrow \quad N > 34.$$

A rational person should switch if they are in position 35 or later, and should stay put if they are in positions 1 through 33. The person at position 34 can either stay or switch, it doesn't change their expected position. So, the answer is 35.

Answer: 35

846. Unknown Gunmen II

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

First recall the one-bullet version, where A and B each have one bullet. Suppose A plans to shoot at position t . If B waits, A wins with probability t , because if A misses, B can simply wait until position 1 and hit for sure. On the other hand, B could shoot just before t . If B shoots at position $y < t$, then B hits with probability y^2 , so A wins with probability $1 - y^2$. As $y \rightarrow t^-$, this becomes $1 - t^2$. Therefore, in equilibrium, A chooses t so that $t = 1 - t^2$. Thus

$$t^2 + t - 1 = 0,$$

so the one-bullet value for A is

$$r = \frac{\sqrt{5} - 1}{2}.$$

Now return to the actual game, where A has two bullets and B has one bullet. Suppose A plans to fire his first bullet at position a , with $a < r$. If B has not fired before a , then A shoots at a . A hits with probability a , in which case A wins immediately. If A misses, which happens with probability $1 - a$, then A and B each have one bullet left, and the game becomes the one-bullet subgame. Since $a < r$, A can still wait until position r , so the continuation value is r . Therefore A's payoff if B waits is $a + (1 - a)r$.

But B can also shoot just before A's planned first shot. If B shoots at position $y < a$, then B hits with probability y^2 , and A loses. If B misses, then B has no bullets left, while A still has two bullets, so A can eventually wait until position 1 and win for sure. Hence A's payoff is $1 - y^2$. As $y \rightarrow a^-$, B can force A's payoff down to $1 - a^2$.

At the optimal choice of a , these two possible payoffs must balance:

$$a + (1 - a)r = 1 - a^2.$$

Using $r = \frac{\sqrt{5}-1}{2}$, and also $r^2 = 1 - r$, we get

$$a + (1 - a)r = 1 - a^2,$$

Since $1 - r = r^2$, this becomes

$$a^2 + r^2 a - r^2 = 0.$$

Solving the quadratic equation gives

$$a = \frac{-r^2 + \sqrt{r^4 + 4r^2}}{2}.$$

Substituting $r = \frac{\sqrt{5}-1}{2}$, we get $a \approx 0.4558867801$.

Therefore the probability that A wins is $V = 1 - a^2$. So $V \approx 1 - (0.4558867801)^2$, which gives $V \approx 0.79217$. Thus A wins with probability 0.79217.

Answer: 0.79217

847. Match Fixing

[[Question ↑](#) | [Hint ↑](#) | [↑](#)]

Suppose you choose the number x . The other three numbers are chosen from the 49 remaining numbers. Once the four numbers are sorted as $a < b < c < d$, the extreme team has sum $a + d$, while the middle team has sum $b + c$. Your payoff is positive if your own team has the larger sum, and negative if your team has to pay.

If all three other numbers are larger than x , then x is the smallest number. By symmetry of the three larger numbers, the expected payoff in this case is 0. Similarly, if all three other numbers are smaller than x , then x is the largest number, and the expected payoff is again 0. So the only useful cases are when x lies between the other numbers.

First suppose exactly one of the other numbers is below x , and two are above x . Then x is the lower middle

number, so your team is the middle team. If the lower number is u , and the two upper numbers are $v < w$, your payoff is $x + v - u - w$.

The expected value of u is $x/2$, and for two numbers chosen from $x + 1, \dots, 50$, the expected gap $w - v$ is $(51 - x)/3$. Therefore the conditional expected payoff in this case is $\frac{x}{2} - \frac{51-x}{3}$. The probability of this case is $\frac{(x-1)\binom{50-x}{2}}{\binom{49}{3}}$.

Now suppose exactly two of the other numbers are below x , and one is above x . Then x is the upper middle number. If the two lower numbers are $u < v$, and the upper number is w , your payoff is $x + v - u - w$.

Here the expected gap $v - u$ among two chosen numbers below x is $x/3$, and the expected value of the upper number w is $(x + 51)/2$. So the conditional expected payoff is $x - \frac{x+51}{2} + \frac{x}{3} = \frac{5x}{6} - \frac{51}{2}$. The probability of this case is $\frac{\binom{x-1}{2}\binom{50-x}{1}}{\binom{49}{3}}$. Thus the expected payoff is

$$E(x) = \frac{(x-1)\binom{50-x}{2}\left(\frac{x}{2} - \frac{51-x}{3}\right) + \binom{x-1}{2}(50-x)\left(\frac{5x}{6} - \frac{51}{2}\right)}{\binom{49}{3}}.$$

This simplifies to

$$E(x) = -\frac{23(x-50)(x-1)(2x-51)}{55272}.$$

To maximize this, treat x temporarily as a real variable and differentiate:

$$E'(x) = -\frac{23(6x^2 - 306x + 2701)}{55272}.$$

Setting this equal to 0, the two critical points are

$$x = \frac{51}{2} \pm \frac{49\sqrt{3}}{6}.$$

The larger one is the local maximum, and it lies between 39 and 40. Since x must be an integer, we compare the nearby values:

$$E(39) = \frac{43263}{9212}, \quad E(40) = \frac{43355}{9212}.$$

Since $E(40) > E(39)$, the best choice is 40.

Answer: 40

848. Extra Information I

[[Question](#) ↑ | [Hint](#) ↑ | ↑]

Let A use a cutoff strategy: keep if his number is at least t , and redraw if his number is below t . We will determine t by making A indifferent at the cutoff.

First suppose B observes that A redraw. Then A's final number is just a fresh uniform $[0, 1]$ number. If B keeps a value b , B wins with probability b . If B redraws, B's new value is also uniform, so B wins with probability $\frac{1}{2}$. Therefore, after seeing A redraw, B keeps exactly when $b \geq \frac{1}{2}$.

So B's winning probability after A redraws is

$$\int_0^{1/2} \frac{1}{2} db + \int_{1/2}^1 b db = \frac{1}{4} + \frac{3}{8} = \frac{5}{8}.$$

Thus, if A redraws, B wins with probability $\frac{5}{8}$.

Now suppose B observes that A kept. Since A keeps only numbers at least t , B knows that A's number is uniformly distributed on $[t, 1]$. If B keeps a value b , then for $b \in [t, 1]$, his winning probability is $\frac{b-t}{1-t}$.

If B redraws, his new number is uniform, so his winning probability is

$$\mathbb{E}[1 - X] = 1 - \frac{1+t}{2} = \frac{1-t}{2},$$

where $X \sim U[t, 1]$ is A's kept number.

Therefore, after seeing A keep, B keeps if

$$\frac{b-t}{1-t} \geq \frac{1-t}{2}.$$

So B's cutoff after observing that A kept is

$$b \geq t + \frac{(1-t)^2}{2} = \frac{1+t^2}{2}.$$

Call this cutoff k . Thus $k = \frac{1+t^2}{2}$.

Now consider A with actual value a . If A keeps, then B redraws whenever $b < k$, which happens with probability k , and then B beats A with probability $1 - a$. If $b \geq k$, B keeps, and B beats A exactly when $b > a$.

At the marginal point $a = t$, since $t < k$, A's loss probability from keeping is $k(1-t) + (1-k) = 1 - kt$. At the optimal cutoff, A must be indifferent between keeping and redrawing. Redrawing gives B win probability $\frac{5}{8}$, so

$$\begin{aligned} 1 - kt &= \frac{5}{8}. \\ \implies kt &= \frac{3}{8}. \end{aligned}$$

Using $k = \frac{1+t^2}{2}$, we get $t \cdot \frac{1+t^2}{2} = \frac{3}{8}$, so $t^3 + t = \frac{3}{4}$. The unique solution in $[0, 1]$ is $t \approx 0.56736$. Then

$$k = \frac{1+t^2}{2} \approx 0.66095.$$

So the optimal strategies are: A keeps iff his initial value is at least 0.56736. If B sees A redraw, B keeps iff his own value is at least $\frac{1}{2}$. If B sees A keep, B keeps iff his own value is at least 0.66095.

Finally, compute B's total winning probability. With probability t , A redraws, giving B probability $\frac{5}{8}$. If A keeps a value $a \in [t, 1]$, then B's win probability is

$$1 - ka \quad \text{for } t \leq a < k,$$

and

$$(1+k)(1-a) \quad \text{for } k \leq a \leq 1.$$

Therefore

$$P(B \text{ wins}) = \frac{5t}{8} + \int_t^k (1 - ka)da + \int_k^1 (1+k)(1-a)da.$$

Substituting $k = \frac{1+t^2}{2}$ and $t^3 + t = \frac{3}{4}$, this simplifies to

$$P(B \text{ wins}) = \frac{5}{8} + \frac{t^2}{8} - \frac{9t}{32}.$$

With $t \approx 0.56736$,

$$P(B \text{ wins}) \approx 0.50567.$$

So under optimal play, B wins with probability 0.50567.

Answer: 0.50567